FM Exam Notes

Evenson Auguste

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Contents

Chapter 1

Interest Theory

1.1 Interest Rate Definitions

Perspective in Financial Transactions

Every financial transaction can be viewed from two different perspectives: the **lender of the capital** and the **borrower of the capital**.

For example, in a savings account, the depositor is the lender and the bank is the borrower. In a car loan, the bank is the lender and the car buyer is the borrower.

1.1.1 Amount Function

Assume that, given the original principal invested, the accumulated value at any point in time can be determined. Let $A(t)$ represent the accumulated value at time t for an original investment of k.

Properties of $A(t)$

1. $A(0) = k$.
2. $A(t)$ is generally increasing.
3. If interest accrues continuously, then the function will be continuous.

The word **accrue** means to increase in value or amount as time passes.

The amount of interest earned during period n is

$$I_n = A(n) - A(n-1).$$

Amount Function

The **amount function** $A(t)$ represents the amount of money in a fund at time t.

If an initial amount $A(0)$ is invested, then $A(t)$ denotes the accumulated value at time t.

Interest Earned During a Period

Interest Earned During a Period

Let $I(t)$ be the amount of interest earned during the t-th period. Then

$$I(t) = A(t) - A(t-1).$$

This means that the interest earned during period t is the increase in the fund from time $t-1$ to time t.

Accumulation Function

The **accumulation function**, denoted by $a(t)$, is a special case of the amount function where the original investment is one unit.

Thus, $a(t)$ represents the accumulated value at time t of an original investment of 1.

Recall that $A(t)$ is the accumulated value at time t of an original investment of k. Therefore,

$$A(t) = k\,a(t).$$

The second and third properties of $A(t)$ also hold for $a(t)$. In particular, $a(t)$ is generally increasing, and if interest accrues continuously, then $a(t)$ is continuous.

As we will see in later lessons, $A(t)$ and $a(t)$ can often be used interchangeably, depending on whether we are working with a general initial investment k or with one unit invested.

1.1.2 Effective Rate of Interest

Effective Rate of Interest

The **effective rate of interest** during the t-th period, denoted by i_t, is the ratio of interest earned during the period to the amount in the fund at the beginning of the period.

Effective Rate of Interest

$$i_t = \frac{I(t)}{A(t-1)} = \frac{A(t) - A(t-1)}{A(t-1)}.$$

Equivalently, the fund multiplies by

$$1 + i_t$$

over the

$$t$$

-th period.

Effective Rates

The term **effective** is used for rates of interest in which interest is paid once per measurement period.

This is in contrast with **nominal** rates, which will be discussed later.

We can accumulate money from time $n-1$ to time n by multiplying by the one-period accumulation factor.

By definition,

$$i_n = \frac{A(n) - A(n-1)}{A(n-1)}.$$

Therefore,

$$\begin{aligned} i_n &= \frac{A(n)}{A(n-1)} - 1, \\ 1 + i_n &= \frac{A(n)}{A(n-1)}. \end{aligned}$$

Thus, to move money forward from time $n-1$ to time n, we multiply by

$$1 + i_n.$$

Since $a(t)$ is the accumulated value at time t of one unit invested at time 0, we also have

$$\frac{A(n)}{A(n-1)} = \frac{a(n)}{a(n-1)}.$$

Hence,

$$1 + i_n = \frac{a(n)}{a(n-1)}.$$

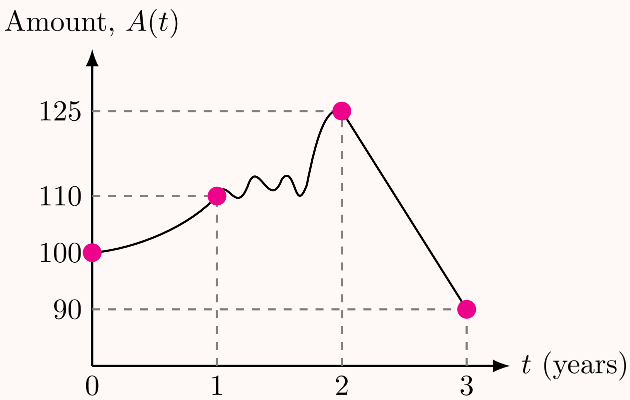
So money can be accumulated from time $n-1$ to time n by multiplying by either

$$1 + i_n \qquad \text{or} \qquad \frac{a(n)}{a(n-1)}.$$

Effective Interest Rates from an Amount Function Graph

Suppose the amount function passes through the following points:

$$A(0) = 100, \qquad A(1) = 110, \qquad A(2) = 125, \qquad A(3) = 90.$$



The effective interest rate for the t-th year is

$$i_t = \frac{A(t) - A(t-1)}{A(t-1)}.$$

Therefore,

$$i_1 = \frac{A(1) - A(0)}{A(0)} = \frac{110 - 100}{100} = 10\%,$$

$$i_2 = \frac{A(2) - A(1)}{A(1)} = \frac{125 - 110}{110} = 13.6364\%,$$

$$i_3 = \frac{A(3) - A(2)}{A(2)} = \frac{90 - 125}{125} = -28\%.$$

An infinite number of amount functions can produce these three effective rates. For example, the function could be smooth during the first year, irregular during the second year, and linear during the third year. Regardless of what happens inside each year, any amount function passing through these four points produces the same three effective interest rates.

If the effective interest rate is constant and equal to i, then

Constant Effective Interest Rate

$$A(t) = A(0)(1+i)^t.$$

Until now, we assumed that the initial investment occurs at time 0. Suppose instead that the initial investment occurs at time k, where $k \leq t$. Then the deposit earns interest from time k to time t.

Amount Function from Time k to Time t

$$A(t) = A(k)(1+i)^{t-k}.$$

This equation remains valid even when $k > t$.

Example 1.1.1

Sam invested \$500 in a bank. After two years, the fund increased to \$561.80.
What was the annual effective interest rate offered by the bank?

Solution
Define t in terms of years. The initial amount is \$500, and the accumulated amount after two years is \$561.80. Thus,

$$A(0) = 500, \qquad A(2) = 561.80.$$

Using the constant effective interest rate formula,

$$A(t) = A(0)(1+i)^t,$$

we get

$$\begin{aligned}
A(2) &= A(0)(1+i)^2, \\
561.80 &= 500(1+i)^2, \\
(1+i)^2 &= \frac{561.80}{500}, \\
(1+i)^2 &= 1.1236, \\
1+i &= \sqrt{1.1236}, \\
i &= \sqrt{1.1236}-1, \\
i &= 0.06.
\end{aligned}$$

Therefore, the annual effective interest rate was

$$\boxed{6\%}.$$

Exam Remark: Certificate of Deposit

This kind of investment is common with a certificate of deposit, usually abbreviated as CD. A certificate of deposit is a type of savings account offered by banks. It is an agreement to deposit funds without making withdrawals for a specified length of time. In return, the bank credits the deposit with interest.

For exam purposes, if a question involves a certificate of deposit, simply treat it as a regular deposit into a savings account.

Example 1.1.2

Beth invests \$300 in a fund earning a 1% monthly effective interest rate.
How much is in her fund at the end of eight months?

Solution
Since we are given the monthly effective interest rate, define t in terms of months.

$$A(8) = A(0)(1+i)^8$$

Thus,

$$\begin{aligned}
A(8) &= 300(1.01)^8 \\
&= 324.8570.
\end{aligned}$$

Therefore, Beth has

$$\boxed{\$324.86}$$

at the end of eight months.

Example 1.1.3

James invests \$100 in a bank. The fund earns an annual effective interest rate of 10%.
Calculate the fund accumulation at the end of six months.

Solution

We are given an annual effective interest rate, but we need to calculate the amount at the end of six months.
Define t in terms of years. Six months is half a year, so

$$t = 0.5.$$

We are given

$$A(0) = 100, \qquad i = 10\%, \qquad t = 0.5.$$

Therefore,

$$\begin{aligned} A(0.5) &= 100(1 + 0.10)^{1/2} \\ &= 104.88. \end{aligned}$$

The fund accumulation is

$$\boxed{\$104.88}$$

at the end of six months.

Example 1.1.4

Toby wants to have \$500 in one year.
How much does he need to invest today if the account credits a 7% annual effective interest rate?

Solution
We are not given the initial investment. Instead, we are given the amount at the end of one year.
Using

$$A(1) = A(0)(1 + i),$$

we solve for $A(0)$:

$$\begin{aligned} A(0) &= \frac{A(1)}{1.07} \\ &= \frac{500}{1.07} \\ &= 467.2897. \end{aligned}$$

Therefore, Toby needs to invest

$$\boxed{\$467.29}$$

today.

Exam Remark: Constant Effective Interest Rate

Most exam problems assume a constant effective interest rate. Thus, if a problem states that the annual effective interest rate is 5%, assume it is constant each year.
Accumulate using the factor

$$(1 + i)^t,$$

where t is the number of years the money is invested.
Observe that this is exponential growth, an example of compound interest.

Exercise 1.1.1

Given

$$A(t) = 10(1.08)^t.$$

Find i_3 and i_7.

Solution

The effective rate of interest for year 3 is

$$i_3 = \frac{A(3) - A(2)}{A(2)}.$$

Using $A(t) = 10(1.08)^t$, we get

$$\begin{aligned} i_3 &= \frac{10(1.08)^3 - 10(1.08)^2}{10(1.08)^2} \\ &= \frac{10(1.08)^2\left[(1.08) - 1\right]}{10(1.08)^2} \\ &= 1.08 - 1 \\ &= 0.08. \end{aligned}$$

Thus,

$$i_3 = 8\%.$$

Similarly, the effective rate of interest for year 7 is

$$i_7 = \frac{A(7) - A(6)}{A(6)}.$$

Therefore,

$$\begin{aligned} i_7 &= \frac{10(1.08)^7 - 10(1.08)^6}{10(1.08)^6} \\ &= \frac{10(1.08)^6\left[(1.08) - 1\right]}{10(1.08)^6} \\ &= 1.08 - 1 \\ &= 0.08. \end{aligned}$$

Thus,

$$i_7 = 8\%.$$

More generally, for year n,

$$i_n = \frac{A(n) - A(n-1)}{A(n-1)}.$$

Using the same amount function,

$$\begin{aligned} i_n &= \frac{10(1.08)^n - 10(1.08)^{n-1}}{10(1.08)^{n-1}} \\ &= \frac{10(1.08)^{n-1}\left[(1.08) - 1\right]}{10(1.08)^{n-1}} \\ &= 1.08 - 1 \\ &= 0.08. \end{aligned}$$

This is compound interest. The effective rate of interest is constant each year, and this is the interest assumption used in almost every FM problem.

$$\boxed{i_3 = i_7 = 8\%}$$

Exercise 1.1.2

An investment of \$1,000 is made into a fund at time $t = 0$. The fund develops the following balances over the next 4 years:

t	0	1	2	3	4
$A(t)$	1,000	1,060	1,113	1,146	1,181

If \$5,000 is invested in the same fund at time 1, find the amount of interest earned on this deposit between time 2 and time 3.

Solution
Since the new deposit is made into the same fund at time 1, its value at later times grows in proportion to the original fund values after time 1.
Let $A^*(t)$ be the accumulated value at time t of the \$5,000 deposited at time 1.
At time 2, we have

$$A^*(2) = 5{,}000\frac{A(2)}{A(1)}.$$

Thus,

$$A^*(2) = 5{,}000\frac{1{,}113}{1{,}060} = 5{,}250.$$

At time 3, we have

$$A^*(3) = 5{,}000\frac{A(3)}{A(1)}.$$

Thus,

$$A^*(3) = 5{,}000\frac{1{,}146}{1{,}060} = 5{,}405.66.$$

Therefore, the interest earned between time 2 and time 3 is

$$I_3^* = A^*(3) - A^*(2).$$

So,

$$I_3^* = 5{,}405.66 - 5{,}250 = 155.66.$$

Therefore, the amount of interest earned on this deposit between time 2 and time 3 is

$$\boxed{\$155.66}.$$

Exercise 1.1.3

You are given the accumulation function

$$a(t) = 2t^2 + 3t + 1.$$

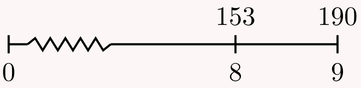
Determine i_9, the effective interest rate during the ninth year.

Solution
We can find the effective interest rate in year 9 by determining the increase in the accumulation function between years 8 and 9, and then dividing by the value of the accumulation function in year 8.
First,

$$a(8) = 2(8)^2 + 3(8) + 1 = 153,$$

and

$$a(9) = 2(9)^2 + 3(9) + 1 = 190.$$



The ninth year is the period from time 8 to time 9. Therefore,

$$i_9 = \frac{a(9) - a(8)}{a(8)}.$$

Substituting the values gives

$$i_9 = \frac{190 - 153}{153}.$$

Thus,

$$i_9 = \frac{37}{153} = 0.24183.$$

Therefore,

$$\boxed{i_9 = 24.183\%}.$$

Exercise 1.1.4

Gertrude deposits \$10,000 in a bank. During the first year, the bank credits an annual effective rate of interest i.
During the second year, the bank credits an annual effective rate of interest $i - 5\%$.
At the end of two years, she has \$12,093.75 in the bank.
What would Gertrude have in the bank at the end of three years, if the annual effective rate of interest were $i + 9\%$ for each of the three years?

Solution
We first construct an equation that lets us determine the value of i using the final balance and the rates in the first two years. From there, we can set up the equation for the accumulation over three years at $i + 9\%$.
At the end of two years,

$$10{,}000(1 + i)(1 + i - 0.05) = 12{,}093.75.$$

Equivalently,

$$10{,}000(1 + i)(0.95 + i) = 12{,}093.75.$$

Expanding,

$$10{,}000(0.95 + 1.95i + i^2) = 12{,}093.75.$$

Therefore,

$$10{,}000i^2 + 19{,}500i - 2{,}593.75 = 0.$$

We solve this equation using the quadratic formula:

$$i = \frac{-19{,}500 \pm \sqrt{19{,}500^2 - 4(10{,}000)(-2{,}593.75)}}{2(10{,}000)}.$$

The positive solution is

$$i = 0.125.$$

Now, if the interest rate were $i + 9\%$ for each of the three years, then

$$i + 0.09 = 0.215.$$

Thus, the accumulated value after three years would be

$$10{,}000(1.215)^3 = 17{,}936.13.$$

Therefore, Gertrude would have

$$\boxed{\$17{,}936.13}.$$

Exercise 1.1.5

[SOA 22; SOA-IT.027] Bruce and Robbie each open up new bank accounts at time 0. Bruce deposits \$100 into his bank account, and Robbie deposits \$50 into his. Each account earns the same annual effective interest rate.
The amount of interest earned in Bruce's account during the 11th year is equal to X. The amount of interest earned in Robbie's account during the 17th year is also equal to X.
Calculate X.

Solution
The amount of interest earned in any account is the difference between its value at the end of a period and its value at the beginning of the period.
The amount of interest earned in Bruce's account during the 11th year is

$$100(1+i)^{11} - 100(1+i)^{10} = X.$$

The amount of interest earned in Robbie's account during the 17th year is

$$50(1+i)^{17} - 50(1+i)^{16} = X.$$

Setting the two expressions equal gives

$$100(1+i)^{11} - 100(1+i)^{10} = 50(1+i)^{17} - 50(1+i)^{16}.$$

Factor both sides:

$$100(1+i)^{10}\big((1+i)-1\big) = 50(1+i)^{16}\big((1+i)-1\big),$$
$$100(1+i)^{10}i = 50(1+i)^{16}i.$$

Since the interest rate is positive, divide by $50(1+i)^{10}i$:

$$2 = (1+i)^6.$$

Thus,

$$1+i = 2^{1/6},$$

and

$$i = 2^{1/6} - 1 = 0.122462.$$

Now substitute this value into Bruce's interest equation:

$$X = 100(1.122462)^{11} - 100(1.122462)^{10}.$$

Therefore,

$$X = 38.88.$$

Thus,

$$\boxed{X = \$38.88}.$$

1.1.3 Present Value & Accumulated Value

Present Value and Accumulated Value

The **present value** is the value at an earlier time of an amount to be received or paid later.

The **accumulated value** is the value at a later time of an amount invested or paid earlier.

In general, moving forward in time corresponds to **accumulating**, while moving backward in time corresponds to **discounting**.

BA II Plus TVM Function Examples 1.1.1 to 1.1.4 can be solved using the TVM function on the BA II Plus. You may refer to the Calculator Notes for more details.

Reproduce the results from each example by entering the following:

Example 1.1.1	Example 1.1.2	Example 1.1.3	Example 1.1.4
$N = 2$	$N = 8$	$N = 0.5$	$N = 1$
$PV = 500$	$I/Y = 1$	$I/Y = 10$	$I/Y = 7$
$PMT = 0$	$PV = 300$	$PV = 100$	$PMT = 0$
$FV = -561.80$	$PMT = 0$	$PMT = 0$	$FV = -500$
$CPT\ I/Y = 6$	$CPT\ FV = -324.85$	$CPT\ FV = -104.88$	$CPT\ PV = 467.29$

When entering data into the TVM, cash inflows and cash outflows must have opposite signs. Cash inflow and cash outflow is relative (an inflow to an investor is an outflow to a banker and vice versa). So, choose one perspective and remain consistent.

In these examples, the PV and FV represented opposite cash flows. We chose a positive sign for PV and a negative sign for FV. The need to be consistent will become evident when we have a third type of cash flow: payments. The sign for the payment must be consistent with the definitions used for the PV and FV.

Discount Factor

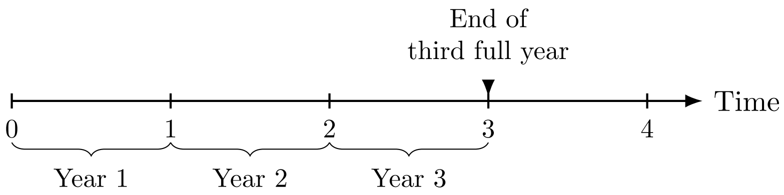
Going back one year requires dividing by $1 + i$, or equivalently multiplying by

$$\frac{1}{1+i}.$$

This process is called **discounting** for one year of interest.
Because we discount frequently, we use a separate symbol, v, called the **discount factor**:

$$v = \frac{1}{1+i}.$$

When creating a time diagram, the time values represent the end of each period. Thus, the value 3 on a time diagram represents the end of the third full year. This will be the standard convention for all time diagrams in these notes.



Exercise 1.1.6

Fund A is invested at an annual effective interest rate of 3%.

Fund B is invested at an annual effective interest rate of 2.5%.

At the end of 20 years, the total in the two funds is $10,000. At the end of 31 years, the amount in Fund A is twice the amount in Fund B.

Calculate the total in the two funds at the end of 10 years.

Solution

Let A and B be the amounts initially invested in Fund A and Fund B, respectively.
At the end of 20 years, we have

$$A(1.03)^{20} + B(1.025)^{20} = 10{,}000. \tag{1}$$

At the end of 31 years, the amount in Fund A is twice the amount in Fund B, so

$$A(1.03)^{31} = 2B(1.025)^{31}. \tag{2}$$

From (2),

$$\begin{aligned} A(1.03)^{31} &= 2B(1.025)^{31}, \\ A &= \frac{2B(1.025)^{31}}{(1.03)^{31}}. \end{aligned}$$

From (1),

$$\begin{aligned} \frac{2B(1.025)^{31}}{(1.03)^{31}}(1.03)^{20} + B(1.025)^{20} &= 10{,}000, \\ B\left[2\left(\frac{1.025^{31}}{1.03^{11}}\right) + 1.025^{20}\right] &= 10{,}000. \end{aligned}$$

Thus,

$$B = 2{,}107.4648.$$

Then

$$A = \frac{2B(1.025)^{31}}{1.03^{31}} = 3{,}624.7344.$$

The total of the two funds in ten years is

$$A(1.03)^{10} + B(1.025)^{10} = 7{,}569.07.$$

Therefore, the total in the two funds at the end of 10 years is

$$\boxed{\$7{,}569.07}.$$

Exercise 1.1.7

Bob invests X in an account that pays interest at an annual effective rate of 8%.

Ten years later, he withdraws X.

Ten years after that, he withdraws $1,000, which exhausts the account.

Calculate X.

Solution
We accumulate all cash flows to time 20.
The initial deposit X accumulates for 20 years, so its accumulated value at time 20 is

$$X(1.08)^{20}.$$

The withdrawal of X at time 10 accumulates for 10 years to time 20, so its accumulated value at time 20 is

$$X(1.08)^{10}.$$

Since the final withdrawal of $1,000 exhausts the account at time 20, we have

$$X(1.08)^{20} - X(1.08)^{10} = 1{,}000.$$

Therefore,

$$\begin{aligned} X\left[(1.08)^{20} - (1.08)^{10}\right] &= 1{,}000, \\ X &= \frac{1{,}000}{(1.08)^{20} - (1.08)^{10}}, \\ X &= 399.6751. \end{aligned}$$

Thus,

$$\boxed{X = \$399.68}.$$

1.1.4 Effective Rate of Discount

Effective Rate of Discount

The **effective rate of discount** during the t-th period, denoted by d_t, is the ratio of interest earned during the period to the amount in the fund at the end of the period.

Effective Rate of Discount

$$d_t = \frac{I(t)}{A(t)} = \frac{A(t) - A(t-1)}{A(t)}.$$

Equivalently, the fund multiplies by

$$1 - d_t$$

when moving one period backward in time (discounting). For a constant effective discount rate

$$d$$

, the corresponding accumulation/discount relationship is summarized by the identities in the formula box below. We usually use effective discount rates to calculate the present value. However, we can also use effective discount rates to calculate an accumulated value. Convert between accumulation and discount factors simply by negating the exponent. Recall that $(1+i)$ accumulates and $(1+i)^{-1}$ discounts. Likewise, $(1-d)$ discounts, so $(1-d)^{-1}$ accumulates.

Relationship Between i and d

$$1+i = \frac{1}{1-d}.$$

$$d = \frac{i}{1+i}, \qquad i = \frac{d}{1-d}.$$

Intuitively, the effective discount rate measures the reduction applied to a future amount in order to express it as a present value. While an interest rate grows money forward in time, a discount rate brings money backward in time. Thus, if $(1+i)$ is an accumulation factor, then $(1+i)^{-1}$ is a discount factor. Similarly, since $(1-d)$ discounts one period, its reciprocal $(1-d)^{-1}$ accumulates one period.

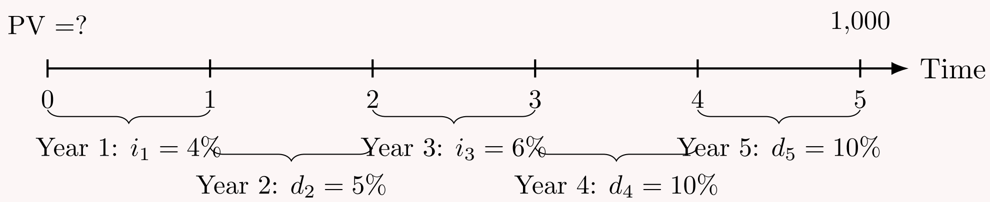
Exercise 1.1.8

John can earn the following rates for the next five years:

- Year 1: effective rate of interest = 4%;
- Year 2: effective rate of discount = 5%;
- Year 3: effective rate of interest = 6%;
- Years 4 and 5: effective rate of discount = 10%.

John needs to have \$1,000 in 5 years. How much should he invest now?

Solution
We need the present value at time 0 of \$1,000 paid at time 5.



To move backward through a year with an effective interest rate i, we divide by $1+i$. To move backward through a year with an effective discount rate d, we multiply by $1-d$. Starting from time 5, we discount year by year:

$$PV = 1{,}000(1-0.10)(1-0.10)(1.06)^{-1}(1-0.05)(1.04)^{-1}.$$

Thus,

$$PV = 1{,}000(0.90)(0.90)\frac{1}{1.06}(0.95)\frac{1}{1.04}.$$

Therefore,

$$PV = 698.02.$$

John should invest

$$\boxed{\$698.02}$$

now.

Exercise 1.1.9

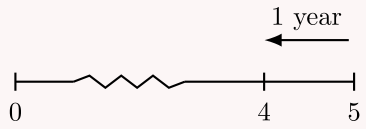
Given

$$a(t) = 1.08^t,$$

find v_5.

Solution

The quantity v_5 is the discount factor for year 5. It moves money backward from time 5 to time 4.



By definition,

$$v_5 = \frac{a(4)}{a(5)}.$$

Since $a(t) = 1.08^t$, we have

$$v_5 = \frac{1.08^4}{1.08^5}.$$

Therefore,

$$v_5 = 1.08^{-1} = 0.92593.$$

Thus,

$$\boxed{v_5 = 0.92593}.$$

Historical note

Discounting originated from the practical problem of valuing future payments today. In commerce and banking, a merchant could sell a future promise of payment to a bank before its maturity, receiving less than the promised amount. The difference was called the discount.

This idea later became a fundamental tool in finance and actuarial science, where discount rates are used to convert future cash flows into present values, especially for loans, bonds, annuities, insurance benefits, pensions, and reserves.

Accumulation from Time k to Time t

If the initial investment occurs at time k, with $k \leq t$, then the amount accumulates from time k to time t according to

$$A(t) = A(k)(1-d)^{-(t-k)}.$$

Here, $(1-d)^{-1}$ is the one-period accumulation factor associated with the effective discount rate d.
If $k > t$, then the exponent becomes positive, and the formula represents discounting from time k back to time t rather than accumulating.

Example 1.1.5

Sam invests \$200 in a bank. The fund earns an annual effective rate of discount of 9.0909%. What will the fund value be at the end of five years?

Solution
Define t in terms of years. We have

$$A(0) = 200, \qquad d = 9.0909\%, \qquad t = 5.$$

Since $(1-d)^{-1}$ is the one-period accumulation factor associated with an effective discount rate, the accumulated value after five years is

$$A(5) = 200(1 - 0.090909)^{-5} = 322.1018.$$

Therefore, the fund accumulates to

$$\boxed{\$322.10}$$

at the end of five years.

Example 1.1.6

Fund A offers a monthly effective discount rate of 1%. Jack deposited \$350 into Fund A four years ago. Now, he withdraws all his funds.
How much does he withdraw?

Solution
Use t in terms of months. Since there are 12 months in a year, four years correspond to 48 months.

$$A(0) = 350, \qquad d = 1\%, \qquad t = 48.$$

Since $(1-d)^{-1}$ is the one-period accumulation factor associated with an effective discount rate, the accumulated value after 48 months is

$$A(48) = 350(1 - 0.01)^{-48} = 566.9943.$$

Therefore, Jack withdraws

$$\boxed{\$566.99}$$

from the fund.

Since

$$v = \frac{1}{1+i},$$

we also have

$$v = \frac{1}{1+i} = 1 - d.$$

Therefore, given any one of i, v, or d, we can derive the other two using these relationships.

Starting from

$$\frac{1}{1+i} = 1 - d,$$

we obtain

$$\begin{aligned} d &= 1 - \frac{1}{1+i} \\ &= \frac{1+i}{1+i} - \frac{1}{1+i} \\ &= \frac{i}{1+i}. \end{aligned}$$

Since

$$v = \frac{1}{1+i},$$

we can also write

$$d = iv.$$

Finally, from

$$d = \frac{i}{1+i},$$

we solve for i:

$$\begin{aligned} d(1+i) &= i, \\ d + di &= i, \\ d &= i(1-d), \\ i &= \frac{d}{1-d}. \end{aligned}$$

Relationships Between i, d, and v

$$v = \frac{1}{1+i} = 1 - d.$$

$$d = \frac{i}{1+i} = iv, \qquad i = \frac{d}{1-d}.$$

Example 1.1.7

Given an 8% annual effective interest rate, calculate the equivalent annual effective discount rate, d, and the discount factor, v.

Solution
The effective interest rate is 8%. Thus,

$$i = 0.08.$$

Recall that the effective discount rate is

$$d = \frac{i}{1+i}.$$

Therefore,

$$\begin{aligned} d &= \frac{0.08}{1+0.08} \\ &= \frac{0.08}{1.08} \\ &= 0.07407. \end{aligned}$$

Now, the discount factor is

$$v = \frac{1}{1+i}.$$

Thus,

$$\begin{aligned} v &= \frac{1}{1+0.08} \\ &= \frac{1}{1.08} \\ &= 0.9259. \end{aligned}$$

Therefore,

$$\boxed{d = 7.407\%} \qquad \text{and} \qquad \boxed{v = 0.9259}.$$

Exercise 1.1.10

A deposit of X is made into a fund which pays an annual effective interest rate of 6% for 10 years.

At the same time, $\frac{X}{2}$ is deposited into another fund which pays an annual effective rate of discount of d for 10 years.

The amounts of interest earned over the 10 years are equal for both funds.

Calculate d.

Solution

The amount of interest earned is the difference between the initial amount and the amount at the end of 10 years.
For the first fund, the accumulated value after 10 years is

$$X(1.06)^{10}.$$

Thus, the interest earned in the first fund is

$$X(1.06)^{10} - X.$$

For the second fund, the annual effective rate of discount is d. Therefore, the accumulation factor over 10 years is

$$(1-d)^{-10}.$$

The accumulated value of the second fund after 10 years is

$$\frac{X}{2}(1-d)^{-10}.$$

Thus, the interest earned in the second fund is

$$\frac{X}{2}(1-d)^{-10} - \frac{X}{2}.$$

Since the amounts of interest earned are equal, we have

$$X(1.06)^{10} - X = \frac{X}{2}(1-d)^{-10} - \frac{X}{2}.$$

Divide both sides by X:

$$1.06^{10} - 1 = \frac{(1-d)^{-10}}{2} - \frac{1}{2}.$$

Rearranging gives

$$(1-d)^{-10} = 2.581695.$$

Therefore,

$$1 - d = (2.581695)^{-1/10}.$$

Hence,

$$d = 1 - (2.581695)^{-1/10}.$$

So,

$$d = 0.09049.$$

Therefore, the annual effective rate of discount is

$$\boxed{9.049\%}.$$

Exercise 1.1.11

[SOA 16; SOA-IT.020] David can receive one of the following two payment streams:

i. 100 at time 0, 200 at time n years, and 300 at time $2n$ years;

ii. 600 at time 10 years.

At an annual effective interest rate of i, the present values of the two streams are equal. Given $v^n = 0.76$, calculate i.

A. 3.5%

B. 4.0%

C. 4.5%

D. 5.0%

E. 5.5%

Solution

We discount each payment to its present value by multiplying by the appropriate power of the discount factor v.

The present value of the first payment stream is

$$100 + 200v^n + 300v^{2n}.$$

The present value of the second payment stream is

$$600v^{10}.$$

Since the present values are equal,

$$600v^{10} = 100 + 200v^n + 300v^{2n}.$$

Using $v^n = 0.76$, we get

$$v^{2n} = (v^n)^2 = (0.76)^2.$$

Therefore,

$$600v^{10} = 100 + 200(0.76) + 300(0.76)^2,$$
$$600v^{10} = 100 + 152 + 173.28,$$
$$600v^{10} = 425.28.$$

Thus,

$$v^{10} = \frac{425.28}{600} = 0.7088.$$

Since

$$v = \frac{1}{1+i},$$

we have

$$v^{10} = (1+i)^{-10}.$$

Hence,

$$(1+i)^{-10} = 0.7088.$$

Solving for i,

$$1 + i = (0.7088)^{-1/10},$$

so

$$i = (0.7088)^{-1/10} - 1.$$

Therefore,

$$i = 0.035.$$

Thus,

$$\boxed{i = 3.5\%}.$$

Exercise 1.1.12

The amount of interest paid on a loan of X for one year is \$108. The equivalent amount of discount paid on a loan of X for one year is \$100.
Determine X.

A. Less than \$1,220

B. At least \$1,220, but less than \$1,260

C. At least \$1,260, but less than \$1,300

D. At least \$1,300, but less than \$1,340

E. At least \$1,340

Solution
This question relies on correctly writing the expressions for the amount of interest paid and the amount of discount paid on the same cash flow.
The amount of interest paid is

$$Xi = 108.$$

The amount of discount paid is

$$Xd = 100.$$

Recall that

$$d = \frac{i}{1+i}.$$

Equivalently,

$$i = \frac{d}{1-d}.$$

Using $Xi = 108$, we have

$$X\left(\frac{d}{1-d}\right) = 108.$$

Since $Xd = 100$, this becomes

$$\frac{Xd}{1-d} = 108,$$

so

$$\frac{100}{1-d} = 108.$$

Therefore,

$$\begin{aligned} 100 &= 108(1-d), \\ 100 &= 108 - 108d, \\ 108d &= 8, \\ d &= \frac{8}{108}. \end{aligned}$$

Now use $Xd = 100$:

$$X\left(\frac{8}{108}\right) = 100.$$

Thus,

$$X = 100\left(\frac{108}{8}\right) = 1{,}350.$$

Therefore,

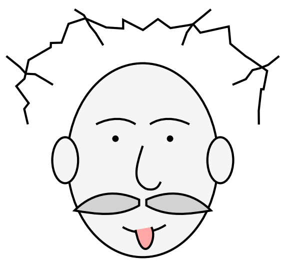
$$\boxed{X = 1{,}350}.$$

The correct choice is $\boxed{\text{E}}$.

1.1.5 Simple Interest and Compound Interest

Compound Interest

"Compound interest is the eighth wonder of the world. He who understands it, earns it; he who doesn't, pays it."



A playful nod to a famous quote
often attributed to Albert Einstein.

Compound Interest

Compound interest means that interest earned also earns interest.

In other words, interest is credited to the balance, and future interest is calculated on the new accumulated balance.

Compound interest is defined using an effective interest rate. Unless stated otherwise, the rate of interest is assumed to be the same each period.

If an initial amount P accumulates at a constant effective interest rate i per period, then

$$A(t) = P(1+i)^t.$$

Compound Interest

Suppose \$100 is borrowed at a compound interest rate of 10% per annum.
Find the loan balance after 3 years.

Solution
The initial loan balance is

$$100.$$

During the first year, the interest earned is

$$0.10(100) = 10.$$

Thus, the balance after one year is

$$100 + 10 = 110.$$

During the second year, interest is earned on the new balance:

$$0.10(110) = 11.$$

Thus, the balance after two years is

$$110 + 11 = 121.$$

During the third year, interest is earned on the balance 121:

$$0.10(121) = 12.1.$$

Thus, the balance after three years is

$$121 + 12.1 = 133.1.$$

Alternatively, we can accumulate directly:

$$100(1.10)(1.10)(1.10) = 100(1.10)^3 = 133.1.$$

Therefore, the loan balance after 3 years is

$$\boxed{\$133.10}.$$

Introduction to the BA II Plus Calculator

The BA II Plus calculator is an essential tool for Exam FM. Before solving time value of money problems, make sure the calculator is correctly set up.

Basic Calculator Setup

Before working on problems, check the following settings:

- **Reset the calculator:** press $\boxed{\text{2ND}}$, then $\boxed{+/-}$, then $\boxed{\text{ENTER}}$.
- **Calculation mode:** choose either chain mode or algebraic operating system mode.
- **Payments and compounding settings:** check that P/Y and C/Y are set correctly.
- **Time value of money keys:** understand the meaning of N, I/Y, PV, PMT, and FV.

For most of our early FM problems, we will usually take

$$P/Y = 1 \qquad \text{and} \qquad C/Y = 1,$$

unless the problem states otherwise.

Before starting a new TVM problem, it is good practice to clear the TVM worksheet:

$$\boxed{\text{2ND}} \quad \boxed{\text{CLR TVM}}.$$

TVM Keys

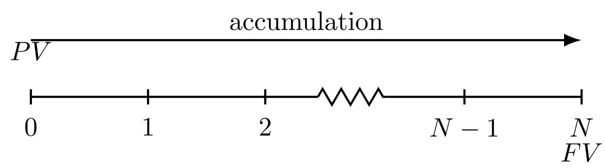
The main TVM keys are:

Key	Meaning
N	number of periods
I/Y	effective rate of interest per period, entered as a percentage
PV	present value
PMT	payment per period
FV	future value, or accumulated value

For now, we will mostly use N, I/Y, PV, and FV. Payments PMT will become important later when we study annuities.

TVM Timeline

A basic TVM problem can be represented by the following time diagram:



In a basic compound interest problem with no payments,

$$FV = PV(1+i)^N.$$

Equivalently,

$$PV = FV(1+i)^{-N}.$$

On the BA II Plus:

Input	**Meaning**
N	number of periods
I/Y	interest rate per period, entered as a percentage
PV	present value
PMT	usually 0 for now
FV	future value

The calculator uses a cash flow sign convention. Money paid out and money received must have opposite signs. For example, if PV is entered as positive, then FV will usually be computed as negative, and vice versa.

Example: Computing a Future Value

Suppose \$500 is invested for 6 years at an annual effective interest rate of 5%.

On the BA II Plus, enter:

N	6
I/Y	5
PV	-500
PMT	0
FV	CPT

The calculator returns

$$FV = 670.05.$$

This agrees with the formula

$$FV = 500(1.05)^6 = 670.05.$$

Exercise 1.1.13

At a certain rate of compound interest, 1 grows to 3 in x years, 3 grows to 14 in y years, and 1 grows to 21 in z years.
Determine what 5 grows to in $z - x - y$ years.

Solution

We can write each given fact as an accumulation factor equation using $(1+i)^t$.
Since 1 grows to 3 in x years,

$$(1+i)^x = 3.$$

Since 3 grows to 14 in y years,

$$3(1+i)^y = 14,$$

so

$$(1+i)^y = \frac{14}{3}.$$

Since 1 grows to 21 in z years,

$$(1+i)^z = 21.$$

We want the accumulated value of 5 after $z-x-y$ years:

$$5(1+i)^{z-x-y}.$$

Rewrite this as

$$5(1+i)^{z-x-y} = \frac{5(1+i)^z}{(1+i)^x(1+i)^y}.$$

Substitute the known values:

$$5(1+i)^{z-x-y} = \frac{5(21)}{3\left(\frac{14}{3}\right)}.$$

Thus,

$$5(1+i)^{z-x-y} = \frac{105}{14} = \frac{15}{2}.$$

Therefore, 5 grows to

$$\boxed{\frac{15}{2}}.$$

Simple Interest

Simple Interest

Simple interest means that the interest rate applies only to the original investment.
In other words, interest is not earned on previously earned interest.
If an initial amount P is invested at a simple interest rate i per period, then the accumulated value at time t is

$$A(t) = P(1+it).$$

Equivalently, the accumulation function is

$$a(t) = 1 + it.$$

Simple Interest

Suppose \$100 is invested at simple interest of 10% per annum.
Find the accumulated value after 3 years.

Solution
The original investment is

$$100.$$

Under simple interest, the interest is calculated only on the original principal.
During the first year, the interest earned is

$$0.10(100) = 10.$$

Thus, the balance after one year is

$$100 + 10 = 110.$$

During the second year, the interest earned is again

$$0.10(100) = 10.$$

Thus, the balance after two years is

$$110 + 10 = 120.$$

During the third year, the interest earned is again

$$0.10(100) = 10.$$

Thus, the balance after three years is

$$120 + 10 = 130.$$

Alternatively, we can compute the accumulated value directly:

$$100 + 0.10(100)(3) = 130.$$

Therefore, the accumulated value after 3 years is

$$\boxed{\$130}.$$

Under simple interest, the amount of interest earned each period is constant.

Under compound interest, the effective interest rate each period is constant.

This distinction is important. With simple interest, the interest is always based on the original principal. With compound interest, the interest is based on the accumulated balance, so previously earned interest also earns interest.

For a positive interest rate i, the simple interest accumulation function is

$$a_{\text{simple}}(t) = 1 + it,$$

while the compound interest accumulation function is

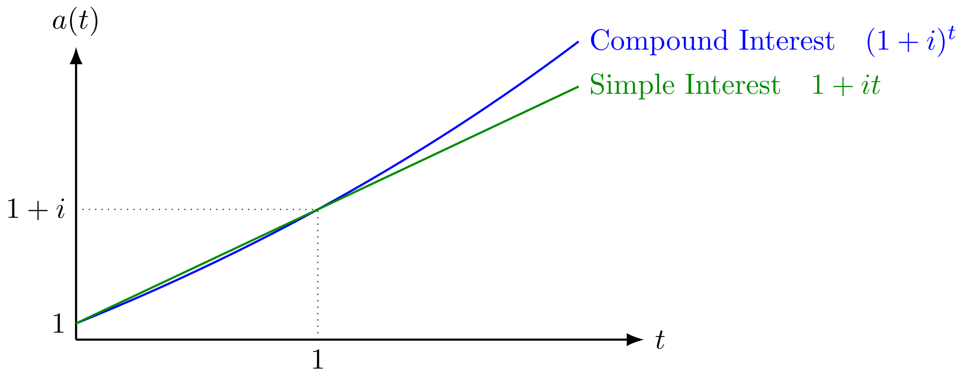
$$a_{\text{compound}}(t) = (1 + i)^t.$$

Both accumulation functions agree at time 0 and time 1:

$$a_{\text{simple}}(0) = a_{\text{compound}}(0) = 1,$$

and

$$a_{\text{simple}}(1) = a_{\text{compound}}(1) = 1 + i.$$



For less than one year, simple interest produces a larger accumulated value than compound interest:

$$1 + it > (1 + i)^t, \qquad 0 < t < 1.$$

Therefore, for periods shorter than one year, a borrower prefers compound interest of i rather than simple interest of i.

For more than one year, compound interest produces a larger accumulated value than simple interest:

$$(1 + i)^t > 1 + it, \qquad t > 1.$$

Therefore, for periods longer than one year, a lender prefers compound interest of i rather than simple interest of i.

Annual Effective Rates Under Simple Interest

Suppose \$100 is invested at simple interest of 10% per annum.
Then the accumulated values are

$$A(1) = 110, \qquad A(2) = 120, \qquad A(3) = 130.$$

Thus, the annual effective rates of interest are:

$$i_1 = \frac{110 - 100}{100} = 10\%,$$
$$i_2 = \frac{120 - 110}{110} = 9.09\%,$$
$$i_3 = \frac{130 - 120}{120} = 8.33\%.$$

Therefore, under simple interest, the annual effective rate of interest decreases each year.

For simple interest of i per annum, the amount of interest earned each year is constant and equal to Pi, where P is the original principal.

At the beginning of year n, the accumulated value is

$$P\big(1 + i(n - 1)\big).$$

Therefore, the effective interest rate during year n is

$$i_n = \frac{Pi}{P\big(1 + i(n - 1)\big)}.$$

After cancelling P, we obtain the following formula.

Effective Interest Rate Under Simple Interest

For simple interest of i per annum, the effective interest rate during year n is

$$i_n = \frac{i}{1 + i(n-1)}.$$

Thus, under simple interest, the annual effective rate of interest decreases each year.

Simple interest is sometimes used for partial years.

Simple Interest for Partial Years

Given $i = 0.10$, suppose simple interest is used only for partial years.
Find the accumulated value of \$100 five and a half years from now.

Solution
First, accumulate for the first 5 full years using compound interest:

$$100(1.10)^5 = 161.05.$$

Then use simple interest for the last half-year.
The simple interest factor for half a year is

$$1 + 0.10(0.5).$$

Therefore, the balance after 5.5 years is

$$161.05\big(1 + 0.10(0.5)\big) = 169.10.$$

Thus, the accumulated value is

$$\boxed{\$169.10}.$$

If we used compound interest for the entire 5.5 years, the balance would be

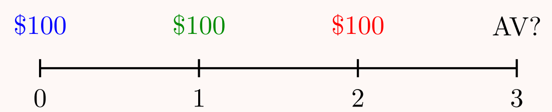
$$100(1.10)^{5.5} = 168.91.$$

▲ **Important.** Under simple interest, every cash flow is treated separately with its own "time 0." This means each deposit earns simple interest based on the length of time since that specific deposit was made.

Simple Interest with Multiple Deposits

Suppose \$100 is deposited at the beginning of each year for 3 years. If the deposits earn simple interest of 10% per annum, what is the accumulated value at the end of 3 years?

Solution
The deposits occur at times 0, 1, and 2. We want the accumulated value at time 3.



Since simple interest is used, each deposit is accumulated separately.

Payment	Accumulated Value at Time 3
1	$100\big(1+0.10(3)\big)=130$
2	$100\big(1+0.10(2)\big)=120$
3	$100\big(1+0.10(1)\big)=110$
Total	$130+120+110=360$

Therefore, the accumulated value at the end of 3 years is

$$\boxed{\$360}.$$

Deposits with a Given Accumulation Function

Suppose \$100 is deposited at the beginning of each year for 3 years.
Given

$$a(t)=1+0.1t,$$

what is the accumulated value at the end of 3 years?

Solution
The deposits occur at times 0, 1, and 2. We want the accumulated value at time 3.

First compute the values of the accumulation function:

$$a(1)=1+0.1(1)=1.1,$$

$$a(2)=1+0.1(2)=1.2,$$

and

$$a(3)=1+0.1(3)=1.3.$$

Since an accumulation function is given, each deposit is accumulated to time 3 using the ratio

$$\frac{a(3)}{a(k)},$$

where k is the time at which the deposit is made.
Therefore,

$$AV=100a(3)+100\frac{a(3)}{a(1)}+100\frac{a(3)}{a(2)}.$$

Substituting the values gives

$$AV=100(1.3)+100\frac{1.3}{1.1}+100\frac{1.3}{1.2}.$$

Thus,

$$AV=356.52.$$

Therefore, the accumulated value at the end of 3 years is

$$\boxed{\$356.52}.$$

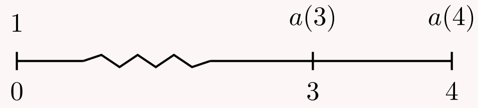
Exercise 1.1.14

A loan is made for five years at a simple interest rate of 12% per annum.
What is the equivalent annual effective rate of discount during the fourth year of the loan?

Solution
Let the original loan amount be 1.
Since the loan uses simple interest at 12% per annum, the accumulation function is

$$a(t) = 1 + 0.12t.$$

The fourth year is the period from time 3 to time 4.



The loan balances at times 3 and 4 are

$$a(3) = 1 + 0.12(3) = 1.36,$$

and

$$a(4) = 1 + 0.12(4) = 1.48.$$

The effective rate of discount during the fourth year is the interest earned during the fourth year divided by the balance at the end of the fourth year:

$$d_4 = \frac{a(4) - a(3)}{a(4)}.$$

Therefore,

$$d_4 = \frac{1.48 - 1.36}{1.48} = 0.08108.$$

Thus, the equivalent annual effective rate of discount during the fourth year is

$$\boxed{8.108\%}.$$

Exercise 1.1.15

At a rate of simple interest i, 10 will accumulate to 15 after x years.
What will 20 accumulate to at a simple interest rate of $2i$ after $5x$ years?

Solution
Under simple interest, the accumulated value is given by

$$A(t) = P(1 + it).$$

We are given that 10 accumulates to 15 after x years at simple interest rate i. Therefore,

$$10(1 + ix) = 15.$$

Dividing by 10, we get

$$1 + ix = 1.5.$$

Thus,

$$ix = 0.5.$$

Now we want the accumulated value of 20 at simple interest rate $2i$ after $5x$ years.
Using the simple interest formula,

$$A = 20\big(1 + (2i)(5x)\big).$$

Simplify:

$$A = 20(1 + 10ix).$$

Since $ix = 0.5$,

$$A = 20(1 + 10(0.5)).$$

Therefore,

$$A = 20(6) = 120.$$

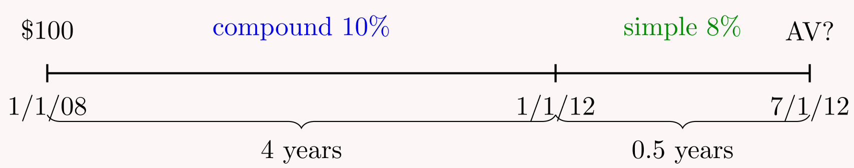
Thus, 20 accumulates to

$$\boxed{120}.$$

Exercise 1.1.16

An investment earns 10% compound interest for each complete year and 8% simple interest for each partial year.
A \$100 investment is made on January 1, 2008.
What is the accumulated value of the investment on July 1, 2012?

Solution
From January 1, 2008, to January 1, 2012, there are 4 complete years.
From January 1, 2012, to July 1, 2012, there is 0.5 of a year.



First, accumulate the investment for 4 complete years using compound interest:

$$100(1.10)^4.$$

Then, for the last half-year, use simple interest at 8%:

$$1 + 0.08(0.5).$$

Therefore,

$$AV = 100(1.10)^4\big(1 + 0.08(0.5)\big).$$

Thus,

$$AV = 100(1.10)^4(1.04) = 152.2664.$$

Therefore, the accumulated value on July 1, 2012, is

$$\boxed{\$152.27}.$$

Exercise 1.1.17

An investor puts \$100 into Fund X and \$100 into Fund Y.
Fund Y earns compound interest at the annual rate of $j > 0$, and Fund X earns simple interest at the annual rate of $1.05j$.
At the end of 2 years, the amount in Fund Y is equal to the amount in Fund X.
Calculate the amount in Fund Y at the end of 5 years.

A. 150

B. 153

C. 157

D. 161

E. 165

Solution
Since we are given information about the accumulated values, we begin by determining the appropriate accumulation functions.
For Fund X, which earns simple interest at rate $1.05j$, the accumulation function is

$$a_X(t) = 1 + 1.05jt.$$

For Fund Y, which earns compound interest at rate j, the accumulation function is

$$a_Y(t) = (1 + j)^t.$$

At the end of 2 years, the two funds have the same amount. Since both initial deposits are \$100, we have

$$100a_X(2) = 100a_Y(2).$$

Canceling 100, this becomes

$$a_X(2) = a_Y(2).$$

Substitute the accumulation functions:

$$1 + 1.05j(2) = (1 + j)^2.$$

Thus,

$$1 + 2.1j = (1 + j)^2.$$

Expanding the right-hand side gives

$$1 + 2.1j = 1 + 2j + j^2.$$

Therefore,

$$j^2 = 0.1j.$$

Since $j > 0$, we divide by j:

$$j = 0.1.$$

Now calculate the amount in Fund Y at the end of 5 years:

$$A_Y(5) = 100a_Y(5).$$

Thus,

$$A_Y(5) = 100(1.10)^5.$$

Therefore,

$$A_Y(5) = 161.05.$$

So the amount in Fund Y at the end of 5 years is approximately

$$\boxed{161.05}.$$

The closest answer is $\boxed{\text{D}}$.

Exercise 1.1.18

Simple interest of $i = 0.04$ is being credited to a fund.
In which year is this equivalent to an effective rate of 2.5%?

A. 14

B. 15

C. 16

D. 17

E. Never

Solution

We begin by finding the accumulation function. Since the fund earns simple interest of 4%, we have

$$a(t) = 1 + 0.04t.$$

The effective rate of interest earned during year t is

$$i_t = \frac{a(t) - a(t-1)}{a(t-1)}.$$

Substituting $a(t) = 1 + 0.04t$, we get

$$i_t = \frac{(1 + 0.04t) - \big(1 + 0.04(t-1)\big)}{1 + 0.04(t-1)}.$$

Simplify the numerator:

$$(1 + 0.04t) - \big(1 + 0.04(t-1)\big) = 0.04.$$

Thus,

$$i_t = \frac{0.04}{1 + 0.04(t-1)}.$$

Since

$$1 + 0.04(t-1) = 0.96 + 0.04t,$$

we have

$$i_t = \frac{0.04}{0.96 + 0.04t}.$$

Now set this equal to 2.5%:

$$0.025 = \frac{0.04}{0.96 + 0.04t}.$$

Solving,

$$\begin{aligned} 0.025(0.96 + 0.04t) &= 0.04, \\ 0.024 + 0.001t &= 0.04, \\ 0.001t &= 0.016, \\ t &= 16. \end{aligned}$$

Therefore, the simple interest rate of 4% is equivalent to an effective rate of 2.5% during year

$$\boxed{16}.$$

The correct choice is $\boxed{\text{C}}$.

1.1.6 Nominal Rate of Interest

Our primary focus has been calculating annual effective interest rates. But, we can calculate effective interest rates for any time period.

Suppose

$$A(0) = 100 \qquad \text{and} \qquad A(1) = 110.$$

Thus, the annual effective rate of interest is 10%. Assume this rate is constant effective for the entire year.

We want to find the effective interest rate for the first six months.

Define this six-month effective rate as j. Since j is an effective rate, it is the interest earned during the period divided by the beginning balance of the period:

$$j = \frac{A(0.5) - A(0)}{A(0)}.$$

We can calculate $A(0.5)$ by accumulating \$100 for half a year at a 10% annual effective rate:

$$A(0.5) = 100(1.10)^{0.5} = 104.8809.$$

Thus,

$$j = \frac{104.8809 - 100}{100} = 0.048809.$$

So the effective interest rate for the first six months is

$$j = 4.8809\%.$$

Now consider the last six months of the year. By definition, the effective interest rate from time 0.5 to time 1 is

$$\frac{A(1) - A(0.5)}{A(0.5)} = \frac{110 - 104.8809}{104.8809} = 0.048809.$$

Therefore, the effective rate for the last six months is also

$$4.8809\%.$$

This is expected because the annual effective interest rate is constant over the year. The same six-month growth factor applies to any six-month interval inside the year.

Equivalent Six-Month and Annual Effective Rates

The annual effective rate

$$i = 10\%$$

and the six-month effective rate

$$j = 4.8809\%$$

are equivalent because they produce the same accumulation over one full year.
Using the annual effective rate,

$$A(1) = 100(1 + i) = 100(1.10) = 110.$$

Using the six-month effective rate twice,

$$A(1) = 100(1 + j)^2.$$

Since $j = 0.048809$, we get

$$A(1) = 100(1.048809)^2 = 110.$$

Thus, the rates $i = 10\%$ annually effective and $j = 4.8809\%$ effective every six months are equivalent rates.

There are two common ways to describe the six-month effective rate $j = 4.8809\%$ as an annual rate.

First, we can convert it to an annual effective rate:

$$(1 + j)^2 - 1.$$

This gives

$$(1.048809)^2 - 1 = 0.10.$$

So the equivalent annual effective rate is

$$10\%.$$

Second, we can multiply the six-month rate by the number of six-month periods in one year:

$$2j = 2(0.048809) = 0.097618.$$

This gives

$$9.7618\%.$$

This second rate is not an effective annual rate. It is called a **nominal rate**.

Nominal Rate of Interest

A **nominal rate of interest** is an annual rate stated "in name only."
It is obtained by multiplying the effective rate per conversion period by the number of conversion periods in one year.
If interest is convertible m times per year, and the effective rate per conversion period is j, then the nominal annual rate convertible m times per year is

$$i^{(m)} = mj.$$

Equivalently,

$$j = \frac{i^{(m)}}{m}.$$

The notation $i^{(m)}$ does not mean that i is raised to the power m. The superscript (m) indicates the number of conversion periods per year.

Nominal Annual Rate Convertible m Times per Year

The notation $i^{(m)}$ represents a **nominal annual rate** convertible, or compounded, m times per year.

Nominal rates are interest rates "in name only." In other words, a nominal rate cannot be used directly as an effective rate. It must first be converted to an effective rate for the appropriate conversion period.

If $i^{(m)}$ is the nominal annual rate convertible m times per year, then the effective rate of interest per $\frac{1}{m}$-th of a year is

$$\frac{i^{(m)}}{m}.$$

Equivalent Rates

Let $i^{(m)}$ be the nominal annual rate of interest convertible m times per year.
The effective rate per conversion period is

$$\frac{i^{(m)}}{m}.$$

The equivalent annual effective rate i satisfies

$$1 + i = \left(1 + \frac{i^{(m)}}{m}\right)^m.$$

Equivalently,

$$i = \left(1 + \frac{i^{(m)}}{m}\right)^m - 1.$$

More generally, after t years,

$$A(t) = A(0)\left(1 + \frac{i^{(m)}}{m}\right)^{mt}.$$

Normally, the first step when given a nominal rate is to convert the nominal rate to an effective rate per conversion period:

$$\frac{i^{(m)}}{m}.$$

Then we accumulate using the appropriate number of conversion periods.

Nominal Rate Convertible Monthly

Assume that a nominal rate compounded twelve times per year is 12%.
What is the equivalent annual effective rate?

Solution
Since interest is compounded twelve times per year, we have

$$m = 12.$$

The nominal annual rate is

$$i^{(12)} = 0.12.$$

The monthly effective rate is therefore

$$\frac{i^{(12)}}{12} = \frac{0.12}{12} = 0.01.$$

Thus, the monthly effective rate is

$$1\%.$$

To find the equivalent annual effective rate, accumulate over all 12 months:

$$1 + i = (1.01)^{12}.$$

Therefore,

$$i = (1.01)^{12} - 1.$$

So

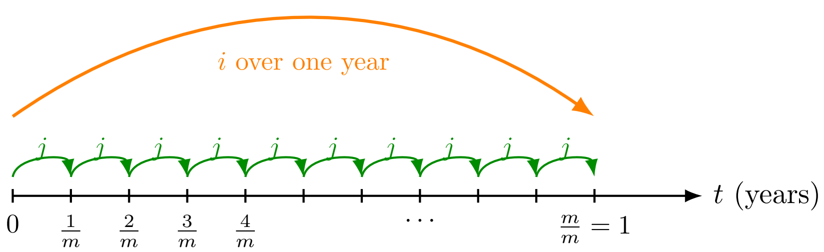
$$i = 0.126825.$$

Thus, the equivalent annual effective rate is

$$\boxed{12.68\%}.$$

General Interpretation

Assume there are m compounding periods in one year.



The symbol j represents the effective rate during one conversion period. Since there are m

conversion periods in one year, we have

$$j = \frac{i^{(m)}}{m}.$$

Accumulating $1 for one year using the conversion-period rate gives

$$(1 + j)^m.$$

Accumulating $1 for one year using the nominal rate convertible m times per year gives

$$\left(1 + \frac{i^{(m)}}{m}\right)^m.$$

Accumulating $1 for one year using the annual effective rate gives

$$1 + i.$$

Since all three represent the same one-year accumulation, we have

$$(1 + j)^m = \left(1 + \frac{i^{(m)}}{m}\right)^m = 1 + i.$$

Again, assume a 12% nominal rate convertible monthly. Since interest is convertible monthly,

$$m = 12.$$

If

$$i^{(12)} = 12\%,$$

then

$$j = \frac{0.12}{12} = 0.01.$$

The annual effective rate is determined by

$$1 + i = (1.01)^{12}.$$

Equivalently,

$$(1 + 0.01)^{12} = \left(1 + \frac{0.12}{12}\right)^{12} = 1.1268.$$

Thus, $1 invested for one year at all three equivalent rates produces the same accumulated value:

$$\$1.1268.$$

Nominal rates are expressed using a variety of terms. The following expressions have the same meaning:

- a nominal rate of 12%, compounded monthly;
- an annual nominal rate of 12%, compounded monthly;
- a nominal annual rate of 12%, compounded monthly;
- a nominal rate of 12%, convertible monthly.

The terms **compounded** and **convertible** are used interchangeably in this context.

Important Convention

A nominal rate is not directly used as an effective annual rate.
When given $i^{(m)}$, first divide by m to get the effective rate per conversion period:

$$\frac{i^{(m)}}{m}.$$

Then use the number of conversion periods to accumulate or discount.

Converting Between $i^{(m)}$ and $i^{(n)}$

Suppose we want to convert between two nominal rates:

$$i^{(m)} \qquad \text{and} \qquad i^{(n)}.$$

The key idea is to accumulate $1 for one year using both rates.

Using the nominal rate $i^{(m)}$, the one-year accumulation factor is

$$\left(1+\frac{i^{(m)}}{m}\right)^m.$$

Using the nominal rate $i^{(n)}$, the one-year accumulation factor is

$$\left(1+\frac{i^{(n)}}{n}\right)^n.$$

Since the two rates are equivalent, these accumulation factors must be equal:

$$\left(1+\frac{i^{(n)}}{n}\right)^n = \left(1+\frac{i^{(m)}}{m}\right)^m.$$

Taking the n-th root of both sides gives

$$1+\frac{i^{(n)}}{n} = \left(1+\frac{i^{(m)}}{m}\right)^{m/n}.$$

Therefore,

$$\frac{i^{(n)}}{n} = \left(1+\frac{i^{(m)}}{m}\right)^{m/n} - 1.$$

Multiplying by n, we obtain the conversion formula:

Converting Between Nominal Rates

If $i^{(m)}$ and $i^{(n)}$ are equivalent nominal interest rates, then

$$i^{(n)} = n\left[\left(1+\frac{i^{(m)}}{m}\right)^{m/n} - 1\right].$$

This formula converts a nominal rate convertible m times per year into an equivalent nominal rate convertible n times per year.

Nominal Interest Rate Compounded Monthly

Sam invests \$200 in a bank. The fund earns a nominal interest rate of 12%, compounded monthly.
What is the fund value five years after the time of the investment?

Solution
Since there are 12 months in each year, we have

$$A(0) = 200, \qquad m = 12, \qquad i^{(12)} = 12\%, \qquad t = 5.$$

The monthly effective interest rate is

$$\frac{i^{(12)}}{12} = \frac{0.12}{12} = 0.01.$$

There are

$$12(5) = 60$$

monthly compounding periods in five years.
Therefore,

$$A(5) = 200\left(1+\frac{0.12}{12}\right)^{12(5)}.$$

Thus,

$$A(5) = 200(1.01)^{60} = 363.3393.$$

So the fund value after five years is

$$\boxed{\$363.34}.$$

We can also convert the nominal interest rate to an annual effective interest rate first:

$$1 + i = \left(1+\frac{0.12}{12}\right)^{12}.$$

Hence,

$$i = \left(1+\frac{0.12}{12}\right)^{12} - 1 = 0.126825.$$

Then

$$A(5) = 200(1.126825)^{5} = 363.3393.$$

This confirms that a 12% nominal interest rate compounded monthly is equivalent to a 12.6825% annual effective interest rate.

Nominal Interest Rate Compounded Quarterly

Sam invested \$100 in a bank four years ago. The fund earned a nominal interest rate of $x\%$, compounded quarterly.
Today, he closes his account and withdraws \$180.
Calculate the following two rates that Sam earned:

- x, the nominal annual interest rate compounded quarterly;
- i, the equivalent annual effective interest rate.

Solution
Since there are four quarters in each year, we have

$$A(0) = 100, \qquad A(4) = 180, \qquad m = 4, \qquad t = 4.$$

Let r be the nominal annual rate expressed as a decimal. Since the problem asks for $x\%$, we will have

$$x = 100r.$$

Because the nominal rate is compounded quarterly, the quarterly effective interest rate is

$$\frac{r}{4}.$$

Over 4 years, there are

$$4(4) = 16$$

quarterly compounding periods.
Thus,

$$100\left(1+\frac{r}{4}\right)^{16} = 180.$$

Dividing by 100, we get

$$\left(1+\frac{r}{4}\right)^{16} = 1.8.$$

Taking the sixteenth root,

$$1+\frac{r}{4} = 1.8^{1/16}.$$

Therefore,

$$\frac{r}{4} = 1.8^{1/16} - 1 = 0.03742.$$

Thus,

$$r = 4(0.03742) = 0.149679.$$

Since $x = 100r$, we obtain

$$x = 14.9679\%.$$

The effective quarterly interest rate is

$$\frac{r}{4} = 3.742\%.$$

Now calculate the equivalent annual effective interest rate:

$$1+i = \left(1+\frac{r}{4}\right)^{4}.$$

Thus,

$$i = \left(1 + \frac{0.149679}{4}\right)^4 - 1.$$

Therefore,

$$i = 0.158292.$$

So the equivalent annual effective interest rate is

$$i = 15.8292\%.$$

Hence,

$$\boxed{x = 14.9679\%} \qquad \text{and} \qquad \boxed{i = 15.8292\%}.$$

Annual Effective Rate and Semiannual Conversion

James invests \$100 in a bank. The fund earns a 10% annual effective interest rate. What is the fund value at the end of six months?

Solution

This is similar to a previous example, but here we solve it by converting the annual effective rate into an equivalent semiannual effective rate.
There are two six-month periods in one year, so

$$m = 2.$$

We are given

$$A(0) = 100, \qquad i = 10\%, \qquad m = 2.$$

Let $i^{(2)}$ be the nominal annual interest rate convertible semiannually. Since it must be equivalent to the annual effective rate $i = 10\%$, we have

$$\left(1 + \frac{i^{(2)}}{2}\right)^2 = 1.10.$$

Taking the square root gives

$$1 + \frac{i^{(2)}}{2} = 1.10^{1/2}.$$

Thus,

$$\frac{i^{(2)}}{2} = 1.10^{1/2} - 1 = 0.04881.$$

Therefore, the effective interest rate for six months is

$$4.881\%.$$

Now accumulate \$100 for one six-month period:

$$A\left(\frac{1}{2}\right) = 100(1.04881).$$

Thus,

$$A\left(\frac{1}{2}\right) = 104.88.$$

Therefore, the fund value at the end of six months is

$$\boxed{\$104.88}.$$

Converting an Annual Effective Rate to Nominal Rates

Calculate $i^{(12)}$ and $i^{(4)}$, given that the annual effective interest rate is

$$i = 10\%.$$

Solution
We use the relationship between an annual effective interest rate i and a nominal interest rate $i^{(m)}$ convertible m times per year:

$$1 + i = \left(1 + \frac{i^{(m)}}{m}\right)^m.$$

Since $i = 10\%$, we have

$$1 + i = 1.10.$$

For the annual nominal interest rate compounded monthly, $m = 12$. Thus,

$$\left(1 + \frac{i^{(12)}}{12}\right)^{12} = 1.10.$$

Taking the twelfth root gives

$$1 + \frac{i^{(12)}}{12} = 1.10^{1/12}.$$

Therefore,

$$\frac{i^{(12)}}{12} = 1.10^{1/12} - 1.$$

So

$$i^{(12)} = 12\left(1.10^{1/12} - 1\right).$$

Thus,

$$i^{(12)} = 0.095690.$$

Hence,

$$\boxed{i^{(12)} = 9.5690\%}.$$

For the annual nominal interest rate compounded quarterly, $m = 4$. Thus,

$$\left(1 + \frac{i^{(4)}}{4}\right)^4 = 1.10.$$

Taking the fourth root gives

$$1 + \frac{i^{(4)}}{4} = 1.10^{1/4}.$$

Therefore,

$$i^{(4)} = 4\left(1.10^{1/4} - 1\right).$$

Thus,

$$i^{(4)} = 0.096455.$$

Hence,

$$\boxed{i^{(4)} = 9.6455\%}.$$

Solving for Nominal Rates

Nominal interest rates are common on Exam FM.
Given an annual effective interest rate i, we can solve for the equivalent nominal rate $i^{(m)}$ convertible m times per year.
Starting from

$$1+i=\left(1+\frac{i^{(m)}}{m}\right)^m,$$

take the m-th root:

$$(1+i)^{1/m}=1+\frac{i^{(m)}}{m}.$$

Therefore,

$$\frac{i^{(m)}}{m}=(1+i)^{1/m}-1.$$

Multiplying by m, we obtain

$$i^{(m)}=m\left[(1+i)^{1/m}-1\right].$$

You may memorize this formula, or simply use the previous strategy of investing \$1 for one year at both the nominal rate and the annual effective rate.
Memorizing the formula is slightly faster, but it requires one more formula to remember.

Exercise 1.1.19

You deposit \$100 into a fund that accumulates at an annual effective interest rate of 7%.
Four years later, the interest rate changes to an annual nominal rate of 8%, compounded quarterly.
How many years will it take for your deposit to double?

A. 9.00

B. 9.25

C. 9.33

D. 9.50

E. 10.00

Solution
First, check whether the fund doubles before the interest rate changes.
After four years, the fund value is

$$100(1.07)^4=131.0796.$$

Since

$$131.0796<200,$$

the fund does not double during the first four years.
Let n be the total number of years needed for the deposit to double.
After year 4, the fund earns a nominal annual interest rate of 8%, compounded quarterly.
Therefore, the effective quarterly interest rate is

$$\frac{0.08}{4}=0.02.$$

From time 4 to time n, there are

$$4(n-4)$$

quarters.
Thus, the doubling equation is

$$100(1.07)^4(1.02)^{4(n-4)} = 200.$$

Dividing by 100, we get

$$(1.07)^4(1.02)^{4(n-4)} = 2.$$

Therefore,

$$(1.02)^{4(n-4)} = \frac{2}{(1.07)^4}.$$

Taking natural logarithms,

$$4(n-4)\ln(1.02) = \ln\left(\frac{2}{(1.07)^4}\right).$$

So

$$n = 4 + \frac{\ln\left(\frac{2}{(1.07)^4}\right)}{4\ln(1.02)}.$$

Thus,

$$n = 9.3340.$$

Therefore, it will take approximately

$$\boxed{9.33}$$

years for the deposit to double.
The correct choice is $\boxed{\text{C}}$.

Exercise 1.1.20

On January 1, 1980, Jack deposited \$1,000 into Bank X to earn interest at the rate of j per annum compounded semiannually.
On January 1, 1985, he transferred his account to Bank Y to earn interest at the rate of k per annum compounded quarterly.
On January 1, 1988, the balance at Bank Y is \$1,990.76.
If Jack could have earned interest at the rate of k per annum compounded quarterly from January 1, 1980, through January 1, 1988, his balance would have been \$2,203.76.
Calculate the ratio

$$\frac{k}{j}.$$

A. 1.25

B. 1.30

C. 1.35

D. 1.40

E. 1.45

Solution

First, calculate k.
From January 1, 1980, to January 1, 1988, there are 8 years. Since the rate k is compounded quarterly, there are

$$4(8) = 32$$

quarterly compounding periods.
Thus,

$$1{,}000\left(1+\frac{k}{4}\right)^{32} = 2{,}203.76.$$

Dividing by 1,000, we get

$$\left(1+\frac{k}{4}\right)^{32} = 2.20376.$$

Taking the thirty-second root,

$$1+\frac{k}{4} = 1.025.$$

Therefore,

$$\frac{k}{4} = 0.025,$$

so

$$k = 0.10.$$

Now calculate j.
From January 1, 1980, to January 1, 1985, there are 5 years. Since the rate j is compounded semiannually, there are

$$2(5) = 10$$

semiannual compounding periods.
From January 1, 1985, to January 1, 1988, there are 3 years. Since the rate k is compounded quarterly, there are

$$4(3) = 12$$

quarterly compounding periods.
Thus,

$$1{,}000\left(1+\frac{j}{2}\right)^{10}\left(1+\frac{k}{4}\right)^{12} = 1{,}990.76.$$

Since

$$1+\frac{k}{4} = 1.025,$$

we have

$$1{,}000\left(1+\frac{j}{2}\right)^{10}(1.025)^{12} = 1{,}990.76.$$

Dividing by 1,000,

$$\left(1+\frac{j}{2}\right)^{10}(1.025)^{12} = 1.99076.$$

Therefore,

$$\left(1+\frac{j}{2}\right)^{10} = \frac{1.99076}{(1.025)^{12}}.$$

So

$$\left(1+\frac{j}{2}\right)^{10} = 1.48024.$$

Taking the tenth root,

$$1 + \frac{j}{2} = 1.04.$$

Hence,

$$\frac{j}{2} = 0.04,$$

so

$$j = 0.08.$$

Therefore,

$$\frac{k}{j} = \frac{0.10}{0.08} = 1.25.$$

Thus,

$$\boxed{\frac{k}{j} = 1.25}.$$

The correct choice is $\boxed{\text{A}}$.

Exercise 1.1.21

Jennifer deposits \$1,000 into a bank account.
The bank credits interest at a nominal annual rate of i, convertible semiannually, for the first 7 years and a nominal annual rate of $2i$, convertible quarterly, for all years thereafter.
The accumulated amount in the account at the end of 5 years is X.
The accumulated amount in the account at the end of 10.5 years is \$1,980.
Calculate X.

A. 1200

B. 1225

C. 1250

D. 1275

E. 1300

Solution
For a nominal annual rate i convertible semiannually, the effective rate per half-year is

$$\frac{i}{2}.$$

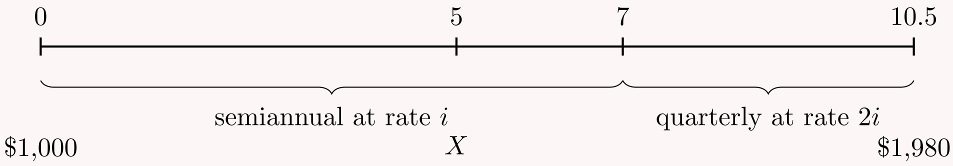
For a nominal annual rate $2i$ convertible quarterly, the effective rate per quarter is

$$\frac{2i}{4} = \frac{i}{2}.$$

Thus, both accumulation periods use the same periodic factor

$$1 + \frac{i}{2}.$$

We visualize the timeline as follows:



To find X, we first express it in terms of i.
From time 0 to time 5, there are

$$2(5) = 10$$

semiannual conversion periods.
Therefore,

$$X = 1{,}000\left(1 + \frac{i}{2}\right)^{10}.$$

Now use the accumulated value at time 10.5.
From time 0 to time 7, there are

$$2(7) = 14$$

semiannual conversion periods.
From time 7 to time 10.5, there are 3.5 years, or

$$4(3.5) = 14$$

quarterly conversion periods.
Since both periods use the same periodic factor $1 + \frac{i}{2}$, the total accumulation from time 0 to time 10.5 is

$$1{,}000\left(1 + \frac{i}{2}\right)^{14}\left(1 + \frac{2i}{4}\right)^{14} = 1{,}980.$$

Since

$$\frac{2i}{4} = \frac{i}{2},$$

we get

$$1{,}000\left(1 + \frac{i}{2}\right)^{28} = 1{,}980.$$

Thus,

$$\left(1 + \frac{i}{2}\right)^{28} = 1.98.$$

Taking the 28-th root,

$$1 + \frac{i}{2} = 1.98^{1/28}.$$

Therefore,

$$1 + \frac{i}{2} = 1.024696.$$

Now substitute this into the expression for X:

$$X = 1{,}000(1.024696)^{10}.$$

Thus,

$$X = 1{,}276.30.$$

The closest answer is

$$\boxed{1275}.$$

Therefore, the correct choice is $\boxed{\text{D}}$.

Exercise 1.1.22

Brian and Jennifer each take out a loan of X.
Jennifer will repay her loan by making one payment of \$800 at the end of year 10.
Brian will repay his loan by making one payment of \$1,120 at the end of year 10.
The nominal annual rate convertible semiannually being charged to Jennifer is exactly one-half the nominal annual rate convertible semiannually being charged to Brian.
Calculate X.

A. 562

B. 565

C. 568

D. 571

E. 574

Solution
Let i be the nominal annual rate convertible semiannually charged to Jennifer.
Then the nominal annual rate convertible semiannually charged to Brian is $2i$.
Since the loans are repaid at the end of 10 years and interest is convertible semiannually, there are

$$2(10) = 20$$

conversion periods.
For Jennifer, the effective semiannual rate is

$$\frac{i}{2}.$$

For Brian, the effective semiannual rate is

$$\frac{2i}{2} = i.$$

Since both loans have the same present value X, we equate the present values of the two repayment amounts:

$$800\left(1 + \frac{i}{2}\right)^{-20} = 1120(1 + i)^{-20}.$$

Divide both sides by 800:

$$\left(1 + \frac{i}{2}\right)^{-20} = 1.4(1 + i)^{-20}.$$

Rewrite $1 + \frac{i}{2}$ as $\frac{2+i}{2}$:

$$\left(\frac{2 + i}{2}\right)^{-20} = 1.4(1 + i)^{-20}.$$

This gives

$$\frac{(2 + i)^{-20}}{2^{-20}} = 1.4(1 + i)^{-20}.$$

Therefore,

$$\left(\frac{2 + i}{1 + i}\right)^{-20} = 1.4(2^{-20}).$$

Taking the $-\frac{1}{20}$-th power gives

$$\frac{2+i}{1+i} = 2(1.4)^{1/20}.$$

Thus,

$$\frac{2+i}{1+i} = 1.9666.$$

Solving,

$$\begin{aligned} 2+i &= 1.9666(1+i), \\ 2+i &= 1.9666 + 1.9666i, \\ 0.0334 &= 0.9666i, \\ i &= 0.034554. \end{aligned}$$

Now use Brian's repayment amount to find X:

$$X = 1120(1+i)^{-20}.$$

Substituting $i = 0.034554$, we get

$$X = 1120(1.034554)^{-20}.$$

Therefore,

$$X = 567.75.$$

The closest answer is

$$\boxed{568}.$$

Thus, the correct choice is $\boxed{\text{C}}$.

Exercise 1.1.23

Brian and Jennifer each take out a loan of X.
Jennifer will repay her loan by making one payment of \$800 at the end of year 10.
Brian will repay his loan by making one payment of \$1,120 at the end of year 10.
The nominal annual rate convertible semiannually being charged to Jennifer is exactly one-half the nominal annual rate convertible semiannually being charged to Brian.
Calculate X.

A. 562

B. 565

C. 568

D. 571

E. 574

Solution
Let i be the nominal annual rate convertible semiannually charged to Jennifer.
Then the nominal annual rate convertible semiannually charged to Brian is $2i$.
Since the loans are repaid at the end of 10 years and interest is convertible semiannually, there are

$$2(10) = 20$$

conversion periods.
For Jennifer, the effective semiannual rate is

$$\frac{i}{2}.$$

For Brian, the effective semiannual rate is

$$\frac{2i}{2} = i.$$

Since both loans have the same present value X, we equate the present values of the two repayment amounts:

$$800\left(1+\frac{i}{2}\right)^{-20} = 1120(1+i)^{-20}.$$

Divide both sides by 800:

$$\left(1+\frac{i}{2}\right)^{-20} = 1.4(1+i)^{-20}.$$

Rewrite $1+\frac{i}{2}$ as $\frac{2+i}{2}$:

$$\left(\frac{2+i}{2}\right)^{-20} = 1.4(1+i)^{-20}.$$

This gives

$$\frac{(2+i)^{-20}}{2^{-20}} = 1.4(1+i)^{-20}.$$

Therefore,

$$\left(\frac{2+i}{1+i}\right)^{-20} = 1.4(2^{-20}).$$

Taking the $-\frac{1}{20}$-th power gives

$$\frac{2+i}{1+i} = 2(1.4)^{1/20}.$$

Thus,

$$\frac{2+i}{1+i} = 1.9666.$$

Solving,

$$\begin{aligned} 2+i &= 1.9666(1+i), \\ 2+i &= 1.9666 + 1.9666i, \\ 0.0334 &= 0.9666i, \\ i &= 0.034554. \end{aligned}$$

Now use Brian's repayment amount to find X:

$$X = 1120(1+i)^{-20}.$$

Substituting $i = 0.034554$, we get

$$X = 1120(1.034554)^{-20}.$$

Therefore,

$$X = 567.75.$$

The closest answer is

$$\boxed{568}.$$

Thus, the correct choice is $\boxed{\text{C}}$.

Exercise 1.1.24

[SOA 3; SOA-IT.003] Eric deposits \$100 into a savings account at time 0, which pays interest at an annual nominal rate of i, compounded semiannually.
Mike deposits \$200 into a different savings account at time 0, which pays simple interest at an annual rate of i.
Eric and Mike earn the same amount of interest during the last 6 months of the 8th year.
Calculate i.

A. 9.06%

B. 9.26%

C. 9.46%

D. 9.66%

E. 9.86%

Solution
We need to equate the interest amounts earned during the last 6 months of the 8th year. This period runs from time 7.5 to time 8.
For Eric, the annual nominal rate i is compounded semiannually. Therefore, the effective semiannual rate is

$$\frac{i}{2}.$$

The account value at time 8 is

$$100\left(1+\frac{i}{2}\right)^{16},$$

and the account value at time 7.5 is

$$100\left(1+\frac{i}{2}\right)^{15}.$$

Thus, the interest earned by Eric during the last 6 months of the 8th year is

$$100\left(1+\frac{i}{2}\right)^{16}-100\left(1+\frac{i}{2}\right)^{15}.$$

For Mike, the account earns simple interest at annual rate i. His accumulated value at time t is

$$200(1+it).$$

Therefore, the interest earned by Mike from time 7.5 to time 8 is

$$200(1+8i)-200(1+7.5i).$$

Since the two interest amounts are equal,

$$100\left(1+\frac{i}{2}\right)^{16}-100\left(1+\frac{i}{2}\right)^{15}=200(1+8i)-200(1+7.5i).$$

Factor the left-hand side and simplify the right-hand side:

$$100\left(1+\frac{i}{2}\right)^{15}\left(\frac{i}{2}\right)=100i.$$

Since $i > 0$, divide both sides by $100i$:

$$\frac{1}{2}\left(1+\frac{i}{2}\right)^{15} = 1.$$

Thus,

$$\left(1+\frac{i}{2}\right)^{15} = 2.$$

Taking the 15-th root gives

$$1+\frac{i}{2} = 2^{1/15}.$$

Therefore,

$$\frac{i}{2} = 2^{1/15} - 1.$$

So

$$i = 2\left(2^{1/15} - 1\right).$$

Hence,

$$i = 0.0946.$$

Thus,

$$\boxed{i = 9.46\%}.$$

The correct choice is $\boxed{\text{C}}$.

1.1.7 Nominal Rate of Discount

Let us review some equivalent rates we have already calculated assuming

$$i = 10\%.$$

The equivalent annual effective rate of discount is

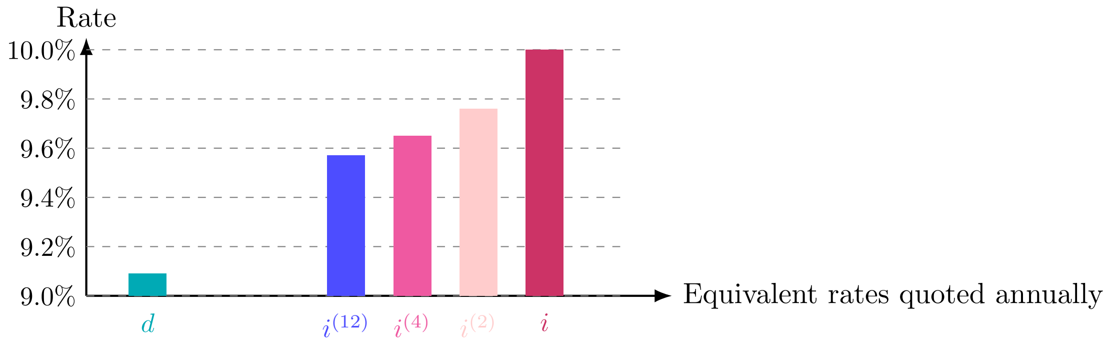
$$d = \frac{i}{1+i} = \frac{0.10}{1.10} = 0.090909.$$

Thus,

$$d = 9.0909\%.$$

We also computed several equivalent nominal rates of interest:

m	$i^{(m)}$
2	9.76%
4	9.65%
12	9.57%



We have

$$d = 9.0909\%,$$

while the nominal rates of interest lie between d and i. We can fill this gap using **nominal rates of discount**.

Before defining nominal discount rates, let k be the effective discount rate for $\frac{1}{m}$ of a year.

If d is the equivalent annual effective discount rate, then discounting \$1 for one full year must give the same present value whether we use d once or k over m equal subperiods.

Therefore,

$$(1 - d) = (1 - k)^m.$$

This follows the same logic as the relationship between a periodic effective interest rate and an annual effective interest rate.

Nominal Rate of Discount

The **nominal annual rate of discount** convertible m times per year is denoted by

$$d^{(m)}.$$

It is defined as m times the effective discount rate per conversion period.
If k is the effective discount rate for $\frac{1}{m}$ of a year, then

$$d^{(m)} = mk.$$

Equivalently,

$$k = \frac{d^{(m)}}{m}.$$

Using this notation, the one-period discount factor is

$$1 - k = 1 - \frac{d^{(m)}}{m}.$$

Since there are m conversion periods in one year, the annual discount factor is

$$\left(1 - \frac{d^{(m)}}{m}\right)^m.$$

This must equal the annual effective discount factor $1 - d$. Hence,

Equivalent Nominal and Effective Discount Rates

If $d^{(m)}$ is the nominal annual rate of discount convertible m times per year, and d is the equivalent annual effective discount rate, then

$$\left(1 - \frac{d^{(m)}}{m}\right)^m = 1 - d.$$

Equivalently,

$$d = 1 - \left(1 - \frac{d^{(m)}}{m}\right)^m.$$

The structure is parallel to nominal rates of interest:

$$\left(1 + \frac{i^{(m)}}{m}\right)^m = 1 + i,$$

but for discount rates we use subtraction:

$$\left(1 - \frac{d^{(m)}}{m}\right)^m = 1 - d.$$

This sign difference is important. Interest accumulates forward, while discounting moves values backward.

Nominal Discount Rates Equivalent to $d = 9.0909\%$

With

$$d = 9.0909\%,$$

solve for the nominal discount rates when $m = 2$, $m = 4$, and $m = 12$.

Solution
The nominal discount rate $d^{(m)}$ satisfies

$$\left(1 - \frac{d^{(m)}}{m}\right)^m = 1 - d.$$

Solving for $d^{(m)}$, we get

$$1 - \frac{d^{(m)}}{m} = (1 - d)^{1/m}.$$

Thus,

$$\frac{d^{(m)}}{m} = 1 - (1 - d)^{1/m}.$$

Therefore,

$$d^{(m)} = m\left[1 - (1 - d)^{1/m}\right].$$

For $m = 2$,

$$d^{(2)} = 2\left[1 - (1 - 0.090909)^{1/2}\right].$$

Thus,

$$d^{(2)} = 9.3075\%.$$

For $m = 4$,

$$d^{(4)} = 4\left[1 - (1 - 0.090909)^{1/4}\right].$$

Thus,

$$d^{(4)} = 9.4184\%.$$

For $m = 12$,

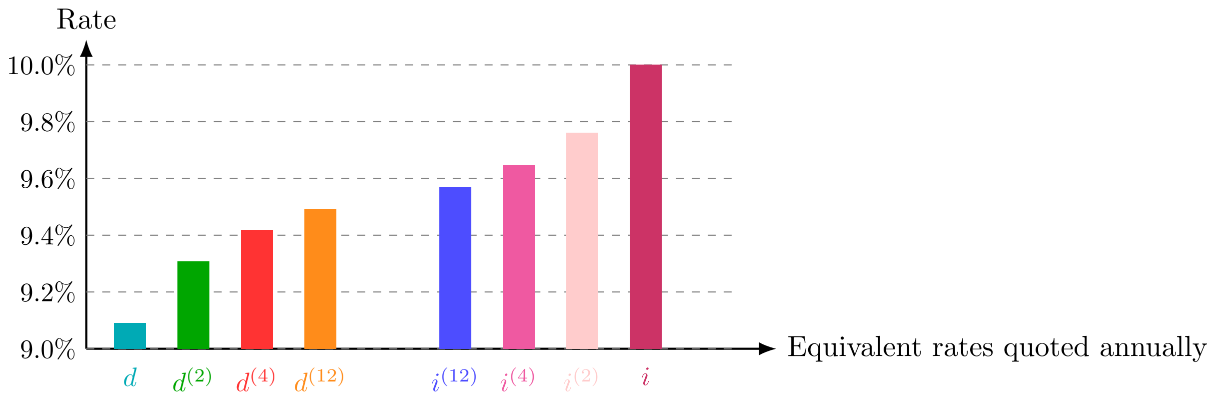
$$d^{(12)} = 12\left[1 - (1 - 0.090909)^{1/12}\right].$$

Thus,

$$d^{(12)} = 9.4933\%.$$

Let us complete the chart of equivalent rates quoted annually.

Rate Type	**Notation**	**Value**
Annual effective discount	d	9.0909%
Nominal discount convertible semiannually	$d^{(2)}$	9.3075%
Nominal discount convertible quarterly	$d^{(4)}$	9.4184%
Nominal discount convertible monthly	$d^{(12)}$	9.4933%
Nominal interest convertible monthly	$i^{(12)}$	9.5690%
Nominal interest convertible quarterly	$i^{(4)}$	9.6455%
Nominal interest convertible semiannually	$i^{(2)}$	9.7618%
Annual effective interest	i	10.0000%



Notice that i is the largest value and d is the smallest value.

Also, as the compounding frequency increases, the nominal discount rates and nominal interest rates move closer together.

In the limit, these two families of rates meet at the force of interest, which will be introduced later.

Converting Between $d^{(m)}$ and $d^{(n)}$

Suppose we want to convert between two nominal rates of discount:

$$d^{(m)} \qquad \text{and} \qquad d^{(n)}.$$

The key idea is to discount \$1 for one year using both rates.

Using the nominal discount rate $d^{(m)}$, the one-year discount factor is

$$\left(1 - \frac{d^{(m)}}{m}\right)^m.$$

Using the nominal discount rate $d^{(n)}$, the one-year discount factor is

$$\left(1 - \frac{d^{(n)}}{n}\right)^n.$$

Since the two rates are equivalent, these one-year discount factors must be equal:

$$\left(1 - \frac{d^{(n)}}{n}\right)^n = \left(1 - \frac{d^{(m)}}{m}\right)^m.$$

Taking the n-th root of both sides gives

$$1 - \frac{d^{(n)}}{n} = \left(1 - \frac{d^{(m)}}{m}\right)^{m/n}.$$

Therefore,

$$\frac{d^{(n)}}{n} = 1 - \left(1 - \frac{d^{(m)}}{m}\right)^{m/n}.$$

Multiplying by n, we obtain the conversion formula.

Converting Between Nominal Discount Rates

If $d^{(m)}$ and $d^{(n)}$ are equivalent nominal rates of discount, then

$$d^{(n)} = n\left[1 - \left(1 - \frac{d^{(m)}}{m}\right)^{m/n}\right].$$

This formula converts a nominal discount rate convertible m times per year into an equivalent nominal discount rate convertible n times per year.

Exercise 1.1.25

John invests \$1,000 in a fund which earns interest during the first year at a nominal rate of K convertible quarterly.
During the second year, the fund earns interest at a nominal discount rate of K convertible quarterly.
At the end of the second year, the fund has accumulated to \$1,173.54.
Calculate K.

A. 0.064

B. 0.068

C. 0.072

D. 0.076

E. 0.080

Solution
During the first year, the fund earns a nominal interest rate K convertible quarterly. Thus, the effective quarterly interest rate is

$$\frac{K}{4}.$$

The accumulation factor for the first year is therefore

$$\left(1+\frac{K}{4}\right)^4.$$

During the second year, the fund earns a nominal discount rate K convertible quarterly. Thus, the effective quarterly discount rate is

$$\frac{K}{4}.$$

The accumulation factor corresponding to a discount rate of $\frac{K}{4}$ per quarter is

$$\left(1-\frac{K}{4}\right)^{-1}.$$

Therefore, the accumulation factor for the second year is

$$\left(1-\frac{K}{4}\right)^{-4}.$$

The accumulated value after two years is

$$1{,}000\left(1+\frac{K}{4}\right)^4\left(1-\frac{K}{4}\right)^{-4}=1{,}173.54.$$

Divide by 1,000:

$$\left(1+\frac{K}{4}\right)^4\left(1-\frac{K}{4}\right)^{-4}=1.17354.$$

Equivalently,

$$\left(\frac{1+\frac{K}{4}}{1-\frac{K}{4}}\right)^4=1.17354.$$

Taking the fourth root gives

$$\frac{1+\frac{K}{4}}{1-\frac{K}{4}}=1.17354^{1/4}.$$

Let

$$r=1.17354^{1/4}.$$

Then

$$1+\frac{K}{4}=r\left(1-\frac{K}{4}\right).$$

Solving for K,

$$\frac{K}{4} = \frac{r-1}{r+1}.$$

Thus,

$$K = 4\left(\frac{1.17354^{1/4} - 1}{1.17354^{1/4} + 1}\right).$$

Therefore,

$$K = 0.0800.$$

Hence,

$$\boxed{K = 0.080}.$$

The correct choice is $\boxed{\text{E}}$.

Relationship Between Nominal Interest and Nominal Discount

If $i^{(n)}$ is a nominal annual rate of interest convertible n times per year, and $d^{(m)}$ is an equivalent nominal annual rate of discount convertible m times per year, then

$$\left(1 + \frac{i^{(n)}}{n}\right)^n = \left(1 - \frac{d^{(m)}}{m}\right)^{-m}.$$

Both sides represent the accumulated value of \$1 after one year.

Exercise 1.1.26

Express $i^{(9)}$ as a function of $d^{(3)}$.

A. $9\left(1 - \frac{d^{(3)}}{3}\right)^{1/3} - 9$

B. $9\left(1 - \frac{d^{(3)}}{3}\right)^{-1/3} - 9$

C. $9\left(1 + \frac{d^{(3)}}{3}\right)^{-1/3} - 9$

D. $9\left(1 - \frac{d^{(3)}}{3}\right)^{1/3} + 9$

E. $9\left(1 - \frac{d^{(3)}}{3}\right)^{-1/3} + 9$

Solution

We rewrite the equivalent annual accumulation factor using both nominal rates.
For the nominal interest rate $i^{(9)}$, the annual accumulation factor is

$$\left(1 + \frac{i^{(9)}}{9}\right)^9.$$

For the nominal discount rate $d^{(3)}$, the annual discount factor is

$$\left(1-\frac{d^{(3)}}{3}\right)^{3}.$$

Therefore, the corresponding annual accumulation factor is

$$\left(1-\frac{d^{(3)}}{3}\right)^{-3}.$$

Since the two rates are equivalent, we set the annual accumulation factors equal:

$$\left(1+\frac{i^{(9)}}{9}\right)^{9}=\left(1-\frac{d^{(3)}}{3}\right)^{-3}.$$

Taking the ninth root gives

$$1+\frac{i^{(9)}}{9}=\left(1-\frac{d^{(3)}}{3}\right)^{-1/3}.$$

Thus,

$$\frac{i^{(9)}}{9}=\left(1-\frac{d^{(3)}}{3}\right)^{-1/3}-1.$$

Multiplying by 9, we obtain

$$i^{(9)}=9\left[\left(1-\frac{d^{(3)}}{3}\right)^{-1/3}-1\right].$$

Equivalently,

$$i^{(9)}=9\left(1-\frac{d^{(3)}}{3}\right)^{-1/3}-9.$$

Therefore, the correct choice is $\boxed{\text{B}}$.

Exercise 1.1.27

At time $t=0$, John deposits \$1,000 into a fund which credits interest at a nominal interest rate of 10%, compounded semiannually.
At the same time, he deposits P into a different fund which credits interest at a nominal discount rate of 6%, compounded monthly.
At time $t=20$, the amounts in each fund are equal.
What is the annual effective interest rate earned on the total deposits, $1000+P$, over the 20-year period?

A. 0.0744

B. 0.0754

C. 0.0764

D. 0.0774

E. 0.0784

Solution
This is a two-part problem.
First, we use the information that the two funds have the same accumulated value at time 20 to solve for P.
For the first fund, the nominal interest rate is 10%, compounded semiannually. The effective semiannual interest rate is

$$\frac{0.10}{2} = 0.05.$$

Over 20 years, there are

$$20(2) = 40$$

semiannual periods. Therefore, the accumulated value of the first fund at time 20 is

$$1000(1.05)^{40}.$$

For the second fund, the nominal discount rate is 6%, compounded monthly. The effective monthly discount rate is

$$\frac{0.06}{12} = 0.005.$$

A discount rate of 0.005 per month corresponds to a monthly accumulation factor of

$$(1-0.005)^{-1}.$$

Over 20 years, there are

$$20(12) = 240$$

monthly periods. Therefore, the accumulated value of the second fund at time 20 is

$$P(1-0.005)^{-240}.$$

Since the two accumulated values are equal at time 20,

$$1000(1.05)^{40} = P(1-0.005)^{-240}.$$

Solving for P,

$$P = 1000(1.05)^{40}(1-0.005)^{240}.$$

Thus,

$$P = 2114.03.$$

Now let i be the annual effective interest rate earned on the total deposits $1000+P$ over the 20-year period.
At time 20, the two funds have equal accumulated values, so the total accumulated value is

$$2\left[1000(1.05)^{40}\right].$$

Therefore,

$$(1000+P)(1+i)^{20} = 2\left[1000(1.05)^{40}\right].$$

Using $P = 2114.03$, we get

$$(1000+2114.03)(1+i)^{20} = 2\left[1000(1.05)^{40}\right].$$

Hence,

$$1 + i = \left(\frac{2\left[1000(1.05)^{40}\right]}{1000 + 2114.03} \right)^{1/20}.$$

Thus,

$$i = \left(\frac{2\left[1000(1.05)^{40}\right]}{1000 + 2114.03} \right)^{1/20} - 1.$$

Therefore,

$$i = 0.07836.$$

So the annual effective interest rate is approximately

$$\boxed{0.0784}.$$

The correct choice is $\boxed{\text{E}}$.

1.1.8 Accumulation Function

Recall that an amount function measures the growth of a fund, assuming a time-0 deposit and no new money is added or withdrawn.

If the fund earns a constant effective interest rate i, then

$$A(t) = A(0)(1 + i)^t.$$

Accumulation Function

The **accumulation function**, denoted by $a(t)$, is the amount to which a deposit of \$1 at time 0 accumulates by time t.
If interest is credited at a constant effective rate i, then

$$a(t) = 1(1 + i)^t = (1 + i)^t.$$

The accumulation function depends only on the interest pattern, not on the size of the initial deposit.

Relationship Between Amount Function and Accumulation Function

For any accumulation function $a(t)$, the amount function satisfies

$$A(t) = A(0)\, a(t).$$

This relationship is true for any accumulation function, not only for constant effective interest.

Doubling Time

Suppose we want to find the time required for an investment of $A(0)$ to double, assuming interest is credited at a constant effective rate i.

We want

$$A(t) = 2A(0).$$

Using the amount function,
$$A(t) = A(0)a(t),$$
we get
$$2A(0) = A(0)a(t).$$
Divide both sides by $A(0)$:
$$a(t) = 2.$$
For constant effective interest,
$$a(t) = (1+i)^t.$$
Thus,
$$(1+i)^t = 2.$$
Taking natural logarithms,
$$t\ln(1+i) = \ln 2.$$
Therefore,
$$t = \frac{\ln 2}{\ln(1+i)}.$$

Doubling Time Under Constant Effective Interest

If an investment earns a constant effective interest rate i, then the time required for the investment to double is
$$t = \frac{\ln 2}{\ln(1+i)}.$$

We can make some observations.

1. The time required for an investment to double does not depend on the initial investment. It only depends on the accumulation function.
2. The accumulation function and the amount function grow at the same rate. The only difference is their starting value.
3. If we want to measure the time required for an investment to triple, we replace $\ln 2$ by $\ln 3$.

Accumulation Function and Amount Function

Josh invests \$300 into a fund earning a 6% annual effective interest rate.
Express the accumulation function and the amount function in terms of the number of years t.

Solution
Recall the amount function formula for a constant effective interest rate:
$$A(t) = A(0)(1+i)^t.$$
Here,
$$A(0) = 300 \qquad \text{and} \qquad i = 0.06.$$
Therefore,
$$A(t) = 300(1.06)^t.$$

The accumulation function is the amount to which \$1 accumulates by time t. Hence,

$$a(t) = (1.06)^t.$$

Thus,

$$\boxed{A(t) = 300(1.06)^t} \quad \text{and} \quad \boxed{a(t) = (1.06)^t}.$$

Exercise 1.1.28

You are given the following amount function:

$$A(t) = t^2 + 100.$$

Suppose \$50 was invested at $t = 4$.
What would be the accumulated value of the \$50 at $t = 6$?

A. 53.8

B. 57.2

C. 58.6

D. 61.5

E. 66.0

Solution
Since the deposit is made at time $t = 4$, we do not use $A(6)$ directly.
Instead, we use the growth factor from time 4 to time 6:

$$\frac{A(6)}{A(4)}.$$

First, compute

$$A(4) = 4^2 + 100 = 116,$$

and

$$A(6) = 6^2 + 100 = 136.$$

Therefore, the accumulated value at $t = 6$ is

$$50 \cdot \frac{A(6)}{A(4)}.$$

Substituting the values gives

$$50 \cdot \frac{136}{116} = 58.6207.$$

Thus, the accumulated value is approximately

$$\boxed{58.6}.$$

The correct choice is $\boxed{\text{C}}$.

Exercise 1.1.29

Gertrude deposits \$10,000 in a bank. During the first year, the bank credits an annual effective rate of interest i. During the second year, the bank credits an annual effective rate of interest $i - 5\%$. At the end of two years, she has \$12,093.75 in the bank.
What would Gertrude have in the bank at the end of three years, if the annual effective rate of interest were $i + 9\%$ for each of the three years?

A. 16,851

B. 17,196

C. 17,499

D. 17,936

E. 18,113

Solution
Our equation of value is

$$10{,}000(1 + i)(1 + i - 0.05) = 12{,}093.75.$$

Solving for i, we get

$$(1 + i)(1 + i - 0.05) = 1.209375.$$

Thus,

$$(1 + i)(0.95 + i) = 1.209375.$$

Expanding,

$$0.95 + i + 0.95i + i^2 = 1.209375.$$

Therefore,

$$i^2 + 1.95i - 0.259375 = 0.$$

Using the quadratic formula,

$$i = \frac{-1.95 \pm \sqrt{1.95^2 - 4(1)(-0.259375)}}{2}.$$

The positive solution is

$$i = 0.125.$$

So

$$i = 12.5\%.$$

Now the new annual effective rate is

$$i + 9\% = 12.5\% + 9\% = 21.5\%.$$

The accumulated value after three years at this rate is

$$10{,}000(1.215)^3.$$

Thus,

$$10{,}000(1.215)^3 = 17{,}936.13.$$

Therefore, Gertrude would have approximately

$$\boxed{\$17{,}936}.$$

The correct choice is $\boxed{\text{D}}$.

1.1.9 Summary

Recall that an **amount function** measures the growth of a fund, assuming a time-0 deposit and no new deposits or withdrawals afterward.

The amount function is denoted by
$$A(t),$$
where $A(t)$ is the amount in the fund at time t.

If the initial fund is \$1, then the amount function becomes a special case called the **accumulation function**. The accumulation function is denoted by

$$a(t).$$

Thus, $a(t)$ measures the amount to which \$1 invested at time 0 accumulates by time t.

The relationship between the amount function and the accumulation function is

$$A(t) = A(0)a(t).$$

Effective Interest and Discount Rates

The effective interest rate during period t is defined by

$$i_t = \frac{I(t)}{A(t-1)} = \frac{A(t) - A(t-1)}{A(t-1)}.$$

The effective discount rate during period t is defined by

$$d_t = \frac{I(t)}{A(t)} = \frac{A(t) - A(t-1)}{A(t)}.$$

Main Amount Functions

If the rates are constant, the main amount functions are summarized below.

Interest Measure	Amount Function
Effective Interest Rate	$A(t) = A(0)(1+i)^t$
Effective Discount Rate	$A(t) = A(0)(1-d)^{-t}$
Nominal Interest Rate	$A(t) = A(0)\left(1+\frac{i^{(m)}}{m}\right)^{mt}$
Nominal Discount Rate	$A(t) = A(0)\left(1-\frac{d^{(m)}}{m}\right)^{-mt}$

Since all four expressions describe equivalent ways to accumulate money over time, we have the following equivalence:

$$(1+i)^t = (1-d)^{-t} = \left(1+\frac{i^{(m)}}{m}\right)^{mt} = \left(1-\frac{d^{(m)}}{m}\right)^{-mt}.$$

Discount Factor

The discount factor v is the factor used to discount one period:

$$v = (1+i)^{-1}.$$

Since

$$d = \frac{i}{1+i},$$

we also have

$$v = 1-d.$$

Therefore,

$$\boxed{v = \frac{1}{1+i} = 1-d}.$$

1.2 Force of Interest

1.2.1 What Is the Force of Interest?

All the interest measures introduced so far are useful for measuring interest over specified intervals of time, such as one year, six months, one quarter, or one month.

However, we also need a way to measure the **instantaneous intensity of interest** at an exact moment in time. In other words, we want to measure interest over infinitesimally small intervals of time.

This measure is called the **force of interest**.

Force of Interest

The **force of interest** at time t, denoted by δ_t, measures the instantaneous intensity of interest at exact time t.
It is expressed as a rate per measurement period, usually per year unless stated otherwise.

Thus, while an effective interest rate measures growth over a finite interval, the force of interest measures the rate of growth at a single instant.

- δ_t measures the intensity of interest at exact time t.
- δ_t is usually expressed as an annualized rate.
- If the force of interest is constant, we write it simply as δ.

Consider a fund with balance $A(t)$ at time t.

Assume that the only force acting on the fund is interest. Then the rate of change of $A(t)$ is measured by the slope of the amount function.

Since the slope of a curve is found by taking the first derivative, the instantaneous rate of change of the fund is

$$A'(t).$$

However, $A'(t)$ alone is not sufficient to measure the intensity of interest, because it depends on the amount invested.

For example, if \$1,000 and \$500 are invested in the same fund, the rate of change of the \$1,000 investment will be twice as large as the rate of change of the \$500 investment. But interest is not operating with twice the intensity. The fund is simply twice as large.

To measure the intensity of interest independently of the amount invested, we divide by the current fund balance $A(t)$.

Force of Interest

The force of interest at time t is defined by

$$\delta_t = \frac{A'(t)}{A(t)}.$$

Since

$$A(t) = A(0)a(t),$$

we also have

$$A'(t) = A(0)a'(t).$$

Therefore,

$$\delta_t = \frac{A'(t)}{A(t)} = \frac{A(0)a'(t)}{A(0)a(t)} = \frac{a'(t)}{a(t)}.$$

Thus,

$$\boxed{\delta_t = \frac{A'(t)}{A(t)} = \frac{a'(t)}{a(t)}}.$$

This formula says that the force of interest is the instantaneous relative rate of growth of the fund.

No External Cash Flows

When we say that the only force acting on the fund is interest, we mean that the fund changes only because interest is being credited.
There are no additional deposits, withdrawals, fees, taxes, or other external cash flows.
This assumption allows us to interpret $A'(t)$ as the instantaneous growth caused by interest alone. Therefore,

$$\delta_t = \frac{A'(t)}{A(t)}$$

measures the pure instantaneous intensity of interest.

1.2.2 Accumulation Function from the Force of Interest

We know that the force of interest satisfies

$$\delta_t = \frac{a'(t)}{a(t)}.$$

Since

$$\frac{a'(t)}{a(t)} = \frac{d}{dt} \ln a(t),$$

we can write

$$\delta_t = \frac{d}{dt} \ln a(t).$$

Integrating both sides from 0 to n, we obtain

$$\int_0^n \delta_s \, ds = \int_0^n \frac{d}{ds} \ln a(s) \, ds.$$

Therefore,

$$\int_0^n \delta_s \, ds = \ln a(n) - \ln a(0).$$

Since $a(0) = 1$, we have

$$\ln a(0) = \ln 1 = 0.$$

Thus,

$$\int_0^n \delta_s \, ds = \ln a(n).$$

Taking the exponential of both sides gives the standard accumulation-function formula (see the formula box below).

Accumulation Function from the Force of Interest

If δ_t is the force of interest at time t, then the accumulation function from time 0 to time n is

$$a(n) = \exp\left(\int_0^n \delta_t \, dt\right).$$

In words, to accumulate from time 0 to time n, we add up the force of interest over the interval $[0, n]$, then take the exponential.

The value n does not have to be an integer.

1.2.3 Accumulation Factor Between Two Times

Let $a(t_1, t_2)$ denote the accumulation factor from time t_1 to time t_2.

In other words, to accumulate from time t_1 to time t_2, we multiply by $a(t_1, t_2)$.

By definition,

$$a(t_1, t_2) = \frac{a(t_2)}{a(t_1)}.$$

Using the force of interest formula,

$$a(t_2) = \exp\left(\int_0^{t_2} \delta_s \, ds\right),$$

and

$$a(t_1) = \exp\left(\int_0^{t_1} \delta_s \, ds\right).$$

Therefore,

$$a(t_1, t_2) = \frac{\exp\left(\int_0^{t_2} \delta_s \, ds\right)}{\exp\left(\int_0^{t_1} \delta_s \, ds\right)}.$$

Using the rule $\frac{e^A}{e^B} = e^{A-B}$, we get

$$a(t_1, t_2) = \exp\left(\int_0^{t_2} \delta_s \, ds - \int_0^{t_1} \delta_s \, ds\right).$$

Thus, the accumulation factor between two times can be expressed in terms of the force of interest (see the formula box below).

Accumulation Factor Between Two Times

The accumulation factor from time t_1 to time t_2 is

$$a(t_1, t_2) = \frac{a(t_2)}{a(t_1)} = \exp\left(\int_{t_1}^{t_2} \delta_t \, dt\right).$$

In words, to accumulate from time t_1 to time t_2, we add up the force of interest from t_1 to t_2, then take the exponential.

Interpreting the Force of Interest

The force of interest is like the **instantaneous speed of growth** of a fund.
The derivative

$$A'(t)$$

does not represent the moment when the amount changes. It represents the **speed at which the amount is changing at time** t.
This is similar to physics:

$$\text{position } s(t) \quad \longrightarrow \quad \text{velocity } s'(t).$$

In interest theory:

$$\text{amount } A(t) \quad \longrightarrow \quad \text{instantaneous growth speed } A'(t).$$

However, $A'(t)$ is measured in dollars per year and depends on the size of the fund. A fund with \$1,000 will usually grow by more dollars than a fund with \$500, even if both funds earn interest at the same intensity.
To remove the effect of the fund size, we divide by the current amount $A(t)$:

$$\delta_t = \frac{A'(t)}{A(t)}.$$

Thus,

$$A'(t) = \text{instantaneous growth speed in dollars},$$

while

$$\delta_t = \frac{A'(t)}{A(t)} = \text{instantaneous relative growth rate}.$$

So the force of interest measures the instantaneous intensity of interest, independently of the amount invested.

Accumulating with a Force of Interest

If

$$\delta_t = 0.15\sqrt{t},$$

and an amount of \$5,000 is invested at time $t = 1$, what is the accumulated value at time $t = 4$?

Solution
To accumulate from time 1 to time 4, we use

$$a(1,4) = \exp\left(\int_1^4 \delta_t \, dt\right).$$

Therefore, the accumulated value is

$$5{,}000a(1,4) = 5{,}000 \exp\left(\int_1^4 \delta_t \, dt\right).$$

Substitute

$$\delta_t = 0.15\sqrt{t} = 0.15t^{1/2}.$$

Then

$$5{,}000a(1,4) = 5{,}000 \exp\left(\int_1^4 0.15t^{1/2} \, dt\right).$$

Compute the integral:

$$\int_1^4 0.15t^{1/2}\,dt = \left[\frac{0.15}{1.5}t^{1.5}\right]_1^4.$$

Since

$$\frac{0.15}{1.5} = 0.10,$$

we get

$$\left[0.10t^{1.5}\right]_1^4 = 0.10\left(4^{1.5} - 1^{1.5}\right).$$

Now,

$$4^{1.5} = 4^{3/2} = 8,$$

so

$$0.10(8-1) = 0.70.$$

Thus,

$$5{,}000a(1,4) = 5{,}000e^{0.70}.$$

Therefore,

$$5{,}000e^{0.70} = 10{,}068.76.$$

The accumulated value at time $t = 4$ is

$$\boxed{\$10{,}068.76}.$$

1.2.4 Discount Function

The accumulation function $a(t)$ accumulates cash flows from time 0 to time t.

The **discount function** discounts cash flows from time t back to time 0. It is given by

$$\frac{1}{a(t)}.$$

Since

$$a(t) = \exp\left(\int_0^t \delta_s\,ds\right),$$

the discount function is

$$\frac{1}{a(t)} = \frac{1}{\exp\left(\int_0^t \delta_s\,ds\right)}.$$

Therefore, the discount function can also be written in terms of the force of interest (see the formula box below).

Discount Function

If δ_t is the force of interest, then the discount function from time t to time 0 is

$$\frac{1}{a(t)} = \exp\left(-\int_0^t \delta_s\,ds\right).$$

Equivalently, using the notation $a^{-1}(t)$,

$$a^{-1}(t) = \frac{1}{a(t)} = \exp\left(-\int_0^t \delta_s\,ds\right).$$

To accumulate from time 0 to time t, we integrate the force of interest over $[0, t]$ and exponentiate. To discount, we do the same but with a negative sign in the exponent.

Notation

In this section, $a^{-1}(t)$ means the reciprocal of the accumulation function:

$$a^{-1}(t) = \frac{1}{a(t)}.$$

This is the financial mathematics notation for the discount function.

Let $a^{-1}(t_1, t_2)$ denote the discount factor from time t_2 back to time t_1.

In other words, to discount a cash flow from time t_2 to time t_1, we multiply by $a^{-1}(t_1, t_2)$.

Since $a(t_1, t_2)$ is the accumulation factor from time t_1 to time t_2, the discount factor is its reciprocal:

$$a^{-1}(t_1, t_2) = \frac{1}{a(t_1, t_2)}.$$

Using the force of interest, we know that

$$a(t_1, t_2) = \exp\left(\int_{t_1}^{t_2} \delta_s \, ds\right).$$

Therefore,

$$a^{-1}(t_1, t_2) = \frac{1}{\exp\left(\int_{t_1}^{t_2} \delta_s \, ds\right)}.$$

Thus,

$$a^{-1}(t_1, t_2) = \exp\left(-\int_{t_1}^{t_2} \delta_s \, ds\right).$$

Discount Factor Between Two Times

The discount factor from time t_2 back to time t_1 is

$$a^{-1}(t_1, t_2) = \frac{1}{a(t_1, t_2)} = \exp\left(-\int_{t_1}^{t_2} \delta_t \, dt\right).$$

In words, to discount from time t_2 to time t_1, add up the force of interest from t_1 to t_2, take the negative, and then take the exponential.

Exercise 1.2.1

On July 1, 1984, a person invested \$1,000 in a fund for which the force of interest at time t is given by

$$\delta_t = \frac{3 + 2t}{50},$$

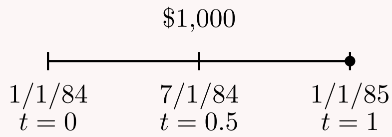
where t is the number of years since January 1, 1984.
Determine the accumulated value of the investment on January 1, 1985.

Solution
Since t is measured from January 1, 1984, we have:

$$\text{July 1, 1984} \quad \longrightarrow \quad t = 0.5,$$

and

$$\text{January 1, 1985} \quad \longrightarrow \quad t = 1.$$



To accumulate from time 0.5 to time 1, we use

$$a(0.5, 1) = \exp\left(\int_{0.5}^{1} \delta_s \, ds\right).$$

Therefore, the accumulated value is

$$1000a(0.5, 1) = 1000 \exp\left(\int_{0.5}^{1} \delta_s \, ds\right).$$

Substitute

$$\delta_s = \frac{3 + 2s}{50}.$$

Then

$$1000a(0.5, 1) = 1000 \exp\left(\int_{0.5}^{1} \frac{3 + 2s}{50} \, ds\right).$$

Compute the integral:

$$\int_{0.5}^{1} \frac{3 + 2s}{50} \, ds = \left[\frac{1}{50}(3s + s^2)\right]_{0.5}^{1}.$$

Thus,

$$\left[\frac{1}{50}(3s + s^2)\right]_{0.5}^{1} = \frac{1}{50}\left[(3(1) + 1^2) - (3(0.5) + (0.5)^2)\right].$$

So

$$\frac{1}{50}[4 - 1.75] = \frac{2.25}{50} = 0.045.$$

Therefore,

$$1000a(0.5, 1) = 1000e^{0.045}.$$

Hence,

$$1000e^{0.045} = 1046.03.$$

The accumulated value on January 1, 1985, is

$$\boxed{\$1{,}046.03}.$$

Exercise 1.2.2

X is deposited into a savings account at time $t = 0$. No other amounts are deposited. The force of interest for the fund is

$$\delta_t = \frac{t}{30}.$$

The balance of the fund after 10 years is $12,500.
Determine X.

Solution

The accumulated value at time 10 is

$$12{,}500.$$

Since X was deposited at time 0, we have

$$12{,}500 = X \exp\left(\int_0^{10} \delta_t \, dt\right).$$

Substitute

$$\delta_t = \frac{t}{30}.$$

Then

$$12{,}500 = X \exp\left(\int_0^{10} \frac{t}{30} \, dt\right).$$

Solving for X,

$$X = 12{,}500 \exp\left(-\int_0^{10} \frac{t}{30} \, dt\right).$$

Compute the integral:

$$\int_0^{10} \frac{t}{30} \, dt = \left[\frac{t^2}{60}\right]_0^{10}.$$

Thus,

$$\left[\frac{t^2}{60}\right]_0^{10} = \frac{100}{60} = \frac{5}{3}.$$

Therefore,

$$X = 12{,}500 e^{-5/3}.$$

Hence,

$$X = 2{,}360.95.$$

So the initial deposit was

$$\boxed{\$2{,}360.95}.$$

Exercise 1.2.3

\$1,000 is deposited into a savings account at time $t = 0$. No other amounts are deposited. The accumulated amount in the account at time t is given by

$$A(t) = 1000\left(1 + \frac{2t}{35}\right)^2.$$

Determine the force of interest at time $t = 32.5$.

Solution

Recall that the force of interest is

$$\delta_t = \frac{A'(t)}{A(t)}.$$

We are given

$$A(t) = 1000\left(1 + \frac{2t}{35}\right)^2.$$

Differentiate:

$$A'(t) = 1000 \cdot 2\left(1 + \frac{2t}{35}\right)\left(\frac{2}{35}\right).$$

Thus,

$$A'(t) = \frac{4000}{35}\left(1 + \frac{2t}{35}\right).$$

Therefore,

$$\delta_t = \frac{\frac{4000}{35}\left(1 + \frac{2t}{35}\right)}{1000\left(1 + \frac{2t}{35}\right)^2}.$$

Cancel common factors:

$$\delta_t = \frac{\frac{4}{35}}{1 + \frac{2t}{35}}.$$

Now evaluate at $t = 32.5$:

$$\delta_{32.5} = \frac{\frac{4}{35}}{1 + \frac{2(32.5)}{35}}.$$

Since

$$2(32.5) = 65,$$

we get

$$\delta_{32.5} = \frac{\frac{4}{35}}{1 + \frac{65}{35}}.$$

Thus,

$$\delta_{32.5} = \frac{\frac{4}{35}}{\frac{100}{35}} = \frac{4}{100} = 0.04.$$

Therefore, the force of interest at time $t = 32.5$ is

$$\boxed{0.04}.$$

Exercise 1.2.4

A deposit of 1 will accumulate to 2.7183 in 10 years with a force of interest

$$\delta_t = \begin{cases} kt, & 0 < t < 5, \\ 0.04kt^2, & 5 < t \leq 10. \end{cases}$$

Determine k.

Solution

We use the accumulation formula under a force of interest:

$$a(10) = \exp\left(\int_0^{10} \delta_t\, dt\right).$$

Since the force of interest is defined piecewise, split the accumulation into two parts:

$$a(10) = a(0,5)a(5,10).$$

Thus,

$$2.7183 = \exp\left(\int_0^5 kt\, dt\right)\exp\left(\int_5^{10} 0.04kt^2\, dt\right).$$

Compute the first integral:

$$\int_0^5 kt\, dt = k\left[\frac{t^2}{2}\right]_0^5 = \frac{25}{2}k = 12.5k.$$

Compute the second integral:

$$\int_5^{10} 0.04kt^2\,dt = 0.04k\left[\frac{t^3}{3}\right]_5^{10}.$$

Therefore,

$$\int_5^{10} 0.04kt^2\,dt = 0.04k\left(\frac{10^3-5^3}{3}\right).$$

Since

$$10^3 - 5^3 = 1000 - 125 = 875,$$

we get

$$0.04k\left(\frac{875}{3}\right) = \frac{35}{3}k.$$

Thus,

$$2.7183 = \exp\left(12.5k + \frac{35}{3}k\right).$$

Since

$$12.5 = \frac{25}{2},$$

we have

$$\frac{25}{2}k + \frac{35}{3}k = \frac{145}{6}k.$$

So

$$2.7183 = e^{\frac{145}{6}k}.$$

Taking natural logarithms,

$$\ln(2.7183) = \frac{145}{6}k.$$

Therefore,

$$k = \frac{6\ln(2.7183)}{145}.$$

Hence,

$$k = 0.0414.$$

Thus,

$$\boxed{k = 0.0414}.$$

Exercise 1.2.5

Given two accumulation functions:

$$a_1(t) = 1 + 0.08t$$

and

$$a_2(t) = 1.05^t.$$

At what time t will the forces of interest be equal for the two functions?

Solution
We solve this by applying the definition of the force of interest in terms of the accumulation function:

$$\delta_t = \frac{a'(t)}{a(t)}.$$

For the first accumulation function,

$$a_1(t) = 1 + 0.08t.$$

Thus,

$$a_1'(t) = 0.08.$$

Therefore,

$$\delta_{1,t} = \frac{a_1'(t)}{a_1(t)} = \frac{0.08}{1 + 0.08t}.$$

For the second accumulation function,

$$a_2(t) = 1.05^t.$$

Thus,

$$a_2'(t) = 1.05^t \ln(1.05).$$

Therefore,

$$\delta_{2,t} = \frac{a_2'(t)}{a_2(t)} = \frac{1.05^t \ln(1.05)}{1.05^t} = \ln(1.05).$$

Set the two forces of interest equal:

$$\frac{0.08}{1 + 0.08t} = \ln(1.05).$$

Solving for t,

$$1 + 0.08t = \frac{0.08}{\ln(1.05)}.$$

Thus,

$$0.08t = \frac{0.08}{\ln(1.05)} - 1.$$

Therefore,

$$t = \frac{\frac{0.08}{\ln(1.05)} - 1}{0.08}.$$

Hence,

$$t = 7.99.$$

So the forces of interest are equal at approximately

$$\boxed{t = 8}.$$

Exercise 1.2.6

A fund earns interest at a force of interest

$$\delta_t = kt.$$

A deposit of \$100 at time $t = 0$ will grow to \$250 at the end of 5 years. Determine k.

A. $0.08 \ln(1.5)$

B. 0.06

C. $0.08\ln(2.5)$

D. 0.08

E. $\ln(2.5)$

Solution
Recall that the accumulation factor for a given force of interest is

$$\exp\left(\int \delta_t \, dt\right).$$

Since the deposit is made at time 0 and accumulated to time 5, we have

$$100\exp\left(\int_0^5 kt \, dt\right) = 250.$$

Compute the integral:

$$\int_0^5 kt \, dt = \left[\frac{kt^2}{2}\right]_0^5.$$

Thus,

$$\int_0^5 kt \, dt = \frac{k(5)^2}{2} = 12.5k.$$

Therefore,

$$100\exp(12.5k) = 250.$$

Divide both sides by 100:

$$\exp(12.5k) = 2.5.$$

Taking natural logarithms gives

$$12.5k = \ln(2.5).$$

Hence,

$$k = \frac{\ln(2.5)}{12.5}.$$

Since

$$\frac{1}{12.5} = 0.08,$$

we obtain

$$k = 0.08\ln(2.5).$$

Therefore,

$$\boxed{k = 0.08\ln(2.5)}.$$

The correct choice is $\boxed{\text{C}}$.

Exercise 1.2.7

In Fund X, money accumulates at a force of interest

$$\delta_t = 0.01t + 0.1, \qquad 0 \le t \le 20.$$

In Fund Y, money accumulates at an annual effective interest rate i.
An amount of 1 is invested in each of Fund X and Fund Y for 20 years. The value of Fund

X at the end of 20 years is equal to the value of Fund Y at the end of 20 years. Calculate the value of Fund Y at the end of 1.5 years.

A. $e^{0.1}$

B. $e^{0.2}$

C. $e^{0.3}$

D. $e^{0.4}$

E. $e^{0.5}$

Solution
First, calculate the accumulated value of Fund X after 20 years.
Since Fund X accumulates with force of interest

$$\delta_t = 0.01t + 0.1,$$

its accumulation factor over 20 years is

$$\exp\left(\int_0^{20} (0.01t + 0.1)\, dt\right).$$

Compute the integral:

$$\int_0^{20} (0.01t + 0.1)\, dt = \left[0.005t^2 + 0.1t\right]_0^{20}.$$

Thus,

$$\left[0.005t^2 + 0.1t\right]_0^{20} = 0.005(20)^2 + 0.1(20).$$

Therefore,

$$0.005(400) + 2 = 2 + 2 = 4.$$

So the value of Fund X at the end of 20 years is

$$e^4.$$

Fund Y accumulates at an annual effective interest rate i. Since an amount of 1 is invested, its value after 20 years is

$$(1 + i)^{20}.$$

The two fund values are equal at time 20, so

$$(1 + i)^{20} = e^4.$$

Now we want the value of Fund Y at time 1.5:

$$(1 + i)^{1.5}.$$

Using

$$(1 + i)^{20} = e^4,$$

we get

$$(1 + i)^{1.5} = \left((1 + i)^{20}\right)^{1.5/20}.$$

Substitute $(1+i)^{20} = e^4$:

$$(1+i)^{1.5} = (e^4)^{1.5/20}.$$

Thus,

$$(1+i)^{1.5} = e^{4(1.5/20)}.$$

Since

$$4\left(\frac{1.5}{20}\right) = 0.3,$$

we obtain

$$(1+i)^{1.5} = e^{0.3}.$$

Therefore, the value of Fund Y at the end of 1.5 years is

$$\boxed{e^{0.3}}.$$

The correct choice is $\boxed{\text{C}}$.

Shortcut: Force of Interest to Accumulation Function

Sometimes it is easy to look at a given force of interest and immediately write down the accumulation function.

Finding the Accumulation Function by Inspection

Given

$$\delta_t = \frac{2t}{t^2+1},$$

find $a(t)$.

Solution
Recall that

$$\delta_t = \frac{a'(t)}{a(t)}.$$

Here,

$$\delta_t = \frac{2t}{t^2+1}.$$

Since

$$\frac{d}{dt}(t^2+1) = 2t,$$

we can recognize that

$$a(t) = t^2+1.$$

Indeed,

$$\frac{a'(t)}{a(t)} = \frac{2t}{t^2+1} = \delta_t.$$

So the accumulation function is

$$\boxed{a(t) = t^2+1}.$$

We could also find $a(t)$ using the usual force of interest formula:

$$a(t) = \exp\left(\int_0^t \delta_s\, ds\right).$$

Substitute

$$\delta_s = \frac{2s}{s^2+1}.$$

Then

$$a(t) = \exp\left(\int_0^t \frac{2s}{s^2+1}\,ds\right).$$

Since

$$\int \frac{2s}{s^2+1}\,ds = \ln(s^2+1),$$

we get

$$a(t) = \exp\left(\left[\ln(s^2+1)\right]_0^t\right).$$

Thus,

$$a(t) = \exp\left(\ln(t^2+1) - \ln(0^2+1)\right).$$

Since

$$\ln(1) = 0,$$

we have

$$a(t) = \exp\left(\ln(t^2+1)\right).$$

Therefore,

$$a(t) = t^2 + 1.$$

Avoid This Trap

Given

$$\delta_t = \frac{2t}{t^2+8},$$

find $a(t)$.
It may be tempting to write

$$a(t) = t^2 + 8,$$

because

$$\frac{d}{dt}(t^2+8) = 2t.$$

However, this is not correct because an accumulation function must satisfy

$$a(0) = 1.$$

But if

$$a(t) = t^2 + 8,$$

then

$$a(0) = 8,$$

which is impossible for an accumulation function.
Instead, rewrite the force of interest as

$$\delta_t = \frac{2t}{t^2+8} = \frac{\frac{2t}{8}}{\frac{t^2+8}{8}} = \frac{\frac{1}{4}t}{\frac{1}{8}t^2+1}.$$

Now observe that

$$\frac{d}{dt}\left(\frac{1}{8}t^2+1\right) = \frac{1}{4}t.$$

Therefore, the correct accumulation function is

$$a(t) = \frac{1}{8}t^2 + 1.$$

Indeed,

$$\frac{a'(t)}{a(t)} = \frac{\frac{1}{4}t}{\frac{1}{8}t^2 + 1} = \frac{2t}{t^2 + 8}.$$

Thus,

$$\boxed{a(t) = \frac{1}{8}t^2 + 1}.$$

Scaling the Force of Interest

Suppose a force of interest δ_t produces an accumulation function $a(t)$.

Now suppose we multiply the force of interest by a constant factor k, so that

$$\delta_t^* = k\delta_t.$$

We want to find the new accumulation function $a^*(t)$.

Using the force of interest formula,

$$a^*(t) = \exp\left(\int_0^t \delta_s^*\, ds\right).$$

Substitute $\delta_s^* = k\delta_s$:

$$a^*(t) = \exp\left(\int_0^t k\delta_s\, ds\right).$$

Since k is constant,

$$a^*(t) = \exp\left(k\int_0^t \delta_s\, ds\right).$$

Therefore,

$$a^*(t) = \left[\exp\left(\int_0^t \delta_s\, ds\right)\right]^k.$$

But

$$a(t) = \exp\left(\int_0^t \delta_s\, ds\right).$$

Thus,

$$a^*(t) = [a(t)]^k.$$

Scaling the Force of Interest

If a force of interest δ_t has accumulation function $a(t)$, then the scaled force of interest

$$\delta_t^* = k\delta_t$$

has accumulation function

$$a^*(t) = [a(t)]^k.$$

Using the Scaling Shortcut

Given

$$\delta_t = \frac{4}{1+t},$$

find $a(t)$.

Solution
We can write

$$\delta_t = 4\left(\frac{1}{1+t}\right).$$

Now observe that

$$\frac{1}{1+t}$$

is the force of interest corresponding to the accumulation function

$$a_0(t) = 1 + t.$$

Indeed,

$$\frac{a_0'(t)}{a_0(t)} = \frac{1}{1+t}.$$

Since the given force of interest is 4 times this force, the accumulation function is

$$a(t) = [a_0(t)]^4.$$

Therefore,

$$a(t) = (1+t)^4.$$

Thus,

$$\boxed{a(t) = (1+t)^4}.$$

Compound Interest and Constant Force of Interest

Suppose the accumulation function is

$$a(t) = (1+i)^t.$$

Then the force of interest is

$$\delta_t = \frac{a'(t)}{a(t)}.$$

Since

$$a'(t) = (1+i)^t \ln(1+i),$$

we get

$$\delta_t = \frac{(1+i)^t \ln(1+i)}{(1+i)^t}.$$

Therefore,

$$\delta_t = \ln(1+i).$$

Since this expression does not depend on t, the force of interest is constant. We write

$$\delta = \ln(1+i).$$

Equivalently,

$$e^{\delta} = 1 + i,$$

so

$$i = e^{\delta} - 1.$$

Constant Force of Interest

If the force of interest is constant and equal to δ, then

$$a(t) = e^{\delta t}.$$

The accumulation factor from time t_1 to time t_2 is

$$a(t_1, t_2) = e^{\delta(t_2 - t_1)}.$$

The discount factor from time t_2 back to time t_1 is

$$a^{-1}(t_1, t_2) = e^{-\delta(t_2 - t_1)}.$$

Also, the annual effective interest rate equivalent to a constant force of interest δ is

$$i = e^{\delta} - 1.$$

Accumulation with a Constant Force of Interest

If $\delta = 0.05$ and an amount of \$5,000 is invested at time $t = 1$, what is the accumulated value at time $t = 4$?

Solution
Since the force of interest is constant, the accumulation factor from time 1 to time 4 is

$$a(1, 4) = e^{0.05(4-1)}.$$

Thus, the accumulated value is

$$5{,}000e^{0.05(3)}.$$

Therefore,

$$5{,}000e^{0.15} = 5{,}809.17.$$

So the accumulated value at time $t = 4$ is

$$\boxed{\$5{,}809.17}.$$

Discounting with a Constant Force of Interest

If $\delta = 0.08$, find the present value of \$1,000 to be paid in 4.25 years.

Solution
Since the payment is made at time 4.25, we discount it back to time 0.
The discount factor is

$$e^{-0.08(4.25)}.$$

Thus, the present value is

$$1{,}000e^{-0.08(4.25)}.$$

Therefore,

$$1{,}000e^{-0.34} = 711.77.$$

So the present value is

$$\boxed{\$711.77}.$$

Simple Interest and Force of Interest

Suppose the accumulation function is simple interest:

$$a(t) = 1 + it.$$

Then the force of interest is

$$\delta_t = \frac{a'(t)}{a(t)}.$$

Since

$$a'(t) = i,$$

we get

$$\delta_t = \frac{i}{1+it}.$$

Thus, under simple interest,

$$\boxed{\delta_t = \frac{i}{1+it}}.$$

This force of interest is not constant. It decreases over time because the denominator $1 + it$ increases as t increases.

Simple Interest and Force of Interest

Under simple interest, the amount of interest earned each period is constant, but the force of interest is not constant.
Indeed,

$$\delta_t = \frac{i}{1+it},$$

so the instantaneous intensity of interest decreases over time.

Terminology Warning

Suppose 5,000 is invested at time $t = 1$. Find the accumulated value at time $t = 4$, given

$$\delta_t = \frac{0.08}{1+0.08t}.$$

Solution
The force of interest

$$\delta_t = \frac{0.08}{1+0.08t}$$

corresponds to the accumulation function

$$a(t) = 1 + 0.08t.$$

Since the deposit is made at time $t = 1$, we accumulate from time 1 to time 4. Therefore, the correct accumulation factor is

$$a(1,4) = \frac{a(4)}{a(1)}.$$

Thus,

$$5{,}000a(1,4) = 5{,}000\frac{a(4)}{a(1)}.$$

Substitute $a(t) = 1 + 0.08t$:

$$5{,}000\frac{a(4)}{a(1)} = 5{,}000\frac{1 + 0.08(4)}{1 + 0.08(1)}.$$

Therefore,

$$5{,}000\frac{1.32}{1.08} = 6{,}111.11.$$

So, using the given force of interest, the accumulated value is

$$\boxed{6{,}111.11}.$$

However, if the problem explicitly says that the deposit earns **simple interest at a rate of** 8%, then the deposit has its own time 0.
In that case, the deposit is invested for

$$4 - 1 = 3$$

years, so the accumulated value is

$$5{,}000\,(1 + 0.08(3))\,.$$

Thus,

$$5{,}000(1.24) = 6{,}200.$$

So under simple interest stated directly for the cash flow, the accumulated value is

$$\boxed{6{,}200}.$$

Important Distinction

If a problem gives a force of interest or an accumulation function, use the accumulation factor between the actual deposit time and the target time:

$$a(t_1, t_2) = \frac{a(t_2)}{a(t_1)}.$$

But if a problem explicitly says **simple interest**, each cash flow is treated separately with its own time 0.
This distinction matters because the two methods can produce different results.

Exercise 1.2.8

At a force of interest $\delta = 0.05$, the following payments have the same present value:

(i) X at the end of year 5 plus $2X$ at the end of year 10;

(ii) Y at the end of year 14.

Calculate

$$\frac{Y}{X}.$$

Solution

Since the force of interest is constant, the discount factor for a payment made at time t is

$$e^{-\delta t}.$$

Here,

$$\delta = 0.05.$$

The present value of the first payment stream is

$$Xe^{-0.05(5)} + 2Xe^{-0.05(10)}.$$

The present value of the second payment stream is

$$Ye^{-0.05(14)}.$$

Since the two present values are equal,

$$Xe^{-0.05(5)} + 2Xe^{-0.05(10)} = Ye^{-0.05(14)}.$$

Thus,

$$Xe^{-0.25} + 2Xe^{-0.50} = Ye^{-0.70}.$$

Factor out X:

$$X\left(e^{-0.25} + 2e^{-0.50}\right) = Ye^{-0.70}.$$

Divide both sides by $Xe^{-0.70}$:

$$\frac{Y}{X} = \frac{e^{-0.25} + 2e^{-0.50}}{e^{-0.70}}.$$

Therefore,

$$\frac{Y}{X} = e^{0.45} + 2e^{0.20}.$$

Numerically,

$$\frac{Y}{X} = 4.0111.$$

Hence,

$$\boxed{\frac{Y}{X} = 4.0111}.$$

Exercise 1.2.9

On January 1, 1997, Kelly deposits X into a bank account. The account is credited with simple interest at a rate of 10% per year.

On the same date, Tara deposits X into a different bank account. The account is credited interest using a force of interest

$$\delta_t = \frac{2t}{t^2 + k}.$$

From the end of the 4th year until the end of the 8th year, both accounts earn the same dollar amount of interest.

Calculate k.

Solution
For Kelly, the account earns simple interest at a rate of 10%. Therefore, Kelly's accumulation function is

$$a_K(t) = 1 + 0.1t.$$

For Tara, we are given

$$\delta_t = \frac{2t}{t^2 + k}.$$

We want an accumulation function $a_T(t)$ such that

$$\frac{a_T'(t)}{a_T(t)} = \frac{2t}{t^2 + k}.$$

Since

$$\frac{d}{dt}(t^2 + k) = 2t,$$

we have

$$\int \frac{2t}{t^2 + k}\, dt = \ln(t^2 + k).$$

Thus,

$$a_T(t) = C(t^2 + k).$$

Because an accumulation function must satisfy

$$a_T(0) = 1,$$

we get

$$1 = Ck.$$

Therefore,

$$C = \frac{1}{k}.$$

So Tara's accumulation function is

$$a_T(t) = \frac{t^2 + k}{k} = \frac{t^2}{k} + 1.$$

The dollar amount of interest earned from the end of the 4th year to the end of the 8th year is equal for Kelly and Tara.
Since both deposited the same amount X, we can cancel X and compare the changes in their accumulation functions:

$$a_K(8) - a_K(4) = a_T(8) - a_T(4).$$

For Kelly,

$$a_K(8) - a_K(4) = (1 + 0.1(8)) - (1 + 0.1(4)).$$

Thus,

$$a_K(8) - a_K(4) = 1.8 - 1.4 = 0.4.$$

For Tara,

$$a_T(8) - a_T(4) = \left(\frac{8^2}{k} + 1\right) - \left(\frac{4^2}{k} + 1\right).$$

Therefore,

$$a_T(8) - a_T(4) = \frac{64}{k} - \frac{16}{k} = \frac{48}{k}.$$

Now set the two interest amounts equal:

$$0.4 = \frac{48}{k}.$$

Solving,

$$k = \frac{48}{0.4} = 120.$$

Therefore,

$$\boxed{k = 120}.$$

Exercise 1.2.10

If the force of interest is

$$\delta_t = \frac{2t}{1+t^2},$$

what is the effective rate of discount during year 4?

A. Less than 0.43

B. At least 0.43, but less than 0.53

C. At least 0.53, but less than 0.63

D. At least 0.63, but less than 0.73

E. At least 0.73

Solution

We use the definition of force of interest:

$$\delta_t = \frac{a'(t)}{a(t)}.$$

Observe that the numerator $2t$ is the derivative of the denominator $1+t^2$. Therefore, by inspection,

$$a(t) = 1 + t^2.$$

Indeed,

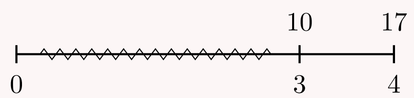
$$\frac{a'(t)}{a(t)} = \frac{2t}{1+t^2} = \delta_t.$$

Now compute the accumulation function at the beginning and end of year 4. Year 4 is the period from time 3 to time 4.

$$a(3) = 1 + 3^2 = 10,$$

and

$$a(4) = 1 + 4^2 = 17.$$



Recall that the effective rate of discount is the interest earned during the period divided by the final value.
Therefore,

$$d_4 = \frac{a(4) - a(3)}{a(4)}.$$

Substituting the values,

$$d_4 = \frac{17 - 10}{17}.$$

Thus,

$$d_4 = \frac{7}{17} = 0.4118.$$

So

$$d_4 \approx 0.412.$$

Therefore, the effective rate of discount during year 4 is less than 0.43.
The correct choice is $\boxed{\text{A}}$.

Exercise 1.2.11

Fund X starts with \$1,000 and accumulates with a force of interest

$$\delta_t = \frac{1}{15 - t}, \qquad 0 \le t \le 15.$$

Fund Y starts with \$1,000 and accumulates with an interest rate of 8% per annum compounded semiannually for the first three years and an effective interest rate of i per annum thereafter.
Fund X equals Fund Y at the end of four years.
Calculate i.

A. 0.0750

B. 0.0775

C. 0.0800

D. 0.0825

E. 0.0850

Solution
We begin by finding the accumulation function for Fund X.
The force of interest is

$$\delta_t = \frac{1}{15 - t}.$$

Notice that

$$\frac{1}{15 - t} = -\frac{-1}{15 - t}.$$

Since

$$\frac{d}{dt}(15 - t) = -1,$$

we can write

$$\delta_t = -\frac{d}{dt}\ln(15 - t).$$

Thus,

$$\int_0^t \delta_s \, ds = \int_0^t \frac{1}{15-s} \, ds.$$

Compute the integral:

$$\int_0^t \frac{1}{15-s} \, ds = \left[-\ln(15-s)\right]_0^t.$$

Therefore,

$$\int_0^t \frac{1}{15-s} \, ds = -\ln(15-t) + \ln(15).$$

So

$$a_X(t) = \exp\left(\ln\left(\frac{15}{15-t}\right)\right).$$

Hence,

$$a_X(t) = \frac{15}{15-t}.$$

Equivalently,

$$a_X(t) = \left(1 - \frac{t}{15}\right)^{-1}.$$

Now compute the value of Fund X at time 4:

$$1000a_X(4) = 1000\left(1 - \frac{4}{15}\right)^{-1}.$$

Thus,

$$1000a_X(4) = 1000\left(\frac{11}{15}\right)^{-1} = 1000\left(\frac{15}{11}\right).$$

Therefore,

$$1000a_X(4) = 1363.64.$$

Now consider Fund Y.
For the first three years, Fund Y earns a nominal annual interest rate of 8%, compounded semiannually. Therefore, the effective semiannual interest rate is

$$\frac{0.08}{2} = 0.04.$$

Over three years, there are

$$3(2) = 6$$

semiannual periods. Thus, after three years, Fund Y has value

$$1000(1.04)^6.$$

During the fourth year, Fund Y earns an annual effective interest rate i. Therefore, at time 4, Fund Y has value

$$1000(1.04)^6(1+i).$$

Since Fund X equals Fund Y at time 4, we have

$$1000(1.04)^6(1+i) = 1000\left(\frac{15}{11}\right).$$

Cancel 1000:

$$(1.04)^6(1+i) = \frac{15}{11}.$$

Solving for i,

$$1+i = \frac{15}{11(1.04)^6}.$$

Thus,

$$i = \frac{15}{11(1.04)^6} - 1.$$

Therefore,

$$i = 0.0777.$$

The closest answer is

$$\boxed{0.0775}.$$

The correct choice is $\boxed{\text{B}}$.

Exercise 1.2.12

A fund starts with a zero balance at time 0. The fund accumulates with a varying force of interest

$$\delta_t = \frac{2t}{t^2+1}, \qquad t > 0.$$

A deposit of $100,000 is made at time 2.
Calculate the number of years from the time of deposit for the fund to double.

A. 0.5

B. 1.0

C. 1.5

D. 2.0

E. 2.5

Solution
We first identify the accumulation function.
The force of interest is

$$\delta_t = \frac{2t}{t^2+1}.$$

Notice that the numerator $2t$ is the derivative of the denominator t^2+1. Therefore,

$$\delta_t = \frac{a'(t)}{a(t)}$$

suggests that

$$a(t) = t^2 + 1.$$

Indeed,

$$\frac{a'(t)}{a(t)} = \frac{2t}{t^2+1} = \delta_t.$$

The deposit is made at time 2. We want to know when the deposit of $100,000 doubles to $200,000.

Let t be the time at which the fund doubles. Since the deposit is made at time 2, we accumulate from time 2 to time t using

$$\frac{a(t)}{a(2)}.$$

Thus,

$$100{,}000\frac{a(t)}{a(2)} = 200{,}000.$$

Cancel 100,000:

$$\frac{a(t)}{a(2)} = 2.$$

Since

$$a(t) = t^2 + 1$$

and

$$a(2) = 2^2 + 1 = 5,$$

we get

$$\frac{t^2 + 1}{5} = 2.$$

Therefore,

$$t^2 + 1 = 10.$$

Thus,

$$t^2 = 9.$$

Since $t > 0$,

$$t = 3.$$

The fund doubles at time $t = 3$. Since the deposit was made at time 2, the number of years from the time of deposit is

$$3 - 2 = 1.$$

Therefore,

$$\boxed{1.0}.$$

The correct choice is $\boxed{\text{B}}$.

Exercise 1.2.13

A deposit of 100 is made into a fund at time $t = 0$.
The fund pays interest at a nominal annual rate of discount d, compounded quarterly, for the first two years.
Beginning at time $t = 2$, interest is credited at a force of interest

$$\delta_t = \frac{1}{t+1}.$$

At time $t = 5$, the accumulated value of the fund is 260.
Calculate d.

A. 12.7%

B. 12.9%

C. 13.1%

D. 13.3%

E. 13.5%

Solution
For the first two years, the fund earns a nominal annual rate of discount d, compounded quarterly.
Therefore, the effective quarterly discount rate is

$$\frac{d}{4}.$$

Since there are 4 quarters per year, the first two years contain

$$2(4) = 8$$

quarters.
Thus, the accumulation factor for the first two years is

$$\left(1 - \frac{d}{4}\right)^{-8}.$$

Beginning at time $t = 2$, the force of interest is

$$\delta_t = \frac{1}{t+1}.$$

We observe that this corresponds to the accumulation function

$$a(t) = t + 1,$$

because

$$\frac{a'(t)}{a(t)} = \frac{1}{t+1} = \delta_t.$$

Therefore, the accumulation factor from time 2 to time 5 is

$$\frac{a(5)}{a(2)}.$$

Now,

$$a(2) = 2 + 1 = 3$$

and

$$a(5) = 5 + 1 = 6.$$

Hence,

$$\frac{a(5)}{a(2)} = \frac{6}{3} = 2.$$

The accumulated value at time 5 is 260, so

$$100\left(1 - \frac{d}{4}\right)^{-8} \frac{a(5)}{a(2)} = 260.$$

Substitute $\frac{a(5)}{a(2)} = 2$:

$$100\left(1-\frac{d}{4}\right)^{-8}(2) = 260.$$

Thus,

$$\left(1-\frac{d}{4}\right)^{-8} = 1.3.$$

Taking both sides to the power $-1/8$,

$$1-\frac{d}{4} = 1.3^{-1/8}.$$

Therefore,

$$\frac{d}{4} = 1 - 1.3^{-1/8}.$$

So

$$d = 4\left(1 - 1.3^{-1/8}\right).$$

Hence,

$$d = 0.12905.$$

Thus,

$$d \approx 12.9\%.$$

The correct choice is $\boxed{\text{B}}$.

1.3 Solution of Problems in Interest

1.3.1 The Basic Problem

There are very few basic principles in the theory of interest.

Future sections will introduce more complex financial transactions, such as annuities, loans, bonds, duration, and immunization. They will also develop simplifying formulas for these transactions.

However, the goal is not to memorize many formulas. Most formulas in this text will be derived from the basic principles introduced in this section.

In practice, most problems in interest can be solved by returning to a small set of fundamental ideas:

1. identify the relevant cash flows;
2. place the cash flows on a timeline;
3. choose a comparison date;
4. accumulate or discount each cash flow to that date;
5. write an equation of value.

The central principle is simple: values can be compared only when they are measured at the same point in time.

Thus, the main skill in interest theory is not memorizing formulas. It is learning how to move money across time consistently.

Interest problems usually involve four main quantities:

1. the principal originally invested;
2. the length of the investment period;
3. the rate, or force, of interest or discount;
4. the accumulated value of the principal at the end of the investment period.

The basic idea is simple: if any three of these quantities are known, then the fourth can usually be found.

For example, under a constant effective interest rate, the relationship is

$$A(t) = P(1+i)^t,$$

where P is the principal, t is the length of the investment, i is the effective interest rate, and $A(t)$ is the accumulated value.

Thus, depending on the information given, we can solve for:

$$P, \qquad t, \qquad i, \qquad \text{or} \qquad A(t).$$

The Four Main Quantities

Most basic interest problems are built around the same four quantities:

principal, time, rate, accumulated value.

Given any three, we can solve for the fourth by writing an equation of value.

Interest problems can usually be viewed from two perspectives: the **borrower**'s perspective and the **lender**'s perspective.

From either perspective, the mathematical problem is the same. However, the wording may change depending on the point of view.

- A **borrower** pays interest.
- A **lender** is credited interest.

For example, if a borrower takes out a loan, the borrower sees the interest as an amount that must be paid. The lender, on the other hand, sees the same interest as an amount earned.

Thus, the cash flows are the same, but their interpretation depends on the perspective.

Borrower and Lender Perspectives

The borrower and the lender describe the same transaction from opposite points of view. A payment made by the borrower is a payment received by the lender. Therefore, when solving interest problems, focus on the timing and amount of each cash flow, not only on the wording.

1.3.2 Equations of Value

The **time value of money** is a fundamental principle in the theory of interest.

It states that the value of an amount of money depends on the time at which the money is paid or received.

More precisely, the value of a payment depends on:

- the time elapsed since the money was paid in the past; or
- the time that will elapse before the money is paid in the future.

Therefore, two or more amounts of money payable at different times cannot be compared directly.

Before comparing them, we must accumulate or discount each amount to a common date.

Comparison Date

The **comparison date** is the common date to which all cash flows are accumulated or discounted before they are compared.
It is also called the **valuation date**.

Equation of Value

An **equation of value** is an equation obtained by accumulating or discounting each payment to the same comparison date.
Once all payments are measured at the same point in time, they can be compared.

The basic principle is:

$$\boxed{\text{Cash flows can only be compared at the same point in time.}}$$

Thus, when solving interest problems, the usual strategy is:

1. identify all cash flows;
2. place them on a timeline;
3. choose a comparison date;
4. move every cash flow to that comparison date;
5. write the equation of value.

Choosing the Comparison Date

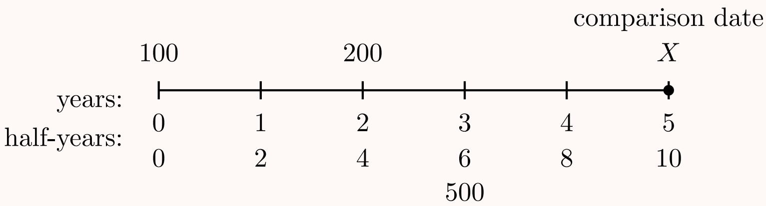
Under compound interest, any comparison date may be chosen.
Different choices may produce different-looking equations, but they lead to the same answer as long as each cash flow is moved correctly to the chosen comparison date.
For this reason, it is usually best to choose the comparison date that makes the equation simplest.

Equation of Value with a Nominal Rate

A company needs to borrow 500 in 3 years. In exchange, it agrees to pay 100 now, 200 in 2 years, and X in 5 years.
If the nominal rate of interest is 6% convertible semiannually, determine X.

Solution
The nominal annual interest rate is 6%, convertible semiannually. Therefore, the effective rate per half-year is

$$\frac{0.06}{2} = 0.03.$$



We choose time 5 as the comparison date.
The payment of 100 at time 0 accumulates for 10 half-years:

$$100(1.03)^{10}.$$

The payment of 200 at time 2 accumulates for 6 half-years:

$$200(1.03)^{6}.$$

The payment X is already at time 5, so it remains X.
The amount borrowed, 500, is received at time 3. Accumulated to time 5, it becomes

$$500(1.03)^{4}.$$

Thus, the equation of value at time 5 is

$$100(1.03)^{10} + 200(1.03)^{6} + X = 500(1.03)^{4}.$$

Solving for X,

$$X = 500(1.03)^{4} - 100(1.03)^{10} - 200(1.03)^{6}.$$

Therefore,

$$X = 189.55.$$

Hence,

$$\boxed{X = 189.55}.$$

Equivalently, we could choose time 0 as the comparison date. The equation of value would be

$$100 + 200v_{0.03}^{4} + Xv_{0.03}^{10} = 500v_{0.03}^{6},$$

where

$$v_{0.03} = \frac{1}{1.03}.$$

This gives the same result:

$$\boxed{X = 189.55}.$$

Exercise 1.3.1

Brown will receive 1000 at the end of 10 years.
In return, he will pay 200 now and another payment X in 5 years.
At a 10% annual effective interest rate, what should the size of Brown's second payment be?

A. 250.00

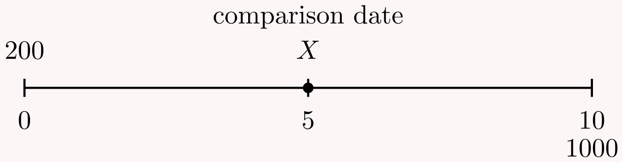
B. 275.50

C. 298.82

D. 322.10

E. 350.00

Solution



We choose time 5 as the comparison date.
The payment of 200 at time 0 accumulates for 5 years:

$$200(1.10)^5.$$

The payment X is already at time 5, so it remains X.
The payment of 1000 at time 10 must be discounted back 5 years:

$$1000v_{0.10}^5,$$

where

$$v_{0.10} = \frac{1}{1.10}.$$

Thus, the equation of value at time 5 is

$$200(1.10)^5 + X = 1000v_{0.10}^5.$$

Since

$$v_{0.10}^5 = (1.10)^{-5},$$

we have

$$200(1.10)^5 + X = 1000(1.10)^{-5}.$$

Solving for X,

$$X = 1000(1.10)^{-5} - 200(1.10)^5.$$

Therefore,

$$X = 298.82.$$

Hence,

$$\boxed{X = 298.82}.$$

The correct choice is $\boxed{\text{C}}$.

Equivalently, we could choose time 0 as the comparison date:

$$200 + Xv_{0.10}^{5} = 1000v_{0.10}^{10}.$$

Or we could choose time 10:

$$200(1.10)^{10} + X(1.10)^{5} = 1000.$$

All three equations lead to the same value:

$$\boxed{X = 298.82}.$$

Exercise 1.3.2

You are given the force of interest

$$\delta_t = \frac{2}{1+t}.$$

A payment of 300 at the end of 3 years and 600 at the end of 6 years has the same present value as a payment of 200 at the end of 2 years and X at the end of 5 years.
Calculate X.

Solution
Since the problem says the two payment streams have the same **present value**, the comparison date is time 0.
First, find the accumulation function from the force of interest:

$$a(t) = \exp\left(\int_0^t \delta_s \, ds\right).$$

Substitute

$$\delta_s = \frac{2}{1+s}.$$

Then

$$a(t) = \exp\left(\int_0^t \frac{2}{1+s} \, ds\right).$$

Compute the integral:

$$\int_0^t \frac{2}{1+s} \, ds = \left[2\ln(1+s)\right]_0^t.$$

Thus,

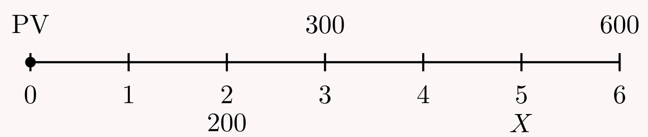
$$a(t) = \exp\left(2\ln(1+t) - 2\ln(1)\right).$$

Since $\ln(1) = 0$,

$$a(t) = \exp\left(2\ln(1+t)\right).$$

Therefore,

$$a(t) = (1+t)^2.$$



The present value of a payment at time t is found by dividing by $a(t)$.
Thus, the equation of value at time 0 is

$$\frac{300}{a(3)} + \frac{600}{a(6)} = \frac{200}{a(2)} + \frac{X}{a(5)}.$$

Since

$$a(t) = (1+t)^2,$$

we have

$$a(2) = 3^2 = 9, \qquad a(3) = 4^2 = 16,$$
$$a(5) = 6^2 = 36, \qquad a(6) = 7^2 = 49.$$

Therefore,

$$\frac{300}{16} + \frac{600}{49} = \frac{200}{9} + \frac{X}{36}.$$

Solving for X,

$$\frac{X}{36} = \frac{300}{16} + \frac{600}{49} - \frac{200}{9}.$$

Thus,

$$X = 36\left(\frac{300}{16} + \frac{600}{49} - \frac{200}{9}\right).$$

Therefore,

$$X = 315.82.$$

Hence,

$$\boxed{X = 315.82}.$$

Exercise 1.3.3

Harry borrows 3500 from Mary. He agrees to repay the loan with annual payments, made at the end of each year, of 500, 1000, 1500, and X.
Mary charges Harry a rate of discount of 10% convertible four times per year.
Determine the amount of the final payment.

A. 1626.12

B. 1656.12

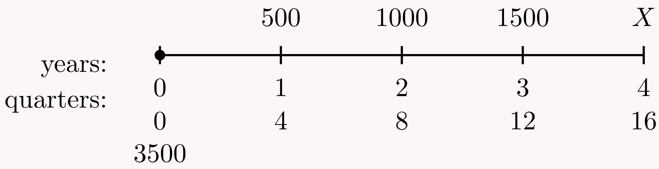
C. 1686.12

D. 1716.12

E. 1746.12

Solution
The nominal annual rate of discount is 10%, convertible four times per year. Therefore, the effective discount rate per quarter is

$$\frac{0.10}{4} = 0.025.$$



We choose time 0 as the comparison date.
Since the quarterly discount rate is 0.025, the discount factor per quarter is

$$1 - 0.025.$$

The payment of 500 at the end of year 1 is discounted for 4 quarters:

$$500(1 - 0.025)^4.$$

The payment of 1000 at the end of year 2 is discounted for 8 quarters:

$$1000(1 - 0.025)^8.$$

The payment of 1500 at the end of year 3 is discounted for 12 quarters:

$$1500(1 - 0.025)^{12}.$$

The final payment X at the end of year 4 is discounted for 16 quarters:

$$X(1 - 0.025)^{16}.$$

Thus, the equation of value at time 0 is

$$3500 = 500(1 - 0.025)^4 + 1000(1 - 0.025)^8 + 1500(1 - 0.025)^{12} + X(1 - 0.025)^{16}.$$

Solving for X,

$$X = \frac{3500 - 500(1 - 0.025)^4 - 1000(1 - 0.025)^8 - 1500(1 - 0.025)^{12}}{(1 - 0.025)^{16}}.$$

Therefore,

$$X = 1686.12.$$

Hence,

$$\boxed{X = 1686.12}.$$

The correct choice is $\boxed{\text{C}}$.

Equivalently, we can first compute the annual effective discount rate:

$$d = 1 - \left(1 - \frac{0.10}{4}\right)^4.$$

Thus,

$$d = 1 - (0.975)^4 = 0.09631.$$

Then the equation of value can also be written as

$$3500 = 500(1 - d) + 1000(1 - d)^2 + 1500(1 - d)^3 + X(1 - d)^4.$$

This gives the same final payment, up to rounding:

$$\boxed{X = 1686.12}.$$

Exercise 1.3.4

Peter deposits 400 into a bank account at time $t = 0$. During the first year, the bank credits interest at a nominal rate of 10% compounded semiannually.
Peter makes an additional deposit of 42 into his bank account at time $t = 1$. During the second year, the bank credits interest at a force of interest

$$\delta_t = \frac{1}{k+t}.$$

The total amount in Peter's account at time $t = 2$ is 552.
Calculate k.

A. 5

B. 6

C. 7

D. 8

E. 9

Solution
During the first year, the nominal annual interest rate is 10%, compounded semiannually. Therefore, the effective semiannual rate is

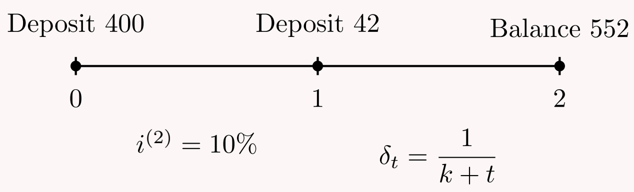
$$\frac{0.10}{2} = 0.05.$$

Thus, the initial deposit of 400 accumulates to time 1 as

$$400(1.05)^2 = 441.$$

At time 1, Peter deposits an additional 42. Therefore, the balance just after time 1 is

$$441 + 42 = 483.$$



Now we need the accumulation factor from time 1 to time 2. We can find it in two equivalent ways.

Method 1: Using the integral of the force of interest
The accumulation factor from time 1 to time 2 is

$$\exp\left(\int_1^2 \frac{1}{k+t}\,dt\right).$$

Compute the integral:

$$\int_1^2 \frac{1}{k+t}\,dt = [\ln(k+t)]_1^2.$$

Thus,

$$\int_1^2 \frac{1}{k+t}\,dt = \ln(k+2) - \ln(k+1).$$

Therefore,

$$\exp\left(\ln(k+2) - \ln(k+1)\right) = \frac{k+2}{k+1}.$$

Method 2: Using the accumulation function

Since

$$\delta_t = \frac{1}{k+t},$$

we can identify an accumulation function by inspection.
Indeed,

$$a(t) = \frac{k+t}{k} = 1 + \frac{t}{k}$$

satisfies

$$\frac{a'(t)}{a(t)} = \frac{\frac{1}{k}}{1+\frac{t}{k}} = \frac{1}{k+t} = \delta_t.$$

So the accumulation factor from time 1 to time 2 is

$$a(1,2) = \frac{a(2)}{a(1)}.$$

Now

$$a(1) = 1 + \frac{1}{k} = \frac{k+1}{k},$$

and

$$a(2) = 1 + \frac{2}{k} = \frac{k+2}{k}.$$

Therefore,

$$\frac{a(2)}{a(1)} = \frac{\frac{k+2}{k}}{\frac{k+1}{k}} = \frac{k+2}{k+1}.$$

Both methods give the same accumulation factor:

$$a(1,2) = \frac{k+2}{k+1}.$$

Now use the final balance:

$$483\left(\frac{k+2}{k+1}\right) = 552.$$

Thus,

$$\frac{k+2}{k+1} = \frac{552}{483}.$$

Since

$$\frac{552}{483} = \frac{8}{7},$$

we have

$$\frac{k+2}{k+1} = \frac{8}{7}.$$

Cross-multiply:

$$7(k+2) = 8(k+1).$$

Therefore,

$$7k + 14 = 8k + 8.$$

Hence,

$$k = 6.$$

The correct choice is

$$\boxed{\text{B}}.$$

Exercise 1.3.5

A loan of 1000 is made at an interest rate of 12% compounded quarterly.
The loan is to be repaid with three payments: 400 at the end of the first year, 800 at the end of the fifth year, and the balance X at the end of the tenth year.
Calculate the amount of the final payment.

A. 587

B. 658

C. 737

D. 777

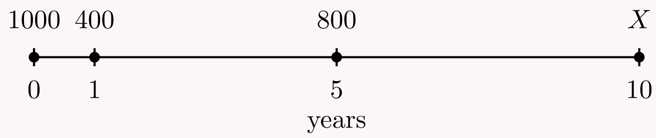
E. 812

Solution
The nominal annual interest rate is 12%, compounded quarterly. Therefore, the effective quarterly interest rate is

$$\frac{0.12}{4} = 0.03.$$

Let

$$v_{0.03} = \frac{1}{1.03}.$$



We choose time 0 as the comparison date.
The payment of 400 at the end of year 1 is discounted for 4 quarters:

$$400v_{0.03}^4.$$

The payment of 800 at the end of year 5 is discounted for 20 quarters:

$$800v_{0.03}^{20}.$$

The final payment X at the end of year 10 is discounted for 40 quarters:

$$Xv_{0.03}^{40}.$$

Thus, the equation of value at time 0 is

$$1000 = 400v_{0.03}^4 + 800v_{0.03}^{20} + Xv_{0.03}^{40}.$$

Solving for X,

$$Xv_{0.03}^{40} = 1000 - 400v_{0.03}^{4} - 800v_{0.03}^{20}.$$

Therefore,

$$X = \frac{1000 - 400v_{0.03}^{4} - 800v_{0.03}^{20}}{v_{0.03}^{40}}.$$

Substituting

$$v_{0.03} = \frac{1}{1.03},$$

we get

$$X = 657.84.$$

Thus, the final payment is approximately

$$\boxed{658}.$$

The correct choice is $\boxed{\text{B}}$.

Equivalently, we could choose time 10 as the comparison date:

$$1000(1.03)^{40} = 400(1.03)^{36} + 800(1.03)^{20} + X.$$

This gives the same result:

$$\boxed{X = 657.84}.$$

Exercise 1.3.6

[SOA 86; SOA-IT.105] A bank agrees to lend 10,000 now and X three years from now in exchange for a single repayment of 75,000 at the end of 10 years.
The bank charges interest at an annual effective rate of 6% for the first 5 years and at a force of interest

$$\delta_t = \frac{1}{t+1}, \qquad t \geq 5.$$

Calculate X.

A. 23,500

B. 24,000

C. 24,500

D. 25,000

E. 25,500

Solution

For $t \geq 5$, the force of interest is

$$\delta_t = \frac{1}{t+1}.$$

We recognize that

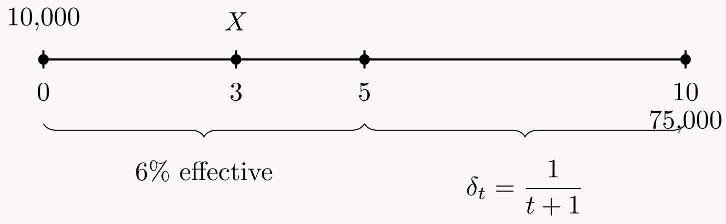
$$\delta_t = \frac{a'(t)}{a(t)}$$

corresponds to

$$a(t) = 1 + t.$$

Indeed,

$$\frac{a'(t)}{a(t)} = \frac{1}{1+t} = \delta_t.$$



We set up an equation of value at time 0.
The amount 10,000 is already at time 0.
The amount X, lent at time 3, has present value

$$Xv_{0.06}^3,$$

where

$$v_{0.06} = \frac{1}{1.06}.$$

The repayment of 75,000 occurs at time 10. First discount it from time 10 to time 5 using the accumulation function ratio:

$$\frac{a(5)}{a(10)}.$$

Since

$$a(t) = 1 + t,$$

we have

$$a(5) = 6$$

and

$$a(10) = 11.$$

Thus,

$$\frac{a(5)}{a(10)} = \frac{6}{11}.$$

Then discount from time 5 to time 0 using the annual effective rate 6%:

$$v_{0.06}^5.$$

Therefore, the present value of the repayment is

$$75{,}000\left(\frac{6}{11}\right)v_{0.06}^5.$$

The equation of value at time 0 is

$$10{,}000 + Xv_{0.06}^3 = 75{,}000\left(\frac{6}{11}\right)v_{0.06}^5.$$

Solving for X,

$$Xv_{0.06}^3 = 75{,}000\left(\frac{6}{11}\right)v_{0.06}^5 - 10{,}000.$$

Therefore,

$$X = \frac{75{,}000\left(\frac{6}{11}\right) v_{0.06}^5 - 10{,}000}{v_{0.06}^3}.$$

Substituting

$$v_{0.06} = \frac{1}{1.06},$$

we get

$$X = 24{,}498.79.$$

Thus,

$$X \approx 24{,}500.$$

The correct choice is $\boxed{\text{C}}$.

1.3.3 Unknown Time

Using Logarithms

When the unknown quantity is time, the equation of value often contains an exponent. To isolate the unknown time, we use logarithms.

Solving for Time Using Logarithms

Find the length of time necessary for 1000 to accumulate to 2000 if the yield rate is an annual effective rate of 5%.

Solution
We use the compound interest formula:

$$A = P(1+i)^n.$$

Here,

$$P = 1000, \qquad A = 2000, \qquad i = 0.05.$$

Thus,

$$1000(1.05)^n = 2000.$$

Divide both sides by 1000:

$$1.05^n = 2.$$

To solve for n, take natural logarithms on both sides:

$$\ln(1.05^n) = \ln(2).$$

Using the logarithm rule

$$\ln(a^n) = n\ln(a),$$

we get

$$n\ln(1.05) = \ln(2).$$

Therefore,

$$n = \frac{\ln(2)}{\ln(1.05)}.$$

Hence,

$$n = 14.2067.$$

So the time required is approximately

$$\boxed{14.21 \text{ years}}.$$

Why Logarithms Appear

Logarithms are used when the unknown variable appears in an exponent.
For example, in

$$(1+i)^n = \frac{A}{P},$$

the unknown n is an exponent. Taking logarithms allows us to bring n down:

$$n \ln(1+i) = \ln\left(\frac{A}{P}\right).$$

Thus,

$$n = \frac{\ln(A/P)}{\ln(1+i)}.$$

Finding an Unknown Comparison Time

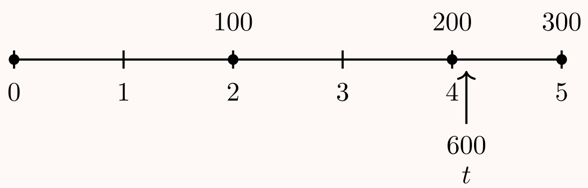
Payments of 100, 200, and 300 are due at the ends of years 2, 4, and 5, respectively. Assuming a nominal annual rate of 6% convertible monthly, find the point in time at which a payment of 600 would be equivalent.

Solution
The nominal annual interest rate is 6%, convertible monthly. Therefore, the effective monthly interest rate is

$$\frac{0.06}{12} = 0.005.$$

Let

$$v_{0.005} = \frac{1}{1.005}.$$



We choose time 0 as the comparison date.
The present value of the single payment 600 made at time t is

$$600v_{0.005}^{12t}.$$

The present value of the three payments is

$$100v_{0.005}^{24} + 200v_{0.005}^{48} + 300v_{0.005}^{60}.$$

Therefore, the equation of value is

$$600v_{0.005}^{12t} = 100v_{0.005}^{24} + 200v_{0.005}^{48} + 300v_{0.005}^{60}.$$

Divide both sides by 600:

$$v_{0.005}^{12t} = \frac{100v_{0.005}^{24} + 200v_{0.005}^{48} + 300v_{0.005}^{60}}{600}.$$

Thus,

$$v_{0.005}^{12t} = 0.780916513.$$

Now take natural logarithms:

$$\ln\left(v_{0.005}^{12t}\right) = \ln(0.780916513).$$

Using

$$\ln(a^b) = b\ln(a),$$

we get

$$12t\ln(v_{0.005}) = \ln(0.780916513).$$

Therefore,

$$t = \frac{\ln(0.780916513)}{12\ln(v_{0.005})}.$$

Hence,

$$t = 4.132.$$

So the equivalent payment of 600 should be made at approximately

$$\boxed{4.132 \text{ years}}.$$

Method of Equated Time

Not in all the textbooks on the FM syllabus. Exam question unlikely

The **method of equated time** gives a quick approximation for the time at which several payments can be replaced by one equivalent payment.

This method is not always emphasized in every FM textbook, and exam questions using it directly are less common. Still, it is useful for intuition.

Suppose cash flows

$$s_1, s_2, \ldots, s_n$$

are paid at times

$$t_1, t_2, \ldots, t_n,$$

respectively.

We want to find a time t such that a single payment

$$s_1 + s_2 + \cdots + s_n$$

made at time t is equivalent to the payments made separately.

Using discount factors, the exact equation of value is

$$(s_1 + s_2 + \cdots + s_n)v^t = s_1v^{t_1} + s_2v^{t_2} + \cdots + s_nv^{t_n}.$$

The method of equated time approximates t by a weighted average of the payment times (formula box below).

Method of Equated Time

The approximate equated time is

$$\boxed{\bar{t} = \frac{s_1t_1 + s_2t_2 + \cdots + s_nt_n}{s_1 + s_2 + \cdots + s_n}}.$$

In words, $\bar{t}$ is the weighted average time of the payments, where each time is weighted by the size of the corresponding payment.

Approximation Warning

The method of equated time is an approximation, not an exact equation of value in general. Under compound interest, the exact time t satisfies

$$(s_1 + s_2 + \cdots + s_n)v^t = s_1v^{t_1} + s_2v^{t_2} + \cdots + s_nv^{t_n}.$$

The approximate time $\bar{t}$ is usually easier to compute, but it may differ from the exact value. It can be shown that, under positive compound interest,

$$\bar{t} \geq t.$$

The approximation is usually better when interest rates are reasonable and the payment times are not too spread out.

Using the Method of Equated Time

Payments of 100, 200, and 300 are due at the ends of years 2, 4, and 5, respectively. Assuming a nominal annual rate of 6% convertible monthly, find the approximate point in time at which a payment of 600 would be equivalent using the method of equated time.

Solution
The method of equated time approximates the equivalent time by taking the weighted average of the payment times, where the weights are the payment amounts.
Thus,

$$\bar{t} = \frac{100(2) + 200(4) + 300(5)}{100 + 200 + 300}.$$

Compute the numerator:

$$100(2) + 200(4) + 300(5) = 200 + 800 + 1500 = 2500.$$

Compute the denominator:

$$100 + 200 + 300 = 600.$$

Therefore,

$$\bar{t} = \frac{2500}{600} = 4.1667.$$

So the method of equated time gives

$$\boxed{\bar{t} = 4.167}.$$

In the previous example, the exact value was

$$t = 4.132.$$

Thus, the method of equated time gives a close approximation, but it is not exact.

Rule of 72

The **Rule of 72** is a quick approximation for the time required for an investment to double.

It is not emphasized in all FM textbooks, and a direct exam question on it is unlikely. Still, it is useful for building intuition.

Suppose an investment grows at an annual effective interest rate i. To find the doubling time n, we solve

$$1(1+i)^n = 2.$$

Thus,

$$(1+i)^n = 2.$$

Taking natural logarithms,

$$\ln\left((1+i)^n\right) = \ln 2.$$

Using the logarithm rule,

$$n\ln(1+i) = \ln 2.$$

Therefore,

$$n = \frac{\ln 2}{\ln(1+i)}.$$

Since

$$\ln 2 \approx 0.693147,$$

we can write

$$n = \frac{0.693147}{\ln(1+i)}.$$

Now rewrite this expression as

$$n = \frac{0.693147}{i} \cdot \frac{i}{\ln(1+i)}.$$

For reasonable interest rates, the second factor

$$\frac{i}{\ln(1+i)}$$

is close to 1. For example, if $i = 0.08$, then

$$\frac{i}{\ln(1+i)} = \frac{0.08}{\ln(1.08)} \approx 1.0395.$$

Thus,

$$n \approx \frac{0.693147}{i}(1.0395) \approx \frac{0.72}{i}.$$

Rule of 72

If i is an annual effective interest rate written as a decimal, then the approximate doubling time is

$$\boxed{n \approx \frac{0.72}{i}}.$$

Equivalently, if the interest rate is written as a percentage $r\%$, then

$$\boxed{n \approx \frac{72}{r}}.$$

For example, at an annual effective interest rate of 8%,

$$n \approx \frac{72}{8} = 9.$$

The exact doubling time is

$$n = \frac{\ln 2}{\ln(1.08)} = 9.006.$$

So the Rule of 72 gives a very close approximation:

$$\boxed{n \approx 9 \text{ years}}.$$

Rule of 72 is an Approximation

The Rule of 72 is not exact. The exact formula for the doubling time is

$$n = \frac{\ln 2}{\ln(1+i)}.$$

The approximation

$$n \approx \frac{0.72}{i}$$

works best for common annual effective interest rates, especially rates near 6% to 10%.

Rule of 72 Approximation

Review the previous example using the Rule of 72.
Find the length of time necessary for 1000 to accumulate to 2000 if the yield rate is an annual effective rate of 5%.

Approximate solution using the Rule of 72
Since the interest rate is 5%, the Rule of 72 gives

$$n \approx \frac{72}{5}.$$

Thus,

$$n \approx 14.4.$$

So the approximate doubling time is

$$\boxed{14.4 \text{ years}}.$$

The exact value from the logarithm formula was

$$n = \frac{\ln 2}{\ln(1.05)} = 14.21.$$

Thus, the Rule of 72 gives a close approximation.

Rate of Interest	Rule of 72	Exact Value
4%	$72/4 = 18$	$\ln 2/\ln(1.04) = 17.67$
8%	$72/8 = 9$	$\ln 2/\ln(1.08) = 9.01$
12%	$72/12 = 6$	$\ln 2/\ln(1.12) = 6.12$
16%	$72/16 = 4.5$	$\ln 2/\ln(1.16) = 4.67$
20%	$72/20 = 3.6$	$\ln 2/\ln(1.20) = 3.80$

Exercise 1.3.7

A loan of 74.93 is to be repaid in two installments of 50 each.
The effective annual rate of interest is 3%. The first installment is due in 6 years.
In how many years is the second installment due?

A. 10

B. 12

C. 14

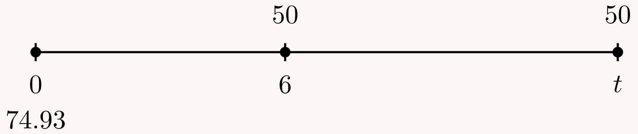
D. 16

E. 18

Solution
Let

$$v_{0.03} = \frac{1}{1.03}.$$

We choose time 0 as the comparison date.



The present value of the first installment is

$$50v_{0.03}^6.$$

The present value of the second installment is

$$50v_{0.03}^t.$$

Thus, the equation of value at time 0 is

$$74.93 = 50v_{0.03}^6 + 50v_{0.03}^t.$$

Solve for $v_{0.03}^t$:

$$50v_{0.03}^t = 74.93 - 50v_{0.03}^6.$$

Therefore,

$$v_{0.03}^t = \frac{74.93 - 50v_{0.03}^6}{50}.$$

Substitute $v_{0.03} = 1.03^{-1}$:

$$v_{0.03}^t = \frac{74.93 - 50(1.03)^{-6}}{50}.$$

Thus,

$$v_{0.03}^t = 0.661157.$$

Taking natural logarithms:

$$\ln\left(v_{0.03}^t\right) = \ln(0.661157).$$

Using $\ln(a^t) = t\ln(a)$, we get

$$t\ln(v_{0.03}) = \ln(0.661157).$$

Therefore,

$$t = \frac{\ln(0.661157)}{\ln(v_{0.03})}.$$

Since

$$v_{0.03} = \frac{1}{1.03},$$

we have

$$t = \frac{\ln(0.661157)}{\ln(1.03^{-1})}.$$

Hence,

$$t = 14.$$

The second installment is due in

$$\boxed{14 \text{ years}}.$$

The correct choice is $\boxed{\text{C}}$.

Exercise 1.3.8

Fund A accumulates at a nominal annual rate of 12%, convertible monthly.
Fund B accumulates with a force of interest

$$\delta_t = \frac{t}{6},$$

for all t.
At time $t = 0$, 1 is deposited in each fund. Let $T > 0$ be the time at which the two funds are equal.
Determine T.

A. 1.25

B. 1.36

C. 1.43

D. 1.50

E. 1.62

Solution

For Fund A, the nominal annual rate is 12%, convertible monthly. Therefore, the effective monthly interest rate is

$$\frac{0.12}{12} = 0.01.$$

Since there are $12t$ months in t years, Fund A's accumulation function is

$$a_A(t) = (1.01)^{12t}.$$

For Fund B, the force of interest is

$$\delta_t = \frac{t}{6}.$$

Thus, Fund B's accumulation function is

$$a_B(t) = \exp\left(\int_0^t \frac{s}{6}\, ds\right).$$

Compute the integral:

$$\int_0^t \frac{s}{6}\, ds = \left[\frac{s^2}{12}\right]_0^t = \frac{t^2}{12}.$$

Therefore,

$$a_B(t) = \exp\left(\frac{t^2}{12}\right).$$

The two funds are equal at time T, so

$$a_A(T) = a_B(T).$$

Hence,

$$(1.01)^{12T} = \exp\left(\frac{T^2}{12}\right).$$

Take natural logarithms:

$$\ln\left((1.01)^{12T}\right) = \ln\left(\exp\left(\frac{T^2}{12}\right)\right).$$

Thus,

$$12T\ln(1.01) = \frac{T^2}{12}.$$

Since $T > 0$, divide both sides by T:

$$12\ln(1.01) = \frac{T}{12}.$$

Therefore,

$$T = 144\ln(1.01).$$

Hence,

$$T = 1.4328.$$

So the two funds are equal at approximately

$$\boxed{T = 1.43}.$$

The correct choice is $\boxed{\text{C}}$.

Exercise 1.3.9

The present value of a payment of 1004 at the end of T months is equal to the present value of 314 after 1 month, 271 after 18 months, and 419 after 24 months.
The effective annual interest rate is 5%. Calculate T.

A. 14

B. 15

C. 16

D. 17

E. 18

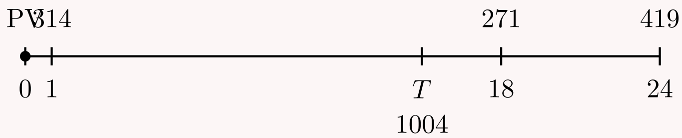
Solution

Let

$$v_{0.05} = \frac{1}{1.05}.$$

Since the interest rate is annual effective, a payment due after m months is discounted by

$$v_{0.05}^{m/12}.$$



The present value of the payment 1004 at time T months is

$$1004v_{0.05}^{T/12}.$$

The present value of the other payments is

$$314v_{0.05}^{1/12} + 271v_{0.05}^{18/12} + 419v_{0.05}^{24/12}.$$

Thus, the equation of value at time 0 is

$$1004v_{0.05}^{T/12} = 314v_{0.05}^{1/12} + 271v_{0.05}^{18/12} + 419v_{0.05}^{24/12}.$$

Divide both sides by 1004:

$$v_{0.05}^{T/12} = \frac{314v_{0.05}^{1/12} + 271v_{0.05}^{18/12} + 419v_{0.05}^{24/12}}{1004}.$$

Thus,

$$v_{0.05}^{T/12} = 0.940835.$$

Taking natural logarithms:

$$\frac{T}{12}\ln(v_{0.05}) = \ln(0.940835).$$

Therefore,

$$T = 12 \cdot \frac{\ln(0.940835)}{\ln(v_{0.05})}.$$

Since

$$v_{0.05} = \frac{1}{1.05},$$

we get

$$T = 14.99.$$

Thus,

$$T \approx 15.$$

The correct choice is $\boxed{\text{B}}$.

Remark.
Since

$$314 + 271 + 419 = 1004,$$

we can estimate T using the method of equated time:

$$\frac{\overline{T}}{12} = \frac{314(1/12) + 271(18/12) + 419(24/12)}{314 + 271 + 419}.$$

Equivalently, in months,

$$\overline{T} = \frac{314(1) + 271(18) + 419(24)}{1004}.$$

Therefore,

$$\overline{T} = 15.2.$$

This is close to the exact value $T = 14.99$, and it also points to answer choice 15.

Exercise 1.3.10

A loan of 7,493 will be repaid in two installments of 5,000 each.
The second installment is due in 14 years. If the annual effective rate of interest is 3%, when is the first installment due?

A. Less than 5.5 years

B. At least 5.5 years, but less than 6.5 years

C. At least 6.5 years, but less than 7.5 years

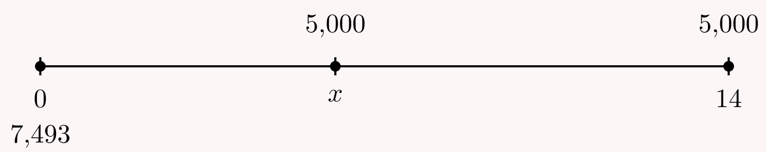
D. At least 7.5 years, but less than 8.5 years

E. At least 8.5 years

Solution
Let

$$v_{0.03} = \frac{1}{1.03}.$$

We choose time 0 as the comparison date.



Let x be the time at which the first installment is due.
The present value of the second installment is

$$5{,}000v_{0.03}^{14}.$$

The present value of the first installment is

$$5{,}000v_{0.03}^{x}.$$

Thus, the equation of value at time 0 is

$$7{,}493 = 5{,}000v_{0.03}^{14} + 5{,}000v_{0.03}^{x}.$$

Using

$$v_{0.03} = (1.03)^{-1},$$

we can write

$$7{,}493 = 5{,}000(1.03)^{-14} + 5{,}000(1.03)^{-x}.$$

Subtract the present value of the second installment:

$$7{,}493 - 5{,}000(1.03)^{-14} = 5{,}000(1.03)^{-x}.$$

Numerically,

$$4{,}187.411 = 5{,}000(1.03)^{-x}.$$

Divide by 5,000:

$$(1.03)^{-x} = 0.837482.$$

Take natural logarithms:

$$\ln\left((1.03)^{-x}\right) = \ln(0.837482).$$

Thus,

$$-x\ln(1.03) = \ln(0.837482).$$

Therefore,

$$x = -\frac{\ln(0.837482)}{\ln(1.03)}.$$

Hence,

$$x = 6.$$

So the first installment is due in 6 years.
The correct choice is

$$\boxed{\text{B}}.$$

Exercise 1.3.11

Jim borrows 5,000 from a bank now, an additional 3,000 one year from now, and an additional 2,000 five years from now.
At what point in time t would a single payment of 10,000 be equivalent at a nominal rate of interest of 12% convertible monthly?

A. Less than 0.9 years

B. At least 0.9 years, but less than 1.0 years

C. At least 1.0 years, but less than 1.1 years

D. At least 1.1 years, but less than 1.2 years

E. At least 1.2 years

Solution
The nominal annual interest rate is 12%, convertible monthly. Therefore, the effective monthly interest rate is

$$\frac{0.12}{12} = 0.01.$$

Let

$$v_{0.01} = \frac{1}{1.01}$$

be the monthly discount factor.

We choose time 0 as the comparison date.
The present value of the three amounts borrowed is

$$5{,}000 + 3{,}000 v_{0.01}^{12} + 2{,}000 v_{0.01}^{60}.$$

The present value of a single payment of 10,000 at time t years is

$$10{,}000 v_{0.01}^{12t}.$$

Thus, the equation of value is

$$10{,}000 v_{0.01}^{12t} = 5{,}000 + 3{,}000 v_{0.01}^{12} + 2{,}000 v_{0.01}^{60}.$$

Divide both sides by 10,000:

$$v_{0.01}^{12t} = \frac{5{,}000 + 3{,}000 v_{0.01}^{12} + 2{,}000 v_{0.01}^{60}}{10{,}000}.$$

Substituting $v_{0.01} = 1/1.01$, we get

$$v_{0.01}^{12t} = 0.87632469.$$

Taking natural logarithms:

$$\ln\left(v_{0.01}^{12t}\right) = \ln(0.87632469).$$

Therefore,

$$12t \ln(v_{0.01}) = \ln(0.87632469).$$

Solving for t,

$$t = \frac{\ln(0.87632469)}{12 \ln(v_{0.01})}.$$

Since
$$v_{0.01} = \frac{1}{1.01},$$
we obtain
$$t = 1.1056.$$
Thus,
$$t \approx 1.106.$$
The correct choice is
$$\boxed{\text{D}}.$$

Equivalently, we can use the annual effective discount factor
$$v = \left(\frac{1}{1.01}\right)^{12}.$$
Then the equation of value becomes
$$10{,}000v^t = 5{,}000 + 3{,}000v + 2{,}000v^5.$$
This gives the same result:
$$\boxed{t = 1.1056}.$$

Exercise 1.3.12

John is 30 years old. He will receive two payments of 2,500 each.
The first payment will be an unknown number of years in the future. The second payment will be five years after the first payment.
At an annual effective interest rate of 5%, the present value of the two payments is 2,607.
Determine at which age John will receive the second payment.

A. Less than 40

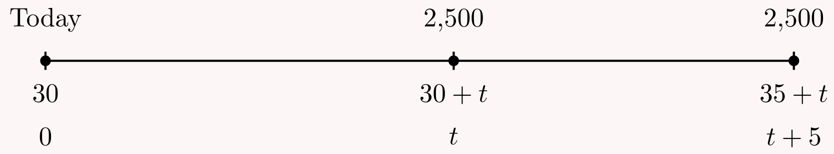
B. At least 40, but less than 45

C. At least 45, but less than 50

D. At least 50, but less than 55

E. At least 55

Solution
Let t be the number of years from today until the first payment.
Then the second payment is received at time
$$t + 5.$$
Since John is currently 30 years old, his age when he receives the second payment will be
$$30 + t + 5 = 35 + t.$$
Let
$$v_{0.05} = \frac{1}{1.05}.$$



The present value of the first payment is

$$2{,}500v_{0.05}^{t}.$$

The present value of the second payment is

$$2{,}500v_{0.05}^{t+5}.$$

Therefore, the equation of value at time 0 is

$$2{,}500v_{0.05}^{t} + 2{,}500v_{0.05}^{t+5} = 2{,}607.$$

Factor out $2{,}500v_{0.05}^{t}$:

$$2{,}500v_{0.05}^{t}\left(1 + v_{0.05}^{5}\right) = 2{,}607.$$

Thus,

$$v_{0.05}^{t} = \frac{2{,}607}{2{,}500\left(1 + v_{0.05}^{5}\right)}.$$

Substituting

$$v_{0.05} = \frac{1}{1.05},$$

we get

$$v_{0.05}^{t} = 0.5847.$$

Taking natural logarithms:

$$t\ln(v_{0.05}) = \ln(0.5847).$$

Therefore,

$$t = \frac{\ln(0.5847)}{\ln(v_{0.05})}.$$

Hence,

$$t \approx 11.$$

So the second payment is received approximately

$$t + 5 = 16$$

years from today.
Since John is currently 30, his age at the second payment is

$$30 + 16 = 46.$$

Therefore, John will receive the second payment at age

$$\boxed{46}.$$

The correct choice is

$$\boxed{\text{C}}.$$

Single Payment

We begin with the simplest unknown-rate problem: one amount accumulates to one future value.

Solving for an Unknown Nominal Rate

An amount of 1000 accumulates to 1500 in 5 years.
Determine $i^{(4)}$, the nominal annual interest rate convertible quarterly.

Solution
Let

$$j = \frac{i^{(4)}}{4}$$

be the effective interest rate per quarter.
Since there are 4 quarters per year, 5 years contain

$$5(4) = 20$$

quarters.
Thus,

$$1000(1+j)^{20} = 1500.$$

Divide both sides by 1000:

$$(1+j)^{20} = 1.5.$$

Take the 20th root:

$$1 + j = (1.5)^{1/20}.$$

Therefore,

$$j = (1.5)^{1/20} - 1.$$

Numerically,

$$j = 0.02048.$$

Since

$$i^{(4)} = 4j,$$

we get

$$i^{(4)} = 4(0.02048) = 0.08192.$$

Hence,

$$\boxed{i^{(4)} = 0.08192}.$$

Equivalently,

$$\boxed{i^{(4)} = 8.192\%}.$$

We can also compute this using a financial calculator:

$$\boxed{\begin{array}{rcl} N & = & 20 \\ PV & = & -1000 \\ PMT & = & 0 \\ FV & = & 1500 \\ \hline CPT\ I/Y & = & 2.048\% \end{array}}$$

The calculator gives the quarterly effective interest rate:

$$j = 2.048\%.$$

Therefore,

$$i^{(4)} = 4(2.048\%) = 8.192\%.$$

Multiple Payments

We now consider unknown-rate problems involving more than one future payment.

Solving for an Unknown Rate with Multiple Payments

At an annual effective interest rate i, the following are equivalent:

(i) 5,000 now;

(ii) 3,000 at the end of 4 years and 2,500 at the end of 8 years.

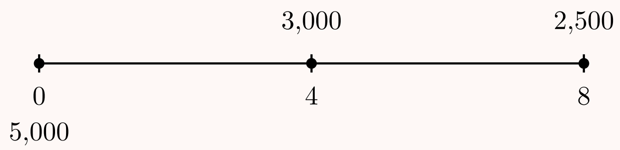
Determine i.

Solution

Let

$$v_i = \frac{1}{1+i}.$$

We choose time 0 as the comparison date.



The present value of 3,000 paid at time 4 is

$$3{,}000v_i^4.$$

The present value of 2,500 paid at time 8 is

$$2{,}500v_i^8.$$

Since these payments are equivalent to 5,000 now, the equation of value is

$$5{,}000 = 3{,}000v_i^4 + 2{,}500v_i^8.$$

Rearrange:

$$2{,}500v_i^8 + 3{,}000v_i^4 - 5{,}000 = 0.$$

Let

$$x = v_i^4.$$

Then

$$v_i^8 = (v_i^4)^2 = x^2.$$

Substitute into the equation:

$$2{,}500x^2 + 3{,}000x - 5{,}000 = 0.$$

Divide both sides by 1,000:

$$2.5x^2 + 3x - 5 = 0.$$

Using the quadratic formula,

$$x = \frac{-3 \pm \sqrt{3^2 - 4(2.5)(-5)}}{2(2.5)}.$$

Thus,

$$x = 0.93622915 \qquad \text{or} \qquad x = -2.13622915.$$

Since

$$x = v_i^4$$

must be positive, we keep

$$x = 0.93622915.$$

Therefore,

$$v_i^4 = 0.93622915.$$

Taking the fourth root,

$$v_i = (0.93622915)^{1/4}.$$

Since

$$v_i = \frac{1}{1+i},$$

we have

$$1 + i = \frac{1}{v_i}.$$

Thus,

$$i = \frac{1}{(0.93622915)^{1/4}} - 1.$$

Therefore,

$$i = 0.01661.$$

Hence,

$$\boxed{i = 0.01661}.$$

Equivalently,

$$\boxed{i = 1.661\%}.$$

When a Quadratic Is Not Available

It is not always possible to solve an unknown-rate problem using a quadratic equation. In the previous example, the substitution

$$x = v_i^4$$

worked because the payment times were 4 and 8, so the powers became x and x^2. However, in many problems, the exponents do not line up so neatly. In those cases, we may need to use one of the following methods:

- numerical trial and error;
- a financial calculator;
- the cash flow function;

- a spreadsheet or numerical solver.

For level payments, we can often use the time value of money function. We will study that method in the annuities chapter.

Using the Cash Flow Function

Rework the previous example using the cash flow function.
At an annual effective interest rate i, the following are equivalent:

(i) 5,000 now;

(ii) 3,000 at the end of 4 years and 2,500 at the end of 8 years.

Determine i.

Solution
The cash flows occur at times 0, 4, and 8. Since the payments are spaced 4 years apart, we can treat one calculator period as 4 years.
The cash flow entries are:

$$CF_0 = -5{,}000,$$
$$C01 = 3{,}000, \qquad F01 = 1,$$
$$C02 = 2{,}500, \qquad F02 = 1.$$

Using the cash flow function, the internal rate of return is

$$IRR = 6.811457479\%.$$

This is the effective rate per 4-year period. Therefore,

$$1 + i = (1.06811457479)^{1/4}.$$

Thus,

$$i = (1.06811457479)^{1/4} - 1.$$

Therefore,

$$i = 0.01661.$$

Hence,

$$\boxed{i = 1.661\%}.$$

This matches the value found using the quadratic method.

Exercise 1.3.13

At a constant force of interest δ, the present value of 500 in 3.25 years is 400.
Determine δ.

A. 0.05866

B. 0.06366

C. 0.06866

D. 0.07366

E. 0.07866

Solution
Under a constant force of interest δ, the present value of a payment F due at time t is

$$PV = Fe^{-\delta t}.$$

Here,

$$PV = 400, \qquad F = 500, \qquad t = 3.25.$$

Thus,

$$500e^{-3.25\delta} = 400.$$

Divide both sides by 500:

$$e^{-3.25\delta} = 0.8.$$

Take natural logarithms:

$$-3.25\delta = \ln(0.8).$$

Therefore,

$$\delta = -\frac{\ln(0.8)}{3.25}.$$

Hence,

$$\delta = 0.06866.$$

So

$$\boxed{\delta = 0.06866}.$$

The correct choice is $\boxed{\text{C}}$.

Alternatively, we can use a financial calculator.

$$\boxed{\begin{array}{rcl} N & = & 3.25 \\ PV & = & -400 \\ PMT & = & 0 \\ FV & = & 500 \\ \hline CPT\ I/Y & = & 7.10715\% \end{array}}$$

The calculator gives the annual effective interest rate

$$i = 0.0710715.$$

Convert this effective interest rate to a constant force of interest:

$$\delta = \ln(1+i).$$

Therefore,

$$\delta = \ln(1.0710715) = 0.06866.$$

This confirms the same answer:

$$\boxed{\delta = 0.06866}.$$

Exercise 1.3.14

David can receive one of the following two payment streams:

(i) 100 at time 0, 200 at time n, and 300 at time $2n$;

(ii) 600 at time 10.

At an annual effective interest rate of i, the present values of the two streams are equal. Given

$$v^n = 0.75941,$$

determine i.

A. 0.02511

B. 0.03011

C. 0.03511

D. 0.04011

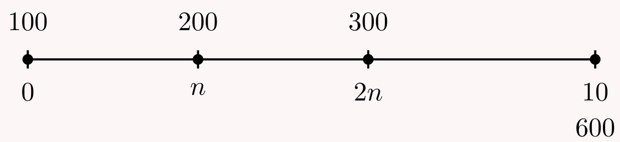
E. 0.04511

Solution

Let

$$v = \frac{1}{1+i}.$$

We choose time 0 as the comparison date.



The present value of the first stream is

$$100 + 200v^n + 300v^{2n}.$$

The present value of the second stream is

$$600v^{10}.$$

Since the present values are equal,

$$600v^{10} = 100 + 200v^n + 300v^{2n}.$$

We are given

$$v^n = 0.75941.$$

Therefore,

$$v^{2n} = (v^n)^2 = (0.75941)^2.$$

Substitute into the equation of value:

$$600v^{10} = 100 + 200(0.75941) + 300(0.75941)^2.$$

Thus,

$$v^{10} = \frac{100 + 200(0.75941) + 300(0.75941)^2}{600}.$$

Hence,

$$v^{10} = 0.708155107.$$

Taking the tenth root,

$$v = (0.708155107)^{1/10}.$$

Since

$$v = \frac{1}{1+i},$$

we have

$$1 + i = \frac{1}{v}.$$

Therefore,

$$i = \frac{1}{(0.708155107)^{1/10}} - 1.$$

So

$$i = 0.03511.$$

Thus,

$$\boxed{i = 0.03511}.$$

The correct choice is

$$\boxed{\text{C}}.$$

Exercise 1.3.15

Mike receives cash flows of 100 today, 200 in one year, and 100 in two years.
The present value of these cash flows is 364.46 at an annual effective rate of interest i.
Calculate i.

A. 0.06

B. 0.08

C. 0.10

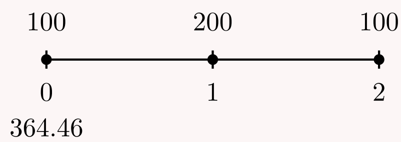
D. 0.12

E. 0.14

Solution
Let

$$v = \frac{1}{1+i}.$$

We choose time 0 as the comparison date.



The present value equation is

$$364.46 = 100 + 200v + 100v^2.$$

Rearrange:

$$100v^2 + 200v - 264.46 = 0.$$

Using the quadratic formula,

$$v = \frac{-200 \pm \sqrt{200^2 - 4(100)(-264.46)}}{2(100)}.$$

Thus,

$$v = 0.909085$$

or a negative value.
Since v must be positive, we keep

$$v = 0.909085.$$

Now use

$$v = \frac{1}{1+i}.$$

Therefore,

$$1 + i = \frac{1}{v}.$$

So

$$i = \frac{1}{0.909085} - 1.$$

Hence,

$$i = 0.10.$$

Thus,

$$\boxed{i = 10\%}.$$

The correct choice is $\boxed{\text{C}}$.

Alternatively, we can use the cash flow function.
Since the cash flow of 100 occurs at time 0, subtract it from the present value:

$$364.46 - 100 = 264.46.$$

Using one valid sign convention:

$$CF_0 = -264.46,$$

$$C01 = 200, \qquad F01 = 1,$$

$$C02 = 100, \qquad F02 = 1.$$

Then

$$IRR = 10\%.$$

This confirms

$$\boxed{i = 0.10}.$$

Exercise 1.3.16

John borrows 1,000 from Jane at an annual effective rate of interest i.
He agrees to pay back 1,000 after six years and 1,366.87 after another six years.
Three years after his first payment, John repays the outstanding balance.
What is the amount of John's second payment?

A. 1,020

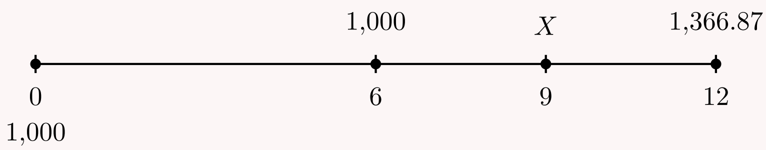
B. 1,027

C. 1,048

D. 1,073

E. 1,094

Solution
Let
$$v = \frac{1}{1+i}.$$
Under the original repayment plan, John pays 1,000 at time 6 and 1,366.87 at time 12.



First, use the original repayment plan to find i.
The equation of value at time 0 is
$$1{,}000 = 1{,}000v^6 + 1{,}366.87v^{12}.$$
Rearrange:
$$1{,}366.87v^{12} + 1{,}000v^6 - 1{,}000 = 0.$$
Let
$$x = v^6.$$
Then
$$v^{12} = x^2.$$
So the equation becomes
$$1{,}366.87x^2 + 1{,}000x - 1{,}000 = 0.$$
Using the quadratic formula,
$$x = \frac{-1{,}000 \pm \sqrt{1{,}000^2 - 4(1{,}366.87)(-1{,}000)}}{2(1{,}366.87)}.$$
We discard the negative solution, since $x = v^6 > 0$. Thus,
$$v^6 = 0.56447.$$
This gives
$$i = 0.10.$$

Now John makes his first payment of 1,000 at time 6, then repays the outstanding balance three years later, at time 9.
At time 9, the only remaining scheduled payment is 1,366.87 due at time 12. Therefore, the outstanding balance at time 9 is the value at time 9 of that future payment:

$$X = 1{,}366.87v^3.$$

Since $i = 0.10$,

$$v = \frac{1}{1.10}.$$

Therefore,

$$X = 1{,}366.87(1.10)^{-3}.$$

Thus,

$$X = 1{,}026.95.$$

So John's second payment is approximately

$$\boxed{1{,}027}.$$

The correct choice is

$$\boxed{\text{B}}.$$

Exercise 1.3.17

At an annual effective interest rate i, with $i > 0$, the following are all equal:

(i) the present value of 10,000 at the end of 6 years;

(ii) the sum of the present values of 6,000 at the end of year t and 56,000 at the end of year $2t$;

(iii) 5,000 immediately.

Calculate the present value of a payment of 8,000 at the end of year $t + 3$, using the same annual effective interest rate.

A. 1,330

B. 1,415

C. 1,600

D. 1,775

E. 2,000

Solution
Let

$$v = \frac{1}{1+i}.$$

Since the present value of 10,000 at the end of 6 years is equal to 5,000 immediately, we have

$$10{,}000v^6 = 5{,}000.$$

Thus,

$$v^6 = \frac{1}{2}.$$

Therefore,

$$v^3 = \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}}.$$

Equivalently,

$$i = \left(\frac{1}{v^6}\right)^{1/6} - 1 = 2^{1/6} - 1 = 0.122462.$$

Now use the equality between the second payment stream and 5,000 immediately:

$$5{,}000 = 6{,}000v^t + 56{,}000v^{2t}.$$

Let

$$x = v^t.$$

Then

$$v^{2t} = x^2.$$

So the equation becomes

$$5{,}000 = 6{,}000x + 56{,}000x^2.$$

Divide by 1,000:

$$5 = 6x + 56x^2.$$

Rearrange:

$$56x^2 + 6x - 5 = 0.$$

Using the quadratic formula,

$$x = \frac{-6 \pm \sqrt{6^2 - 4(56)(-5)}}{2(56)}.$$

Thus,

$$x = \frac{-6 \pm \sqrt{1156}}{112}.$$

Since

$$\sqrt{1156} = 34,$$

we get

$$x = \frac{-6 + 34}{112} = \frac{28}{112} = \frac{1}{4}.$$

We discard the negative solution, since $x = v^t > 0$. Therefore,

$$v^t = \frac{1}{4}.$$

We want the present value of 8,000 at time $t + 3$:

$$8{,}000v^{t+3}.$$

Rewrite:

$$8{,}000v^{t+3} = 8{,}000v^t v^3.$$

Substitute

$$v^t = \frac{1}{4} \qquad \text{and} \qquad v^3 = \frac{1}{\sqrt{2}}.$$

Then

$$8{,}000v^{t+3} = 8{,}000\left(\frac{1}{4}\right)\left(\frac{1}{\sqrt{2}}\right).$$

Thus,

$$8{,}000v^{t+3} = 2{,}000 \cdot \frac{1}{\sqrt{2}}.$$

Therefore,

$$8{,}000v^{t+3} = 1{,}414.21.$$

So the present value is approximately

$$\boxed{1{,}415}.$$

The correct choice is

$$\boxed{\text{B}}.$$

Exercise 1.3.18

At time $t = 0$, Donald puts 1,000 into a fund crediting interest at a nominal annual rate of i, compounded semiannually.
At time $t = 2$, Lewis puts 1,000 into a different fund crediting interest at a force of interest

$$\delta_t = \frac{1}{5+t},$$

for all t.
At time $t = 16$, the amounts in each fund will be equal.
Calculate i.

A. 6.9%

B. 7.0%

C. 7.1%

D. 7.2%

E. 7.3%

Solution
For Lewis, we first identify the accumulation function from the force of interest.
Since

$$\delta_t = \frac{1}{5+t},$$

we can write

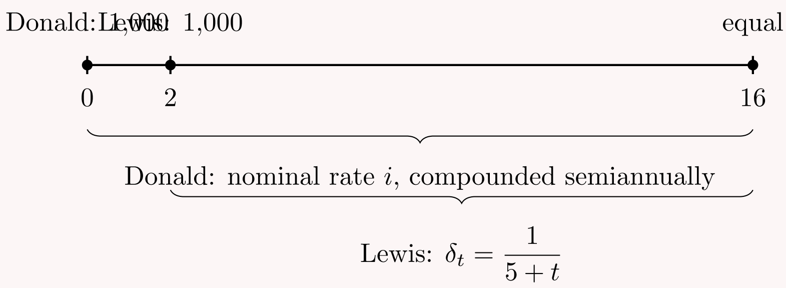
$$\delta_t = \frac{\frac{1}{5}}{1+\frac{t}{5}}.$$

Thus, an accumulation function is

$$a(t) = 1 + \frac{t}{5}.$$

Indeed,

$$\frac{a'(t)}{a(t)} = \frac{\frac{1}{5}}{1+\frac{t}{5}} = \frac{1}{5+t} = \delta_t.$$



Donald accumulates from time 0 to time 16. Since his nominal annual rate is i, compounded semiannually, the effective semiannual rate is

$$\frac{i}{2}.$$

There are

$$16(2) = 32$$

semiannual periods. Therefore, Donald's accumulated value at time 16 is

$$1{,}000\left(1+\frac{i}{2}\right)^{32}.$$

Lewis accumulates from time 2 to time 16. His accumulation factor is

$$\frac{a(16)}{a(2)}.$$

Now,

$$a(16) = 1 + \frac{16}{5} = 4.2$$

and

$$a(2) = 1 + \frac{2}{5} = 1.4.$$

Thus,

$$\frac{a(16)}{a(2)} = \frac{4.2}{1.4} = 3.$$

Since the two funds are equal at time 16,

$$1{,}000\left(1+\frac{i}{2}\right)^{32} = 1{,}000\frac{a(16)}{a(2)}.$$

So

$$\left(1+\frac{i}{2}\right)^{32} = 3.$$

Taking the 32nd root:

$$1+\frac{i}{2} = 3^{1/32}.$$

Therefore,

$$\frac{i}{2} = 3^{1/32} - 1.$$

Hence,

$$i = 2\left(3^{1/32} - 1\right).$$

Thus,

$$i = 0.06986.$$

So

$$i \approx 7.0\%.$$

The correct choice is

$$\boxed{\text{B}}.$$

Chapter 2

Annuities

2.1 Basic Annuities

Sum of a Geometric Series

Before deriving annuity formulas, we recall the sum of a finite geometric series.

Consider the sum

$$S = a + ar + ar^2 + \cdots + ar^{n-1}.$$

Multiply both sides by r:

$$rS = ar + ar^2 + ar^3 + \cdots + ar^n.$$

Now subtract the second equation from the first:

$$S - rS = \left(a + ar + ar^2 + \cdots + ar^{n-1}\right) - \left(ar + ar^2 + \cdots + ar^{n-1} + ar^n\right).$$

All middle terms cancel, leaving

$$S - rS = a - ar^n.$$

Thus,

$$S(1 - r) = a - ar^n.$$

Therefore, if $r \neq 1$,

$$S = \frac{a - ar^n}{1 - r}.$$

Equivalently,

$$\boxed{a + ar + ar^2 + \cdots + ar^{n-1} = \frac{a(1 - r^n)}{1 - r}}.$$

Finite Geometric Series

For $r \neq 1$,

$$\boxed{a + ar + ar^2 + \cdots + ar^{n-1} = \frac{a(1 - r^n)}{1 - r}}.$$

In words, the sum of a geometric series is

$$\frac{\text{first term} - \text{next term after the last term}}{1 - \text{common ratio}}.$$

Special Case

The formula above assumes that $r \neq 1$.
If $r = 1$, then every term is equal to a, so

$$S = a + a + \cdots + a = na.$$

Using the Geometric Series Formula

Given $i = 0.05$, find

$$v^2 + v^4 + \cdots + v^{12}.$$

Solution
Since

$$v = \frac{1}{1+i},$$

we have

$$v = \frac{1}{1.05}.$$

The expression

$$v^2 + v^4 + \cdots + v^{12}$$

is a finite geometric series.
The first term is

$$a = v^2,$$

the common ratio is

$$r = v^2,$$

and the next term after the last term is

$$v^{14}.$$

Using the geometric series formula,

$$S = \frac{\text{first term} - \text{next term after the last term}}{1 - \text{common ratio}},$$

we get

$$v^2 + v^4 + \cdots + v^{12} = \frac{v^2 - v^{14}}{1 - v^2}.$$

Substitute $v = 1.05^{-1}$:

$$\frac{v^2 - v^{14}}{1 - v^2} = \frac{1.05^{-2} - 1.05^{-14}}{1 - 1.05^{-2}}.$$

Therefore,

$$v^2 + v^4 + \cdots + v^{12} = 4.32.$$

Hence,

$$\boxed{4.32}.$$

A More Complicated Geometric Series

Given $i = 0.08$ and $r = 0.05$, find

$$(1+i)^{20} + (1+i)^{25}(1+r)^3 + (1+i)^{30}(1+r)^6 + \cdots + (1+i)^{55}(1+r)^{21}.$$

Solution
This is still a geometric series. The first term is

$$a = (1+i)^{20}.$$

To move from one term to the next, the exponent of $(1+i)$ increases by 5, and the exponent of $(1+r)$ increases by 3.
Therefore, the common ratio is

$$q = (1+i)^5(1+r)^3.$$

The last term is

$$(1+i)^{55}(1+r)^{21}.$$

The next term after the last term is

$$(1+i)^{60}(1+r)^{24}.$$

Using the geometric series formula,

$$S = \frac{\text{first term} - \text{next term after the last term}}{1 - \text{common ratio}},$$

we get

$$S = \frac{(1+i)^{20} - (1+i)^{60}(1+r)^{24}}{1-(1+i)^5(1+r)^3}.$$

Substitute $i = 0.08$ and $r = 0.05$:

$$S = \frac{1.08^{20} - (1.08)^{60}(1.05)^{24}}{1-(1.08)^5(1.05)^3}.$$

Therefore,

$$S = 459.25.$$

Hence,

$$\boxed{459.25}.$$

2.1.1 Annuity-Immediate

What is an Annuity-Immediate?

Annuity

An **annuity** is a series of payments made at equal time intervals.
Common examples include rent payments, mortgage payments, car payments, paychecks, and pension benefits.

Annuity-Immediate

An **annuity-immediate** is an annuity whose payments are made at the **end** of each payment period.
For example, a 10-year annuity-immediate with annual payments has payments at times

$$1, 2, 3, \ldots, 10.$$

There is no payment at time 0.

Present Value of an Annuity-Immediate

Given $i = 0.05$, find the present value of a 10-year annuity-immediate with annual payments of 100.

Solution
Since this is an annuity-immediate, the payments occur at the end of each year.
Thus, the payments of 100 occur at times

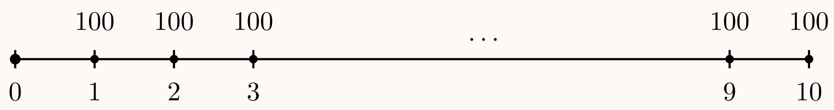
$$1, 2, 3, \ldots, 10.$$

Let

$$v = \frac{1}{1+i}.$$

Since $i = 0.05$,

$$v = \frac{1}{1.05}.$$



The present value is the sum of the present values of all payments:

$$PV = 100v + 100v^2 + \cdots + 100v^{10}.$$

This is a geometric series. The first term is

$$100v,$$

the common ratio is

$$v,$$

and the next term after the last term is

$$100v^{11}.$$

Using the geometric series formula,

$$PV = \frac{100v - 100v^{11}}{1 - v}.$$

Substitute

$$v = \frac{1}{1.05}.$$

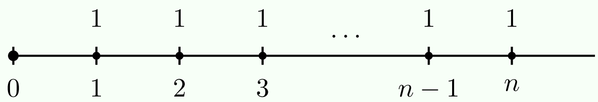
Then
$$PV = \frac{100(1.05)^{-1} - 100(1.05)^{-11}}{1 - (1.05)^{-1}}.$$
Therefore,
$$PV = 772.17.$$
Hence,
$$\boxed{PV = 772.17}.$$

Present Value

We now derive the present value formula for an annuity-immediate from the geometric series formula introduced above.

Present Value of an Annuity-Immediate

At an annual effective interest rate i, the present value of an n-year annuity-immediate with annual payments of 1 is denoted by
$$a_{\overline{n}|}.$$
The payments are made at the end of each year, so they occur at times
$$1, 2, 3, \ldots, n.$$



Since the payment at time k has present value v^k, the present value is
$$PV = v + v^2 + \cdots + v^n.$$
This is a finite geometric series with first term v, common ratio v, and next term after the last term v^{n+1}. Therefore,
$$PV = \frac{v - v^{n+1}}{1 - v}.$$
Factor v in the numerator:
$$PV = \frac{v(1 - v^n)}{1 - v}.$$
Now recall that
$$v = \frac{1}{1+i}.$$
Thus,
$$1 - v = 1 - \frac{1}{1+i} = \frac{i}{1+i} = iv.$$
Therefore,
$$PV = \frac{v(1 - v^n)}{iv}.$$
Cancel v:
$$PV = \frac{1 - v^n}{i}.$$
Hence,
$$\boxed{a_{\overline{n}|} = \frac{1 - v^n}{i}}.$$

Meaning of $a_{\overline{n}|}$

The symbol

$$a_{\overline{n}|}$$

represents the present value of n payments of 1, paid at the end of each period. If each payment is R instead of 1, then the present value is

$$Ra_{\overline{n}|}.$$

So the coefficient of $a_{\overline{n}|}$ is the amount of each payment.

Present Value of a Basic Annuity-Immediate

Given $i = 0.05$, find the present value of a 10-year annuity-immediate with annual payments of 100.

Solution
A 10-year annuity-immediate has payments at times

$$1, 2, 3, \ldots, 10.$$

Since each payment is 100, the present value is

$$PV = 100a_{\overline{10}|}.$$

Using

$$a_{\overline{n}|} = \frac{1 - v^n}{i},$$

we get

$$PV = 100\left(\frac{1 - v^{10}}{0.05}\right).$$

Since

$$v = \frac{1}{1.05},$$

we have

$$PV = 100\left(\frac{1 - (1.05)^{-10}}{0.05}\right).$$

Therefore,

$$PV = 772.17.$$

Hence,

$$\boxed{PV = 772.17}.$$

Where $a_{\overline{n}|}$ Is Measured

The symbol

$$a_{\overline{n}|}$$

represents the value of n payments of 1, measured **one period before the first payment**. So if payments occur at times

$$1, 2, \ldots, n,$$

then $a_{\overline{n}|}$ is measured at time 0.

If payments occur at times

$$4, 5, 6, 7,$$

then $a_{\overline{4}|}$ is first measured at time 3, one period before the first payment. To find the value at time 0, we must discount by v^3.

A Shifted Annuity-Immediate

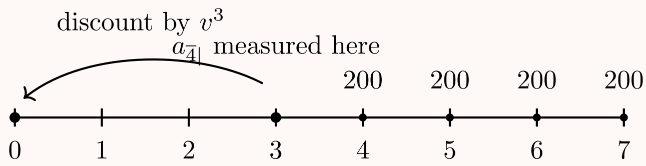
Given $i = 0.10$, find the present value of payments of 200 at times 4, 5, 6, and 7.

Solution
The payments are

$$200, 200, 200, 200$$

at times

$$4, 5, 6, 7.$$



The four payments form an annuity-immediate of 4 payments, but the first payment occurs at time 4.
Therefore,

$$200a_{\overline{4}|}$$

is measured at time 3, one period before the first payment.
To find the present value at time 0, we discount by v^3:

$$PV = 200a_{\overline{4}|}v^3.$$

Using

$$a_{\overline{4}|} = \frac{1 - v^4}{i},$$

we get

$$PV = 200\left(\frac{1 - v^4}{0.10}\right)v^3.$$

Since

$$v = \frac{1}{1.10},$$

we have

$$PV = 200\left(\frac{1 - (1.10)^{-4}}{0.10}\right)(1.10)^{-3}.$$

Therefore,

$$PV = 476.31.$$

Hence,

$$\boxed{PV = 476.31}.$$

Alternatively, we can view the payments at times $4, 5, 6, 7$ as the difference between a 7-year annuity-immediate and a 3-year annuity-immediate:

$$PV = 200a_{\overline{7}|} - 200a_{\overline{3}|}.$$

This gives the same result:

$$\boxed{PV = 476.31}.$$

Alternative Derivation of $a_{\overline{n}|}$

We can derive the present value formula for an annuity-immediate using a simple investment argument.
Suppose you invest 1 in a fund that pays an annual effective interest rate i.
At the end of each year, you withdraw only the interest earned during that year, leaving the principal 1 in the fund. After n years, you withdraw the principal 1.

The original investment has present value 1.
The interest payments form an n-year annuity-immediate with payments of i, so their present value is

$$i a_{\overline{n}|}.$$

The final withdrawal of the principal 1 at time n has present value

$$v^n.$$

Using an equation of value at time 0, we get

$$i a_{\overline{n}|} + v^n = 1.$$

Therefore,

$$i a_{\overline{n}|} = 1 - v^n.$$

Dividing both sides by i, we obtain

$$a_{\overline{n}|} = \frac{1 - v^n}{i}.$$

Thus,

$$\boxed{a_{\overline{n}|} = \frac{1 - v^n}{i}}.$$

Rate per Payment Period

In the formula

$$a_{\overline{n}|} = \frac{1 - v^n}{i},$$

the rate i must be interpreted as the **effective interest rate per payment period**.
Similarly, v is the discount factor per payment period:

$$v = \frac{1}{1+i}.$$

For example, if payments are monthly, then i must be the effective monthly interest rate, not the annual effective rate.

Monthly Payments

Given an annual effective interest rate of

$$i = 0.061677812,$$

find the present value of a 5-year annuity-immediate with monthly payments of 100.

Solution
Since the payments are monthly, we need the effective monthly interest rate. Let j be the effective monthly rate.
Then

$$1 + j = (1.061677812)^{1/12}.$$

Thus,

$$j = (1.061677812)^{1/12} - 1.$$

Therefore,

$$j = 0.005.$$

The number of monthly payments is

$$5(12) = 60.$$

So the present value is

$$PV = 100a_{\overline{60}|j}.$$

Using the annuity-immediate formula,

$$a_{\overline{60}|j} = \frac{1 - v_j^{60}}{j},$$

where

$$v_j = \frac{1}{1+j}.$$

Thus,

$$PV = 100\left(\frac{1 - v_j^{60}}{j}\right).$$

Substitute $j = 0.005$:

$$PV = 100\left(\frac{1 - (1.005)^{-60}}{0.005}\right).$$

Therefore,

$$PV = 5{,}172.56.$$

Hence,

$$\boxed{PV = 5{,}172.56}.$$

Exercise 2.1.1

A payment of 1 is to be made each year for 22 years.
The effective annual rate of interest is 8.25%. If the first payment is one year from now, the present value of these payments is 10.
Calculate the present value if the first payment is 22 years from now.

A. 1.69

B. 1.79

C. 1.89

D. 1.99

E. 2.09

Solution
If the first payment is one year from now, the payments occur at times

$$1, 2, \ldots, 22.$$

We are given that the present value of this annuity-immediate is

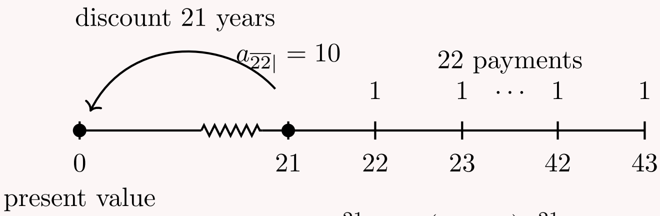
$$a_{\overline{22}|} = 10.$$

If the first payment is instead 22 years from now, then the payments occur at times

$$22, 23, \ldots, 43.$$

This is still a 22-payment annuity-immediate, but its value is measured one period before the first payment. Therefore, the value of the annuity at time 21 is

$$a_{\overline{22}|} = 10.$$



$$PV = 10v^{21} = 10(1.0825)^{-21}$$

Therefore, the present value at time 0 is

$$PV = 10v^{21}.$$

Since

$$i = 0.0825,$$

we have

$$v = \frac{1}{1.0825}.$$

Thus,

$$PV = 10(1.0825)^{-21}.$$

Therefore,

$$PV = 1.89.$$

Hence,

$$\boxed{PV = 1.89}.$$

The correct choice is

$$\boxed{\text{C}}.$$

Exercise 2.1.2

John invests 100,000 into a fund earning an annual effective interest rate of 4%.
John will make withdrawals of X at the end of every 5 years for a total of 10 withdrawals.
After the last withdrawal, the fund's balance will be 0.
Calculate X.

A. 23,213.09

B. 24,213.09

C. 25,213.09

D. 26,213.09

E. 27,213.09

Solution
The withdrawals are made every 5 years. Therefore, we should use the effective interest rate per 5-year period.
Let j be the effective rate per 5 years. Since the annual effective interest rate is 4%,

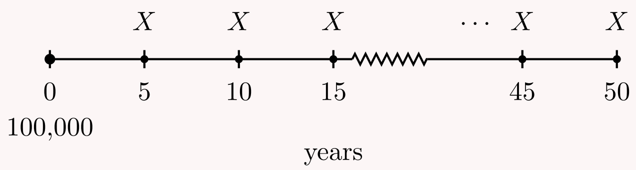
$$1 + j = (1.04)^5.$$

Thus,

$$j = (1.04)^5 - 1.$$

So

$$j = 0.216652902.$$



The withdrawals form an annuity-immediate with 10 payments of X, where each payment period is 5 years.
Thus, the present value of the withdrawals is

$$Xa_{\overline{10}|j}.$$

Since the initial fund is 100,000, the equation of value at time 0 is

$$100{,}000 = Xa_{\overline{10}|j}.$$

Using

$$a_{\overline{10}|j} = \frac{1 - v_j^{10}}{j},$$

we get

$$100{,}000 = X\left(\frac{1 - v_j^{10}}{j}\right),$$

where

$$v_j = \frac{1}{1 + j}.$$

Since
$$j = 0.216652902,$$
we have
$$v_j = \frac{1}{1.216652902}.$$
Therefore,
$$100{,}000 = X\left(\frac{1-(1.216652902)^{-10}}{0.216652902}\right).$$
Solving for X,
$$X = \frac{100{,}000}{\frac{1-(1.216652902)^{-10}}{0.216652902}}.$$
Thus,
$$X = 25{,}213.09.$$
Hence,
$$\boxed{X = 25{,}213.09}.$$
The correct choice is
$$\boxed{\text{C}}.$$

Exercise 2.1.3

A 20-year annuity pays 100 every other year beginning at the end of the second year, with additional payments of 300 each at the ends of years 3, 9, and 15.
The effective annual interest rate is 4%.
Calculate the present value of the annuity.

A. 1,310

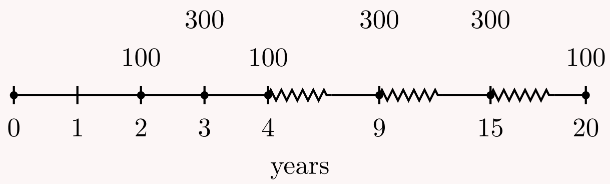
B. 1,340

C. 1,370

D. 1,400

E. 1,430

Solution
Let
$$v_{0.04} = \frac{1}{1.04}.$$
The annuity pays 100 every other year beginning at time 2. Therefore, the regular payments occur at times
$$2, 4, 6, \ldots, 20.$$
There are 10 regular payments.



For the payments of 100, the payment period is 2 years. Therefore, we use the effective interest rate per 2-year period.
Let j be the effective rate per 2 years. Then

$$1 + j = (1.04)^2.$$

Thus,

$$j = (1.04)^2 - 1 = 0.0816.$$

The present value of the regular payments of 100 is

$$100a_{\overline{10}|j}.$$

The additional payments of 300 occur at times 3, 9, and 15. Their present value is

$$300v_{0.04}^3 + 300v_{0.04}^9 + 300v_{0.04}^{15}.$$

Therefore, the total present value is

$$PV = 100a_{\overline{10}|j} + 300v_{0.04}^3 + 300v_{0.04}^9 + 300v_{0.04}^{15}.$$

Using

$$a_{\overline{10}|j} = \frac{1 - v_j^{10}}{j},$$

we get

$$PV = 100\left(\frac{1 - (1.0816)^{-10}}{0.0816}\right) + 300(1.04)^{-3} + 300(1.04)^{-9} + 300(1.04)^{-15}.$$

Thus,

$$PV = 1{,}310.25.$$

So the present value is approximately

$$\boxed{1{,}310}.$$

The correct choice is

$$\boxed{\text{A}}.$$

Exercise 2.1.4

On January 1, 1985, Marc has the following two options for repaying a loan:

(i) sixty monthly payments of 100, commencing February 1, 1985;

(ii) a single payment of 6,000 at the end of K months.

Interest is at a nominal annual rate of 12%, compounded monthly. The two options have the same present value.
Determine K.

A. 29.0

B. 29.5

C. 30.0

D. 30.5

E. 31.0

Solution
The nominal annual interest rate is 12%, compounded monthly. Therefore, the effective monthly interest rate is
$$j = \frac{0.12}{12} = 0.01.$$
Let
$$v_{0.01} = \frac{1}{1.01}.$$
The first option consists of 60 monthly payments of 100, beginning one month from January 1, 1985. Therefore, its present value is
$$100a_{\overline{60}|0.01}.$$
The second option is a single payment of 6,000 at the end of K months. Its present value is
$$6{,}000v_{0.01}^{K}.$$
Since the two options have the same present value,
$$100a_{\overline{60}|0.01} = 6{,}000v_{0.01}^{K}.$$
Using
$$a_{\overline{60}|0.01} = \frac{1 - v_{0.01}^{60}}{0.01},$$
we get
$$100\left(\frac{1 - v_{0.01}^{60}}{0.01}\right) = 6{,}000v_{0.01}^{K}.$$
Thus,
$$v_{0.01}^{K} = \frac{100\left(\frac{1-v_{0.01}^{60}}{0.01}\right)}{6{,}000}.$$
Substituting
$$v_{0.01} = \frac{1}{1.01},$$
we obtain
$$v_{0.01}^{K} = 0.7492506.$$
Taking natural logarithms,
$$K\ln(v_{0.01}) = \ln(0.7492506).$$
Therefore,
$$K = \frac{\ln(0.7492506)}{\ln(v_{0.01})}.$$
Since
$$v_{0.01} = \frac{1}{1.01},$$
we get
$$K = 29.0123.$$
Thus,
$$K \approx 29.0.$$
The correct choice is
$$\boxed{\text{A}}.$$

Exercise 2.1.5

An annuity pays 2 at the end of each year for 18 years.
Another annuity pays 2.5 at the end of each year for 9 years.
At an effective annual interest rate of i, where $0 < i < 1$, the present values of both annuities are equal.
Calculate i.

A. 14%

B. 17%

C. 20%

D. 23%

E. 26%

Solution
Let
$$v = \frac{1}{1+i}.$$
The present value of the first annuity is
$$2a_{\overline{18}|}.$$
The present value of the second annuity is
$$2.5a_{\overline{9}|}.$$
Since the present values are equal,
$$2a_{\overline{18}|} = 2.5a_{\overline{9}|}.$$
Using
$$a_{\overline{n}|} = \frac{1-v^n}{i},$$
we get
$$2\left(\frac{1-v^{18}}{i}\right) = 2.5\left(\frac{1-v^9}{i}\right).$$
Since $i > 0$, multiply both sides by i:
$$2(1-v^{18}) = 2.5(1-v^9).$$
Expand:
$$2-2v^{18} = 2.5-2.5v^9.$$
Rearrange:
$$2v^{18} - 2.5v^9 + 0.5 = 0.$$
Let
$$x = v^9.$$
Then
$$v^{18} = x^2.$$

So the equation becomes

$$2x^2 - 2.5x + 0.5 = 0.$$

Using the quadratic formula,

$$x = \frac{2.5 \pm \sqrt{2.5^2 - 4(2)(0.5)}}{2(2)}.$$

Thus,

$$x = 1 \qquad \text{or} \qquad x = 0.25.$$

Since

$$x = v^9,$$

we have

$$v^9 = 1 \qquad \text{or} \qquad v^9 = 0.25.$$

The solution $v^9 = 1$ would imply $v = 1$, hence $i = 0$. But the problem states that $0 < i < 1$, so we discard this solution.
Therefore,

$$v^9 = 0.25.$$

Taking the ninth root:

$$v = (0.25)^{1/9}.$$

Since

$$v = \frac{1}{1+i},$$

we have

$$1 + i = \frac{1}{v}.$$

Thus,

$$i = \frac{1}{(0.25)^{1/9}} - 1.$$

Therefore,

$$i = 0.16653.$$

So

$$i \approx 16.653\%.$$

The closest answer choice is

$$\boxed{17\%}.$$

The correct choice is

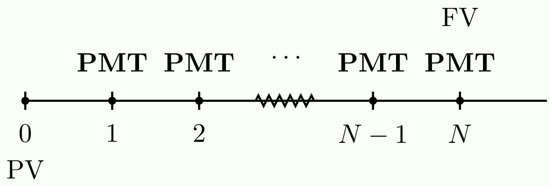
$$\boxed{\text{B}}.$$

Useful Tools

The annuity formulas are the main tool, but financial calculator functions can also be useful for checking answers quickly.

TVM Function

The TVM function can be used for level-payment annuity problems.
For an annuity-immediate, the standard TVM timeline is:



Here:

$$N = \text{number of payment periods},$$

$$I/Y = \text{effective rate of interest per payment period},$$

$$PMT = \text{level payment},$$

$$PV = \text{present value},$$

and

$$FV = \text{future value}.$$

Calculator Convention

In TVM calculations, I/Y must match the payment period.
For example, if payments are monthly, then I/Y must be the effective monthly interest rate, not the annual effective interest rate.
Also, most financial calculators require opposite signs for money paid out and money received. For example, if PV is entered as negative, then the computed PMT or FV may appear positive.

Using the TVM Function

Rework the previous example using the TVM function.
Given an annual effective interest rate of

$$i = 0.061677812,$$

find the present value of a 5-year annuity-immediate with monthly payments of 100.

Solution
Since the payments are monthly, we use the effective monthly interest rate.
Let j be the effective monthly rate. Then

$$j = (1.061677812)^{1/12} - 1.$$

Thus,

$$j = 0.005.$$

So the monthly effective rate is

$$0.5\%.$$

The number of payments is

$$N = 5(12) = 60.$$

Using the TVM function:

$$\begin{array}{rcl} N & = & 60 \\ I/Y & = & 0.5 \\ PMT & = & 100 \\ FV & = & 0 \\ \hline CPT\ PV & = & -5{,}172.56 \end{array}$$

The negative sign means that the present value is an amount paid now in exchange for receiving the monthly payments.
Therefore, the present value is

$$\boxed{5{,}172.56}.$$

How to Count Payments

In annuity problems, we often need to count the number of payments before applying a formula.
If payments are made at times

$$t_1, t_1 + h, t_1 + 2h, \ldots, t_n,$$

then the number of payments is

$$\boxed{\frac{t_n - t_1}{h} + 1}.$$

Equivalently, if $t_0 = t_1 - h$ is the "previous term" before the first payment, then

$$\boxed{\frac{t_n - t_0}{h}}.$$

Counting Payments

Payments are made at times

$$1, 2, 3, \ldots, 10.$$

Here, the first payment time is 1, the last payment time is 10, and the spacing is 1.
Thus, the number of payments is

$$\frac{10 - 1}{1} + 1 = 10.$$

Equivalently, the previous term before 1 is 0, so

$$\frac{10 - 0}{1} = 10.$$

Counting Payments with Larger Spacing

Payments are made at times

$$37, 41, 45, \ldots, 121.$$

The spacing between payments is

$$41 - 37 = 4.$$

The previous term before 37 is
$$37 - 4 = 33.$$
Therefore, the number of payments is
$$\frac{121 - 33}{4} = 22.$$
So there are
$$\boxed{22}$$
payments.

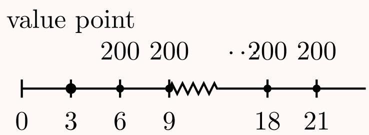
Counting Rule

A fast way to count payments is:
$$\boxed{\text{number of payments} = \frac{\text{last payment time} - \text{previous term}}{\text{spacing}}}.$$
The “previous term” means the time one step before the first payment.
For example, if payments start at time 37 and increase by 4, then the previous term is
$$37 - 4 = 33.$$

Payments Every Three Years under a Constant Force

Given $\delta = 0.05$, find the present value of payments of 200 made at times
$$6, 9, \ldots, 21.$$

Solution
The payments occur every 3 years, beginning at time 6 and ending at time 21.
The previous term before 6 is
$$6 - 3 = 3.$$
Therefore, the number of payments is
$$\frac{21 - 3}{3} = 6.$$



Using first principles
Under a constant force of interest $\delta = 0.05$, the discount factor from time t to time 0 is
$$e^{-0.05t}.$$
Thus, the present value is
$$PV = 200e^{-0.05(6)} + 200e^{-0.05(9)} + \cdots + 200e^{-0.05(21)}.$$

Let

$$v = e^{-0.05}.$$

Then

$$PV = 200(v^6 + v^9 + \cdots + v^{21}).$$

This is a geometric series with first term v^6, common ratio v^3, and next term after the last term v^{24}. Therefore,

$$PV = 200\left(\frac{v^6 - v^{24}}{1 - v^3}\right).$$

Substituting $v = e^{-0.05}$, we get

$$PV = 200\left(\frac{e^{-6(0.05)} - e^{-24(0.05)}}{1 - e^{-3(0.05)}}\right).$$

Thus,

$$PV = 631.23.$$

Using the annuity formula
Since the payments are made every 3 years, use the effective interest rate per 3-year period. Let j be the effective rate per 3 years. Then

$$1 + j = e^{3(0.05)}.$$

Thus,

$$j = e^{0.15} - 1 = 0.161834243.$$

The six payments form an annuity-immediate measured one period before the first payment. Since the first payment is at time 6, the annuity value is measured at time 3. Therefore,

$$PV = 200a_{\overline{6}|j}v_j,$$

where

$$v_j = \frac{1}{1+j} = e^{-0.15}.$$

Hence,

$$PV = 200\left(\frac{1 - v_j^6}{j}\right)v_j.$$

Substituting $j = 0.161834243$, we obtain

$$PV = 631.23.$$

Therefore,

$$\boxed{PV = 631.23}.$$

Grouping Payments

Sometimes it is helpful to split a long annuity into smaller groups of payments.

For example, an annuity with 20 payments can be viewed as two annuities with 10 payments each: one starting now, and one deferred by 10 periods.

Grouping Payments

Given

$$a_{\overline{10}|} = 7.722$$

and

$$v^{10} = 0.614,$$

find

$$a_{\overline{20}|}.$$

Solution

The annuity $a_{\overline{20}|}$ represents the present value of 20 payments of 1, made at times

$$1, 2, \ldots, 20.$$

We split these payments into two groups:

$$1, 2, \ldots, 10$$

and

$$11, 12, \ldots, 20.$$

The first group has present value

$$a_{\overline{10}|}.$$

The second group is also a 10-payment annuity-immediate, but it is deferred by 10 periods. Its value at time 10 is

$$a_{\overline{10}|}.$$

Therefore, its value at time 0 is

$$v^{10} a_{\overline{10}|}.$$

Thus,

$$a_{\overline{20}|} = a_{\overline{10}|} + v^{10} a_{\overline{10}|}.$$

Substitute the given values:

$$a_{\overline{20}|} = 7.722 + 0.614(7.722).$$

Therefore,

$$a_{\overline{20}|} = 12.463.$$

Hence,

$$\boxed{a_{\overline{20}|} = 12.463}.$$

Grouping Annuities

A $2n$-payment annuity-immediate can be split into two groups of n payments:

$$\boxed{a_{\overline{2n}|} = a_{\overline{n}|} + v^n a_{\overline{n}|}}.$$

Similarly,

$$\boxed{a_{\overline{3n}|} = a_{\overline{n}|} + v^n a_{\overline{n}|} + v^{2n} a_{\overline{n}|}}.$$

More generally,

$$a_{\overline{kn}|} = a_{\overline{n}|} + v^n a_{\overline{n}|} + v^{2n} a_{\overline{n}|} + \cdots + v^{(k-1)n} a_{\overline{n}|}.$$

Why the Discount Factor Appears

The symbol $a_{\overline{n}|}$ is measured one period before the first payment.
So the second block of n payments in $a_{\overline{2n}|}$ is not measured at time 0. It is measured at time n.
That is why we discount it by v^n.

Exercise 2.1.6

A fifteen-year annuity-immediate has payments of 500 per year.
The rate of interest is 3.5% for the first nine years and 5% thereafter.
What is the present value of this annuity?

A. 5,365.94

B. 5,515.94

C. 5,665.94

D. 5,815.94

E. 5,965.94

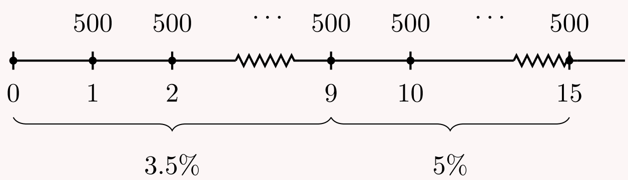
Solution
The first 9 payments occur at times

$$1, 2, \ldots, 9.$$

These payments are valued using the interest rate 3.5%.
The remaining 6 payments occur at times

$$10, 11, \ldots, 15.$$

These payments are valued at time 9 using the interest rate 5%, then discounted back to time 0 using the rate 3.5%.



The present value of the first 9 payments is

$$500 a_{\overline{9}|0.035}.$$

The value at time 9 of the last 6 payments is

$$500 a_{\overline{6}|0.05}.$$

To bring this value back to time 0, discount it for 9 years at 3.5%:

$$500a_{\overline{6}|0.05}v_{0.035}^{9}.$$

Therefore,

$$PV = 500a_{\overline{9}|0.035} + 500a_{\overline{6}|0.05}v_{0.035}^{9}.$$

Using

$$a_{\overline{n}|} = \frac{1 - v^n}{i},$$

we get

$$PV = 500\left(\frac{1 - (1.035)^{-9}}{0.035}\right) + 500\left(\frac{1 - (1.05)^{-6}}{0.05}\right)(1.035)^{-9}.$$

Thus,

$$PV = 5{,}665.94.$$

Hence,

$$\boxed{PV = 5{,}665.94}.$$

The correct choice is

$$\boxed{\text{C}}.$$

Exercise 2.1.7

John invests 100,000 into a fund earning an annual effective interest rate of 4%.
John will make withdrawals of X at the end of every 5 years for a total of 10 withdrawals.
After the last withdrawal, the fund's balance will be 0.
Calculate X.

A. 23,213.09

B. 24,213.09

C. 25,213.09

D. 26,213.09

E. 27,213.09

Solution
In a previous exercise, we solved this problem using the formula for $a_{\overline{n}|}$. Now we solve it using the TVM function.
The withdrawals are made every 5 years, so we use the effective interest rate per 5-year period.
Let j be the effective rate per 5 years. Since the annual effective interest rate is 4%,

$$1 + j = (1.04)^5.$$

Thus,

$$j = (1.04)^5 - 1.$$

So

$$j = 0.216652902.$$

The equation of value is

$$100{,}000 = Xa_{\overline{10}|j}.$$

Using the TVM function, we enter:

$$\boxed{\begin{array}{rcl} N & = & 10 \\ I/Y & = & 100\left((1.04)^5 - 1\right) \\ PV & = & -100{,}000 \\ FV & = & 0 \\ \hline CPT\ PMT & = & 25{,}213.09 \end{array}}$$

Therefore,

$$X = 25{,}213.09.$$

Hence,

$$\boxed{X = 25{,}213.09}.$$

The correct choice is

$$\boxed{\text{C}}.$$

Exercise 2.1.8

Payments are made every 5 years, with the first payment on July 1, 2014 and the last payment on July 1, 2064.
Determine the number of payments.

A. 9

B. 10

C. 11

D. 12

E. 13

Solution
The payments are made every 5 years.
The first payment is in 2014, and the last payment is in 2064. Since the spacing is 5 years, the payment years are

$$2014, 2019, 2024, \ldots, 2064.$$

To count the number of payments, we can use the previous term method.
The term before 2014 is

$$2014 - 5 = 2009.$$

Therefore, the number of payments is

$$\frac{2064 - 2009}{5}.$$

Thus,

$$\frac{2064 - 2009}{5} = \frac{55}{5} = 11.$$

So there are

$$\boxed{11}$$

payments.
The correct choice is

$$\boxed{\text{C}}.$$

Equivalently, we could count using

$$\frac{\text{last payment year} - \text{first payment year}}{\text{spacing}} + 1.$$

This gives

$$\frac{2064 - 2014}{5} + 1 = \frac{50}{5} + 1 = 11.$$

Exercise 2.1.9

John has a 30-year 100,000 mortgage with monthly payments based on a nominal annual interest rate of 12%, convertible monthly.
The first payment is due in one month.
Find X such that if John decided to add X to each monthly payment, starting with the first payment, the term of the mortgage would be reduced to 25 years.

A. Less than 20

B. At least 20, but less than 30

C. At least 30, but less than 40

D. At least 40, but less than 50

E. At least 50

Solution
The nominal annual interest rate is 12%, convertible monthly. Therefore, the effective monthly interest rate is

$$j = \frac{0.12}{12} = 0.01.$$

First, let R be the original monthly payment for the 30-year mortgage.
There are

$$30(12) = 360$$

monthly payments.
Thus,

$$100{,}000 = Ra_{\overline{360}|0.01}.$$

Using

$$a_{\overline{360}|0.01} = \frac{1 - (1.01)^{-360}}{0.01},$$

we get

$$R = \frac{100{,}000}{a_{\overline{360}|0.01}}.$$

Therefore,

$$R = 1{,}028.61.$$

Now suppose John wants to reduce the term to 25 years.
There would be

$$25(12) = 300$$

monthly payments.
The new monthly payment is

$$R + X.$$

Thus,

$$100{,}000 = (R + X)a_{\overline{300}|0.01}.$$

Using

$$a_{\overline{300}|0.01} = \frac{1 - (1.01)^{-300}}{0.01},$$

we get

$$R + X = \frac{100{,}000}{a_{\overline{300}|0.01}}.$$

Therefore,

$$R + X = 1{,}053.22.$$

Since

$$R = 1{,}028.61,$$

we have

$$X = 1{,}053.22 - 1{,}028.61.$$

Thus,

$$X = 24.61.$$

So X is at least 20, but less than 30.
The correct choice is

$$\boxed{\text{B}}.$$

Using the TVM function for the 25-year mortgage:

$$\boxed{\begin{array}{rcl} N & = & 25(12) = 300 \\ I/Y & = & 1 \\ PV & = & -100{,}000 \\ FV & = & 0 \\ \hline CPT\ PMT & = & 1{,}053.22 \end{array}}$$

Therefore,

$$1{,}053.22 = 1{,}028.61 + X,$$

so

$$\boxed{X = 24.61}.$$

Exercise 2.1.10

Eric receives 12,000 from a life insurance policy. He uses the fund to purchase two different annuities, each costing 6,000.
The first annuity is a 24-year annuity-immediate paying K per year to himself.
The second annuity is an 8-year annuity-immediate paying $2K$ per year to his son.
Both annuities are based on an annual effective interest rate of i, where $i > 0$.
Determine i.

A. 6.0%

B. 6.2%

C. 6.4%

D. 6.6%

E. 6.8%

Solution
We begin by writing the information given in the problem.
For the first annuity,

$$Ka_{\overline{24}|i} = 6{,}000.$$

For the second annuity,

$$2Ka_{\overline{8}|i} = 6{,}000.$$

Therefore,

$$Ka_{\overline{8}|i} = 3{,}000.$$

The key idea is to rewrite the 24-year annuity as three deferred 8-year annuities.
We have

$$a_{\overline{24}|i} = a_{\overline{8}|i} + v^8 a_{\overline{8}|i} + v^{16} a_{\overline{8}|i}.$$

Multiplying both sides by K, we get

$$Ka_{\overline{24}|i} = Ka_{\overline{8}|i} + Kv^8 a_{\overline{8}|i} + Kv^{16} a_{\overline{8}|i}.$$

Substitute the known values:

$$6{,}000 = 3{,}000 + 3{,}000v^8 + 3{,}000v^{16}.$$

Divide by 3,000:

$$2 = 1 + v^8 + v^{16}.$$

Thus,

$$v^{16} + v^8 - 1 = 0.$$

Let

$$x = v^8.$$

Then

$$v^{16} = x^2.$$

So the equation becomes

$$x^2 + x - 1 = 0.$$

Using the quadratic formula,

$$x = \frac{-1 \pm \sqrt{1^2 - 4(1)(-1)}}{2(1)}.$$

Thus,

$$x = \frac{-1 \pm \sqrt{5}}{2}.$$

Since $x = v^8 > 0$, we keep the positive solution:

$$x = \frac{-1 + \sqrt{5}}{2}.$$

Therefore,

$$v^8 = \frac{-1 + \sqrt{5}}{2}.$$

Taking the eighth root,

$$v = \left(\frac{-1 + \sqrt{5}}{2}\right)^{1/8}.$$

Since

$$v = \frac{1}{1+i},$$

we have

$$i = \frac{1}{v} - 1.$$

Therefore,

$$i = \left(\frac{-1 + \sqrt{5}}{2}\right)^{-1/8} - 1.$$

Thus,

$$i = 0.061997.$$

So

$$i \approx 6.2\%.$$

The correct choice is

$$\boxed{\text{B}}.$$

Alternative approach using the cash flow worksheet
From

$$6{,}000 = 3{,}000 + 3{,}000v^8 + 3{,}000v^{16},$$

we can subtract 3,000 from both sides:

$$3{,}000 = 3{,}000v^8 + 3{,}000v^{16}.$$

This can be entered in the cash flow worksheet using 8-year periods:

$$CF_0 = -3{,}000,$$

$$C01 = 3{,}000, \qquad F01 = 2.$$

The calculator gives

$$IRR = 61.80339887\%.$$

This is the effective interest rate per 8-year period.
Convert it to an annual effective interest rate:

$$i = (1 + 0.6180339887)^{1/8} - 1.$$

Therefore,

$$i = 0.061997.$$

So again,

$$\boxed{i \approx 6.2\%}.$$

Accumulated Value

We can find the accumulated value of an annuity-immediate by accumulating each payment to the time of the last payment.

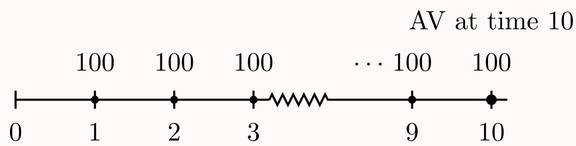
Accumulated Value from First Principles

Given $i = 0.05$, find the accumulated value at time 10 of a 10-year annuity-immediate with annual payments of 100.

Solution
A 10-year annuity-immediate has payments at times

$$1, 2, 3, \ldots, 10.$$

We want the accumulated value at time 10.



The payment at time 1 accumulates for 9 years:

$$100(1.05)^9.$$

The payment at time 2 accumulates for 8 years:

$$100(1.05)^8.$$

Continuing this way, the accumulated value at time 10 is

$$AV = 100(1.05)^9 + 100(1.05)^8 + \cdots + 100(1.05) + 100.$$

Rewrite the terms in increasing powers:

$$AV = 100 + 100(1.05) + 100(1.05)^2 + \cdots + 100(1.05)^9.$$

This is a geometric series with first term 100, common ratio 1.05, and next term after the last term $100(1.05)^{10}$. Therefore,

$$AV = \frac{100 - 100(1.05)^{10}}{1 - 1.05}.$$

Equivalently,

$$AV = 100\left(\frac{1.05^{10}-1}{0.05}\right).$$

Thus,

$$AV = 1{,}257.79.$$

Hence,

$$\boxed{AV = 1{,}257.79}.$$

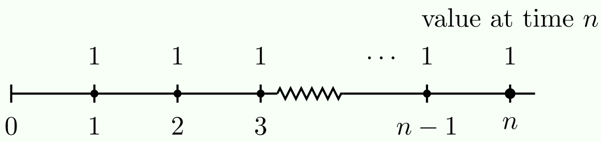
Accumulated Value of an Annuity-Immediate

At an annual effective interest rate i, the accumulated value at the time of the last payment of an n-year annuity-immediate with annual payments of 1 is denoted by

$$s_{\overline{n}|}.$$

The payments occur at times

$$1, 2, 3, \ldots, n.$$



The accumulated value at time n is

$$AV = (1+i)^{n-1} + (1+i)^{n-2} + \cdots + (1+i) + 1.$$

Rewrite the terms in increasing powers:

$$AV = 1 + (1+i) + (1+i)^2 + \cdots + (1+i)^{n-1}.$$

This is a finite geometric series with first term 1, common ratio $1+i$, and next term after the last term $(1+i)^n$. Therefore,

$$AV = \frac{1-(1+i)^n}{1-(1+i)}.$$

Since

$$1-(1+i) = -i,$$

we get

$$AV = \frac{(1+i)^n - 1}{i}.$$

Thus,

$$\boxed{s_{\overline{n}|} = \frac{(1+i)^n - 1}{i}}.$$

Meaning of $s_{\overline{n}|}$

The symbol

$$s_{\overline{n}|}$$

represents the accumulated value of n payments of 1, measured at the time of the last payment.
If each payment is R instead of 1, then the accumulated value is

$$Rs_{\overline{n}|}.$$

So the coefficient of $s_{\overline{n}|}$ is the amount of each payment.

Using the Accumulated Value Formula

Given $i = 0.05$, find the accumulated value at time 10 of a 10-year annuity-immediate with annual payments of 100.

Solution
Since the payments occur at times

$$1, 2, \ldots, 10,$$

and the accumulated value is requested at time 10, we can use

$$s_{\overline{10}|}.$$

Therefore,

$$AV = 100s_{\overline{10}|}.$$

Using the formula,

$$s_{\overline{n}|} = \frac{(1+i)^n - 1}{i},$$

we get

$$AV = 100\left(\frac{1.05^{10} - 1}{0.05}\right).$$

Thus,

$$AV = 1{,}257.79.$$

Hence,

$$\boxed{AV = 1{,}257.79}.$$

Where $s_{\overline{n}|}$ Is Measured

The symbol

$$s_{\overline{n}|}$$

represents the accumulated value of n payments of 1, measured at the time of the last payment.
So if payments occur at times

$$1, 2, \ldots, n,$$

then $s_{\overline{n}|}$ is measured at time n.
If payments occur at times

$$4, 5, 6, 7,$$

then $s_{\overline{4}|}$ is measured at time 7.

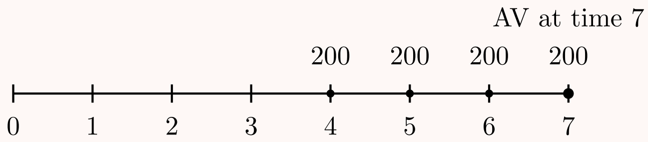
Accumulated Value of a Shifted Annuity-Immediate

Given $i = 0.10$, find the accumulated value at time 7 of payments of 200 at times 4, 5, 6, and 7.

Solution
The payments occur at times

$$4, 5, 6, 7.$$

Since the accumulated value is requested at time 7, which is the time of the last payment, these payments form a 4-payment annuity-immediate.



Therefore,

$$AV = 200 s_{\overline{4}|}.$$

Using the accumulated value formula,

$$s_{\overline{4}|} = \frac{(1+i)^4 - 1}{i}.$$

Substitute $i = 0.10$:

$$AV = 200 \left(\frac{1.10^4 - 1}{0.10} \right).$$

Thus,

$$AV = 928.20.$$

Hence,

$$\boxed{AV = 928.20}.$$

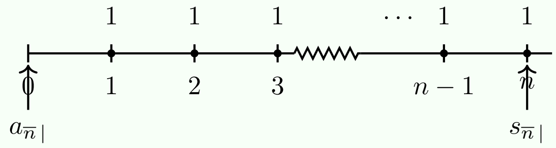
Relationship Between $a_{\overline{n}|}$ and $s_{\overline{n}|}$

We can derive $s_{\overline{n}|}$ from $a_{\overline{n}|}$.
The present value of n payments of 1 is

$$a_{\overline{n}|}.$$

The accumulated value of the same payments at time n is

$$s_{\overline{n}|}.$$



To move the present value at time 0 to the accumulated value at time n, multiply by

$(1+i)^n$:

$$s_{\overline{n}|} = a_{\overline{n}|}(1+i)^n.$$

Using

$$a_{\overline{n}|} = \frac{1-v^n}{i},$$

we get

$$s_{\overline{n}|} = \left(\frac{1-v^n}{i}\right)(1+i)^n.$$

Since

$$v^n = (1+i)^{-n},$$

we have

$$s_{\overline{n}|} = \frac{(1+i)^n - 1}{i}.$$

Thus,

$$\boxed{s_{\overline{n}|} = a_{\overline{n}|}(1+i)^n = \frac{(1+i)^n - 1}{i}}.$$

What Interest Rate To Use

In the formula

$$s_{\overline{n}|} = \frac{(1+i)^n - 1}{i},$$

the rate i must be interpreted as the **effective interest rate per payment period**. For example, if payments are monthly, then the interest rate used in $s_{\overline{n}|}$ must be the effective monthly interest rate.

Accumulated Value with Monthly Payments

Given

$$i^{(4)} = 0.12,$$

find the accumulated value of a 5-year annuity-immediate with monthly payments of 100.

Solution
The notation $i^{(4)} = 0.12$ means a nominal annual interest rate of 12%, compounded quarterly.
Thus, the effective quarterly interest rate is

$$\frac{0.12}{4} = 0.03.$$

Since the payments are monthly, we need the effective monthly interest rate.
Let j be the effective monthly interest rate. Since one quarter contains 3 months,

$$(1+j)^3 = 1.03.$$

Therefore,

$$1 + j = 1.03^{1/3}.$$

So

$$j = 1.03^{1/3} - 1.$$

Thus,

$$j = 0.009901634.$$

The number of monthly payments is

$$5(12) = 60.$$

Since the accumulated value is requested at the time of the last payment, we use

$$s_{\overline{60}|j}.$$

Therefore,

$$AV = 100 s_{\overline{60}|j}.$$

Using the accumulated value formula,

$$s_{\overline{60}|j} = \frac{(1+j)^{60} - 1}{j}.$$

Hence,

$$AV = 100\left(\frac{(1+j)^{60} - 1}{j}\right).$$

Substituting

$$j = 0.009901634,$$

we get

$$AV = 100\left(\frac{(1.009901634)^{60} - 1}{0.009901634}\right).$$

Therefore,

$$AV = 8{,}141.19.$$

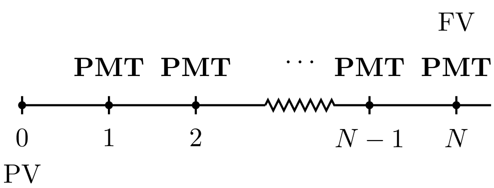
Thus,

$$\boxed{AV = 8{,}141.19}.$$

TVM Function

We can also solve accumulated value annuity problems using the TVM function.

For an annuity-immediate, the standard TVM timeline is:



Here, I/Y is the effective rate of interest per payment period.

Using TVM for Accumulated Value

Rework the previous example using the TVM function.
Given

$$i^{(4)} = 0.12,$$

find the accumulated value of a 5-year annuity-immediate with monthly payments of 100.

Solution

The nominal annual interest rate is 12%, compounded quarterly. Therefore, the effective quarterly interest rate is

$$\frac{0.12}{4} = 0.03.$$

Since the payments are monthly, we need the effective monthly interest rate.
Let j be the effective monthly interest rate. Since one quarter contains 3 months,

$$(1+j)^3 = 1.03.$$

Thus,

$$j = (1.03)^{1/3} - 1.$$

Therefore,

$$j = 0.009901634.$$

Since a financial calculator usually expects I/Y as a percentage, we enter

$$I/Y = 100j.$$

The number of monthly payments is

$$N = 5(12) = 60.$$

Using the TVM function:

$$\boxed{\begin{array}{rcl} N & = & 60 \\ I/Y & = & 100\left((1+0.12/4)^{1/3} - 1\right) \\ PV & = & 0 \\ PMT & = & 100 \\ \hline CPT\ FV & = & -8{,}141.19 \end{array}}$$

The negative sign comes from the calculator's cash-flow convention. Mathematically, the accumulated value is positive:

$$AV = 8{,}141.19.$$

Hence,

$$\boxed{AV = 8{,}141.19}.$$

Grouping Payments

Sometimes it is helpful to split a long accumulated annuity into smaller blocks of payments.

Grouping Payments for Accumulated Values

Consider $2n$ payments of 1, made at times

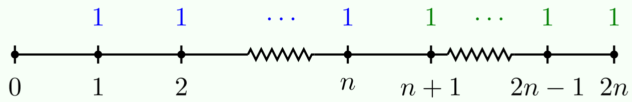
$$1, 2, \ldots, 2n.$$

We can split the payments into two groups:

$$1, 2, \ldots, n$$

and

$$n+1, n+2, \ldots, 2n.$$



We want the accumulated value at time $2n$.
The second group of n payments has accumulated value

$$s_{\overline{n}|}$$

at time $2n$.
The first group of n payments has accumulated value

$$s_{\overline{n}|}$$

at time n. To move it from time n to time $2n$, we accumulate it for another n periods:

$$s_{\overline{n}|}(1+i)^n.$$

Therefore,

$$s_{\overline{2n}|} = s_{\overline{n}|}(1+i)^n + s_{\overline{n}|}.$$

Thus,

$$\boxed{s_{\overline{2n}|} = s_{\overline{n}|}(1+i)^n + s_{\overline{n}|}}.$$

Same Idea as Present Value Grouping

This is the accumulated value version of the present value identity

$$a_{\overline{2n}|} = a_{\overline{n}|} + v^n a_{\overline{n}|}.$$

For present values, later blocks are discounted back.
For accumulated values, earlier blocks are accumulated forward.

Exercise 2.1.11

Grandma decides to set up a college fund for her newborn grandson.
She will deposit 1,335.10 into a savings account every year on his birthday, starting on his first birthday and ending on his 18th birthday.
She hopes to accumulate 50,000 to give to him on his 18th birthday.
What is the minimum annual compound interest rate the savings account must earn in order to reach her goal?

A. 6%

B. 7%

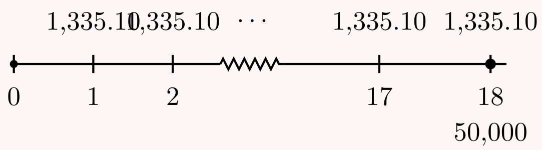
C. 8%

D. 9%

E. 10%

Solution

The deposits are made on birthdays $1, 2, \ldots, 18$. Therefore, there are 18 deposits.
Since the goal is to have 50,000 on the 18th birthday, we need the accumulated value at time 18.



These deposits form an annuity-immediate with 18 payments of 1,335.10.
Thus, the accumulated value equation is

$$1{,}335.10 s_{\overline{18}|i} = 50{,}000.$$

Using

$$s_{\overline{n}|i} = \frac{(1+i)^n - 1}{i},$$

we get

$$1{,}335.10\left(\frac{(1+i)^{18} - 1}{i}\right) = 50{,}000.$$

This equation can be solved using a financial calculator.

$$\boxed{\begin{array}{rcl} N & = & 18 \\ PV & = & 0 \\ PMT & = & -1{,}335.10 \\ FV & = & 50{,}000 \\ \hline CPT\ I/Y & = & 8 \end{array}}$$

Therefore, the savings account must earn at least

$$\boxed{8\%}.$$

The correct choice is

$$\boxed{\text{C}}.$$

Exercise 2.1.12

You are given

$$\delta_t = \frac{1}{1+t}, \qquad 0 \leq t \leq 5.$$

Calculate $s_{\overline{5}|}$.

A. 7.7

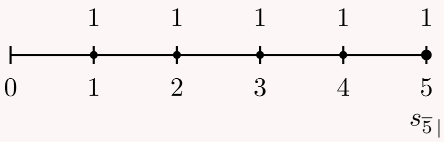
B. 8.2

C. 8.7

D. 9.2

E. 9.7

Solution
We first identify the accumulation function from the force of interest.
Since
$$\delta_t = \frac{1}{1+t},$$
we can recognize that
$$a(t) = 1 + t.$$
Indeed,
$$\frac{a'(t)}{a(t)} = \frac{1}{1+t} = \delta_t.$$
The symbol $s_{\overline{5}|}$ represents the accumulated value at time 5 of five payments of 1, made at times
$$1, 2, 3, 4, 5.$$



A payment made at time k accumulates to time 5 by the factor
$$\frac{a(5)}{a(k)}.$$
Therefore,
$$s_{\overline{5}|} = \frac{a(5)}{a(1)} + \frac{a(5)}{a(2)} + \frac{a(5)}{a(3)} + \frac{a(5)}{a(4)} + 1.$$
Since
$$a(t) = 1 + t,$$
we have
$$a(5) = 6.$$
Thus,
$$s_{\overline{5}|} = \frac{6}{2} + \frac{6}{3} + \frac{6}{4} + \frac{6}{5} + 1.$$
So
$$s_{\overline{5}|} = 3 + 2 + 1.5 + 1.2 + 1.$$
Therefore,
$$s_{\overline{5}|} = 8.7.$$
Hence,
$$\boxed{s_{\overline{5}|} = 8.7}.$$
The correct choice is
$$\boxed{\text{C}}.$$

Exercise 2.1.13

A person age 40 wishes to accumulate a fund for retirement by depositing an amount X at the end of each year into an account paying 4% interest.
At age 65, the person will use the entire account balance to purchase a 15-year annuity-immediate with annual payments of 10,000, based on an annual effective interest rate of 5%.
Find X.

A. 2,292.36

B. 2,392.36

C. 2,492.36

D. 2,592.36

E. 2,692.36

Solution
The person deposits X at the end of each year from age 41 through age 65.
Thus, there are

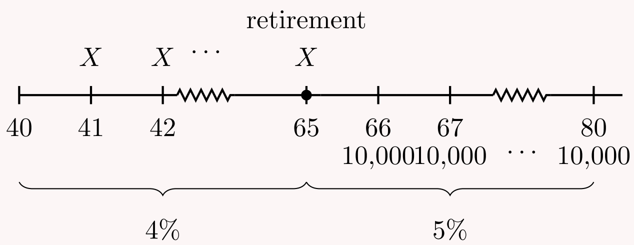
$$65 - 40 = 25$$

deposits.
The accumulated value of these deposits at age 65 is

$$X s_{\overline{25}|0.04}.$$

At age 65, the person purchases a 15-year annuity-immediate paying 10,000 per year. Since the first annuity payment is one year later, at age 66, the value at age 65 is

$$10{,}000 a_{\overline{15}|0.05}.$$



At age 65, the accumulated deposits must equal the cost of the retirement annuity:

$$X s_{\overline{25}|0.04} = 10{,}000 a_{\overline{15}|0.05}.$$

Therefore,

$$X = \frac{10{,}000 a_{\overline{15}|0.05}}{s_{\overline{25}|0.04}}.$$

Using

$$a_{\overline{15}|0.05} = \frac{1 - (1.05)^{-15}}{0.05},$$

and

$$s_{\overline{25}|0.04} = \frac{(1.04)^{25} - 1}{0.04},$$

we get

$$X = \frac{10{,}000\left(\frac{1-(1.05)^{-15}}{0.05}\right)}{\frac{(1.04)^{25}-1}{0.04}}.$$

Thus,

$$X = 2{,}492.36.$$

Hence,

$$\boxed{X = 2{,}492.36}.$$

The correct choice is

$$\boxed{\text{C}}.$$

Using the financial calculator, first find the amount needed at age 65 to purchase the annuity:

$$\boxed{\begin{array}{rcl} N & = & 15 \\ I/Y & = & 5 \\ PMT & = & 10{,}000 \\ FV & = & 0 \\ \hline CPT\ PV & = & -103{,}796.5804 \end{array}}$$

So the fund needed at age 65 is 103,796.5804.
Now solve for the annual deposit X:

$$\boxed{\begin{array}{rcl} FV & = & -103{,}796.5804 \\ N & = & 25 \\ I/Y & = & 4 \\ PV & = & 0 \\ \hline CPT\ PMT & = & 2{,}492.36 \end{array}}$$

Therefore,

$$\boxed{X = 2{,}492.36}.$$

Exercise 2.1.14

At an effective annual interest rate i, you are given:

(i) The present value of an annuity-immediate with annual payments of 1 for n years is 40.

(ii) The present value of an annuity-immediate with annual payments of 1 for $3n$ years is 70.

Calculate the accumulated value of an annuity-immediate with annual payments of 1 for $2n$ years.

A. 180

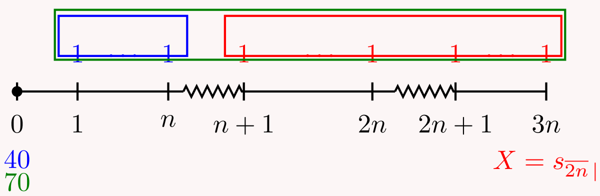
B. 210

C. 240

D. 270

E. 300

Solution
We are given
$$a_{\overline{n}|} = 40$$
and
$$a_{\overline{3n}|} = 70.$$
First, split the $3n$-payment annuity into three blocks of n payments:
$$a_{\overline{3n}|} = a_{\overline{n}|} + v^n a_{\overline{n}|} + v^{2n} a_{\overline{n}|}.$$
Substitute $a_{\overline{n}|} = 40$ and $a_{\overline{3n}|} = 70$:
$$70 = 40 + 40v^n + 40v^{2n}.$$
Let
$$x = v^n.$$
Then
$$70 = 40 + 40x + 40x^2.$$
Subtract 40:
$$30 = 40x + 40x^2.$$
Divide by 40:
$$\frac{3}{4} = x + x^2.$$
Thus,
$$x^2 + x - \frac{3}{4} = 0.$$
Using the quadratic formula,
$$x = \frac{-1 \pm \sqrt{1^2 - 4(1)\left(-\frac{3}{4}\right)}}{2}.$$
Therefore,
$$x = \frac{-1 \pm 2}{2}.$$
Since $x = v^n > 0$, we keep
$$x = \frac{1}{2}.$$
Hence,
$$v^n = \frac{1}{2}.$$



Now let

$$X = s_{\overline{2n}|}.$$

The red block represents $2n$ payments of 1, accumulated to the time of its last payment, time $3n$. Therefore, its value at time $3n$ is

$$X.$$

Its present value at time 0 is

$$Xv^{3n}.$$

Since the present value of the full $3n$-year annuity is 70, and the present value of the first n-year block is 40, we have

$$70 = 40 + Xv^{3n}.$$

Since

$$v^n = \frac{1}{2},$$

we get

$$v^{3n} = \left(\frac{1}{2}\right)^3 = \frac{1}{8}.$$

Thus,

$$70 = 40 + \frac{X}{8}.$$

So

$$30 = \frac{X}{8}.$$

Therefore,

$$X = 240.$$

Hence,

$$\boxed{s_{\overline{2n}|} = 240}.$$

The correct choice is

$$\boxed{\text{C}}.$$

Alternative using the cash flow worksheet
From

$$70 = 40 + 40v^n + 40v^{2n},$$

we can write

$$30 = 40v^n + 40v^{2n}.$$

Using n-year periods, enter:

$$CF_0 = -30, \qquad C01 = 40, \qquad F01 = 2.$$

The calculator gives

$$IRR = 100\%.$$

Thus, the effective interest rate per n-year period is 100%, so

$$v^n = \frac{1}{1+1} = 0.5.$$

This confirms

$$\boxed{s_{\overline{2n}|} = 240}.$$

Exercise 2.1.15

Harriet wishes to accumulate 60,000 in a fund at the end of 25 years.
She plans to deposit 80 into the fund at the end of each of the first 120 months.
She then plans to deposit $80 + X$ into the fund at the end of each of the last 180 months.
Assume the fund earns interest at an annual effective rate of 3.66%.
Determine X.

A. Less than 86

B. At least 86, but less than 88

C. At least 88, but less than 90

D. At least 90, but less than 92

E. At least 92

Solution
The deposits are monthly, so we need the effective monthly interest rate.
Let j be the effective monthly rate. Since the annual effective rate is 3.66%,

$$1 + j = (1.0366)^{1/12}.$$

Thus,

$$j = (1.0366)^{1/12} - 1.$$

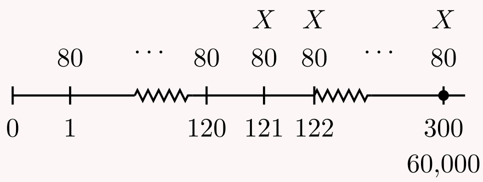
So

$$j \approx 0.003.$$

The target date is the end of 25 years, which is the end of

$$25(12) = 300$$

months.



One efficient way to view the deposits is:

- 80 is deposited every month for all 300 months;
- an extra X is deposited every month for the last 180 months.

Therefore, the accumulated value at month 300 is

$$80s_{\overline{300}|j} + Xs_{\overline{180}|j}.$$

We need this to equal 60,000, so

$$80s_{\overline{300}|j} + Xs_{\overline{180}|j} = 60{,}000.$$

Using
$$s_{\overline{n}|j} = \frac{(1+j)^n - 1}{j},$$
we have
$$80\left(\frac{(1+j)^{300}-1}{j}\right) + X\left(\frac{(1+j)^{180}-1}{j}\right) = 60{,}000.$$
Substituting
$$j = (1.0366)^{1/12} - 1,$$
we solve for X:
$$X = \frac{60{,}000 - 80s_{\overline{300}|j}}{s_{\overline{180}|j}}.$$
Thus,
$$X = 88.85.$$
So X is at least 88, but less than 90.
The correct choice is
$$\boxed{\text{C}}.$$

Equivalently, we could separate the two deposit streams directly:
$$60{,}000 = 80s_{\overline{120}|j}(1+j)^{180} + (80+X)s_{\overline{180}|j}.$$
This gives the same result:
$$\boxed{X = 88.85}.$$

Exercise 2.1.16

Jeff deposits 100 at the end of each year for 13 years into Fund X.
Rachel deposits 100 at the end of each year for 13 years into Fund Y.
Fund X earns an annual effective rate of 15% for the first 5 years and an annual effective rate of 6% thereafter. Fund Y earns an annual effective rate of i.
At the end of 13 years, the accumulated value of Fund X equals the accumulated value of Fund Y.
Calculate i.

A. 6.4%

B. 6.7%

C. 7.0%

D. 7.4%

E. 7.8%

Solution
The key is to accumulate Fund X correctly, because the interest rate changes after 5 years.
The deposits into Fund X can be split into two groups.
The first 5 deposits accumulate at 15% until time 5, then at 6% from time 5 to time 13.
Their accumulated value at time 13 is
$$100s_{\overline{5}|0.15}(1.06)^8.$$

The last 8 deposits occur after the rate has changed to 6%. Their accumulated value at time 13 is

$$100s_{\overline{8}|0.06}.$$

Therefore,

$$AV_X = 100s_{\overline{5}|0.15}(1.06)^8 + 100s_{\overline{8}|0.06}.$$

For Fund Y, the deposits form a 13-year annuity-immediate at rate i, so

$$AV_Y = 100s_{\overline{13}|i}.$$

Since the accumulated values are equal at time 13,

$$100s_{\overline{5}|0.15}(1.06)^8 + 100s_{\overline{8}|0.06} = 100s_{\overline{13}|i}.$$

Using

$$s_{\overline{n}|i} = \frac{(1+i)^n - 1}{i},$$

we get

$$100\left(\frac{1.15^5 - 1}{0.15}\right)(1.06)^8 + 100\left(\frac{1.06^8 - 1}{0.06}\right) = 100\left(\frac{(1+i)^{13} - 1}{i}\right).$$

The left-hand side is

$$2{,}064.38.$$

Thus,

$$100s_{\overline{13}|i} = 2{,}064.38.$$

So

$$s_{\overline{13}|i} = 20.6438.$$

Solving

$$\frac{(1+i)^{13} - 1}{i} = 20.6438$$

gives

$$i = 0.0738.$$

Therefore,

$$i \approx 7.38\%.$$

The closest answer choice is

$$\boxed{7.4\%}.$$

The correct choice is

$$\boxed{\text{D}}.$$

Exercise 2.1.17

Tom has served in the Navy for 21 years and plans to retire sometime in the future.
Every three years, he has received a re-enlistment bonus and deposited a certain dollar amount, X, in an account for when he retires.
After making his current deposit, the value of his account is 180,176.
Assuming an annual effective interest rate of 9%, what were his tri-annual deposits?

A. Less than 10,400

B. At least 10,400, but less than 10,425

C. At least 10,425, but less than 10,450

D. At least 10,450, but less than 10,475

E. At least 10,475

Solution

The deposits are made every 3 years.
Since Tom has served for 21 years and has just made his current deposit, the deposits occur at times

$$3, 6, 9, 12, 15, 18, 21.$$

Thus, there are

$$\frac{21}{3} = 7$$

deposits.
Since deposits are made every 3 years, we use the effective interest rate per 3-year period. Let j be the effective rate per 3 years. Then

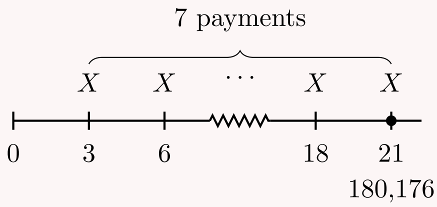
$$1 + j = (1.09)^3.$$

Thus,

$$j = (1.09)^3 - 1.$$

So

$$j = 0.295029.$$



The accumulated value after the current deposit is

$$X s_{\overline{7}|j}.$$

Therefore,

$$X s_{\overline{7}|j} = 180{,}176.$$

Using

$$s_{\overline{7}|j} = \frac{(1 + j)^7 - 1}{j},$$

we get

$$X = \frac{180{,}176}{s_{\overline{7}|j}}.$$

Substitute

$$j = (1.09)^3 - 1.$$

Then

$$X = \frac{180{,}176}{\frac{(1.295029)^7 - 1}{0.295029}}.$$

Thus,

$$X = 10{,}405.$$

Therefore, the tri-annual deposits were

$$\boxed{10{,}405}.$$

The correct choice is

$$\boxed{\text{B}}.$$

2.1.2 Annuity-Due

Present Value

An annuity-due is similar to an annuity-immediate, except that payments are made at the beginning of each payment period.

Annuity-Immediate vs. Annuity-Due

An **annuity-immediate** has payments at the **end** of each time interval.
An **annuity-due** has payments at the **beginning** of each time interval.
For example, a 10-year annuity-immediate has payments at times

$$1, 2, \ldots, 10.$$

A 10-year annuity-due has payments at times

$$0, 1, \ldots, 9.$$

Terminology

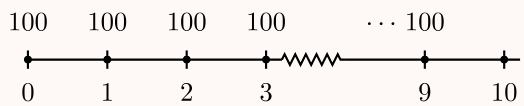
The names can feel counterintuitive.
An annuity-immediate does not begin immediately. Its first payment is one period later.
An annuity-due begins immediately. Its first payment is made at time 0.

Present Value of an Annuity-Due

Given $i = 0.05$, find the present value of a 10-year annuity-due with annual payments of 100.

Solution
Since this is an annuity-due, the payments are made at the beginning of each year.
Therefore, the payments of 100 occur at times

$$0, 1, 2, \ldots, 9.$$



The present value is

$$PV = 100 + 100v + 100v^2 + \cdots + 100v^9.$$

This is a finite geometric series with first term 100, common ratio v, and next term after the last term $100v^{10}$.
Therefore,
$$PV = \frac{100 - 100v^{10}}{1 - v}.$$
Since
$$v = \frac{1}{1.05},$$
we get
$$PV = \frac{100 - 100(1.05)^{-10}}{1 - (1.05)^{-1}}.$$
Thus,
$$PV = 810.78.$$
Hence,
$$\boxed{PV = 810.78}.$$

Why the Names Are Confusing

The name **annuity-immediate** can be misleading.

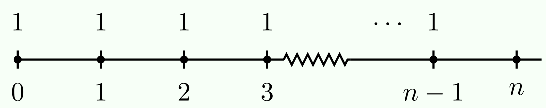
It does not mean that the first payment is made immediately. It means that the annuity is not deferred: it begins in the first payment period. However, because payments are made at the end of each period, the first payment occurs one period later.

An **annuity-due** has payments due at the beginning of each period, so its first payment occurs at time 0.

Thus, for an n-period annuity:
$$\text{annuity-immediate: } 1, 2, \ldots, n,$$
while
$$\text{annuity-due: } 0, 1, \ldots, n-1.$$

Present Value of an Annuity-Due

At an annual effective interest rate i, the present value of an n-year annuity-due with annual payments of 1 is denoted by
$$\ddot{a}_{\overline{n}|}.$$
An annuity-due has payments at the beginning of each period, so the payments occur at times
$$0, 1, 2, \ldots, n-1.$$



The present value is
$$PV = 1 + v + v^2 + \cdots + v^{n-1}.$$

This is a finite geometric series with first term 1, common ratio v, and next term after the last term v^n. Therefore,

$$PV = \frac{1 - v^n}{1 - v}.$$

Recall that the effective discount rate is

$$d = 1 - v.$$

Thus,

$$PV = \frac{1 - v^n}{d}.$$

Therefore,

$$\boxed{\ddot{a}_{\overline{n}|} = \frac{1 - v^n}{d}}.$$

Annuity-Immediate vs. Annuity-Due

Recall that the immediate and due present values share the same numerator

$$1 - v^n$$

, but differ by whether we divide by the effective interest rate

$$i$$

(immediate) or the effective discount rate

$$d$$

(due):

$$a_{\overline{n}|} = \frac{1 - v^n}{i}, \qquad \ddot{a}_{\overline{n}|} = \frac{1 - v^n}{d}.$$

The reason is that an annuity-due starts one period earlier than an annuity-immediate.

Using the Annuity-Due Formula

Given $i = 0.05$, find the present value of a 10-year annuity-due with annual payments of 100.

Solution
A 10-year annuity-due has payments at times

$$0, 1, 2, \ldots, 9.$$

Thus,

$$PV = 100\ddot{a}_{\overline{10}|}.$$

Using

$$\ddot{a}_{\overline{n}|} = \frac{1 - v^n}{d},$$

we get

$$PV = 100\left(\frac{1 - v^{10}}{d}\right).$$

Since

$$v = \frac{1}{1.05}$$

and

$$d = \frac{i}{1+i} = \frac{0.05}{1.05},$$

we have

$$PV = 100\left(\frac{1-(1.05)^{-10}}{0.05/1.05}\right).$$

Therefore,

$$PV = 810.78.$$

Hence,

$$\boxed{PV = 810.78}.$$

Where $\ddot{a}_{\overline{n}|}$ Is Measured

The symbol

$$\ddot{a}_{\overline{n}|}$$

represents the value of n payments of 1, measured at the time of the first payment. So if payments occur at times

$$0, 1, \ldots, n-1,$$

then $\ddot{a}_{\overline{n}|}$ is measured at time 0.
If payments occur at times

$$4, 5, 6, 7,$$

then $\ddot{a}_{\overline{4}|}$ is measured at time 4.

A Shifted Annuity-Due

Given $i = 0.10$, find the present value of payments of 200 at times 4, 5, 6, and 7, using $\ddot{a}_{\overline{n}|}$.

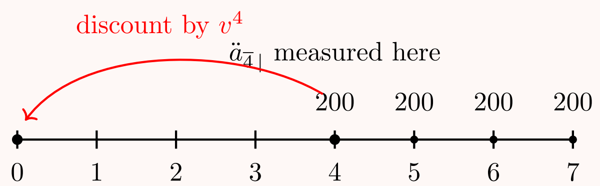
Solution
The payments occur at times

$$4, 5, 6, 7.$$

These four payments form a 4-payment annuity-due. Since an annuity-due is measured at the time of the first payment,

$$200\ddot{a}_{\overline{4}|}$$

is measured at time 4.



To find the present value at time 0, we discount by v^4:

$$PV = 200\ddot{a}_{\overline{4}|}v^4.$$

Using

$$\ddot{a}_{\overline{4}|} = \frac{1-v^4}{d},$$

we get

$$PV = 200\left(\frac{1-v^4}{d}\right)v^4.$$

Since

$$i = 0.10, \qquad v = \frac{1}{1.10}, \qquad d = \frac{0.10}{1.10},$$

we have

$$PV = 200\left(\frac{1-(1.10)^{-4}}{0.10/1.10}\right)(1.10)^{-4}.$$

Therefore,

$$PV = 476.31.$$

Hence,

$$\boxed{PV = 476.31}.$$

In a previous lesson, we found the same present value using the annuity-immediate expression

$$PV = 200a_{\overline{4}|}v^3.$$

Both methods are equivalent.

What Interest Rate To Use

In the formula

$$\ddot{a}_{\overline{n}|} = \frac{1-v^n}{d},$$

the rate d must be interpreted as the **effective discount rate per payment period**.
Also, v must be the discount factor per payment period.
For example, if payments are monthly, then v and d must be based on the effective monthly interest rate.

Monthly Payments with an Annuity-Due

Given an annual effective interest rate of

$$i = 0.061677812,$$

find the present value of a 5-year annuity-due with monthly payments of 100.

Solution
Since the payments are monthly, we need the effective monthly interest rate.
Let j be the effective monthly interest rate. Then

$$1 + j = (1.061677812)^{1/12}.$$

Thus,

$$j = (1.061677812)^{1/12} - 1.$$

So

$$j = 0.005.$$

The number of monthly payments is

$$5(12) = 60.$$

Since this is an annuity-due, the present value is

$$PV = 100\ddot{a}_{\overline{60}|j}.$$

Using the annuity-due formula,

$$\ddot{a}_{\overline{60}|j} = \frac{1 - v_j^{60}}{d_j},$$

where

$$v_j = \frac{1}{1+j}$$

and

$$d_j = \frac{j}{1+j}.$$

Since $j = 0.005$, we have

$$v_j = \frac{1}{1.005}$$

and

$$d_j = \frac{0.005}{1.005}.$$

Therefore,

$$PV = 100\left(\frac{1 - v_j^{60}}{d_j}\right).$$

Substitute the values:

$$PV = 100\left(\frac{1 - (1.005)^{-60}}{0.005/1.005}\right).$$

Thus,

$$PV = 5{,}198.42.$$

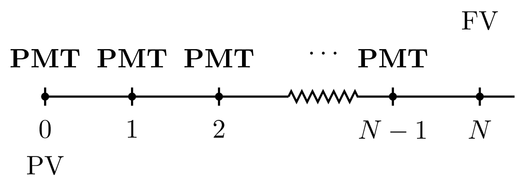
Hence,

$$\boxed{PV = 5{,}198.42}.$$

TVM Function

To value an annuity-due using the TVM function, we need to switch the calculator to **BGN mode**.

In BGN mode, the payments are made at the beginning of each period.



BGN mode only changes the timing of the PMT amounts. It tells the calculator that payments occur at the beginning of each payment period instead of at the end.

Using TVM for an Annuity-Due

Rework the previous example using the TVM function.
Given an annual effective interest rate of

$$i = 0.061677812,$$

find the present value of a 5-year annuity-due with monthly payments of 100.

Solution
Since the payments are monthly, we first convert the annual effective interest rate into an effective monthly interest rate.
Let j be the effective monthly interest rate. Then

$$1 + j = (1.061677812)^{1/12}.$$

Thus,

$$j = (1.061677812)^{1/12} - 1.$$

So

$$j = 0.005.$$

Since a financial calculator usually expects I/Y as a percentage, we enter

$$I/Y = 100j = 0.5.$$

The number of monthly payments is

$$N = 5(12) = 60.$$

Because this is an annuity-due, we use **BGN mode**.

BGN mode

$$\begin{aligned} N &= 60 \\ I/Y &= 100\left((1.061677812)^{1/12} - 1\right) = 0.5 \\ PMT &= 100 \\ FV &= 0 \\ \hline CPT\ PV &= -5{,}198.42 \end{aligned}$$

The negative sign comes from the calculator's cash-flow convention. Mathematically, the present value is positive:

$$PV = 5{,}198.42.$$

Hence,

$$\boxed{PV = 5{,}198.42}.$$

Exercise 2.1.18

At an annual effective interest rate of 8%, the present value of a 30-year annuity-due with annual payments of X is 1,500.
Determine X using:

(1) first principles;

(2) the $\ddot{a}_{\overline{n}|}$ formula;

(3) the TVM function.

A. 113.37

B. 118.37

C. 123.37

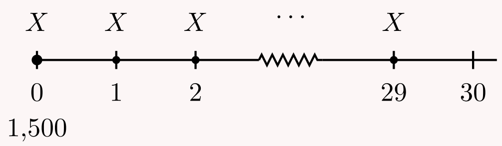
D. 128.37

E. 133.37

Solution

Since this is an annuity-due, the payments occur at the beginning of each year. Therefore, the payments occur at times

$$0, 1, 2, \ldots, 29.$$



Method 1: First principles

The present value is

$$1{,}500 = X + Xv + Xv^2 + \cdots + Xv^{29}.$$

Factor out X:

$$1{,}500 = X(1 + v + v^2 + \cdots + v^{29}).$$

This is a geometric series:

$$1 + v + v^2 + \cdots + v^{29} = \frac{1 - v^{30}}{1 - v}.$$

Thus,

$$1{,}500 = X\left(\frac{1 - v^{30}}{1 - v}\right).$$

Since

$$v = \frac{1}{1.08},$$

we solve for X:

$$X = \frac{1{,}500}{\frac{1-v^{30}}{1-v}}.$$

Therefore,

$$X = 123.37.$$

Method 2: Annuity-due formula

For an annuity-due,

$$\ddot{a}_{\overline{n}|} = \frac{1 - v^n}{d}.$$

Here,

$$d = \frac{i}{1+i} = \frac{0.08}{1.08}.$$

The equation of value is

$$1{,}500 = X\ddot{a}_{\overline{30}|0.08}.$$

Therefore,

$$1{,}500 = X\left(\frac{1-v^{30}}{d}\right).$$

So

$$X = \frac{1{,}500}{\frac{1-v^{30}}{d}}.$$

Thus,

$$X = 123.37.$$

Method 3: TVM function
Because this is an annuity-due, use **BGN mode**.

BGN mode		
N	$=$	30
I/Y	$=$	8
PV	$=$	$-1{,}500$
FV	$=$	0
$CPT\ PMT$	$=$	123.37

Therefore,

$$\boxed{X = 123.37}.$$

The correct choice is

$$\boxed{\text{C}}.$$

Exercise 2.1.19

Given an annual effective discount rate of $d = 0.10$, calculate the present value of 20 payments of 1,000, paid every two years starting immediately.

A. 4,985.36

B. 5,085.36

C. 5,185.36

D. 5,285.36

E. 5,385.36

Solution
Since payments are made every two years, we need the effective rate per two-year payment period.
We are given the annual effective discount rate

$$d = 0.10.$$

Thus, the annual discount factor is

$$v = 1 - d = 0.90.$$

The discount factor over two years is

$$v^2 = (0.90)^2 = 0.81.$$

Therefore, the effective discount rate per two-year period is

$$d_2 = 1 - v^2 = 1 - 0.81 = 0.19.$$

Since the payments start immediately, this is an annuity-due with 20 payments.
Thus, the present value is

$$PV = 1{,}000\ddot{a}_{\overline{20}|}.$$

Using the annuity-due formula,

$$\ddot{a}_{\overline{20}|} = \frac{1 - v^{20}}{d_2},$$

where now $v = 0.81$ is the two-year discount factor.
Therefore,

$$PV = 1{,}000 \left(\frac{1 - (0.81)^{20}}{0.19} \right).$$

Thus,

$$PV = 5{,}185.36.$$

Hence,

$$\boxed{PV = 5{,}185.36}.$$

The correct choice is

$$\boxed{\text{C}}.$$

Equivalently, using the TVM function, first convert to the effective interest rate per two years:

$$j = (1 - d)^{-2} - 1.$$

So

$$j = (0.90)^{-2} - 1.$$

Using BGN mode:

BGN mode		
N	$=$	20
I/Y	$=$	$100\left((1 - 0.10)^{-2} - 1\right)$
PMT	$=$	$1{,}000$
FV	$=$	0
$CPT\ PV$	$=$	$-5{,}185.36$

The negative sign comes from the calculator's cash-flow convention. Mathematically,

$$\boxed{PV = 5{,}185.36}.$$

Exercise 2.1.20

A renter with 1,104 has a one-year lease.
The landlord is willing to accept two payment options:

(1) 1,104 now;

(2) 100 paid at the beginning of each month for 12 months.

What monthly interest rate would be required for the two options to be equivalent?

A. Less than 0.012

B. At least 0.012, but less than 0.013

C. At least 0.013, but less than 0.014

D. At least 0.014, but less than 0.015

E. At least 0.015

Solution
The second option consists of payments made at the beginning of each month. Therefore, it is a monthly annuity-due.
Let j be the effective monthly interest rate.
The present value of the first option is

$$1{,}104.$$

The present value of the second option is

$$100\ddot{a}_{\overline{12}|j}.$$

Since the two options are equivalent,

$$1{,}104 = 100\ddot{a}_{\overline{12}|j}.$$

Using the TVM function, we use **BGN mode** because the payments are made at the beginning of each month.

BGN mode		
N	$=$	12
PV	$=$	$-1{,}104$
PMT	$=$	100
FV	$=$	0
$CPT\ I/Y$	$=$	1.55372

The calculator gives $I/Y = 1.55372$, which is a percentage. Therefore,

$$j = 1.55372\% = 0.0155372.$$

Thus, the required monthly interest rate is

$$\boxed{j = 0.0155372}.$$

So the rate is at least 0.015.
The correct choice is

$$\boxed{\text{E}}.$$

Exercise 2.1.21

The proceeds of a 10,000 death benefit are left on deposit with an insurance company for seven years at an annual effective interest rate of 5%.
The balance at the end of seven years is paid to the beneficiary in 120 equal monthly payments of X, with the first payment made immediately.
During the payout period, interest is credited at an annual effective interest rate of 3%.
Calculate X.

A. 117

B. 118

C. 129

D. 135

E. 158

Solution
We set up an equation of value at time 7.
First, accumulate the death benefit for 7 years at the annual effective interest rate of 5%:

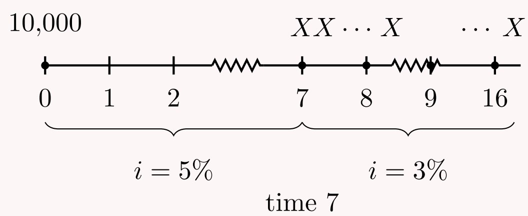
$$10{,}000(1.05)^7.$$

At time 7, the beneficiary receives 120 monthly payments of X, with the first payment made immediately. Therefore, the payout stream is a monthly annuity-due.
Since the payout interest rate is an annual effective rate of 3%, we convert it to an effective monthly rate.
Let j be the effective monthly interest rate. Then

$$1 + j = (1.03)^{1/12}.$$

Thus,

$$j = (1.03)^{1/12} - 1.$$



At time 7, the value of the payout annuity is

$$X\ddot{a}_{\overline{120}|j}.$$

Therefore, the equation of value at time 7 is

$$10{,}000(1.05)^7 = X\ddot{a}_{\overline{120}|j}.$$

Using the annuity-due formula,

$$\ddot{a}_{\overline{120}|j} = \frac{1 - v_j^{120}}{d_j},$$

where

$$v_j = \frac{1}{1+j}$$

and

$$d_j = \frac{j}{1+j}.$$

Thus,

$$X = \frac{10{,}000(1.05)^7}{\ddot{a}_{\overline{120}|j}}.$$

Substituting

$$j = (1.03)^{1/12} - 1,$$

we obtain

$$X = 135.27.$$

Therefore,

$$\boxed{X = 135.27}.$$

The closest answer choice is

$$\boxed{135}.$$

The correct choice is

$$\boxed{\text{D}}.$$

Accumulated Value

We can find the accumulated value of an annuity-due by accumulating each payment to the valuation date.

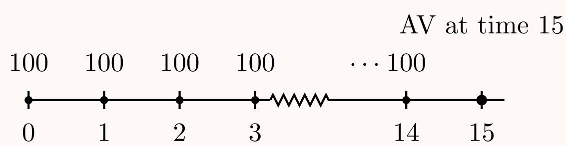
Accumulated Value of an Annuity-Due from First Principles

Given $i = 0.08$, find the accumulated value at time 15 of a 15-year annuity-due with annual payments of 100.

Solution
A 15-year annuity-due has payments at the beginning of each year. Therefore, the payments occur at times

$$0, 1, 2, \ldots, 14.$$

The accumulated value is requested at time 15.



The payment at time 0 accumulates for 15 years:

$$100(1.08)^{15}.$$

The payment at time 1 accumulates for 14 years:

$$100(1.08)^{14}.$$

Continuing this way, the accumulated value at time 15 is

$$AV = 100(1.08)^{15} + 100(1.08)^{14} + \cdots + 100(1.08).$$

Rewrite the terms in increasing powers:

$$AV = 100(1.08) + 100(1.08)^2 + \cdots + 100(1.08)^{15}.$$

This is a geometric series with first term 100(1.08), common ratio 1.08, and next term after the last term $100(1.08)^{16}$.
Thus,

$$AV = \frac{100(1.08) - 100(1.08)^{16}}{1 - 1.08}.$$

Equivalently,

$$AV = 100(1.08)\left(\frac{(1.08)^{15} - 1}{0.08}\right).$$

Therefore,

$$AV = 2{,}932.43.$$

Hence,

$$\boxed{AV = 2{,}932.43}.$$

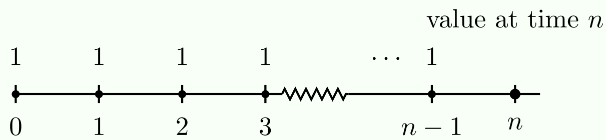
Accumulated Value of an Annuity-Due

At an annual effective interest rate i, the accumulated value at time n of an n-year annuity-due with annual payments of 1 is denoted by

$$\ddot{s}_{\overline{n}|}.$$

An annuity-due has payments at the beginning of each period, so the payments occur at times

$$0, 1, 2, \ldots, n-1.$$



The accumulated value at time n is

$$AV = (1+i)^n + (1+i)^{n-1} + \cdots + (1+i).$$

This is a finite geometric series with first term $(1+i)$, common ratio $(1+i)$, and next term after the last term $(1+i)^{n+1}$.
Therefore,

$$AV = \frac{(1+i) - (1+i)^{n+1}}{1 - (1+i)}.$$

Since

$$1 - (1+i) = -i,$$

we get

$$AV = \frac{(1+i)^{n+1} - (1+i)}{i}.$$

Factoring out $(1+i)$,

$$AV = (1+i)\left(\frac{(1+i)^n - 1}{i}\right).$$

Since

$$d = \frac{i}{1+i},$$

we have

$$\frac{1}{d} = \frac{1+i}{i}.$$

Thus,

$$AV = \frac{(1+i)^n - 1}{d}.$$

Therefore,

$$\boxed{\ddot{s}_{\overline{n}|} = \frac{(1+i)^n - 1}{d}}.$$

Annuity-Immediate vs. Annuity-Due

Recall that the accumulated value of an annuity-immediate is

$$s_{\overline{n}|} = \frac{(1+i)^n - 1}{i}.$$

For an annuity-due, the accumulated value is

$$\ddot{s}_{\overline{n}|} = \frac{(1+i)^n - 1}{d}.$$

So the numerator is the same, but the denominator changes from i to d.
This is not just a trick: it happens because an annuity-due payment stream is shifted one period earlier than an annuity-immediate payment stream.
Equivalently,

$$\ddot{s}_{\overline{n}|} = (1+i)s_{\overline{n}|}.$$

Using the Accumulated Value Formula for an Annuity-Due

Rework the previous example using the $\ddot{s}_{\overline{n}|}$ formula.
Given $i = 0.08$, find the accumulated value at time 15 of a 15-year annuity-due with annual payments of 100.

Solution
A 15-year annuity-due has payments at times

$$0, 1, 2, \ldots, 14.$$

The accumulated value is requested at time 15, one period after the last payment.
Thus,

$$AV = 100\ddot{s}_{\overline{15}|0.08}.$$

Using the formula

$$\ddot{s}_{\overline{n}|} = \frac{(1+i)^n - 1}{d},$$

we get

$$AV = 100\left(\frac{1.08^{15}-1}{d}\right).$$

Since

$$d = \frac{i}{1+i} = \frac{0.08}{1.08},$$

we have

$$AV = 100\left(\frac{1.08^{15}-1}{0.08/1.08}\right).$$

Therefore,

$$AV = 2{,}932.43.$$

Hence,

$$\boxed{AV = 2{,}932.43}.$$

Where $\ddot{s}_{\overline{n}|}$ Is Measured

The symbol

$$\ddot{s}_{\overline{n}|}$$

represents the accumulated value of n payments of 1, measured one period after the last payment.
So if payments occur at times

$$0, 1, \ldots, n-1,$$

then $\ddot{s}_{\overline{n}|}$ is measured at time n.
If payments occur at times

$$4, 5, 6, 7,$$

then $\ddot{s}_{\overline{4}|}$ is measured at time 8.

A Shifted Annuity-Due Accumulated Value

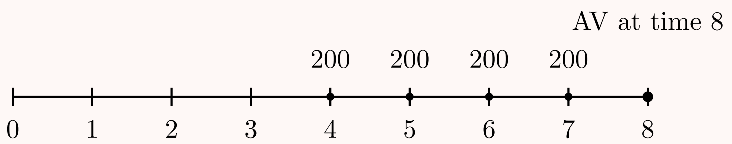
Given $i = 0.10$, find the accumulated value at time 8 of payments of 200 at times 4, 5, 6, and 7.

Solution
The payments occur at times

$$4, 5, 6, 7.$$

These four payments form a 4-payment annuity-due.
Since the accumulated value of an annuity-due is measured one period after the last payment, the accumulated value at time 8 is

$$200\ddot{s}_{\overline{4}|}.$$



Therefore,

$$AV = 200\ddot{s}_{\overline{4}|}.$$

Using

$$\ddot{s}_{\overline{n}|} = \frac{(1+i)^n - 1}{d},$$

we get

$$AV = 200\left(\frac{1.10^4 - 1}{d}\right).$$

Since

$$d = \frac{i}{1+i} = \frac{0.10}{1.10},$$

we have

$$AV = 200\left(\frac{1.10^4 - 1}{0.10/1.10}\right).$$

Thus,

$$AV = 1{,}021.02.$$

Hence,

$$\boxed{AV = 1{,}021.02}.$$

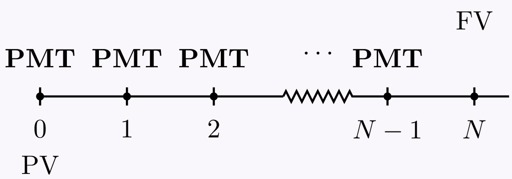
What Interest Rate to Use and TVM Function

In the formula

$$\ddot{s}_{\overline{n}|} = \frac{(1+i)^n - 1}{d},$$

the rate i must be interpreted as the **effective interest rate per payment period**. Similarly, d must be interpreted as the **effective discount rate per payment period**. For example, if payments are monthly, then i and d must be monthly rates.

To value an annuity-due using the TVM function, we need to switch the calculator to **BGN mode**.
In BGN mode, payments occur at the beginning of each payment period.



BGN mode only changes the timing of the PMT amounts.
It tells the calculator that payments occur at the beginning of each payment period instead of at the end.

Exercise 2.1.22

Given $d = 0.04$, find the accumulated value at time 20 of an annuity with payments of 250 every two years.
The first payment is made immediately and the last payment is made 18 years from now.
Solve using:

(1) the $\ddot{s}_{\overline{n}|}$ formula;

(2) the TVM function.

A. 3,925.61

B. 3,975.61

C. 4,025.61

D. 4,075.61

E. 4,125.61

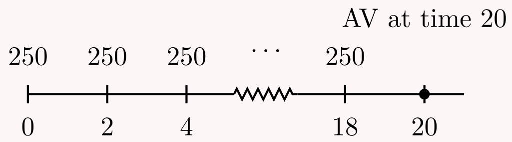
Solution
The payments are made every 2 years, starting immediately.
Thus, the payments occur at times

$$0, 2, 4, \ldots, 18.$$

There are

$$\frac{18-0}{2} + 1 = 10$$

payments.
Since the accumulated value is requested at time 20, which is one two-year period after the last payment, this is an annuity-due accumulated value problem.



Because payments are made every 2 years, we need the effective interest rate per 2-year period.
We are given the annual effective discount rate

$$d = 0.04.$$

Thus, the annual discount factor is

$$v = 1 - d = 0.96.$$

The two-year discount factor is

$$v^2 = (0.96)^2.$$

Therefore, the effective interest rate per 2-year period is

$$j = (0.96)^{-2} - 1.$$

So

$$j = (1 - 0.04)^{-2} - 1.$$

Using the annuity-due accumulated value formula,

$$\ddot{s}_{\overline{n}|j} = \frac{(1+j)^n - 1}{d_j},$$

where

$$d_j = \frac{j}{1+j},$$

we get

$$AV = 250\ddot{s}_{\overline{10}|j}.$$

Therefore,

$$AV = 250\left(\frac{(1+j)^{10}-1}{j/(1+j)}\right).$$

Substituting

$$j = (1-0.04)^{-2}-1,$$

we obtain

$$AV = 4{,}025.61.$$

Hence,

$$\boxed{AV = 4{,}025.61}.$$

The correct choice is

$$\boxed{\text{C}}.$$

TVM function
Since the first payment is made immediately, use **BGN mode**.

BGN mode		
N	$=$	10
I/Y	$=$	$100\left((1-0.04)^{-2}-1\right)$
PV	$=$	0
PMT	$=$	-250
$CPT\ FV$	$=$	$4{,}025.61$

Thus,

$$\boxed{AV = 4{,}025.61}.$$

Exercise 2.1.23

Bill deposits X into a savings account every six months.
His first deposit is made today.
The savings account earns a nominal annual interest rate convertible semiannually of

$$i^{(2)} = 0.06.$$

Bill's account balance immediately before his 25th deposit is 1,772.96.
Determine X.

A. 40

B. 45

C. 50

D. 55

E. 60

Solution
Since deposits are made every six months, there are two deposits per year.
The nominal annual interest rate convertible semiannually is

$$i^{(2)} = 0.06.$$

Therefore, the effective interest rate per six-month period is

$$j = \frac{0.06}{2} = 0.03.$$

Bill makes his first deposit today. Since deposits are made every six months, the deposit times are

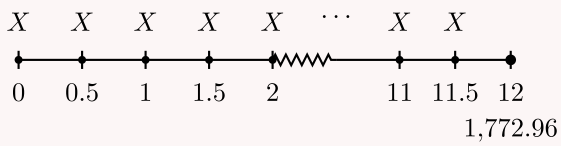
$$0, 0.5, 1, 1.5, \ldots.$$

The 25th deposit occurs at time

$$\frac{24}{2} = 12.$$

We are given the account balance immediately before the 25th deposit. Therefore, the balance is measured at time 12, but before the deposit at time 12 is made.
Thus, only the first 24 deposits are included.
These deposits occur at times

$$0, 0.5, 1, \ldots, 11.5.$$



The first 24 deposits form an annuity-due, accumulated one payment period after the last included deposit.
Therefore,

$$X\ddot{s}_{\overline{24}|0.03} = 1{,}772.96.$$

Using

$$\ddot{s}_{\overline{n}|j} = \frac{(1+j)^n - 1}{d_j},$$

where

$$d_j = \frac{j}{1+j},$$

we get

$$\ddot{s}_{\overline{24}|0.03} = \frac{(1.03)^{24} - 1}{0.03/1.03}.$$

Thus,

$$X = \frac{1{,}772.96}{\ddot{s}_{\overline{24}|0.03}}.$$

So

$$X = \frac{1{,}772.96}{\frac{(1.03)^{24}-1}{0.03/1.03}}.$$

Therefore,

$$X = 50.$$

Hence,

$$\boxed{X = 50}.$$

The correct choice is

$$\boxed{\text{C}}.$$

Using the TVM function, use **BGN mode** because the first deposit is made immediately.

BGN mode		
N	$=$	24
I/Y	$=$	3
PV	$=$	0
FV	$=$	1,772.96
$CPT\ PMT$	$=$	-50

The negative sign comes from the calculator's cash-flow convention. Mathematically,

$$\boxed{X = 50}.$$

Exercise 2.1.24

John wishes to accumulate 15,000 in a trust fund at the end of 10 years.
He will make 10 annual deposits of 1,000, with each annual deposit made at the beginning of the year.
In order to accumulate 15,000 in the trust fund, John needs to increase each of the last 2 payments by X.
The deposits earn interest at an effective annual rate of 5%.
Determine X.

A. Less than 850

B. At least 850, but less than 950

C. At least 950, but less than 1,050

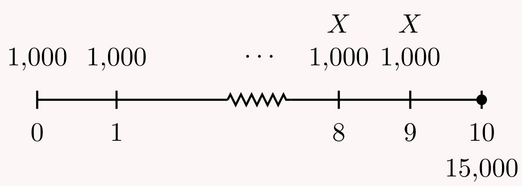
D. At least 1,050, but less than 1,150

E. At least 1,150

Solution
The deposits are made at the beginning of each year, so the 10 regular deposits of 1,000 occur at times

$$0, 1, 2, \ldots, 9.$$

The target accumulated value is at time 10.
The last two deposits occur at times 8 and 9. These are the deposits that are increased by X.



We can split the deposits into two streams:

- 1,000 deposited at the beginning of each year for 10 years;
- an additional X deposited at times 8 and 9.

The accumulated value at time 10 of the regular deposits is

$$1{,}000\ddot{s}_{\overline{10}|0.05}.$$

The accumulated value at time 10 of the two extra deposits is

$$X\ddot{s}_{\overline{2}|0.05}.$$

Therefore, the equation of value at time 10 is

$$1{,}000\ddot{s}_{\overline{10}|0.05} + X\ddot{s}_{\overline{2}|0.05} = 15{,}000.$$

Using

$$\ddot{s}_{\overline{n}|i} = \frac{(1+i)^n - 1}{d},$$

where

$$d = \frac{i}{1+i},$$

we get

$$\ddot{s}_{\overline{10}|0.05} = \frac{1.05^{10} - 1}{0.05/1.05}$$

and

$$\ddot{s}_{\overline{2}|0.05} = \frac{1.05^{2} - 1}{0.05/1.05}.$$

Thus,

$$1{,}000\left(\frac{1.05^{10} - 1}{0.05/1.05}\right) + X\left(\frac{1.05^{2} - 1}{0.05/1.05}\right) = 15{,}000.$$

Solving for X,

$$X = \frac{15{,}000 - 1{,}000\ddot{s}_{\overline{10}|0.05}}{\ddot{s}_{\overline{2}|0.05}}.$$

Substitute the values:

$$X = \frac{15{,}000 - 1{,}000\left(\frac{1.05^{10}-1}{0.05/1.05}\right)}{\frac{1.05^{2}-1}{0.05/1.05}}.$$

Therefore,

$$X = 833.08.$$

Hence,

$$\boxed{X = 833.08}.$$

The correct choice is

$$\boxed{\text{A}}.$$

Equivalently, we can write the equation by separating the first 8 regular deposits from the last 2 increased deposits:

$$1{,}000\ddot{s}_{\overline{8}|0.05}(1.05)^2 + (1{,}000 + X)\ddot{s}_{\overline{2}|0.05} = 15{,}000.$$

This gives the same result:

$$\boxed{X = 833.08}.$$

Exercise 2.1.25

[SOA 2; SOA-IT.002] Kathryn deposits 100 into an account at the beginning of each 4-year period for 40 years.
The account credits interest at an annual effective interest rate of i.
The accumulated amount in the account at the end of 40 years is X, which is 5 times the accumulated amount in the account at the end of 20 years.
Calculate X.

A. 4,695

B. 5,070

C. 5,445

D. 5,820

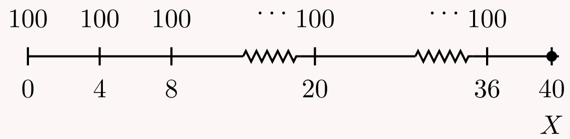
E. 6,195

Solution
The deposits are made at the beginning of each 4-year period.
Therefore, the deposits occur at times

$$0, 4, 8, \ldots, 36.$$

There are 10 deposits in total.
Let j be the effective interest rate per 4-year period. Then

$$1 + j = (1 + i)^4.$$



At time 20, there have been 5 deposits, at times

$$0, 4, 8, 12, 16.$$

Since these deposits are made at the beginning of each 4-year period, their accumulated value at time 20 is

$$100\ddot{s}_{\overline{5}|j}.$$

At time 40, there have been 10 deposits, at times

$$0, 4, 8, \ldots, 36.$$

Thus, the accumulated value at time 40 is

$$100\ddot{s}_{\overline{10}|j}.$$

We are given that the accumulated value at time 40 is 5 times the accumulated value at time 20. Therefore,

$$100\ddot{s}_{\overline{10}|j} = 5\left(100\ddot{s}_{\overline{5}|j}\right).$$

So

$$\ddot{s}_{\overline{10}|j} = 5\ddot{s}_{\overline{5}|j}.$$

Using

$$\ddot{s}_{\overline{n}|j} = \frac{(1+j)^n - 1}{d_j},$$

where

$$d_j = \frac{j}{1+j},$$

we get

$$\frac{(1+j)^{10} - 1}{d_j} = 5\left(\frac{(1+j)^5 - 1}{d_j}\right).$$

Cancel d_j:

$$(1+j)^{10} - 1 = 5\left((1+j)^5 - 1\right).$$

Rearrange:

$$(1+j)^{10} - 1 = 5(1+j)^5 - 5.$$

Thus,

$$(1+j)^{10} - 5(1+j)^5 + 4 = 0.$$

Let

$$y = (1+j)^5.$$

Then

$$y^2 - 5y + 4 = 0.$$

Factor:

$$(y-1)(y-4) = 0.$$

So

$$y = 1 \qquad \text{or} \qquad y = 4.$$

We reject $y = 1$, since it would imply $j = 0$. Therefore,

$$(1+j)^5 = 4.$$

Thus,

$$1 + j = 4^{1/5}.$$

So

$$j = 4^{1/5} - 1.$$

Now compute X:

$$X = 100\ddot{s}_{\overline{10}|j}.$$

Using

$$\ddot{s}_{\overline{10}|j} = \frac{(1+j)^{10} - 1}{d_j},$$

and

$$(1+j)^5 = 4,$$

we have

$$(1+j)^{10} = 16.$$

Also,

$$d_j = 1 - \frac{1}{1+j} = 1 - 4^{-1/5}.$$

Therefore,

$$X = 100\left(\frac{16-1}{1-4^{-1/5}}\right).$$

So

$$X = 6{,}195.$$

Hence,

$$\boxed{X = 6{,}195}.$$

The correct choice is

$$\boxed{\text{E}}.$$

2.1.3 Annuity-Immediate vs. Annuity-Due

The labels *annuity-immediate* and *annuity-due* describe the timing of the payments, but they do not force us to use only one type of notation.

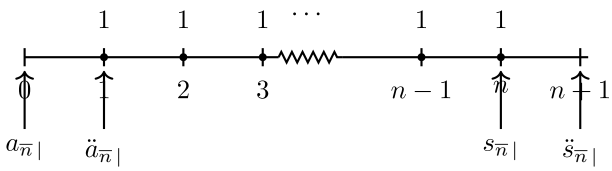
Just because an annuity is an annuity-immediate does not mean we must always value it using

$$a_{\overline{n}|} \qquad \text{or} \qquad s_{\overline{n}|}.$$

Likewise, just because an annuity is an annuity-due does not mean we must always value it using

$$\ddot{a}_{\overline{n}|} \qquad \text{or} \qquad \ddot{s}_{\overline{n}|}.$$

The best approach is to draw a time diagram and choose the notation that values the payments at the most convenient time.



Where Each Notation Measures the Value

For n payments of 1, the four standard annuity symbols measure the value at different points in time:

$a_{\overline{n}|}$ is the value of n payments of 1, one period before the first payment.

$\ddot{a}_{\overline{n}|}$ is the value of n payments of 1, at the time of the first payment.

$s_{\overline{n}|}$ is the value of n payments of 1, at the time of the last payment.

$\ddot{s}_{\overline{n}|}$ is the value of n payments of 1, one period after the last payment.

Main Strategy

Do not memorize these formulas as separate objects only.
Instead, remember where each symbol places the valuation date. Once the valuation date is clear, you can move the value forward or backward using powers of $1+i$ or v.

Which Is Larger: $a_{\overline{n}|}$ or $\ddot{a}_{\overline{n}|}$?

Both $a_{\overline{n}|}$ and $\ddot{a}_{\overline{n}|}$ represent the present value of n payments of 1. The difference is the timing of the payments.
For $a_{\overline{n}|}$, the payments occur at times

$$1, 2, \ldots, n.$$

For $\ddot{a}_{\overline{n}|}$, the payments occur at times

$$0, 1, \ldots, n-1.$$

Thus, the annuity-due payments are received one period earlier. Therefore,

$$\ddot{a}_{\overline{n}|}$$

is larger.
More precisely, the conversion between annuity-immediate and annuity-due present values is summarized in the formula box below.

Useful Relations Between Immediate and Due Annuities

For present values:

$$a_{\overline{n}|} = v\ddot{a}_{\overline{n}|}.$$

Therefore,

$$\boxed{\ddot{a}_{\overline{n}|} = (1+i)a_{\overline{n}|}}.$$

Also,

$$\ddot{a}_{\overline{n}|} = 1 + v + v^2 + \cdots + v^{n-1}.$$

Since

$$a_{\overline{n-1}|} = v + v^2 + \cdots + v^{n-1},$$

we have

$$\boxed{\ddot{a}_{\overline{n}|} = 1 + a_{\overline{n-1}|}}.$$

For accumulated values:

$$\ddot{s}_{\overline{n}|} = (1+i)s_{\overline{n}|}.$$

So

$$\boxed{\ddot{s}_{\overline{n}|} = s_{\overline{n}|}(1+i)}.$$

Also,

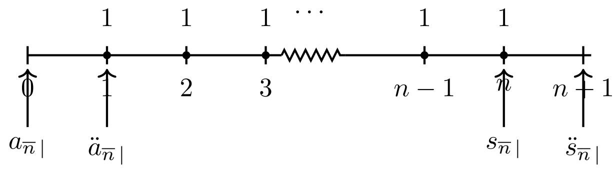
$$s_{\overline{n+1}|} = 1 + (1+i) + \cdots + (1+i)^n.$$

Since

$$\ddot{s}_{\overline{n}|} = (1+i) + (1+i)^2 + \cdots + (1+i)^n,$$

we get

$$\boxed{\ddot{s}_{\overline{n}|} = s_{\overline{n+1}|} - 1}.$$



Exercise 2.1.26

Given

$$a_{\overline{8}|} = 10,$$

find

$$\ddot{a}_{\overline{9}|}.$$

A. 9

B. 10

C. 11

D. 12

E. 13

Solution
Recall that

$$a_{\overline{8}|}$$

is the value of 8 payments of 1, one period before the first payment.
Thus, it represents payments of 1 at times

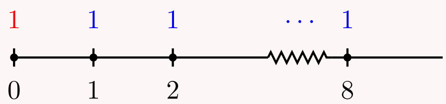
$$1, 2, \ldots, 8.$$

Also,

$$\ddot{a}_{\overline{9}|}$$

is the value of 9 payments of 1, measured at the time of the first payment.
Thus, it represents payments of 1 at times

$$0, 1, 2, \ldots, 8.$$



Therefore, $\ddot{a}_{\overline{9}|}$ is just $a_{\overline{8}|}$ plus the extra payment of 1 at time 0:

$$\ddot{a}_{\overline{9}|} = 1 + a_{\overline{8}|}.$$

Since

$$a_{\overline{8}|} = 10,$$

we get

$$\ddot{a}_{\overline{9}|} = 1 + 10.$$

Thus,

$$\boxed{\ddot{a}_{\overline{9}|} = 11}.$$

The correct choice is

$$\boxed{\text{C}}.$$

Exercise 2.1.27

On June 1, 1970, a company begins making equal annual deposits into a fund to provide 15,000 a year for five successive years with which to retire a bond issue.
The fund earns interest at an effective annual rate of 3%.
The last deposit into the fund is made on June 1, 1980.
The first bonds are redeemed on June 1, 1980.
Calculate the amount of the annual deposit.

A. 5,024.49

B. 5,274.49

C. 5,524.49

D. 5,774.49

E. 6,024.49

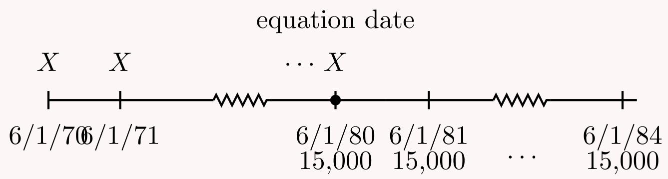
Solution
Let X be the annual deposit.
The company makes deposits from June 1, 1970 through June 1, 1980. Therefore, the deposits occur at times

$$0, 1, 2, \ldots, 10.$$

There are 11 deposits.
The bond redemptions are 15,000 per year for five successive years, starting on June 1, 1980. Therefore, the payments occur at times

$$10, 11, 12, 13, 14.$$

We use June 1, 1980 as the equation-of-value date.



The accumulated value of the deposits at June 1, 1980 is

$$X s_{\overline{11}|0.03}.$$

The value at June 1, 1980 of the bond redemption payments is

$$15{,}000 \ddot{a}_{\overline{5}|0.03}.$$

Thus, the equation of value is

$$X s_{\overline{11}|0.03} = 15{,}000 \ddot{a}_{\overline{5}|0.03}.$$

Using

$$s_{\overline{11}|0.03} = \frac{1.03^{11} - 1}{0.03}$$

and

$$\ddot{a}_{\overline{5}|0.03} = \frac{1-v^5}{d},$$

where

$$v = \frac{1}{1.03} \qquad \text{and} \qquad d = \frac{0.03}{1.03},$$

we get

$$X = \frac{15{,}000\ddot{a}_{\overline{5}|0.03}}{s_{\overline{11}|0.03}}.$$

Therefore,

$$X = 5{,}524.49.$$

Hence,

$$\boxed{X = 5{,}524.49}.$$

The correct choice is

$$\boxed{\text{C}}.$$

Using the TVM function, first find the fund needed on June 1, 1980 to provide the five bond redemptions.
Since the first redemption occurs immediately, use **BGN mode**.

BGN mode		
N	=	5
I/Y	=	3
PMT	=	15,000
FV	=	0
$CPT\ PV$	=	−70,756.476

Now find the annual deposits needed to accumulate 70,756.476 by June 1, 1980.
There are 11 deposits, and this time we use ordinary END mode because

$$X s_{\overline{11}|0.03}$$

matches an annuity-immediate accumulation pattern.

END mode		
N	=	11
I/Y	=	3
PV	=	0
FV	=	−70,756.476
$CPT\ PMT$	=	5,524.49

Thus,

$$\boxed{X = 5{,}524.49}.$$

Exercise 2.1.28

Which of the following does **not** accurately reflect the relationship between immediate and due, present and accumulated value annuities?

1. $\ddot{a}_{\overline{n}|} = (1+i)a_{\overline{n}|}$
2. $\ddot{a}_{\overline{n}|} = 1 + a_{\overline{n-1}|}$
3. $\ddot{s}_{\overline{n}|} = (1+i)s_{\overline{n}|}$
4. $\ddot{s}_{\overline{n}|} = s_{\overline{n+1}|} - 1$
5. $s_{\overline{n}|} = a_{\overline{n}|}(1+i)^{n-1}$

A. 1

B. 2

C. 3

D. 4

E. 5

Solution
We check each identity.
First,

$$\ddot{a}_{\overline{n}|} = (1+i)a_{\overline{n}|}$$

is correct because an annuity-due is shifted one period earlier than an annuity-immediate.
Also,

$$\ddot{a}_{\overline{n}|} = 1 + a_{\overline{n-1}|}$$

is correct, since the annuity-due has one payment at time 0, followed by $n-1$ payments at times $1, 2, \ldots, n-1$.
Next,

$$\ddot{s}_{\overline{n}|} = (1+i)s_{\overline{n}|}$$

is correct because the annuity-due accumulated value is one period after the corresponding annuity-immediate accumulated value.
Also,

$$\ddot{s}_{\overline{n}|} = s_{\overline{n+1}|} - 1$$

is correct. Indeed,

$$s_{\overline{n+1}|} = 1 + (1+i) + \cdots + (1+i)^n,$$

while

$$\ddot{s}_{\overline{n}|} = (1+i) + (1+i)^2 + \cdots + (1+i)^n.$$

Thus,

$$\ddot{s}_{\overline{n}|} = s_{\overline{n+1}|} - 1.$$

Finally, the relationship between present value and accumulated value is

$$s_{\overline{n}|} = a_{\overline{n}|}(1+i)^n.$$

Therefore,

$$s_{\overline{n}|} = a_{\overline{n}|}(1+i)^{n-1}$$

is not correct.
The incorrect statement is statement 5.
Hence, the correct choice is

$$\boxed{\text{E}}.$$

2.1.4 Basic Annuities: Miscellaneous

This section collects useful techniques for annuity problems that do not fit neatly into a single basic formula.

The main idea is that most annuity problems can be solved by drawing a time diagram, choosing a valuation date, and breaking the cash flows into simpler pieces.

Fission Method

The fission method is useful when a payment stream is difficult to value directly, but can be replaced by an equivalent stream of smaller regular payments.

The idea is to split each large payment into several smaller payments whose accumulated value equals the original payment.

Fission Method

Given

$$a_{\overline{20}|i} = 12.46221$$

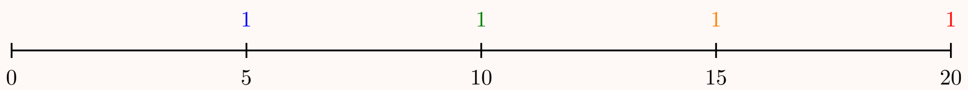
and

$$s_{\overline{5}|i} = 5.52563,$$

where i is the effective annual interest rate, find the present value of 4 payments of 1 paid every 5 years, with the first payment 5 years from now.

Solution
The payments of 1 occur at times

$$5, 10, 15, 20.$$



We do not need to solve for i. Instead, we replace each payment of 1 by a block of five annual payments.
Since

$$s_{\overline{5}|i} = 5.52563,$$

a payment of

$$\frac{1}{s_{\overline{5}|i}}$$

made annually for 5 years has accumulated value 1 at the end of the fifth year, because

$$\frac{1}{s_{\overline{5}|i}} s_{\overline{5}|i} = 1.$$

Therefore, each payment of 1 at times $5, 10, 15, 20$ can be replaced by five annual payments of

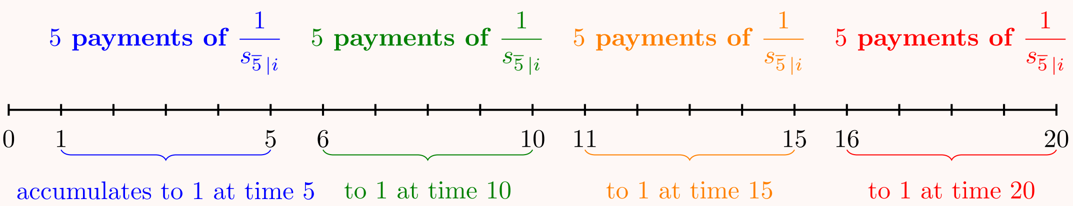
$$\frac{1}{s_{\overline{5}|i}}.$$

Thus, the original payment stream is equivalent to 20 annual payments of

$$\frac{1}{s_{\overline{5}|i}}$$

made at times

$$1, 2, \dots, 20.$$



The present value of 20 annual payments of

$$\frac{1}{s_{\overline{5}|i}}$$

is

$$PV = \frac{1}{s_{\overline{5}|i}} a_{\overline{20}|i}.$$

Substitute the given values:

$$PV = \frac{12.46221}{5.52563}.$$

Therefore,

$$PV = 2.255.$$

Hence,

$$\boxed{PV = 2.255}.$$

Deferred Annuities

Deferred Annuity

A **deferred annuity** is an annuity whose payments do not start in the first payment period.
Instead, the first payment is delayed by a certain number of periods.

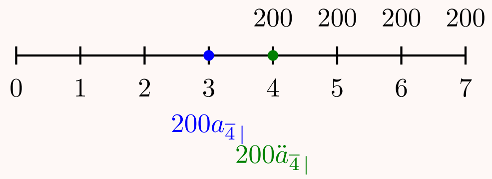
Deferred annuities appear when a payment stream begins later on the timeline.

The main idea is simple: first value the annuity at its natural valuation date, then discount that value back to the desired date.

A Deferred Annuity

Given $i = 0.10$, find the present value of payments of 200 at times 4, 5, 6, and 7.

Solution
The payments occur at times
$$4, 5, 6, 7.$$



There are two natural ways to describe this payment stream.

Method 1: Deferred annuity-immediate
The payments at times $4, 5, 6, 7$ form a 4-payment annuity-immediate valued at time 3. Thus, the value at time 3 is
$$200a_{\overline{4}|}.$$

To bring this value back to time 0, discount by v^3:
$$PV = 200a_{\overline{4}|}v^3.$$

In deferred annuity notation,
$$PV = 200\,{}_{3|}a_{\overline{4}|}.$$

Now compute:
$$PV = 200\left(\frac{1-v^4}{i}\right)v^3.$$

Since
$$i = 0.10 \qquad \text{and} \qquad v = \frac{1}{1.10},$$

we get
$$PV = 200\left(\frac{1-(1.10)^{-4}}{0.10}\right)(1.10)^{-3}.$$

Thus,
$$PV = 476.31.$$

Method 2: Deferred annuity-due
The same payments can also be viewed as a 4-payment annuity-due valued at time 4. Thus, the value at time 4 is
$$200\ddot{a}_{\overline{4}|}.$$

To bring this value back to time 0, discount by v^4:
$$PV = 200\ddot{a}_{\overline{4}|}v^4.$$

In deferred annuity notation,
$$PV = 200\,{}_{4|}\ddot{a}_{\overline{4}|}.$$

Both methods give the same value:
$$\boxed{PV = 476.31}.$$

How to Read Deferred Annuity Notation

The notation

$$ {}_{m|}a_{\overline{n}|} $$

(or

$$ {}_{m|}\ddot{a}_{\overline{n}|} $$

) represents an annuity deferred for

$$ m $$

periods; its value at time

$$ 0 $$

is obtained by discounting the corresponding non-deferred annuity value back

$$ m $$

periods (see formula box below).
The key is to locate the natural valuation date of the annuity.

Deferred Annuities in Terms of Non-Deferred Annuities

Consider an m-year deferred annuity-immediate with n payments of 1.
The payments occur at times

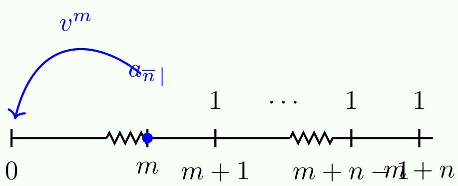
$$ m+1, m+2, \ldots, m+n. $$

The value one period before the first payment is

$$ a_{\overline{n}|}. $$

Since this value is located at time m, the present value at time 0 is

$$ {}_{m|}a_{\overline{n}|} = v^m a_{\overline{n}|}. $$



Therefore,

$$ \boxed{{}_{m|}a_{\overline{n}|} = v^m a_{\overline{n}|}}. $$

Another way to find the same present value is to subtract two non-deferred annuities.
The annuity

$$ a_{\overline{m+n}|} $$

contains payments at times

$$ 1, 2, \ldots, m, m+1, \ldots, m+n. $$

The annuity

$$ a_{\overline{m}|} $$

contains payments at times

$$1, 2, \ldots, m.$$

Subtracting removes the first m payments and leaves only the deferred payments:

$$_{m|}a_{\overline{n}|} = a_{\overline{m+n}|} - a_{\overline{m}|}.$$

Thus,

$$\boxed{_{m|}a_{\overline{n}|} = v^m a_{\overline{n}|} = a_{\overline{m+n}|} - a_{\overline{m}|}}.$$

Similarly, for an m-year deferred annuity-due,

$$\boxed{_{m|}\ddot{a}_{\overline{n}|} = v^m \ddot{a}_{\overline{n}|} = \ddot{a}_{\overline{m+n}|} - \ddot{a}_{\overline{m}|}}.$$

Do Not Memorize These Blindly

These relationships come directly from the time diagram.
The idea is simple:

- $v^m a_{\overline{n}|}$ discounts the annuity value from time m back to time 0;
- $a_{\overline{m+n}|} - a_{\overline{m}|}$ subtracts the unwanted first m payments.

Both methods describe the same deferred payment stream.

Accumulated Value More Than One Period After the Last Payment

Sometimes the accumulated value is requested several periods after the last payment.

In that case, first identify where the standard annuity symbol is naturally measured, then accumulate to the required valuation date.

Accumulated Value More Than One Period After the Last Payment

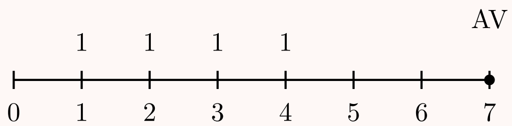
What is the accumulated value of a 4-year annuity-immediate, 3 years after the last payment?

Solution
A 4-year annuity-immediate has payments at times

$$1, 2, 3, 4.$$

The last payment is at time 4. Therefore, 3 years after the last payment is time

$$4 + 3 = 7.$$



The most direct way is to use

$$s_{\overline{4}|},$$

which gives the accumulated value at the time of the last payment, namely time 4.

Since we need the value at time 7, we accumulate for 3 more years:

$$AV = s_{\overline{4}|}(1+i)^3.$$

There are several equivalent ways to write the same accumulated value:

$$\boxed{AV = s_{\overline{4}|}(1+i)^3}$$

$$\boxed{AV = \ddot{s}_{\overline{4}|}(1+i)^2}$$

$$\boxed{AV = a_{\overline{4}|}(1+i)^7}$$

$$\boxed{AV = \ddot{a}_{\overline{4}|}(1+i)^6}$$

We can also obtain the same value by subtraction:

$$\boxed{AV = s_{\overline{7}|} - s_{\overline{3}|}}$$

and

$$\boxed{AV = \ddot{s}_{\overline{6}|} - \ddot{s}_{\overline{2}|}}.$$

Why These Expressions Are Equivalent

Each expression values the same four payments at time 7.
For example,

$$s_{\overline{4}|}$$

is measured at time 4, so we multiply by $(1+i)^3$.
Also,

$$\ddot{s}_{\overline{4}|}$$

is measured one period after the last payment, namely time 5, so we multiply by $(1+i)^2$.
Similarly,

$$a_{\overline{4}|}$$

is measured one period before the first payment, namely time 0, so we multiply by $(1+i)^7$.
Finally,

$$\ddot{a}_{\overline{4}|}$$

is measured at the time of the first payment, namely time 1, so we multiply by $(1+i)^6$.

Current Value

Current Value

The **current value** of an annuity is the value of the annuity at a time during the payment stream.
The valuation date is not necessarily time 0, and it is not necessarily the final payment date.

The method is always the same: choose the valuation date, then move every payment to that date.

Payments before the valuation date are accumulated forward, and payments after the valuation date are discounted backward.

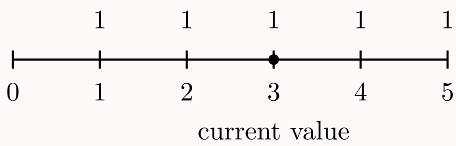
Current Value

What is the current value at time 3 of a 5-year annuity-immediate with annual payments of 1?

Solution
A 5-year annuity-immediate has payments at times

$$1, 2, 3, 4, 5.$$

We want the value at time 3.



At time 3, the payments at times $1, 2, 3$ are accumulated to time 3, while the payments at times $4, 5$ are discounted back to time 3.
One way to write the current value is

$$CV_3 = s_{\overline{3}|} + a_{\overline{2}|}.$$

Indeed,

$$s_{\overline{3}|}$$

values the payments at times $1, 2, 3$ at time 3, and

$$a_{\overline{2}|}$$

values the payments at times $4, 5$ at time 3.
Thus,

$$\boxed{CV_3 = s_{\overline{3}|} + a_{\overline{2}|}}.$$

There are many equivalent ways to write the same value:

$$\boxed{CV_3 = \ddot{s}_{\overline{2}|} + \ddot{a}_{\overline{3}|}}$$

$$\boxed{CV_3 = a_{\overline{5}|}(1+i)^3}$$

$$\boxed{CV_3 = \ddot{a}_{\overline{5}|}(1+i)^2}$$

$$\boxed{CV_3 = s_{\overline{5}|}v^2}$$

$$\boxed{CV_3 = \ddot{s}_{\overline{5}|}v^3}$$

All of these expressions represent the same five payments valued at time 3.

Current Value Strategy

A current value problem is not a new type of annuity formula.
It is simply an equation-of-value problem where the valuation date is somewhere inside the timeline.
The steps are:

1. Draw the timeline.
2. Mark the valuation date.
3. Split the payments if needed.
4. Move every cash flow to the valuation date.

Exercise 2.1.29

An annuity makes 5 payments of 200 every two years starting today. You are given:

$$\ddot{a}_{\overline{2}|} = 1.943$$

and

$$\ddot{s}_{\overline{10}|} = 13.972.$$

Find the accumulated value of the annuity two years after the last payment.

A. 1,238.19

B. 1,338.19

C. 1,438.19

D. 1,538.19

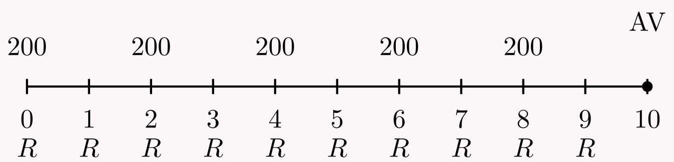
E. 1,638.19

Solution
The annuity makes payments of 200 every two years starting today. Therefore, the payments occur at times

$$0, 2, 4, 6, 8.$$

The last payment is at time 8. Two years after the last payment is time

$$10.$$



$$R = \frac{200}{\ddot{a}_{\overline{2}|}}$$

We use the fission method.
A payment of 200 every two years can be replaced by two annual payments of R, where

$$R\ddot{a}_{\overline{2}|} = 200.$$

Thus,

$$R = \frac{200}{\ddot{a}_{\overline{2}|}}.$$

Therefore, the original payment stream is equivalent to 10 annual payments of

$$\frac{200}{\ddot{a}_{\overline{2}|}}$$

made at times

$$0, 1, 2, \ldots, 9.$$

Since the accumulated value is required at time 10, we use

$$\ddot{s}_{\overline{10}|}.$$

Thus,

$$AV = \left(\frac{200}{\ddot{a}_{\overline{2}|}}\right) \ddot{s}_{\overline{10}|}.$$

Substitute the given values:

$$AV = \left(\frac{200}{1.943}\right)(13.972).$$

Therefore,

$$AV = 1{,}438.19.$$

Hence,

$$\boxed{AV = 1{,}438.19}.$$

The correct choice is

$$\boxed{\text{C}}.$$

Exercise 2.1.30

Annuities A and B have the following payments:

End of Year	Annuity A	Annuity B
1–5	0	K
6–10	2	0
11–15	1	K

Annuities A and B have equal present values at an annual effective interest rate i such that

$$v^5 = \frac{1}{3}.$$

Calculate K.

A. 0.5

B. 0.6

C. 0.7

D. 0.8

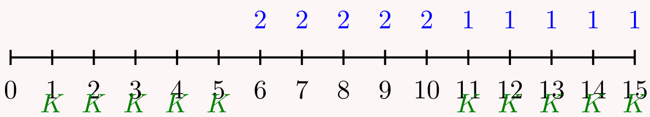
E. 0.9

Solution
Let

$$q = v^5.$$

We are given

$$q = \frac{1}{3}.$$



The present value of Annuity A is

$$2a_{\overline{5}|}v^5 + a_{\overline{5}|}v^{10}.$$

The present value of Annuity B is

$$Ka_{\overline{5}|} + Ka_{\overline{5}|}v^{10}.$$

Since the two present values are equal,

$$2a_{\overline{5}|}v^5 + a_{\overline{5}|}v^{10} = Ka_{\overline{5}|} + Ka_{\overline{5}|}v^{10}.$$

Factor out $a_{\overline{5}|}$:

$$a_{\overline{5}|}\left(2v^5 + v^{10}\right) = Ka_{\overline{5}|}\left(1 + v^{10}\right).$$

Cancel $a_{\overline{5}|}$:

$$2v^5 + v^{10} = K(1 + v^{10}).$$

Since

$$v^5 = \frac{1}{3},$$

we have

$$v^{10} = \left(v^5\right)^2 = \left(\frac{1}{3}\right)^2 = \frac{1}{9}.$$

Therefore,

$$2\left(\frac{1}{3}\right) + \frac{1}{9} = K\left(1 + \frac{1}{9}\right).$$

So

$$\frac{2}{3} + \frac{1}{9} = K\left(\frac{10}{9}\right).$$

Thus,

$$\frac{7}{9} = K\left(\frac{10}{9}\right).$$

Solving for K,

$$K = \frac{7}{10}.$$

Therefore,

$$\boxed{K = 0.7}.$$

The correct choice is

$$\boxed{\text{C}}.$$

Exercise 2.1.31

You are given that X is the current value at the end of year 2 of a 20-year annuity-due of 1 per annum.
The effective annual interest rate for year t is

$$\frac{1}{8+t}.$$

Calculate X.

A. $\displaystyle\sum_{t=9}^{18} \frac{10}{t}$

B. $\displaystyle\sum_{t=9}^{28} \frac{11}{t}$

C. $\displaystyle\sum_{t=10}^{29} \frac{10}{t}$

D. $\displaystyle\sum_{t=10}^{29} \frac{11}{t}$

E. $\displaystyle\sum_{t=11}^{30} \frac{10}{t}$

Solution
A 20-year annuity-due has payments of 1 at times

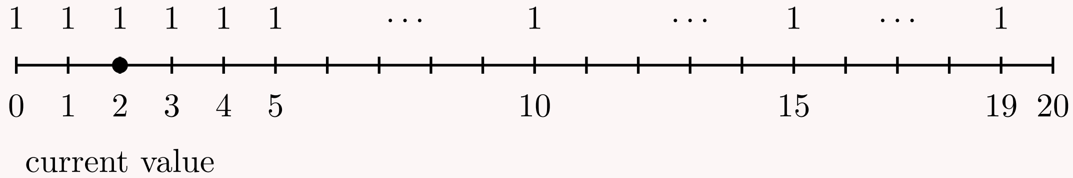
$$0, 1, 2, \ldots, 19.$$

We want the current value at the end of year 2, that is, at time 2.
The effective interest rate for year t is

$$i_t = \frac{1}{8+t}.$$

Thus, the accumulation factor for year t is

$$1 + i_t = 1 + \frac{1}{8+t} = \frac{9+t}{8+t}.$$



Now value each payment at time 2.
The payment at time 0 accumulates for years 1 and 2:

$$\left(1+\frac{1}{9}\right)\left(1+\frac{1}{10}\right) = \frac{10}{9} \cdot \frac{11}{10} = \frac{11}{9}.$$

The payment at time 1 accumulates for year 2:

$$1 + \frac{1}{10} = \frac{11}{10}.$$

The payment at time 2 is already at the valuation date:

$$1 = \frac{11}{11}.$$

The payment at time 3 is discounted one year back to time 2:

$$\left(1 + \frac{1}{11}\right)^{-1} = \left(\frac{12}{11}\right)^{-1} = \frac{11}{12}.$$

The payment at time 4 is discounted two years back:

$$\left(1 + \frac{1}{11}\right)^{-1}\left(1 + \frac{1}{12}\right)^{-1} = \frac{11}{12} \cdot \frac{12}{13} = \frac{11}{13}.$$

Continuing this pattern, the current value is

$$X = \frac{11}{9} + \frac{11}{10} + \frac{11}{11} + \frac{11}{12} + \cdots + \frac{11}{28}.$$

Therefore,

$$X = \sum_{t=9}^{28} \frac{11}{t}.$$

Hence,

$$\boxed{X = \sum_{t=9}^{28} \frac{11}{t}}.$$

The correct choice is

$$\boxed{\text{B}}.$$

Exercise 2.1.32

Mr. Smith wants to give his son an annuity of 5,000 per year starting on his 21st birthday, which will be increased to 10,000 per year on his 25th birthday, with the final payment on his 30th birthday.
What is the present value of that annuity on his son's 10th birthday if the effective annual interest rate is 5%?

A. Less than 35,500

B. At least 35,500, but less than 36,000

C. At least 36,000, but less than 36,500

D. At least 36,500, but less than 37,000

E. At least 37,000

Solution

We value the annuity on the son's 10th birthday.
The payments are:

$$5{,}000 \quad \text{at ages } 21, 22, 23, 24,$$

and

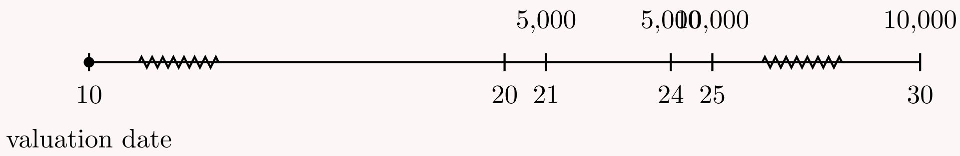
$$10{,}000 \quad \text{at ages } 25, 26, 27, 28, 29, 30.$$

The effective annual interest rate is

$$i = 0.05.$$

Thus,

$$v = \frac{1}{1.05}.$$



One direct approach is to split the payments into two deferred annuities.
The 5,000 payments at ages 21 through 24 form a 4-payment annuity-immediate valued one year before the first payment, at age 20.
So their value at age 10 is

$$5{,}000 a_{\overline{4}|0.05} v^{10}.$$

The 10,000 payments at ages 25 through 30 form a 6-payment annuity-immediate valued one year before the first payment, at age 24.
So their value at age 10 is

$$10{,}000 a_{\overline{6}|0.05} v^{14}.$$

Therefore,

$$PV = 5{,}000 a_{\overline{4}|0.05} v^{10} + 10{,}000 a_{\overline{6}|0.05} v^{14}.$$

Using

$$a_{\overline{n}|i} = \frac{1 - v^n}{i},$$

we get

$$PV = 5{,}000 \left(\frac{1 - v^4}{0.05} \right) v^{10} + 10{,}000 \left(\frac{1 - v^6}{0.05} \right) v^{14}.$$

Substituting $v = 1/1.05$,

$$PV = 36{,}520.22.$$

Thus,

$$\boxed{PV = 36{,}520.22}.$$

The correct choice is

$$\boxed{\text{D}}.$$

Another elegant method is to rewrite the cash flows as

$$10{,}000 a_{\overline{20}|0.05} - 5{,}000 a_{\overline{14}|0.05} - 5{,}000 a_{\overline{10}|0.05}.$$

Indeed, $10{,}000 a_{\overline{20}|}$ gives payments of 10,000 from ages 11 to 30. Then subtracting

$$5{,}000 a_{\overline{14}|}$$

removes 5,000 from ages 11 to 24, and subtracting

$$5{,}000a_{\overline{10}|}$$

removes another 5,000 from ages 11 to 20.
This leaves exactly 5,000 from ages 21 to 24, and 10,000 from ages 25 to 30.
So

$$PV = 10{,}000a_{\overline{20}|0.05} - 5{,}000a_{\overline{14}|0.05} - 5{,}000a_{\overline{10}|0.05}.$$

This gives the same result:

$$PV = 36{,}520.22.$$

Exercise 2.1.33

A loan of 8,000 is to be repaid in 36 equal monthly installments, with the first one paid six months after the loan is made.
The nominal annual interest rate is 10% compounded quarterly.
Determine the amount of the monthly payment.

A. Less than 258

B. At least 258, but less than 262

C. At least 262, but less than 266

D. At least 266, but less than 270

E. At least 270

Solution
Let X be the monthly payment.
Since the payments are monthly, we first convert the nominal annual rate compounded quarterly into an effective monthly rate.
The quarterly effective rate is

$$\frac{0.10}{4} = 0.025.$$

Let j be the effective monthly interest rate. Since one quarter contains 3 months,

$$(1 + j)^3 = 1.025.$$

Thus,

$$j = (1.025)^{1/3} - 1.$$

Equivalently,

$$j = \left(1 + \frac{0.10}{4}\right)^{4/12} - 1.$$

So

$$j = 0.008264838.$$

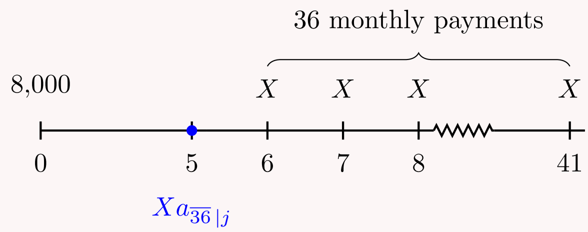
The first payment is made 6 months after the loan is made. Therefore, the 36 monthly payments occur at months

$$6, 7, 8, \ldots, 41.$$

A 36-payment annuity-immediate with first payment at month 6 is naturally valued one month before the first payment, namely at month 5.

Therefore, the present value at month 0 is

$$Xa_{\overline{36}|j}v_j^5.$$



The equation of value at the loan date is

$$8{,}000 = Xa_{\overline{36}|j}v_j^5.$$

Using

$$a_{\overline{36}|j} = \frac{1 - v_j^{36}}{j},$$

we solve for X:

$$X = \frac{8{,}000}{a_{\overline{36}|j}v_j^5}.$$

Substituting

$$j = 0.008264838,$$

we obtain

$$X = 268.66.$$

Therefore,

$$\boxed{X = 268.66}.$$

The correct choice is

$$\boxed{\text{D}}.$$

Exercise 2.1.34

Eloise plans to accumulate 100,000 at the end of 42 years.
She makes the following deposits:

(i) X at the beginning of years 1 through 14;

(ii) no deposits at the beginning of years 15 through 32;

(iii) Y at the beginning of years 33 through 42.

The annual effective interest rate is 7%. Also,

$$X - Y = 100.$$

Calculate Y.

A. 479

B. 499

C. 519

D. 539

E. 559

Solution
The deposits of X are made at the beginning of years 1 through 14. Therefore, they occur at times

$$0, 1, 2, \ldots, 13.$$

The deposits of Y are made at the beginning of years 33 through 42. Therefore, they occur at times

$$32, 33, \ldots, 41.$$

The target accumulated value is at time 42.
Since

$$X - Y = 100,$$

we have

$$X = Y + 100.$$

One way to represent the first set of deposits is to value them at time 0 using $\ddot{a}_{\overline{14}|}$, then accumulate that value to time 42.
Thus, the accumulated value at time 42 of the first set of deposits is

$$X\ddot{a}_{\overline{14}|0.07}(1.07)^{42}.$$

The last 10 deposits of Y occur at times $32, 33, \ldots, 41$. Their accumulated value at time 42 is

$$Y\ddot{s}_{\overline{10}|0.07}.$$

Therefore, the equation of value at time 42 is

$$X\ddot{a}_{\overline{14}|0.07}(1.07)^{42} + Y\ddot{s}_{\overline{10}|0.07} = 100{,}000.$$

Since $X = Y + 100$, we get

$$(Y + 100)\ddot{a}_{\overline{14}|0.07}(1.07)^{42} + Y\ddot{s}_{\overline{10}|0.07} = 100{,}000.$$

Solving gives

$$Y \approx 479.17.$$

Therefore,

$$\boxed{Y \approx 479}.$$

The correct choice is

$$\boxed{\text{A}}.$$

Equivalently, we can accumulate the first set of deposits first to time 14, then for another 28 years.
This gives

$$(Y + 100)\ddot{s}_{\overline{14}|0.07}(1.07)^{28} + Y\ddot{s}_{\overline{10}|0.07} = 100{,}000.$$

This is the same equation and gives the same result:

$$\boxed{Y \approx 479}.$$

2.1.5 Perpetuities

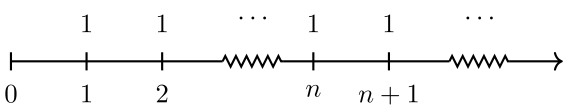
Perpetuity-Immediate

Perpetuity

A **perpetuity** is an annuity whose payments continue forever.
A **perpetuity-immediate** is a perpetuity with payments at the end of each period forever.

Consider a perpetuity-immediate with payments of 1 at times

$$1, 2, 3, \ldots.$$



The present value of this perpetuity is denoted by

$$a_{\overline{\infty}|}.$$

Since the payments are made at times $1, 2, 3, \ldots$, we have

$$a_{\overline{\infty}|} = v + v^2 + v^3 + \cdots.$$

Multiplying both sides by v,

$$va_{\overline{\infty}|} = v^2 + v^3 + v^4 + \cdots.$$

Subtracting the second equation from the first gives

$$a_{\overline{\infty}|} - va_{\overline{\infty}|} = v.$$

Therefore,

$$(1 - v)a_{\overline{\infty}|} = v.$$

So

$$a_{\overline{\infty}|} = \frac{v}{1 - v}.$$

Since

$$v = \frac{1}{1 + i},$$

we get

$$a_{\overline{\infty}|} = \frac{\frac{1}{1+i}}{1 - \frac{1}{1+i}}.$$

Simplifying,

$$a_{\overline{\infty}|} = \frac{\frac{1}{1+i}}{\frac{i}{1+i}} = \frac{1}{i}.$$

Thus,

$$\boxed{a_{\overline{\infty}|} = \frac{1}{i}}.$$

Perpetuity-Immediate Formula

For a perpetuity-immediate with payments of R each period, the present value is

$$PV = \frac{R}{i}.$$

This works only when $i > 0$. If $i = 0$, the present value of infinitely many positive payments is infinite.

Faster Derivations of $a_{\overline{\infty}|}$

A faster derivation uses the finite annuity-immediate formula:

$$a_{\overline{n}|} = \frac{1 - v^n}{i}.$$

As $n \to \infty$, if $i > 0$, then

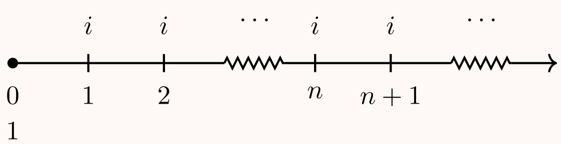
$$v^n \to 0.$$

Therefore,

$$a_{\overline{\infty}|} = \lim_{n\to\infty} a_{\overline{n}|} = \lim_{n\to\infty} \frac{1 - v^n}{i} = \frac{1}{i}.$$

Investment Interpretation of a Perpetuity-Immediate

Suppose we invest 1 now in a fund earning an effective interest rate i per period.
At the end of each period, we withdraw only the interest earned during that period.
Since the principal remains equal to 1, the fund generates payments of i forever.



The present value of the stream of interest payments is

$$ia_{\overline{\infty}|}.$$

But this stream is generated by an initial investment of 1. Therefore,

$$1 = ia_{\overline{\infty}|}.$$

Solving,

$$a_{\overline{\infty}|} = \frac{1}{i}.$$

Hence,

$$\boxed{a_{\overline{\infty}|} = \frac{1}{i}}.$$

Interest Rate to Use

In the perpetuity-immediate formula

$$a_{\overline{\infty}|} = \frac{1}{i},$$

the rate i must be the effective interest rate per payment period.
If payments are monthly, use the effective monthly interest rate.
If payments are quarterly, use the effective quarterly interest rate.

Perpetuity with Payments Every Two Years

Given $i = 0.10$, find the present value of a perpetuity with payments of 300 at times

$$2, 4, 6, \ldots.$$

Solution
The payments are made every two years, so we need the effective interest rate per 2-year payment period.
Let j be the effective interest rate per 2 years. Then

$$1 + j = (1.10)^2.$$

Therefore,

$$j = (1.10)^2 - 1.$$

So

$$j = 0.21.$$

The payments occur at times

$$2, 4, 6, \ldots,$$

so this is a perpetuity-immediate with respect to 2-year periods.
Thus,

$$PV = 300a_{\overline{\infty}|j}.$$

Using

$$a_{\overline{\infty}|j} = \frac{1}{j},$$

we get

$$PV = 300\left(\frac{1}{0.21}\right).$$

Therefore,

$$PV = \frac{300}{0.21} = 1{,}428.57.$$

Hence,

$$\boxed{PV = 1{,}428.57}.$$

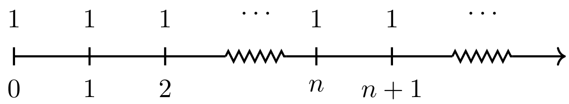
Perpetuity-Due

Perpetuity-Due

A **perpetuity-due** is an annuity with payments at the beginning of each period forever.

Consider a perpetuity-due with payments of 1 at times

$$0, 1, 2, 3, \ldots .$$



The present value of this perpetuity-due is denoted by

$$\ddot{a}_{\overline{\infty}|}.$$

Since the payments are made at times $0, 1, 2, \ldots$, we have

$$\ddot{a}_{\overline{\infty}|} = 1 + v + v^2 + v^3 + \cdots .$$

But

$$a_{\overline{\infty}|} = v + v^2 + v^3 + \cdots .$$

Therefore,

$$\ddot{a}_{\overline{\infty}|} = 1 + a_{\overline{\infty}|}.$$

Since

$$a_{\overline{\infty}|} = \frac{1}{i},$$

we get

$$\ddot{a}_{\overline{\infty}|} = 1 + \frac{1}{i}.$$

Thus,

$$\ddot{a}_{\overline{\infty}|} = \frac{1+i}{i}.$$

Since

$$d = \frac{i}{1+i},$$

we have

$$\frac{1}{d} = \frac{1+i}{i}.$$

Therefore,

$$\boxed{\ddot{a}_{\overline{\infty}|} = \frac{1}{d}}.$$

Perpetuity-Due Formula

For a perpetuity-due with payments of R each period, the present value can be written in any of the equivalent forms summarized below.

Immediate vs. Due Perpetuities

For a perpetuity-immediate,

$$a_{\overline{\infty}|} = \frac{1}{i}.$$

For a perpetuity-due,

$$\ddot{a}_{\overline{\infty}|} = \frac{1}{d}.$$

Since the perpetuity-due payment stream is shifted one period earlier,

$$\ddot{a}_{\overline{\infty}|} = (1+i)a_{\overline{\infty}|}.$$

Indeed,

$$(1+i)a_{\overline{\infty}|} = (1+i)\frac{1}{i} = \frac{1+i}{i} = \frac{1}{d}.$$

Exercise 2.1.35

Deposits of 1,000 are placed into a fund at the beginning of each year for 30 years.
At the end of the 40th year, annual withdrawals commence and continue forever.
Interest is at an effective annual rate of 5%.
Calculate the annual withdrawal.

A. 5,211.09

B. 5,311.09

C. 5,411.09

D. 5,511.09

E. 5,611.09

Solution

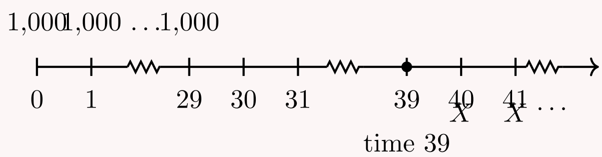
The deposits are made at the beginning of each year for 30 years, so they occur at times

$$0, 1, 2, \ldots, 29.$$

The annual withdrawals start at the end of the 40th year, so the withdrawals occur at times

$$40, 41, 42, \ldots.$$

Let X be the annual withdrawal.



A convenient equation of value is at time 39, one year before the first withdrawal.
The deposits at times $0, 1, \ldots, 29$ have accumulated value at time 39:

$$1{,}000 s_{\overline{30}|0.05}(1.05)^{10}.$$

The withdrawals at times $40, 41, 42, \ldots$ form a perpetuity-immediate valued at time 39:

$$X a_{\overline{\infty}|0.05}.$$

Therefore,

$$1{,}000 s_{\overline{30}|0.05}(1.05)^{10} = X a_{\overline{\infty}|0.05}.$$

Using

$$a_{\overline{\infty}|0.05} = \frac{1}{0.05},$$

we get

$$1{,}000 s_{\overline{30}|0.05}(1.05)^{10} = X\left(\frac{1}{0.05}\right).$$

Thus,

$$X = 1{,}000 s_{\overline{30}|0.05}(1.05)^{10}(0.05).$$

Using

$$s_{\overline{30}|0.05} = \frac{1.05^{30} - 1}{0.05},$$

we obtain

$$X = 5{,}411.09.$$

Hence,

$$\boxed{X = 5{,}411.09}.$$

The correct choice is

$$\boxed{\text{C}}.$$

Equivalently, we could use an equation of value at time 40.
At time 40, the withdrawals form a perpetuity-due:

$$X \ddot{a}_{\overline{\infty}|0.05}.$$

The deposits have accumulated value

$$1{,}000 \ddot{a}_{\overline{30}|0.05}(1.05)^{40}.$$

So another valid equation is

$$1{,}000 \ddot{a}_{\overline{30}|0.05}(1.05)^{40} = X \ddot{a}_{\overline{\infty}|0.05}.$$

This gives the same result:

$$\boxed{X = 5{,}411.09}.$$

Exercise 2.1.36

A perpetuity of 1 each year, with the first payment due immediately, has a present value of 25 at an annual effective interest rate i.
The owner is considering exchanging it now for another perpetuity with the first payment

due immediately and subsequent payments due at two-year intervals.
What should the payment of the second perpetuity be in order to keep the same interest rate i and the same present value?

A. 1.76

B. 1.86

C. 1.96

D. 2.06

E. 2.16

Solution
The first perpetuity pays 1 immediately and then 1 each year forever.
Thus, it is a perpetuity-due with annual payments of 1. Its present value is

$$\ddot{a}_{\overline{\infty}|i} = 25.$$

For a perpetuity-due,

$$\ddot{a}_{\overline{\infty}|i} = \frac{1}{d}.$$

Therefore,

$$\frac{1}{d} = 25.$$

So

$$d = 0.04.$$

The annual discount factor is

$$v = 1 - d = 0.96.$$

The second perpetuity pays X immediately and then every two years forever.
Thus, we need the effective discount rate per two-year period.
Let e be the effective discount rate per two years. Then

$$e = 1 - v^2.$$

Since

$$v = 0.96,$$

we get

$$e = 1 - (0.96)^2.$$

Thus,

$$e = 0.0784.$$

The second perpetuity is a perpetuity-due with payments every two years. Therefore, its present value is

$$\frac{X}{e}.$$

We want this present value to remain equal to 25. Hence,

$$\frac{X}{e} = 25.$$

So

$$X = 25e.$$

Substitute $e = 0.0784$:

$$X = 25(0.0784).$$

Therefore,

$$X = 1.96.$$

Hence,

$$\boxed{X = 1.96}.$$

The correct choice is

$$\boxed{\text{C}}.$$

Exercise 2.1.37

John's estate is to be divided into three equal parts and invested, to be paid out as follows:

- John's two children will each receive their share in 20 level annual payments beginning one year after John's death.
- Charity Q will receive its share as equal annual payments in perpetuity beginning 21 years after John's death.

Charity Q's annual payment is twice the annual payment for one child.
Determine the effective annual interest rate at which the estate is invested.

A. 4.6%

B. 5.1%

C. 5.6%

D. 6.1%

E. 6.6%

Solution
The phrase "divided into three equal parts" means that the present value of each party's payments is equal.
Let X be the annual payment for one child.
Then Charity Q's annual payment is

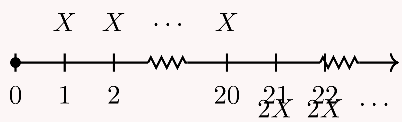
$$2X.$$

Each child receives payments at times

$$1, 2, \ldots, 20.$$

Charity Q receives payments at times

$$21, 22, 23, \ldots.$$



The present value of one child's payments is

$$X a_{\overline{20}|}.$$

Charity Q's payments form a perpetuity-immediate beginning at time 21. Its value one period before the first payment, at time 20, is

$$2X a_{\overline{\infty}|}.$$

Discounting this value back 20 years gives

$$2X a_{\overline{\infty}|} v^{20}.$$

Since the present values are equal,

$$X a_{\overline{20}|} = 2X a_{\overline{\infty}|} v^{20}.$$

Cancel X:

$$a_{\overline{20}|} = 2a_{\overline{\infty}|} v^{20}.$$

Using

$$a_{\overline{20}|} = \frac{1 - v^{20}}{i}$$

and

$$a_{\overline{\infty}|} = \frac{1}{i},$$

we get

$$\frac{1 - v^{20}}{i} = 2\left(\frac{1}{i}\right) v^{20}.$$

Cancel $1/i$:

$$1 - v^{20} = 2v^{20}.$$

Thus,

$$1 = 3v^{20}.$$

So

$$v^{20} = \frac{1}{3}.$$

Taking reciprocals,

$$(1 + i)^{20} = 3.$$

Therefore,

$$i = 3^{1/20} - 1.$$

Numerically,

$$i = 0.0565.$$

Thus,

$$\boxed{i \approx 5.65\%}.$$

The closest choice is

$$\boxed{\text{C}}.$$

Exercise 2.1.38

Perpetuity A pays 100 at the end of each year.
Perpetuity B pays 25 at the end of each quarter.
The present value of Perpetuity A at the annual effective interest rate i is 2,000.
What is the present value of Perpetuity B at the same annual effective interest rate i?

A. Less than 2,020

B. At least 2,020, but less than 2,025

C. At least 2,025, but less than 2,030

D. At least 2,030, but less than 2,035

E. At least 2,035

Solution
First, find the annual effective interest rate i using Perpetuity A.
Since Perpetuity A pays 100 at the end of each year, it is a perpetuity-immediate. Therefore,

$$100a_{\overline{\infty}|i} = 2{,}000.$$

Using

$$a_{\overline{\infty}|i} = \frac{1}{i},$$

we get

$$100\left(\frac{1}{i}\right) = 2{,}000.$$

Thus,

$$i = \frac{100}{2{,}000} = 0.05.$$

So

$$i = 5\%.$$

Now find the effective interest rate per quarter.
Let j be the effective quarterly interest rate. Since the annual effective interest rate is 5%,

$$(1+j)^4 = 1.05.$$

Therefore,

$$j = (1.05)^{1/4} - 1.$$

So

$$j = 0.012272234.$$

Perpetuity B pays 25 at the end of each quarter, so it is a perpetuity-immediate with quarterly payments.
Therefore,

$$PV_B = 25a_{\overline{\infty}|j}.$$

Using

$$a_{\overline{\infty}|j} = \frac{1}{j},$$

we get

$$PV_B = 25\left(\frac{1}{j}\right).$$

Thus,

$$PV_B = 25\left(\frac{1}{0.012272234}\right).$$

Therefore,

$$PV_B = 2{,}037.12.$$

Hence,

$$\boxed{PV_B = 2{,}037.12}.$$

The correct choice is

$$\boxed{\text{E}}.$$

Exercise 2.1.39

The death benefit on a life insurance policy can be paid in any of the following ways, each of which has the same present value as the death benefit:

i. a perpetuity of 120 at the end of each month;

ii. 365.47 at the end of each month for n years;

iii. a payment of 17,866.32 at the end of n years.

Calculate the amount of the death benefit.

A. 8,000

B. 9,000

C. 10,000

D. 12,000

E. 15,000

Solution

Let j be the effective monthly interest rate.
Since all three payment options have the same present value, we first equate options (i) and (ii).
Option (i) is a monthly perpetuity-immediate of 120, so its present value is

$$120a_{\overline{\infty}|j}.$$

Option (ii) is a monthly annuity-immediate of 365.47 for n years. Since there are $12n$ monthly payments, its present value is

$$365.47a_{\overline{12n}|j}.$$

Thus,

$$120a_{\overline{\infty}|j} = 365.47a_{\overline{12n}|j}.$$

Using

$$a_{\overline{\infty}|j} = \frac{1}{j}$$

and

$$a_{\overline{12n}|j} = \frac{1 - v_j^{12n}}{j},$$

we get

$$120\left(\frac{1}{j}\right) = 365.47\left(\frac{1 - v_j^{12n}}{j}\right).$$

Cancel $\frac{1}{j}$:

$$120 = 365.47\left(1 - v_j^{12n}\right).$$

Therefore,

$$1 - v_j^{12n} = \frac{120}{365.47}.$$

So

$$v_j^{12n} = 1 - \frac{120}{365.47}.$$

Thus,

$$v_j^{12n} = 0.67166.$$

Now use option (iii). The death benefit is equal to the present value of the single payment:

$$17{,}866.32 v_j^{12n}.$$

Therefore,

$$\text{Death benefit} = 17{,}866.32(0.67166).$$

So

$$\text{Death benefit} = 12{,}000.$$

Hence,

$$\boxed{12{,}000}.$$

The correct choice is

$$\boxed{\text{D}}.$$

Exercise 2.1.40

Deposits of 1,000 are placed into a fund at the beginning of each year for 30 years.
At the end of the 40th year, annual payments commence and continue forever.
Interest is at an effective rate of 5%.
Calculate the annual payment.

A. 5,440

B. 5,430

C. 5,420

D. 5,410

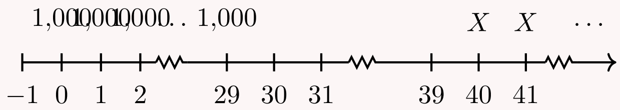
E. 5,400

Solution
Let X be the annual payment.
The deposits are made at the beginning of each year for 30 years, so they occur at times

$$0, 1, 2, \ldots, 29.$$

The annual payments begin at the end of the 40th year, so they occur at times

$$40, 41, 42, \ldots.$$



A convenient equation of value is at time -1.
At time -1, the deposits form an annuity-immediate with 30 payments:

$$1{,}000 a_{\overline{30}|0.05}.$$

The annual payments form a perpetuity-immediate beginning at time 40. Its value one period before the first payment, at time 39, is

$$X a_{\overline{\infty}|0.05}.$$

Discounting this value back 40 years to time -1, we get

$$X a_{\overline{\infty}|0.05} v^{40}.$$

Thus,

$$1{,}000 a_{\overline{30}|0.05} = X a_{\overline{\infty}|0.05} v^{40}.$$

Using

$$a_{\overline{\infty}|0.05} = \frac{1}{0.05}$$

and

$$a_{\overline{30}|0.05} = \frac{1 - v^{30}}{0.05},$$

we solve for X.
Equivalently, using time 39, the equation is

$$1{,}000 s_{\overline{30}|0.05} (1.05)^{10} = X a_{\overline{\infty}|0.05}.$$

So

$$1{,}000 s_{\overline{30}|0.05} (1.05)^{10} = X \left(\frac{1}{0.05} \right).$$

Therefore,

$$X = 1{,}000 s_{\overline{30}|0.05} (1.05)^{10} (0.05).$$

Since

$$s_{\overline{30}|0.05} = \frac{1.05^{30} - 1}{0.05},$$

we get

$$X = 1{,}000 \left(\frac{1.05^{30} - 1}{0.05} \right) (1.05)^{10} (0.05).$$

Thus,

$$X = 5{,}411.09.$$

The closest choice is

$$\boxed{5{,}410}.$$

The correct choice is

$$\boxed{\text{D}}.$$

2.1.6 Annuities Payable m-thly

m-thly annuity

An m-thly annuity is an annuity with m payments per year.
For example:

$$m = 2 \quad \text{semi-annual payments,}$$

$$m = 4 \quad \text{quarterly payments,}$$

$$m = 12 \quad \text{monthly payments.}$$

Two valuation approaches

There are two common approaches for valuing an annuity payable m-thly.

1. Use the effective interest rate per payment period.
2. Use the annual effective interest rate and convert the m-thly payments into equivalent annual payments.

The first approach is usually the most direct. The second approach is often called the fusion method.

Number of payments

If an annuity makes m payments per year for n years, then the total number of payments is

$$mn.$$

Each payment period has length

$$\frac{1}{m}$$

years.

Effective Rate Per Payment Period Method

The first method for valuing an annuity payable m-thly is to use the effective interest rate per payment period.

If the annual effective interest rate is i, first convert it to the effective rate per payment period j (defined in the formula box below), then value the annuity as an ordinary annuity with mn payments.

Present value using the payment-period rate

For an n-year annuity-immediate with m payments per year, each payment being R, the present value is

$$PV = Ra_{\overline{mn}|j},$$

where

$$j = (1+i)^{1/m} - 1.$$

Equivalently,

$$PV = R\left(\frac{1 - v_j^{mn}}{j}\right),$$

where

$$v_j = \frac{1}{1+j}.$$

Monthly annuity-immediate

Given an annual effective interest rate

$$i = 0.061677812,$$

find the present value of a 5-year annuity-immediate with monthly payments of 100. Since payments are monthly, we have

$$m = 12.$$

Let j be the effective monthly interest rate. Then

$$j = (1+i)^{1/12} - 1.$$

Substituting the given value of i,

$$j = (1.061677812)^{1/12} - 1 = 0.005.$$

The number of payments is

$$5 \times 12 = 60.$$

Therefore,

$$PV = 100a_{\overline{60}|0.005}.$$

Using the annuity-immediate formula,

$$PV = 100\left(\frac{1 - v^{60}}{j}\right),$$

where

$$v = \frac{1}{1.005}.$$

Hence,

$$PV = 100\left(\frac{1 - 1.005^{-60}}{0.005}\right).$$

Therefore,

$$\boxed{PV = 5172.56}.$$

Common mistake

When the annual effective interest rate is given, do not divide it by 12 to obtain the monthly rate. The correct monthly effective rate is

$$j = (1+i)^{1/12} - 1.$$

Fusion Method

The fusion method consists of grouping the m-thly payments made during each year into one equivalent annual payment.

Instead of valuing each payment separately using the rate per payment period, we first accumulate all payments made during one year to the end of that year. Then we value the resulting annual annuity using the annual effective interest rate.

Fusion method

Suppose an annuity-immediate pays R at the end of each m-th of a year for n years. Let

$$j = (1+i)^{1/m} - 1$$

be the effective interest rate per payment period.
The accumulated value at the end of one year of the m payments is

$$Rs_{\overline{m}|j}.$$

Therefore, the present value is

$$PV = Rs_{\overline{m}|j}a_{\overline{n}|i}.$$

Since

$$s_{\overline{m}|j} = \frac{(1+j)^m - 1}{j} = \frac{i}{j},$$

we obtain

$$PV = R\left(\frac{i}{j}\right)a_{\overline{n}|i}.$$

If $i^{(m)}$ denotes the nominal interest rate convertible m-thly, then

$$j = \frac{i^{(m)}}{m},$$

and so

$$PV = Rm\left(\frac{i}{i^{(m)}}\right)a_{\overline{n}|i}.$$

Fusion method for monthly payments

Given an annual effective interest rate

$$i = 0.061677812,$$

find the present value of a 5-year annuity-immediate with monthly payments of 100.
Since payments are monthly, we have

$$m = 12.$$

Let j be the effective monthly interest rate. Then

$$j = (1+i)^{1/12} - 1.$$

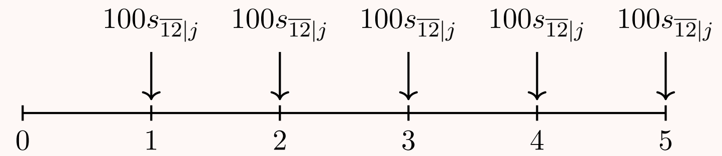
Thus,

$$j = (1.061677812)^{1/12} - 1 = 0.005.$$

Equivalently,

$$i^{(12)} = 12j = 12(0.005) = 0.06.$$

We now group the 12 monthly payments in each year into one equivalent payment at the end of the year.



The present value is therefore

$$PV = 100 s_{\overline{12}|j} a_{\overline{5}|i}.$$

Using the formula for an accumulated annuity-immediate,

$$s_{\overline{12}|j} = \frac{(1+j)^{12} - 1}{j}.$$

Hence,

$$PV = 100 \left(\frac{(1.005)^{12} - 1}{0.005} \right) a_{\overline{5}|0.061677812}.$$

Since

$$(1.005)^{12} - 1 = 0.061677812,$$

we get

$$PV = 100 \left(\frac{0.061677812}{0.005} \right) a_{\overline{5}|0.061677812}.$$

Equivalently,

$$PV = 1200 \left(\frac{i}{i^{(12)}} \right) a_{\overline{5}|i}.$$

Substituting the values,

$$PV = 1200 \left(\frac{0.061677812}{0.06} \right) a_{\overline{5}|0.061677812}.$$

Now,

$$a_{\overline{5}|0.061677812} = \frac{1 - (1.061677812)^{-5}}{0.061677812}.$$

Therefore,

$$PV = 1200 \left(\frac{0.061677812}{0.06} \right) \left(\frac{1 - (1.061677812)^{-5}}{0.061677812} \right).$$

Thus,

$$\boxed{PV = 5172.56}.$$

Key idea

In the fusion method, the coefficient of the annuity factor is the total annual payment. Here, the monthly payment is 100, so the total annual payment is

$$12(100) = 1200.$$

That is why the final expression is written as

$$PV = 1200a^{(12)}_{\overline{5}|i}.$$

Standard m-thly Annuity Factors

In actuarial notation, an m-thly annuity factor usually assumes that the total payment per year is 1.

Therefore, if payments are made m times per year, each payment is

$$\frac{1}{m}.$$

For example, a monthly annuity with total annual payment 1 has monthly payments of

$$\frac{1}{12}.$$

Annuity-immediate payable m-thly

The present value of an n-year annuity-immediate payable m-thly is denoted by

$$a^{(m)}_{\overline{n}|}.$$

It represents n years of payments of $\frac{1}{m}$, made at the end of each $\frac{1}{m}$-th of a year.

Let

$$j$$

be the effective interest rate per payment period. If $i^{(m)}$ is the nominal interest rate convertible m-thly, then

$$j = \frac{i^{(m)}}{m}.$$

During each year, the accumulated value at the end of the year of the m payments of $\frac{1}{m}$ is

$$\frac{1}{m}s_{\overline{m}|j}.$$

Therefore, by the fusion method,

$$a^{(m)}_{\overline{n}|} = \left(\frac{1}{m}s_{\overline{m}|j}\right)a_{\overline{n}|i}.$$

Since

$$s_{\overline{m}|j} = \frac{(1+j)^m - 1}{j},$$

we have

$$a_{\overline{n}|}^{(m)} = \left(\frac{1}{m} \cdot \frac{(1+j)^m - 1}{j}\right) a_{\overline{n}|i}.$$

But

$$(1+j)^m - 1 = i,$$

and

$$mj = i^{(m)}.$$

Thus,

$$a_{\overline{n}|}^{(m)} = \frac{i}{i^{(m)}} a_{\overline{n}|i}.$$

Since

$$a_{\overline{n}|i} = \frac{1 - v^n}{i},$$

we obtain the standard formula

$$\boxed{a_{\overline{n}|}^{(m)} = \frac{1 - v^n}{i^{(m)}}.}$$

Standard m-thly annuity factors

For an annual effective interest rate i, let

$$v = \frac{1}{1+i}.$$

The standard annuity-immediate factor payable m-thly is

$$a_{\overline{n}|}^{(m)} = \frac{i}{i^{(m)}} a_{\overline{n}|i} = \frac{1 - v^n}{i^{(m)}}.$$

The standard annuity-due factor payable m-thly is

$$\ddot{a}_{\overline{n}|}^{(m)} = \frac{d}{d^{(m)}} \ddot{a}_{\overline{n}|i} = \frac{1 - v^n}{d^{(m)}}.$$

The accumulated value of an annuity-immediate payable m-thly is

$$s_{\overline{n}|}^{(m)} = \frac{i}{i^{(m)}} s_{\overline{n}|i} = \frac{(1+i)^n - 1}{i^{(m)}}.$$

The accumulated value of an annuity-due payable m-thly is

$$\ddot{s}_{\overline{n}|}^{(m)} = \frac{d}{d^{(m)}} \ddot{s}_{\overline{n}|i} = \frac{(1+i)^n - 1}{d^{(m)}}.$$

When to use m-thly notation

Use the m-thly annuity notation when the problem gives m-thly annuity factors or when the answer choices are written using m-thly annuity factors.
Otherwise, it is often simpler to find the effective rate per payment period and value the annuity directly.

Exercise 2.1.41

Dave receives twenty payments of 500 every 6 months, with the first payment immediately. He immediately deposits each payment into an account earning an annual effective interest rate of i.
Which of the following expressions represents the value of the account at time 10?

A. $500\ddot{s}^{(2)}_{\overline{10}|i}$ B. $500\ddot{s}^{(2)}_{\overline{20}|i}$

C. $1000\ddot{s}^{(2)}_{\overline{10}|i}$ D. $1000\ddot{s}^{(2)}_{\overline{20}|i}$

Solution.
Payments are made every 6 months, so there are

$$m = 2$$

payments per year.
Since Dave receives twenty payments every 6 months, the payments cover

$$\frac{20}{2} = 10$$

years.
The first payment is made immediately, so this is an annuity-due payable semiannually. In m-thly annuity notation, the coefficient represents the total annual payment. Since each semiannual payment is 500, the total annual payment is

$$2(500) = 1000.$$

Therefore, the accumulated value at time 10 is

$$1000\ddot{s}^{(2)}_{\overline{10}|i}.$$

Thus, the correct answer is

$$\boxed{\text{C. } 1000\ddot{s}^{(2)}_{\overline{10}|i}}.$$

Equivalently, if j is the effective interest rate per 6 months, then

$$j = (1+i)^{1/2} - 1.$$

The account value can also be written as

$$500\ddot{s}_{\overline{20}|j}.$$

This is equivalent to

$$1000\ddot{s}^{(2)}_{\overline{10}|i}.$$

Exercise 2.1.42

Given

$$\ddot{a}^{(4)}_{\overline{3}|} = 2.7732$$

and

$$v^3 = 0.8396,$$

what is the present value of 12 quarterly payments of 100, with the first payment one quarter from now?

Solution.

The payments are quarterly, so

$$m = 4.$$

There are 12 quarterly payments, so the payment period covers

$$\frac{12}{4} = 3$$

years.

Since each quarterly payment is 100, the total annual payment is

$$4(100) = 400.$$

The payments occur at times

$$0.25,\ 0.50,\ 0.75,\ \ldots,\ 3.$$

Thus, this is an annuity-immediate payable quarterly, and the present value is

$$PV = 400a_{\overline{3}|}^{(4)}.$$

However, we are given the annuity-due factor

$$\ddot{a}_{\overline{3}|}^{(4)}.$$

The quarterly annuity-due has payments from time 0 to time 2.75, while the quarterly annuity-immediate has payments from time 0.25 to time 3.

Therefore,

$$a_{\overline{3}|}^{(4)} = \ddot{a}_{\overline{3}|}^{(4)} - \frac{1}{4} + \frac{1}{4}v^3.$$

Multiplying by the total annual payment 400, we get

$$PV = 400a_{\overline{3}|}^{(4)}.$$

So,

$$PV = 400\left(\ddot{a}_{\overline{3}|}^{(4)} - \frac{1}{4} + \frac{1}{4}v^3\right).$$

Equivalently,

$$PV = 400\ddot{a}_{\overline{3}|}^{(4)} - 100 + 100v^3.$$

Substituting the given values,

$$PV = 400(2.7732) - 100 + 100(0.8396).$$

Thus,

$$PV = 1109.28 - 100 + 83.96.$$

Therefore,

$$\boxed{PV = 1093.24}.$$

Alternatively, since the annuity-immediate is the annuity-due shifted one quarter later,

$$a_{\overline{3}|}^{(4)} = v^{1/4}\ddot{a}_{\overline{3}|}^{(4)}.$$

Since

$$v^{1/4} = (v^3)^{1/12},$$

we also have

$$PV = 400(2.7732)(0.8396)^{1/12} = 1093.24.$$

Exercise 2.1.43

John made a deposit of 1000 into a fund at the beginning of each year for 20 years.
At the end of 20 years, he began making semiannual withdrawals of 3000 at the beginning of each six months, with a smaller final withdrawal to exhaust the fund. The fund earned an annual effective interest rate of 8.16%.
Calculate the amount of the final withdrawal.

A. 561 B. 1226

C. 1430 D. 1488

E. 2240

Solution.
The annual effective interest rate is

$$i = 0.0816.$$

The effective 6-month interest rate is

$$j = (1.0816)^{1/2} - 1.$$

Since

$$1.0816 = (1.04)^2,$$

we get

$$j = 0.04.$$

John deposits 1000 at the beginning of each year for 20 years. Thus, the deposits are made at times

$$0, 1, 2, \ldots, 19.$$

At time 19.5, the accumulated value of the deposits is

$$1000 s_{\overline{20}|0.0816}(1.0816)^{1/2}.$$

Using

$$s_{\overline{20}|0.0816} = \frac{(1.0816)^{20} - 1}{0.0816},$$

we obtain

$$1000 s_{\overline{20}|0.0816}(1.0816)^{1/2} = 48444.38.$$

Now value the withdrawals at time 19.5. Since the first withdrawal of 3000 is made at time 20, it is one half-year after time 19.5. Therefore, relative to time 19.5, the withdrawals form an annuity-immediate with interest rate

$$j = 0.04.$$

Let n be the number of full semiannual withdrawals of 3000 that would exhaust the fund. Then

$$48444.38 = 3000 a_{\overline{n}|0.04}.$$

Solving for n,

$$a_{\overline{n}|0.04} = \frac{48444.38}{3000} = 16.14813.$$

Since

$$a_{\overline{n}|0.04} = \frac{1 - (1.04)^{-n}}{0.04},$$

we have

$$16.14813 = \frac{1-(1.04)^{-n}}{0.04}.$$

Thus,

$$n \approx 26.47.$$

This means John can make 26 full withdrawals of 3000, followed by one smaller final withdrawal X six months later.
Therefore, at time 19.5,

$$48444.38 = 3000a_{\overline{26}|0.04} + X(1.04)^{-27}.$$

Now,

$$3000a_{\overline{26}|0.04} = 3000\left(\frac{1-(1.04)^{-26}}{0.04}\right) = 47948.31.$$

Hence,

$$48444.38 = 47948.31 + X(1.04)^{-27}.$$

So,

$$X(1.04)^{-27} = 496.07.$$

Therefore,

$$X = 496.07(1.04)^{27}.$$

Thus,

$$X = 1430.35.$$

The closest answer is

$$\boxed{\text{C. } 1430}.$$

2.1.7 Increasing Annuities: Arithmetic

Increasing Annuity-Immediate Present Value

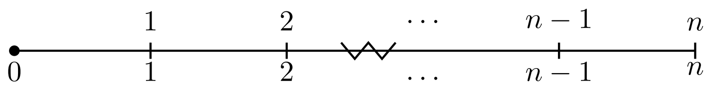
An increasing annuity-immediate is an annuity whose payments increase by a fixed amount each period, with payments made at the end of each period.

We first consider the standard increasing annuity-immediate with payments

$$1, 2, 3, \ldots, n$$

at times

$$1, 2, 3, \ldots, n.$$



Increasing annuity-immediate

The present value of an increasing annuity-immediate with payments

$$1, 2, 3, \ldots, n$$

at the end of each period is denoted by

$$(Ia)_{\overline{n}|i}.$$

It is defined by

$$(Ia)_{\overline{n}|i} = v + 2v^2 + 3v^3 + \cdots + nv^n.$$

Present value of an increasing annuity-immediate

The present value of the increasing annuity-immediate is

$$\boxed{(Ia)_{\overline{n}|i} = \frac{\ddot{a}_{\overline{n}|i} - nv^n}{i}}$$

where

$$v = \frac{1}{1+i}.$$

Derivation (sketch).

Starting from the definition of

$$(Ia)_{\overline{n}|i}$$

and using the standard "multiply by

$$1+i$$

and subtract" trick, we obtain

$$(Ia)_{\overline{n}|i} = \frac{\ddot{a}_{\overline{n}|i} - nv^n}{i},$$

which matches the formula box above. Dividing by i, we obtain

$$\boxed{(Ia)_{\overline{n}|i} = \frac{\ddot{a}_{\overline{n}|i} - nv^n}{i}}.$$

Formula to memorize

The formula

$$(Ia)_{\overline{n}|i} = \frac{\ddot{a}_{\overline{n}|i} - nv^n}{i}$$

is one of the main formulas for arithmetic increasing annuities.
It applies when the payments are

$$1, 2, 3, \ldots, n$$

and the first payment occurs one period from now.

Using the TVM Function

The formula for the present value of an increasing annuity-immediate is

$$(Ia)_{\overline{n}|i} = \frac{\ddot{a}_{\overline{n}|i} - nv^n}{i}.$$

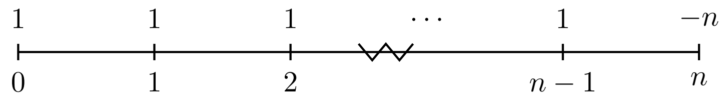
The numerator

$$\ddot{a}_{\overline{n}|i} - nv^n$$

can be interpreted as the present value of the following cash flows:

$$1, 1, 1, \ldots, 1, -n$$

where the payments of 1 occur from time 0 to time $n-1$, and the payment $-n$ occurs at time n.



Therefore, the numerator can be calculated in one step using the TVM function.

BA II Plus setup

Use **BGN mode**, because the payments of 1 start immediately at time 0.

$$\begin{aligned} N &= n, \\ I/Y &= 100i, \\ PMT &= 1, \\ FV &= -n, \\ CPT\ PV. & \end{aligned}$$

The calculator gives the present value of

$$\ddot{a}_{\overline{n}|i} - nv^n.$$

Then change the sign if necessary and divide by i:

$$(Ia)_{\overline{n}|i} = \frac{\ddot{a}_{\overline{n}|i} - nv^n}{i}.$$

Important calculator convention

On the BA II Plus, I/Y is entered as a percentage. Therefore, if the effective interest rate is $i = 0.06$, enter

$$I/Y = 6,$$

not 0.06.

Increasing annuity-immediate

Given

$$i = 0.05,$$

find the present value of payments

$$10, 20, \ldots, 90$$

at times

$$1, 2, \ldots, 9$$

respectively.

Solution.

The payments increase arithmetically:

$$10, 20, \ldots, 90.$$

We can factor out 10:

$$10, 20, \ldots, 90 = 10(1, 2, \ldots, 9).$$

Therefore, the present value is

$$PV = 10(Ia)_{\overline{9}|0.05}.$$

Using the formula for an increasing annuity-immediate,

$$(Ia)_{\overline{n}|i} = \frac{\ddot{a}_{\overline{n}|i} - nv^n}{i},$$

we get

$$(Ia)_{\overline{9}|0.05} = \frac{\ddot{a}_{\overline{9}|0.05} - 9v^9}{0.05}.$$

Thus,

$$PV = 10\left(\frac{\ddot{a}_{\overline{9}|0.05} - 9v^9}{0.05}\right).$$

With

$$v = \frac{1}{1.05},$$

this gives

$$PV = 332.35.$$

Therefore,

$$\boxed{PV = 332.35}.$$

Using the BA II Plus.
To compute the numerator

$$\ddot{a}_{\overline{9}|0.05} - 9v^9,$$

use **BGN mode**:

$$\begin{aligned} N &= 9, \\ I/Y &= 5, \\ PMT &= 1, \\ FV &= -9, \\ CPT\ PV &= -1.661732513. \end{aligned}$$

Change the sign:

$$1.661732513.$$

Then divide by $i = 0.05$:

$$\frac{1.661732513}{0.05} = 33.23465027.$$

Finally, multiply by 10:

$$10(33.23465027) = 332.3465027.$$

Therefore,

$$\boxed{PV = 332.35}.$$

Calculator reminder

For $(Ia)_{\overline{n}|i}$, do not forget to use **BGN mode** when computing the numerator

$$\ddot{a}_{\overline{n}|i} - nv^n.$$

The calculator gives the numerator only. You must still divide by i, then multiply by the payment scale if the payments are not $1, 2, \ldots, n$.

Increasing Annuity-Due Present Value

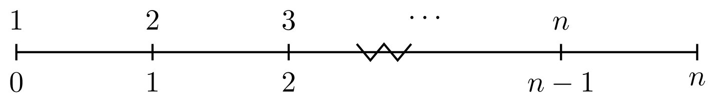
An increasing annuity-due is an annuity whose payments increase by a fixed amount each period, with payments made at the beginning of each period.

We first consider the standard increasing annuity-due with payments

$$1, 2, 3, \ldots, n$$

at times

$$0, 1, 2, \ldots, n-1.$$



Increasing annuity-due

The present value of an increasing annuity-due with payments

$$1, 2, 3, \ldots, n$$

at the beginning of each period is denoted by

$$(I\ddot{a})_{\overline{n}|i}.$$

It is defined by

$$(I\ddot{a})_{\overline{n}|i} = 1 + 2v + 3v^2 + \cdots + nv^{n-1}.$$

The payments are the same as those of the increasing annuity-immediate, but each payment is made one period earlier.

Therefore,

$$(I\ddot{a})_{\overline{n}|i} = (1+i)(Ia)_{\overline{n}|i}.$$

Using the formula

$$(Ia)_{\overline{n}|i} = \frac{\ddot{a}_{\overline{n}|i} - nv^n}{i},$$

we get

$$(I\ddot{a})_{\overline{n}|i} = (1+i)\left(\frac{\ddot{a}_{\overline{n}|i} - nv^n}{i}\right).$$

Since

$$d = \frac{i}{1+i},$$

we have

$$\frac{i}{1+i} = d.$$

Thus,

$$(I\ddot{a})_{\overline{n}|i} = \frac{\ddot{a}_{\overline{n}|i} - nv^n}{d}.$$

Present value of an increasing annuity-due

The present value of the increasing annuity-due is

$$\boxed{(I\ddot{a})_{\overline{n}|i} = \frac{\ddot{a}_{\overline{n}|i} - nv^n}{d}}$$

where

$$v = \frac{1}{1+i} \qquad \text{and} \qquad d = \frac{i}{1+i}.$$

Relation with the increasing annuity-immediate

The increasing annuity-due payments occur one period earlier than the increasing annuity-immediate payments.
Therefore,

$$(I\ddot{a})_{\overline{n}|i} = (1+i)(Ia)_{\overline{n}|i}.$$

This is why the formula divides by d instead of i:

$$(Ia)_{\overline{n}|i} = \frac{\ddot{a}_{\overline{n}|i} - nv^n}{i}, \qquad (I\ddot{a})_{\overline{n}|i} = \frac{\ddot{a}_{\overline{n}|i} - nv^n}{d}.$$

Increasing Annuity-Immediate Accumulated Value

The accumulated value of an n-year increasing annuity-immediate is denoted by

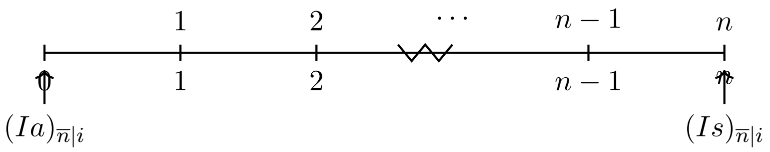
$$(Is)_{\overline{n}|i}.$$

It represents the accumulated value at time n of payments

$$1, 2, 3, \ldots, n$$

made at times

$$1, 2, 3, \ldots, n.$$



Increasing annuity-immediate accumulated value

The accumulated value of the increasing annuity-immediate is

$$(Is)_{\overline{n}|i} = 1(1+i)^{n-1} + 2(1+i)^{n-2} + \cdots + (n-1)(1+i) + n.$$

Since accumulated value at time n is obtained by multiplying the present value by $(1+i)^n$, we have

$$(Is)_{\overline{n}|i} = (Ia)_{\overline{n}|i}(1+i)^n.$$

Using

$$(Ia)_{\overline{n}|i} = \frac{\ddot{a}_{\overline{n}|i} - nv^n}{i},$$

we get

$$(Is)_{\overline{n}|i} = \left(\frac{\ddot{a}_{\overline{n}|i} - nv^n}{i}\right)(1+i)^n.$$

Now,

$$\ddot{a}_{\overline{n}|i}(1+i)^n = \ddot{s}_{\overline{n}|i},$$

and

$$nv^n(1+i)^n = n.$$

Therefore,

$$(Is)_{\overline{n}|i} = \frac{\ddot{s}_{\overline{n}|i} - n}{i}.$$

Accumulated value of an increasing annuity-immediate

The accumulated value at time n is

$$\boxed{(Is)_{\overline{n}|i} = \frac{\ddot{s}_{\overline{n}|i} - n}{i}}$$

where

$$\ddot{s}_{\overline{n}|i} = (1+i)s_{\overline{n}|i}.$$

Avoid this calculator trap

Do not try to use the previous TVM setup for $(Ia)_{\overline{n}|i}$ and then press $CPT\ FV$.
In that setup, the FV register is already used for the value $-n$. If you press $CPT\ FV$, the calculator simply replaces the FV register instead of giving the accumulated value of the increasing annuity.
For accumulated value, use the formula

$$(Is)_{\overline{n}|i} = \frac{\ddot{s}_{\overline{n}|i} - n}{i}.$$

Increasing Annuity-Due Accumulated Value

The accumulated value of an n-year increasing annuity-due is denoted by

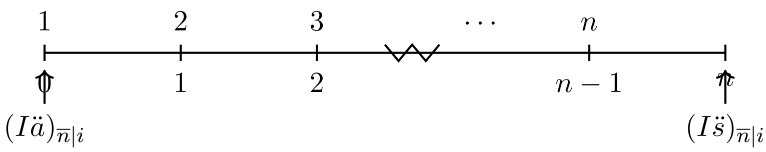
$$(I\ddot{s})_{\overline{n}|i}.$$

It represents the accumulated value at time n of payments

$$1, 2, 3, \ldots, n$$

made at times

$$0, 1, 2, \ldots, n-1.$$



Increasing annuity-due accumulated value

The accumulated value of the increasing annuity-due is

$$(I\ddot{s})_{\overline{n}|i} = 1(1+i)^n + 2(1+i)^{n-1} + 3(1+i)^{n-2} + \cdots + n(1+i).$$

Since accumulated value at time n is obtained by multiplying the present value by $(1+i)^n$, we have

$$(I\ddot{s})_{\overline{n}|i} = (I\ddot{a})_{\overline{n}|i}(1+i)^n.$$

Using

$$(I\ddot{a})_{\overline{n}|i} = \frac{\ddot{a}_{\overline{n}|i} - nv^n}{d},$$

we get

$$(I\ddot{s})_{\overline{n}|i} = \left(\frac{\ddot{a}_{\overline{n}|i} - nv^n}{d}\right)(1+i)^n.$$

Now,

$$\ddot{a}_{\overline{n}|i}(1+i)^n = \ddot{s}_{\overline{n}|i},$$

and

$$nv^n(1+i)^n = n.$$

Therefore,

$$(I\ddot{s})_{\overline{n}|i} = \frac{\ddot{s}_{\overline{n}|i} - n}{d}.$$

Accumulated value of an increasing annuity-due

The accumulated value at time n is

$$\boxed{(I\ddot{s})_{\overline{n}|i} = \frac{\ddot{s}_{\overline{n}|i} - n}{d}}$$

where

$$d = \frac{i}{1+i}.$$

Comparison with the immediate case

For the increasing annuity-immediate, we have

$$(Is)_{\overline{n}|i} = \frac{\ddot{s}_{\overline{n}|i} - n}{i}.$$

For the increasing annuity-due, the payments occur one period earlier, so the accumulated value is larger:

$$(I\ddot{s})_{\overline{n}|i} = \frac{\ddot{s}_{\overline{n}|i} - n}{d}.$$

The denominator changes from i to d, just as it does when comparing level annuity-immediate and level annuity-due formulas.

Increasing Perpetuity

An increasing perpetuity can be viewed as an increasing annuity-immediate where

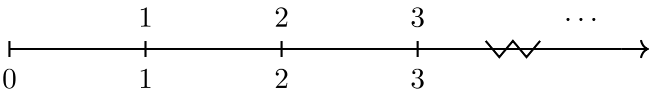
$$n \to \infty.$$

We first consider payments

$$1, 2, 3, \ldots$$

made at times

$$1, 2, 3, \ldots$$



Increasing perpetuity-immediate

The present value of an increasing perpetuity-immediate is denoted by

$$(Ia)_{\overline{\infty}|i}.$$

It is defined by

$$(Ia)_{\overline{\infty}|i} = v + 2v^2 + 3v^3 + 4v^4 + \cdots.$$

To derive the formula, start with

$$(Ia)_{\overline{\infty}|i} = v + 2v^2 + 3v^3 + 4v^4 + \cdots.$$

Multiply both sides by v:

$$v(Ia)_{\overline{\infty}|i} = v^2 + 2v^3 + 3v^4 + 4v^5 + \cdots.$$

Subtract:

$$(1-v)(Ia)_{\overline{\infty}|i} = v + v^2 + v^3 + v^4 + \cdots.$$

The right-hand side is the present value of a level perpetuity-immediate of 1, so

$$v + v^2 + v^3 + \cdots = a_{\overline{\infty}|i} = \frac{1}{i}.$$

Therefore,

$$(Ia)_{\overline{\infty}|i} = \frac{a_{\overline{\infty}|i}}{1-v}.$$

Since

$$a_{\overline{\infty}|i} = \frac{1}{i} \qquad \text{and} \qquad 1 - v = d,$$

we get

$$(Ia)_{\overline{\infty}|i} = \frac{1}{id}.$$

Equivalently, since

$$d = \frac{i}{1+i},$$

we also have

$$(Ia)_{\overline{\infty}|i} = \frac{1+i}{i^2}.$$

Increasing perpetuity-immediate

The present value of an increasing perpetuity-immediate is

$$\boxed{(Ia)_{\overline{\infty}|i} = \frac{1}{id}}$$

or equivalently,

$$\boxed{(Ia)_{\overline{\infty}|i} = \frac{1+i}{i^2}}.$$

Increasing perpetuity-due

The present value of an increasing perpetuity-due is denoted by

$$(I\ddot{a})_{\overline{\infty}|i}.$$

It represents payments

$$1, 2, 3, \ldots$$

made at times

$$0, 1, 2, \ldots$$

respectively.

The increasing perpetuity-due payments occur one period earlier than the increasing perpetuity-immediate payments. Therefore,

$$(I\ddot{a})_{\overline{\infty}|i} = (1+i)(Ia)_{\overline{\infty}|i}.$$

Using

$$(Ia)_{\overline{\infty}|i} = \frac{1+i}{i^2},$$

we get

$$(I\ddot{a})_{\overline{\infty}|i} = \frac{(1+i)^2}{i^2}.$$

Since

$$d = \frac{i}{1+i},$$

this becomes

$$(I\ddot{a})_{\overline{\infty}|i} = \frac{1}{d^2}.$$

Increasing perpetuity-due

The present value of an increasing perpetuity-due is

$$\boxed{(I\ddot{a})_{\overline{\infty}|i} = \frac{1}{d^2}}.$$

Payment scale

If the payments are

$$R, 2R, 3R, \ldots,$$

then the present value is R times the standard factor.
Thus,

$$PV = R(Ia)_{\overline{\infty}|i} = \frac{R}{id}$$

for an increasing perpetuity-immediate, and

$$PV = R(I\ddot{a})_{\overline{\infty}|i} = \frac{R}{d^2}$$

for an increasing perpetuity-due.

Exercise 2.1.44

Olga buys a five-year increasing annuity for X. Olga will receive 2 at the end of the first month, 4 at the end of the second month, and for each month thereafter the payment increases by 2.
The nominal interest rate is 9% convertible quarterly.
Calculate X.
Solution.
The payments are monthly for 5 years, so the total number of payments is

$$5 \times 12 = 60.$$

The payments are

$$2, 4, 6, \ldots, 120.$$

We can factor out 2:

$$2, 4, 6, \ldots, 120 = 2(1, 2, 3, \ldots, 60).$$

Therefore,

$$X = 2(Ia)_{\overline{60}|j},$$

where j is the effective monthly interest rate.
The nominal interest rate is 9% convertible quarterly, so the effective quarterly rate is

$$\frac{0.09}{4} = 0.0225.$$

Since one quarter contains 3 months, the effective monthly rate j satisfies

$$(1 + j)^3 = 1.0225.$$

Thus,

$$j = (1.0225)^{1/3} - 1.$$

Hence,

$$j = 0.0074444275.$$

Using the increasing annuity-immediate formula,

$$(Ia)_{\overline{n}|j} = \frac{\ddot{a}_{\overline{n}|j} - nv_j^n}{j},$$

we get

$$(Ia)_{\overline{60}|j} = \frac{\ddot{a}_{\overline{60}|j} - 60v_j^{60}}{j}.$$

Therefore,

$$X = 2\left(\frac{\ddot{a}_{\overline{60}|j} - 60v_j^{60}}{j}\right).$$

Using the BA II Plus to compute the numerator

$$\ddot{a}_{\overline{60}|j} - 60v_j^{60},$$

use **BGN mode**:

$$\begin{aligned} N &= 60, \\ I/Y &= 100j = 0.74444275, \\ PMT &= 1, \\ FV &= -60, \\ CPT\ PV &= -10.15873987. \end{aligned}$$

Change the sign:

$$10.15873987.$$

Then divide by j:

$$\frac{10.15873987}{0.0074444275} = 1364.607296.$$

Finally, multiply by 2:

$$X = 2(1364.607296) = 2729.214593.$$

Therefore,

$$\boxed{X = 2729.21}.$$

Exercise 2.1.45

Smith makes deposits to a savings account on July 1 of each year. The first deposit was made on July 1, 1972, and the last deposit will be made on July 1, 1982.
The first deposit was 1000, and each subsequent deposit increases by 100. The account is credited with an effective annual interest rate of 4%.
Determine the value of Smith's account as of July 2, 1982.
Solution.
Let time 0 be July 1, 1971. Then the deposits occur at times

$$1, 2, \ldots, 11.$$

The deposits are

$$1000, 1100, 1200, \ldots, 2000.$$

We decompose the payment stream as

$$1000, 1100, 1200, \ldots, 2000 = 900(1, 1, 1, \ldots, 1) + 100(1, 2, 3, \ldots, 11).$$

The level part contributes

$$900s_{\overline{11}|0.04}$$

at time 11.

The increasing part has payments

$$100, 200, 300, \ldots, 1100$$

at times $1, 2, \ldots, 11$. Its present value at time 0 is

$$100(Ia)_{\overline{11}|0.04}.$$

Therefore, its accumulated value at time 11 is

$$100(Ia)_{\overline{11}|0.04}(1.04)^{11}.$$

Thus, the value of Smith's account is

$$AV = 900s_{\overline{11}|0.04} + 100(Ia)_{\overline{11}|0.04}(1.04)^{11}.$$

Using

$$s_{\overline{11}|0.04} = \frac{1.04^{11} - 1}{0.04},$$

and

$$(Ia)_{\overline{11}|0.04} = \frac{\ddot{a}_{\overline{11}|0.04} - 11v^{11}}{0.04},$$

where

$$v = \frac{1}{1.04},$$

we obtain

$$AV = 900s_{\overline{11}|0.04} + 100(Ia)_{\overline{11}|0.04}(1.04)^{11}.$$

Therefore,

$$AV = 19702.23.$$

Hence, the value of Smith's account as of July 2, 1982 is

$$\boxed{19702.23}.$$

Exercise 2.1.46

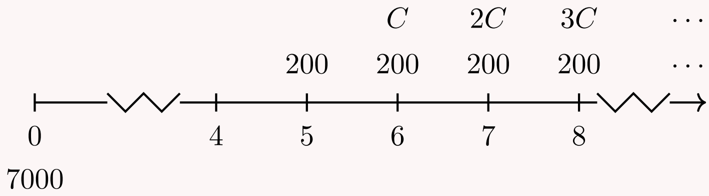
Jones purchased a perpetuity today for 7000. He will receive the first annual payment of 200 five years from now. The second annual payment will be 200 plus an amount C. Each subsequent payment will be the prior amount plus an additional constant amount C.
If the effective interest rate is 4%, find C.
Solution.
The first payment is 200 at time 5. The second payment is $200 + C$ at time 6, the third payment is $200 + 2C$ at time 7, and so on.
We decompose the payment stream into two parts:

$$200, 200, 200, \ldots$$

starting at time 5, and

$$C, 2C, 3C, \ldots$$

starting at time 6.



Let

$$i = 0.04.$$

The level perpetuity of 200 begins at time 5. Its value one period before the first payment, at time 4, is

$$200a_{\overline{\infty}|0.04}.$$

Therefore, its present value at time 0 is

$$200a_{\overline{\infty}|0.04}v^4.$$

The increasing part has payments

$$C, 2C, 3C, \ldots$$

starting at time 6. This is an increasing perpetuity-immediate valued at time 5. Therefore, its value at time 5 is

$$C(Ia)_{\overline{\infty}|0.04}.$$

Its present value at time 0 is

$$C(Ia)_{\overline{\infty}|0.04}v^5.$$

Thus, the equation of value at time 0 is

$$7000 = 200a_{\overline{\infty}|0.04}v^4 + C(Ia)_{\overline{\infty}|0.04}v^5.$$

Now,

$$a_{\overline{\infty}|0.04} = \frac{1}{0.04} = 25.$$

Also,

$$d = \frac{i}{1+i} = \frac{0.04}{1.04},$$

and

$$(Ia)_{\overline{\infty}|0.04} = \frac{1}{id} = \frac{1}{0.04\left(\frac{0.04}{1.04}\right)}.$$

Hence,

$$7000 = 200a_{\overline{\infty}|0.04}v^4 + C(Ia)_{\overline{\infty}|0.04}v^5.$$

$$7000 = 200\left(\frac{1}{0.04}\right)(1.04)^{-4} + C\left(\frac{1}{0.04\left(\frac{0.04}{1.04}\right)}\right)(1.04)^{-5}.$$

Solving for C,

$$C = 5.10.$$

Therefore,

$$\boxed{C = 5.10}.$$

Exercise 2.1.47

Jones purchased a perpetuity today for 7000. He will receive the first annual payment of 200 five years from now. The second annual payment will be 200 plus an amount C. Each subsequent payment will be the prior amount plus an additional constant amount C.
If the effective annual interest rate is 4%, find C.

A. Less than 6
B. At least 6, but less than 7
C. At least 7, but less than 8
D. At least 8, but less than 9
E. At least 9

Solution.
The payments are

$$200, \quad 200 + C, \quad 200 + 2C, \quad 200 + 3C, \ldots$$

at times

$$5, 6, 7, 8, \ldots$$

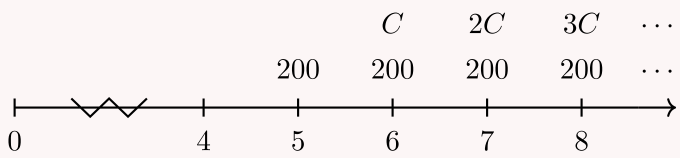
We decompose the cash flow into two parts:

$$200, 200, 200, 200, \ldots$$

starting at time 5, and

$$C, 2C, 3C, \ldots$$

starting at time 6.



We write an equation of value at time 4.
The purchase price accumulated to time 4 is

$$7000(1.04)^4.$$

At time 4, the level perpetuity of 200 has value

$$200a_{\overline{\infty}|0.04}.$$

The increasing part starts with C at time 6. Therefore, at time 4, it has value

$$C(Ia)_{\overline{\infty}|0.04}v.$$

Thus,

$$7000(1.04)^4 = 200a_{\overline{\infty}|0.04} + C(Ia)_{\overline{\infty}|0.04}v.$$

Now,

$$a_{\overline{\infty}|0.04} = \frac{1}{0.04} = 25.$$

Also,

$$d = \frac{0.04}{1.04},$$

so

$$(Ia)_{\overline{\infty}|0.04} = \frac{1}{id} = \frac{1}{0.04\left(\frac{0.04}{1.04}\right)} = 650.$$

Since

$$v = \frac{1}{1.04},$$

we get

$$(Ia)_{\overline{\infty}|0.04} v = 650\left(\frac{1}{1.04}\right) = 625.$$

Therefore,

$$7000(1.04)^4 = 200(25) + 625C.$$

So,

$$8189.01 = 5000 + 625C.$$

Hence,

$$625C = 3189.01,$$

and

$$C = 5.10.$$

Therefore,

$$C < 6.$$

The correct answer is

$$\boxed{\text{A. Less than 6}}.$$

Exercise 2.1.48

Sally buys a house. She has limited initial funds, so she agrees to make 360 monthly payments as follows:

1. The first payment is to be 500, with each subsequent payment increasing by 10.
2. The first payment is due one month after the date of the loan.
3. The nominal annual interest rate is 6% compounded monthly.

Determine how much Sally borrowed.

A. Less than 200,000

B. At least 200,000, but less than 230,000

C. At least 230,000, but less than 260,000

D. At least 260,000, but less than 290,000

E. At least 290,000

Solution.
The nominal annual interest rate is 6% compounded monthly, so the effective monthly interest rate is

$$j = \frac{0.06}{12} = 0.005.$$

The payments are

$$500, 510, 520, \ldots$$

for a total of 360 monthly payments.

We decompose the payment stream as

$$500, 510, 520, \ldots = 490(1, 1, 1, \ldots, 1) + 10(1, 2, 3, \ldots, 360).$$

Therefore, the amount borrowed is the present value

$$PV = 490a_{\overline{360}|0.005} + 10(Ia)_{\overline{360}|0.005}.$$

Now,

$$a_{\overline{360}|0.005} = \frac{1 - v^{360}}{0.005},$$

where

$$v = \frac{1}{1.005}.$$

Also,

$$(Ia)_{\overline{360}|0.005} = \frac{\ddot{a}_{\overline{360}|0.005} - 360v^{360}}{0.005}.$$

Using these formulas,

$$490a_{\overline{360}|0.005} = 81727.89,$$

and

$$10(Ia)_{\overline{360}|0.005} = 215700.96.$$

Thus,

$$PV = 81727.89 + 215700.96.$$

Therefore,

$$PV = 297428.85.$$

So Sally borrowed approximately

$$\boxed{297429}.$$

The correct answer is

$$\boxed{\text{E. At least 290,000}}.$$

Alternative expression.
Another equivalent way to write the present value is

$$PV = 500a_{\overline{360}|0.005} + v \cdot 10(Ia)_{\overline{359}|0.005}.$$

This works because the first 500 is treated as part of a level annuity, and the increasing part

$$10, 20, 30, \ldots$$

starts one month later.

Exercise 2.1.49

Sally takes out a mortgage with the following repayment schedule:

- 100 at the beginning of each month in year 1,
- 200 at the beginning of each month in year 2,
- $\cdots$
- $100n$ at the beginning of each month in year n,

- $\cdots$
- 1500 at the beginning of each month in year 15.

Sally will pay interest at an effective annual rate of 7%.
Determine the present value of Sally's loan payments at the beginning of year 1, just before she makes her first 100 payment.

A. Less than 75,500
B. At least 75,500, but less than 76,500
C. At least 76,500, but less than 77,500
D. At least 77,500, but less than 78,500
E. At least 78,500

Solution.
This problem becomes easier if we first roll up the monthly payments in each year into one accumulated value at the end of that year.
Let j be the effective monthly interest rate. Since the annual effective interest rate is 7%,

$$j = (1.07)^{1/12} - 1.$$

During year 1, Sally pays 100 at the beginning of each month. The accumulated value of these payments at the end of year 1 is

$$100\ddot{s}_{\overline{12}|j}.$$

During year 2, Sally pays 200 at the beginning of each month. The accumulated value of these payments at the end of year 2 is

$$200\ddot{s}_{\overline{12}|j}.$$

Continuing in the same way, during year 15, the accumulated value at the end of year 15 is

$$1500\ddot{s}_{\overline{12}|j}.$$

Therefore, after rolling up each year's monthly payments, the annual cash flows are

$$100\ddot{s}_{\overline{12}|j}, \quad 200\ddot{s}_{\overline{12}|j}, \quad \ldots, \quad 1500\ddot{s}_{\overline{12}|j}$$

at times

$$1, 2, \ldots, 15.$$

This is an increasing annuity-immediate with annual effective interest rate 7%. Hence,

$$PV = 100\ddot{s}_{\overline{12}|j}(Ia)_{\overline{15}|0.07}.$$

Now,

$$j = (1.07)^{1/12} - 1.$$

Using this monthly effective rate,

$$\ddot{s}_{\overline{12}|j} = 12.45029715.$$

Also,

$$(Ia)_{\overline{15}|0.07} = 61.55396701.$$

Therefore,

$$PV = 100(12.45029715)(61.55396701).$$

Thus,

$$PV = 76636.52.$$

So the present value is approximately

$$\boxed{76637}.$$

The correct answer is

$$\boxed{\text{C. At least 76,500, but less than 77,500}}.$$

Exercise 2.1.50

A scholarship fund is to be established by making annual deposits as follows:

Time	Deposit
1	1000
2	$1000 + 500$
3	$1000 + 500(2)$
$\vdots$	$\vdots$
t	$1000 + 500(t-1)$
$\vdots$	$\vdots$
10	$1000 + 500(9)$

Beginning at time $t = 11$, the fund will pay one scholarship of amount X annually into perpetuity. Assuming $i = 0.12$, what is the largest value of X that the fund can provide?

A. Less than 5,500

B. At least 5,500, but less than 6,000

C. At least 6,000, but less than 6,500

D. At least 6,500, but less than 7,000

E. At least 7,000

Solution.
We write an equation of value at time 10.
The deposits are

$$1000,\ 1500,\ 2000, \ldots, 5500$$

at times

$$1, 2, 3, \ldots, 10.$$

We decompose the deposits as

$$1000, 1500, 2000, \ldots, 5500 = 1000(1, 1, \ldots, 1) + 500(0, 1, 2, \ldots, 9).$$

The level part accumulates to time 10 as

$$1000 s_{\overline{10}|0.12}.$$

The increasing part starts after time 1, with payments

$$500, 1000, 1500, \ldots, 4500$$

at times $2, 3, \ldots, 10$. Its accumulated value at time 10 is

$$500(Is)_{\overline{9}|0.12}.$$

Thus, the accumulated value of all deposits at time 10 is

$$1000s_{\overline{10}|0.12} + 500(Is)_{\overline{9}|0.12}.$$

Beginning at time 11, the fund pays X annually forever. At time 10, this perpetuity has present value

$$Xa_{\overline{\infty}|0.12}.$$

Therefore,

$$1000s_{\overline{10}|0.12} + 500(Is)_{\overline{9}|0.12} = Xa_{\overline{\infty}|0.12}.$$

Now,

$$s_{\overline{10}|0.12} = \frac{1.12^{10} - 1}{0.12} = 17.548735.$$

Also,

$$(Is)_{\overline{9}|0.12} = \frac{\ddot{s}_{\overline{9}|0.12} - 9}{0.12} = 62.906126.$$

Hence,

$$1000(17.548735) + 500(62.906126) = Xa_{\overline{\infty}|0.12}.$$

So,

$$49001.80 = X\left(\frac{1}{0.12}\right).$$

Therefore,

$$X = 49001.80(0.12).$$

Thus,

$$X = 5880.22.$$

The largest scholarship payment is approximately

$$\boxed{5880}.$$

The correct answer is

$$\boxed{\text{B. At least 5,500, but less than 6,000}}.$$

2.1.8 Decreasing Annuities: Arithmetic

Decreasing Annuity-Immediate Present Value

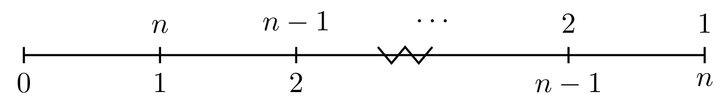
A decreasing annuity-immediate is an annuity whose payments decrease by a fixed amount each period, with payments made at the end of each period.

We first consider the standard decreasing annuity-immediate with payments

$$n, n-1, n-2, \ldots, 2, 1$$

at times

$$1, 2, 3, \ldots, n-1, n.$$



Decreasing annuity-immediate

The present value of a decreasing annuity-immediate with payments

$$n, n-1, n-2, \ldots, 2, 1$$

at the end of each period is denoted by

$$(Da)_{\overline{n}|i}.$$

It is defined by

$$(Da)_{\overline{n}|i} = nv + (n-1)v^2 + (n-2)v^3 + \cdots + 2v^{n-1} + v^n.$$

Derivation.

Start with

$$(Da)_{\overline{n}|i} = nv + (n-1)v^2 + (n-2)v^3 + \cdots + 2v^{n-1} + v^n.$$

Multiply both sides by $1+i$. Since

$$(1+i)v = 1,$$

we get

$$(1+i)(Da)_{\overline{n}|i} = n + (n-1)v + (n-2)v^2 + \cdots + 2v^{n-2} + v^{n-1}.$$

Now subtract the first equation from the second:

$$(1+i)(Da)_{\overline{n}|i} - (Da)_{\overline{n}|i} = n - v - v^2 - \cdots - v^{n-1} - v^n.$$

Therefore, rewriting the geometric sums in terms of

$$a_{\overline{n}|i}$$

gives the closed form for

$$(Da)_{\overline{n}|i}$$

summarized in the formula box below.

Present value of a decreasing annuity-immediate

The present value of the decreasing annuity-immediate is

$$\boxed{(Da)_{\overline{n}|i} = \frac{n - a_{\overline{n}|i}}{i}}$$

where

$$a_{\overline{n}|i} = \frac{1-v^n}{i} \qquad \text{and} \qquad v = \frac{1}{1+i}.$$

Formula to memorize

The formula

$$(Da)_{\overline{n}|i} = \frac{n - a_{\overline{n}|i}}{i}$$

applies when the payments are

$$n, n-1, n-2, \ldots, 1$$

and the first payment occurs one period from now.
This is the decreasing annuity-immediate counterpart of

$$(Ia)_{\overline{n}|i} = \frac{\ddot{a}_{\overline{n}|i} - nv^n}{i}.$$

Interpreting the coefficient

For the standard decreasing annuity-immediate,

$$(Da)_{\overline{n}|i} = \frac{n - a_{\overline{n}|i}}{i},$$

the payments are

$$n, n-1, n-2, \ldots, 2, 1.$$

If the factor is multiplied by a coefficient R, then

$$R(Da)_{\overline{n}|i}$$

represents the present value of payments

$$Rn,\ R(n-1),\ R(n-2), \ldots, 2R,\ R.$$

Thus, the coefficient R is the amount by which each payment decreases. The first payment is nR, and the last payment is R.

Interpreting $30(Da)_{\overline{5}|i}$

The expression

$$30(Da)_{\overline{5}|i}$$

represents the present value of a 5-payment decreasing annuity-immediate where the payments decrease by 30 each period.
Since $n = 5$ and $R = 30$, the payments are

$$30(5),\ 30(4),\ 30(3),\ 30(2),\ 30(1).$$

Therefore, the payments are

$$150,\ 120,\ 90,\ 60,\ 30.$$

Hence,

$$30(Da)_{\overline{5}|i} = 150v + 120v^2 + 90v^3 + 60v^4 + 30v^5.$$

Common mistake

The coefficient in front of $(Da)_{\overline{n}|i}$ is not the first payment.
For example,

$$30(Da)_{\overline{5}|i}$$

does not start with a payment of 30. It starts with

$$5(30) = 150.$$

The coefficient is the constant decrease between consecutive payments.

Decreasing annuity-immediate

An annuity-immediate makes annual payments of

$$1500,\ 1450,\ 1400, \ldots,\ 900.$$

Given

$$i = 0.06,$$

find the present value of this annuity.
Solution.
First, count the number of payments.
The payments decrease by 50 each year. Let n be the number of payments. Then

$$1500 - 50(n-1) = 900.$$

Thus,

$$50(n-1) = 600,$$

so

$$n = 13.$$

The payment stream is

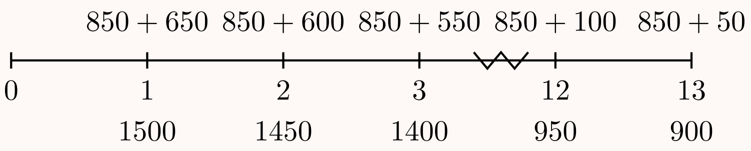
$$1500,\ 1450,\ 1400, \ldots,\ 900.$$

We can decompose it as a level annuity plus a decreasing annuity:

$$1500,\ 1450,\ 1400, \ldots,\ 900 = 850(1,1,1,\ldots,1) + 50(13,12,11,\ldots,1).$$

Therefore,

$$PV = 850a_{\overline{13}|0.06} + 50(Da)_{\overline{13}|0.06}.$$



Now use

$$(Da)_{\overline{n}|i} = \frac{n - a_{\overline{n}|i}}{i}.$$

Hence,

$$(Da)_{\overline{13}|0.06} = \frac{13 - a_{\overline{13}|0.06}}{0.06}.$$

Also,

$$a_{\overline{13}|0.06} = \frac{1-v^{13}}{0.06}, \qquad v = \frac{1}{1.06}.$$

Therefore,

$$PV = 850a_{\overline{13}|0.06} + 50(Da)_{\overline{13}|0.06}.$$

Substituting the values gives

$$PV = 10980.88.$$

Thus,

$$\boxed{PV = 10980.88}.$$

Using the BA II Plus.

We have

$$PV = 50(Da)_{\overline{13}|0.06} + 850a_{\overline{13}|0.06}.$$

Recall that

$$(Da)_{\overline{n}|i} = \frac{n - a_{\overline{n}|i}}{i}.$$

First compute

$$a_{\overline{13}|0.06}$$

using the TVM function:

$$\begin{aligned} N &= 13, \\ I/Y &= 6, \\ PMT &= 1, \\ FV &= 0, \\ CPT\ PV &= -8.852682963. \end{aligned}$$

Therefore,

$$a_{\overline{13}|0.06} = 8.852682963.$$

Now compute the decreasing annuity factor:

$$(Da)_{\overline{13}|0.06} = \frac{13 - 8.852682963}{0.06}.$$

Thus,

$$(Da)_{\overline{13}|0.06} = 69.12195062.$$

So the decreasing part is

$$50(Da)_{\overline{13}|0.06} = 50(69.12195062) = 3456.097531.$$

The level annuity part is

$$850a_{\overline{13}|0.06} = 850(8.852682963) = 7524.780518.$$

Therefore,

$$PV = 3456.097531 + 7524.780518.$$

Hence,

$$PV = 10980.878049.$$

So,

$$\boxed{PV = 10980.88}.$$

Solving a decreasing annuity using an increasing annuity

An annuity-immediate makes annual payments of

$$1500,\ 1450,\ 1400, \ldots,\ 900.$$

Given

$$i = 0.06,$$

find the present value of this annuity.
Solution.
The payments decrease by 50 each year. First, count the number of payments:

$$1500 - 50(n-1) = 900.$$

Thus,

$$50(n-1) = 600,$$

so

$$n = 13.$$

We now rewrite the decreasing payment stream as a level annuity minus an increasing annuity:

$$1500,\ 1450,\ 1400, \ldots,\ 900 = 1550(1, 1, 1, \ldots, 1) - 50(1, 2, 3, \ldots, 13).$$

Indeed,

$$1550 - 50 = 1500,$$

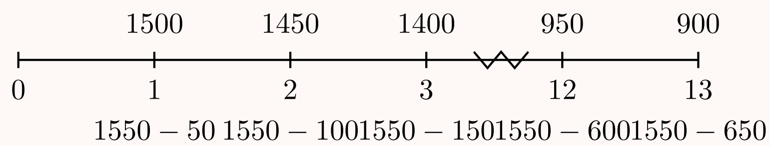
$$1550 - 100 = 1450,$$

and

$$1550 - 650 = 900.$$

Therefore,

$$PV = 1550a_{\overline{13}|0.06} - 50(Ia)_{\overline{13}|0.06}.$$



Using

$$a_{\overline{13}|0.06} = \frac{1 - v^{13}}{0.06}, \qquad v = \frac{1}{1.06},$$

we get

$$a_{\overline{13}|0.06} = 8.852682963.$$

Also,

$$(Ia)_{\overline{13}|0.06} = \frac{\ddot{a}_{\overline{13}|0.06} - 13v^{13}}{0.06}.$$

This gives

$$(Ia)_{\overline{13}|0.06} = 54.815610854.$$

Therefore,

$$PV = 1550(8.852682963) - 50(54.815610854).$$

So,

$$PV = 13721.658592 - 2740.780543.$$

Hence,

$$PV = 10980.878049.$$

Thus,

$$\boxed{PV = 10980.88}.$$

Decreasing Annuity-Due Present Value

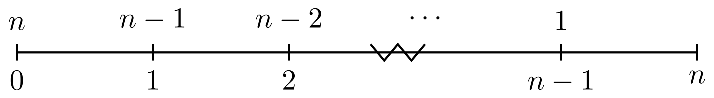
A decreasing annuity-due is an annuity whose payments decrease by a fixed amount each period, with payments made at the beginning of each period.

We first consider the standard decreasing annuity-due with payments

$$n, n-1, n-2, \ldots, 2, 1$$

at times

$$0, 1, 2, \ldots, n-2, n-1.$$



Decreasing annuity-due

The present value of a decreasing annuity-due with payments

$$n, n-1, n-2, \ldots, 2, 1$$

at the beginning of each period is denoted by

$$(D\ddot{a})_{\overline{n}|i}.$$

It is defined by

$$(D\ddot{a})_{\overline{n}|i} = n + (n-1)v + (n-2)v^2 + \cdots + 2v^{n-2} + v^{n-1}.$$

The decreasing annuity-due has the same payments as the decreasing annuity-immediate, but each payment occurs one period earlier.

Therefore,

$$(D\ddot{a})_{\overline{n}|i} = (1+i)(Da)_{\overline{n}|i}.$$

Using

$$(Da)_{\overline{n}|i} = \frac{n - a_{\overline{n}|i}}{i},$$

we get

$$(D\ddot{a})_{\overline{n}|i} = (1+i)\left(\frac{n - a_{\overline{n}|i}}{i}\right).$$

Since

$$d = \frac{i}{1+i},$$

we have

$$(D\ddot{a})_{\overline{n}|i} = \frac{n - a_{\overline{n}|i}}{d}.$$

Present value of a decreasing annuity-due

The present value of the decreasing annuity-due is

$$\boxed{(D\ddot{a})_{\overline{n}|i} = \frac{n - a_{\overline{n}|i}}{d}}$$

where

$$a_{\overline{n}|i} = \frac{1-v^n}{i}, \qquad v = \frac{1}{1+i}, \qquad d = \frac{i}{1+i}.$$

Useful comparison of annuity formulas

$$a_{\overline{n}|i} = \frac{1-v^n}{i}, \qquad \ddot{a}_{\overline{n}|i} = \frac{1-v^n}{d}.$$

$$(Ia)_{\overline{n}|i} = \frac{\ddot{a}_{\overline{n}|i} - nv^n}{i}, \qquad (I\ddot{a})_{\overline{n}|i} = \frac{\ddot{a}_{\overline{n}|i} - nv^n}{d}.$$

$$(Da)_{\overline{n}|i} = \frac{n - a_{\overline{n}|i}}{i}, \qquad (D\ddot{a})_{\overline{n}|i} = \frac{n - a_{\overline{n}|i}}{d}.$$

Immediate versus due

For decreasing annuities, the same pattern holds as for level and increasing annuities. The immediate version divides by i:

$$(Da)_{\overline{n}|i} = \frac{n - a_{\overline{n}|i}}{i}.$$

The due version divides by d:

$$(D\ddot{a})_{\overline{n}|i} = \frac{n - a_{\overline{n}|i}}{d}.$$

This happens because the annuity-due payments occur one period earlier.

Decreasing Annuities Accumulated Value

The accumulated value of an n-year decreasing annuity-immediate is denoted by

$$(Ds)_{\overline{n}|i}.$$

It represents the accumulated value at time n of payments

$$n, n-1, n-2, \ldots, 2, 1$$

made at times

$$1, 2, 3, \ldots, n-1, n.$$

Accumulated value of a decreasing annuity-immediate

The accumulated value of the decreasing annuity-immediate is

$$(Ds)_{\overline{n}|i} = n(1+i)^{n-1} + (n-1)(1+i)^{n-2} + \cdots + 2(1+i) + 1.$$

Since accumulated value at time n is obtained by multiplying the present value by $(1+i)^n$, we have

$$(Ds)_{\overline{n}|i} = (Da)_{\overline{n}|i}(1+i)^n.$$

Using

$$(Da)_{\overline{n}|i} = \frac{n - a_{\overline{n}|i}}{i},$$

we get

$$(Ds)_{\overline{n}|i} = \left(\frac{n - a_{\overline{n}|i}}{i}\right)(1+i)^n.$$

Now,

$$a_{\overline{n}|i}(1+i)^n = s_{\overline{n}|i}.$$

Therefore,

$$(Ds)_{\overline{n}|i} = \frac{n(1+i)^n - s_{\overline{n}|i}}{i}.$$

Accumulated value of a decreasing annuity-immediate

The accumulated value at time n is

$$\boxed{(Ds)_{\overline{n}|i} = \frac{n(1+i)^n - s_{\overline{n}|i}}{i}}$$

where

$$s_{\overline{n}|i} = \frac{(1+i)^n - 1}{i}.$$

The accumulated value of an n-year decreasing annuity-due is denoted by

$$(D\ddot{s})_{\overline{n}|i}.$$

It represents the accumulated value at time n of payments

$$n, n-1, n-2, \ldots, 2, 1$$

made at times

$$0, 1, 2, \ldots, n-2, n-1.$$

Since the due payments occur one period earlier than the immediate payments,

$$(D\ddot{s})_{\overline{n}|i} = (1+i)(Ds)_{\overline{n}|i}.$$

Thus,

$$(D\ddot{s})_{\overline{n}|i} = (1+i)\left(\frac{n(1+i)^n - s_{\overline{n}|i}}{i}\right).$$

Since

$$d = \frac{i}{1+i},$$

we obtain

$$(D\ddot{s})_{\overline{n}|i} = \frac{n(1+i)^n - s_{\overline{n}|i}}{d}.$$

Accumulated value of a decreasing annuity-due

The accumulated value at time n is

$$\boxed{(D\ddot{s})_{\overline{n}|i} = \frac{n(1+i)^n - s_{\overline{n}|i}}{d}}$$

where

$$d = \frac{i}{1+i}.$$

Practical method

Instead of memorizing these accumulated value formulas, it is often easier to first find the present value using

$$(Da)_{\overline{n}|i} = \frac{n - a_{\overline{n}|i}}{i}$$

or

$$(D\ddot{a})_{\overline{n}|i} = \frac{n - a_{\overline{n}|i}}{d},$$

and then accumulate to the desired time.

Exercise 2.1.51

Jane receives a ten-year increasing annuity-immediate paying 100 the first year and increasing by 100 each year thereafter.
Mary receives a ten-year decreasing annuity-immediate paying X the first year and decreasing by $X/10$ each year thereafter.
At an effective interest rate of 5%, both annuities have the same present value. Calculate X.
Solution.
Jane's payments are

$$100,\ 200,\ 300, \ldots,\ 1000.$$

Therefore, Jane's present value is

$$100(Ia)_{\overline{10}|0.05}.$$

Mary's payments are

$$X,\ \frac{9X}{10},\ \frac{8X}{10}, \ldots,\ \frac{X}{10}.$$

This can be written as

$$\frac{X}{10}(10, 9, 8, \ldots, 1).$$

Therefore, Mary's present value is

$$\frac{X}{10}(Da)_{\overline{10}|0.05}.$$

Since both present values are equal,

$$100(Ia)_{\overline{10}|0.05} = \frac{X}{10}(Da)_{\overline{10}|0.05}.$$

Solving for X,

$$X = \frac{1000(Ia)_{\overline{10}|0.05}}{(Da)_{\overline{10}|0.05}}.$$

We now compute the two annuity factors.
For the increasing annuity-immediate,

$$(Ia)_{\overline{n}|i} = \frac{\ddot{a}_{\overline{n}|i} - nv^n}{i}.$$

Using the BA II Plus in **BGN mode** to compute the numerator

$$\ddot{a}_{\overline{10}|0.05} - 10v^{10},$$

enter

$$\begin{aligned} N &= 10, \\ I/Y &= 5, \\ PMT &= 1, \\ FV &= -10, \\ CPT\ PV &= -1.96868914. \end{aligned}$$

Change the sign:

$$1.96868914.$$

Then divide by 0.05:

$$(Ia)_{\overline{10}|0.05} = \frac{1.96868914}{0.05} = 39.3737828.$$

Thus,

$$100(Ia)_{\overline{10}|0.05} = 100(39.3737828) = 3937.37828.$$

For the decreasing annuity-immediate,

$$(Da)_{\overline{n}|i} = \frac{n - a_{\overline{n}|i}}{i}.$$

First compute

$$a_{\overline{10}|0.05}.$$

Using the BA II Plus in ordinary mode, enter

$$\begin{aligned} N &= 10, \\ I/Y &= 5, \\ PMT &= 1, \\ FV &= 0, \\ CPT\ PV &= -7.721734929. \end{aligned}$$

Therefore,

$$a_{\overline{10}|0.05} = 7.721734929.$$

So,

$$(Da)_{\overline{10}|0.05} = \frac{10 - 7.721734929}{0.05}.$$

Hence,

$$(Da)_{\overline{10}|0.05} = 45.56530142.$$

Now substitute into the equation:

$$3937.37828 = \frac{X}{10}(45.56530142).$$

Thus,

$$X = \frac{10(3937.37828)}{45.56530142}.$$

Therefore,

$$X = 864.12.$$

So,

$$\boxed{X = 864.12}.$$

Exercise 2.1.52

For a decreasing annuity-due with annual payments:

1. The first payment is 120.
2. Each subsequent payment decreases by 10.
3. The last payment is 40.
4. The annual effective rate of interest is 5%.

Find the accumulated value of the annuity payments one year after the last payment.
Solution.
The payments are

$$120,\ 110,\ 100, \ldots,\ 40.$$

Since the payments decrease by 10, we first count the number of payments:

$$120 - 10(n-1) = 40.$$

Thus,

$$10(n-1) = 80,$$

so

$$n = 9.$$

Since this is an annuity-due, the payments occur at times

$$0, 1, 2, \ldots, 8.$$

The accumulated value is required one year after the last payment, which is time 9.
We decompose the payment stream as

$$120,\ 110,\ 100, \ldots,\ 40 = 30(1, 1, 1, \ldots, 1) + 10(9, 8, 7, \ldots, 1).$$

Therefore, the accumulated value at time 9 is

$$AV = 30\ddot{s}_{\overline{9}|0.05} + 10(D\ddot{s})_{\overline{9}|0.05}.$$

Using the formula

$$(D\ddot{s})_{\overline{n}|i} = \frac{n(1+i)^n - s_{\overline{n}|i}}{d},$$

where

$$d = \frac{i}{1+i},$$

we get

$$(D\ddot{s})_{\overline{9}|0.05} = \frac{9(1.05)^9 - s_{\overline{9}|0.05}}{0.05/1.05}.$$

Thus,

$$AV = 10\left(\frac{9(1.05)^9 - s_{\overline{9}|0.05}}{0.05/1.05}\right) + 30\ddot{s}_{\overline{9}|0.05}.$$

Therefore,

$$AV = 963.77.$$

So the accumulated value one year after the last payment is

$$\boxed{963.77}.$$

Alternative method.
Instead of using accumulated value formulas directly, we can value the payments one period before the first payment, at time -1, using annuity-immediate formulas.
At time -1, the present value is

$$10(Da)_{\overline{9}|0.05} + 30a_{\overline{9}|0.05}.$$

Using

$$(Da)_{\overline{n}|i} = \frac{n - a_{\overline{n}|i}}{i},$$

we obtain

$$10(Da)_{\overline{9}|0.05} + 30a_{\overline{9}|0.05} = 591.67.$$

Now accumulate this value from time -1 to time 9, which is 10 years:

$$591.67(1.05)^{10} = 963.77.$$

Thus,

$$\boxed{AV = 963.77}.$$

Exercise 2.1.53

For an annuity-immediate with monthly payments:

1. The monthly payments for each year are level.
2. The monthly payments for the first year are 100.
3. Each subsequent year the monthly payments decrease by 10. For example, in year two the monthly payments are 90, in year three the monthly payments are 80, and so on.
4. There are ten years worth of payments.

5. The annual effective interest rate is 8%.

Find the present value of this annuity.
Solution.
Let j be the effective monthly interest rate. Since the annual effective interest rate is 8%, we have

$$j = (1.08)^{1/12} - 1.$$

During year 1, the annuity pays 100 at the end of each month. The accumulated value at the end of year 1 is

$$100s_{\overline{12}|j}.$$

During year 2, the monthly payment is 90, so the accumulated value at the end of year 2 is

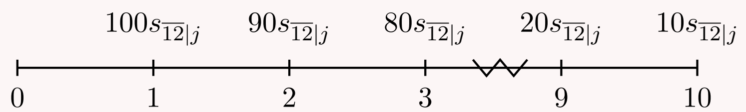
$$90s_{\overline{12}|j}.$$

Continuing in the same way, the annual rolled-up cash flows are

$$100s_{\overline{12}|j},\ 90s_{\overline{12}|j},\ 80s_{\overline{12}|j}, \dots,\ 10s_{\overline{12}|j}$$

at times

$$1, 2, 3, \dots, 10.$$



Factor out $10s_{\overline{12}|j}$:

$$100s_{\overline{12}|j},\ 90s_{\overline{12}|j}, \dots,\ 10s_{\overline{12}|j} = 10s_{\overline{12}|j}(10, 9, \dots, 1).$$

Therefore, the present value is

$$PV = 10s_{\overline{12}|j}(Da)_{\overline{10}|0.08}.$$

Now,

$$s_{\overline{12}|j} = \frac{(1+j)^{12} - 1}{j} = \frac{1.08 - 1}{j}.$$

Also,

$$(Da)_{\overline{10}|0.08} = \frac{10 - a_{\overline{10}|0.08}}{0.08}.$$

Using

$$j = (1.08)^{1/12} - 1,$$

we obtain

$$10s_{\overline{12}|j}(Da)_{\overline{10}|0.08} = 5113.31.$$

Therefore,

$$\boxed{PV = 5113.31}.$$

Exercise 2.1.54

An annuity-immediate pays 10 at the ends of years 1 and 2, 9 at the ends of years 3 and 4, etc., with payments decreasing by 1 every second year, until nothing is paid.
The effective annual rate of interest is 5%.
Calculate the present value of this annuity-immediate.

A. 71 B. 78

C. 84 D. 88

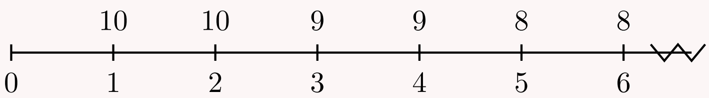
E. 94

Solution.
The payments are

$$10, 10, 9, 9, 8, 8, \ldots, 1, 1$$

at times

$$1, 2, 3, 4, 5, 6, \ldots, 19, 20.$$



Let j be the effective interest rate per two years. Then

$$j = (1.05)^2 - 1 = 0.1025.$$

Now split the payments into two streams.
The payments at the ends of years $2, 4, 6, \ldots, 20$ are

$$10, 9, 8, \ldots, 1.$$

Using two-year periods, their present value is

$$(Da)_{\overline{10}|0.1025}.$$

The payments at the ends of years $1, 3, 5, \ldots, 19$ are also

$$10, 9, 8, \ldots, 1.$$

They occur one year earlier than the corresponding payments at years $2, 4, 6, \ldots, 20$. Therefore, their present value is

$$1.05(Da)_{\overline{10}|0.1025}.$$

Hence, the total present value is

$$PV = (Da)_{\overline{10}|0.1025} + 1.05(Da)_{\overline{10}|0.1025}.$$

Thus,

$$PV = 2.05(Da)_{\overline{10}|0.1025}.$$

Using

$$(Da)_{\overline{n}|i} = \frac{n - a_{\overline{n}|i}}{i},$$

we get

$$(Da)_{\overline{10}|0.1025} = \frac{10 - a_{\overline{10}|0.1025}}{0.1025}.$$

This gives

$$(Da)_{\overline{10}|0.1025} = 38.2482.$$

Therefore,

$$PV = 2.05(38.2482).$$

So,

$$PV = 78.41.$$

The closest answer is

$$\boxed{\text{B. } 78}.$$

Exercise 2.1.55

François purchases a 10-year annuity-immediate with annual payments of $10X$.
Jacques purchases a 10-year decreasing annuity-immediate which also makes annual payments. The payment at the end of year 1 is equal to 50. At the end of year 2, and at the end of each year through year 10, each subsequent payment is reduced from the previous year's payment by an amount equal to X.
At an annual effective interest rate of 7.072%, both annuities have the same present value.
Calculate X, where $X < 5$.

A. 3.29 B. 3.39

C. 3.49 D. 3.59

E. 3.69

Solution.
Let

$$i = 0.07072 \qquad \text{and} \qquad v = \frac{1}{1+i}.$$

François receives $10X$ at the end of each year for 10 years. Therefore, François's present value is

$$10Xa_{\overline{10}|i}.$$

Jacques receives payments

$$50,\ 50 - X,\ 50 - 2X, \ldots,\ 50 - 9X.$$

We can decompose Jacques's payments as a level annuity of 50, minus an increasing annuity of X starting one year after the first payment.
Thus, Jacques's present value is

$$50a_{\overline{10}|i} - X(Ia)_{\overline{9}|i}v.$$

Since both present values are equal,

$$10Xa_{\overline{10}|i} = 50a_{\overline{10}|i} - X(Ia)_{\overline{9}|i}v.$$

Move the terms involving X to the left:

$$10Xa_{\overline{10}|i} + X(Ia)_{\overline{9}|i}v = 50a_{\overline{10}|i}.$$

Factor out X:

$$X\left(10a_{\overline{10}|i} + (Ia)_{\overline{9}|i}v\right) = 50a_{\overline{10}|i}.$$

Therefore,

$$X = \frac{50a_{\overline{10}|i}}{10a_{\overline{10}|i} + (Ia)_{\overline{9}|i}v}.$$

Using $i = 0.07072$, we get

$$a_{\overline{10}|i} = 7.000265,$$

and

$$(Ia)_{\overline{9}|i} = 29.536439.$$

Therefore,

$$X = \frac{50(7.000265)}{10(7.000265) + 29.536439v}.$$

Since

$$v = \frac{1}{1.07072},$$

we obtain

$$X = 3.5866.$$

Thus,

$$X \approx 3.59.$$

The correct answer is

$$\boxed{\text{D. } 3.59}.$$

Exercise 2.1.56

You are given an annuity-immediate paying 10 for 10 years, then decreasing by one per year for nine years, and paying one per year thereafter, forever. The annual effective rate of interest is 4%.
Calculate the present value of this annuity.

A. 119 B. 121

C. 123 D. 125

E. 127

Solution.
The payment stream is

$$10, 10, \ldots, 10$$

for the first 10 years, then

$$9, 8, 7, \ldots, 1$$

for the next 9 years, and then

$$1, 1, 1, \ldots$$

forever.
We can represent the annuity as the sum of three separate streams:

1. a level annuity of 10 for the first 10 years;

2. a 10-year deferred decreasing annuity for the next 9 years;

3. a level perpetuity of 1 deferred for 19 years.

Let

$$i = 0.04 \qquad \text{and} \qquad v = \frac{1}{1.04}.$$

The present value of the first stream is

$$10a_{\overline{10}|0.04}.$$

The decreasing payments $9, 8, \ldots, 1$ occur at times $11, 12, \ldots, 19$. Their value at time 10 is

$$(Da)_{\overline{9}|0.04}.$$

Thus, their present value at time 0 is

$$v^{10}(Da)_{\overline{9}|0.04}.$$

The perpetuity of 1 begins at time 20. Its value at time 19 is

$$a_{\overline{\infty}|0.04}.$$

Therefore, its present value at time 0 is

$$v^{19}a_{\overline{\infty}|0.04}.$$

Hence,

$$PV = 10a_{\overline{10}|0.04} + v^{10}(Da)_{\overline{9}|0.04} + v^{19}a_{\overline{\infty}|0.04}.$$

Now,

$$a_{\overline{10}|0.04} = \frac{1 - v^{10}}{0.04},$$

$$(Da)_{\overline{9}|0.04} = \frac{9 - a_{\overline{9}|0.04}}{0.04},$$

and

$$a_{\overline{\infty}|0.04} = \frac{1}{0.04}.$$

Substituting,

$$PV = 10a_{\overline{10}|0.04} + v^{10}(Da)_{\overline{9}|0.04} + v^{19}a_{\overline{\infty}|0.04}.$$

This gives

$$PV = 119.40.$$

The closest answer is

$$\boxed{\text{A. } 119}.$$

2.2 Varying Annuities

2.2.1 Geometric Varying Annuities

Part 1

Geometric Annuity-Immediate

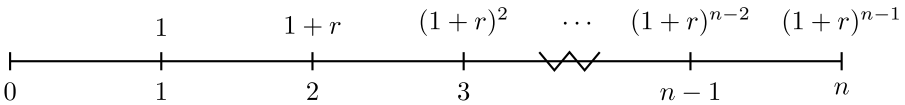
A geometric annuity-immediate is an annuity whose payments increase by a constant percentage each period.

Consider the following annuity-immediate with payments

$$1, \quad 1+r, \quad (1+r)^2, \quad \ldots, \quad (1+r)^{n-2}, \quad (1+r)^{n-1}$$

at times

$$1, 2, 3, \ldots, n-1, n.$$



Geometric annuity-immediate

We denote the present value of this annuity by

$$(Ga)_{\overline{n}|i,r}.$$

This notation is useful for these notes, but it is not standard actuarial notation.

The present value is

$$(Ga)_{\overline{n}|i,r} = v_i + (1+r)v_i^2 + (1+r)^2 v_i^3 + \cdots + (1+r)^{n-1} v_i^n.$$

Factor out v_i:

$$(Ga)_{\overline{n}|i,r} = v_i \left[1 + (1+r)v_i + (1+r)^2 v_i^2 + \cdots + (1+r)^{n-1} v_i^{n-1}\right].$$

This is a finite geometric series with common ratio

$$(1+r)v_i = \frac{1+r}{1+i}.$$

Therefore, summing the geometric series and substituting

$$v_i = \frac{1}{1+i}$$

yields the closed form summarized in the formula box below.

Present value of a geometric annuity-immediate

If $i \neq r$, then

$$\boxed{(Ga)_{\overline{n}|i,r} = \frac{1 - \left(\frac{1+r}{1+i}\right)^n}{i-r}}$$

where i is the effective interest rate and r is the payment growth rate.

Coefficient interpretation

The coefficient in front of the geometric annuity factor is the amount of the first payment. If the payments are

$$P, \quad P(1+r), \quad P(1+r)^2, \quad \ldots, \quad P(1+r)^{n-1},$$

then the present value is

$$PV = P(Ga)_{\overline{n}|i,r}.$$

Formula to memorize

The main formula is

$$(Ga)_{\overline{n}|i,r} = \frac{1 - \left(\frac{1+r}{1+i}\right)^n}{i - r}.$$

This formula assumes $i \neq r$.

Geometric annuity-immediate

For an annuity-immediate with annual payments:

1. The first payment is 1000.
2. Each subsequent payment increases by 8%.
3. The last payment is $1000(1.08)^{15}$.
4. The annual effective rate of interest is 6%.

Find the present value of this annuity.

Solution.

The payments are

$$1000,\ 1000(1.08),\ 1000(1.08)^2, \ldots,\ 1000(1.08)^{15}.$$

The exponents are

$$0, 1, 2, \ldots, 15,$$

so the number of payments is

$$16.$$

This is a geometric annuity-immediate with first payment 1000, interest rate

$$i = 0.06,$$

and payment growth rate

$$r = 0.08.$$

Therefore,

$$PV = 1000(Ga)_{\overline{16}|0.06,0.08}.$$

Using the formula

$$(Ga)_{\overline{n}|i,r} = \frac{1 - \left(\frac{1+r}{1+i}\right)^n}{i - r},$$

we get

$$PV = 1000\left(\frac{1 - \left(\frac{1.08}{1.06}\right)^{16}}{0.06 - 0.08}\right).$$

Thus,

$$PV = 17430.48.$$

Therefore,

$$\boxed{PV = 17430.48}.$$

Alternative computation.
The present value can also be written directly as a finite geometric series:

$$PV = 1000(1.06)^{-1} + 1000(1.08)(1.06)^{-2} + \cdots + 1000(1.08)^{15}(1.06)^{-16}.$$

Using the finite geometric series formula,

$$PV = \frac{1000(1.06)^{-1} - 1000(1.08)^{16}(1.06)^{-17}}{1 - (1.08)(1.06)^{-1}}.$$

Hence,

$$PV = 17430.48.$$

Accumulated Value of a Geometric Annuity-Immediate

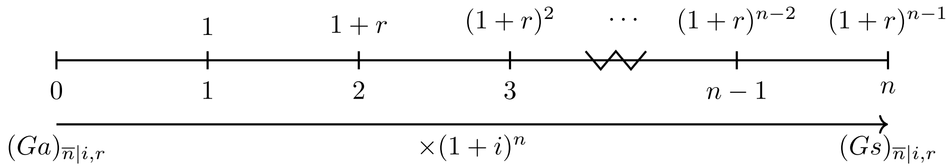
To find the accumulated value of a geometric annuity-immediate, first find the present value and then accumulate it to time n.

We consider the same payment stream:

$$1, \quad 1+r, \quad (1+r)^2, \quad \ldots, \quad (1+r)^{n-2}, \quad (1+r)^{n-1}$$

at times

$$1, 2, 3, \ldots, n-1, n.$$



Geometric annuity-immediate accumulated value

We denote the accumulated value at time n by

$$(Gs)_{\overline{n}|i,r}.$$

It is obtained by accumulating the present value:

$$(Gs)_{\overline{n}|i,r} = (Ga)_{\overline{n}|i,r}(1+i)^n.$$

Using

$$(Ga)_{\overline{n}|i,r} = \frac{1 - \left(\frac{1+r}{1+i}\right)^n}{i - r},$$

we get

$$(Gs)_{\overline{n}|i,r} = \left(\frac{1 - \left(\frac{1+r}{1+i}\right)^n}{i - r}\right)(1+i)^n.$$

Simplifying,

$$(Gs)_{\overline{n}|i,r} = \frac{(1+i)^n - (1+r)^n}{i - r}.$$

Accumulated value of a geometric annuity-immediate

If $i \neq r$, then

$$\boxed{(Gs)_{\overline{n}|i,r} = \frac{(1+i)^n - (1+r)^n}{i-r}}$$

where i is the effective interest rate and r is the payment growth rate.

Coefficient interpretation

If the first payment is P, then the payments are

$$P, \quad P(1+r), \quad P(1+r)^2, \quad \ldots, \quad P(1+r)^{n-1}.$$

The accumulated value at time n is

$$AV = P(Gs)_{\overline{n}|i,r}.$$

Special case

The formula

$$(Gs)_{\overline{n}|i,r} = \frac{(1+i)^n - (1+r)^n}{i-r}$$

assumes $i \neq r$.
If $i = r$, then

$$(Gs)_{\overline{n}|i,i} = n(1+i)^{n-1}.$$

Geometric Annuity-Due

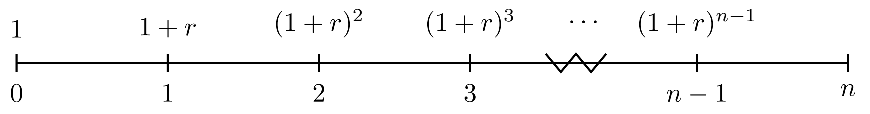
A geometric annuity-due is an annuity whose payments increase by a constant percentage each period, with payments made at the beginning of each period.

Consider the following annuity-due with payments

$$1, \quad 1+r, \quad (1+r)^2, \quad \ldots, \quad (1+r)^{n-1}$$

at times

$$0, 1, 2, \ldots, n-1.$$



Geometric annuity-due

We denote the present value of this annuity by

$$(G\ddot{a})_{\overline{n}|i,r}.$$

This notation is useful for these notes, but it is not standard actuarial notation.

The geometric annuity-due has the same payments as the geometric annuity-immediate, but each payment occurs one period earlier.

Therefore,

$$(G\ddot{a})_{\overline{n}|i,r} = (1+i)(Ga)_{\overline{n}|i,r}.$$

Using

$$(Ga)_{\overline{n}|i,r} = \frac{1 - \left(\frac{1+r}{1+i}\right)^n}{i-r},$$

we get

$$(G\ddot{a})_{\overline{n}|i,r} = (1+i)\left(\frac{1 - \left(\frac{1+r}{1+i}\right)^n}{i-r}\right).$$

Hence,

$$(G\ddot{a})_{\overline{n}|i,r} = \frac{1 - \left(\frac{1+r}{1+i}\right)^n}{\dfrac{i-r}{1+i}}.$$

Present value of a geometric annuity-due

If $i \neq r$, then

$$\boxed{(G\ddot{a})_{\overline{n}|i,r} = \frac{1 - \left(\frac{1+r}{1+i}\right)^n}{\dfrac{i-r}{1+i}}}$$

Equivalently,

$$\boxed{(G\ddot{a})_{\overline{n}|i,r} = (1+i)(Ga)_{\overline{n}|i,r}}.$$

Equivalent annuity-due interpretation

Let

$$v_j = \frac{1+r}{1+i}.$$

Then

$$d_j = 1 - v_j = 1 - \frac{1+r}{1+i} = \frac{i-r}{1+i}.$$

Therefore,

$$(G\ddot{a})_{\overline{n}|i,r} = \frac{1 - v_j^n}{d_j}.$$

This has the same form as

$$\ddot{a}_{\overline{n}|j},$$

where the effective rate j satisfies

$$v_j = \frac{1}{1+j}.$$

Since

$$\frac{1}{1+j} = \frac{1+r}{1+i},$$

we get

$$j = \frac{i-r}{1+r}.$$

Coefficient interpretation

The coefficient in front of the geometric annuity-due factor is the first payment.
If the payments are

$$P, \quad P(1+r), \quad P(1+r)^2, \quad \ldots, \quad P(1+r)^{n-1},$$

with the first payment made immediately, then

$$PV = P(G\ddot{a})_{\overline{n}|i,r}.$$

Special case

The formula

$$(G\ddot{a})_{\overline{n}|i,r} = \frac{1 - \left(\frac{1+r}{1+i}\right)^n}{\dfrac{i-r}{1+i}}$$

assumes $i \neq r$.
If $i = r$, then each discounted payment is equal to 1, so

$$(G\ddot{a})_{\overline{n}|i,i} = n.$$

Geometric annuity-due

For an annuity-due with annual payments:

1. The first payment is 1000.
2. Each subsequent payment increases by 6%.
3. The last payment is $1000(1.06)^{15}$.
4. The annual effective rate of interest is 8%.

Find the present value of this annuity.
Solution.
The payments are

$$1000, \ 1000(1.06), \ 1000(1.06)^2, \ldots, \ 1000(1.06)^{15}.$$

Since the first payment is made immediately, this is a geometric annuity-due.
The exponents are

$$0, 1, 2, \ldots, 15,$$

so the number of payments is

$$16.$$

Here,

$$i = 0.08 \qquad \text{and} \qquad r = 0.06.$$

Therefore,

$$PV = 1000(G\ddot{a})_{\overline{16}|0.08,0.06}.$$

Using

$$(G\ddot{a})_{\overline{n}|i,r} = (1+i)(Ga)_{\overline{n}|i,r},$$

we get

$$PV = 1000(1.08)(Ga)_{\overline{16}|0.08,0.06}.$$

Now use

$$(Ga)_{\overline{n}|i,r} = \frac{1 - \left(\frac{1+r}{1+i}\right)^n}{i - r}.$$

Thus,

$$PV = 1000(1.08)\left(\frac{1 - \left(\frac{1.06}{1.08}\right)^{16}}{0.08 - 0.06}\right).$$

Therefore,

$$PV = 13958.76.$$

Hence,

$$\boxed{PV = 13958.76}.$$

Alternative solution 1.
We can rewrite the geometric annuity-due as a level annuity-due under an adjusted effective rate j.
Let

$$j = \frac{i - r}{1 + r}.$$

Then

$$j = \frac{0.08 - 0.06}{1.06} = 0.0188679245.$$

Therefore,

$$1000(G\ddot{a})_{\overline{16}|0.08,0.06} = 1000\ddot{a}_{\overline{16}|j}.$$

So,

$$PV = 1000\ddot{a}_{\overline{16}|0.0188679245}.$$

This gives

$$PV = 13958.76.$$

Alternative solution 2.
We can also compute the present value directly as a finite geometric series:

$$PV = 1000 + 1000(1.06)(1.08)^{-1} + 1000(1.06)^2(1.08)^{-2} + \cdots + 1000(1.06)^{15}(1.08)^{-15}.$$

The common ratio is

$$\frac{1.06}{1.08}.$$

Thus,

$$PV = \frac{1000 - 1000(1.06)^{16}(1.08)^{-16}}{1 - (1.06)(1.08)^{-1}}.$$

Therefore,

$$PV = 13958.76.$$

Accumulated Value of a Geometric Annuity-Due

To find the accumulated value of a geometric annuity-due, we use the same idea as for the annuity-immediate: first find the present value, then accumulate it to time n.

The accumulated value at time n is denoted by

$$(G\ddot{s})_{\overline{n}|i,r}.$$

Since
$$(G\ddot{a})_{\overline{n}|i,r} = (1+i)(Ga)_{\overline{n}|i,r},$$
we have
$$(G\ddot{s})_{\overline{n}|i,r} = (G\ddot{a})_{\overline{n}|i,r}(1+i)^n.$$
Therefore,
$$(G\ddot{s})_{\overline{n}|i,r} = (1+i)(Ga)_{\overline{n}|i,r}(1+i)^n.$$
Hence,
$$(G\ddot{s})_{\overline{n}|i,r} = (Ga)_{\overline{n}|i,r}(1+i)^{n+1}.$$
Using
$$(Ga)_{\overline{n}|i,r} = \frac{1-\left(\frac{1+r}{1+i}\right)^n}{i-r},$$
we get
$$(G\ddot{s})_{\overline{n}|i,r} = \left(\frac{1-\left(\frac{1+r}{1+i}\right)^n}{i-r}\right)(1+i)^{n+1}.$$
Simplifying,
$$(G\ddot{s})_{\overline{n}|i,r} = \frac{(1+i)^{n+1}-(1+i)(1+r)^n}{i-r}.$$

Accumulated value of a geometric annuity-due

If $i \neq r$, then
$$\boxed{(G\ddot{s})_{\overline{n}|i,r} = (G\ddot{a})_{\overline{n}|i,r}(1+i)^n}$$
or equivalently,
$$\boxed{(G\ddot{s})_{\overline{n}|i,r} = (Ga)_{\overline{n}|i,r}(1+i)^{n+1}}$$
and
$$\boxed{(G\ddot{s})_{\overline{n}|i,r} = \frac{(1+i)^{n+1}-(1+i)(1+r)^n}{i-r}}.$$

Practical method

For a geometric annuity-due, it is often easier to remember
$$(G\ddot{s})_{\overline{n}|i,r} = (G\ddot{a})_{\overline{n}|i,r}(1+i)^n.$$
That is, find the present value first, then accumulate it to time n.

Special case

The formula
$$(G\ddot{s})_{\overline{n}|i,r} = \frac{(1+i)^{n+1}-(1+i)(1+r)^n}{i-r}$$
assumes $i \neq r$.
If $i = r$, then each payment accumulates to the same value at time n, so
$$(G\ddot{s})_{\overline{n}|i,i} = n(1+i)^n.$$

Exercise 2.2.1

Annual deposits are made into a fund at the beginning of each year for 10 years. The first 5 deposits are 100 each, and deposits increase by 7% per year thereafter.
If the fund earns an annual effective rate of 5%, find the accumulated value at the end of 10 years.

Solution.
The deposits are made at the beginning of each year, so they occur at times

$$0, 1, 2, \ldots, 9.$$

The first 5 deposits are 100, so the deposits at times $0, 1, 2, 3, 4$ are all 100. After that, the deposits increase by 7% per year:

$$100(1.07),\ 100(1.07)^2, \ldots,\ 100(1.07)^5.$$

A convenient valuation date is time 3.
At time 3, the first four deposits, made at times $0, 1, 2, 3$, have accumulated value

$$100 s_{\overline{4}|0.05}.$$

The remaining six deposits, made at times $4, 5, \ldots, 9$, form a geometric annuity-immediate from the perspective of time 3. The first payment is 100, the interest rate is

$$i = 0.05,$$

and the growth rate is

$$r = 0.07.$$

Therefore, their value at time 3 is

$$100(Ga)_{\overline{6}|0.05,0.07}.$$

Hence, the value at time 3 is

$$100 s_{\overline{4}|0.05} + 100(Ga)_{\overline{6}|0.05,0.07}.$$

Using

$$s_{\overline{4}|0.05} = \frac{1.05^4 - 1}{0.05} = 4.310125,$$

and

$$(Ga)_{\overline{6}|0.05,0.07} = \frac{1 - \left(\frac{1.07}{1.05}\right)^6}{0.05 - 0.07} = 5.993404737,$$

we get

$$100 s_{\overline{4}|0.05} + 100(Ga)_{\overline{6}|0.05,0.07} = 1030.35.$$

Now accumulate this value from time 3 to time 10, which is 7 years:

$$AV = 1030.35(1.05)^7.$$

Therefore,

$$AV = 1449.81.$$

Thus, the accumulated value at the end of 10 years is

$$\boxed{1449.81}.$$

Exercise 2.2.2

Stan elects to receive his retirement benefit over 20 years at the rate of 2000 per month, beginning one month from now. The monthly benefit increases by 5% each year.
At a nominal rate of 6% convertible monthly, calculate the present value of the retirement benefit.

Solution.

The nominal annual interest rate is 6% convertible monthly, so the effective monthly interest rate is

$$j = \frac{0.06}{12} = 0.005.$$

The effective annual interest rate is

$$i = (1+j)^{12} - 1.$$

Thus,

$$i = (1.005)^{12} - 1 = 0.0616778.$$

During the first year, Stan receives 2000 at the end of each month. The accumulated value of these 12 payments at the end of the first year is

$$2000 s_{\overline{12}|j}.$$

During the second year, the monthly payment increases by 5%, so the accumulated value at the end of the second year is

$$2000(1.05) s_{\overline{12}|j}.$$

Continuing in the same way, the rolled-up annual cash flows are

$$2000 s_{\overline{12}|j}, \quad 2000(1.05) s_{\overline{12}|j}, \quad 2000(1.05)^2 s_{\overline{12}|j}, \quad \ldots, \quad 2000(1.05)^{19} s_{\overline{12}|j}.$$

These occur at times

$$1, 2, 3, \ldots, 20.$$

Therefore, the present value is

$$PV = 2000 s_{\overline{12}|j} (Ga)_{\overline{20}|i,0.05}.$$

Now,

$$s_{\overline{12}|j} = \frac{(1+j)^{12} - 1}{j}.$$

Since

$$j = 0.005,$$

we have

$$s_{\overline{12}|0.005} = \frac{(1.005)^{12} - 1}{0.005} = 12.33556237.$$

Also,

$$(Ga)_{\overline{20}|i,0.05} = \frac{1 - \left(\frac{1.05}{1+i}\right)^{20}}{i - 0.05}.$$

Using

$$i = 0.0616778,$$

we get

$$(Ga)_{\overline{20}|0.0616778,0.05} = \frac{1 - \left(\frac{1.05}{1.0616778}\right)^{20}}{0.0616778 - 0.05}.$$

Thus,

$$PV = 2000(12.33556237)\left(\frac{1-\left(\frac{1.05}{1.0616778}\right)^{20}}{0.0616778-0.05}\right).$$

Therefore,

$$PV = 419253.25.$$

So the present value of the retirement benefit is

$$\boxed{419253.25}.$$

Exercise 2.2.3

Smith receives annual dividends which began one year after an investment of 10,000 fifteen years ago. Dividends have increased 2% every year. The dividend Smith received today was 659.74.
What is the accumulated value today of Smith's dividends assuming an effective annual interest rate of 5.06%?

A. Less than 12,240
B. At least 12,240, but less than 12,300
C. At least 12,300, but less than 12,360
D. At least 12,360, but less than 12,420
E. At least 12,420

Solution.
Let k be the first dividend, paid one year after the investment.
Since the investment was made 15 years ago and the first dividend was paid one year after the investment, the dividends occur at times

$$1, 2, \ldots, 15.$$

The dividends increase by 2% each year, so the dividend received today at time 15 is

$$k(1.02)^{14}.$$

We are given

$$k(1.02)^{14} = 659.74.$$

Therefore,

$$k = \frac{659.74}{(1.02)^{14}}.$$

Thus,

$$k = 500.$$

So the dividends are

$$500,\ 500(1.02),\ 500(1.02)^2, \ldots, 500(1.02)^{14}.$$

We need the accumulated value today, at time 15. This is the accumulated value of a geometric annuity-immediate with

$$n = 15, \qquad i = 0.0506, \qquad r = 0.02.$$

Hence,

$$AV = 500(Gs)_{\overline{15}|0.0506,0.02}.$$

Using
$$(Gs)_{\overline{n}|i,r} = \frac{(1+i)^n - (1+r)^n}{i-r},$$
we get
$$AV = 500\left(\frac{(1.0506)^{15} - (1.02)^{15}}{0.0506 - 0.02}\right).$$
Therefore,
$$AV = 12270.44.$$
The accumulated value today is approximately
$$\boxed{12270}.$$
The correct answer is
$$\boxed{\text{B. At least 12,240, but less than 12,300}}.$$

Exercise 2.2.4

Mary is to receive an annuity with 30 annual payments. The first payment of 1000 is due immediately, and each successive payment is 5% less than the payment for the preceding year.
Interest is 12% compounded annually. Determine the present value of the annuity.

A. Less than 6,400
B. At least 6,400, but less than 6,500
C. At least 6,500, but less than 6,600
D. At least 6,600, but less than 6,700
E. At least 6,700

Solution.
The first payment is made immediately, so this is a geometric annuity-due.
The payments are
$$1000, \quad 1000(0.95), \quad 1000(0.95)^2, \quad \ldots, \quad 1000(0.95)^{29}.$$
They occur at times
$$0, 1, 2, \ldots, 29.$$
The annual effective interest rate is
$$i = 0.12.$$
Since each payment is 5% less than the previous payment, the growth rate is
$$r = -0.05.$$
Thus,
$$1 + r = 0.95.$$
The present value is
$$PV = 1000(G\ddot{a})_{\overline{30}|0.12,-0.05}.$$
Using
$$(G\ddot{a})_{\overline{n}|i,r} = (1+i)(Ga)_{\overline{n}|i,r},$$
we get
$$PV = 1000(1.12)(Ga)_{\overline{30}|0.12,-0.05}.$$

Now,

$$(Ga)_{\overline{n}|i,r} = \frac{1 - \left(\frac{1+r}{1+i}\right)^n}{i - r}.$$

Therefore,

$$PV = 1000(1.12)\left(\frac{1 - \left(\frac{0.95}{1.12}\right)^{30}}{0.12 - (-0.05)}\right).$$

Since

$$0.12 - (-0.05) = 0.17,$$

we have

$$PV = 1000(1.12)\left(\frac{1 - \left(\frac{0.95}{1.12}\right)^{30}}{0.17}\right).$$

Thus,

$$PV = 6541.04.$$

The present value is approximately

$$\boxed{6541}.$$

The correct answer is

$$\boxed{\text{C. At least 6,500, but less than 6,600}}.$$

Exercise 2.2.5

Bill is due to retire in 20 years, at which time he will start collecting pension benefits. His employer is scheduled to begin funding Bill's pension benefits with the first annual payment due at the end of year 1 and the last annual payment due at the end of year 20. The initial employer's contribution is 2000 and will increase by 3% each year. Employer contributions will earn interest at an annual effective interest rate of 7%.
Bill's employer can elect alternative funding: annual contributions beginning at the end of year 6 with the last contribution at the end of year 20. These annual contributions begin at 1000 and increase by Q per year. Under this alternative funding, the employer's contributions will earn an annual effective interest rate of 8%.
Determine Q so that both funding options have the same accumulated value at the end of year 20.

A. Less than 440

B. At least 440, but less than 480

C. At least 480, but less than 520

D. At least 520, but less than 560

E. At least 560

Solution.
We compare the accumulated values at the end of year 20.
First funding option.
The first option consists of annual payments

$$2000,\ 2000(1.03),\ 2000(1.03)^2, \ldots, 2000(1.03)^{19}$$

at times

$$1, 2, 3, \ldots, 20.$$

These payments earn an annual effective interest rate of 7%. Therefore, the accumulated value at time 20 is

$$AV_1 = 2000(Gs)_{\overline{20}|0.07,0.03}.$$

Using

$$(Gs)_{\overline{n}|i,r} = \frac{(1+i)^n - (1+r)^n}{i-r},$$

we get

$$AV_1 = 2000\left(\frac{1.07^{20} - 1.03^{20}}{0.07 - 0.03}\right).$$

Thus,

$$AV_1 = 103178.66.$$

Second funding option.
The second option has payments from time 6 through time 20, so there are

$$20 - 6 + 1 = 15$$

payments.
The payments are

$$1000,\ 1000 + Q,\ 1000 + 2Q, \ldots, 1000 + 14Q.$$

We decompose this into a level annuity plus an increasing annuity:

$$1000(1, 1, \ldots, 1) + Q(0, 1, 2, \ldots, 14).$$

At time 20, the level part has accumulated value

$$1000 s_{\overline{15}|0.08}.$$

The increasing part has accumulated value

$$Q(Is)_{\overline{14}|0.08}.$$

Therefore,

$$AV_2 = 1000 s_{\overline{15}|0.08} + Q(Is)_{\overline{14}|0.08}.$$

Set the two accumulated values equal:

$$AV_1 = AV_2.$$

Hence,

$$2000\left(\frac{1.07^{20} - 1.03^{20}}{0.07 - 0.03}\right) = 1000 s_{\overline{15}|0.08} + Q(Is)_{\overline{14}|0.08}.$$

Now,

$$s_{\overline{15}|0.08} = 27.15211393,$$

so

$$1000 s_{\overline{15}|0.08} = 27152.11.$$

Also,

$$(Is)_{\overline{14}|0.08} = \frac{\ddot{s}_{\overline{14}|0.08} - 14}{0.08} = 151.9014241.$$

Thus,

$$103178.66 = 27152.11 + 151.9014241Q.$$

So,

$$151.9014241Q = 76026.55.$$

Therefore,

$$Q = 500.50.$$

The correct answer is

$$\boxed{\text{C. At least 480, but less than 520}}.$$

Part 2

Special Case $r = i$

For a geometric annuity, the usual formulas

$$(Ga)_{\overline{n}|i,r} = \frac{1 - \left(\frac{1+r}{1+i}\right)^n}{i - r}$$

and

$$(Gs)_{\overline{n}|i,r} = \frac{(1+i)^n - (1+r)^n}{i - r}$$

cannot be used directly when

$$r = i,$$

because the denominator becomes zero.

We therefore need separate formulas for the special case $r = i$.

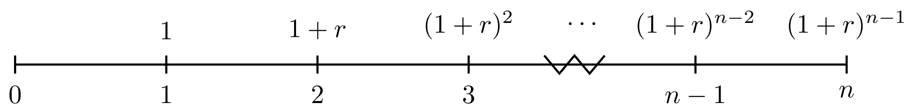
Geometric Annuity-Immediate

Consider the geometric annuity-immediate with payments

$$1, \quad 1+r, \quad (1+r)^2, \quad \ldots, \quad (1+r)^{n-1}$$

at times

$$1, 2, 3, \ldots, n.$$



When $i = r$, each discounted payment is the same:

$$(1+r)^{k-1} v_r^k = (1+r)^{k-1}(1+r)^{-k} = v_r.$$

Therefore,

$$(Ga)_{\overline{n}|r,r} = v_r + v_r + \cdots + v_r.$$

Since there are n payments,

$$(Ga)_{\overline{n}|r,r} = n v_r.$$

Special case $r = i$: geometric annuity-immediate

For a geometric annuity-immediate, if $r = i$, then

$$\boxed{(Ga)_{\overline{n}|r,r} = nv_r}$$

where

$$v_r = \frac{1}{1+r}.$$

The accumulated value at time n is

$$(Gs)_{\overline{n}|r,r} = (Ga)_{\overline{n}|r,r}(1+r)^n.$$

Thus,

$$(Gs)_{\overline{n}|r,r} = nv_r(1+r)^n.$$

Since

$$v_r = \frac{1}{1+r},$$

we obtain

$$(Gs)_{\overline{n}|r,r} = n(1+r)^{n-1}.$$

Special case $r = i$: accumulated value immediate

For a geometric annuity-immediate, if $r = i$, then

$$\boxed{(Gs)_{\overline{n}|r,r} = n(1+r)^{n-1}}.$$

Geometric Annuity-Due

For the geometric annuity-due, the payments are the same, but each payment occurs one period earlier.

Therefore,

$$(G\ddot{a})_{\overline{n}|r,r} = (1+r)(Ga)_{\overline{n}|r,r}.$$

Using

$$(Ga)_{\overline{n}|r,r} = nv_r,$$

we get

$$(G\ddot{a})_{\overline{n}|r,r} = (1+r)nv_r.$$

Since

$$(1+r)v_r = 1,$$

we have

$$(G\ddot{a})_{\overline{n}|r,r} = n.$$

Special case $r = i$: geometric annuity-due

For a geometric annuity-due, if $r = i$, then

$$\boxed{(G\ddot{a})_{\overline{n}|r,r} = n}.$$

The accumulated value at time n is

$$(G\ddot{s})_{\overline{n}|r,r} = (G\ddot{a})_{\overline{n}|r,r}(1+r)^n.$$

Therefore,

$$(G\ddot{s})_{\overline{n}|r,r} = n(1+r)^n.$$

Special case $r = i$: accumulated value due

For a geometric annuity-due, if $r = i$, then

$$\boxed{(G\ddot{s})_{\overline{n}|r,r} = n(1+r)^n}.$$

Why this special case matters

When $r = i$, the growth of the payments exactly offsets the interest discounting.
For the annuity-immediate, each discounted payment equals

$$v_r.$$

For the annuity-due, each discounted payment equals

$$1.$$

This is why the formulas become much simpler.

Summary of the special case $r = i$

$$\boxed{(Ga)_{\overline{n}|r,r} = nv_r}$$

$$\boxed{(Gs)_{\overline{n}|r,r} = n(1+r)^{n-1}}$$

$$\boxed{(G\ddot{a})_{\overline{n}|r,r} = n}$$

$$\boxed{(G\ddot{s})_{\overline{n}|r,r} = n(1+r)^n}$$

Special case $r = i$

An annuity provides for 12 annual payments. The first payment is 100, paid at the end of the first year, and each subsequent payment is 5% more than the one preceding it.
Calculate the present value of this annuity if

$$i = 0.05.$$

Solution.
The payments are

$$100, \quad 100(1.05), \quad 100(1.05)^2, \quad \dots, \quad 100(1.05)^{11}.$$

They are paid at times

$$1, 2, 3, \dots, 12.$$

This is a geometric annuity-immediate with

$$n = 12, \qquad i = 0.05, \qquad r = 0.05.$$

Since

$$r = i,$$

we use the special-case formula

$$(Ga)_{\overline{n}|r,r} = nv_r.$$

Here,

$$v_r = \frac{1}{1.05}.$$

Therefore,

$$PV = 100(Ga)_{\overline{12}|0.05,0.05}.$$

Using the special-case formula,

$$PV = 100(12)(1.05)^{-1}.$$

Thus,

$$PV = 1142.86.$$

Therefore,

$$\boxed{PV = 1142.86}.$$

Geometric Perpetuity

Recall that for a geometric annuity-immediate,

$$(Ga)_{\overline{n}|i,r} = \frac{1 - \left(\frac{1+r}{1+i}\right)^n}{i - r}.$$

To find the present value of a geometric perpetuity-immediate, we let

$$n \to \infty.$$

Thus,

$$(Ga)_{\overline{\infty}|i,r} = \lim_{n\to\infty} \frac{1 - \left(\frac{1+r}{1+i}\right)^n}{i - r}.$$

The behavior of the limit depends on the value of

$$\frac{1+r}{1+i}.$$

If

$$r < i,$$

then

$$\frac{1+r}{1+i} < 1,$$

and therefore

$$\lim_{n\to\infty} \left(\frac{1+r}{1+i}\right)^n = 0.$$

Hence,

$$(Ga)_{\overline{\infty}|i,r} = \frac{1}{i-r}.$$

If

$$r \geq i,$$

then the present value does not converge to a finite value.

Geometric perpetuity-immediate

For a geometric perpetuity-immediate with first payment 1, interest rate i, and growth rate r,

$$\boxed{(Ga)_{\overline{\infty}|i,r} = \frac{1}{i-r}}$$

provided that

$$r < i.$$

If

$$r \geq i,$$

the present value is not finite.

Geometric Perpetuity-Due

For a geometric perpetuity-due, the payments are the same as the geometric perpetuity-immediate, but each payment occurs one period earlier.

Therefore,

$$(G\ddot{a})_{\overline{\infty}|i,r} = (1+i)(Ga)_{\overline{\infty}|i,r}.$$

Using

$$(Ga)_{\overline{\infty}|i,r} = \frac{1}{i-r},$$

we get

$$(G\ddot{a})_{\overline{\infty}|i,r} = \frac{1+i}{i-r}.$$

Geometric perpetuity-due

For a geometric perpetuity-due with first payment 1, interest rate i, and growth rate r,

$$\boxed{(G\ddot{a})_{\overline{\infty}|i,r} = \frac{1+i}{i-r}}$$

provided that

$$r < i.$$

If

$$r \geq i,$$

the present value is not finite.

Convergence condition

A geometric perpetuity has a finite present value only if the payment growth rate is smaller than the interest rate:
$$r < i.$$
If the payments grow as fast as, or faster than, the interest rate, then the discounted payments do not shrink enough for the infinite sum to converge.

Coefficient interpretation

If the first payment is P, then the present value of a geometric perpetuity-immediate is
$$PV = P(Ga)_{\overline{\infty}|i,r} = \frac{P}{i-r},$$
provided that $r < i$.
For a geometric perpetuity-due, the present value is
$$PV = P(G\ddot{a})_{\overline{\infty}|i,r} = \frac{P(1+i)}{i-r},$$
provided that $r < i$.

Geometric perpetuity-due

An annuity provides payments forever. The first payment is 100, paid at the beginning of the first year, and each subsequent payment is 5% more than the one preceding it.
Calculate the present value of this annuity if
$$i = 0.08.$$

Solution.
The first payment is paid at the beginning of the first year, so this is a geometric perpetuity-due.
The payments are
$$100, \quad 100(1.05), \quad 100(1.05)^2, \quad \ldots$$
at times
$$0, 1, 2, \ldots$$
The interest rate is
$$i = 0.08,$$
and the growth rate is
$$r = 0.05.$$
Since
$$r < i,$$
the present value is finite.
For a geometric perpetuity-due, we use
$$(G\ddot{a})_{\overline{\infty}|i,r} = \frac{1+i}{i-r}.$$
Therefore,
$$PV = 100(G\ddot{a})_{\overline{\infty}|0.08,0.05}.$$

Substituting the values,

$$PV = 100\left(\frac{1.08}{0.08-0.05}\right).$$

Thus,

$$PV = 100\left(\frac{1.08}{0.03}\right).$$

Therefore,

$$PV = 3600.$$

So,

$$\boxed{PV = 3600}.$$

Exercise 2.2.6

Smith will make deposits to a fund each January 1 and July 1 from 1985 through 1995. Each July 1, the level deposit is increased by 10.25% over the previous level.
The effective annual interest rate is 10.25%. The fund balance on December 31, 1995 will be 15000.
Calculate the amount of Smith's initial deposit on January 1, 1985.
Solution.
Let

$$X$$

be Smith's initial deposit on January 1, 1985.
The effective annual interest rate is

$$i = 0.1025.$$

Since the deposit level increases each July 1 by 10.25%, the January deposits are

$$X,\ X(1.1025),\ X(1.1025)^2,\ldots,X(1.1025)^{10}.$$

The July deposits are

$$X(1.1025),\ X(1.1025)^2,\ldots,X(1.1025)^{11}.$$

There are 11 January deposits and 11 July deposits.
We value the deposits at January 1, 1985.
For the January deposits, the growth rate equals the interest rate. Therefore, this is the special case $r = i$ for a geometric annuity-due:

$$(G\ddot{a})_{\overline{11}|0.1025,0.1025} = 11.$$

Thus, the value of the January deposits at January 1, 1985 is

$$X(G\ddot{a})_{\overline{11}|0.1025,0.1025} = 11X.$$

For the July deposits, the first July deposit is

$$1.1025X,$$

and it occurs one half-year after January 1, 1985.
At July 1, 1985, the value of the July deposits is

$$1.1025X(G\ddot{a})_{\overline{11}|0.1025,0.1025}.$$

Discounting this value back one half-year gives

$$1.1025X(G\ddot{a})_{\overline{11}|0.1025,0.1025}v^{1/2}.$$

Since

$$v = \frac{1}{1.1025},$$

we have

$$v^{1/2} = (1.1025)^{-1/2}.$$

Thus, the value at January 1, 1985 of all deposits is

$$11X + 1.1025X(11)(1.1025)^{-1/2}.$$

Since

$$(1.1025)^{1/2} = 1.05,$$

this becomes

$$11X + 11X(1.05) = 22.55X.$$

Now accumulate this value from January 1, 1985 to December 31, 1995. This is 11 years, so

$$22.55X(1.1025)^{11} = 15000.$$

Therefore,

$$X = \frac{15000}{22.55(1.1025)^{11}}.$$

Hence,

$$X = 227.39.$$

So the initial deposit is

$$\boxed{227.39}.$$

Exercise 2.2.7

A perpetuity pays $2X$ one year from today. The annual payments increase by 5% per year thereafter. The effective annual interest rate on this perpetuity is 6%. The present value is 32400.
A second perpetuity pays Y one year from now, and the annual payments increase by an amount of X per year thereafter. The effective annual interest rate on this perpetuity is i, and the present value is 24000.
A third perpetuity pays Y per year, with the first payment one year from now. The effective annual interest rate on this perpetuity is i, and the present value is 4000.
What is Y?
Solution.
First consider the first perpetuity. It is a geometric perpetuity-immediate with first payment $2X$, interest rate 6%, and growth rate 5%.
Thus,

$$2X(Ga)_{\overline{\infty}|0.06,0.05} = 32400.$$

Since

$$(Ga)_{\overline{\infty}|i,r} = \frac{1}{i-r},$$

we have

$$(Ga)_{\overline{\infty}|0.06,0.05} = \frac{1}{0.06 - 0.05} = 100.$$

Therefore,

$$2X(100) = 32400.$$

So,

$$200X = 32400,$$

and

$$X = 162.$$

Now consider the third perpetuity. It is a level perpetuity-immediate with payment Y and interest rate i. Its present value is

$$Ya_{\overline{\infty}|i} = 4000.$$

Since

$$a_{\overline{\infty}|i} = \frac{1}{i},$$

we get

$$\frac{Y}{i} = 4000.$$

Thus,

$$Y = 4000i.$$

Now consider the second perpetuity. Its payments are

$$Y, \quad Y + X, \quad Y + 2X, \quad Y + 3X, \ldots$$

at times

$$1, 2, 3, 4, \ldots$$

We decompose this into two parts:

$$Y, Y, Y, Y, \ldots$$

and

$$0, X, 2X, 3X, \ldots$$

The level part has present value

$$Ya_{\overline{\infty}|i}.$$

The increasing part starts one year later than a standard increasing perpetuity-immediate, so its present value is

$$X(Ia)_{\overline{\infty}|i}v_i.$$

Therefore,

$$Ya_{\overline{\infty}|i} + X(Ia)_{\overline{\infty}|i}v_i = 24000.$$

Using the third perpetuity, we already know that

$$Ya_{\overline{\infty}|i} = 4000.$$

Hence,

$$4000 + X(Ia)_{\overline{\infty}|i}v_i = 24000.$$

So,

$$X(Ia)_{\overline{\infty}|i} v_i = 20000.$$

For an increasing perpetuity-immediate,

$$(Ia)_{\overline{\infty}|i} = \frac{1}{id},$$

where

$$d = \frac{i}{1+i}.$$

Thus,

$$(Ia)_{\overline{\infty}|i} v_i = \frac{1}{id} \cdot v_i.$$

Since

$$v_i = \frac{1}{1+i} \qquad \text{and} \qquad d = \frac{i}{1+i},$$

we have

$$\frac{v_i}{id} = \frac{1}{i^2}.$$

Therefore,

$$X(Ia)_{\overline{\infty}|i} v_i = \frac{X}{i^2}.$$

So,

$$\frac{X}{i^2} = 20000.$$

Using

$$X = 162,$$

we get

$$\frac{162}{i^2} = 20000.$$

Thus,

$$i^2 = \frac{162}{20000} = 0.0081.$$

Therefore,

$$i = 0.09.$$

Finally, since

$$Y = 4000i,$$

we obtain

$$Y = 4000(0.09) = 360.$$

Thus,

$$\boxed{Y = 360}.$$

Exercise 2.2.8

An annuity provides for 10 annual payments. The first of these payments is 100, and each subsequent payment is 10% higher than the one preceding it.
Find the present value, to the nearest 50, of this annuity at the time one year prior to the first payment if $i = 10\%$.

A. 800 B. 850

C. 900 D. 950

E. 1000

Solution.
The payments are

$$100, \quad 100(1.10), \quad 100(1.10)^2, \quad \ldots, \quad 100(1.10)^9.$$

There are

$$n = 10$$

payments.
The payment growth rate is

$$r = 0.10,$$

and the effective interest rate is

$$i = 0.10.$$

Thus,

$$r = i.$$

Since the first payment is one year after the valuation date, this is a geometric annuity-immediate.
For the special case $r = i$, we use

$$(Ga)_{\overline{n}|i,i} = nv_i.$$

Therefore,

$$PV = 100(Ga)_{\overline{10}|0.10,0.10}.$$

So,

$$PV = 100(10)v.$$

Since

$$v = \frac{1}{1.10},$$

we get

$$PV = 100(10)\left(\frac{1}{1.10}\right).$$

Thus,

$$PV = 909.09.$$

To the nearest 50, this is

$$900.$$

Therefore, the correct answer is

$$\boxed{\text{C. } 900}.$$

Detailed verification.
Writing out the discounted cash flows gives

$$PV = 100v + 100(1.10)v^2 + 100(1.10)^2v^3 + \cdots + 100(1.10)^9v^{10}.$$

Since

$$v = (1.10)^{-1},$$

each term becomes

$$100(1.10)^{-1}.$$

There are 10 terms, so

$$PV = 10 \cdot 100(1.10)^{-1} = 909.09.$$

Exercise 2.2.9

Mary purchases an increasing annuity-immediate for 50,000 that makes twenty annual payments as follows:

1. $P, 2P, \ldots, 10P$ in years 1 through 10;
2. $10P(1.05), 10P(1.05)^2, \ldots, 10P(1.05)^{10}$ in years 11 through 20.

The annual effective interest rate is 7% for the first 10 years and 5% thereafter. Calculate P.

A. 564 B. 574

C. 584 D. 594

E. 604

Solution.
We split the annuity into two parts.
For years 1 through 10, the payments are

$$P, 2P, \ldots, 10P.$$

This is an arithmetic increasing annuity-immediate. Since the interest rate for the first 10 years is 7%, the present value of this part is

$$P(Ia)_{\overline{10}|0.07}.$$

For years 11 through 20, the payments are

$$10P(1.05), 10P(1.05)^2, \ldots, 10P(1.05)^{10}.$$

At time 10, this is a geometric annuity-immediate with first payment

$$10P(1.05),$$

growth rate

$$r = 0.05,$$

and interest rate

$$i = 0.05.$$

Since

$$r = i,$$

we use the special-case formula

$$(Ga)_{\overline{n}|i,i} = nv_i.$$

Thus, the value at time 10 of the second part is

$$10P(1.05)(Ga)_{\overline{10}|0.05,0.05}.$$

Using the special-case formula,

$$10P(1.05)(Ga)_{\overline{10}|0.05,0.05} = 10P(1.05)(10v_{0.05}).$$

Since

$$v_{0.05} = \frac{1}{1.05},$$

we get

$$10P(1.05)(10v_{0.05}) = 100P.$$

This value is at time 10. Discounting it back to time 0 at 7%, we get

$$100Pv_{0.07}^{10}.$$

Therefore, the equation of value at time 0 is

$$50{,}000 = P(Ia)_{\overline{10}|0.07} + 100Pv_{0.07}^{10}.$$

Factor out P:

$$50{,}000 = P\left[(Ia)_{\overline{10}|0.07} + 100v_{0.07}^{10}\right].$$

Now,

$$(Ia)_{\overline{10}|0.07} = 34.73913325,$$

and

$$100v_{0.07}^{10} = 50.83492921.$$

Thus,

$$50{,}000 = P(34.73913325 + 50.83492921).$$

So,

$$50{,}000 = 85.57406246P.$$

Therefore,

$$P = 584.29.$$

The closest answer is

$$\boxed{\text{C. } 584}.$$

Exercise 2.2.10

[SOA 11; SOA-IT.014]
Mike buys a perpetuity-immediate with varying annual payments. During the first 5 years, the payment is constant and equal to 10. Beginning in year 6, the payments start to increase. For year 6 and all future years, the payment in that year is $k\%$ larger than the payment in the year immediately preceding that year, where $k < 9.2$.
At an annual effective interest rate of 9.2%, the perpetuity has a present value of 167.50.
Calculate k.

A. 4.0 B. 4.2

C. 4.4 D. 4.6

E. 4.8

Solution.
Let j be the decimal representation of $k\%$. Thus,

$$j = \frac{k}{100}.$$

The annual effective interest rate is

$$i = 0.092.$$

The first five payments are level:

$$10,\ 10,\ 10,\ 10,\ 10.$$

Their present value is

$$10a_{\overline{5}|0.092}.$$

Beginning in year 6, the payments form a geometric perpetuity-immediate. The first payment of this geometric perpetuity is

$$10(1+j),$$

and it occurs at time 6. Therefore, its value at time 5 is

$$\frac{10(1+j)}{0.092-j}.$$

Discounting this value back to time 0, we multiply by

$$v_{0.092}^5.$$

Thus, the equation of value is

$$167.50 = 10a_{\overline{5}|0.092} + 10(1+j)\left(\frac{1}{0.092-j}\right)v_{0.092}^5.$$

Using

$$v_{0.092} = \frac{1}{1.092},$$

this equation simplifies to

$$20 = \frac{1+j}{0.092-j}.$$

Now solve for j:

$$20(0.092-j) = 1+j.$$

So,

$$1.84 - 20j = 1+j.$$

Therefore,

$$0.84 = 21j.$$

Thus,

$$j = 0.040.$$

Since j is the decimal representation of $k\%$,

$$k = 4.0.$$

The correct answer is

$$\boxed{\text{A. } 4.0}.$$

2.2.2 Continuous Annuities

Part 1

What is a Continuous Annuity?

A continuous annuity is an annuity with payments made continuously over time.

It is useful as a theoretical tool and also as an approximation for annuities payable with very high frequency, such as daily payments.

For m-thly annuities, the total amount paid during one year is 1, but the payments are split into m smaller payments.

Annuity	Payments
$a_{\overline{n}\vert}$	1 every year
$a_{\overline{n}\vert}^{(2)}$	$\frac{1}{2}$ every 6 months
$a_{\overline{n}\vert}^{(4)}$	$\frac{1}{4}$ every 3 months
$a_{\overline{n}\vert}^{(12)}$	$\frac{1}{12}$ every month
$a_{\overline{n}\vert}^{(365)}$	$\frac{1}{365}$ every day

As the payment frequency becomes extremely large, we obtain a continuous annuity.

$$a_{\overline{n}|}^{(\infty)} \equiv \bar{a}_{\overline{n}|}.$$

Continuous annuity

A continuous annuity pays continuously over time.
Instead of being given a payment amount per period, we are given a **rate of payment** per year.
If the payment rate at time t is $f(t)$, then the amount paid between time t and time $t+dt$ is

$$f(t)\,dt.$$

For the standard continuous annuity $\bar{a}_{\overline{n}|}$, the rate of payment is 1 per year.

Therefore, the amount paid during year 1 is

$$\int_0^1 1\,dt = 1.$$

The amount paid during year 2 is

$$\int_1^2 1\,dt = 1.$$

More generally, for any year k,

$$\int_{k-1}^{k} 1\,dt = 1.$$

Thus, $\bar{a}_{\overline{n}|}$ has a total payment of 1 per year, just like

$$a_{\overline{n}|}, \qquad a_{\overline{n}|}^{(2)}, \qquad a_{\overline{n}|}^{(4)}, \qquad a_{\overline{n}|}^{(12)}, \qquad a_{\overline{n}|}^{(365)}.$$

Key idea

For discrete annuities, we usually think in terms of payment amounts.
For continuous annuities, we think in terms of a payment rate.
If the rate is 1 per year, then the annuity pays a total of 1 over each year:

$$\int_t^{t+1} 1\,ds = 1.$$

Present Value of a Continuous Annuity

Recall the formulas for discrete annuities:

$$a_{\overline{n}|} = \frac{1-v^n}{i}, \qquad \ddot{a}_{\overline{n}|} = \frac{1-v^n}{d}, \qquad a^{(m)}_{\overline{n}|} = \frac{1-v^n}{i^{(m)}}.$$

For a continuous annuity, the limiting denominator is the force of interest δ. Thus, we expect

$$\bar{a}_{\overline{n}|} = \frac{1-v^n}{\delta}.$$

Present value of a continuous annuity

The symbol

$$\bar{a}_{\overline{n}|}$$

represents the present value of n years of payments made continuously at a rate of 1 per year.

The payment rate is 1. Therefore, the amount paid between time t and time $t + dt$ is

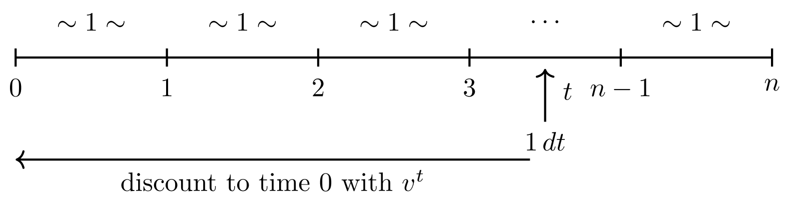
$$1\,dt.$$

The present value of this infinitesimal payment is

$$v^t\,dt.$$

Since payments occur continuously from time 0 to time n, the present value is

$$\bar{a}_{\overline{n}|} = \int_0^n v^t\,dt.$$



Now compute the integral:

$$\bar{a}_{\overline{n}|} = \int_0^n v^t\,dt.$$

Since

$$v = \frac{1}{1+i},$$

we have

$$v^t = e^{t\ln v}.$$

Therefore,

$$\int_0^n v^t\,dt = \left[\frac{v^t}{\ln v}\right]_0^n.$$

Thus, simplifying the geometric-series expression and using $\delta = \ln(1+i)$ yields the standard closed form for $\bar{a}_{\overline{n}|}$ given in the formula box below.

Present value of a continuous annuity

The present value of a continuous annuity paying at rate 1 per year for n years is

$$\boxed{\bar{a}_{\overline{n}|} = \frac{1 - v^n}{\delta}}$$

where

$$v = \frac{1}{1+i} \qquad \text{and} \qquad \delta = \ln(1+i).$$

Continuous payment interpretation

The annuity does not pay 1 at each moment.
Instead, it pays at a rate of 1 per year. Over a small time interval from t to $t + dt$, the payment is

$$1\,dt.$$

That small payment is discounted by v^t, so its present value is

$$v^t\,dt.$$

Alternative Derivation

An alternate way to derive the continuous annuity formula is to take the limit as

$$m \to \infty$$

of the m-thly annuity factor.

Recall that

$$a^{(m)}_{\overline{n}|} = \frac{1 - v^n}{i^{(m)}}.$$

Therefore,

$$\bar{a}_{\overline{n}|} = \lim_{m\to\infty} a^{(m)}_{\overline{n}|}.$$

Thus,

$$\bar{a}_{\overline{n}|} = \lim_{m\to\infty} \frac{1 - v^n}{i^{(m)}}.$$

Since

$$\lim_{m\to\infty} i^{(m)} = \delta,$$

we obtain

$$\bar{a}_{\overline{n}|} = \frac{1 - v^n}{\delta}.$$

Continuous annuity as a limit

The continuous annuity factor can be viewed as the limit of the m-thly annuity factor:

$$\boxed{\bar{a}_{\overline{n}|} = \lim_{m\to\infty} a^{(m)}_{\overline{n}|} = \frac{1 - v^n}{\delta}}$$

where

$$\delta = \ln(1+i).$$

Continuous annuity present value

The annual effective rate of interest is 5%.
Find the present value of an annuity of 25 per year payable continuously for 15 years.
Solution.
The annuity pays continuously at a rate of

$$25$$

per year for 15 years.
The annual effective interest rate is

$$i = 0.05.$$

Therefore,

$$\delta = \ln(1.05).$$

The present value is

$$PV = 25\bar{a}_{\overline{15}|0.05}.$$

Using

$$\bar{a}_{\overline{n}|} = \frac{1 - v^n}{\delta},$$

we get

$$PV = 25\left(\frac{1-(1.05)^{-15}}{\ln(1.05)}\right).$$

Thus,

$$PV = 265.93.$$

Therefore,

$$\boxed{PV = 265.93}.$$

Accumulated Value of a Continuous Annuity

We can derive the accumulated value of a continuous annuity directly by integration.

For a continuous annuity paying at rate 1 per year from time 0 to time n, the amount paid between time t and time $t + dt$ is

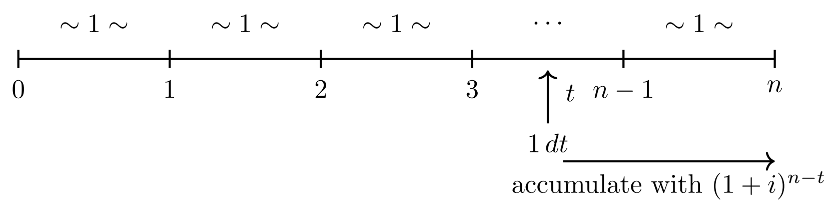
$$1\,dt.$$

To accumulate this payment to time n, we multiply by

$$(1+i)^{n-t}.$$

Thus, the accumulated value is

$$\bar{s}_{\overline{n}|} = \int_0^n (1+i)^{n-t}\,dt.$$



Now compute:

$$\bar{s}_{\overline{n}|} = \int_0^n (1+i)^{n-t}\,dt.$$

Factor out $(1+i)^n$:

$$\bar{s}_{\overline{n}|} = (1+i)^n \int_0^n (1+i)^{-t}\,dt.$$

But

$$\int_0^n (1+i)^{-t}\,dt = \bar{a}_{\overline{n}|}.$$

Therefore,

$$\bar{s}_{\overline{n}|} = (1+i)^n \bar{a}_{\overline{n}|}.$$

Using

$$\bar{a}_{\overline{n}|} = \frac{1-v^n}{\delta},$$

we get

$$\bar{s}_{\overline{n}|} = (1+i)^n \left(\frac{1-v^n}{\delta}\right).$$

Since

$$(1+i)^n v^n = 1,$$

we obtain

$$\bar{s}_{\overline{n}|} = \frac{(1+i)^n - 1}{\delta}.$$

Equivalently, since

$$1+i = e^{\delta},$$

we have

$$\bar{s}_{\overline{n}|} = \frac{e^{\delta n} - 1}{\delta}.$$

Accumulated value of a continuous annuity

The accumulated value at time n of a continuous annuity paying at rate 1 per year is

$$\boxed{\bar{s}_{\overline{n}|} = \frac{(1+i)^n - 1}{\delta}}$$

or equivalently,

$$\boxed{\bar{s}_{\overline{n}|} = \frac{e^{\delta n} - 1}{\delta}}.$$

Comparison with discrete annuities

The continuous accumulated value formula fits the same pattern as the ordinary accumulated value formulas:

$$s_{\overline{n}|} = \frac{(1+i)^n - 1}{i}, \qquad \ddot{s}_{\overline{n}|} = \frac{(1+i)^n - 1}{d}, \qquad \bar{s}_{\overline{n}|} = \frac{(1+i)^n - 1}{\delta}.$$

Continuous annuity accumulated value

A ten-year annuity pays 50 per year payable continuously.
Given

$$\delta = 0.04,$$

find the accumulated value at the end of the ten years.
Solution.
The annuity pays continuously at a rate of

$$50$$

per year for 10 years.
Since

$$\delta = 0.04,$$

we use

$$\bar{s}_{\overline{n}|} = \frac{e^{\delta n} - 1}{\delta}.$$

Thus,

$$AV = 50\bar{s}_{\overline{10}|}.$$

So,

$$AV = 50\left(\frac{e^{10(0.04)} - 1}{0.04}\right).$$

Therefore,

$$AV = 614.78.$$

Hence,

$$\boxed{AV = 614.78}.$$

Relationship Between $\bar{a}_{\overline{n}|}$ and $a_{\overline{n}|}$

The quickest way to derive the relationship is to compare the formulas:

$$\bar{a}_{\overline{n}|} = \frac{1 - v^n}{\delta}.$$

Since

$$a_{\overline{n}|} = \frac{1 - v^n}{i},$$

we can write

$$\bar{a}_{\overline{n}|} = \frac{1 - v^n}{\delta} = \frac{i}{\delta} \cdot \frac{1 - v^n}{i}.$$

Therefore,

$$\bar{a}_{\overline{n}|} = \frac{i}{\delta} a_{\overline{n}|}.$$

Relationship between continuous and annual annuities

$$\boxed{\bar{a}_{\overline{n}|} = \frac{i}{\delta} a_{\overline{n}|}}$$

where

$$\delta = \ln(1 + i).$$

Interpretation.

The factor

$$\frac{i}{\delta}$$

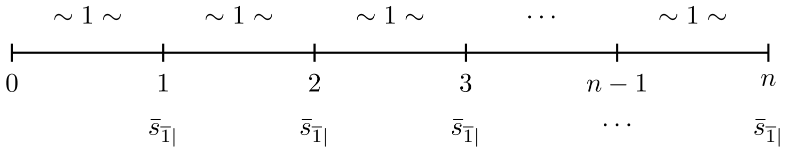
represents the accumulated value at the end of one year of payments made continuously at a rate of 1 per year.

Indeed,

$$\bar{s}_{\overline{1}|} = \frac{(1+i)^1 - 1}{\delta} = \frac{i}{\delta}.$$

So instead of valuing continuous payments directly, we can think of each year of continuous payments as being fused into one end-of-year payment of

$$\bar{s}_{\overline{1}|} = \frac{i}{\delta}.$$



Thus,

$$\bar{a}_{\overline{n}|} = \bar{s}_{\overline{1}|} a_{\overline{n}|}.$$

Since

$$\bar{s}_{\overline{1}|} = \frac{i}{\delta},$$

we get again

$$\bar{a}_{\overline{n}|} = \frac{i}{\delta} a_{\overline{n}|}.$$

Fusing continuous payments

The factor

$$\frac{i}{\delta}$$

plays the same role as the factor

$$\frac{i}{i^{(m)}}$$

in the m-thly annuity formula

$$a^{(m)}_{\overline{n}|} = \frac{i}{i^{(m)}} a_{\overline{n}|}.$$

Both factors represent the idea of fusing smaller payments within each year into one equivalent end-of-year payment.

Continuous annuity using $\bar{a}_{\overline{n}|} = \frac{i}{\delta} a_{\overline{n}|}$

The annual effective rate of interest is 5%.
Find the present value of an annuity of 25 per year payable continuously for 15 years.
Solution.
We are given

$$i = 0.05.$$

The force of interest is

$$\delta = \ln(1+i) = \ln(1.05).$$

Instead of using
$$\bar{a}_{\overline{15}|} = \frac{1-v^{15}}{\delta},$$
we use the relationship
$$\bar{a}_{\overline{n}|} = \frac{i}{\delta} a_{\overline{n}|}.$$
Therefore,
$$25\bar{a}_{\overline{15}|0.05} = 25\left(\frac{0.05}{\ln(1.05)}\right) a_{\overline{15}|0.05}.$$
Using
$$a_{\overline{15}|0.05} = \frac{1-(1.05)^{-15}}{0.05},$$
we get
$$25\bar{a}_{\overline{15}|0.05} = 25\left(\frac{0.05}{\ln(1.05)}\right)\left(\frac{1-(1.05)^{-15}}{0.05}\right).$$
Thus,
$$PV = 265.93.$$
Therefore,
$$\boxed{PV = 265.93}.$$

Using the BA II Plus.
Since
$$\bar{a}_{\overline{15}|0.05} = \frac{0.05}{\ln(1.05)} a_{\overline{15}|0.05},$$
we can treat the continuous annuity as an ordinary annuity-immediate with an adjusted annual payment.
The adjusted payment is
$$25\left(\frac{0.05}{\ln(1.05)}\right).$$
So on the calculator, enter
$$\begin{aligned} N &= 15, \\ I/Y &= 5, \\ PMT &= 25\left(\frac{0.05}{\ln(1.05)}\right), \\ FV &= 0, \\ CPT\ PV &= -265.9259867. \end{aligned}$$
Therefore,
$$PV = 265.93.$$

When to use each formula

If you are given i, it is often convenient to use
$$\bar{a}_{\overline{n}|} = \frac{i}{\delta} a_{\overline{n}|}, \qquad \delta = \ln(1+i).$$
If you are given δ, it is usually faster to use
$$\bar{a}_{\overline{n}|} = \frac{1-v^n}{\delta}.$$

Continuous Perpetuity

The symbol

$$\bar{a}_{\overline{\infty}|}$$

represents the present value of a continuous perpetuity with a rate of payment of 1 per year.

Recall that

$$\bar{a}_{\overline{n}|} = \frac{1 - v^n}{\delta}.$$

To obtain the continuous perpetuity formula, let

$$n \to \infty.$$

Thus,

$$\bar{a}_{\overline{\infty}|} = \lim_{n\to\infty} \bar{a}_{\overline{n}|}.$$

So,

$$\bar{a}_{\overline{\infty}|} = \lim_{n\to\infty} \frac{1 - v^n}{\delta}.$$

Since

$$0 < v < 1,$$

we have

$$\lim_{n\to\infty} v^n = 0.$$

Therefore,

$$\bar{a}_{\overline{\infty}|} = \frac{1 - 0}{\delta}.$$

Hence,

$$\bar{a}_{\overline{\infty}|} = \frac{1}{\delta}.$$

Present value of a continuous perpetuity

The present value of a continuous perpetuity paying at rate 1 per year is

$$\boxed{\bar{a}_{\overline{\infty}|} = \frac{1}{\delta}}$$

where

$$\delta = \ln(1 + i).$$

Comparison with ordinary perpetuities

The continuous perpetuity formula fits the same pattern as the ordinary perpetuity formulas:

$$a_{\overline{\infty}|} = \frac{1}{i}, \qquad \ddot{a}_{\overline{\infty}|} = \frac{1}{d}, \qquad \bar{a}_{\overline{\infty}|} = \frac{1}{\delta}.$$

So the denominator tells us how the payments are made:

$$i \quad \text{for annuity-immediate,}$$

$$d \quad \text{for annuity-due,}$$

$$\delta \quad \text{for continuous annuity.}$$

Payment rate interpretation

If the continuous perpetuity pays at rate R per year, then its present value is

$$PV = R\bar{a}_{\overline{\infty}|} = \frac{R}{\delta}.$$

For example, if the payment rate is 25 per year, then

$$PV = \frac{25}{\delta}.$$

Exercise 2.2.11

Do these on your calculator without pen or paper.

Given	Calculate	Answer
$i = 0.04$	$\bar{a}_{\overline{10}|}$	8.272
$\delta = 0.06$	$\bar{a}_{\overline{20}|}$	11.647
$i = 0.10$	$\bar{s}_{\overline{15}|}$	33.336
$\delta = 0.08$	$\bar{s}_{\overline{25}|}$	79.863

Solution.
For continuous annuities, use

$$\bar{a}_{\overline{n}|} = \frac{1 - v^n}{\delta}, \qquad \bar{s}_{\overline{n}|} = \frac{(1+i)^n - 1}{\delta}.$$

If i is given, then

$$\delta = \ln(1+i).$$

If δ is given, then

$$1 + i = e^{\delta}.$$

1. Given $i = 0.04$, calculate $\bar{a}_{\overline{10}|}$.
First,

$$\delta = \ln(1.04).$$

Then

$$\bar{a}_{\overline{10}|} = \frac{1 - (1.04)^{-10}}{\ln(1.04)}.$$

Thus,

$$\bar{a}_{\overline{10}|} = 8.272.$$

2. Given $\delta = 0.06$, calculate $\bar{a}_{\overline{20}|}$.
Since

$$v^n = e^{-\delta n},$$

we have

$$\bar{a}_{\overline{20}|} = \frac{1 - e^{-0.06(20)}}{0.06}.$$

Thus,

$$\bar{a}_{\overline{20}|} = 11.647.$$

3. Given $i = 0.10$, calculate $\bar{s}_{\overline{15}|}$.
First,

$$\delta = \ln(1.10).$$

Then

$$\bar{s}_{\overline{15}|} = \frac{(1.10)^{15} - 1}{\ln(1.10)}.$$

Thus,

$$\bar{s}_{\overline{15}|} = 33.336.$$

4. Given $\delta = 0.08$, calculate $\bar{s}_{\overline{25}|}$.
Since

$$1 + i = e^{\delta},$$

we have

$$(1+i)^n = e^{\delta n}.$$

Therefore,

$$\bar{s}_{\overline{25}|} = \frac{e^{0.08(25)} - 1}{0.08}.$$

Thus,

$$\bar{s}_{\overline{25}|} = 79.863.$$

Exercise 2.2.12

A continuous perpetuity has a rate of payment of 10 per year for the first 5 years and 20 per year thereafter.
If the annual effective rate of interest is 8%, find the present value of this perpetuity.
Solution.
The annual effective interest rate is

$$i = 0.08.$$

Therefore, the force of interest is

$$\delta = \ln(1.08).$$

For the first 5 years, the perpetuity pays continuously at a rate of 10 per year. The present value of this part is

$$10\bar{a}_{\overline{5}|}.$$

After time 5, the perpetuity pays continuously at a rate of 20 per year forever. The value at time 5 of this continuous perpetuity is

$$20\bar{a}_{\overline{\infty}|}.$$

Discounting this value back to time 0, we get

$$20\bar{a}_{\overline{\infty}|}v^5.$$

Thus,

$$PV = 10\bar{a}_{\overline{5}|} + 20\bar{a}_{\overline{\infty}|}v^5.$$

Using

$$\bar{a}_{\overline{5}|} = \frac{1 - v^5}{\delta}, \qquad \bar{a}_{\overline{\infty}|} = \frac{1}{\delta},$$

we have

$$PV = 10\left(\frac{1 - v^5}{\delta}\right) + 20\left(\frac{1}{\delta}\right)v^5.$$

Since
$$v = \frac{1}{1.08} \qquad \text{and} \qquad \delta = \ln(1.08),$$
we get
$$PV = 10\left(\frac{1-1.08^{-5}}{\ln(1.08)}\right) + 20\left(\frac{1}{\ln(1.08)}\right)1.08^{-5}.$$
Therefore,
$$PV = 218.37.$$
So the present value is
$$\boxed{218.37}.$$
Alternative method.
We can also view the perpetuity as a continuous perpetuity of 20 per year forever, minus a continuous temporary annuity of 10 per year for the first 5 years.
Thus,
$$PV = 20\bar{a}_{\overline{\infty}|} - 10\bar{a}_{\overline{5}|}.$$
So,
$$PV = 20\left(\frac{1}{\ln(1.08)}\right) - 10\left(\frac{1-1.08^{-5}}{\ln(1.08)}\right).$$
Therefore,
$$PV = 218.37.$$

Exercise 2.2.13

The present value of a continuous annuity of 1 per year for a certain number of years is 4.
The continuously compounded rate of interest is 12.5%.
What is the accumulated value of a continuous annuity of 1 per year for twice that number of years?
Solution.
Let the unknown number of years be n.
Since the annuity is payable continuously at a rate of 1 per year, its present value is
$$\bar{a}_{\overline{n}|}.$$
We are given
$$\bar{a}_{\overline{n}|} = 4.$$
The continuously compounded rate of interest is
$$\delta = 0.125.$$
Using
$$\bar{a}_{\overline{n}|} = \frac{1-v^n}{\delta},$$
we get
$$\frac{1-v^n}{0.125} = 4.$$
Thus,
$$1 - v^n = 0.5.$$
Therefore,
$$v^n = 0.5.$$

Since
$$v^n = (1+i)^{-n},$$
we have
$$(1+i)^n = 0.5^{-1} = 2.$$
We need the accumulated value of a continuous annuity of 1 per year for $2n$ years. This is
$$\bar{s}_{\overline{2n}|} = \frac{(1+i)^{2n} - 1}{\delta}.$$
But
$$(1+i)^n = 2,$$
so
$$(1+i)^{2n} = 2^2 = 4.$$
Therefore,
$$\bar{s}_{\overline{2n}|} = \frac{4-1}{0.125}.$$
Thus,
$$\bar{s}_{\overline{2n}|} = \frac{3}{0.125} = 24.$$
Hence, the accumulated value is
$$\boxed{24}.$$

Exercise 2.2.14

You are given an effective annual rate of interest $i = 5\%$. Which of the following values are greater than or equal to 20?

1. The present value of an annuity of 1 per year payable continuously for 50 years.
2. The accumulated value in 5 years of annual payments starting at 5 and decreasing by 1 per year. Payments start one year from now.
3. The present value of a perpetuity paying 1 every 5 years and increasing by 1 each payment. Payments start five years from now.

A. (1) B. (2)
C. (3) D. (1) and (2)
E. None of (1), (2), or (3)

Solution.
We are given
$$i = 0.05, \qquad v = \frac{1}{1.05}, \qquad \delta = \ln(1.05).$$
Value 1.
The present value of an annuity of 1 per year payable continuously for 50 years is
$$\bar{a}_{\overline{50}|} = \frac{1 - v^{50}}{\delta}.$$

Thus,

$$\bar{a}_{\overline{50}|} = \frac{1-(1.05)^{-50}}{\ln(1.05)}.$$

Therefore,

$$\bar{a}_{\overline{50}|} = 18.71.$$

Since

$$18.71 < 20,$$

value 1 is not greater than or equal to 20.

Value 2.

The payments are

$$5,\ 4,\ 3,\ 2,\ 1$$

at times

$$1, 2, 3, 4, 5.$$

The accumulated value at time 5 is a decreasing annuity-immediate accumulated value:

$$(Ds)_{\overline{5}|0.05}.$$

Using

$$(Ds)_{\overline{n}|i} = (Da)_{\overline{n}|i}(1+i)^n = \frac{n - a_{\overline{n}|i}}{i}(1+i)^n,$$

we get

$$(Ds)_{\overline{5}|0.05} = \frac{5 - a_{\overline{5}|0.05}}{0.05}(1.05)^5.$$

Thus,

$$(Ds)_{\overline{5}|0.05} = 17.12.$$

Since

$$17.12 < 20,$$

value 2 is not greater than or equal to 20.

Value 3.

The payments are

$$1,\ 2,\ 3, \ldots$$

at times

$$5, 10, 15, \ldots$$

Let j be the effective interest rate per 5 years. Then

$$j = (1.05)^5 - 1.$$

So,

$$j = 0.27628.$$

This is an increasing perpetuity-immediate measured in 5-year periods. Therefore,

$$PV = (Ia)_{\overline{\infty}|j}.$$

For an increasing perpetuity-immediate,

$$(Ia)_{\overline{\infty}|j} = \frac{1}{jd},$$

where

$$d = \frac{j}{1+j}.$$

Thus,

$$(Ia)_{\overline{\infty}|j} = \frac{1+j}{j^2}.$$

Substituting

$$j = 0.27628,$$

we get

$$PV = 16.72.$$

Since

$$16.72 < 20,$$

value 3 is not greater than or equal to 20.
Therefore, none of the three values are greater than or equal to 20.
The correct answer is

$$\boxed{\text{E. None of } (1), (2), \text{ or } (3)}.$$

Exercise 2.2.15

An annuity has the following series of payments:

1. 1000 per year for the first 4 years payable continuously.
2. 1000 starting at the end of the 5th year, increasing by X per year at the end of years 6 through 11.
3. 3000 per year at the beginning of years 13 through 15.

The nominal rate of interest is 10% compounded twice per year. The accumulated value of the payments at the end of the 15th year is 50000.
Determine X.

A. Less than 295
B. At least 295, but less than 305
C. At least 305, but less than 315
D. At least 315, but less than 325
E. At least 325

Solution.
The nominal rate is 10% compounded twice per year, so the annual effective rate is

$$i = \left(1 + \frac{0.10}{2}\right)^2 - 1.$$

Thus,

$$i = 0.1025.$$

Let

$$v = \frac{1}{1.1025}.$$

We write an equation of value at time 0. Since the accumulated value at time 15 is 50000, its present value is

$$50000v^{15}.$$

First payment stream.
The first payment stream is 1000 per year payable continuously for the first 4 years. Its present value is

$$1000\bar{a}_{\overline{4}|0.1025}.$$

Second payment stream.
The second stream pays 1000 at the end of year 5, then

$$1000 + X,\ 1000 + 2X, \ldots, 1000 + 6X$$

at the ends of years $6, 7, \ldots, 11$.
The level part is 1000 paid at the ends of years 5 through 11. This is a 7-payment annuity-immediate deferred 4 years:

$$1000a_{\overline{7}|0.1025}v^4.$$

The increasing part is

$$X, 2X, \ldots, 6X$$

paid at the ends of years 6 through 11. This is an increasing annuity-immediate deferred 5 years:

$$X(Ia)_{\overline{6}|0.1025}v^5.$$

Third payment stream.
The third stream pays 3000 at the beginning of years 13 through 15, which means payments at times

$$12, 13, 14.$$

Equivalently, this is a 3-payment annuity-immediate starting one year after time 11, so its present value is

$$3000a_{\overline{3}|0.1025}v^{11}.$$

Therefore, the equation of value is

$$1000\bar{a}_{\overline{4}|0.1025} + 1000a_{\overline{7}|0.1025}v^4 + X(Ia)_{\overline{6}|0.1025}v^5 + 3000a_{\overline{3}|0.1025}v^{11} = 50000v^{15}.$$

Substituting the annuity values gives

$$3311.74 + 4387.60 + 8.53894X + 1419.80 = 11568.87.$$

Thus,

$$8.53894X = 2449.73.$$

Therefore,

$$X = 286.89.$$

So,

$$\boxed{X \approx 286.89}.$$

The correct answer is

$$\boxed{\text{A. Less than 295}}.$$

Part 2

Generalized Continuous Annuity

In Part 1, we showed that for a level continuous annuity with a constant interest rate,

$$\bar{a}_{\overline{n}|} = \int_0^n v^t \, dt.$$

This formula assumes two things:

1. the payment rate is constant;
2. the force of interest is constant.

We now generalize the formula by allowing both the payment rate and the force of interest to vary with time.

Consider an n-year continuous annuity where:

$$f(t)$$

is the rate of payment at time t, and

$$\delta_t$$

is the force of interest at time t.

The amount paid between time t and time $t + dt$ is

$$f(t) \, dt.$$

Accumulation function

Let

$$a(t)$$

denote the accumulation function from time 0 to time t. In other words, $a(t)$ is the accumulated value at time t of 1 invested at time 0.
If the force of interest is δ_s, then

$$a(t) = \exp\left(\int_0^t \delta_s \, ds\right).$$

Present value.

The small payment made at time t is

$$f(t) \, dt.$$

To discount this payment back to time 0, we divide by the accumulation function $a(t)$. Therefore, the present value of this small payment is

$$\frac{f(t)}{a(t)} \, dt.$$

Adding all payments from time 0 to time n, we get

$$PV = \int_0^n \frac{f(t)}{a(t)} \, dt.$$

Since

$$a(t) = \exp\left(\int_0^t \delta_s \, ds\right),$$

we have

$$\frac{1}{a(t)} = \exp\left(-\int_0^t \delta_s\,ds\right).$$

Thus,

$$PV = \int_0^n f(t)\exp\left(-\int_0^t \delta_s\,ds\right)dt.$$

Present value of a generalized continuous annuity

For a continuous annuity with payment rate $f(t)$ and force of interest δ_t,

$$\boxed{PV = \int_0^n f(t)\exp\left(-\int_0^t \delta_s\,ds\right)dt}$$

Accumulated value.

Now suppose we want the accumulated value at time n.

A payment made at time t must be accumulated from time t to time n.

Let

$$a(t,n)$$

denote the accumulation factor from time t to time n. Then

$$a(t,n) = \exp\left(\int_t^n \delta_s\,ds\right).$$

Therefore, the accumulated value of the small payment $f(t)\,dt$ is

$$f(t)a(t,n)\,dt.$$

Adding all payments from time 0 to time n, we get

$$AV = \int_0^n f(t)a(t,n)\,dt.$$

Using the force of interest, this becomes

$$AV = \int_0^n f(t)\exp\left(\int_t^n \delta_s\,ds\right)dt.$$

Accumulated value of a generalized continuous annuity

For a continuous annuity with payment rate $f(t)$ and force of interest δ_t,

$$\boxed{AV = \int_0^n f(t)\exp\left(\int_t^n \delta_s\,ds\right)dt}$$

How to read the present value formula

The formula
$$PV = \int_0^n \frac{f(t)}{a(t)}\,dt$$
means:
$$\text{present value} = \int \frac{\text{small payment at time } t}{\text{accumulation from 0 to } t}.$$
The small payment is
$$f(t)\,dt.$$
The discount factor is
$$\frac{1}{a(t)}.$$
So the present value of the small payment is
$$\frac{f(t)}{a(t)}\,dt.$$

Connection with the simpler formula

If the payment rate is constant,
$$f(t) = 1,$$
and the force of interest is constant,
$$\delta_t = \delta,$$
then
$$\int_0^t \delta_s\,ds = \delta t.$$
Therefore,
$$PV = \int_0^n e^{-\delta t}\,dt.$$
Since
$$e^{-\delta t} = v^t,$$
we recover the formula from Part 1:
$$\bar{a}_{\overline{n}|} = \int_0^n v^t\,dt.$$

Continuously Increasing Annuity

Consider a continuously increasing annuity for n years with continuous payments.

The rate of payment at time t is
$$f(t) = t.$$

Assume that the force of interest is constant and equal to
$$\delta.$$

Then the amount paid between time t and time $t + dt$ is
$$t\,dt.$$

The present value is

$$PV = \int_0^n te^{-\delta t}\,dt.$$

We denote this present value by

$$(\bar{I}\bar{a})_{\overline{n}|}.$$

To compute the integral, use integration by parts.

Let

$$u = t, \qquad dv = e^{-\delta t}\,dt.$$

Then

$$du = dt, \qquad v = -\frac{e^{-\delta t}}{\delta}.$$

Therefore,

$$\int_0^n te^{-\delta t}\,dt = \left[-\frac{te^{-\delta t}}{\delta}\right]_0^n + \frac{1}{\delta}\int_0^n e^{-\delta t}\,dt.$$

So,

$$\int_0^n te^{-\delta t}\,dt = -\frac{ne^{-\delta n}}{\delta} + \frac{1}{\delta}\int_0^n e^{-\delta t}\,dt.$$

But

$$e^{-\delta n} = v^n,$$

and

$$\int_0^n e^{-\delta t}\,dt = \bar{a}_{\overline{n}|}.$$

Thus,

$$(\bar{I}\bar{a})_{\overline{n}|} = -\frac{nv^n}{\delta} + \frac{\bar{a}_{\overline{n}|}}{\delta}.$$

Hence,

$$(\bar{I}\bar{a})_{\overline{n}|} = \frac{\bar{a}_{\overline{n}|} - nv^n}{\delta}.$$

Present value of a continuously increasing annuity

For a continuously increasing annuity with payment rate

$$f(t) = t$$

from time 0 to time n, the present value is

$$\boxed{(\bar{I}\bar{a})_{\overline{n}|} = \frac{\bar{a}_{\overline{n}|} - nv^n}{\delta}}$$

where

$$\bar{a}_{\overline{n}|} = \frac{1 - v^n}{\delta}.$$

The accumulated value at time n is obtained by accumulating the present value:

$$(\bar{I}\bar{s})_{\overline{n}|} = (\bar{I}\bar{a})_{\overline{n}|}(1+i)^n.$$

Using

$$(\bar{I}\bar{a})_{\overline{n}|} = \frac{\bar{a}_{\overline{n}|} - nv^n}{\delta},$$

we get

$$(\bar{I}\bar{s})_{\overline{n}|} = \frac{\bar{s}_{\overline{n}|} - n}{\delta}.$$

Accumulated value of a continuously increasing annuity

For a continuously increasing annuity with payment rate

$$f(t) = t$$

from time 0 to time n, the accumulated value at time n is

$$\boxed{(\bar{I}\bar{s})_{\overline{n}|} = \frac{\bar{s}_{\overline{n}|} - n}{\delta}}$$

where

$$\bar{s}_{\overline{n}|} = \frac{(1+i)^n - 1}{\delta}.$$

Comparison with discrete increasing annuities

The continuous formulas mirror the ordinary increasing annuity formulas:

$$(Ia)_{\overline{n}|} = \frac{\ddot{a}_{\overline{n}|} - nv^n}{i}, \qquad (\bar{I}\bar{a})_{\overline{n}|} = \frac{\bar{a}_{\overline{n}|} - nv^n}{\delta}.$$

Similarly,

$$(Is)_{\overline{n}|} = \frac{\ddot{s}_{\overline{n}|} - n}{i}, \qquad (\bar{I}\bar{s})_{\overline{n}|} = \frac{\bar{s}_{\overline{n}|} - n}{\delta}.$$

Interpretation

The payment rate is not constant. At time t, the annuity pays at rate t per year.
So early payments are small, and later payments are larger. The present value is found by adding all infinitesimal discounted payments:

$$\int_0^n te^{-\delta t}\,dt.$$

Continuously Decreasing Annuity

Consider a continuous annuity whose rate of payment starts at n and decreases linearly to 0 by time n.

Thus, the rate of payment at time t is

$$f(t) = n - t.$$

Assume that the force of interest is constant and equal to

$$\delta.$$

Then the amount paid between time t and time $t + dt$ is

$$(n-t)\,dt.$$

The present value is therefore

$$PV = \int_0^n (n-t)e^{-\delta t}\,dt.$$

We denote this present value by

$$(\bar{D}\bar{a})_{\overline{n}|}.$$

Derivation.

We can write

$$\int_0^n (n-t)e^{-\delta t}\,dt = n\int_0^n e^{-\delta t}\,dt - \int_0^n te^{-\delta t}\,dt.$$

The first integral is

$$\int_0^n e^{-\delta t}\,dt = \bar{a}_{\overline{n}|}.$$

The second integral is the present value of a continuously increasing annuity:

$$\int_0^n te^{-\delta t}\,dt = (\bar{I}\bar{a})_{\overline{n}|}.$$

Therefore,

$$(\bar{D}\bar{a})_{\overline{n}|} = n\bar{a}_{\overline{n}|} - (\bar{I}\bar{a})_{\overline{n}|}.$$

Using

$$(\bar{I}\bar{a})_{\overline{n}|} = \frac{\bar{a}_{\overline{n}|} - nv^n}{\delta},$$

we obtain

$$(\bar{D}\bar{a})_{\overline{n}|} = n\bar{a}_{\overline{n}|} - \frac{\bar{a}_{\overline{n}|} - nv^n}{\delta}.$$

This simplifies to

$$(\bar{D}\bar{a})_{\overline{n}|} = \frac{n - \bar{a}_{\overline{n}|}}{\delta}.$$

Present value of a continuously decreasing annuity

For a continuously decreasing annuity with payment rate

$$f(t) = n - t$$

from time 0 to time n, the present value is

$$\boxed{(\bar{D}\bar{a})_{\overline{n}|} = \frac{n - \bar{a}_{\overline{n}|}}{\delta}}$$

where

$$\bar{a}_{\overline{n}|} = \frac{1 - v^n}{\delta}.$$

This mirrors the ordinary decreasing annuity formula:

$$(Da)_{\overline{n}|} = \frac{n - a_{\overline{n}|}}{i}.$$

For the continuous version, we replace

$$a_{\overline{n}|} \quad \text{by} \quad \bar{a}_{\overline{n}|},$$

and

$$i \quad \text{by} \quad \delta.$$

Thus,

$$(Da)_{\overline{n}|} = \frac{n - a_{\overline{n}|}}{i} \qquad \Longrightarrow \qquad (\bar{D}\bar{a})_{\overline{n}|} = \frac{n - \bar{a}_{\overline{n}|}}{\delta}.$$

The accumulated value at time n is

$$(\bar{D}\bar{s})_{\overline{n}|} = (\bar{D}\bar{a})_{\overline{n}|}(1+i)^n.$$

Using

$$(\bar{D}\bar{a})_{\overline{n}|} = \frac{n - \bar{a}_{\overline{n}|}}{\delta},$$

we get

$$(\bar{D}\bar{s})_{\overline{n}|} = \frac{n(1+i)^n - \bar{s}_{\overline{n}|}}{\delta}.$$

Accumulated value of a continuously decreasing annuity

For a continuously decreasing annuity with payment rate

$$f(t) = n - t$$

from time 0 to time n, the accumulated value at time n is

$$\boxed{(\bar{D}\bar{s})_{\overline{n}|} = \frac{n(1+i)^n - \bar{s}_{\overline{n}|}}{\delta}}$$

where

$$\bar{s}_{\overline{n}|} = \frac{(1+i)^n - 1}{\delta}.$$

Interpretation

The payment rate starts at n and decreases continuously to 0.
At time 0, the payment rate is

$$n.$$

At time t, the payment rate is

$$n - t.$$

At time n, the payment rate is

$$0.$$

So this is the continuous analogue of a decreasing annuity.

Exercise 2.2.16

Given

$$\delta = 0.05,$$

calculate:

1. $(\bar{I}\bar{a})_{\overline{10}|}$
2. $(\bar{D}\bar{a})_{\overline{10}|}$
3. $(\bar{I}\bar{a})_{\overline{\infty}|}$

Solution.
Since
$$\delta = 0.05,$$
we have
$$v^t = e^{-\delta t}.$$
In particular,
$$v^{10} = e^{-0.05(10)} = e^{-0.5}.$$
Also,
$$\bar{a}_{\overline{10}|} = \frac{1 - v^{10}}{\delta}.$$
Thus,
$$\bar{a}_{\overline{10}|} = \frac{1 - e^{-0.5}}{0.05}.$$

1. Continuously increasing annuity
The present value of a continuously increasing annuity is
$$(\bar{I}\bar{a})_{\overline{n}|} = \frac{\bar{a}_{\overline{n}|} - nv^n}{\delta}.$$
Therefore,
$$(\bar{I}\bar{a})_{\overline{10}|} = \frac{\bar{a}_{\overline{10}|} - 10v^{10}}{0.05}.$$
Substitute
$$\bar{a}_{\overline{10}|} = \frac{1 - v^{10}}{0.05}.$$
Then
$$(\bar{I}\bar{a})_{\overline{10}|} = \frac{\dfrac{1 - v^{10}}{0.05} - 10v^{10}}{0.05}.$$
Using
$$v^{10} = e^{-0.5},$$
we get
$$(\bar{I}\bar{a})_{\overline{10}|} = 36.08.$$
Thus,
$$\boxed{(\bar{I}\bar{a})_{\overline{10}|} = 36.08}.$$

2. Continuously decreasing annuity
The present value of a continuously decreasing annuity is
$$(\bar{D}\bar{a})_{\overline{n}|} = \frac{n - \bar{a}_{\overline{n}|}}{\delta}.$$
Therefore,
$$(\bar{D}\bar{a})_{\overline{10}|} = \frac{10 - \bar{a}_{\overline{10}|}}{0.05}.$$

Substitute

$$\bar{a}_{\overline{10}|} = \frac{1 - v^{10}}{0.05}.$$

Then

$$(\bar{D}\bar{a})_{\overline{10}|} = \frac{10 - \dfrac{1 - v^{10}}{0.05}}{0.05}.$$

Using

$$v^{10} = e^{-0.5},$$

we get

$$(\bar{D}\bar{a})_{\overline{10}|} = 42.61.$$

Thus,

$$\boxed{(\bar{D}\bar{a})_{\overline{10}|} = 42.61}.$$

3. Continuously increasing perpetuity

The present value of a continuously increasing perpetuity is

$$(\bar{I}\bar{a})_{\overline{\infty}|} = \frac{1}{\delta^2}.$$

Therefore,

$$(\bar{I}\bar{a})_{\overline{\infty}|} = \frac{1}{0.05^2}.$$

Thus,

$$(\bar{I}\bar{a})_{\overline{\infty}|} = 400.$$

So,

$$\boxed{(\bar{I}\bar{a})_{\overline{\infty}|} = 400}.$$

Exercise 2.2.17

Payments are made to an account at a continuous rate of

$$8k + tk,$$

where

$$0 \leq t \leq 10.$$

Interest is credited at a force of interest

$$\delta_t = \frac{1}{8 + t}.$$

After 10 years, the account is worth 20000. Calculate k.

Solution.

The payment rate at time t is

$$f(t) = 8k + tk = k(8 + t).$$

We are given the force of interest

$$\delta_t = \frac{1}{8 + t}.$$

The accumulation function from time 0 to time t is

$$a(t) = \exp\left(\int_0^t \delta_s \, ds\right).$$

Thus,

$$a(t) = \exp\left(\int_0^t \frac{1}{8+s} \, ds\right).$$

Compute the integral:

$$\int_0^t \frac{1}{8+s} \, ds = [\ln(8+s)]_0^t = \ln(8+t) - \ln(8).$$

Therefore,

$$a(t) = \exp\left(\ln\left(\frac{8+t}{8}\right)\right).$$

So,

$$a(t) = \frac{8+t}{8} = 1 + \frac{t}{8}.$$

The accumulated value at time 10 is

$$AV = \int_0^{10} f(t)a(t,10) \, dt,$$

where

$$a(t,10) = \frac{a(10)}{a(t)}.$$

Now,

$$a(10) = 1 + \frac{10}{8} = \frac{18}{8}.$$

Also,

$$a(t) = 1 + \frac{t}{8} = \frac{8+t}{8}.$$

Hence,

$$a(t,10) = \frac{\frac{18}{8}}{\frac{8+t}{8}} = \frac{18}{8+t}.$$

Therefore,

$$20000 = \int_0^{10} k(8+t)\left(\frac{18}{8+t}\right) \, dt.$$

The factor $(8+t)$ cancels:

$$20000 = \int_0^{10} 18k \, dt.$$

Thus,

$$20000 = 18k(10).$$

So,

$$20000 = 180k.$$

Therefore,

$$k = \frac{20000}{180}.$$

Hence,

$$k = 111.11.$$

So,

$$\boxed{k = 111.11}.$$

Exercise 2.2.18

You are given:

1. The force of interest at time t is kt^3.
2. R is the present value of a four-year continuously increasing annuity which has a rate of payment of mt^3 at time t.

Calculate R.

A. $\dfrac{k - me^{-4k}}{k}$ B. $\dfrac{k - me^{-64k}}{k}$

C. $\dfrac{1 - me^{-4k}}{k}$ D. $\dfrac{1 - me^{-64k}}{k}$

E. $\dfrac{m(1 - e^{-64k})}{k}$

Solution.
The payment rate at time t is

$$f(t) = mt^3.$$

The force of interest is

$$\delta_t = kt^3.$$

The present value of a generalized continuous annuity is

$$R = \int_0^4 f(t) \exp\left(-\int_0^t \delta_s \, ds\right) dt.$$

Substitute

$$f(t) = mt^3 \qquad \text{and} \qquad \delta_s = ks^3.$$

Then

$$R = \int_0^4 mt^3 \exp\left(-\int_0^t ks^3 \, ds\right) dt.$$

Compute the inner integral:

$$\int_0^t ks^3 \, ds = k\left[\frac{s^4}{4}\right]_0^t = \frac{kt^4}{4}.$$

Therefore,

$$R = \int_0^4 mt^3 e^{-kt^4/4} \, dt.$$

Now let

$$u = \frac{t^4}{4}.$$

Then

$$du = t^3 \, dt.$$

When $t = 0$,

$$u = 0.$$

When $t = 4$,

$$u = \frac{4^4}{4} = 64.$$

Thus,

$$R = \int_0^{64} me^{-ku}\,du.$$

Now integrate:

$$R = m\left[-\frac{e^{-ku}}{k}\right]_0^{64}.$$

So,

$$R = -\frac{m}{k}e^{-64k} + \frac{m}{k}.$$

Therefore,

$$R = \frac{m(1-e^{-64k})}{k}.$$

The correct answer is

$$\boxed{\text{E. } \frac{m(1-e^{-64k})}{k}}.$$

2.2.3 Annuity Tricks

Double-Dots Cancel

If you have annuity-due factors on both sides of an equation, then you can replace those annuity-due factors with the corresponding annuity-immediate factors.

For example, suppose

$$X\ddot{a}_{\overline{n}|} = Y\ddot{a}_{\overline{m}|}.$$

Using

$$\ddot{a}_{\overline{n}|} = \frac{1-v^n}{d}, \qquad \ddot{a}_{\overline{m}|} = \frac{1-v^m}{d},$$

we get

$$X\left(\frac{1-v^n}{d}\right) = Y\left(\frac{1-v^m}{d}\right).$$

Now multiply both sides by

$$\frac{d}{i}.$$

Then

$$X\left(\frac{1-v^n}{d}\right)\frac{d}{i} = Y\left(\frac{1-v^m}{d}\right)\frac{d}{i}.$$

So,

$$X\left(\frac{1-v^n}{i}\right) = Y\left(\frac{1-v^m}{i}\right).$$

Therefore,

$$Xa_{\overline{n}|} = Ya_{\overline{m}|}.$$

Double-dots cancel

If the same type of annuity-due factor appears on both sides of an equation, the double dots can often be removed.

For example,

$$X\ddot{a}_{\overline{n}|} = Y\ddot{a}_{\overline{m}|}$$

is equivalent to

$$Xa_{\overline{n}|} = Ya_{\overline{m}|}.$$

This works because every annuity-due factor contains the same adjustment factor relative to the corresponding annuity-immediate factor.

$$\ddot{a}_{\overline{n}|} = (1+i)a_{\overline{n}|}.$$

This also works for ratios.

For example,

$$\frac{\ddot{a}_{\overline{20}|}}{\ddot{s}_{\overline{5}|}} = \frac{a_{\overline{20}|}}{s_{\overline{5}|}}.$$

Indeed,

$$\ddot{a}_{\overline{20}|} = (1+i)a_{\overline{20}|},$$

and

$$\ddot{s}_{\overline{5}|} = (1+i)s_{\overline{5}|}.$$

Thus,

$$\frac{\ddot{a}_{\overline{20}|}}{\ddot{s}_{\overline{5}|}} = \frac{(1+i)a_{\overline{20}|}}{(1+i)s_{\overline{5}|}} = \frac{a_{\overline{20}|}}{s_{\overline{5}|}}.$$

When the trick is safe

The double-dots cancel only when the same due adjustment appears on both sides or both numerator and denominator.
It is safe in expressions like

$$X\ddot{a}_{\overline{n}|} = Y\ddot{a}_{\overline{m}|}$$

or

$$\frac{\ddot{a}_{\overline{n}|}}{\ddot{s}_{\overline{m}|}}.$$

It is not safe if only one side has a double dot.

Upper (m)'s Cancel

If you have m-thly annuity factors on both sides of an equation, then you can replace those m-thly annuity factors with the corresponding annual annuity factors.

For example, suppose

$$Xa^{(m)}_{\overline{p}|} = Ys^{(m)}_{\overline{q}|}.$$

Using

$$a^{(m)}_{\overline{p}|} = \frac{1-v^p}{i^{(m)}}, \qquad s^{(m)}_{\overline{q}|} = \frac{(1+i)^q-1}{i^{(m)}},$$

we get

$$X\left(\frac{1-v^p}{i^{(m)}}\right) = Y\left(\frac{(1+i)^q-1}{i^{(m)}}\right).$$

Now multiply both sides by

$$\frac{i^{(m)}}{i}.$$

Then

$$X\left(\frac{1-v^p}{i^{(m)}}\right)\frac{i^{(m)}}{i} = Y\left(\frac{(1+i)^q-1}{i^{(m)}}\right)\frac{i^{(m)}}{i}.$$

So,

$$X\left(\frac{1-v^p}{i}\right) = Y\left(\frac{(1+i)^q-1}{i}\right).$$

Therefore,

$$Xa_{\overline{p}|} = Ys_{\overline{q}|}.$$

Upper (m)'s cancel

If the same m-thly adjustment appears on both sides of an equation, the upper (m)'s can often be removed.
For example,

$$Xa^{(m)}_{\overline{p}|} = Ys^{(m)}_{\overline{q}|}$$

is equivalent to

$$Xa_{\overline{p}|} = Ys_{\overline{q}|}.$$

This works because every m-thly annuity factor contains the same adjustment factor:

$$a^{(m)}_{\overline{n}|} = \frac{i}{i^{(m)}}a_{\overline{n}|}, \qquad s^{(m)}_{\overline{n}|} = \frac{i}{i^{(m)}}s_{\overline{n}|}.$$

This also works for ratios.

For example,

$$\frac{\ddot{a}^{(12)}_{\overline{20}|}}{\ddot{s}^{(12)}_{\overline{5}|}} = \frac{a_{\overline{20}|}}{s_{\overline{5}|}}.$$

Indeed, the upper (12)'s cancel because both factors have the same 12-thly adjustment, and the double dots cancel because both factors are annuity-due factors.

More explicitly,

$$\ddot{a}^{(12)}_{\overline{20}|} = \frac{d}{d^{(12)}}\ddot{a}_{\overline{20}|}, \qquad \ddot{s}^{(12)}_{\overline{5}|} = \frac{d}{d^{(12)}}\ddot{s}_{\overline{5}|}.$$

Therefore,

$$\frac{\ddot{a}^{(12)}_{\overline{20}|}}{\ddot{s}^{(12)}_{\overline{5}|}} = \frac{\frac{d}{d^{(12)}}\ddot{a}_{\overline{20}|}}{\frac{d}{d^{(12)}}\ddot{s}_{\overline{5}|}} = \frac{\ddot{a}_{\overline{20}|}}{\ddot{s}_{\overline{5}|}}.$$

Then the double dots cancel:

$$\frac{\ddot{a}_{\overline{20}|}}{\ddot{s}_{\overline{5}|}} = \frac{a_{\overline{20}|}}{s_{\overline{5}|}}.$$

Thus,

$$\boxed{\frac{\ddot{a}^{(12)}_{\overline{20}|}}{\ddot{s}^{(12)}_{\overline{5}|}} = \frac{a_{\overline{20}|}}{s_{\overline{5}|}}}.$$

When the trick is safe

The upper (m)'s cancel only when the same payment frequency appears on both sides or in both numerator and denominator.
It is safe in expressions like

$$X a_{\overline{p}|}^{(m)} = Y s_{\overline{q}|}^{(m)}$$

or

$$\frac{a_{\overline{p}|}^{(m)}}{s_{\overline{q}|}^{(m)}}.$$

It is not safe if only one side has an upper (m).

$a_{\overline{2n}|}/a_{\overline{n}|}$ Trick

A useful ratio trick is

$$\frac{a_{\overline{2n}|}}{a_{\overline{n}|}} = 1 + v^n.$$

Indeed,

$$\frac{a_{\overline{2n}|}}{a_{\overline{n}|}} = \frac{\frac{1-v^{2n}}{i}}{\frac{1-v^n}{i}}.$$

The i's cancel:

$$\frac{a_{\overline{2n}|}}{a_{\overline{n}|}} = \frac{1-v^{2n}}{1-v^n}.$$

Now use the identity

$$1 - v^{2n} = (1-v^n)(1+v^n).$$

Therefore,

$$\frac{a_{\overline{2n}|}}{a_{\overline{n}|}} = \frac{(1-v^n)(1+v^n)}{1-v^n}.$$

Thus,

$$\boxed{\frac{a_{\overline{2n}|}}{a_{\overline{n}|}} = 1 + v^n}.$$

Ratio trick

Whenever the numerator has twice the term of the denominator, remember:

$$\frac{a_{\overline{2n}|}}{a_{\overline{n}|}} = 1 + v^n.$$

The same idea also works after cancelling common double-dot factors:

$$\frac{\ddot{a}_{\overline{2n}|}}{\ddot{a}_{\overline{n}|}} = \frac{a_{\overline{2n}|}}{a_{\overline{n}|}} = 1 + v^n.$$

Using the $a_{\overline{2n}|}/a_{\overline{n}|}$ trick

Given

$$\ddot{a}_{\overline{20}|} = 13.0853 \qquad \text{and} \qquad \ddot{a}_{\overline{10}|} = 8.1078,$$

find i.

Solution.

Since both factors are annuity-due factors, the double dots cancel in the ratio:

$$\frac{\ddot{a}_{\overline{20}|}}{\ddot{a}_{\overline{10}|}} = \frac{a_{\overline{20}|}}{a_{\overline{10}|}}.$$

Now use the ratio trick with $n = 10$:

$$\frac{a_{\overline{20}|}}{a_{\overline{10}|}} = 1 + v^{10}.$$

Therefore,

$$\frac{13.0853}{8.1078} = 1 + v^{10}.$$

So,

$$v^{10} = \frac{13.0853}{8.1078} - 1.$$

Thus,

$$v^{10} = 0.6139.$$

Taking the tenth root,

$$v = 0.95238.$$

Since

$$v = \frac{1}{1+i},$$

we get

$$1 + i = \frac{1}{0.95238}.$$

Therefore,

$$i = 0.05.$$

So,

$$\boxed{i = 0.05}.$$

Pyramid Annuity

Consider an annuity-immediate with payments

$$1,\ 2,\ 3, \ldots, n-1,\ n,\ n-1, \ldots, 3,\ 2,\ 1.$$

This is called a pyramid annuity because the payments increase up to n, then decrease back to 1.

For example, when $n = 5$, the payments are

$$1,\ 2,\ 3,\ 4,\ 5,\ 4,\ 3,\ 2,\ 1.$$

A useful way to understand this annuity is to decompose it into several level annuities.

For $n = 5$, we can write

$$1,\ 2,\ 3,\ 4,\ 5,\ 4,\ 3,\ 2,\ 1$$

as the sum of five annuities with payments of 1:

$$\begin{array}{ccccccccc}
1 & 1 & 1 & 1 & 1 & & & & \\
 & 1 & 1 & 1 & 1 & 1 & & & \\
 & & 1 & 1 & 1 & 1 & 1 & & \\
 & & & 1 & 1 & 1 & 1 & 1 & \\
 & & & & 1 & 1 & 1 & 1 & 1 \\
\hline
1 & 2 & 3 & 4 & 5 & 4 & 3 & 2 & 1
\end{array}$$

Each row is a 5-payment annuity-immediate, but each row is shifted one period later than the previous row.

Therefore, by viewing the pyramid as a sum of shifted level annuities, the present value simplifies to a product of an annuity-immediate and annuity-due factor; the general result is stated in the formula box below.

Pyramid annuity-immediate

For an annuity-immediate with payments

$$1,\ 2, \ldots, n-1,\ n,\ n-1, \ldots, 2,\ 1,$$

the present value is

$$\boxed{PV_{\text{pyramid-immediate}} = a_{\overline{n}|}\ddot{a}_{\overline{n}|}}.$$

Pyramid annuity-due.

If the same pyramid pattern starts immediately at time 0, then the annuity is a pyramid annuity-due.

The payments are

$$1,\ 2, \ldots, n-1,\ n,\ n-1, \ldots, 2,\ 1$$

at times

$$0, 1, \ldots, 2n-2.$$

By the same decomposition, each row is now an annuity-due. Therefore,

$$PV_{\text{pyramid-due}} = \ddot{a}_{\overline{n}|}\left(1 + v + v^2 + \cdots + v^{n-1}\right).$$

Since

$$1 + v + v^2 + \cdots + v^{n-1} = \ddot{a}_{\overline{n}|},$$

we get

$$PV_{\text{pyramid-due}} = \ddot{a}_{\overline{n}|}\ddot{a}_{\overline{n}|}.$$

Pyramid annuity-due

For a pyramid annuity-due with payments

$$1,\ 2, \ldots, n-1,\ n,\ n-1, \ldots, 2,\ 1,$$

the present value is

$$\boxed{PV_{\text{pyramid-due}} = \ddot{a}^2_{\overline{n}|}}.$$

Memory trick

A pyramid annuity is just a stack of level annuities.
For an annuity-immediate:

$$\boxed{PV_{\text{pyramid-immediate}} = a_{\overline{n}|}\ddot{a}_{\overline{n}|}}$$

For an annuity-due:

$$\boxed{PV_{\text{pyramid-due}} = \ddot{a}_{\overline{n}|}^{2}}$$

The first factor values one row. The second factor values the shifts of the rows.

Upside Pyramid Annuity

Now consider the upside-down pyramid annuity-immediate:

$$n,\ n-1,\ n-2,\dots,2,\ 1,\ 2,\dots,n-2,\ n-1,\ n.$$

For example, when $n = 5$, the payments are

$$5,\ 4,\ 3,\ 2,\ 1,\ 2,\ 3,\ 4,\ 5.$$

A useful trick is to compare this upside pyramid with the ordinary pyramid:

$$1,\ 2,\ 3,\ 4,\ 5,\ 4,\ 3,\ 2,\ 1.$$

Adding the two payment streams gives a level annuity:

$$\begin{array}{ccccccccc} 5 & 4 & 3 & 2 & 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 3 & 4 & 5 & 4 & 3 & 2 & 1 \\ \hline 6 & 6 & 6 & 6 & 6 & 6 & 6 & 6 & 6 \end{array}$$

Therefore, the upside pyramid can be written as

$$\text{upside pyramid} = \text{level annuity of } 6 - \text{ordinary pyramid}.$$

Thus, for $n = 5$,

$$PV = 6a_{\overline{9}|} - a_{\overline{5}|}\ddot{a}_{\overline{5}|}.$$

More generally, the upside pyramid has $2n-1$ payments, and the corresponding level annuity has payment $n+1$. Hence,

$$PV_{\text{upside pyramid}} = (n+1)a_{\overline{2n-1}|} - a_{\overline{n}|}\ddot{a}_{\overline{n}|}.$$

Upside pyramid annuity-immediate

For an annuity-immediate with payments

$$n,\ n-1,\dots,2,\ 1,\ 2,\dots,n-1,\ n,$$

the present value is

$$\boxed{PV_{\text{upside pyramid}} = (n+1)a_{\overline{2n-1}|} - a_{\overline{n}|}\ddot{a}_{\overline{n}|}}.$$

Memory trick

An upside pyramid is easier to value by subtracting an ordinary pyramid from a level annuity.
For $n = 5$:

$$\boxed{PV = 6a_{\overline{9}|} - a_{\overline{5}|}\ddot{a}_{\overline{5}|}}.$$

In general:

$$\boxed{PV = (n+1)a_{\overline{2n-1}|} - a_{\overline{n}|}\ddot{a}_{\overline{n}|}}.$$

0% **Test**

A useful way to check annuity formulas is to test the special case

$$i = 0.$$

If

$$i = 0,$$

then

$$v = \frac{1}{1+i} = 1.$$

Therefore, discounting and accumulating have no effect.

For level annuities, every payment of 1 simply keeps its value. Hence,

$$a_{\overline{n}|} = v + v^2 + \cdots + v^n = 1 + 1 + \cdots + 1 = n.$$

Similarly,

$$\ddot{a}_{\overline{n}|} = 1 + v + v^2 + \cdots + v^{n-1} = 1 + 1 + \cdots + 1 = n.$$

Also,

$$s_{\overline{n}|} = a_{\overline{n}|}(1+i)^n = n(1)^n = n,$$

and

$$\ddot{s}_{\overline{n}|} = \ddot{a}_{\overline{n}|}(1+i)^n = n(1)^n = n.$$

Thus, when $i = 0$,

$$\boxed{a_{\overline{n}|} = \ddot{a}_{\overline{n}|} = s_{\overline{n}|} = \ddot{s}_{\overline{n}|} = n}.$$

For increasing annuities, the payments are

$$1, 2, 3, \ldots, n.$$

Since there is no discounting,

$$(Ia)_{\overline{n}|} = v + 2v^2 + 3v^3 + \cdots + nv^n.$$

When $v = 1$, this becomes

$$(Ia)_{\overline{n}|} = 1 + 2 + 3 + \cdots + n.$$

Therefore,

$$(Ia)_{\overline{n}|} = \frac{n(n+1)}{2}.$$

The same value applies to the increasing accumulated value, the decreasing present value, and the decreasing accumulated value:

$$(Ia)_{\overline{n}|} = (Is)_{\overline{n}|} = (Da)_{\overline{n}|} = (Ds)_{\overline{n}|} = \frac{n(n+1)}{2}.$$

The same is true for the annuity-due versions:

$$(I\ddot{a})_{\overline{n}|} = (I\ddot{s})_{\overline{n}|} = (D\ddot{a})_{\overline{n}|} = (D\ddot{s})_{\overline{n}|} = \frac{n(n+1)}{2}.$$

The 0% test

When $i = 0$, all discounting and accumulating disappear.
So every formula should reduce to a simple sum of payments.
For level annuities:

$$\boxed{a_{\overline{n}|} = \ddot{a}_{\overline{n}|} = s_{\overline{n}|} = \ddot{s}_{\overline{n}|} = n}$$

For increasing or decreasing arithmetic annuities:

$$\boxed{(Ia)_{\overline{n}|} = (Is)_{\overline{n}|} = (Da)_{\overline{n}|} = (Ds)_{\overline{n}|} = \frac{n(n+1)}{2}}$$

and

$$\boxed{(I\ddot{a})_{\overline{n}|} = (I\ddot{s})_{\overline{n}|} = (D\ddot{a})_{\overline{n}|} = (D\ddot{s})_{\overline{n}|} = \frac{n(n+1)}{2}}.$$

Warning

Some formulas have denominators like

$$i, \qquad d, \qquad \delta.$$

If $i = 0$, these denominators become 0, so you should not plug directly into those formulas. Instead, use the cash-flow interpretation or take the limiting value.

Exercise 2.2.19

An 11-year annuity has a series of payments

$$1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1$$

with the first payment made at the end of the second year. The present value of this annuity is 25 at an interest rate i.
A 12-year annuity has a series of payments

$$1, 2, 3, 4, 5, 6, 6, 5, 4, 3, 2, 1$$

with the first payment made at the end of the first year.
Calculate the present value of the 12-year annuity at interest rate i.
Solution.
The 11-year annuity has payments

$$1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1$$

at times

$$2, 3, 4, \ldots, 12.$$

This is a deferred pyramid annuity with peak 6. A pyramid annuity-immediate with peak 6 has present value

$$a_{\overline{6}|}\ddot{a}_{\overline{6}|}.$$

Since this annuity is deferred one year, its present value is

$$va_{\overline{6}|}\ddot{a}_{\overline{6}|}.$$

We are given that this value is 25, so

$$va_{\overline{6}|}\ddot{a}_{\overline{6}|} = 25.$$

Using

$$\ddot{a}_{\overline{6}|} = (1+i)a_{\overline{6}|},$$

we get

$$v\ddot{a}_{\overline{6}|} = v(1+i)a_{\overline{6}|}.$$

Since

$$v(1+i) = 1,$$

we have

$$v\ddot{a}_{\overline{6}|} = a_{\overline{6}|}.$$

Therefore,

$$va_{\overline{6}|}\ddot{a}_{\overline{6}|} = a^2_{\overline{6}|}.$$

Thus,

$$a^2_{\overline{6}|} = 25.$$

Since annuity factors are positive,

$$a_{\overline{6}|} = 5.$$

Now compare the 12-year annuity with the 11-year annuity.
The 12-year annuity has payments

$$1, 2, 3, 4, 5, 6, 6, 5, 4, 3, 2, 1$$

at times

$$1, 2, 3, \ldots, 12.$$

The 11-year annuity contributes

$$0, 1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1$$

at times

$$1, 2, 3, \ldots, 12.$$

The difference is a level annuity-immediate of 1 for 6 years:

$$1, 1, 1, 1, 1, 1$$

at times

$$1, 2, 3, 4, 5, 6.$$

The present value of this extra stream is

$$a_{\overline{6}|}.$$

Therefore, the present value of the 12-year annuity is

$$25 + a_{\overline{6}|}.$$

Since

$$a_{\overline{6}|} = 5,$$

we get

$$PV = 25 + 5 = 30.$$

Thus,

$$\boxed{30}.$$

Exercise 2.2.20

A fund is built with annual payments increasing by 1 from 1 to 10 and then decreasing by 1 to 0. The first payment of 1 is made today.
If the fund is used to purchase a 10-year level annuity with the first payment at 20 years from today, what is the amount of the level payment?
Assume an annual effective rate of interest of 4%.

A. Less than 16
B. At least 16, but less than 17
C. At least 17, but less than 18
D. At least 18, but less than 19
E. At least 19

Solution.
The fund payments are

$$1, 2, 3, \ldots, 10, 9, 8, \ldots, 2, 1$$

with the first payment made today.
Thus, the payments occur at times

$$0, 1, 2, \ldots, 18.$$

This is a pyramid annuity-due with peak 10. Therefore, its present value at time 0 is

$$\ddot{a}_{\overline{10}|}\ddot{a}_{\overline{10}|} = \ddot{a}^2_{\overline{10}|}.$$

The fund is used to purchase a 10-year level annuity with the first payment at time 20. The level annuity payments occur at times

$$20, 21, \ldots, 29.$$

It is convenient to value both sides at time 19, one period before the first level payment. Accumulating the fund value from time 0 to time 19, we get

$$\ddot{a}^2_{\overline{10}|}(1+i)^{19}.$$

Rewrite this as

$$\ddot{a}^2_{\overline{10}|}(1+i)^{19} = \left(\ddot{a}_{\overline{10}|}(1+i)^{10}\right)\left(\ddot{a}_{\overline{10}|}(1+i)^{9}\right).$$

Now,

$$\ddot{a}_{\overline{10}|}(1+i)^{10} = \ddot{s}_{\overline{10}|},$$

and

$$\ddot{a}_{\overline{10}|}(1+i)^9 = s_{\overline{10}|}.$$

Therefore, the accumulated value of the fund at time 19 is

$$\ddot{s}_{\overline{10}|}s_{\overline{10}|}.$$

Let X be the level payment of the 10-year annuity. At time 19, the present value of the level annuity is

$$Xa_{\overline{10}|}.$$

Therefore,

$$\ddot{s}_{\overline{10}|}s_{\overline{10}|} = Xa_{\overline{10}|}.$$

So,

$$X = \frac{\ddot{s}_{\overline{10}|}s_{\overline{10}|}}{a_{\overline{10}|}}.$$

Using $i = 0.04$, we calculate

$$a_{\overline{10}|} = 8.1109,$$

$$s_{\overline{10}|} = 12.0061,$$

and

$$\ddot{s}_{\overline{10}|} = 12.4864.$$

Thus,

$$X = \frac{12.4864(12.0061)}{8.1109}.$$

Therefore,

$$X = 18.48.$$

So the level payment is

$$\boxed{18.48}.$$

The correct answer is

$$\boxed{\text{D. At least 18, but less than 19}}.$$

Exercise 2.2.21

Hilary receives payments of X at the end of each year for n years. The present value of her annuity is 493.
Ned receives payments of $3X$ at the end of each year for $2n$ years. The present value of his annuity is 2748.
Both present values are calculated at the same annual effective interest rate.
Determine v^n.

A. 0.86 B. 0.87

C. 0.88 D. 0.89

E. 0.90

Solution.
Hilary's present value is

$$Xa_{\overline{n}|} = 493.$$

Ned's present value is

$$3Xa_{\overline{2n}|} = 2748.$$

Divide Ned's equation by Hilary's equation:

$$\frac{3Xa_{\overline{2n}|}}{Xa_{\overline{n}|}} = \frac{2748}{493}.$$

Cancel X:

$$3\frac{a_{\overline{2n}|}}{a_{\overline{n}|}} = \frac{2748}{493}.$$

Now use the annuity ratio trick:

$$\frac{a_{\overline{2n}|}}{a_{\overline{n}|}} = 1 + v^n.$$

Therefore,

$$3(1 + v^n) = \frac{2748}{493}.$$

So,

$$1 + v^n = \frac{2748}{3(493)}.$$

Thus,

$$v^n = \frac{2748}{1479} - 1.$$

Hence,

$$v^n = 0.8580.$$

The closest answer is

$$\boxed{\text{A. } 0.86}.$$

Exercise 2.2.22

An annuity pays 1 now, 2 a year from now, 3 two years from now, etc., until a maximum payment of 5 is made, after which time the payments decrease by 1 per year until there are no further payments.
Find the present value.

A. $\left(\ddot{a}_{\overline{5}|}\right)^2$ B. $5\ddot{a}_{\overline{5}|}$

C. $\left(a_{\overline{5}|}\right)^2$ D. $5a_{\overline{5}|}$

E. $\ddot{a}_{\overline{5}|}a_{\overline{5}|}$

Solution.
The payments are

$$1, 2, 3, 4, 5, 4, 3, 2, 1.$$

Since the first payment is made now, the payments occur at times

$$0, 1, 2, 3, 4, 5, 6, 7, 8.$$

This is a pyramid annuity-due with peak 5.
Recall that a pyramid annuity-immediate with peak n has present value

$$a_{\overline{n}|}\ddot{a}_{\overline{n}|}.$$

Here, because the first payment is made immediately, the whole pyramid is shifted one period earlier. Therefore, the present value is

$$a_{\overline{5}|}\ddot{a}_{\overline{5}|}(1+i).$$

Using

$$\ddot{a}_{\overline{5}|} = (1+i)a_{\overline{5}|},$$

we get

$$a_{\overline{5}|}(1+i) = \ddot{a}_{\overline{5}|}.$$

Thus,

$$a_{\overline{5}|}\ddot{a}_{\overline{5}|}(1+i) = \ddot{a}_{\overline{5}|}\ddot{a}_{\overline{5}|}.$$

Therefore,

$$PV = \left(\ddot{a}_{\overline{5}|}\right)^2.$$

The correct answer is

$$\boxed{\text{A. } \left(\ddot{a}_{\overline{5}|}\right)^2}.$$

Chapter 3

Loans

Ca peut

3.1 Amortization Schedules

3.1.1 Amortization Schedule – Part 1

Amortization Method

Amortization method

Under the amortization method, each loan payment first pays the interest due for the period.
Anything left after paying interest reduces the outstanding loan balance.

In other words, each loan payment is split into interest and principal (see the formula box below).

The interest paid covers the interest due on the outstanding balance. The principal paid reduces the loan balance.

Payment decomposition

For a payment R_t made at time t,

$$\boxed{R_t = I_t + P_t}$$

where I_t is the interest paid and P_t is the principal paid.

Building an amortization schedule

A loan at an annual effective interest rate of 10% is repaid with three annual payments of 500.
Calculate the loan amount L, then build the amortization schedule.
Solution.
The loan amount is the present value of the three payments:

$$L = 500a_{\overline{3}|0.10}.$$

Thus,

$$L = 500\left(\frac{1-(1.10)^{-3}}{0.10}\right).$$

Therefore,

$$L = 1243.43.$$

Time 1.
The interest due after one year is

$$1243.43(0.10) = 124.34.$$

If the borrower paid only 124.34, the loan balance would remain 1243.43.
However, the borrower pays 500, so the extra amount reduces the loan balance.
The principal paid is

$$500 - 124.34 = 375.66.$$

The new loan balance after the first payment is

$$1243.43 - 375.66 = 867.77.$$

$$\boxed{OB_1 = 867.77}$$

Time 2.
The interest due during the second year is based on the outstanding balance after the first payment:

$$867.77(0.10) = 86.78.$$

The principal paid is

$$500 - 86.78 = 413.22.$$

The new loan balance after the second payment is

$$867.77 - 413.22 = 454.55.$$

$$\boxed{OB_2 = 454.55}$$

Time 3.
The interest due during the third year is

$$454.55(0.10) = 45.46.$$

The principal paid is

$$500 - 45.46 = 454.54.$$

The new loan balance after the third payment is

$$454.55 - 454.54 = 0.01.$$

The final loan balance should be 0. The small balance of 0.01 is due to rounding.

$$\boxed{OB_3 \approx 0}$$

Amortization schedule.

Time	Loan Payment	Interest Paid	Principal Paid	Loan Balance
0				1243.43
1	500.00	124.34	375.66	867.77
2	500.00	86.78	413.22	454.55
3	500.00	45.46	454.54	0.01
Total	1500.00	256.58	1243.42	

This table is called the **amortization schedule**.
The total principal paid equals the original loan amount, up to rounding:

$$375.66 + 413.22 + 454.54 = 1243.42.$$

The total interest paid equals total loan payments minus the original loan amount:

$$1500.00 - 1243.43 = 256.57.$$

Because of rounding, the table gives

$$256.58.$$

Meaning of amortization

The word "amortize" comes from the Old French word *amortir*, which means to reduce to the point of death.
In a loan context, amortization means that the loan balance is gradually reduced until it disappears.

Retrospective Method

We can find loan balances by looking at what has happened from the loan issue date up to time t.

This is called the **retrospective method**.

For example, after one payment:

$$OB_1 = 1243.43(1.10) - 500.$$

Thus,

$$OB_1 = 867.77.$$

After two payments:

$$OB_2 = (1243.43(1.10) - 500)\,(1.10) - 500.$$

Equivalently,

$$OB_2 = 1243.43(1.10)^2 - 500s_{\overline{2}|0.10}.$$

Therefore,

$$OB_2 = 454.55.$$

Retrospective loan balance

For a level-payment loan with original loan amount L, payment R, and interest rate i, the outstanding balance after the t-th payment is

$$\boxed{OB_t = L(1+i)^t - Rs_{\overline{t}|i}}.$$

Prospective Method

We can also find loan balances by looking forward at the remaining payments.

This is called the **prospective method**.

After the first payment, there are two payments of 500 remaining. Therefore,

$$OB_1 = 500a_{\overline{2}|0.10}.$$

Thus,

$$OB_1 = 867.77.$$

After the second payment, there is one payment of 500 remaining. Therefore,

$$OB_2 = 500v.$$

Thus,

$$OB_2 = 454.55.$$

Prospective loan balance

For a level-payment loan with payment R, interest rate i, and n total payments, the outstanding balance after the t-th payment is

$$\boxed{OB_t = Ra_{\overline{n-t}|i}}.$$

Retrospective versus prospective

The retrospective method looks at the accumulated value of what has already happened:

$$OB_t = \text{accumulated loan} - \text{accumulated payments}.$$

The prospective method looks at the present value of what remains:

$$OB_t = \text{present value of future payments}.$$

Both methods give the same loan balance.

AMORT Worksheet

For a loan with level payments, the BA II Plus can perform amortization calculations using the `2ND AMORT` worksheet.

For the previous example:

$$\begin{aligned} N &= 3, \\ I/Y &= 10, \\ PMT &= 500, \\ FV &= 0. \end{aligned}$$

Then:

$$CPT\ PV = -1243.425995.$$

The negative sign is a calculator cash-flow convention. The loan amount is

$$1243.43.$$

To find the amortization details for payment 1, use:

$$P1 = 1, \qquad P2 = 1.$$

The calculator gives:

$$BAL = -867.768595,$$
$$PRN = 375.6574005,$$
$$INT = 124.3425995.$$

Thus:

$$OB_1 = 867.77, \qquad P_1 = 375.66, \qquad I_1 = 124.34.$$

To summarize payments 1 through 3, use:

$$P1 = 1, \qquad P2 = 3.$$

The calculator gives approximately:

$$BAL \approx 0,$$
$$PRN = 1243.425995,$$
$$INT = 256.5740045.$$

Thus, over the whole loan:

$$\text{total principal paid} = 1243.43,$$

and

$$\text{total interest paid} = 256.57.$$

Calculator sign convention

On the BA II Plus, balances may appear with a negative sign because of the calculator's cash-flow convention.
For loan amortization, focus on the magnitude of the balance, principal, and interest.

Notation

It is useful to introduce notation for amortization schedules.

$$\begin{aligned}
t &: \text{ time, usually in years unless stated otherwise,}\\
R_t &: \text{ payment amount at time } t,\\
I_t &: \text{ amount of interest paid in } R_t,\\
P_t &: \text{ amount of principal paid in } R_t,\\
OB_t &: \text{ outstanding loan balance after the payment at time } t,\\
L &: \text{ original loan amount.}
\end{aligned}$$

The loan amount is the present value of all payments:

$$L = \sum_{\text{all } t} R_t v^t.$$

The interest paid at time t is the previous outstanding balance times the interest rate:

$$I_t = OB_{t-1} i.$$

The principal paid at time t is the payment minus the interest paid:

$$P_t = R_t - I_t.$$

The outstanding balance after payment t is the previous outstanding balance minus the principal paid:

$$OB_t = OB_{t-1} - P_t.$$

Amortization formulas

The basic amortization formulas are

$$\boxed{I_t = OB_{t-1} i}$$

$$\boxed{P_t = R_t - I_t}$$

$$\boxed{OB_t = OB_{t-1} - P_t}$$

and

$$\boxed{L = \sum_{\text{all } t} R_t v^t}.$$

Exercise 3.1.1

A loan is repaid with 20 annual payments of 1000, with the first payment one year from now.
The interest rate is 6%.
Find the loan balance after the 15th payment using:

1. the prospective method;
2. the retrospective method;
3. the BA II Plus AMORT worksheet.

Solution.
The loan is repaid with 20 level annual payments of

$$1000.$$

The annual effective interest rate is

$$i = 0.06.$$

We want the outstanding loan balance immediately after the 15th payment, denoted

$$OB_{15}.$$

Part 1: Prospective method.
The prospective method looks forward at the remaining payments.
After the 15th payment, there are

$$20 - 15 = 5$$

payments remaining.
Therefore, the outstanding balance after the 15th payment is the present value at time 15 of the remaining 5 payments:

$$OB_{15} = 1000a_{\overline{5}|0.06}.$$

Using

$$a_{\overline{5}|0.06} = \frac{1-(1.06)^{-5}}{0.06},$$

we get

$$OB_{15} = 1000\left(\frac{1-(1.06)^{-5}}{0.06}\right).$$

Thus,

$$OB_{15} = 4212.36.$$

So,

$$\boxed{OB_{15} = 4212.36}.$$

Part 2: Retrospective method.
The retrospective method looks backward from the beginning of the loan to time 15.
First, find the original loan amount:

$$L = 1000a_{\overline{20}|0.06}.$$

Using

$$a_{\overline{20}|0.06} = \frac{1-(1.06)^{-20}}{0.06},$$

we get

$$L = 11469.92.$$

At time 15, the accumulated value of the original loan is

$$11469.92(1.06)^{15}.$$

The accumulated value of the first 15 payments is

$$1000s_{\overline{15}|0.06}.$$

Therefore,

$$OB_{15} = 11469.92(1.06)^{15} - 1000s_{\overline{15}|0.06}.$$

Using

$$s_{\overline{15}|0.06} = \frac{(1.06)^{15}-1}{0.06},$$

we get

$$OB_{15} = 4212.36.$$

Thus,

$$\boxed{OB_{15} = 4212.36}.$$

Part 3: BA II Plus AMORT worksheet.
First enter the loan information:

$$\begin{aligned} N &= 20, \\ I/Y &= 6, \\ PMT &= 1000, \\ FV &= 0. \end{aligned}$$

Then compute the present value:

$$CPT\ PV = -11469.92122.$$

The loan amount is therefore

$$L = 11469.92.$$

Now use the AMORT worksheet:

$$\begin{aligned} P1 &= 15, \\ P2 &= 15. \end{aligned}$$

The calculator gives

$$BAL = -4212.363786.$$

The negative sign is due to the calculator cash-flow convention. Therefore,

$$OB_{15} = 4212.36.$$

Hence,

$$\boxed{OB_{15} = 4212.36}.$$

Exercise 3.1.2

A loan of 4724.80 is repaid in annual payments of 1000, 1500, and 3000, with the first payment one year from now.
The interest rate is 5% for year 1, 10% for year 2, and 5% for year 3.
Build the amortization schedule for this loan.

Solution.

The initial loan balance is

$$OB_0 = 4724.80.$$

Year 1.

The interest rate for year 1 is 5%. Therefore,

$$I_1 = 4724.80(0.05) = 236.24.$$

The first payment is 1000, so the principal paid is

$$P_1 = 1000 - 236.24 = 763.76.$$

Thus, the outstanding balance after the first payment is

$$OB_1 = 4724.80 - 763.76 = 3961.04.$$

Year 2.

The interest rate for year 2 is 10%. Therefore,

$$I_2 = 3961.04(0.10) = 396.10.$$

The second payment is 1500, so the principal paid is

$$P_2 = 1500 - 396.10 = 1103.90.$$

Thus, the outstanding balance after the second payment is

$$OB_2 = 3961.04 - 1103.90 = 2857.14.$$

Year 3.
The interest rate for year 3 is 5%. Therefore,

$$I_3 = 2857.14(0.05) = 142.86.$$

The third payment is 3000, so the principal paid is

$$P_3 = 3000 - 142.86 = 2857.14.$$

Thus, the outstanding balance after the third payment is

$$OB_3 = 2857.14 - 2857.14 = 0.$$

Therefore, the amortization schedule is:

t	i_t	R_t	I_t	P_t	OB_t
0					4724.80
1	5%	1000.00	236.24	763.76	3961.04
2	10%	1500.00	396.10	1103.90	2857.14
3	5%	3000.00	142.86	2857.14	0.00

The loan is fully repaid after the third payment.

Exercise 3.1.3

On December 31, 1984, Smith borrowed 5000 to be repaid in four years with level payments made at the end of every quarter.
The first payment was made on March 31, 1985.
The effective annual interest rate was 4%.
What was the amount of interest paid in 1986?
Solution.
The loan is repaid over 4 years with quarterly payments, so the number of payments is

$$4(4) = 16.$$

The effective annual interest rate is 4%. Therefore, the effective quarterly interest rate is

$$j = (1.04)^{1/4} - 1.$$

Thus,

$$j = 0.0098534.$$

Let R be the quarterly payment. The loan equation is

$$5000 = Ra_{\overline{16}|j}.$$

Therefore,

$$R = \frac{5000}{a_{\overline{16}|j}}.$$

Using

$$j = (1.04)^{1/4} - 1,$$

we get

$$R = 339.3144271.$$

The first four payments are made in 1985:

$$1, 2, 3, 4.$$

The next four payments are made in 1986:

$$5, 6, 7, 8.$$

Thus, we need the total interest paid in payments 5 through 8.
Using the BA II Plus AMORT worksheet, enter:

$$\begin{aligned} N &= 16, \\ I/Y &= 100\left((1.04)^{1/4} - 1\right), \\ PV &= -5000, \\ FV &= 0. \end{aligned}$$

Then compute:

$$CPT\ PMT = 339.3144271.$$

Now use the AMORT worksheet with

$$P1 = 5, \qquad P2 = 8.$$

The calculator gives

$$INT = 132.7094727.$$

Therefore, the amount of interest paid in 1986 is

$$\boxed{132.71}.$$

Exercise 3.1.4

Todd has a 100000 25-year mortgage with a 12% nominal interest rate convertible monthly. The first payment is due one month after the mortgage is taken out. Twelve years after taking out the mortgage, after making his 144th payment, he refinances with a new nominal interest rate of 8%, again convertible monthly.
The new mortgage will be paid off on the same date as the original one.
Calculate the difference in the monthly mortgage payment after refinancing.

A. Less than 170
B. At least 170, but less than 190
C. At least 190, but less than 210
D. At least 210, but less than 230
E. At least 230

Solution.
The original mortgage is for 25 years with monthly payments, so the total number of payments is

$$25(12) = 300.$$

The original nominal annual interest rate is 12%, convertible monthly. Therefore, the original monthly interest rate is

$$j_1 = \frac{0.12}{12} = 0.01.$$

Let X be the original monthly mortgage payment. Then

$$100000 = Xa_{\overline{300}|0.01}.$$

Thus,

$$X = \frac{100000}{a_{\overline{300}|0.01}}.$$

Using

$$a_{\overline{300}|0.01} = \frac{1-(1.01)^{-300}}{0.01},$$

we get

$$X = 1053.22.$$

After 12 years, Todd has made

$$12(12) = 144$$

payments.
Therefore, the number of payments remaining is

$$300 - 144 = 156.$$

The outstanding balance immediately after the 144th payment is the present value of the remaining 156 payments at the original monthly rate:

$$OB_{144} = 1053.22a_{\overline{156}|0.01}.$$

Thus,

$$OB_{144} = 83018.22.$$

Todd refinances this balance at a new nominal annual interest rate of 8%, convertible monthly. Therefore, the new monthly interest rate is

$$j_2 = \frac{0.08}{12}.$$

Let Y be the new monthly payment. Since the new mortgage will be paid off on the same date as the original one, there are still 156 payments remaining.
Thus,

$$83018.22 = Ya_{\overline{156}|j_2}.$$

So,

$$Y = \frac{83018.22}{a_{\overline{156}|0.08/12}}.$$

Using

$$a_{\overline{156}|0.08/12} = \frac{1-(1+0.08/12)^{-156}}{0.08/12},$$

we get

$$Y = 857.64.$$

Therefore, the difference in monthly mortgage payments is

$$1053.22 - 857.64 = 195.58.$$

Thus, the correct answer is

$$\boxed{\text{C. At least 190, but less than 210}}.$$

Exercise 3.1.5

A 3000 loan at 18% per year is being repaid by 10 equal annual payments, the first payment due one year after the loan is made.
Determine how much the interest part of the sixth payment is.

A. Less than 300
B. At least 300, but less than 320
C. At least 320, but less than 340
D. At least 340, but less than 360
E. At least 360

Solution.
First, find the level annual payment R.
The loan is repaid with 10 equal annual payments at an annual effective interest rate of 18%, so

$$3000 = Ra_{\overline{10}|0.18}.$$

Thus,

$$R = \frac{3000}{a_{\overline{10}|0.18}}.$$

Using

$$a_{\overline{10}|0.18} = \frac{1-(1.18)^{-10}}{0.18},$$

we get

$$R = 667.543924.$$

The interest part of the sixth payment is the interest due during year 6.
Therefore, we need the outstanding balance immediately after the fifth payment, OB_5.
Using the prospective method, after the fifth payment there are

$$10 - 5 = 5$$

payments remaining.
Hence,

$$OB_5 = Ra_{\overline{5}|0.18}.$$

So,

$$OB_5 = 667.543924a_{\overline{5}|0.18}.$$

Using

$$a_{\overline{5}|0.18} = \frac{1-(1.18)^{-5}}{0.18},$$

we get

$$OB_5 = 2087.524.$$

The interest part of the sixth payment is

$$I_6 = OB_5(0.18).$$

Thus,

$$I_6 = 2087.524(0.18) = 375.754.$$

Therefore,

$$I_6 = 375.75.$$

So the correct answer is

$$\boxed{\text{E. At least 360}}.$$

BA II Plus AMORT method.
Enter:

$$\begin{aligned} N &= 10, \\ I/Y &= 18, \\ PV &= 3000, \\ FV &= 0. \end{aligned}$$

Then compute:

$$CPT\ PMT = -667.543924.$$

Now use the AMORT worksheet:

$$P1 = 6, \qquad P2 = 6.$$

The calculator gives

$$INT = -375.7543226.$$

The negative sign is due to the calculator cash-flow convention. The amount of interest paid is the magnitude:

$$\boxed{375.75}.$$

Exercise 3.1.6

Kevin takes out a 10-year loan of L, which he repays by the amortization method at an annual effective interest rate of i.
Kevin makes payments of 1000 at the end of each year.
The total amount of interest repaid during the life of the loan is also equal to L.
Calculate the amount of interest repaid during the first year of the loan.

A. 725 B. 750

C. 755 D. 760

E. 765

Solution.
Kevin makes 10 annual payments of 1000, so the total amount paid over the life of the loan is

$$10(1000) = 10000.$$

The total interest paid is equal to total payments minus the original loan amount:

$$\text{Total interest} = 10000 - L.$$

We are told that the total interest paid is also equal to L. Therefore,

$$10000 - L = L.$$

Thus,

$$2L = 10000,$$

and so

$$L = 5000.$$

Now use the loan equation:

$$5000 = 1000a_{\overline{10}|i}.$$

Therefore,

$$a_{\overline{10}|i} = 5.$$

So i satisfies

$$\frac{1-(1+i)^{-10}}{i} = 5.$$

Using a financial calculator:

$$N = 10, \qquad PV = 5000, \qquad PMT = -1000, \qquad FV = 0.$$

Then

$$CPT\ I/Y = 15.09841448.$$

Thus,

$$i = 0.1509841448.$$

The interest paid during the first year is based on the original loan balance:

$$I_1 = Li.$$

Therefore,

$$I_1 = 5000(0.1509841448).$$

So,

$$I_1 = 754.92.$$

The closest answer is

$$\boxed{\text{C. } 755}.$$

3.1.2 Amortization Schedule – Part 2

Level Payments

The most common loan type is one with level loan payments R and a level interest rate i.

Consider a loan of amount L, repaid with n level payments of R, where the first payment is made one period after the loan is issued.

The equation of value at time 0 is

$$L = Ra_{\overline{n}|}.$$

After the payment at time t, the outstanding loan balance can be found by either the prospective or retrospective method; the two equivalent formulas are summarized in the formula box below.

Outstanding balance for level payments

For a level-payment loan with original loan amount L, payment R, interest rate i, and n payments, the outstanding balance after payment t is

$$\boxed{OB_t = Ra_{\overline{n-t}|}}$$

by the prospective method, and

$$\boxed{OB_t = L(1+i)^t - Rs_{\overline{t}|}}$$

by the retrospective method.

Interest and Principal in the t-th Payment

The interest paid in the payment at time t is

$$I_t = OB_{t-1}i.$$

For level payments,

$$OB_{t-1} = Ra_{\overline{n-t+1|}}.$$

Therefore,

$$I_t = Ra_{\overline{n-t+1|}}i.$$

Using

$$a_{\overline{n-t+1|}} = \frac{1-v^{n-t+1}}{i},$$

we get

$$I_t = R\left(\frac{1-v^{n-t+1}}{i}\right)i.$$

Thus,

$$I_t = R(1-v^{n-t+1}).$$

The principal paid in the payment at time t is

$$P_t = R - I_t.$$

Hence,

$$P_t = R - R(1-v^{n-t+1}).$$

So,

$$P_t = Rv^{n-t+1}.$$

Principal paid in the t-th payment

For a level-payment loan,

$$\boxed{P_t = Rv^{n-t+1}}$$

where R is the level payment, n is the total number of payments, and t is the payment number.

Important interpretation

The formula

$$P_t = Rv^{n-t+1}$$

is not a present value calculation.
It is a shortcut formula for the amount of principal repaid in the t-th payment of a level-payment loan.

Level-payment loan calculations

A loan of 100000 is to be repaid in 30 equal annual payments starting one year from now. If the annual effective interest rate is 5%, calculate the following without using the AMORT worksheet:

1. OB_9
2. P_9
3. I_9

Solution.
First, find the level annual payment R.
The equation of value at time 0 is

$$100000 = Ra_{\overline{30}|0.05}.$$

Thus,

$$R = \frac{100000}{a_{\overline{30}|0.05}}.$$

Using

$$a_{\overline{30}|0.05} = \frac{1-(1.05)^{-30}}{0.05},$$

we get

$$R = 6505.14.$$

Part 1: Find OB_9.
Using the prospective method, after the 9th payment there are

$$30 - 9 = 21$$

payments remaining.
Therefore,

$$OB_9 = 6505.14a_{\overline{21}|0.05}.$$

Thus,

$$OB_9 = 83403.39.$$

Using the retrospective method,

$$OB_9 = 100000(1.05)^9 - 6505.14s_{\overline{9}|0.05}.$$

Thus,

$$OB_9 = 83403.48.$$

The small difference is due to rounding the payment R.

$$\boxed{OB_9 \approx 83403.4}$$

Part 2: Find P_9.
For a level-payment loan,

$$P_t = Rv^{n-t+1}.$$

Here,

$$n = 30, \qquad t = 9.$$

Therefore,

$$P_9 = 6505.14v^{30-9+1}.$$

So,

$$P_9 = 6505.14v^{22}.$$

Since
$$v = \frac{1}{1.05},$$
we get
$$P_9 = 6505.14(1.05)^{-22}.$$
Thus,
$$P_9 = 2223.78.$$

$$\boxed{P_9 = 2223.78}$$

Part 3: Find I_9.
The payment is split into interest and principal:
$$R = I_9 + P_9.$$
Thus,
$$I_9 = R - P_9.$$
Therefore,
$$I_9 = 6505.14 - 2223.78.$$
So,
$$I_9 = 4281.36.$$

$$\boxed{I_9 = 4281.36}$$

Using the BA II Plus AMORT worksheet

A loan of 100000 is to be repaid in 30 equal annual payments starting one year from now. If the annual effective interest rate is 5%, calculate the following using the AMORT worksheet:

1. OB_9
2. P_9
3. I_9
4. P_{10-15}
5. I_{10-15}
6. Total interest paid over the life of the loan

Solution.
First enter the loan information:
$$\begin{aligned} N &= 30, \\ I/Y &= 5, \\ PV &= -100000, \\ FV &= 0. \end{aligned}$$
Then compute:
$$CPT\ PMT = 6505.143508.$$

For payment 9, use the AMORT worksheet with

$$P1 = 9, \qquad P2 = 9.$$

The calculator gives:

$$BAL = -83403.4383,$$

$$PRN = 2223.78247,$$

$$INT = 4281.361038.$$

Thus,

$$OB_9 = 83403.44,$$

$$P_9 = 2223.78,$$

and

$$I_9 = 4281.36.$$

For payments 10 through 15, use

$$P1 = 10, \qquad P2 = 15.$$

The calculator gives:

$$PRN = 15882.2732,$$

$$INT = 23148.58785.$$

Thus,

$$P_{10-15} = 15882.27,$$

and

$$I_{10-15} = 23148.59.$$

For the total interest over the life of the loan, use

$$P1 = 1, \qquad P2 = 30.$$

The calculator gives:

$$INT = 95154.30524.$$

So the total interest paid is

$$95154.31.$$

Alternatively, the total interest can be calculated as

$$\text{total payments} - \text{original loan amount}.$$

Thus,

$$30(6505.14) - 100000 = 95154.20.$$

The small difference is due to rounding the payment.

Why Memorize the Principal Formula?

The formula

$$P_t = Rv^{n-t+1}$$

is useful because sometimes the problem gives information about the principal or interest part of a specific payment.

It is also faster for problems that ask when the principal part of the payment exceeds the interest part.

First payment where principal exceeds interest

A loan of 100000 is to be repaid in 30 equal annual payments starting one year from now. If the annual effective interest rate is 5%, find the first payment in which the principal payment exceeds the interest payment.
Solution.
We want
$$P_t > I_t.$$
Since
$$I_t = R - P_t,$$
this becomes
$$P_t > R - P_t.$$
Thus,
$$2P_t > R.$$
Using
$$P_t = Rv^{n-t+1},$$
we get
$$2Rv^{n-t+1} > R.$$
Cancel R:
$$2v^{n-t+1} > 1.$$
Here,
$$n = 30.$$
Therefore,
$$2v^{31-t} > 1.$$
So,
$$v^{31-t} > \frac{1}{2}.$$
Taking logarithms,
$$(31-t)\ln(v) > \ln\left(\frac{1}{2}\right).$$
Since
$$0 < v < 1,$$
we have
$$\ln(v) < 0.$$
Therefore, when dividing by $\ln(v)$, the inequality reverses:
$$31 - t < \frac{\ln(1/2)}{\ln(v)}.$$
Using
$$v = \frac{1}{1.05},$$
we get
$$31 - t < 14.2067.$$

Thus,

$$t > 16.7933.$$

Therefore, the first integer value of t is

$$t = 17.$$

So the first payment in which the principal exceeds the interest is the

$$\boxed{\text{17th payment}}.$$

Compounding Principal Payments

For level payments, recall that

$$P_t = Rv^{n-t+1}.$$

Therefore,

$$P_1 = Rv^n.$$

The second principal payment is

$$P_2 = Rv^{n-1}.$$

But

$$Rv^{n-1} = Rv^n v^{-1}.$$

Since

$$v^{-1} = 1 + i,$$

we have

$$P_2 = P_1(1 + i).$$

Similarly,

$$P_3 = P_2(1 + i) = P_1(1 + i)^2.$$

More generally,

$$P_{t+k} = P_t(1 + i)^k.$$

Principal payments compound at the loan interest rate

For a level-payment loan,

$$\boxed{P_{t+k} = P_t(1 + i)^k}$$

where P_t is the principal part of the payment at time t.

We can also write the original loan amount in terms of the first principal payment:

$$L = P_1 + P_2 + \cdots + P_n.$$

Since the principal payments grow geometrically,

$$L = P_1 + P_1(1 + i) + \cdots + P_1(1 + i)^{n-1}.$$

Thus,

$$L = P_1 \left[1 + (1 + i) + \cdots + (1 + i)^{n-1}\right].$$

Therefore,

$$L = P_1 s_{\overline{n}|}.$$

Loan amount from first principal payment

For a level-payment loan,

$$\boxed{L = P_1 s_{\overline{n}|}}$$

where P_1 is the principal part of the first payment.

Finding the loan interest rate from principal and interest payments

For a loan with level payments of 1018.52, the principal repaid in the 5th payment is 297.30, and the interest paid in the 16th payment is 325.33.
Find the loan interest rate.
Solution.
We are given

$$R = 1018.52,$$

$$P_5 = 297.30,$$

and

$$I_{16} = 325.33.$$

Since

$$R = P_{16} + I_{16},$$

we have

$$P_{16} = R - I_{16}.$$

Thus,

$$P_{16} = 1018.52 - 325.33.$$

So,

$$P_{16} = 693.19.$$

Principal payments compound at the loan interest rate:

$$P_{16} = P_5(1+i)^{16-5}.$$

Therefore,

$$693.19 = 297.30(1+i)^{11}.$$

So,

$$(1+i)^{11} = \frac{693.19}{297.30}.$$

Taking the 11th root:

$$1 + i = \left(\frac{693.19}{297.30}\right)^{1/11}.$$

Therefore,

$$i = 0.08.$$

Thus, the loan interest rate is

$$\boxed{8\%}.$$

Non-Level Payments

For non-level payments, use first principles.

This means we should build the balance directly from the payment pattern rather than relying blindly on level-payment formulas.

Non-level payments

A loan has payments of

$$10, 9, 8, \ldots, 1$$

starting one year from now.
The loan interest rate is 5%.
Determine the following:

1. OB_3
2. I_4
3. P_4

Solution.
The payment pattern is:

t	1	2	3	4	5	6	7	8	9	10
R_t	10	9	8	7	6	5	4	3	2	1

Part 1: Find OB_3.
After the payment at time 3, the remaining payments are

$$7, 6, 5, 4, 3, 2, 1.$$

So OB_3 is the present value at time 3 of this decreasing annuity.
Thus,

$$OB_3 = (Da)_{\overline{7}|0.05}.$$

Using the decreasing annuity formula,

$$(Da)_{\overline{n}|} = \frac{n - a_{\overline{n}|}}{i},$$

we get

$$OB_3 = \frac{7 - a_{\overline{7}|0.05}}{0.05}.$$

Therefore,

$$OB_3 = 24.27.$$

$$\boxed{OB_3 = 24.27}$$

Part 2: Find I_4.
The interest paid in the fourth payment is based on the outstanding balance after the third payment:

$$I_4 = OB_3(0.05).$$

Thus,

$$I_4 = 24.27(0.05) = 1.21.$$

$$\boxed{I_4 = 1.21}$$

Part 3: Find P_4.
The fourth payment is

$$R_4 = 7.$$

Therefore,

$$P_4 = R_4 - I_4.$$

So,

$$P_4 = 7 - 1.21.$$

Thus,

$$P_4 = 5.79.$$

$$\boxed{P_4 = 5.79}$$

Level versus non-level payments

For level-payment loans, formulas such as

$$OB_t = Ra_{\overline{n-t}|}$$

and

$$P_t = Rv^{n-t+1}$$

are very efficient.
For non-level-payment loans, focus on the actual remaining cash flows and use the basic ideas:

$$OB_t = \text{present value of remaining payments},$$

$$I_t = OB_{t-1}i,$$

and

$$P_t = R_t - I_t.$$

Exercise 3.1.7

A loan is repaid with 10 annual payments of 1295.05, with the first payment one year after the loan is issued.
The loan rate is an annual effective rate of 5%.
The interest portion of the n-th payment is 328.67.
Determine n.
Solution.
To avoid confusion, let N be the total number of payments and let t be the payment number we want to find.
Here,

$$N = 10.$$

The level annual payment is

$$R = 1295.05.$$

The annual effective interest rate is

$$i = 0.05,$$

so

$$v = \frac{1}{1.05}.$$

We are given that the interest portion of the t-th payment is

$$I_t = 328.67.$$

Since

$$R = I_t + P_t,$$

the principal portion of the t-th payment is

$$P_t = 1295.05 - 328.67.$$

Thus,

$$P_t = 966.38.$$

For a level-payment loan, the principal portion of the t-th payment is

$$P_t = Rv^{N-t+1}.$$

Therefore,

$$966.38 = 1295.05v^{10-t+1}.$$

So,

$$966.38 = 1295.05v^{11-t}.$$

Substituting

$$v = \frac{1}{1.05},$$

we get

$$966.38 = 1295.05(1.05)^{-(11-t)}.$$

Solving gives

$$t = 5.$$

Therefore, the interest portion of 328.67 occurs in the 5th payment.
Hence,

$$\boxed{n = 5}.$$

Exercise 3.1.8

Joe negotiates an 8-year loan which requires him to pay 1200 per month for the first 4 years and 1500 per month for the remaining years.
The interest rate is 13%, convertible monthly, and the first payment is due in one month. Determine the amount of principal in the 17th payment.
Solution.
The nominal annual interest rate is 13%, convertible monthly. Therefore, the monthly interest rate is

$$j = \frac{0.13}{12}.$$

The loan lasts 8 years, so the total number of monthly payments is

$$8(12) = 96.$$

The first 4 years correspond to the first

$$4(12) = 48$$

monthly payments.
Thus, payments 1 through 48 are 1200, and payments 49 through 96 are 1500.
We want the amount of principal in the 17th payment.
The principal portion of the 17th payment is

$$P_{17} = R_{17} - I_{17}.$$

Since the 17th payment is still in the first 48 payments,

$$R_{17} = 1200.$$

To find I_{17}, we need the outstanding balance immediately after the 16th payment, denoted OB_{16}.
After the 16th payment, there are

$$96 - 16 = 80$$

payments remaining.
One way to value the remaining payments is to imagine that all 80 remaining payments are 1500, and then subtract 300 from the first $48 - 16 = 32$ payments.
Therefore,

$$OB_{16} = 1500a_{\overline{80}|j} - 300a_{\overline{32}|j}.$$

Substituting

$$j = \frac{0.13}{12},$$

we get

$$OB_{16} = 1500a_{\overline{80}|0.13/12} - 300a_{\overline{32}|0.13/12}.$$

Thus,

$$OB_{16} = 71911.10.$$

The interest portion of the 17th payment is

$$I_{17} = OB_{16}j.$$

So,

$$I_{17} = 71911.10\left(\frac{0.13}{12}\right).$$

Therefore,

$$I_{17} = 779.04.$$

Finally,

$$P_{17} = R_{17} - I_{17}.$$

Thus,

$$P_{17} = 1200 - 779.04.$$

So,

$$P_{17} = 420.96.$$

Therefore, the amount of principal in the 17th payment is

$$\boxed{420.96}.$$

Exercise 3.1.9

Larry is repaying a loan with payments of 2500 at the end of every two years.
If the amount of interest in the fourth installment is 2458, find the amount of principal in the seventh installment.
Assume an annual effective interest rate of 13%.
Solution.
Payments are made every two years, so we first find the effective interest rate per two-year payment period.

$$j = (1.13)^2 - 1.$$

Thus,

$$j = 0.2769.$$

Each installment is

$$2500.$$

We are told that the interest portion of the fourth installment is

$$I_4 = 2458.$$

Therefore, the principal portion of the fourth installment is

$$P_4 = 2500 - 2458.$$

So,

$$P_4 = 42.$$

For a level-payment loan, principal payments grow at the effective interest rate per payment period.
Thus,

$$P_{t+k} = P_t(1 + j)^k.$$

We want P_7. Since the seventh installment is three payment periods after the fourth installment,

$$P_7 = P_4(1 + j)^3.$$

Substitute $P_4 = 42$ and $1 + j = 1.2769$:

$$P_7 = 42(1.2769)^3.$$

Therefore,

$$P_7 = 87.44.$$

Thus, the amount of principal in the seventh installment is

$$\boxed{87.44}.$$

Exercise 3.1.10

A loan is repaid in ten equal annual installments with the first installment paid one year after the loan is made.
The effective annual interest rate is 4%.
The total amount of principal repaid in the fifth, sixth, and seventh payments combined is 6083.

What is the total amount of interest paid in the second, third, and fourth payments combined?

A. Less than 2000
B. At least 2000, but less than 2020
C. At least 2020, but less than 2040
D. At least 2040, but less than 2060
E. At least 2060

Solution.
For a level-payment loan, the principal portions of the payments grow at the loan interest rate.
Since the effective annual interest rate is 4%, we have

$$P_{t+1} = P_t(1.04).$$

Thus,

$$P_5 = P_1(1.04)^4,$$
$$P_6 = P_1(1.04)^5,$$

and

$$P_7 = P_1(1.04)^6.$$

We are given

$$P_5 + P_6 + P_7 = 6083.$$

Therefore,

$$P_1(1.04)^4 + P_1(1.04)^5 + P_1(1.04)^6 = 6083.$$

Factor out P_1:

$$P_1\left[(1.04)^4 + (1.04)^5 + (1.04)^6\right] = 6083.$$

Solving gives

$$P_1 = 1665.74.$$

The original loan amount is the sum of all principal repayments:

$$L = P_1 + P_2 + \cdots + P_{10}.$$

Since the principal payments grow geometrically,

$$L = P_1 s_{\overline{10}|0.04}.$$

Thus,

$$L = 1665.74 s_{\overline{10}|0.04}.$$

Using

$$s_{\overline{10}|0.04} = \frac{(1.04)^{10} - 1}{0.04},$$

we get

$$L = 20000.$$

Now find the level payment R. Since the loan is repaid with 10 equal annual payments,

$$20000 = Ra_{\overline{10}|0.04}.$$

Therefore,

$$R = \frac{20000}{a_{\overline{10}|0.04}}.$$

This gives

$$R = 2465.82.$$

We want the total interest paid in the second, third, and fourth payments:

$$I_2 + I_3 + I_4.$$

Since

$$I_t = R - P_t,$$

we have

$$I_2 + I_3 + I_4 = 3R - (P_2 + P_3 + P_4).$$

Now,

$$P_2 = P_1(1.04),$$

$$P_3 = P_1(1.04)^2,$$

and

$$P_4 = P_1(1.04)^3.$$

Therefore,

$$I_2 + I_3 + I_4 = 3(2465.82) - 1665.74\left[(1.04) + (1.04)^2 + (1.04)^3\right].$$

Thus,

$$I_2 + I_3 + I_4 = 1989.44.$$

So the correct answer is

$$\boxed{\text{A. Less than 2000}}.$$

Exercise 3.1.11

Sam borrowed 1000 on January 1, 1993, to be repaid by level payments every two years beginning January 1, 1995, at an effective annual rate of interest of 9%.
The amount of interest in the fourth installment is 177.72.
Determine the amount of principal in the sixth installment.

A. Less than 30

B. At least 30, but less than 35

C. At least 35, but less than 40

D. At least 40, but less than 45

E. At least 45

Solution.
Payments are made every two years, so we work with the effective two-year interest rate.
The annual effective interest rate is 9%, so the effective rate per two-year payment period is

$$j = (1.09)^2 - 1.$$

Thus,

$$j = 0.1881.$$

Let R be the level installment payment.
The fourth installment is paid at time 8, so the interest portion of the fourth installment is based on the outstanding balance immediately after the third installment.

Using the retrospective method, the outstanding balance after the third installment is

$$OB_3 = 1000(1+j)^3 - Rs_{\overline{3}|j}.$$

Therefore, the interest portion of the fourth installment is

$$I_4 = OB_3 j.$$

We are given

$$I_4 = 177.72.$$

So,

$$177.72 = \left(1000(1+j)^3 - Rs_{\overline{3}|j}\right) j.$$

Substitute $j = 0.1881$:

$$177.72 = \left(1000(1.1881)^3 - Rs_{\overline{3}|0.1881}\right)(0.1881).$$

Solving gives

$$R = 203.43.$$

The principal portion of the fourth installment is

$$P_4 = R - I_4.$$

Thus,

$$P_4 = 203.43 - 177.72 = 25.71.$$

For a level-payment loan, the principal portions grow at the effective interest rate per payment period.
Therefore,

$$P_6 = P_4(1+j)^2.$$

So,

$$P_6 = 25.71(1.1881)^2.$$

Hence,

$$P_6 = 36.29.$$

Therefore, the amount of principal in the sixth installment is

$$\boxed{36.29}.$$

The correct answer is

$$\boxed{\text{C. At least 35, but less than 40}}.$$

Exercise 3.1.12

Donald takes out a loan to be repaid with annual payments of 500 at the end of each year for $2n$ years.
The annual effective interest rate is 4.94%.
The sum of the interest paid in year 1 plus the interest paid in year $n+1$ is equal to 720.
Calculate the amount of interest paid in year 10.

A. 318 B. 335

C. 364 D. 376

E. 388

Solution.
The loan is repaid with level annual payments of

$$R = 500.$$

The total number of payments is

$$2n.$$

For a level-payment amortized loan, the principal portion of the t-th payment is

$$P_t = Rv^{2n-t+1}.$$

Therefore,

$$P_t = 500v^{2n-t+1}.$$

Since each payment is split into interest and principal,

$$I_t = R - P_t.$$

Thus,

$$I_t = 500 - 500v^{2n-t+1}.$$

We are given that

$$I_1 + I_{n+1} = 720.$$

Now,

$$I_1 = 500 - 500v^{2n},$$

and

$$I_{n+1} = 500 - 500v^n.$$

Therefore,

$$720 = \left(500 - 500v^{2n}\right) + \left(500 - 500v^n\right).$$

So,

$$720 = 1000 - 500v^{2n} - 500v^n.$$

Rearranging,

$$500v^{2n} + 500v^n = 280.$$

Divide by 500:

$$v^{2n} + v^n = 0.56.$$

Let

$$x = v^n.$$

Then

$$x^2 + x = 0.56.$$

So,

$$x^2 + x - 0.56 = 0.$$

Using the quadratic formula,

$$x = \frac{-1 + \sqrt{1^2 - 4(1)(-0.56)}}{2}.$$

Thus,

$$x = 0.4.$$

Therefore,

$$v^n = 0.4.$$

We need the interest paid in year 10:

$$I_{10} = 500 - 500v^{2n-10+1}.$$

So,

$$I_{10} = 500 - 500v^{2n-9}.$$

Now,

$$v^{2n-9} = v^{2n}v^{-9}.$$

Since

$$v^n = 0.4,$$

we have

$$v^{2n} = (v^n)^2 = (0.4)^2 = 0.16.$$

Also,

$$v^{-9} = (1+i)^9.$$

Since

$$i = 0.0494,$$

we get

$$v^{-9} = (1.0494)^9.$$

Therefore,

$$I_{10} = 500 - 500(0.16)(1.0494)^9.$$

Thus,

$$I_{10} = 376.53.$$

The closest answer is

$$\boxed{\text{D. } 376}.$$

3.1.3 Drop Payment vs. Balloon Payment

Sometimes a loan with level payments has a final payment that is smaller or larger than the regular level payment.

Drop payment and balloon payment

If the last payment is smaller than the regular level payment, it is called a **drop payment**. If the last payment is larger than the regular level payment, it is called a **balloon payment**.

The idea is simple:

$$\text{drop payment} < \text{regular level payment}$$

and

$$\text{balloon payment} > \text{regular level payment}.$$

Example: Drop Payment

Drop payment

A 3000 loan at an annual effective interest rate of 6% is repaid with annual payments of 750, plus a smaller final drop payment.
Determine the amount of the drop payment.
Solution.
First, find how many full payments of 750 would be needed if all payments were level.
We solve

$$750a_{\overline{n}|0.06} = 3000.$$

Thus,

$$a_{\overline{n}|0.06} = 4.$$

Solving gives

$$n = 4.71.$$

This means that four full payments of 750 are made, followed by a smaller drop payment at time 5.
Let the drop payment be X. The timeline is

t	0	1	2	3	4	5
Payment		750	750	750	750	X

Using an equation of value at time 0,

$$3000 = 750a_{\overline{4}|0.06} + Xv^5.$$

Therefore,

$$Xv^5 = 3000 - 750a_{\overline{4}|0.06}.$$

So,

$$X = \frac{3000 - 750a_{\overline{4}|0.06}}{v^5}.$$

Using

$$v = \frac{1}{1.06},$$

we get

$$X = 536.86.$$

Thus, the drop payment is

$$\boxed{536.86}.$$

Alternative method.
A useful method is to find the outstanding balance after the fourth regular payment.
Using the retrospective method,

$$OB_4 = 3000(1.06)^4 - 750s_{\overline{4}|0.06}.$$

Thus,

$$OB_4 = 506.47.$$

The drop payment is made one year later, so it must pay off this balance with one year of interest:

$$X = OB_4(1.06).$$

Therefore,

$$X = 506.47(1.06) = 536.86.$$

So again,

$$\boxed{X = 536.86}.$$

Example: Balloon Payment

Balloon payment

A 3000 loan at an annual effective interest rate of 6% is repaid with three annual payments of 750, plus a balloon payment at the end of year 4.
Determine the amount of the balloon payment.
Solution.
Let the balloon payment be Y.
The cash flows are

t	0	1	2	3	4
Payment		750	750	750	Y

Using an equation of value at time 0,

$$3000 = 750a_{\overline{3}|0.06} + Yv^4.$$

Therefore,

$$Yv^4 = 3000 - 750a_{\overline{3}|0.06}.$$

So,

$$Y = \frac{3000 - 750a_{\overline{3}|0.06}}{v^4}.$$

Using

$$v = \frac{1}{1.06},$$

we get

$$Y = 1256.47.$$

Thus, the balloon payment is

$$\boxed{1256.47}.$$

Alternative method.
Suppose the borrower paid 750 for four years.
The outstanding balance after the fourth regular payment would be

$$3000(1.06)^4 - 750s_{\overline{4}|0.06}.$$

From the previous example, this balance is

$$506.47.$$

Since the borrower only makes three regular payments before the final payment, the final payment must include the regular payment of 750 plus the remaining balance of 506.47. Therefore,

$$Y = 750 + 506.47.$$

So,

$$Y = 1256.47.$$

Thus,

$$\boxed{Y = 1256.47}.$$

How to recognize the situation

If the problem says payments are made until a final payment smaller than the regular payment, use a **drop payment**.
If the problem says regular payments are made for a fixed number of periods and then a larger final payment is made, use a **balloon payment**.

Practical method

For both drop payments and balloon payments, a very reliable method is:

$$\boxed{\text{Find the outstanding balance just before the final payment.}}$$

Then accumulate that balance to the final payment date if necessary.

Exercise 3.1.13

A 1000 loan at 3.5% annual effective rate of interest can be repaid by either:

1. 11 annual payments of 100 plus a final balloon payment, or
2. 12 annual payments of 100 plus a final drop payment.

Determine the difference between the balloon payment and the drop payment.
Solution.
Suppose that 100 is paid for 12 years.
The outstanding balance immediately after the 12th payment is

$$OB_{12} = 1000(1.035)^{12} - 100s_{\overline{12}|0.035}.$$

Using

$$s_{\overline{12}|0.035} = \frac{(1.035)^{12} - 1}{0.035},$$

we get

$$OB_{12} = 50.87.$$

Balloon payment.
Under the balloon payment option, the borrower makes 11 payments of 100, followed by a final balloon payment at time 12.
If a regular payment of 100 were made at time 12, the remaining balance after that payment would be 50.87.
Therefore, the balloon payment must include both the regular payment of 100 and the remaining balance of 50.87.

Thus,

$$\text{Balloon payment} = 100 + 50.87.$$

So,

$$\text{Balloon payment} = 150.87.$$

Drop payment.
Under the drop payment option, the borrower makes 12 payments of 100, followed by a smaller final drop payment.
After the 12th payment, the outstanding balance is

$$50.87.$$

This balance must be accumulated one more year to the final drop payment date.
Therefore,

$$\text{Drop payment} = 50.87(1.035).$$

So,

$$\text{Drop payment} = 52.65.$$

The difference between the balloon payment and the drop payment is

$$150.87 - 52.65 = 98.22.$$

Thus, the difference is

$$\boxed{98.22}.$$

Exercise 3.1.14

For the previous exercise, calculate the absolute difference between the total interest paid in the two repayment options.
Solution.
Recall from the previous exercise:

$$\text{Balloon payment} = 150.87,$$

and

$$\text{Drop payment} = 52.65.$$

The total interest paid is equal to

$$\text{total interest} = \text{total loan payments} - \text{loan amount}.$$

Balloon payment option.
Under the balloon payment option, the borrower makes 11 payments of 100, plus a balloon payment of 150.87.
Thus, the total amount paid is

$$100(11) + 150.87.$$

Therefore, the total interest paid is

$$100(11) + 150.87 - 1000 = 250.87.$$

Drop payment option.

Under the drop payment option, the borrower makes 12 payments of 100, plus a drop payment of 52.65.
Thus, the total amount paid is

$$100(12) + 52.65.$$

Therefore, the total interest paid is

$$100(12) + 52.65 - 1000 = 252.65.$$

The absolute difference between the total interest paid in the two options is

$$|252.65 - 250.87| = 1.78.$$

Thus, the absolute difference is

$$\boxed{1.78}.$$

Exercise 3.1.15

A loan of 10000 will be paid back in N months.
Payments of 250 will be made at the end of each month for the first $N-1$ months. At the end of the N-th month, a smaller payment of P will be made.
Assuming an interest rate of 12% per annum convertible monthly, what is the amount of the last payment P?

A. $P \leq 80$　B. $80 < P \leq 90$
C. $90 < P \leq 100$　D. $100 < P \leq 110$
E. $110 < P$

Solution.
The nominal annual interest rate is 12%, convertible monthly. Therefore, the monthly interest rate is

$$i = \frac{0.12}{12} = 0.01.$$

First, determine how many full payments of 250 are needed.
We solve

$$250 a_{\overline{n}|0.01} = 10000.$$

Thus,

$$a_{\overline{n}|0.01} = 40.$$

Using

$$a_{\overline{n}|0.01} = \frac{1 - v^n}{0.01},$$

we get

$$40 = \frac{1 - v^n}{0.01}.$$

Therefore,

$$0.40 = 1 - v^n,$$

so

$$v^n = 0.60.$$

Solving gives

$$n = 51.33.$$

This means that 51 full payments of 250 are made, followed by a smaller final payment at month 52.
Thus, the equation of value at time 0 is

$$250a_{\overline{51}|0.01} + Pv^{52} = 10000.$$

Solving for P,

$$Pv^{52} = 10000 - 250a_{\overline{51}|0.01}.$$

Hence,

$$P = \frac{10000 - 250a_{\overline{51}|0.01}}{v^{52}}.$$

Using

$$v = \frac{1}{1.01},$$

we get

$$P = 84.67.$$

Therefore,

$$80 < P \leq 90.$$

The correct answer is

$$\boxed{\text{B. } 80 < P \leq 90}.$$

Exercise 3.1.16

Sally takes out a 7000 loan. The original terms of the loan require 60 equal monthly payments starting at the end of the first month at a nominal annual interest rate of 8% convertible monthly.
The lender offers an alternative payment scheme: Sally would make no payments for the first 5 months, followed by the same monthly payment as the original terms of the loan. She will still repay the entire loan in 60 months by making an extra payment at the end of the last month.
Determine the amount of the extra payment to be made at the end of the 60th month.

A. Less than 700
B. At least 700, but less than 800
C. At least 800, but less than 900
D. At least 900, but less than 1000
E. At least 1000

Solution.
The nominal annual interest rate is 8%, convertible monthly. Therefore, the monthly interest rate is

$$i = \frac{0.08}{12}.$$

First, find the original monthly payment. Let this payment be X.
The original loan is repaid with 60 equal monthly payments, so

$$7000 = Xa_{\overline{60}|0.08/12}.$$

Thus,

$$X = \frac{7000}{a_{\overline{60}|0.08/12}}.$$

Using

$$a_{\overline{60}|0.08/12} = \frac{1 - \left(1 + \frac{0.08}{12}\right)^{-60}}{0.08/12},$$

we get

$$X = 141.935.$$

Under the alternative payment scheme, Sally makes no payments for the first 5 months. Then she makes payments of 141.935 from month 6 through month 60. This gives

$$60 - 5 = 55$$

regular monthly payments.
Let Y be the extra payment made at the end of month 60.
Now write an equation of value at time 60.
The accumulated value of the original loan at time 60 is

$$7000\left(1 + \frac{0.08}{12}\right)^{60}.$$

The accumulated value of the 55 regular payments from months 6 through 60 is

$$141.935 s_{\overline{55}|0.08/12}.$$

The extra payment Y is also made at time 60.
Therefore,

$$7000\left(1 + \frac{0.08}{12}\right)^{60} = 141.935 s_{\overline{55}|0.08/12} + Y.$$

Solving for Y,

$$Y = 7000\left(1 + \frac{0.08}{12}\right)^{60} - 141.935 s_{\overline{55}|0.08/12}.$$

Thus,

$$Y = 1036.47.$$

Therefore, the extra payment is

$$\boxed{1036.47}.$$

The correct answer is

$$\boxed{\text{E. At least 1000}}.$$

Exercise 3.1.17

A 10000 loan is being paid off by annual payments of 2000 plus a smaller final payment. If the annual effective rate of interest is 15%, and the first payment is made one year after the time of the loan, find the amount of interest, X, contained in the fifth payment.

A. $X \leq 800$ B. $800 < X \leq 900$

C. $900 < X \leq 1000$ D. $1000 < X \leq 1100$

E. $1100 < X$

Solution.
The annual effective interest rate is

$$i = 0.15.$$

We want the amount of interest contained in the fifth payment.
The interest portion of the fifth payment is based on the outstanding balance immediately after the fourth payment.
So we first find OB_4.
Using the retrospective method,

$$OB_4 = 10000(1.15)^4 - 2000s_{\overline{4}|0.15}.$$

Using

$$s_{\overline{4}|0.15} = \frac{(1.15)^4 - 1}{0.15},$$

we get

$$OB_4 = 7503.3125.$$

The interest contained in the fifth payment is

$$I_5 = OB_4(0.15).$$

Thus,

$$I_5 = 7503.3125(0.15).$$

Therefore,

$$I_5 = 1125.50.$$

So,

$$X = 1125.50.$$

The correct answer is

$$\boxed{\text{E. } 1100 < X}.$$

Chapter 4

Bonds

4.1 Bonds and Other Securities

4.1.1 Types of Securities

A **security** is a financial instrument that represents a financial claim.

In this section, we introduce the main types of securities that appear in interest theory and financial mathematics.

- Bonds
- Preferred stock
- Common stock
- Money market instruments
- Mutual funds
- Derivative securities

How Corporations Raise Capital

Corporations usually raise capital in two main ways:

1. **Issue debt**
   - The corporation borrows money from investors.
   - A bond is the most common debt security.
2. **Issue equity**
   - Investors buy an ownership interest in the company.
   - There is no guaranteed return.
   - Common stock is the most common equity security.

Debt versus equity

Debt usually creates a contractual obligation to pay interest and repay principal.
Equity represents ownership in the company, but the return is not guaranteed.

Bonds

Bond

A **bond** is a debt security issued by a corporation or government.
The bond issuer borrows money from the bondholder and promises to make regular interest payments, called **coupons**, and to repay a redemption amount at maturity.

Bonds are issued by corporations and governments.

A bond typically has:

- a maturity date;
- coupon payments;
- a redemption amount paid at maturity.

Bond maturities are often between 1 and 30 years.

Technically, fixed-income securities may be classified by maturity:

- maturity of one year or less: **bill**;
- maturity greater than one year and up to ten years: **note**;
- maturity greater than ten years: **bond**.

Interest payments are tax deductible for the corporation.

In bankruptcy, bondholders are paid before stockholders.

Priority of claims

Bondholders have a stronger claim than stockholders because bonds are debt obligations.
Stockholders are residual claimants, meaning they are paid only after debt obligations have been met.

Types of Bonds

Common types of bonds include:

- **Fixed-rate bonds**: the interest rate is fixed over the life of the bond.
- **Floating-rate bonds**: the interest rate is allowed to fluctuate.
- **Zero-coupon bonds**: all interest is effectively paid at maturity, with no coupon payments during the life of the bond.

- **Callable bonds**: the bond issuer has the right to redeem the bond before maturity.
- **Convertible bonds**: the bondholder has the option to convert the bond into a fixed number of shares of common stock.

Default Risk

Default risk

Default risk is the risk that the bond issuer will not be able to make coupon payments or repay principal.

Government bonds are often treated as having very low default risk.

The yield on a bond with no default risk is called the **risk-free rate**.

Investment-grade bonds have a low expected default risk.

Junk bonds, also called high-yield bonds, have much greater default risk and therefore usually offer a higher yield to compensate investors for the extra risk.

Secured bonds have a senior claim over unsecured bonds, also called debentures.

Risk and yield

In general, investors demand a higher yield when default risk is higher.
This is why high-yield bonds usually have higher promised yields than investment-grade bonds.

Preferred Stock

Preferred stock

Preferred stock is a hybrid security between debt and equity.

Preferred stock has features of both bonds and common stock.

It usually pays fixed dividends, similar to bond coupons, but the dividends are paid at the discretion of the company.

Preferred dividends must be paid before common stock dividends.

Preferred stock usually has no maturity date, but it may sometimes be callable.

Preferred stockholders usually do not have voting rights.

In bankruptcy or liquidation, preferred stockholders are paid after bondholders but before common stockholders.

Common Stock

Common stock

Common stock represents an ownership share in the issuing company.

Common stock is a capital investment with no guaranteed return.

It is sometimes called simply **stock** or **equity**.

The owner of a share of stock may receive a share of the company's profits in the form of dividends.

Dividends are usually paid quarterly, but the amount of the dividend is at the discretion of the company and is often based on earnings.

Common stockholders have a residual claim. This means they only get paid after the obligations to debt holders and preferred stockholders have been met.

Common stockholders usually have voting rights.

Money Market Instruments

Money market instruments

Money market instruments are short-term, liquid, low-risk financial instruments.

Money market instruments are cash-like investments.

They are usually:

- short-term;
- liquid, meaning easy to buy and sell;
- relatively low-risk.

Examples include:

- Treasury bills;
- certificates of deposit;
- commercial paper.

Treasury Bills

Treasury bills, often called **T-bills**, are short-term debt securities issued by the government.

They usually have a maturity of less than one year.

The government pays the face amount at the end of the term.

T-bills do not make coupon payments.

Face amounts are usually round values, such as 100, 1000, or 10000.

T-bills are often quoted on a simple discount yield basis, using a 360-day year.

Treasury bill price

If n is the number of days to maturity, then

$$\boxed{\text{Price} = \text{Face Amount}\left[1 - \text{discount yield} \cdot \frac{n}{360}\right]}$$

Exam note

T-bill pricing is included here as background knowledge only.
It is not central to the current Exam FM syllabus.

Certificates of Deposit

A **certificate of deposit**, or CD, is a time deposit with a bank that cannot be withdrawn before maturity without restrictions or penalties.

At the end of the term, the investor receives principal plus interest.

The interest rate may be fixed or may vary over the term of the deposit.

Commercial Paper

Commercial paper is a short-term unsecured debt note issued by a corporation.

The term is usually less than 270 days, often one to two months.

Large companies sometimes use commercial paper to raise money instead of borrowing from a bank.

Mutual Funds

Mutual fund

A **mutual fund** pools together deposits from many investors and places those deposits under the control of a professional money manager.

Each investor receives a pro-rata share of the fund's income, capital gains, and losses.

Mutual funds invest in many different financial instruments.

There are two common strategies:

- **Active strategy**: the fund manager tries to outperform the market.
- **Passive strategy**: the fund manager tries to match a broad market index, such as the S&P 500.

Some mutual funds charge a **load**, which is a commission.

They also usually charge a management fee, often expressed as a percentage of assets under management.

Derivative Securities

Derivative security

A **derivative security** is a financial instrument whose value is based on the value of another financial variable or asset.

Derivative securities can be based on:

- stock prices;
- interest rates;
- market indices;
- exchange rates;
- commodities.

Examples include:

- options;
- futures;
- swaps.

Exercise 4.1.1

Name two ways corporations raise capital.
Solution.
Corporations usually raise capital in two main ways:

1. **Issue debt**, such as bonds.
2. **Issue equity**, such as common stock.

Debt means the corporation borrows money from investors and usually promises to make interest payments and repay principal.
Equity means investors buy an ownership interest in the company, but the return is not guaranteed.
Therefore, the two ways are

$$\boxed{\text{issue debt and issue equity.}}$$

Exercise 4.1.2

Company XYZ issues common stock, preferred stock, and bonds.
Rank the degree of security from most secure to least secure for the three financial instruments.
Solution.
The degree of security depends on the priority of claims.
In general, bondholders are paid before stockholders.
Preferred stockholders are paid after bondholders, but before common stockholders.
Common stockholders have the residual claim, meaning they are paid only after the obligations to bondholders and preferred stockholders have been met.
Therefore, from most secure to least secure, the ranking is:

1. **Bonds**
2. **Preferred stock**
3. **Common stock**

Thus,

$$\boxed{\text{Bonds} > \text{Preferred Stock} > \text{Common Stock}}$$

in terms of security.

Exercise 4.1.3

Which of the following correctly orders the length of maturity for the debt securities below, ordered from shortest maturity to longest maturity?

A. Note, Bond, Bill B. Bill, Note, Bond

C. Bond, Bill, Note D. Bill, Bond, Note

E. Note, Bill, Bond

Solution.
Debt securities are often classified by their time to maturity.
A **bill** has the shortest maturity:

$$\text{Bill: maturity of one year or less.}$$

A **note** has an intermediate maturity:

$$\text{Note: maturity greater than one year and up to ten years.}$$

A **bond** has the longest maturity:

$$\text{Bond: maturity greater than ten years.}$$

Therefore, from shortest maturity to longest maturity, the correct order is

$$\text{Bill, Note, Bond.}$$

Thus, the correct answer is

$$\boxed{\text{B. Bill, Note, Bond}}.$$

4.1.2 Price of a Bond

What Is a Bond?

Bond

A **bond** is a type of loan issued by a government or a corporation.
The bond issuer borrows money from investors and promises to make interest payments, called **coupons**, and to repay a redemption amount at maturity.

Most bonds have two main cash-flow components:

1. Regular coupon payments during the life of the bond.
2. A redemption payment at maturity.

A bond can therefore be viewed as an annuity of coupons plus a final lump-sum payment.

Basic bond pricing example

To fund a new road project, the City of New York issues bonds.
Each bond pays coupons of 25 every six months for 10 years and pays 1000 at the end of 10 years.
If the yield rate of the bond is 6% convertible semiannually, calculate the price of the bond.
Solution.
The bond pays coupons every six months for 10 years, so the number of coupon payments is

$$n = 10(2) = 20.$$

The yield rate is 6% convertible semiannually, so the yield rate per coupon period is

$$i = \frac{0.06}{2} = 0.03.$$

The bond price is the present value of the coupon payments plus the present value of the redemption amount:

$$P = 25a_{\overline{20}|0.03} + 1000v_{0.03}^{20}.$$

Thus,

$$P = 925.61.$$

Therefore, the price of the bond is

$$\boxed{925.61}.$$

Notation and Details

We use the following notation for bond pricing.

$$\begin{aligned}
P &: \text{ price of the bond,}\\
F &: \text{ face amount or par value of the bond,}\\
r &: \text{ coupon rate per coupon payment period,}\\
Fr &: \text{ amount of each coupon payment,}\\
C &: \text{ redemption value of the bond,}\\
g &: \text{ modified coupon rate,}\\
i &: \text{ yield rate per coupon payment period,}\\
n &: \text{ number of coupon payments,}\\
K &: \text{ present value of the redemption value.}
\end{aligned}$$

Face amount

The **face amount**, also called the **par value**, is the unit in which the bond is issued.
Common face amounts include 100, 1000, and 10000.
The face amount is used to determine the coupon payment.

The coupon rate is usually stated as an annual nominal rate.

Therefore, we must convert it to a coupon rate per payment period.

For example:

$$\begin{aligned} 6\% \text{ semiannual coupons} &\implies r = \tfrac{0.06}{2} = 0.03, \\ 8\% \text{ quarterly coupons} &\implies r = \tfrac{0.08}{4} = 0.02. \end{aligned}$$

The coupon payment is

$$\boxed{\text{Coupon payment} = Fr.}$$

For example, a 1000 par value bond with 6% semiannual coupons has coupon payment

$$Fr = 1000(0.03) = 30.$$

A 100 par value bond with 8% quarterly coupons has coupon payment

$$Fr = 100(0.02) = 2.$$

Redemption value

The **redemption value** C is the amount paid to the bondholder at maturity.
Unless stated otherwise, assume

$$C = F.$$

The **modified coupon rate** g is defined by

$$Fr = Cg.$$

Therefore,

$$\boxed{g = \frac{Fr}{C}.}$$

The yield rate i, also called the **yield-to-maturity**, is the investor's effective rate of return per coupon payment period.

Important distinction

The coupon rate r determines the coupon payment.
The yield rate i is the discount rate used to price the bond.
They are not necessarily equal.

The present value of the redemption value is denoted by

$$K.$$

Thus,

$$\boxed{K = Cv_i^n.}$$

Basic Bond Price Formula

A bond is priced as the present value of its coupons plus the present value of its redemption value.

The coupon payments form an annuity-immediate.

Therefore, the bond price is given by the basic present-value formula (see the formula box below).

Basic bond price formula

$$\boxed{P = Fra_{\overline{n}|i} + Cv_i^n}$$

where

$$v_i = \frac{1}{1+i}.$$

Since

$$K = Cv_i^n,$$

we can also write

$$\boxed{P = Fra_{\overline{n}|i} + K.}$$

Premium / Discount Formula

Start with the basic formula:

$$P = Fra_{\overline{n}|i} + Cv_i^n.$$

Recall that

$$a_{\overline{n}|i} = \frac{1 - v_i^n}{i}.$$

Therefore,

$$v_i^n = 1 - ia_{\overline{n}|i}.$$

Substitute this into the basic bond price formula:

$$P = Fra_{\overline{n}|i} + C\left(1 - ia_{\overline{n}|i}\right).$$

So,

$$P = Fra_{\overline{n}|i} + C - Cia_{\overline{n}|i}.$$

Therefore,

$$P = C + (Fr - Ci)a_{\overline{n}|i}.$$

Premium / discount formula

$$\boxed{P = C + (Fr - Ci)a_{\overline{n}|i}}$$

Since $Fr = Cg$, this can also be written as

$$P = C + (Cg - Ci)a_{\overline{n}|i}.$$

Factoring out C,

$$\boxed{P = C + C(g - i)a_{\overline{n}|i}.}$$

Premium and discount intuition

If $r > i$, then coupon payments are high relative to the yield rate, so the bond sells at a premium.
If $r < i$, then coupon payments are low relative to the yield rate, so the bond sells at a discount.

Makeham's Formula

Makeham's formula is another bond pricing formula.

It is less central, but it can be useful for some problems.

Start with the basic bond price formula and rewrite it in terms of

$$K = Cv_i^n$$

and the coupon rate

$$g$$

. After algebraic simplification, this yields Makeham's formula (see the formula box below).

Makeham's formula

$$\boxed{P = K + \frac{g}{i}(C - K)}$$

where

$$K = Cv_i^n.$$

Exam note

The basic bond price formula and the premium / discount formula are the most important formulas.
Makeham's formula is useful to recognize, but it is not usually the first formula to use unless the problem is clearly designed for it.

Example: Pricing a Bond Using Three Formulas

Pricing a bond

A 1000 par value ten-year bond has 8.4% semiannual coupons.
The bond is redeemable at 1050.
If the bond is bought to yield 10% convertible semiannually, calculate the price using all three bond formulas.

Solution.

We identify the given information.

$$F = 1000.$$

The annual coupon rate is 8.4%, payable semiannually. Therefore,

$$r = \frac{0.084}{2} = 0.042.$$

The number of coupon payments is

$$n = 10(2) = 20.$$

The redemption value is

$$C = 1050.$$

The yield rate is 10% convertible semiannually. Therefore,

$$i = \frac{0.10}{2} = 0.05.$$

The coupon payment is

$$Fr = 1000(0.042) = 42.$$

Also,

$$Ci = 1050(0.05) = 52.50.$$

The present value of the redemption value is

$$K = 1050v_{0.05}^{20}.$$

Thus,

$$K = 395.73.$$

Finally,

$$g = \frac{Fr}{C} = \frac{42}{1050} = 0.04.$$

Method 1: Basic formula.

$$P = Fra_{\overline{n}|i} + Cv_i^n.$$

Therefore,

$$P = 42a_{\overline{20}|0.05} + 1050v_{0.05}^{20}.$$

Thus,

$$P = 919.15.$$

$$\boxed{P = 919.15}$$

Method 2: Premium / discount formula.

$$P = C + (Fr - Ci)a_{\overline{n}|i}.$$

Therefore,

$$P = 1050 + (42 - 52.50)a_{\overline{20}|0.05}.$$

Thus,

$$P = 919.15.$$

$$\boxed{P = 919.15}$$

Method 3: Makeham's formula.

$$P = K + \frac{g}{i}(C - K).$$

Therefore,

$$P = 395.73 + \frac{0.04}{0.05}(1050 - 395.73).$$

Thus,

$$P = 919.15.$$

$$\boxed{P = 919.15}$$

All three methods give the same bond price.

Exercise 4.1.4

A 1000 par value twenty-year bond has coupons at 5% convertible semiannually.
If the yield rate is 6% convertible semiannually, what is the bond's price?

Solution.

The par value is

$$F = 1000.$$

The annual coupon rate is 5%, convertible semiannually. Therefore, the coupon rate per semiannual period is

$$r = \frac{0.05}{2} = 0.025.$$

Thus, the semiannual coupon payment is

$$Fr = 1000(0.025) = 25.$$

The bond lasts 20 years, and coupons are paid semiannually. Therefore, the number of coupon payments is

$$n = 20(2) = 40.$$

The yield rate is 6%, convertible semiannually. Therefore, the yield rate per coupon period is

$$i = \frac{0.06}{2} = 0.03.$$

Unless stated otherwise, the redemption value is assumed to equal the par value. Hence,

$$C = 1000.$$

Using the basic bond price formula,

$$P = Fra_{\overline{n}|i} + Cv_i^n.$$

Substitute the values:

$$P = 25a_{\overline{40}|0.03} + 1000v_{0.03}^{40}.$$

Therefore,

$$P = 884.43.$$

Thus, the bond's price is

$$\boxed{884.43}.$$

Exercise 4.1.5

A 9% bond with a 1000 par value and coupons payable semiannually is redeemable at maturity for 1100.
At a purchase price of P, the bond yields a nominal annual interest rate of 8%, compounded semiannually, and the present value of the redemption amount is 190.
Determine P.
Solution.
The par value is

$$F = 1000.$$

The annual coupon rate is 9%, payable semiannually. Therefore, the coupon rate per semiannual period is

$$r = \frac{0.09}{2} = 0.045.$$

Thus, the semiannual coupon payment is

$$Fr = 1000(0.045) = 45.$$

The bond is redeemable at maturity for

$$C = 1100.$$

The yield rate is 8%, compounded semiannually. Therefore, the yield rate per coupon period is

$$i = \frac{0.08}{2} = 0.04.$$

We are also given that the present value of the redemption amount is

$$K = 190.$$

Recall that

$$K = Cv_i^n.$$

Thus,

$$190 = 1100v_{0.04}^n.$$

Solving for n,

$$n = 44.77.$$

Since the number of coupon payments must be an integer, and the given value 190 is likely rounded, we use

$$n = 45.$$

Using the basic bond price formula,

$$P = Fra_{\overline{n}|i} + Cv_i^n.$$

Substitute the values:

$$P = 45a_{\overline{45}|0.04} + 1100v_{0.04}^{45}.$$

Therefore,

$$P = 1120.72.$$

Thus,

$$\boxed{P = 1120.72}.$$

Alternative method: Makeham's formula.
The modified coupon rate is

$$g = \frac{Fr}{C}.$$

Thus,

$$g = \frac{45}{1100}.$$

Using Makeham's formula,

$$P = K + \frac{g}{i}(C - K).$$

Substitute

$$K = 190, \qquad g = \frac{45}{1100}, \qquad i = 0.04, \qquad C = 1100.$$

Then

$$P = 190 + \frac{45/1100}{0.04}(1100 - 190).$$

Therefore,

$$P = 1120.68.$$

The small difference from 1120.72 is due to rounding in the given present value $K = 190$. Hence,

$$\boxed{P \approx 1120.72}.$$

Exercise 4.1.6

An insurance company owns a 1000 par value 10% bond with semiannual coupons.
The bond will mature for 1000 at the end of 10 years.
The company decides that an 8-year bond would be preferable. Current yield rates are 7% compounded semiannually.

The company uses the proceeds from the sale of the 10% bond to purchase a 6% bond with semiannual coupons, maturing at par at the end of 8 years.
Calculate the par value of the 8-year bond.

A. 1000 B. 1290

C. 1306 D. 1419

E. 1497

Solution.
First, find the selling price of the 10-year bond.
For the 10-year bond, we have

$$F = 1000,$$

$$C = 1000.$$

The annual coupon rate is 10%, with semiannual coupons. Therefore, the coupon rate per half-year is

$$r = \frac{0.10}{2} = 0.05.$$

Thus, each coupon payment is

$$Fr = 1000(0.05) = 50.$$

The bond has 10 years remaining, with semiannual coupons, so

$$n = 10(2) = 20.$$

The current yield rate is 7% compounded semiannually, so the yield rate per half-year is

$$i = \frac{0.07}{2} = 0.035.$$

Using the basic bond price formula,

$$P = Fra_{\overline{n}|i} + Cv_i^n,$$

we get

$$P = 50a_{\overline{20}|0.035} + 1000v_{0.035}^{20}.$$

Thus,

$$P = 1213.19.$$

The insurance company uses these proceeds to buy an 8-year bond.
Let F be the par value of the new 8-year bond.
Since the new bond matures at par,

$$C = F.$$

The annual coupon rate is 6%, with semiannual coupons. Therefore,

$$r = \frac{0.06}{2} = 0.03.$$

So each coupon payment is

$$Fr = 0.03F.$$

The new bond lasts 8 years, with semiannual coupons, so

$$n = 8(2) = 16.$$

The yield rate is still
$$i = 0.035.$$
The price of the new bond must equal the proceeds from the sale of the old bond:
$$1213.19 = 0.03Fa_{\overline{16}|0.035} + Fv_{0.035}^{16}.$$
Factor out F:
$$1213.19 = F\left(0.03a_{\overline{16}|0.035} + v_{0.035}^{16}\right).$$
Therefore,
$$F = \frac{1213.19}{0.03a_{\overline{16}|0.035} + v_{0.035}^{16}}.$$
Thus,
$$F = 1291.27.$$
The closest answer is
$$\boxed{\text{B. } 1290}.$$

Exercise 4.1.7

A 700 par value five-year 10% bond with semiannual coupons is purchased for 670.60. The present value of the redemption value is 372.05.
Calculate the redemption value.

A. 500 B. 599

C. 606 D. 700

E. 1000

Solution.
The par value is
$$F = 700.$$
The annual coupon rate is 10%, with semiannual coupons. Therefore, the coupon rate per semiannual period is
$$r = \frac{0.10}{2} = 0.05.$$
Thus, the semiannual coupon payment is
$$Fr = 700(0.05) = 35.$$
The bond lasts 5 years with semiannual coupons, so the number of coupon payments is
$$n = 5(2) = 10.$$
We are given the bond price:
$$P = 670.60.$$
We are also given the present value of the redemption value:
$$K = 372.05.$$
Using
$$P = Fra_{\overline{n}|i} + K,$$

we get

$$670.60 = 35a_{\overline{10}|i} + 372.05.$$

Therefore,

$$35a_{\overline{10}|i} = 298.55.$$

So,

$$a_{\overline{10}|i} = 8.53.$$

Solving

$$a_{\overline{10}|i} = 8.53$$

gives

$$i = 0.03.$$

Now use the definition of K:

$$K = Cv_i^n.$$

Thus,

$$372.05 = Cv_{0.03}^{10}.$$

Since

$$v_{0.03}^{10} = (1.03)^{-10},$$

we have

$$372.05 = C(1.03)^{-10}.$$

Therefore,

$$C = 372.05(1.03)^{10}.$$

So,

$$C = 500.$$

The redemption value is

$$\boxed{500}.$$

The correct answer is

$$\boxed{\text{A. } 500}.$$

Exercise 4.1.8

A 1000 par value 10-year bond with semiannual coupons and redeemable at 1100 is purchased at 1135 to yield 12% convertible semiannually.
The first coupon is X. Each subsequent coupon is 4% greater than the preceding coupon. Determine X.

A. 40 B. 42

C. 44 D. 48

E. 50

Solution.
The bond is a 10-year bond with semiannual coupons, so the number of coupon payments is

$$n = 10(2) = 20.$$

The bond yields 12% convertible semiannually, so the yield rate per coupon period is

$$i = \frac{0.12}{2} = 0.06.$$

Thus,

$$v = \frac{1}{1.06}.$$

The redemption value is

$$C = 1100.$$

The first coupon is X, and each subsequent coupon is 4% greater than the preceding coupon. Therefore, the coupons are

$$X, \quad X(1.04), \quad X(1.04)^2, \quad \ldots, \quad X(1.04)^{19}.$$

The price of the bond is the present value of the increasing coupons plus the present value of the redemption value.
Thus,

$$1135 = Xv + X(1.04)v^2 + X(1.04)^2v^3 + \cdots + X(1.04)^{19}v^{20} + 1100v^{20}.$$

Factor out X:

$$1135 = X\left[v + 1.04v^2 + (1.04)^2v^3 + \cdots + (1.04)^{19}v^{20}\right] + 1100v^{20}.$$

The coupon present value is a geometric annuity. Since $i = 0.06$ and the coupon growth rate is 0.04, we have

$$v + 1.04v^2 + \cdots + (1.04)^{19}v^{20} = \frac{1 - \left(\frac{1.04}{1.06}\right)^{20}}{0.06 - 0.04}.$$

Therefore,

$$1135 = X\left(\frac{1 - \left(\frac{1.04}{1.06}\right)^{20}}{0.06 - 0.04}\right) + 1100v_{0.06}^{20}.$$

Solving for X,

$$X = \frac{1135 - 1100v_{0.06}^{20}}{\frac{1 - \left(\frac{1.04}{1.06}\right)^{20}}{0.06 - 0.04}}.$$

Thus,

$$X = 50.$$

The correct answer is

$$\boxed{\text{E. } 50}.$$

4.1.3 Premium and Discount

A bond can be bought at par, at a premium, or at a discount.

Premium and discount

If

$$P > C,$$

then the bond is bought at a **premium**.
If

$$P < C,$$

then the bond is bought at a **discount**.
If

$$P = C,$$

then the bond is **priced at par**.

Here:

$$P = \text{price of the bond}$$

and

$$C = \text{redemption value of the bond.}$$

Important interpretation

A bond bought at a premium is not necessarily a bad deal.
A bond bought at a discount is not necessarily a good deal.
The price reflects the relationship between the coupon rate and the yield rate.

Recall the premium / discount formula:

$$P = C + (Fr - Ci)a_{\overline{n}|i}.$$

The term Ci can be interpreted as the coupon payment that would make the bond price equal to the redemption value.

In other words:

$$Ci = \text{required coupon payment for the bond to be priced at par.}$$

Bond Priced at Par

Bond priced at par

Bond A is a two-year 1000 par value bond with 8% semiannual coupons.
The bond is bought to yield 8% convertible semiannually.
Determine whether the bond is bought at par, at a premium, or at a discount.
Solution.
The redemption value is

$$C = 1000.$$

The yield rate is 8% convertible semiannually, so the yield rate per coupon period is

$$i = \frac{0.08}{2} = 0.04.$$

Therefore,

$$Ci = 1000(0.04) = 40.$$

The coupon rate is also 8% convertible semiannually, so the coupon rate per coupon period is

$$r = \frac{0.08}{2} = 0.04.$$

Thus, the coupon payment is

$$Fr = 1000(0.04) = 40.$$

Since

$$Fr = Ci,$$

the coupon payment is exactly equal to the required payment.
Using the premium / discount formula:

$$P = C + (Fr - Ci)a_{\overline{n}|i}.$$

Here, $n = 2(2) = 4$. Therefore,

$$P = 1000 + (40 - 40)a_{\overline{4}|0.04}.$$

So,

$$P = 1000.$$

Thus,

$$\boxed{P = C}.$$

The bond is

$$\boxed{\text{priced at par}}.$$

Bond Bought at a Premium

Bond bought at a premium

Bond B is a two-year 1000 par value bond with 10% semiannual coupons.
The bond is bought to yield 8% convertible semiannually.
Determine whether the bond is bought at par, at a premium, or at a discount.
Solution.
The redemption value is

$$C = 1000.$$

The yield rate is 8% convertible semiannually, so

$$i = \frac{0.08}{2} = 0.04.$$

Therefore,

$$Ci = 1000(0.04) = 40.$$

The coupon rate is 10% convertible semiannually, so

$$r = \frac{0.10}{2} = 0.05.$$

Thus, the coupon payment is

$$Fr = 1000(0.05) = 50.$$

Since

$$Fr > Ci,$$

the coupon payment is larger than the required payment.
So the bond should sell for more than its redemption value.
Using the premium / discount formula:

$$P = C + (Fr - Ci)a_{\overline{n}|i}.$$

Here,

$$n = 2(2) = 4.$$

Therefore,

$$P = 1000 + (50 - 40)a_{\overline{4}|0.04}.$$

So,

$$P = 1000 + 36.30.$$

Thus,

$$P = 1036.30.$$

Since

$$P > C,$$

the bond is bought at a premium.
The premium is

$$P - C = 1036.30 - 1000.$$

Therefore,

$$\boxed{\text{Premium} = 36.30}.$$

Bond Bought at a Discount

Bond bought at a discount

Bond C is a two-year 1000 par value bond with 5% semiannual coupons.
The bond is bought to yield 8% convertible semiannually.
Determine whether the bond is bought at par, at a premium, or at a discount.
Solution.
The redemption value is

$$C = 1000.$$

The yield rate is 8% convertible semiannually, so

$$i = \frac{0.08}{2} = 0.04.$$

Therefore,
$$Ci = 1000(0.04) = 40.$$
The coupon rate is 5% convertible semiannually, so
$$r = \frac{0.05}{2} = 0.025.$$
Thus, the coupon payment is
$$Fr = 1000(0.025) = 25.$$
Since
$$Fr < Ci,$$
the coupon payment is smaller than the required payment.
So the bond should sell for less than its redemption value.
Using the premium / discount formula:
$$P = C + (Fr - Ci)a_{\overline{n}|i}.$$
Here,
$$n = 2(2) = 4.$$
Therefore,
$$P = 1000 + (25 - 40)a_{\overline{4}|0.04}.$$
So,
$$P = 1000 - 54.45.$$
Thus,
$$P = 945.55.$$
Since
$$P < C,$$
the bond is bought at a discount.
The discount is
$$C - P = 1000 - 945.55.$$
Therefore,
$$\boxed{\text{Discount} = 54.45}.$$

Summary

The three examples can be summarized as follows:

Bond	Ci	Fr	Price	Premium / Discount
A	40	40	1000.00	priced at par
B	40	50	1036.30	premium = 36.30
C	40	25	945.55	discount = 54.45

Best buy?

There is no automatic "best buy" among a par bond, a premium bond, and a discount bond.
If the bonds are priced using the same yield rate, they have the same yield-to-maturity.
The investor should choose the bond whose coupon pattern best matches their needs.

Formulas

Start with the premium / discount formula:

$$P = C + (Fr - Ci)a_{\overline{n}|i}.$$

If

$$Fr > Ci,$$

then the bond is bought at a premium.

The premium is

$$P - C.$$

From the formula,

$$P - C = (Fr - Ci)a_{\overline{n}|i}.$$

Thus,

$$\boxed{P - C = (Fr - Ci)a_{\overline{n}|i}}$$

is the premium.

If

$$Ci > Fr,$$

then the bond is bought at a discount.

The discount is

$$C - P.$$

From the formula,

$$C - P = (Ci - Fr)a_{\overline{n}|i}.$$

Thus,

$$\boxed{C - P = (Ci - Fr)a_{\overline{n}|i}}$$

is the discount.

Premium and discount formulas

If $Fr > Ci$, then

$$\boxed{\text{Premium} = P - C = (Fr - Ci)a_{\overline{n}|i}.}$$

If $Ci > Fr$, then

$$\boxed{\text{Discount} = C - P = (Ci - Fr)a_{\overline{n}|i}.}$$

We can also write the formula in terms of the modified coupon rate g.

Recall that

$$Fr = Cg.$$

So

$$P = C + (Cg - Ci)a_{\overline{n}|i}.$$

Factor out C:

$$P = C + C(g - i)a_{\overline{n}|i}.$$

Therefore:

$$g > i \quad \Longrightarrow \quad \text{bond bought at a premium,}$$

and

$$g < i \quad \Longrightarrow \quad \text{bond bought at a discount.}$$

Coupon rate versus yield rate

A bond is priced at par when the coupon rate equals the yield rate.
A bond is bought at a premium when the coupon rate is greater than the yield rate.
A bond is bought at a discount when the coupon rate is less than the yield rate.

Exercise 4.1.9

True or false?

1. If $r > i$, then the bond must be a premium bond.
2. If $g < i$, then the bond must be a discount bond.
3. If $Ki > Fr$, then the bond must be a discount bond.
4. If $i = r$ and the bond is redeemable at par, then the bond's price is equal to the par amount.

Solution.

1. **False.**

   We cannot determine whether a bond is bought at a premium or at a discount by only comparing r and i.

   The correct comparison is between the actual coupon payment Fr and the required coupon payment Ci, or equivalently between g and i.

   Recall that

   $$g = \frac{Fr}{C}.$$

   The bond is bought at a premium if

   $$Fr > Ci,$$

   or equivalently,

   $$g > i.$$

   Therefore, $r > i$ alone is not enough unless the bond is redeemable at par, so that $C = F$.

2. **True.**

   If
$$g < i,$$
   then
$$Cg < Ci.$$
   Since
$$Fr = Cg,$$
   we get
$$Fr < Ci.$$
   Therefore, the coupon payment is smaller than the required payment, so the bond is bought at a discount.

3. **True.**

   Recall that
$$K = Cv_i^n,$$
   so K is the present value of the redemption amount.

   Since $v_i^n < 1$, we have
$$K < C.$$
   Multiplying by i, we get
$$Ki < Ci.$$
   Now suppose
$$Ki > Fr.$$
   Since
$$Ci > Ki,$$
   it follows that
$$Ci > Fr.$$
   Therefore, the coupon payment is smaller than the required payment, so the bond is a discount bond.

4. **True.**

   If the bond is redeemable at par, then
$$C = F.$$
   If also
$$i = r,$$
   then
$$Ci = Fi = Fr.$$
   Therefore,
$$Fr - Ci = 0.$$

Using the premium / discount formula,

$$P = C + (Fr - Ci)a_{\overline{n}|i},$$

we get

$$P = C.$$

Since $C = F$, the bond's price is equal to the par amount.

Thus, the answers are

$$\boxed{\text{False, True, True, True}}.$$

Exercise 4.1.10

A fifteen-year 1000 bond with quarterly coupons of 25 is redeemable for 1200.
The bond is priced to yield 8% convertible quarterly.
Determine the amount of the premium or discount.

Solution.

The bond has quarterly coupons for 15 years, so the number of coupon payments is

$$n = 15(4) = 60.$$

The yield rate is 8% convertible quarterly, so the yield rate per quarter is

$$i = \frac{0.08}{4} = 0.02.$$

The redemption value is

$$C = 1200.$$

The quarterly coupon payment is given directly:

$$Fr = 25.$$

The required coupon payment for the bond to be priced at par is

$$Ci = 1200(0.02) = 24.$$

Since

$$Fr > Ci,$$

the bond is bought at a premium.
The premium is

$$P - C = (Fr - Ci)a_{\overline{n}|i}.$$

Therefore,

$$\text{Premium} = (25 - 24)a_{\overline{60}|0.02}.$$

Thus,

$$\text{Premium} = 34.76.$$

So the bond is bought at a premium of

$$\boxed{34.76}.$$

Alternative method.

Calculate the price directly:

$$P = 25a_{\overline{60}|0.02} + 1200v_{0.02}^{60}.$$

Thus,

$$P = 1234.76.$$

Since

$$P - C = 1234.76 - 1200 = 34.76,$$

the bond is bought at a premium of

$$\boxed{34.76}.$$

Exercise 4.1.11

For a ten-year bond with semiannual coupons, you are given:

1. $C = 1200$,
2. $r = 0.04$,
3. $g = 0.03$,
4. $i = 0.04$.

Determine the bond premium or discount.

Solution.

The bond has semiannual coupons for 10 years, so the number of coupon payments is

$$n = 10(2) = 20.$$

We are given

$$C = 1200, \qquad g = 0.03, \qquad i = 0.04.$$

Since

$$g < i,$$

the bond is bought at a discount.

Using

$$Fr = Cg,$$

we can write the discount formula as

$$\text{Discount} = (Ci - Fr)a_{\overline{n}|i}.$$

Since $Fr = Cg$, this becomes

$$\text{Discount} = (Ci - Cg)a_{\overline{n}|i}.$$

Factor out C:

$$\text{Discount} = C(i - g)a_{\overline{n}|i}.$$

Substitute the values:

$$\text{Discount} = 1200(0.04 - 0.03)a_{\overline{20}|0.04}.$$

Thus,

$$\text{Discount} = 12a_{\overline{20}|0.04}.$$

Using

$$a_{\overline{20}|0.04} = \frac{1-(1.04)^{-20}}{0.04},$$

we get

$$\text{Discount} = 163.08.$$

Therefore, the bond is bought at a discount of

$$\boxed{163.08}.$$

Exercise 4.1.12

You are given:

1. A 10-year 8% semiannual coupon bond is purchased at a discount of X.
2. A 10-year 9% semiannual coupon bond is purchased at a premium of Y.
3. A 10-year 10% semiannual coupon bond is purchased at a premium of $2X$.
4. All bonds were purchased at the same yield rate and have par values of 1000.

Calculate Y.

A. $\frac{X}{3}$ B. $\frac{2X}{5}$
C. $\frac{X}{2}$ D. $\frac{2X}{3}$
E. X

Solution.
All bonds have the same par value:

$$F = 1000.$$

Since the bonds are redeemed at par, we also have

$$C = 1000.$$

All bonds are 10-year bonds with semiannual coupons, so the number of coupon periods is

$$n = 10(2) = 20.$$

Let i be the yield rate per semiannual period.
The premium / discount formula is

$$P = C + (Fr - Ci)a_{\overline{n}|i}.$$

Therefore,

$$P - C = (Fr - Ci)a_{\overline{n}|i}.$$

For a discount bond,

$$C - P = (Ci - Fr)a_{\overline{n}|i}.$$

Bond i: 8% semiannual coupon bond.
The semiannual coupon payment is

$$Fr = 1000\left(\frac{0.08}{2}\right) = 40.$$

This bond is purchased at a discount of X. Hence,

$$X = (1000i - 40)a_{\overline{20}|i}.$$

Bond ii: 9% **semiannual coupon bond.**
The semiannual coupon payment is

$$Fr = 1000\left(\frac{0.09}{2}\right) = 45.$$

This bond is purchased at a premium of Y. Hence,

$$Y = (45 - 1000i)a_{\overline{20}|i}.$$

Bond iii: 10% **semiannual coupon bond.**
The semiannual coupon payment is

$$Fr = 1000\left(\frac{0.10}{2}\right) = 50.$$

This bond is purchased at a premium of $2X$. Hence,

$$2X = (50 - 1000i)a_{\overline{20}|i}.$$

Now substitute

$$X = (1000i - 40)a_{\overline{20}|i}$$

into

$$2X = (50 - 1000i)a_{\overline{20}|i}.$$

This gives

$$2(1000i - 40)a_{\overline{20}|i} = (50 - 1000i)a_{\overline{20}|i}.$$

Cancel $a_{\overline{20}|i}$:

$$2(1000i - 40) = 50 - 1000i.$$

So,

$$2000i - 80 = 50 - 1000i.$$

Therefore,

$$3000i = 130,$$

and

$$i = \frac{13}{300}.$$

Thus,

$$1000i = \frac{130}{3}.$$

Now calculate Y:

$$Y = (45 - 1000i)a_{\overline{20}|i}.$$

Substitute $1000i = 130/3$:

$$Y = \left(45 - \frac{130}{3}\right)a_{\overline{20}|i}.$$

Since

$$45 = \frac{135}{3},$$

we get

$$Y = \frac{5}{3} a_{\overline{20}|i}.$$

Also,

$$X = (1000i - 40) a_{\overline{20}|i}.$$

Substitute $1000i = 130/3$:

$$X = \left(\frac{130}{3} - 40 \right) a_{\overline{20}|i}.$$

Since

$$40 = \frac{120}{3},$$

we get

$$X = \frac{10}{3} a_{\overline{20}|i}.$$

Therefore,

$$Y = \frac{5}{3} a_{\overline{20}|i} = \frac{1}{2} \left(\frac{10}{3} a_{\overline{20}|i} \right).$$

Hence,

$$Y = \frac{X}{2}.$$

The correct answer is

$$\boxed{\text{C. } \frac{X}{2}}.$$

4.1.4 Bond Amortization

Bond amortization works very much like loan amortization.

A bond is a type of loan. The main difference is terminology.

- Instead of **outstanding balance**, for bonds we say **book value**.
- Instead of **loan payment**, for bonds we say **coupon payment**.
- Instead of **principal payment**, for bonds we say **write-down** or **write-up**, depending on whether the bond is bought at a premium or at a discount.

Book value

The **book value** of a bond at a given time is the present value, at that time, of the remaining bond payments.
It plays the same role as the outstanding balance for a loan.

For a bond bought at a premium, we amortize the premium

$$P - C.$$

The book value starts at the purchase price P, and then decreases toward the redemption value C. This decrease is called a **write-down**.

For a bond bought at a discount, the book value starts below the redemption value C, and then increases toward C. This increase is called a **write-up**.

Write-Down Example

Premium bond amortization

Create an amortization schedule for a two-year 1000 par value bond with 10% semiannual coupons.
The bond is bought to yield 8% convertible semiannually.
Solution.
The semiannual coupon is
$$Fr = 1000\left(\frac{0.10}{2}\right) = 50.$$
The yield rate per half-year is
$$i = \frac{0.08}{2} = 0.04.$$
The number of coupon periods is
$$n = 2(2) = 4.$$
The redemption value is
$$C = 1000.$$
The bond price is
$$P = 50a_{\overline{4}|0.04} + 1000v_{0.04}^4.$$
Thus,
$$P = 1036.30.$$
Since
$$P > C,$$
this is a premium bond.
The initial book value is
$$BV_0 = 1036.30.$$
For each half-year, we compute:
$$I_t = BV_{t-1}i,$$
$$\text{Write-down} = Fr - I_t,$$
and
$$BV_t = BV_{t-1} - \text{Write-down}.$$
For half-year 1:
$$I_1 = 1036.30(0.04) = 41.45,$$
$$\text{Write-down}_1 = 50 - 41.45 = 8.55,$$
$$BV_1 = 1036.30 - 8.55 = 1027.75.$$
For half-year 2:
$$I_2 = 1027.75(0.04) = 41.11,$$
$$\text{Write-down}_2 = 50 - 41.11 = 8.89,$$
$$BV_2 = 1027.75 - 8.89 = 1018.86.$$

For half-year 3:

$$I_3 = 1018.86(0.04) = 40.75,$$

$$\text{Write-down}_3 = 50 - 40.75 = 9.25,$$

$$BV_3 = 1018.86 - 9.25 = 1009.61.$$

For half-year 4:

$$I_4 = 1009.61(0.04) = 40.39,$$

$$\text{Write-down}_4 = 50 - 40.39 = 9.61,$$

$$BV_4 = 1009.61 - 9.61 = 1000.00.$$

The amortization schedule is

Half-Year	Coupon	Interest	Write-Down	Book Value
0				1036.30
1	50	41.45	8.55	1027.75
2	50	41.11	8.89	1018.86
3	50	40.75	9.25	1009.61
4	50	40.39	9.61	1000.00
Total			36.30	

The total write-downs equal the premium:

$$8.55 + 8.89 + 9.25 + 9.61 = 36.30.$$

Also,

$$P - C = 1036.30 - 1000 = 36.30.$$

Thus, the total write-down is exactly the bond premium.

Book value interpretation

Just like the outstanding balance on a loan is the present value of future loan payments, the book value of a bond is the present value of the future bond payments.

Using the AMORT Worksheet

For a bond with level coupons, the BA II Plus AMORT worksheet can be used in the same way as for loans.

For the previous example, the calculator setup is

$$\begin{aligned} N &= 4 \\ I/Y &= 4 \\ PMT &= 50 \\ FV &= 1000 \\ CPT\ PV &= -1036.298952 \end{aligned}$$

For example, to find the values for half-year 2, use

$$P1 = 2$$
$$P2 = 2$$
$$BAL = -1018.860947$$
$$PRN = 8.889963587$$
$$INT = 41.11003641$$

The negative sign on BAL comes from the calculator's cash-flow convention.

Bond Amortization Formulas

The bond amortization formulas are very similar to the loan amortization formulas.

The book value after the coupon payment at time t can be found prospectively:

$$BV_t = Fra_{\overline{n-t}|i} + Cv_i^{n-t}.$$

It can also be found retrospectively:

$$BV_t = P(1+i)^t - Frs_{\overline{t}|i}.$$

It can also be updated recursively:

$$BV_t = BV_{t-1} - P_t,$$

where P_t denotes the amount by which the book value is reduced.

The interest earned during period t is

$$I_t = BV_{t-1}i.$$

The write-down amount is

$$P_t = Fr - I_t.$$

Using the premium / discount structure, we get

$$P_t = (Fr - Ci)v_i^{n-t+1}.$$

Bond amortization formulas

$$\boxed{BV_t = Fra_{\overline{n-t}|i} + Cv_i^{n-t}}$$

$$\boxed{BV_t = P(1+i)^t - Frs_{\overline{t}|i}}$$

$$\boxed{I_t = BV_{t-1}i}$$

$$\boxed{P_t = Fr - I_t = (Fr - Ci)v_i^{n-t+1}}$$

For a premium bond,

$$Fr > Ci,$$

so

$$P_t > 0.$$

The book value decreases. This is a write-down.

For a discount bond,

$$Fr < Ci,$$

so

$$P_t < 0.$$

The book value increases. This is a write-up.

Also, just like for loans, each write-down or write-up grows geometrically by a factor of $1+i$:

$$P_{t+k} = P_t(1+i)^k.$$

Write-Up Example

Discount bond amortization

Create an amortization schedule for a two-year 1000 par value bond with 8% semiannual coupons, redeemable at 1050.
The bond is bought to yield 8% convertible semiannually.
Solution.
The semiannual coupon is

$$Fr = 1000\left(\frac{0.08}{2}\right) = 40.$$

The yield rate per half-year is

$$i = \frac{0.08}{2} = 0.04.$$

The number of coupon periods is

$$n = 2(2) = 4.$$

The redemption value is

$$C = 1050.$$

The bond price is

$$P = 40a_{\overline{4}|0.04} + 1050v_{0.04}^4.$$

Thus,

$$P = 1042.74.$$

Since

$$P < C,$$

this is a discount bond.
The initial book value is

$$BV_0 = 1042.74.$$

For a discount bond, the interest earned each period is greater than the coupon paid. The difference is added to the book value. This is called a write-up.
For half-year 1:

$$I_1 = 1042.74(0.04) = 41.71,$$

$$Fr - I_1 = 40 - 41.71 = -1.71.$$

So the write-up is

$$1.71.$$

Thus,

$$BV_1 = 1042.74 + 1.71 = 1044.45.$$

For half-year 2:

$$I_2 = 1044.45(0.04) = 41.78,$$

$$Fr - I_2 = 40 - 41.78 = -1.78.$$

So the write-up is

$$1.78.$$

Thus,

$$BV_2 = 1044.45 + 1.78 = 1046.23.$$

For half-year 3:

$$I_3 = 1046.23(0.04) = 41.85,$$

$$Fr - I_3 = 40 - 41.85 = -1.85.$$

So the write-up is

$$1.85.$$

Thus,

$$BV_3 = 1046.23 + 1.85 = 1048.08.$$

For half-year 4:

$$I_4 = 1048.08(0.04) = 41.92,$$

$$Fr - I_4 = 40 - 41.92 = -1.92.$$

So the write-up is

$$1.92.$$

Thus,

$$BV_4 = 1048.08 + 1.92 = 1050.00.$$

The amortization schedule is

Half-Year	Coupon	Interest	Write-Up	Book Value
0				1042.74
1	40	41.71	1.71	1044.45
2	40	41.78	1.78	1046.23
3	40	41.85	1.85	1048.08
4	40	41.92	1.92	1050.00
Total			7.26	

The total write-ups equal the discount:

$$1.71 + 1.78 + 1.85 + 1.92 = 7.26.$$

Also,

$$C - P = 1050 - 1042.74 = 7.26.$$

Thus, the total write-up is exactly the bond discount.
The write-ups are in geometric progression. For example,

$$1.71(1.04)^2 \approx 1.85.$$

For the write-up example, the calculator setup is

$$\begin{aligned}
N &= 4 \\
I/Y &= 4 \\
PMT &= 40 \\
FV &= 1050 \\
CPT\ PV &= -1042.74021
\end{aligned}$$

For example, to find the values for half-year 3, use

$$\begin{aligned}
P1 &= 3 \\
P2 &= 3 \\
BAL &= -1048.076923 \\
PRN &= -1.849112426 \\
INT &= 41.84911243
\end{aligned}$$

The negative value of PRN indicates a write-up rather than a write-down.

Exercise 4.1.13

A 1000 par value bond redeemable in 20 years at par with 6% annual coupons is purchased to produce an annual yield of 5%.
The purchaser values the bond for accounting purposes at its amortized value.
Calculate the amortized value of the bond at the end of the 5th year.
Solution.
The bond has par value

$$F = 1000.$$

Since the bond is redeemable at par,

$$C = 1000.$$

The annual coupon rate is 6%, so the annual coupon payment is

$$Fr = 1000(0.06) = 60.$$

The annual yield rate is

$$i = 0.05.$$

The amortized value of the bond at the end of the 5th year is the book value at time 5.
At that time, there are

$$20 - 5 = 15$$

future coupon payments remaining.
The amortized value is the present value at time 5 of the future bond payments.
Therefore,

$$BV_5 = 60a_{\overline{15}|0.05} + 1000v_{0.05}^{15}.$$

Using

$$v_{0.05} = \frac{1}{1.05},$$

we get

$$BV_5 = 1103.80.$$

Therefore, the amortized value of the bond at the end of the 5th year is

$$\boxed{1103.80}.$$

Alternative method: AMORT worksheet.
First calculate the purchase price:

$$P = 60a_{\overline{20}|0.05} + 1000v_{0.05}^{20}.$$

Thus,

$$P = 1124.622103.$$

Using the AMORT worksheet:

$$P1 = 5$$
$$P2 = 5$$
$$BAL = -1103.79658$$

The negative sign is only due to the calculator's cash-flow convention.
Thus,

$$BV_5 = 1103.80.$$

Exercise 4.1.14

A ten-year bond bears semiannual coupons of 4 each and has a redemption value of 100. The bond is purchased to yield 10% compounded semiannually.
Find the amount of increase in the book value at the time of the tenth coupon payment.
Solution.
The bond has semiannual coupons for 10 years, so the number of coupon periods is

$$n = 10(2) = 20.$$

The yield rate is 10% compounded semiannually, so the yield rate per coupon period is

$$i = \frac{0.10}{2} = 0.05.$$

The coupon payment is

$$Fr = 4.$$

The redemption value is

$$C = 100.$$

Since

$$Ci = 100(0.05) = 5,$$

and

$$Fr = 4,$$

we have

$$Fr < Ci.$$

Therefore, this is a discount bond, and its book value increases over time.
The book value after the 9th coupon payment is the present value at time 9 of the remaining payments. There are

$$20 - 9 = 11$$

coupon payments remaining.
Thus,

$$BV_9 = 4a_{\overline{11}|0.05} + 100v_{0.05}^{11}.$$

So,

$$BV_9 = 91.694.$$

The book value after the 10th coupon payment is

$$BV_{10} = 4a_{\overline{10}|0.05} + 100v_{0.05}^{10}.$$

So,

$$BV_{10} = 92.278.$$

Therefore, the increase in book value at the time of the tenth coupon payment is

$$BV_{10} - BV_9 = 92.278 - 91.694.$$

Thus,

$$\boxed{0.584}.$$

Alternative method.
The increase in book value is the write-up.
The interest earned during the 10th period is based on BV_9:

$$I_{10} = BV_9 i.$$

So,

$$I_{10} = 91.694(0.05) = 4.5847.$$

Since the coupon payment is only 4, the book value must increase by

$$4.5847 - 4 = 0.5847.$$

Equivalently,

$$P_{10} = Fr - I_{10} = 4 - 4.5847 = -0.5847.$$

The negative sign means this is a write-up. Therefore, the increase in book value is

$$\boxed{0.584}.$$

Exercise 4.1.15

A bond with a par value of 1000 and 6% semiannual coupons is redeemable for 1100. You are given:

1. The bond is purchased at P to yield 8% convertible semiannually.
2. The amount of principal adjustment for the 16th semiannual period is 5.

Calculate P.
Solution.
The par value is

$$F = 1000.$$

The bond has 6% semiannual coupons, so the coupon rate per half-year is

$$r = \frac{0.06}{2} = 0.03.$$

Thus, the coupon payment is

$$Fr = 1000(0.03) = 30.$$

The redemption value is

$$C = 1100.$$

The bond is purchased to yield 8% convertible semiannually, so the yield rate per half-year is

$$i = \frac{0.08}{2} = 0.04.$$

The required payment is

$$Ci = 1100(0.04) = 44.$$

Since

$$Fr < Ci,$$

this is a discount bond. Therefore, the book value increases over time, and the principal adjustments are negative if we use

$$P_t = Fr - I_t.$$

We are told that the amount of principal adjustment for the 16th semiannual period is 5. Since this is a discount bond,

$$P_{16} = -5.$$

For bond amortization, principal adjustments grow geometrically:

$$P_{t+k} = P_t(1+i)^k.$$

Thus,

$$P_{16} = P_1(1.04)^{15}.$$

Therefore,

$$-5 = P_1(1.04)^{15}.$$

So,

$$P_1 = \frac{-5}{(1.04)^{15}}.$$

Hence,

$$P_1 = -2.7763.$$

Now use the first principal adjustment to find the purchase price P.
During the first semiannual period, the interest earned on the book value is

$$I_1 = P(0.04).$$

The principal adjustment is

$$P_1 = Fr - I_1.$$

Therefore,

$$P_1 = 30 - P(0.04).$$

Substitute $P_1 = -2.7763$:

$$-2.7763 = 30 - 0.04P.$$

Thus,

$$0.04P = 32.7763.$$

So,

$$P = \frac{32.7763}{0.04}.$$

Therefore,

$$P = 819.41.$$

The purchase price is

$$\boxed{819.41}.$$

Exercise 4.1.16

A 30-year 10000 bond that pays 3% annual coupons matures at par.
It is purchased to yield 5% for the first 15 years and 4% thereafter.
Calculate the amount for accumulation of discount for year 8.

A. 78 B. 83

C. 88 D. 93

E. 98

Solution.
The bond has face amount

$$F = 10000.$$

It pays 3% annual coupons, so the annual coupon payment is

$$Fr = 10000(0.03) = 300.$$

The bond matures at par, so

$$C = 10000.$$

We want the accumulation of discount for year 8.
Since year 8 is during the first 15 years, the applicable yield rate for that year is

$$i = 0.05.$$

To find the accumulation of discount for year 8, we first need the book value at time 7.
At time 15, there are 15 annual coupons remaining, and the bond is then valued using the 4% yield rate.
Thus, the book value at time 15 is

$$BV_{15} = 300a_{\overline{15}|0.04} + 10000v_{0.04}^{15}.$$

Now at time 7, there are 8 years until time 15. Therefore,

$$BV_7 = 300a_{\overline{8}|0.05} + BV_{15}v_{0.05}^{8}.$$

Substitute the expression for BV_{15}:

$$BV_7 = 300a_{\overline{8}|0.05} + \left(300a_{\overline{15}|0.04} + 10000v_{0.04}^{15}\right) v_{0.05}^{8}.$$

Thus,

$$BV_7 = 7954.82.$$

The interest due during year 8 is

$$I_8 = BV_7(0.05).$$

So,

$$I_8 = 7954.82(0.05) = 397.74.$$

The coupon payment is only

$$Fr = 300.$$

Therefore, the principal adjustment is

$$P_8 = Fr - I_8.$$

Thus,
$$P_8 = 300 - 397.74 = -97.74.$$
The negative sign means this is an accumulation of discount, or a write-up.
Therefore, the amount of accumulation of discount is
$$97.74.$$
Rounded to the nearest answer choice, this is
$$\boxed{98}.$$
The correct answer is
$$\boxed{\text{E. } 98}.$$

Exercise 4.1.17

Becky buys an n-year 1000 par value bond with 6.5% annual coupons at a price of 825.44.
The price assumes an annual effective yield rate of i.
The total write-up in book value of the bond during the first 2 years after purchase is 23.76.
Calculate i, where $i > 0$.

A. 8.50% B. 8.75%

C. 9.00% D. 9.25%

E. 9.50%

Solution.
The par value is
$$F = 1000.$$
The annual coupon rate is 6.5%, so the annual coupon payment is
$$Fr = 1000(0.065) = 65.$$
The purchase price is
$$P = 825.44.$$
The total write-up in book value during the first 2 years is 23.76. Therefore, the book value at time 2 is
$$BV_2 = 825.44 + 23.76 = 849.20.$$
Now write a value equation at time 0 using the first two coupons and the book value at time 2:
$$825.44 = 65a_{\overline{2}|i} + 849.20v_i^2.$$
Using a financial calculator:
$$\begin{aligned} N &= 2, \\ PV &= -825.44, \\ PMT &= 65, \\ FV &= 849.20. \end{aligned}$$
Then
$$CPT\ I/Y = 9.2502.$$
Therefore,
$$i = 9.2502\%.$$

The closest answer is

$$\boxed{\text{D. } 9.25\%}.$$

4.1.5 Callable Bonds

A **callable bond** is a bond that can be redeemed by the bond issuer before maturity.

This early redemption happens at the issuer's option.

Callable bond

A **callable bond** gives the bond issuer the right to redeem the bond before its maturity date.

The main terminology is:

$$\text{Call price} = \text{price paid by the issuer to redeem the bond before maturity.}$$

$$\text{Call premium} = \text{Call price} - \text{Redemption value.}$$

A **call protection period** is a period during which the bond cannot be called.

Why issue callable bonds?

Callable bonds are useful for bond issuers when interest rates fall.
If rates fall, the issuer can call the old bonds and issue new bonds at a lower yield rate.
This is good for the issuer, but it creates reinvestment risk for the investor.

Callable bonds usually offer higher yields to compensate investors for this extra risk.

The realized yield cannot be known in advance because it depends on when the bond is actually redeemed.

Yield Depends on the Call Date

For a callable bond, the yield depends on the redemption date.

Suppose a bond can be called at different dates. Then each possible redemption date gives a different equation of value and therefore a different yield.

Premium callable bond

Consider a ten-year 100 par value bond with 4% annual coupons.
The bond is callable at par starting 5 years from now.
If the price of the bond is 104.58, calculate the yield if the bond is redeemed in 5 years, 6 years, and so on.
Solution.
The annual coupon is

$$Fr = 100(0.04) = 4.$$

If the bond is redeemed in 5 years, then

$$104.58 = 4a_{\overline{5}|i} + 100v_i^5.$$

Solving gives

$$i = 0.0300.$$

If the bond is redeemed in 6 years, then

$$104.58 = 4a_{\overline{6}|i} + 100v_i^6.$$

Solving gives

$$i = 0.0315.$$

Continuing this process gives:

Redemption Year	Yield
5	0.0300
6	0.0315
7	0.0326
8	0.0334
9	0.0340
10	0.0345

For a premium bond, the worst-case scenario for the investor is the earliest call date.

Discount callable bond

Consider a ten-year 100 par value bond with 4% annual coupons.
The bond is callable at par starting 5 years from now.
If the price of the bond is 95.67, calculate the yield if the bond is redeemed in 5 years, 6 years, and so on.
Solution.
The annual coupon is

$$Fr = 100(0.04) = 4.$$

If the bond is redeemed in 5 years, then

$$95.67 = 4a_{\overline{5}|i} + 100v_i^5.$$

Solving gives

$$i = 0.0500.$$

If the bond is redeemed in 6 years, then

$$95.67 = 4a_{\overline{6}|i} + 100v_i^6.$$

Solving gives

$$i = 0.0485.$$

Continuing this process gives:

Redemption Year	Yield
5	0.0500
6	0.0485
7	0.0474
8	0.0466
9	0.0460
10	0.0455

For a discount bond, the worst-case scenario for the investor is the maturity date.

Calculating the Price of a Callable Bond

Because the yield depends on when the bond is redeemed, we usually cannot calculate one exact price from one exact yield.

Instead, we calculate the **maximum price** the investor can pay while still earning at least a required minimum yield.

Callable bond pricing algorithm

To find the price of a callable bond:

1. Calculate the price of the bond for each possible redemption date using the desired yield rate.
2. Take the minimum of all the prices found.

The reason we take the minimum price is that the investor wants to guarantee a yield not less than the required yield, no matter when the bond is called.

If the call premium is zero, the premium / discount formula is especially useful:

$$P = C + (Fr - Ci)a_{\overline{n}|i}.$$

If

$$Fr - Ci < 0,$$

then the minimum price occurs at the latest redemption date.

If

$$Fr - Ci > 0,$$

then the minimum price occurs at the earliest redemption date.

Worst-case call date

For a premium callable bond, the investor's worst case is usually early redemption.
For a discount callable bond, the investor's worst case is usually later redemption.

Callable Bond Pricing Example

Highest price for a minimum yield

An investor bought a 15-year bond with par value 100000 and 8% semiannual coupons.
The bond is callable at par on any coupon date beginning with the 24th coupon.
Find the highest price paid that will yield a rate not less than

$$i^{(2)} = 10\%.$$

Solution.
The par value is

$$F = 100000.$$

The bond has 8% semiannual coupons, so the coupon rate per half-year is

$$r = \frac{0.08}{2} = 0.04.$$

Thus, each coupon is

$$Fr = 100000(0.04) = 4000.$$

The required yield is

$$i^{(2)} = 10\%.$$

Therefore, the yield rate per half-year is

$$i = \frac{0.10}{2} = 0.05.$$

The bond is callable at par, so

$$C = 100000.$$

The first possible call date is the 24th coupon, and the final maturity date is the 30th coupon.
Thus, possible redemption dates are

$$n = 24, 25, \ldots, 30.$$

For each possible redemption date, the price is

$$P = 4000a_{\overline{n}|0.05} + 100000v_{0.05}^{n}.$$

Using the premium / discount formula,

$$P = C + (Fr - Ci)a_{\overline{n}|i}.$$

Here,

$$Ci = 100000(0.05) = 5000.$$

Therefore,

$$Fr - Ci = 4000 - 5000 = -1000.$$

So

$$P = 100000 - 1000a_{\overline{n}|0.05}.$$

Since $Fr - Ci < 0$, the minimum price occurs at the latest redemption date.
Therefore, use

$$n = 30.$$

Thus,

$$P = 100000 - 1000a_{\overline{30}|0.05}.$$

So,

$$P = 84627.55.$$

Therefore, the highest price the investor can pay while still earning a yield not less than $i^{(2)} = 10\%$ is

$$\boxed{84627.55}.$$

Why take the minimum price?
If the investor instead paid the price calculated using the earliest call date $n = 24$, then

$$P = 100000 - 1000a_{\overline{24}|0.05}.$$

So,

$$P = 86201.36.$$

This price is higher than 84627.55.
However, if the investor pays 86201.36 and the bond is not called until the 30th coupon, then the realized yield y satisfies

$$86201.36 = 4000a_{\overline{30}|y} + 100000v_y^{30}.$$

Solving gives

$$y = 0.04886.$$

This is less than the required semiannual yield of 0.05.
Therefore, paying 86201.36 does not guarantee a yield of at least $i^{(2)} = 10\%$.
That is why we use the minimum price across all possible call dates.

Exercise 4.1.18

An investor bought a 15-year bond with par value of 100000 and 8% semiannual coupons. The bond is callable at par on any coupon date beginning with the 24th coupon.
Find the highest price paid that will yield a rate not less than

$$i^{(2)} = 6\%.$$

Solution.
The par value is

$$F = 100000.$$

The bond has 8% semiannual coupons, so the coupon rate per half-year is

$$r = \frac{0.08}{2} = 0.04.$$

Thus, each coupon payment is

$$Fr = 100000(0.04) = 4000.$$

The desired yield rate is

$$i^{(2)} = 6\%.$$

Therefore, the yield rate per half-year is

$$i = \frac{0.06}{2} = 0.03.$$

The bond is callable at par, so the redemption value at any call date is

$$C = 100000.$$

The bond can be called on any coupon date beginning with the 24th coupon. Since the bond is a 15-year bond with semiannual coupons, the final maturity date is the 30th coupon.
Thus, the possible redemption dates are

$$n = 24, 25, \ldots, 30.$$

For each possible redemption date, the price is

$$P = Fra_{\overline{n}|i} + Cv_i^n.$$

Using the premium / discount formula,

$$P = C + (Fr - Ci)a_{\overline{n}|i}.$$

Here,

$$Ci = 100000(0.03) = 3000.$$

So,

$$Fr - Ci = 4000 - 3000 = 1000.$$

Therefore,

$$P = 100000 + 1000a_{\overline{n}|0.03}, \qquad n = 24, 25, \ldots, 30.$$

Since

$$Fr - Ci > 0,$$

the bond is a premium bond. For a premium callable bond, the worst case for the investor is the earliest call date.
Therefore, the minimum price occurs at

$$n = 24.$$

Thus, the highest price the investor can pay while still earning a yield not less than $i^{(2)} = 6\%$ is

$$P = 100000 + 1000a_{\overline{24}|0.03}.$$

Using

$$a_{\overline{24}|0.03} = \frac{1 - (1.03)^{-24}}{0.03},$$

we get

$$P = 116935.54.$$

Therefore, the highest price paid is

$$\boxed{116935.54}.$$

Exercise 4.1.19

A 1000 par value bond with coupons of 9% payable semiannually was called for 1100 prior to maturity.
The bond was bought for 917.44 immediately after a coupon payment and was held to call.
The nominal yield rate convertible semiannually was 10%.
Calculate the number of years the bond was held.
Solution.
The par value is

$$F = 1000.$$

The coupon rate is 9% payable semiannually, so the coupon rate per half-year is

$$r = \frac{0.09}{2} = 0.045.$$

Thus, each semiannual coupon payment is

$$Fr = 1000(0.045) = 45.$$

The bond was called for

$$C = 1100.$$

The nominal yield rate convertible semiannually is 10%, so the yield rate per half-year is

$$i = \frac{0.10}{2} = 0.05.$$

Let n be the number of half-years the bond was held.
Since the bond was bought immediately after a coupon payment, the first coupon received is one half-year later.
The value equation is

$$917.44 = 45a_{\overline{n}|0.05} + 1100v_{0.05}^{n}.$$

Solving this equation gives

$$n = 50.$$

Thus, the bond was held for

$$\frac{50}{2} = 25$$

years.
Therefore, the number of years the bond was held is

$$\boxed{25}.$$

Exercise 4.1.20

A 1000 par value bond pays annual coupons of 80.
The bond is redeemable at par in 30 years, but is callable any time from the end of the 10th year at 1050.
Based on her desired yield rate, an investor calculates the following potential purchase prices P:

1. Assuming the bond is called at the end of the 10th year, $P = 957$.
2. Assuming the bond is held until maturity, $P = 897$.

The investor buys the bond at the highest price that guarantees she will receive at least her desired yield rate regardless of when the bond is called.
The investor holds the bond for 20 years, after which time the bond is called.
Calculate the annual yield rate the investor earns.
Solution.
For a callable bond, the investor must choose the highest price that guarantees at least the desired yield regardless of when the bond is called.
This means we choose the minimum of the possible purchase prices.
The two possible purchase prices are

$$957$$

and

$$897.$$

Therefore, the investor pays

$$P = 897.$$

The bond is then held for 20 years and called for

$$1050.$$

The annual coupon payment is

$$Fr = 80.$$

Let i be the annual effective yield rate actually earned.
The value equation is

$$897 = 80a_{\overline{20}|i} + 1050v_i^{20}.$$

Solving for i gives

$$i = 0.0924.$$

Therefore, the annual yield rate earned by the investor is

$$\boxed{9.24\%}.$$

Exercise 4.1.21

An investor purchases a 1000 bond redeemable at par that pays 8% semiannual coupons and matures in 10 years.
The bond will yield 7% convertible semiannually to maturity.
If the bond is called in five years, the minimum redemption value the investor needs to realize the same yield is X.
Determine X.

A. 1036 B. 1042

C. 1048 D. 1054

E. 1060

Solution.
A 1000 bond means that the par value is

$$F = 1000.$$

The bond is redeemable at par, so the redemption value at maturity is

$$C = 1000.$$

The bond pays 8% semiannual coupons, so the coupon rate per half-year is

$$r = \frac{0.08}{2} = 0.04.$$

Thus, each semiannual coupon payment is

$$Fr = 1000(0.04) = 40.$$

The yield rate is 7% convertible semiannually, so the yield rate per half-year is

$$i = \frac{0.07}{2} = 0.035.$$

Since the bond matures in 10 years and pays semiannual coupons, the number of coupon periods is

$$n = 10(2) = 20.$$

First, find the original purchase price of the bond.

Using the bond price formula,

$$P = 40a_{\overline{20}|0.035} + 1000v_{0.035}^{20}.$$

Equivalently, using the premium / discount formula:

$$P = C + (Fr - Ci)a_{\overline{20}|i}.$$

Here,

$$Ci = 1000(0.035) = 35.$$

Therefore,

$$P = 1000 + (40 - 35)a_{\overline{20}|0.035}.$$

So,

$$P = 1000 + 5a_{\overline{20}|0.035}.$$

Thus,

$$P = 1071.06.$$

Now suppose the bond is called in five years.
Five years corresponds to

$$5(2) = 10$$

semiannual periods.
The investor wants to realize the same yield of 7% convertible semiannually. Therefore, using the same yield rate $i = 0.035$, we write the value equation

$$1071.06 = 40a_{\overline{10}|0.035} + Xv_{0.035}^{10}.$$

Now,

$$a_{\overline{10}|0.035} = 8.3166$$

and

$$v_{0.035}^{10} = 0.7089.$$

Thus,

$$1071.06 = 40(8.3166) + X(0.7089).$$

So,

$$1071.06 = 332.66 + 0.7089X.$$

Therefore,

$$X = \frac{1071.06 - 332.66}{0.7089}.$$

Hence,

$$X = 1041.58.$$

The closest answer is

$$\boxed{\text{B. } 1042}.$$

Exercise 4.1.22

A 1000 ten-year bond bearing annual coupons of 4% is callable at par at any time after the fifth year.
A purchaser pays a price for the bond that will assure him of a yield of not less than 5%

if it is held to maturity or prior call.
Find:

1. the purchase price to the nearest dollar;

2. the call date most favorable to the purchaser.

A. Purchase price: 923; Call date: 5 years hence
B. Purchase price: 923; Call date: 10 years hence
C. Purchase price: 957; Call date: 5 years hence
D. Purchase price: 957; Call date: 10 years hence
E. Purchase price: 1000; Call date: 10 years hence

Solution.
The par value is
$$F = 1000.$$
The bond pays annual coupons of 4%, so the annual coupon payment is
$$Fr = 1000(0.04) = 40.$$
The desired yield rate is
$$i = 0.05.$$
The bond is callable at par, so the redemption value is
$$C = 1000.$$
If the bond is called or redeemed at time n, its price at a 5% yield is
$$P = 40a_{\overline{n}|0.05} + 1000v_{0.05}^{n}.$$
Using the premium / discount formula,
$$P = C + (Fr - Ci)a_{\overline{n}|i}.$$
Here,
$$Ci = 1000(0.05) = 50.$$
Thus,
$$Fr - Ci = 40 - 50 = -10.$$
Therefore,
$$P = 1000 - 10a_{\overline{n}|0.05}.$$
The possible redemption dates range from year 5 to year 10.
For $n = 5$,
$$P = 1000 - 10a_{\overline{5}|0.05}.$$
Thus,
$$P = 957.$$
For $n = 10$,
$$P = 1000 - 10a_{\overline{10}|0.05}.$$
Thus,
$$P = 923.$$

To guarantee a yield of at least 5%, the purchaser must use the minimum of the possible prices.
Therefore, the purchase price is

$$\boxed{923}.$$

If the purchaser pays 923, the most favorable call date for the purchaser is the earliest possible call date, which is 5 years hence.
This is because the bond is bought at a discount. Receiving the par value earlier gives the investor a higher realized yield.
Thus, the correct answer is

$$\boxed{\text{A. Purchase price: 923; Call date: 5 years hence}}.$$

Exercise 4.1.23

A 1000 par-value 4% bond with semiannual coupons matures at the end of 10 years.
The bond is callable at 1050 on any coupon date from the 3rd through the 6th years, at 1025 from the 7th through the 9th years, and at 1000 during the 10th year.
Find the maximum price that an investor can pay and still be certain of a yield of 5% convertible semiannually.

A. 922 B. 944

C. 959 D. 986

E. 1016

Solution.
The par value is

$$F = 1000.$$

The bond pays 4% semiannual coupons, so the coupon rate per half-year is

$$r = \frac{0.04}{2} = 0.02.$$

Thus, each semiannual coupon payment is

$$Fr = 1000(0.02) = 20.$$

The required yield is 5% convertible semiannually, so the yield rate per half-year is

$$i = \frac{0.05}{2} = 0.025.$$

The maximum price that guarantees the required yield is the minimum price over all possible call dates.
For a call date n semiannual periods from now, the price is

$$P = 20a_{\overline{n}|0.025} + Cv_{0.025}^n,$$

where C is the call price at that time.
The possible call prices are

$$P = \begin{cases} 20a_{\overline{n}|0.025} + 1050v_{0.025}^n, & n = 5, \dots, 12, \\ 20a_{\overline{n}|0.025} + 1025v_{0.025}^n, & n = 13, \dots, 18, \\ 20a_{\overline{n}|0.025} + 1000v_{0.025}^n, & n = 19, 20. \end{cases}$$

We test the relevant endpoints of each call-price interval.
For $n = 5$:

$$P = 20a_{\overline{5}|0.025} + 1050v_{0.025}^{5} = 1021.$$

For $n = 12$:

$$P = 20a_{\overline{12}|0.025} + 1050v_{0.025}^{12} = 985.$$

For $n = 13$:

$$P = 20a_{\overline{13}|0.025} + 1025v_{0.025}^{13} = 963.$$

For $n = 18$:

$$P = 20a_{\overline{18}|0.025} + 1025v_{0.025}^{18} = 944.$$

For $n = 19$:

$$P = 20a_{\overline{19}|0.025} + 1000v_{0.025}^{19} = 925.$$

For $n = 20$:

$$P = 20a_{\overline{20}|0.025} + 1000v_{0.025}^{20} = 922.$$

The smallest of these prices is

$$922.$$

Therefore, the maximum price the investor can pay and still be certain of a yield of 5% convertible semiannually is

$$\boxed{922}.$$

The correct answer is

$$\boxed{\text{A. } 922}.$$

Chapter 5

General Cash Flows, Portfolios, and Asset-Liability Management

5.1 Yield Rates

5.1.1 Net Present Value and Internal Rate of Return

Discounted Cash Flow Analysis

Discounted cash flow analysis

Discounted cash flow analysis is the evaluation of a project to determine whether it adds value to a company.

There are two main methods:

1. Net Present Value (NPV)
2. Internal Rate of Return (IRR)

French terminology

In French,

$$\text{NPV} = \text{Net Present Value} = \text{VAN} = \text{Valeur actuelle nette.}$$

Also,

$$\text{IRR} = \text{Internal Rate of Return} = \text{TRI} = \text{Taux de rendement interne.}$$

Net Present Value (NPV)

For an n-period project, the net present value (NPV) is the present value of the project's net cash flows discounted at the required return r.

Net Present Value

For an n-period project, the net present value is

$$\boxed{NPV = \sum_{t=0}^{n} \frac{(\text{net CF})_t}{(1+r)^t}}$$

where r is the required return, cost of capital, opportunity cost of capital, or benchmark interest rate.

NPV decision rule

If

$$NPV > 0,$$

then the company should do the project.
If choosing between projects, choose the project with the greatest NPV.

Net Present Value

Your company is considering a 5-year project that will require cash investments of 10000 at time 0, 1000 at time 1, and 1000 at time 2.
In return, the company will receive cash flows of 5000 at times $2, 3, 4$, and 5.
If the cost of capital is 5%, determine the net present value of the project.
Solution.
The cash flows are:

Time	0	1	2	3	4	5
Inflows	0	0	5000	5000	5000	5000
Outflows	10000	1000	1000	0	0	0
Net CF	-10000	-1000	4000	5000	5000	5000

Since the cost of capital is 5%, we use

$$v = \frac{1}{1.05}.$$

Thus,

$$NPV = -10000 - 1000v + 4000v^2 + 5000v^3 + 5000v^4 + 5000v^5.$$

Substituting

$$v = \frac{1}{1.05},$$

we get

$$NPV = -10000 - \frac{1000}{1.05} + \frac{4000}{1.05^2} + \frac{5000}{1.05^3} + \frac{5000}{1.05^4} + \frac{5000}{1.05^5}.$$

Therefore,

$$NPV = 5026.07.$$

So,

$$\boxed{NPV = 5026.07}.$$

Since

$$NPV > 0,$$

the project adds value.

Using the BA II Plus.
Use the CF worksheet:

$$\begin{aligned} CF_0 &= -10000, & \\ C01 &= -1000, & F01 = 1, \\ C02 &= 4000, & F02 = 1, \\ C03 &= 5000, & F03 = 3. \end{aligned}$$

Then enter

$$I = 5.$$

Compute:

$$CPT\ NPV = 5026.06816.$$

Thus,

$$NPV = 5026.07.$$

Internal Rate of Return (IRR)

The internal rate of return of a project is the interest rate such that the present value of the cash inflows is equal to the present value of the cash outflows.

In other words, the IRR is the rate that makes the NPV equal to zero.

$$\sum_{t=0}^{n} \frac{(\text{cash inflow})_t}{(1+IRR)^t} = \sum_{t=0}^{n} \frac{(\text{cash outflow})_t}{(1+IRR)^t}.$$

Equivalently,

$$NPV = 0 = \sum_{t=0}^{n} \frac{(\text{net CF})_t}{(1+IRR)^t}.$$

Internal Rate of Return

The internal rate of return, or IRR, is the rate i such that

$$\boxed{\sum_{t=0}^{n} \frac{(\text{net CF})_t}{(1+i)^t} = 0}.$$

IRR decision rule

If

$$IRR > r,$$

then the company should do the project.
If choosing between projects, choose the project with the greatest IRR.

Warning about multiple IRRs

It is possible that the IRR equation yields more than one IRR.
This happens because the IRR equation is usually a polynomial equation, and polynomials can have more than one root.
In an exam question, they would normally clarify which IRR they want.

Internal Rate of Return

Your company is considering a 5-year project that will require cash investments of 10000 at time 0, 1000 at time 1, and 1000 at time 2.
In return, the company will receive cash flows of 5000 at times $2, 3, 4$, and 5.
Calculate the IRR of this project.
Solution.
The cash flows are the same as in the previous example:

Time	0	1	2	3	4	5
Inflows	0	0	5000	5000	5000	5000
Outflows	10000	1000	1000	0	0	0
Net CF	-10000	-1000	4000	5000	5000	5000

Recall that at

$$r = 0.05,$$

we found

$$NPV = 5026.07 > 0.$$

Therefore, we expect the IRR to be greater than 5%.
The IRR solves

$$0 = -10000 - \frac{1000}{1+IRR} + \frac{4000}{(1+IRR)^2} + \frac{5000}{(1+IRR)^3} + \frac{5000}{(1+IRR)^4} + \frac{5000}{(1+IRR)^5}.$$

This equation is not convenient to solve by hand. Use a computer or the BA II Plus calculator.
Using the same CF worksheet as before:

$$\begin{aligned} CF_0 &= -10000, \\ C01 &= -1000, \qquad F01 = 1, \\ C02 &= 4000, \qquad F02 = 1, \\ C03 &= 5000, \qquad F03 = 3. \end{aligned}$$

Then compute:

$$IRR\ CPT = 17.44731389.$$

Thus,

$$IRR = 0.1745.$$

Therefore,

$$\boxed{IRR = 17.45\%}.$$

NPV versus IRR

NPV and IRR answer two different questions.
For NPV, the interest rate is given, and we calculate the value added by the project:

$$\text{given } r, \quad \text{calculate } NPV.$$

For IRR, the NPV is set equal to zero, and we solve for the interest rate:

$$NPV = 0, \quad \text{calculate } IRR.$$

Exercise 5.1.1

A five-year investment project requires an initial investment of 10000 at inception and maintenance expenses at the beginning of each year.
The maintenance expense is 300 for the first year and is anticipated to increase 6% each year thereafter.
Projected annual returns from the project are 3000 at the end of the first year and decrease by 100 per year thereafter.

1. If the cost of capital is 7%, calculate the NPV of the project.

2. Compute the internal rate of return for the project.

Solution.
The initial investment and the first maintenance expense occur at time 0. Therefore, the cash outflow at time 0 is

$$10000 + 300 = 10300.$$

The maintenance expenses occur at the beginning of each year, so they occur at times

$$0, 1, 2, 3, 4.$$

The maintenance expenses are

$$300, \quad 300(1.06), \quad 300(1.06)^2, \quad 300(1.06)^3, \quad 300(1.06)^4.$$

The project returns occur at the ends of years 1 through 5:

$$3000, \quad 2900, \quad 2800, \quad 2700, \quad 2600.$$

Thus, the net cash flows are

Time	0	1	2	3	4	5
Inflows	0	3000	2900	2800	2700	2600
Outflows	10300	300(1.06)	$300(1.06)^2$	$300(1.06)^3$	$300(1.06)^4$	0
Net CF	−10300	2682.00	2562.92	2442.70	2321.26	2600

Part 1: Net Present Value.
The cost of capital is

$$r = 0.07.$$

Therefore,

$$NPV = -10300 + \frac{2682.00}{1.07} + \frac{2562.92}{1.07^2} + \frac{2442.70}{1.07^3} + \frac{2321.26}{1.07^4} + \frac{2600}{1.07^5}.$$

Thus,

$$NPV = 63.70.$$

Rounding to match the calculator output,

$$\boxed{NPV = 63.71}.$$

Since

$$NPV > 0,$$

the project adds value at a 7% cost of capital.
Part 2: Internal Rate of Return.

The IRR is the rate i such that

$$NPV = 0.$$

Thus, i solves

$$0 = -10300 + \frac{2682.00}{1+i} + \frac{2562.92}{(1+i)^2} + \frac{2442.70}{(1+i)^3} + \frac{2321.26}{(1+i)^4} + \frac{2600}{(1+i)^5}.$$

Using the BA II Plus CF and IRR functions, we get

$$IRR = 0.07234.$$

Therefore,

$$\boxed{IRR = 7.234\%}.$$

Using the BA II Plus.
Enter the net cash flows in the CF worksheet:

$$\begin{aligned} CF_0 &= -10300, \\ C01 &= 2682.00, \qquad F01 = 1, \\ C02 &= 2562.92, \qquad F02 = 1, \\ C03 &= 2442.70, \qquad F03 = 1, \\ C04 &= 2321.26, \qquad F04 = 1, \\ C05 &= 2600, \qquad F05 = 1. \end{aligned}$$

For the NPV:

$$I = 7, \qquad CPT\ NPV = 63.71.$$

For the IRR:

$$CPT\ IRR = 7.234\%.$$

Exercise 5.1.2

Jean takes out a loan to be repaid in 120 monthly payments of 3000 each, the first payment being due one month after the date of the loan.
The nominal annual interest rate is 6%, convertible monthly.
Immediately after making the 60th payment, Jean negotiates to pay the bank a lump sum of 150000 to pay off the loan.
Determine the effective annual yield to the lender over the life of the loan.

A. Less than 5.0%
B. At least 5.0%, but less than 5.3%
C. At least 5.3%, but less than 5.6%
D. At least 5.6%, but less than 5.9%
E. At least 5.9%

Solution.
First, compute the original loan amount using the contract interest rate.
The nominal annual interest rate is 6%, convertible monthly, so the monthly rate is

$$j = \frac{0.06}{12} = 0.005.$$

The loan is repaid by 120 monthly payments of 3000, so the loan amount is

$$L = 3000 a_{\overline{120}|0.005}.$$

Thus,

$$L = 3000\left(\frac{1-(1.005)^{-120}}{0.005}\right).$$

Therefore,

$$L = 270220.36.$$

Now we compute the actual yield to the lender.
Jean makes 60 monthly payments of 3000, and then immediately after the 60th payment, he makes a lump sum payment of 150000.
Let j^* be the actual monthly yield to the lender. Then the equation of value at time 0 is

$$270220.36 = 3000a_{\overline{60}|j^*} + 150000(1+j^*)^{-60}.$$

Equivalently,

$$270220.36 = 3000\left(\frac{1-(1+j^*)^{-60}}{j^*}\right) + 150000(1+j^*)^{-60}.$$

Solving numerically gives

$$j^* = 0.00465789.$$

Therefore, the effective annual yield is

$$(1+j^*)^{12} - 1.$$

So,

$$(1.00465789)^{12} - 1 = 0.05735.$$

Thus, the effective annual yield is

$$5.735\%.$$

The correct answer is

$$\boxed{\text{D. At least } 5.6\%, \text{ but less than } 5.9\%}.$$

Exercise 5.1.3

[SOA 19; SOA-IT.023] Project P requires an investment of 4000 today. The investment pays 2000 one year from today and 4000 two years from today.
Project Q requires an investment of X two years from today. The investment pays 2000 today and 4000 one year from today.
The net present values of the two projects are equal at an annual effective interest rate of 10%.
Calculate X.

A. 5400	B. 5420
C. 5440	D. 5460
E. 5480	

Solution.
Let

$$v = \frac{1}{1.10}.$$

The net present value of Project P is

$$NPV_P = -4000 + 2000v + 4000v^2.$$

The net present value of Project Q is

$$NPV_Q = 2000 + 4000v - Xv^2.$$

Since the two net present values are equal, we have

$$-4000 + 2000v + 4000v^2 = 2000 + 4000v - Xv^2.$$

Now substitute

$$v = \frac{1}{1.10}.$$

Then

$$-4000 + \frac{2000}{1.10} + \frac{4000}{1.10^2} = 2000 + \frac{4000}{1.10} - \frac{X}{1.10^2}.$$

Solving for X, we get

$$X = 5460.$$

Therefore, the correct answer is

$$\boxed{\text{D. } 5460}.$$

5.1.2 Reinvestment Rates

Default Assumption

Unless stated otherwise, we assume that interest earned is automatically reinvested at the same interest rate.

This is the assumption we have used so far.

Default reinvestment assumption

When a problem gives one interest rate and does not mention reinvestment, we assume that all interest earned is reinvested at that same rate.

Default reinvestment assumption

A deposit of 100 is made into a fund at time 0 and another deposit of 100 is made at time 1.

If the fund earns an annual effective interest rate of 5%, what is the balance at time 2?

Solution.

The deposits are made at times 0 and 1. Therefore, this is a two-payment annuity-due.

The balance at time 2 is

$$100\ddot{s}_{\overline{2}|0.05}.$$

Thus,

$$100\ddot{s}_{\overline{2}|0.05} = 100\left((1.05)^2 + 1.05\right).$$

Therefore,

$$100\ddot{s}_{\overline{2}|0.05} = 215.25.$$

So the balance at time 2 is

$$\boxed{215.25}.$$

Interest breakdown.
We can also separate the deposits from the interest earned.
The two deposits contribute

$$100 + 100.$$

The interest earned is

$$5 + 5 + 5 + 0.25.$$

Therefore,

$$\text{Balance at time } 2 = 100 + 100 + 5 + 5 + 5 + 0.25.$$

So,

$$\text{Balance at time } 2 = 215.25.$$

Alternative representation.
The two original deposits contribute

$$100(2).$$

The interest payments of 5 behave like an increasing annuity accumulated to time 2:

$$5(Is)_{\overline{2}|0.05}.$$

Thus,

$$100(2) + 5(Is)_{\overline{2}|0.05} = 215.25.$$

Reinvested Rates

Sometimes an investment does not allow automatic reinvestment at the same rate.

For example, bonds have coupon payments that cannot be reinvested in the bond itself. Instead, coupon payments may be reinvested in another fund at a different rate.

Reinvestment rate

A reinvestment rate is the rate earned on interest payments or coupon payments after they are received.
This rate may be different from the yield rate or the rate earned by the original investment.

Reinvestment at a different rate

A deposit of 100 is made into a fund at time 0 and another deposit of 100 is made at time 1.
The fund earns interest at an annual effective interest rate of 5%, payable annually.
Interest payments are reinvested in a separate fund that earns an annual effective interest rate of 3%.
Calculate the total balance of the funds at time 2.
Solution.
The original deposits remain in the 5% fund.
Thus, the principal balance is

$$100 + 100 = 200.$$

The interest payments from the 5% fund are:

$$5 \quad \text{at time } 1,$$

from the time 0 deposit, and

$$5 + 5 \quad \text{at time } 2,$$

from the two deposits.
The interest payment received at time 1 is reinvested at 3% until time 2. Therefore, it accumulates to

$$5(1.03) = 5.15.$$

The two interest payments received at time 2 are already at the valuation date, so they contribute

$$5 + 5 = 10.$$

Therefore, the total balance at time 2 is

$$200 + 5.15 + 10.$$

Thus,

$$\text{Balance at time } 2 = 215.15.$$

So,

$$\boxed{215.15}.$$

Alternative representation.
The original deposits contribute

$$100(2).$$

The interest payments are 5 each and are accumulated at the reinvestment rate 3%. Hence,

$$100(2) + 5(Is)_{\overline{2}|0.03} = 215.15.$$

General Setup

Suppose X is invested at the beginning of each year for n years in a fund earning an annual effective rate i, payable annually.

Interest payments are reinvested in a separate fund earning an annual effective rate j.

We want the total balance of the two funds at time n.

The deposits of X remain in the original fund. Since there are n deposits, their total principal is

$$nX.$$

Each year, each deposit earns interest of

$$iX.$$

These interest payments are reinvested at rate j.

The accumulated value of the interest payments at time n is

$$(iX)(Is)_{\overline{n}|j}.$$

Therefore, the total balance at time n is

$$nX + (iX)(Is)_{\overline{n}|j}.$$

Total balance with reinvested interest

If X is deposited at the beginning of each year for n years, the fund earns interest at rate i payable annually, and interest payments are reinvested at rate j, then the total balance at time n is

$$\boxed{nX + (iX)(Is)_{\overline{n}|j}}.$$

Do not memorize blindly

The formula

$$nX + (iX)(Is)_{\overline{n}|j}$$

is useful, but the process matters more.
Separate the problem into two parts:

$$\text{total balance} = \text{principal still in the original fund} + \text{accumulated value of reinvested interest}.$$

The original deposits contribute

$$nX.$$

The interest payments contribute

$$(iX)(Is)_{\overline{n}|j}.$$

Exercise 5.1.4

Phil invests 10000 now into a fund earning an annual effective rate of 8%, payable annually. The interest payments are reinvested in a separate fund earning an annual effective rate of 6%.
Determine the total value of Phil's investment ten years from now.
Solution.
Phil invests

$$10000$$

at time 0.
The original fund earns interest at an annual effective rate of 8%, payable annually.
Therefore, each year, the interest payment is

$$10000(0.08) = 800.$$

These interest payments are not reinvested in the original 8% fund. Instead, they are reinvested in a separate fund earning 6%.
At time 10, the original principal is still

$$10000.$$

The interest payments of 800 are made at times

$$1, 2, 3, \ldots, 10.$$

Since these payments are reinvested at 6%, their accumulated value at time 10 is

$$800s_{\overline{10}|0.06}.$$

Thus, the total value at time 10 is

$$10000 + 800s_{\overline{10}|0.06}.$$

Using

$$s_{\overline{10}|0.06} = \frac{(1.06)^{10} - 1}{0.06},$$

we get

$$10000 + 800s_{\overline{10}|0.06} = 10000 + 800\left(\frac{(1.06)^{10} - 1}{0.06}\right).$$

Therefore,

$$10000 + 800s_{\overline{10}|0.06} = 20544.64.$$

So, the total value of Phil's investment ten years from now is

$$\boxed{20544.64}.$$

Important

Do not use an increasing annuity factor here.
The interest payments are level:

$$800, 800, \ldots, 800.$$

So the correct accumulated value is

$$800s_{\overline{10}|0.06},$$

not

$$800(Is)_{\overline{10}|0.06}.$$

Exercise 5.1.5

Tiger invests 1000 at the beginning of each year into a fund earning a nominal annual rate of 8% convertible semiannually, payable semiannually.
The interest payments are reinvested in a separate fund earning an annual effective rate of 6%.
Determine the total value of Tiger's investment ten years from now.
Solution.
Tiger deposits

$$1000$$

at the beginning of each year for 10 years.
Thus, the deposits occur at times

$$0, 1, 2, \ldots, 9.$$

Since interest is paid out and reinvested separately, the original principal remains in the original fund.
Therefore, the principal balance at time 10 is

$$10(1000) = 10000.$$

The original fund earns a nominal annual rate of 8% convertible semiannually. Therefore, the semiannual interest rate is

$$\frac{0.08}{2} = 0.04.$$

Each deposit of 1000 earns a semiannual interest payment of

$$1000(0.04) = 40.$$

These interest payments are reinvested in a separate fund earning an annual effective rate of 6%.
Let j be the effective semiannual rate corresponding to the annual effective rate of 6%. Then

$$j = (1.06)^{1/2} - 1.$$

During each year, each active deposit produces two semiannual interest payments of 40. The accumulated value of these two payments at the end of that year is

$$40s_{\overline{2}|j}.$$

At the end of year 1, there is one active deposit. At the end of year 2, there are two active deposits, and so on. At the end of year 10, there are ten active deposits.
Therefore, the annual accumulated interest blocks are proportional to

$$1, 2, 3, \ldots, 10.$$

Their accumulated value at time 10, using the 6% reinvestment rate, is

$$40s_{\overline{2}|j}(Is)_{\overline{10}|0.06}.$$

Hence, the total value at time 10 is

$$10(1000) + 40s_{\overline{2}|j}(Is)_{\overline{10}|0.06}.$$

Using

$$j = (1.06)^{1/2} - 1,$$

we get

$$j = 0.029563.$$

Also,

$$s_{\overline{2}|j} = 2.029563,$$

and

$$(Is)_{\overline{10}|0.06} = 66.1940.$$

Therefore,

$$10(1000) + 40s_{\overline{2}|j}(Is)_{\overline{10}|0.06} = 10000 + 40(2.029563)(66.1940).$$

Thus,

$$\text{Total value} = 15373.80.$$

So,

$$\boxed{15373.80}.$$

Exercise 5.1.6

At time $t = 0$, Rory invests 2000 in a fund earning 8% convertible quarterly, but payable annually.
He reinvests each interest payment in individual separate funds each earning 9% convertible quarterly, but payable annually.
The interest payments from the separate funds are accumulated in a side fund that earns an annual effective rate of 7%.
Determine the total value of all funds at $t = 10$.
Solution.
First, convert the nominal annual rate of 8% convertible quarterly into an annual effective rate.

$$i_1 = \left(1 + \frac{0.08}{4}\right)^4 - 1.$$

Thus,

$$i_1 = 0.08243.$$

The original fund pays interest annually, so the annual interest payment from the original fund is

$$2000(0.08243) = 164.86.$$

These interest payments are made at times

$$1, 2, \ldots, 10.$$

Now convert the nominal annual rate of 9% convertible quarterly into an annual effective rate.

$$i_2 = \left(1 + \frac{0.09}{4}\right)^4 - 1.$$

Thus,

$$i_2 = 0.09308.$$

Each interest payment of 164.86 is reinvested in a separate fund earning $i_2 = 0.09308$, but the interest from these separate funds is paid annually into a side fund.
Therefore, each separate fund produces an annual interest payment of

$$164.86(0.09308) = 15.35.$$

At time 10, the original fund still contains the original principal:

$$2000.$$

The ten separate 9% funds contain the ten reinvested interest payments:

$$10(164.86).$$

The side fund accumulates the annual interest payments from the separate funds.
At time 2, there is one interest payment of 15.35. At time 3, there are two interest payments of 15.35. This continues until time 10, when there are nine interest payments of 15.35.
Thus, the accumulated value in the side fund at time 10 is

$$15.35(Is)_{\overline{9}|0.07}.$$

Therefore, the total value at time 10 is

$$2000 + 10(164.86) + 15.35(Is)_{\overline{9}|0.07}.$$

Using

$$(Is)_{\overline{9}|0.07} = \frac{\ddot{s}_{\overline{9}|0.07} - 9}{0.07},$$

we get

$$2000 + 10(164.86) + 15.35(Is)_{\overline{9}|0.07} = 4485.49.$$

Thus, the total value of all funds at $t = 10$ is

$$\boxed{4485.49}.$$

Exercise 5.1.7

Mary invests 1000 at the end of each year for 5 years at an annual effective rate of 9% and reinvests the interest at an annual effective rate of 9%. At the end of 5 years, her investment has a value of X.
John invests 1000 at the beginning of each year for 5 years at an annual effective rate of 10% and reinvests the interest at an annual effective rate of 8%. At the end of 5 years, his investment has a value of Y.
Calculate $Y - X$.

A. 99 B. 147
C. 327 D. 570
E. 685

Solution.
Mary's investment.
Mary invests 1000 at the end of each year for 5 years.
Since the investment rate and the reinvestment rate are both 9%, this is equivalent to an ordinary accumulated annuity-immediate.
Thus,

$$X = 1000 s_{\overline{5}|0.09}.$$

Using

$$s_{\overline{5}|0.09} = \frac{(1.09)^5 - 1}{0.09},$$

we get

$$X = 1000\left(\frac{(1.09)^5 - 1}{0.09}\right).$$

Therefore,

$$X = 5984.71.$$

John's investment.
John invests 1000 at the beginning of each year for 5 years.
The original principal remains in the 10% fund, so at the end of 5 years the principal contributes

$$5(1000) = 5000.$$

Each deposit earns annual interest of

$$1000(0.10) = 100.$$

Since deposits are made at the beginning of each year, the interest payments form the following increasing pattern:

$$100,\ 200,\ 300,\ 400,\ 500.$$

These interest payments are reinvested at 8%.
Thus, the accumulated value of John's investment is

$$Y = 5000 + 100(Is)_{\overline{5}|0.08}.$$

Using

$$(Is)_{\overline{5}|0.08} = \frac{\ddot{s}_{\overline{5}|0.08} - 5}{0.08},$$

we get

$$Y = 6669.91.$$

Therefore,

$$Y - X = 6669.91 - 5984.71.$$

So,

$$Y - X = 685.20.$$

The closest answer is

$$\boxed{\text{E. } 685}.$$

Exercise 5.1.8

Payments of 1000 are invested at the end of each year for 5 years.
The payments earn interest at an annual effective rate of 10%. The interest can be reinvested at an annual effective rate of 6% in the first 4 years and at an annual effective rate of k thereafter.
The amount in the fund at the end of 5 years is 6090.
Calculate k.

A. 7.5% B. 8.5%

C. 9.5% D. 10.5%

E. 11.5%

Solution.
The payments of 1000 are made at the end of each year for 5 years, so the deposits occur at times

$$1, 2, 3, 4, 5.$$

Since the interest is paid out and reinvested separately, the original principal at time 5 is simply

$$5(1000) = 5000.$$

Therefore, the reinvested interest must account for

$$6090 - 5000 = 1090.$$

Each deposit earns interest at an annual effective rate of 10%, so each deposit produces annual interest of

$$1000(0.10) = 100.$$

The interest payments received before or at time 5 are:

Time	2	3	4	5
Interest	100	200	300	400

The first three interest payments,

$$100, \ 200, \ 300,$$

can be accumulated to time 4 at 6%. Their accumulated value at time 4 is

$$100(Is)_{\overline{3}|0.06}.$$

This amount must then be accumulated one more year from time 4 to time 5 at rate k, giving

$$100(Is)_{\overline{3}|0.06}(1+k).$$

The final interest payment of 400 is received at time 5, so it is already at the valuation date.
Thus, the equation of value at time 5 is

$$6090 = 5000 + 100(Is)_{\overline{3}|0.06}(1+k) + 400.$$

Now compute

$$(Is)_{\overline{3}|0.06} = 1(1.06)^2 + 2(1.06) + 3.$$

Thus,

$$(Is)_{\overline{3}|0.06} = 6.2436.$$

Substitute into the equation:

$$6090 = 5000 + 100(6.2436)(1+k) + 400.$$

So,

$$6090 = 5400 + 624.36(1+k).$$

Therefore,

$$690 = 624.36(1+k).$$

Hence,

$$1 + k = \frac{690}{624.36}.$$

So,

$$k = 0.1051.$$

Thus,

$$k \approx 10.5\%.$$

The correct answer is

$$\boxed{\text{D. } 10.5\%}.$$

Exercise 5.1.9

Amy invests 1000 at an effective annual rate of 14% for 10 years. Interest is payable annually and is reinvested at an annual effective rate of i. At the end of 10 years, the accumulated interest is 2341.08.

Bob invests 150 at the end of each year for 20 years at an annual effective rate of 15%. Interest is payable annually and is reinvested at an annual effective rate of i.
Find Bob's accumulated interest at the end of 20 years.

A. 9000 B. 9010

C. 9020 D. 9030

E. 9040

Solution.
First, use Amy's investment to determine the reinvestment rate i.
Amy invests

$$1000$$

at an effective annual rate of 14%.
Since interest is payable annually, Amy receives annual interest payments of

$$1000(0.14) = 140.$$

These interest payments are reinvested at annual effective rate i.
At the end of 10 years, the accumulated value of the reinvested interest is

$$140s_{\overline{10}|i}.$$

We are given that this accumulated interest is 2341.08, so

$$140s_{\overline{10}|i} = 2341.08.$$

Thus,

$$s_{\overline{10}|i} = \frac{2341.08}{140}.$$

So,

$$s_{\overline{10}|i} = 16.722.$$

Using a financial calculator:

$$N = 10, \qquad PV = 0, \qquad PMT = -140, \qquad FV = 2341.08.$$

Then

$$CPT\ I/Y = 11.$$

Therefore,

$$i = 0.11.$$

Now consider Bob's investment.
Bob invests

$$150$$

at the end of each year for 20 years.
Each deposit earns interest at an annual effective rate of 15%, so each deposit produces annual interest of

$$150(0.15) = 22.5.$$

The interest payments are reinvested at the annual effective rate

$$i = 0.11.$$

The first deposit is made at time 1, so it produces interest payments from time 2 through time 20.
Thus, the accumulated interest payments form an increasing annuity:

$$22.5, \quad 2(22.5), \quad 3(22.5), \quad \ldots, \quad 19(22.5).$$

Therefore, Bob's accumulated interest at the end of 20 years is

$$22.5(Is)_{\overline{19}|0.11}.$$

Using

$$(Is)_{\overline{19}|0.11} = \frac{\ddot{s}_{\overline{19}|0.11} - 19}{0.11},$$

we get

$$22.5(Is)_{\overline{19}|0.11} = 9041.49.$$

The closest answer is

$$\boxed{\text{E. } 9040}.$$

5.1.3 Yield Rates

What is a Yield Rate?

Yield rate

A yield rate, usually denoted by y, is another term for the internal rate of return.
In many exam problems, the terms yield rate and internal rate of return can be used interchangeably.

Unless stated otherwise, we assume that cash inflows are reinvested at the yield rate.

This is the same assumption used for the internal rate of return.

Thus, the yield rate y satisfies

$$\sum_{t=0}^{n} \frac{(\text{cash inflow})_t}{(1+y)^t} = \sum_{t=0}^{n} \frac{(\text{cash outflow})_t}{(1+y)^t}.$$

Equivalently,

$$NPV = 0.$$

Yield rate equation

The yield rate y is the rate that satisfies

$$\boxed{\sum_{t=0}^{n} \frac{(\text{cash inflow})_t}{(1+y)^t} = \sum_{t=0}^{n} \frac{(\text{cash outflow})_t}{(1+y)^t}}$$

or equivalently,

$$\boxed{\sum_{t=0}^{n} \frac{(\text{net CF})_t}{(1+y)^t} = 0}.$$

Default assumption

Unless stated otherwise, cash inflows are assumed to be reinvested at the yield rate.
This is why the standard yield rate equation is the same as the IRR equation.

When the reinvestment rate is different

If we are told that cash inflows are reinvested at a rate other than the yield rate, then we should not blindly use the IRR equation.

Instead, we use the idea:

$$\boxed{\text{what I have} = \text{what I demand}}$$

Here:

$$\text{what I have}$$

means the accumulated value of the cash inflows using the reinvestment rate.

And

$$\text{what I demand}$$

means the accumulated value of the cash outflows using the desired yield rate.

Important distinction

If the reinvestment rate is equal to the yield rate, use the standard IRR equation.
If the reinvestment rate is different from the yield rate, accumulate the inflows at the reinvestment rate and compare them with the accumulated value of the outflows at the yield rate.

Internal Rate of Return vs. Yield Rate

Reinvestment rate equals yield rate

Bill agrees to loan Sally 10000 in exchange for payments of 1295 at the end of each year for 10 years.
What is Bill's yield rate?
Solution.
Since no separate reinvestment rate is given, we assume the payments are reinvested at the yield rate.
Therefore, we use the standard yield equation:

$$10000 = 1295a_{\overline{10}|y}.$$

So,

$$a_{\overline{10}|y} = \frac{10000}{1295}.$$

Solving for y, we get

$$y = 0.05.$$

Thus, Bill's yield rate is

$$\boxed{5\%}.$$

Reinvestment rate different from yield rate

Bill agrees to loan Sally 10000 in exchange for payments of 1295 at the end of each year for 10 years.
Bill can invest Sally's payments at an annual effective rate of 3%.
Determine Bill's annual effective yield rate over the 10 years.
Solution.
Here, the reinvestment rate is not necessarily the same as the yield rate.
The payments of 1295 are reinvested at 3%.
Therefore, the accumulated value of the payments at time 10 is

$$1295 s_{\overline{10}|0.03}.$$

This is what Bill has at time 10.
Now, Bill originally loaned out 10000. If his annual effective yield rate is y, then the accumulated value of his original investment at time 10 should be

$$10000(1+y)^{10}.$$

This is what Bill demands.
Thus, we use

$$\text{what I have} = \text{what I demand}.$$

So,

$$1295 s_{\overline{10}|0.03} = 10000(1+y)^{10}.$$

Using

$$s_{\overline{10}|0.03} = \frac{(1.03)^{10}-1}{0.03},$$

we get

$$1295 s_{\overline{10}|0.03} = 14845.72.$$

Therefore,

$$14845.72 = 10000(1+y)^{10}.$$

So,

$$(1+y)^{10} = 1.484572.$$

Taking the tenth root,

$$1+y = (1.484572)^{1/10}.$$

Thus,

$$y = 0.0403.$$

Therefore, Bill's annual effective yield rate is

$$\boxed{4.03\%}.$$

Why the yield is less than 5%

In the first example, Bill's yield is 5% because the payments are assumed to be reinvested at the yield rate.
In the second example, the payments can only be reinvested at 3%.
Since the reinvestment rate is lower, Bill's actual annual effective yield over the 10 years

decreases to

$$4.03\%.$$

Exercise 5.1.10

Sally lends 10000 to Tim. Tim agrees to pay back the loan over 5 years with monthly payments payable at the end of each month.
Sally can reinvest the monthly payments from Tim in a savings account paying interest at 6%, compounded monthly. The yield rate earned on Sally's investment over the five-year period turned out to be 7.45%, compounded semiannually.
What nominal rate of interest, compounded monthly, did Sally charge Tim on the loan?
Solution.
First, find Tim's monthly loan payment.
Sally can reinvest Tim's monthly payments at a nominal annual rate of 6%, compounded monthly. Therefore, the monthly reinvestment rate is

$$\frac{0.06}{12} = 0.005.$$

The yield rate earned by Sally is 7.45%, compounded semiannually. Therefore, the semi-annual yield rate is

$$\frac{0.0745}{2}.$$

Over 5 years, there are

$$10$$

semiannual periods.
Let X be Tim's monthly payment.
Using the idea

$$\text{what I have} = \text{what I demand},$$

we get

$$X s_{\overline{60}|0.005} = 10000 \left(1 + \frac{0.0745}{2}\right)^{10}.$$

Thus,

$$X = \frac{10000\left(1 + \frac{0.0745}{2}\right)^{10}}{s_{\overline{60}|0.005}}.$$

Using

$$s_{\overline{60}|0.005} = \frac{(1.005)^{60} - 1}{0.005},$$

we get

$$X = 206.62.$$

Now find the loan interest rate.
Let j be the monthly interest rate charged on Tim's loan. Since Tim pays 206.62 monthly for 60 months, the loan equation is

$$10000 = 206.62 a_{\overline{60}|j}.$$

That is,

$$10000 = 206.62 \left(\frac{1 - (1 + j)^{-60}}{j}\right).$$

Solving for j, we get
$$j = 0.007333766.$$
The nominal annual rate of interest compounded monthly is
$$12j.$$
Therefore,
$$12j = 12(0.007333766) = 0.0880.$$
Thus, the nominal annual rate charged on the loan is
$$\boxed{8.8\% \text{ compounded monthly}}.$$

Exercise 5.1.11

An investor purchased a 5-year financial instrument:

1. The investor receives payments of 1000 at the end of each year for 5 years.
2. These payments earn interest at an effective rate of 4% per year. At the end of the year, this interest is reinvested at the rate of 3% per year.

Calculate the purchase price to the investor to produce a yield rate of 4%.
Solution.
Let P be the purchase price.
The investor receives payments of
$$1000$$
at times
$$1, 2, 3, 4, 5.$$
These payments are kept in a fund earning 4%, but the interest earned each year is removed and reinvested at 3%.
At time 5, the original payments contribute
$$5(1000) = 5000.$$
Each payment of 1000 earns annual interest of
$$1000(0.04) = 40.$$
The reinvested interest payments form an increasing annuity:
$$40, \quad 2(40), \quad 3(40), \quad 4(40).$$
These payments are accumulated at 3% to time 5. Therefore, the accumulated value of the reinvested interest is
$$40(Is)_{\overline{4}|0.03}.$$
Thus, the total accumulated value at time 5 is
$$5(1000) + 40(Is)_{\overline{4}|0.03}.$$
Now use the yield condition:
$$\text{what I have} = \text{what I demand}.$$

The investor wants a yield rate of 4%, so the purchase price P must accumulate to

$$P(1.04)^5.$$

Hence,

$$5(1000) + 40(Is)_{\overline{4}|0.03} = P(1.04)^5.$$

Compute

$$(Is)_{\overline{4}|0.03} = 1(1.03)^3 + 2(1.03)^2 + 3(1.03) + 4.$$

Thus,

$$(Is)_{\overline{4}|0.03} = 10.3045.$$

Therefore,

$$5000 + 40(10.3045) = P(1.04)^5.$$

So,

$$5412.18 = P(1.04)^5.$$

Solving for P,

$$P = \frac{5412.18}{(1.04)^5}.$$

Therefore,

$$P = 4448.42.$$

Thus, the purchase price is

$$\boxed{4448.42}.$$

Exercise 5.1.12

Esther invests 100 at the end of each year for 12 years at an annual effective interest rate of i.
The interest payments are reinvested at an annual effective rate of 5%.
The accumulated value at the end of 12 years is 1748.40.
Calculate i.

A. 6% B. 7%

C. 8% D. 9%

E. 10%

Solution.
Esther deposits

$$100$$

at the end of each year for 12 years.
Thus, the deposits occur at times

$$1, 2, \ldots, 12.$$

Since the interest payments are paid out and reinvested separately, the original principal at time 12 is simply

$$100(12) = 1200.$$

Each deposit earns annual interest of

$$100i.$$

The interest payments form an increasing annuity because more deposits are earning interest as time passes.
At time 2, there is one interest payment of $100i$.
At time 3, there are two interest payments of $100i$.
This continues until time 12, when there are eleven interest payments of $100i$.
Therefore, the accumulated value of the reinvested interest at time 12 is

$$100i(Is)_{\overline{11}|0.05}.$$

Hence, the total accumulated value is

$$1200 + 100i(Is)_{\overline{11}|0.05}.$$

We are given

$$1200 + 100i(Is)_{\overline{11}|0.05} = 1748.40.$$

Now compute

$$(Is)_{\overline{11}|0.05}.$$

Using the accumulated increasing annuity formula,

$$(Is)_{\overline{11}|0.05} = \frac{\ddot{s}_{\overline{11}|0.05} - 11}{0.05}.$$

Thus,

$$(Is)_{\overline{11}|0.05} = 78.3425.$$

Substitute:

$$1200 + 100i(78.3425) = 1748.40.$$

So,

$$100i(78.3425) = 548.40.$$

Therefore,

$$i = \frac{548.40}{100(78.3425)}.$$

Thus,

$$i = 0.07.$$

So,

$$i = 7\%.$$

The correct answer is

$$\boxed{\text{B. } 7\%}.$$

Exercise 5.1.13

An investor pays P for an annuity which provides payments of 100 at the beginning of each month for 10 years.
These payments are invested at a nominal annual interest rate of 12% convertible monthly.
Monthly interest payments are reinvested at a nominal annual interest rate of 6% convertible monthly.
The annual yield rate over the 10-year period is 8% effective.
Calculate P.

A. 9600 B. 9650

C. 9700 D. 9750

E. 9800

Solution.
The annuity provides payments of 100 at the beginning of each month for 10 years. Thus, there are

$$10(12) = 120$$

monthly payments.
The payments are invested at a nominal annual interest rate of 12% convertible monthly. Therefore, the monthly interest rate earned by the payments is

$$\frac{0.12}{12} = 0.01.$$

Each payment of 100 earns monthly interest of

$$100(0.01) = 1.$$

The monthly interest payments are reinvested at a nominal annual interest rate of 6% convertible monthly. Therefore, the monthly reinvestment rate is

$$\frac{0.06}{12} = 0.005.$$

At the end of 10 years, the original 120 payments contribute

$$100(120) = 12000.$$

The reinvested interest payments form an increasing annuity:

$$1,\ 2,\ 3, \dots, 120.$$

Thus, the accumulated value of the reinvested interest at time 10 years is

$$(Is)_{\overline{120}|0.005}.$$

Therefore, the total accumulated value of the annuity payments and reinvested interest is

$$100(120) + (Is)_{\overline{120}|0.005}.$$

Now use the yield condition:

$$\text{what I have} = \text{what I demand}.$$

The investor wants an annual effective yield rate of 8% over 10 years, so the purchase price P must accumulate to

$$P(1.08)^{10}.$$

Hence,

$$P(1.08)^{10} = 100(120) + (Is)_{\overline{120}|0.005}.$$

Using

$$(Is)_{\overline{120}|0.005} = \frac{\ddot{s}_{\overline{120}|0.005} - 120}{0.005},$$

we get

$$(Is)_{\overline{120}|0.005} = 8939.75.$$

Thus,

$$P(1.08)^{10} = 12000 + 8939.75.$$

So,

$$P(1.08)^{10} = 20939.75.$$

Therefore,

$$P = \frac{20939.75}{(1.08)^{10}}.$$

Thus,

$$P = 9699.16.$$

The closest answer is

$$\boxed{\text{C. } 9700}.$$

Exercise 5.1.14

Frances borrows 10000 and agrees to make 20 equal payments where the first payment is due in one year.
Each year, she will pay interest at 12% effective on the outstanding principal. The lender wishes to sell the loan to an investor immediately after the loan is made.
Determine the sale price such that the investor will achieve a yield of 15% on this investment.

A. Less than 8400

B. At least 8400, but less than 8500

C. At least 8500, but less than 8600

D. At least 8600, but less than 8700

E. At least 8700

Solution.
First, find Frances's yearly payment.
The loan is repaid with 20 equal annual payments, and the loan interest rate is 12% effective.
Thus,

$$10000 = Ra_{\overline{20}|0.12}.$$

Therefore,

$$R = \frac{10000}{a_{\overline{20}|0.12}}.$$

Using

$$a_{\overline{20}|0.12} = \frac{1-(1.12)^{-20}}{0.12},$$

we get

$$R = 1338.79.$$

Now look at the investor's cash flows.
At time 0, the investor pays the sale price P.
At times $1, 2, \ldots, 20$, the investor receives the annual payments of

$$1338.79.$$

The investor wants a yield rate of 15%. Therefore, the sale price must be the present value of these 20 payments at 15%:

$$P = 1338.79 a_{\overline{20}|0.15}.$$

Using

$$a_{\overline{20}|0.15} = \frac{1-(1.15)^{-20}}{0.15},$$

we get

$$P = 1338.79(6.25933).$$

Thus,

$$P = 8379.93.$$

Therefore, the sale price is

$$\boxed{8379.93}.$$

The correct answer is

$$\boxed{\text{A. Less than 8400}}.$$

5.2 Interest Rate Behavior

5.2.1 Yield Curves

In many earlier problems, we assumed that one interest rate applied to every time period.

In practice, yield rates often depend on the length of the investment. A short-term investment and a long-term investment may have different yield rates, even if the instruments are otherwise similar.

Term structure of interest rates

The **term structure of interest rates** describes how yield rates vary depending on the term, or length, of an investment.
In other words, it shows the relationship between:

$$\text{term of investment} \quad \text{and} \quad \text{yield rate.}$$

For example, a bank might offer different yield rates for certificates of deposit with different terms:

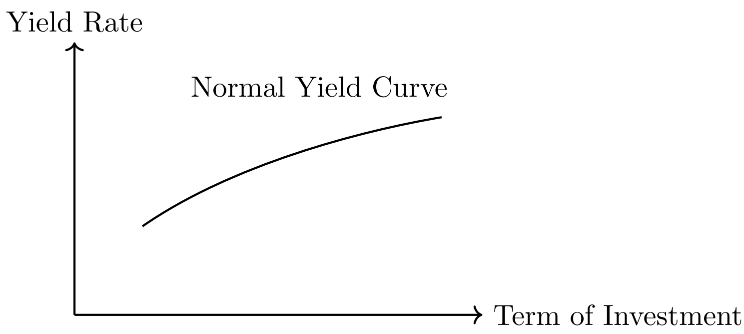
Term	Yield Rate
12 months	0.45%
18 months	1.15%
24 months	2.00%

Here, the yield rate increases as the term of the investment increases.

Yield curve

A **yield curve** is the graphical representation of the term structure of interest rates.
The horizontal axis represents the term of the investment, and the vertical axis represents the yield rate.

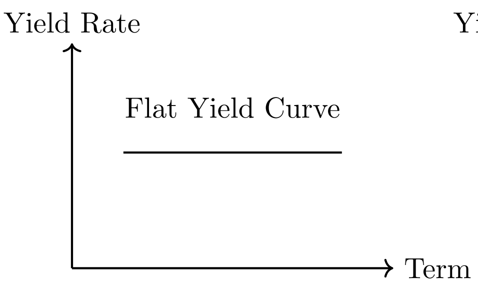
Most of the time, longer investment terms have larger yield rates. This is called a **normal yield curve**.

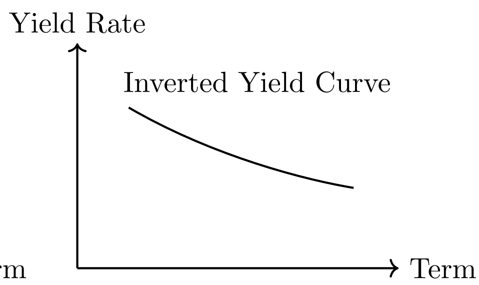


A normal yield curve means that investors require higher yield rates to invest for longer periods.

This can happen because longer-term investments usually involve more uncertainty, such as inflation risk, interest rate risk, and reinvestment risk.

There are two other common shapes of yield curves: flat and inverted.





- A **normal yield curve** slopes upward. Longer terms have higher yield rates.
- A **flat yield curve** is approximately horizontal. Short-term and long-term rates are about the same.
- An **inverted yield curve** slopes downward. Short-term rates are higher than long-term rates.

FM exam interpretation

For most Exam FM problems, unless the problem explicitly gives different rates for different terms, we usually assume a flat yield curve.
That means the same yield rate is used for every maturity.

The important idea is that a yield rate is not always just one number. It may depend on the term of the investment.

When the yield rate changes by term, we must be careful to discount each cash flow using the rate that corresponds to its timing.

Exercise 5.2.1

What is the gist of the phenomenon called *term structure of interest rates*?
Solution.
The *term structure of interest rates* means that interest rates depend on the term, or length, of the investment.
In other words, otherwise similar investments may have different yield rates if their maturities are different.
Thus, the main idea is:

$$\boxed{\text{Interest rates vary by the term of the investment.}}$$

5.2.2 Spot Rates

A **spot rate** is a yield rate that applies from time 0 to one specific future time.

Spot rate

The t-year spot rate, denoted s_t, is the yield rate for a zero-coupon bond that matures in t years.
Unless stated otherwise, s_t is assumed to be an annual effective interest rate.

In other words, s_t is the interest rate from time 0 to time t. A cash flow CF_t paid at time t is discounted using the matching t-year spot rate.

Present value using spot rates

$$\boxed{PV(CF_t) = \frac{CF_t}{(1+s_t)^t}}$$

The important point is that each cash flow is discounted using the spot rate that corresponds to its time.

For example, a cash flow at time 1 uses s_1, while a cash flow at time 2 uses s_2.

Example 1: Pricing a Bond Using Spot Rates

Pricing with spot rates

Given

$$s_1 = 5\% \qquad \text{and} \qquad s_2 = 6\%,$$

calculate the price of a two-year bond with 100 par value and 8% annual coupons.
Solution.
The bond pays annual coupons of 8%, so the annual coupon is

$$100(0.08) = 8.$$

The cash flows are:

Time	Cash Flow
1	8
2	108

The 8 at time 1 is discounted using $s_1 = 5\%$.
The 108 at time 2 is discounted using $s_2 = 6\%$.
Therefore,

$$P = \frac{8}{1.05} + \frac{108}{(1.06)^2}.$$

Thus,

$$P = 103.74.$$

So the bond price is

$$\boxed{103.74}.$$

Spot rates versus one yield rate

With spot rates, each cash flow may be discounted at a different rate.
This differs from the usual bond yield method, where one single yield rate is used for all cash flows.

Example 2: Yield Rate from a Spot-Rate Price

Yield rate of the same bond

Suppose the investor from the previous example buys the bond for 103.74 and holds it to maturity.
Calculate the annual yield rate.
Solution.
The cash flows are still:

Time	Cash Flow
1	8
2	108

Now we want one single annual yield rate i that explains the purchase price.
So we write

$$103.74 = 8a_{\overline{2}|i} + 100v_i^2.$$

Equivalently,

$$103.74 = \frac{8}{1+i} + \frac{108}{(1+i)^2}.$$

Solving gives

$$i = 0.0596.$$

Therefore, the investor's annual yield rate is

$$\boxed{5.96\%}.$$

Key distinction

The spot rates were

$$s_1 = 5\%, \qquad s_2 = 6\%.$$

But the bond's yield rate is

$$i = 5.96\%.$$

The yield rate is a single average-style rate that prices the whole bond. The spot rates are maturity-specific rates.

Bootstrapping Spot Rates

Sometimes spot rates are not given directly. Instead, we infer them from bond prices.

This process is called **bootstrapping**.

Bootstrapping

Bootstrapping is the process of using bond prices to solve for spot rates one maturity at a time.
Usually, we first solve for s_1, then use s_1 to solve for s_2, then use s_1 and s_2 to solve for s_3, and so on.

Bootstrapping s_2

Given the following 100 face amount bonds with annual coupons:

Bond	Maturity	Coupon Rate	Price
A	1	10%	101.85
B	2	10%	102.66

Determine s_2.
Solution.
First, use Bond A to find s_1.
Bond A matures in one year and pays

$$100(0.10) + 100 = 110$$

at time 1.
Thus,

$$101.85 = \frac{110}{1+s_1}.$$

Solving gives

$$1 + s_1 = \frac{110}{101.85}.$$

Therefore,

$$s_1 = 0.08.$$

Now use Bond B to find s_2.
Bond B pays

$$10$$

at time 1, and

$$110$$

at time 2.
The time 1 cash flow is discounted using $s_1 = 8\%$, while the time 2 cash flow is discounted using s_2.
Thus,

$$102.66 = \frac{10}{1.08} + \frac{110}{(1+s_2)^2}.$$

Solving gives

$$s_2 = 0.08523.$$

Therefore,

$$\boxed{s_2 = 8.523\%}.$$

Percentage of Price Method

Sometimes we can avoid recomputing a spot rate by using proportional present values.

Suppose we know the present value of a cash flow A paid at time t. Any other cash flow B paid at the same time has a present value proportional to B/A.

Percentage of price method

If two cash flows occur at the same time t, then

$$\boxed{PV(B) = PV(A)\left(\frac{B}{A}\right)}$$

This works because both cash flows are discounted using the same discount factor.

Using the percentage of price method

You are given that the price of a 5-year zero-coupon bond redeemable at 100 is 95.25.
Find the present value of 10 paid in five years.
Solution.
The cash flow 10 is

$$\frac{10}{100} = 10\%$$

of the zero-coupon bond's cash flow.

Therefore, its present value is 10% of the bond price:

$$PV = 95.25\left(\frac{10}{100}\right).$$

Thus,

$$PV = 9.525.$$

So the present value is

$$\boxed{9.525}.$$

Bootstrapping with the Percentage of Price Method

Reworking the bootstrapping example

Use the percentage of price method to rework the previous bootstrapping example.
The bonds are:

Bond	Maturity	Coupon Rate	Price
A	1	10%	101.85
B	2	10%	102.66

Determine s_2.
Solution.
Bond A has a single cash flow of 110 at time 1, and its price is 101.85.
Bond B has cash flows:

$$10 \quad \text{at time } 1,$$

and

$$110 \quad \text{at time } 2.$$

Since Bond A tells us the present value of 110 at time 1, the present value of 10 at time 1 is

$$101.85\left(\frac{10}{110}\right).$$

Now write the price equation for Bond B:

$$102.66 = 101.85\left(\frac{10}{110}\right) + \frac{110}{(1+s_2)^2}.$$

Solving gives

$$s_2 = 0.08523.$$

Therefore,

$$\boxed{s_2 = 8.523\%}.$$

Exercise 5.2.2

Company XYZ invests 10000 into a project that expects to return 5000 one year after the investment and 7000 two years after the investment.
For XYZ's investment, the one-year spot rate is 10%.
Calculate the two-year spot rate.
Solution.
Let s_2 be the two-year spot rate.
The initial investment is

$$10000.$$

The project returns

$$5000$$

at time 1, and

$$7000$$

at time 2.
The one-year spot rate is

$$s_1 = 0.10.$$

Using spot rates, the present value equation is

$$10000 = \frac{5000}{1.10} + \frac{7000}{(1+s_2)^2}.$$

First,

$$\frac{5000}{1.10} = 4545.45.$$

Thus,

$$10000 = 4545.45 + \frac{7000}{(1+s_2)^2}.$$

So,

$$5454.55 = \frac{7000}{(1+s_2)^2}.$$

Therefore,

$$(1+s_2)^2 = \frac{7000}{5454.55}.$$

Thus,

$$(1+s_2)^2 = 1.28333.$$

Taking the square root:

$$1 + s_2 = \sqrt{1.28333}.$$

So,

$$s_2 = 0.133.$$

Therefore, the two-year spot rate is

$$\boxed{13.3\%}.$$

Exercise 5.2.3

Given the following annual effective spot rates:

$$s_1 = 4.00\%, \qquad s_2 = 4.50\%, \qquad s_3 = 4.75\%, \qquad s_4 = 5.00\%.$$

Calculate

$$100(Ia)_{\overline{4|}}.$$

Solution.
The expression

$$100(Ia)_{\overline{4|}}$$

represents an increasing annuity-immediate with payments

$$100,\ 200,\ 300,\ 400$$

at times $1, 2, 3, 4$, respectively.
Since we are given spot rates, each cash flow must be discounted using the spot rate corresponding to its payment time.
Thus, the payment of 100 at time 1 is discounted using $s_1 = 4.00\%$:

$$\frac{100}{1.04}.$$

The payment of 200 at time 2 is discounted using $s_2 = 4.50\%$:

$$\frac{200}{(1.045)^2}.$$

The payment of 300 at time 3 is discounted using $s_3 = 4.75\%$:

$$\frac{300}{(1.0475)^3}.$$

The payment of 400 at time 4 is discounted using $s_4 = 5.00\%$:

$$\frac{400}{(1.05)^4}.$$

Therefore,

$$100(Ia)_{\overline{4|}} = \frac{100}{1.04} + \frac{200}{(1.045)^2} + \frac{300}{(1.0475)^3} + \frac{400}{(1.05)^4}.$$

So,

$$100(Ia)_{\overline{4|}} = 869.39.$$

Therefore,

$$\boxed{869.39}.$$

Exercise 5.2.4

Given the following prices of zero-coupon bonds with a redemption amount of 105:

Maturity	Price
1	100.96
2	98.02
3	95.39
4	93.29

Calculate the price of a 4-year 100 par value bond with 5% annual coupons.
Solution.

The bond has par value

$$F = 100.$$

The annual coupon rate is 5%, so each annual coupon is

$$100(0.05) = 5.$$

The bond matures in 4 years, so the cash flows are

Time	Cash Flow
1	5
2	5
3	5
4	105

We are given prices of zero-coupon bonds that each pay 105 at maturity.
Using the percentage of price method, the present value of a cash flow at the same time is proportional to the price of the corresponding zero-coupon bond.
For time 1:

$$PV_1 = 100.96\left(\frac{5}{105}\right).$$

For time 2:

$$PV_2 = 98.02\left(\frac{5}{105}\right).$$

For time 3:

$$PV_3 = 95.39\left(\frac{5}{105}\right).$$

For time 4:

$$PV_4 = 93.29\left(\frac{105}{105}\right).$$

Therefore, the bond price is

$$P = 100.96\left(\frac{5}{105}\right) + 98.02\left(\frac{5}{105}\right) + 95.39\left(\frac{5}{105}\right) + 93.29\left(\frac{105}{105}\right).$$

Thus,

$$P = 107.31.$$

Therefore, the price of the bond is

$$\boxed{107.31}.$$

Exercise 5.2.5

Consider the following information on three U.S. Treasury bonds with coupons paid semi-annually:

Bond	Bond Principal	Years to Maturity	Annual Coupon	Bond Price
A	100	0.50	0	95.00
B	100	1.00	9	98.50
C	100	1.50	12	101.50

Using the bootstrap method, calculate the continuously compounded 1.5-year zero rate.

A. 0.1000 B. 0.1062

C. 0.1100 D. 0.1147

E. 0.1200

Solution.
Let

$$\delta_{0.5}, \quad \delta_1, \quad \delta_{1.5}$$

denote the continuously compounded zero rates for maturities 0.5, 1, and 1.5 years.
Since coupons are paid semiannually, the semiannual coupon payment is one-half of the annual coupon.

Bond A.
Bond A has no coupon and matures in 0.5 years. Therefore,

$$100e^{-0.5\delta_{0.5}} = 95.$$

Solving gives

$$\delta_{0.5} = 0.1026.$$

Bond B.
Bond B has an annual coupon of 9, so its semiannual coupon is

$$\frac{9}{2} = 4.5.$$

Its cash flows are 4.5 at time 0.5 and 104.5 at time 1. Thus,

$$4.5e^{-0.5\delta_{0.5}} + 104.5e^{-\delta_1} = 98.50.$$

Using $\delta_{0.5} = 0.1026$, solve for δ_1:

$$\delta_1 = 0.1035.$$

Bond C.
Bond C has an annual coupon of 12, so its semiannual coupon is

$$\frac{12}{2} = 6.$$

Its cash flows are 6 at time 0.5, 6 at time 1, and 106 at time 1.5. Therefore,

$$6e^{-0.5\delta_{0.5}} + 6e^{-\delta_1} + 106e^{-1.5\delta_{1.5}} = 101.50.$$

Substituting the previously found zero rates:

$$6e^{-0.5(0.1026)} + 6e^{-0.1035} + 106e^{-1.5\delta_{1.5}} = 101.50.$$

Solving gives

$$\delta_{1.5} = 0.1062.$$

Therefore, the continuously compounded 1.5-year zero rate is

$$\boxed{0.1062}.$$

The correct answer is

$$\boxed{\text{B. } 0.1062}.$$

Exercise 5.2.6

You are given the following information with respect to a bond:

1. Par value: 1000
2. Term to maturity: 3 years
3. Annual coupon rate: 6%, payable annually

You are also given that the one-year, two-year, and three-year annual spot interest rates are 7%, 8%, and 9%, respectively.
Calculate the value of the bond.

A. 906 B. 926
C. 930 D. 950
E. 1000

Solution.
The par value is

$$F = 1000.$$

The annual coupon rate is 6%, so the annual coupon payment is

$$1000(0.06) = 60.$$

Since the bond matures in 3 years and coupons are paid annually, the cash flows are:

Time	Cash Flow
1	60
2	60
3	1060

Using spot rates, each cash flow must be discounted using the spot rate corresponding to its payment time.
Thus, the bond value is

$$P = \frac{60}{1.07} + \frac{60}{(1.08)^2} + \frac{1060}{(1.09)^3}.$$

Therefore,

$$P = 926.03.$$

The closest answer is

$$\boxed{\text{B. } 926}.$$

5.2.3 Forward Rates

A **forward rate** is an interest rate that applies to an investment made at a future point in time.

In other words, a forward rate answers the following question:

What rate is implied for an investment that starts later and ends even later?

Unless stated otherwise, forward rates are assumed to be annual effective rates.

Forward rate

An m-year forward rate deferred t years is the interest rate that applies from time t to time $t+m$.
We denote this forward rate by

$$f_{t,t+m}.$$

So $f_{t,t+m}$ is the rate earned between time t and time $t+m$.

For example:

$$f_{1,2}$$

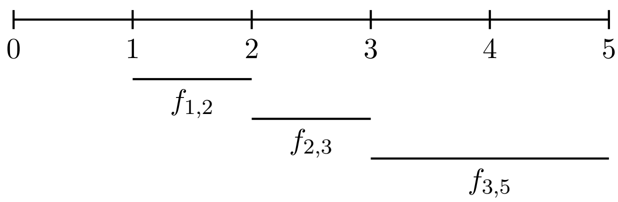
is the one-year forward rate starting in one year.

$$f_{3,5}$$

is the two-year forward rate starting in three years.

$$f_{2,3}$$

is the one-year forward rate starting in two years.



Forward Rates Implied by Spot Rates

Forward rates are implied by spot rates.

Suppose we want the one-year forward rate starting in 4 years. This is

$$f_{4,5}.$$

There are two equivalent ways to accumulate 1 dollar from time 0 to time 5.

First, we can use the 5-year spot rate directly:

$$1(1+s_5)^5.$$

Second, we can accumulate from time 0 to time 4 using s_4, then from time 4 to time 5 using the forward rate $f_{4,5}$:

$$1(1+s_4)^4(1+f_{4,5}).$$

These must be equal:

$$(1+s_4)^4(1+f_{4,5}) = (1+s_5)^5.$$

Solving for $f_{4,5}$, we get

$$1+f_{4,5} = \frac{(1+s_5)^5}{(1+s_4)^4}.$$

Therefore,

$$f_{4,5} = \frac{(1+s_5)^5}{(1+s_4)^4} - 1.$$

One-year forward rate

The one-year forward rate starting at time t is

$$\boxed{f_{t,t+1} = \frac{(1+s_{t+1})^{t+1}}{(1+s_t)^t} - 1}$$

More generally, the m-year forward rate starting at time t satisfies

$$(1+s_t)^t(1+f_{t,t+m})^m = (1+s_{t+m})^{t+m}.$$

Thus,

$$(1+f_{t,t+m})^m = \frac{(1+s_{t+m})^{t+m}}{(1+s_t)^t}.$$

So,

$$f_{t,t+m} = \left[\frac{(1+s_{t+m})^{t+m}}{(1+s_t)^t}\right]^{1/m} - 1.$$

Important idea

You do not need to memorize the formula.
Instead, remember the equation of accumulation:

$$(1+s_t)^t(1+f_{t,t+m})^m = (1+s_{t+m})^{t+m}.$$

This equation says that two equivalent investment strategies must give the same accumulated value.

Locking in a Forward Rate

Forward rates are implied by spot rates, but we can also lock in a forward rate through a borrowing and investing strategy.

Locking in a forward rate

Given

$$s_1 = 0.08 \qquad \text{and} \qquad s_2 = 0.085,$$

1. Calculate the one-year forward rate starting in one year.
2. If you borrow 100 for one year and invest 100 for two years, find your rate of return during year 2.

Solution.
First, calculate $f_{1,2}$.
There are two ways to accumulate 1 dollar from time 0 to time 2:

$$(1 + s_1)(1 + f_{1,2}) = (1 + s_2)^2.$$

Substitute the given spot rates:

$$(1.08)(1 + f_{1,2}) = (1.085)^2.$$

Thus,

$$1 + f_{1,2} = \frac{(1.085)^2}{1.08}.$$

So,

$$f_{1,2} = 0.09.$$

Therefore,

$$\boxed{f_{1,2} = 9\%}.$$

Now suppose you borrow 100 for one year and invest 100 for two years.
At time 0, the two cash flows cancel:

$$+100 - 100 = 0.$$

At time 1, you must repay the one-year loan:

$$100(1.08) = 108.$$

At time 2, the two-year investment grows to

$$100(1.085)^2 = 117.72.$$

Thus, the rate of return from time 1 to time 2 is

$$\frac{117.72}{108} - 1.$$

Therefore,

$$\frac{117.72}{108} - 1 = 0.09.$$

So the rate of return during year 2 is

$$\boxed{9\%}.$$

This is exactly the forward rate $f_{1,2}$.

Interpretation

A forward rate is not just a theoretical number.
It can be locked in by borrowing for the shorter term and investing for the longer term, or by using equivalent market instruments.

Spot Rates in Terms of Forward Rates

We can also write spot rates in terms of forward rates.

For example, suppose we want to write s_3 in terms of the one-year forward rates

$$f_{0,1}, \qquad f_{1,2}, \qquad f_{2,3}.$$

Accumulating 1 dollar from time 0 to time 3 using the 3-year spot rate gives

$$1(1+s_3)^3.$$

Accumulating year by year using forward rates gives

$$1(1+f_{0,1})(1+f_{1,2})(1+f_{2,3}).$$

These must be equal:

$$(1+s_3)^3 = (1+f_{0,1})(1+f_{1,2})(1+f_{2,3}).$$

Therefore,

$$s_3 = [(1+f_{0,1})(1+f_{1,2})(1+f_{2,3})]^{1/3} - 1.$$

More generally, if we have one-year forward rates, then

$$(1+s_n)^n = \prod_{k=0}^{n-1}(1+f_{k,k+1}).$$

So,

$$s_n = \left[\prod_{k=0}^{n-1}(1+f_{k,k+1})\right]^{1/n} - 1.$$

Spot rate from one-year forward rates

$$\boxed{s_n = \left[\prod_{k=0}^{n-1}(1+f_{k,k+1})\right]^{1/n} - 1}$$

Meaning of $f_{0,1}$

The rate $f_{0,1}$ is the one-year forward rate starting immediately.
So it is the same as the one-year spot rate:

$$f_{0,1} = s_1.$$

Exercise 5.2.7

Given the following spot rates:

$$s_1 = 4.00\%, \qquad s_2 = 5.00\%, \qquad s_3 = 5.50\%, \qquad s_4 = 5.75\%, \qquad s_5 = 6.00\%.$$

Calculate the 3-year forward rate, deferred for two years.
Solution.
A 3-year forward rate deferred for two years is the rate from time 2 to time 5.
Therefore, we want

$$f_{2,5}.$$

There are two equivalent ways to accumulate 1 from time 0 to time 5.
First, use the 5-year spot rate:

$$(1 + s_5)^5.$$

Second, use the 2-year spot rate from time 0 to time 2, then the 3-year forward rate from time 2 to time 5:

$$(1 + s_2)^2(1 + f_{2,5})^3.$$

Thus,

$$(1 + s_2)^2(1 + f_{2,5})^3 = (1 + s_5)^5.$$

Substitute

$$s_2 = 0.05 \qquad \text{and} \qquad s_5 = 0.06.$$

Then

$$(1.05)^2(1 + f_{2,5})^3 = (1.06)^5.$$

So,

$$(1 + f_{2,5})^3 = \frac{(1.06)^5}{(1.05)^2}.$$

Taking the cube root:

$$1 + f_{2,5} = \left[\frac{(1.06)^5}{(1.05)^2}\right]^{1/3}.$$

Therefore,

$$f_{2,5} = \left[\frac{(1.06)^5}{(1.05)^2}\right]^{1/3} - 1.$$

Thus,

$$f_{2,5} = 0.06672.$$

Therefore, the 3-year forward rate deferred for two years is

$$\boxed{6.672\%}.$$

Exercise 5.2.8

You are given the following information about the Treasury market. Note that Treasuries always have a par value of 100.

Term	Coupon	Price
1 year	0%	96.62
2 years	0%	x
3 years	0%	88.90

It is known that the one-year forward rate starting in two years is 4.5%.
Calculate x.
Solution.
Since the Treasuries have 0% coupons, they are zero-coupon bonds.
The 3-year Treasury has price 88.90 and pays 100 at time 3.
The one-year forward rate starting in two years is

$$f_{2,3} = 0.045.$$

Let s_2 be the 2-year spot rate.
To accumulate from time 0 to time 3, we can use the 2-year spot rate from time 0 to time 2, then the forward rate from time 2 to time 3.
Thus,

$$88.90 = \frac{100}{(1+s_2)^2(1.045)}.$$

Therefore,

$$(1+s_2)^2 = \frac{100}{88.90(1.045)}.$$

So,

$$(1+s_2)^2 = 1.0764.$$

Taking the square root:

$$1+s_2 = \sqrt{1.0764}.$$

Thus,

$$s_2 = 0.0375.$$

Now the 2-year Treasury price is

$$x = \frac{100}{(1+s_2)^2}.$$

Substitute $s_2 = 0.0375$:

$$x = \frac{100}{(1.0375)^2}.$$

Therefore,

$$x = 92.90.$$

So,

$$\boxed{x = 92.90}.$$

Exercise 5.2.9

You are given the following information about the Treasury market. Note that Treasuries always have a par value of 100.

Term	Coupon	Price
1 year	0%	96.62
2 years	0%	x
3 years	0%	88.90

It is known that the one-year forward rate starting in two years is 4.5%.
Calculate x.
Solution.
Since the Treasuries have 0% coupons, they are zero-coupon bonds.
The 3-year Treasury has price 88.90 and pays 100 at time 3.
The one-year forward rate starting in two years is

$$f_{2,3} = 0.045.$$

Let s_2 be the 2-year spot rate.
To accumulate from time 0 to time 3, we can use the 2-year spot rate from time 0 to time 2, then the forward rate from time 2 to time 3.
Thus,

$$88.90 = \frac{100}{(1+s_2)^2(1.045)}.$$

Therefore,

$$(1+s_2)^2 = \frac{100}{88.90(1.045)}.$$

So,

$$(1+s_2)^2 = 1.0764.$$

Taking the square root:

$$1 + s_2 = \sqrt{1.0764}.$$

Thus,

$$s_2 = 0.0375.$$

Now the 2-year Treasury price is

$$x = \frac{100}{(1+s_2)^2}.$$

Substitute $s_2 = 0.0375$:

$$x = \frac{100}{(1.0375)^2}.$$

Therefore,

$$x = 92.90.$$

So,

$$\boxed{x = 92.90}.$$

Exercise 5.2.10

You are given:

1. the one-year spot rate is 5%;
2. the one-year forward rate starting in one year is 5.5%;
3. the one-year forward rate starting in two years is 6.0%.

Calculate

$$\ddot{a}_{\overline{3}|}.$$

Solution.
An annuity-due with 3 payments has payments of 1 at times

$$0,\ 1,\ 2.$$

Therefore,

$$\ddot{a}_{\overline{3}|} = 1 + PV(1 \text{ at time } 1) + PV(1 \text{ at time } 2).$$

The payment at time 0 is already at the valuation date, so its present value is

$$1.$$

The payment at time 1 is discounted using the one-year spot rate:

$$\frac{1}{1.05}.$$

The payment at time 2 is discounted using the one-year spot rate for year 1 and the one-year forward rate for year 2:

$$\frac{1}{(1.05)(1.055)}.$$

Thus,

$$\ddot{a}_{\overline{3}|} = 1 + \frac{1}{1.05} + \frac{1}{(1.05)(1.055)}.$$

Therefore,

$$\ddot{a}_{\overline{3}|} = 2.855.$$

So,

$$\boxed{2.855}.$$

Exercise 5.2.11

You are given the following information about three bonds:

Bond	Bond Price on Jan. 1, 2001	Jul. 1, 2001	Jul. 1, 2002	Jan. 1, 2003
A	96	100		
B	88		100	
C	125			150

Assuming no arbitrage opportunity exists, calculate the forward rate from July 1, 2002 to January 1, 2003 as a nominal rate convertible semiannually.

A. 0.083 B. 0.087

C. 0.093 D. 0.100

E. 0.112

Solution.
Set January 1, 2001 as time 0.
Then

$$\text{July 1, 2002} = 1.5, \qquad \text{January 1, 2003} = 2.$$

We need the forward rate from time 1.5 to time 2, expressed as a nominal rate convertible semiannually.
Bond B costs 88 at time 0 and pays 100 at time 1.5. Thus, the accumulation factor from time 0 to time 1.5 is

$$a(1.5) = \frac{100}{88}.$$

Bond C costs 125 at time 0 and pays 150 at time 2. Thus, the accumulation factor from time 0 to time 2 is

$$a(2) = \frac{150}{125}.$$

The effective forward rate for the half-year from time 1.5 to time 2 is therefore

$$\frac{a(2)}{a(1.5)} - 1.$$

Substitute:

$$\frac{a(2)}{a(1.5)} = \frac{150/125}{100/88}.$$

So,

$$\frac{a(2)}{a(1.5)} = 1.056.$$

Therefore, the effective half-year forward rate is

$$1.056 - 1 = 0.056.$$

Since the answer is requested as a nominal rate convertible semiannually, we multiply by 2:

$$i^{(2)} = 2(0.056) = 0.112.$$

Therefore, the forward rate is

$$\boxed{0.112}.$$

The correct answer is

$$\boxed{\text{E. } 0.112}.$$

Exercise 5.2.12

The following is a list of prices for zero-coupon bonds with par value of 1000 and varying maturities.

Maturity (Years)	Price of Bond
1	943.40
2	898.47
3	847.62
4	792.16

Calculate the 1-year forward rate, deferred 3 years.

A. 0.057 B. 0.060

C. 0.070 D. 0.080

E. 0.083

Solution.

We are looking for the 1-year forward rate deferred 3 years. This is the forward rate from year 3 to year 4.
Let this forward rate be

$$f_{3,4}.$$

Since the bonds are zero-coupon bonds with par value 1000, the accumulation factor from time 0 to time 3 is

$$\frac{1000}{847.62}.$$

The accumulation factor from time 0 to time 4 is

$$\frac{1000}{792.16}.$$

The 4-year accumulation factor must equal the 3-year accumulation factor multiplied by the 1-year forward accumulation factor from year 3 to year 4. Therefore,

$$\frac{1000}{847.62}(1 + f_{3,4}) = \frac{1000}{792.16}.$$

Thus,

$$1 + f_{3,4} = \frac{1000/792.16}{1000/847.62}.$$

So,

$$1 + f_{3,4} = \frac{847.62}{792.16}.$$

Therefore,

$$f_{3,4} = \frac{847.62}{792.16} - 1.$$

Hence,

$$f_{3,4} = 0.0700.$$

The correct answer is

$$\boxed{\text{C. } 0.070}.$$

Exercise 5.2.13

You are given the following information with respect to a non-callable bond:

- Par amount: 1000
- Term to maturity: 4 years
- Annual coupon rate: 8%, payable annually

The following one-year annual forward interest rates are given:

Time	Scenario X	Scenario Y
0	7%	7%
1	7%	6%
2	8%	7%
3	10%	5%

Each interest rate scenario has an equal probability of occurring.
Calculate the value of the bond, that is, the expected present value of the bond payments.

A. 1000.00 B. 1018.40
C. 1022.80 D. 1030.39
E. 1031.07

Solution.
The annual coupon payment is
$$1000(0.08) = 80.$$
Since the bond matures in 4 years, the cash flows are
$$80, \quad 80, \quad 80, \quad 1080.$$
We discount the cash flows separately under each interest rate scenario, then average the two present values.
Scenario X.
Under Scenario X, the one-year forward rates are
$$7\%, \quad 7\%, \quad 8\%, \quad 10\%.$$
Thus,
$$PV_X = \frac{80}{1.07} + \frac{80}{(1.07)(1.07)} + \frac{80}{(1.07)(1.07)(1.08)} + \frac{1080}{(1.07)(1.07)(1.08)(1.10)}.$$
Therefore,
$$PV_X = 1003.37.$$
Scenario Y.
Under Scenario Y, the one-year forward rates are
$$7\%, \quad 6\%, \quad 7\%, \quad 5\%.$$
Thus,
$$PV_Y = \frac{80}{1.07} + \frac{80}{(1.07)(1.06)} + \frac{80}{(1.07)(1.06)(1.07)} + \frac{1080}{(1.07)(1.06)(1.07)(1.05)}.$$
Therefore,
$$PV_Y = 1058.76.$$
Since the two scenarios are equally likely, the expected present value is
$$\frac{PV_X + PV_Y}{2}.$$
So,
$$\frac{1003.37 + 1058.76}{2} = 1031.07.$$
Therefore, the value of the bond is
$$\boxed{1031.07}.$$
The correct answer is
$$\boxed{\text{E. } 1031.07}.$$

5.2.4 Real Rates of Interest

A **real rate of interest** measures the true growth in purchasing power after adjusting for inflation.

Suppose you have 10000 in a savings account today and the account earns a 1% annual effective interest rate. After one year, you have

$$10000(1.01) = 10100.$$

You have 1% more dollars, but that does not necessarily mean that you can buy 1% more goods and services.

If inflation is 2%, then the amount needed one year later to have the same purchasing power as 10000 today is

$$10000(1.02) = 10200.$$

Since you only have 10100, your purchasing power has decreased.

Measured in today's dollars, your future amount is

$$\frac{10100}{1.02} = 9901.96.$$

Therefore, your real rate of return is

$$\frac{9901.96}{10000} - 1 = -0.0098.$$

So even though the nominal return is positive, the real return is negative.

Real rate of interest

The **real rate of interest** is the interest rate after adjusting for inflation.
It measures the change in purchasing power rather than the change in the number of dollars.

Let

$$i = \text{annual effective interest rate,}$$

$$r = \text{inflation rate,}$$

and

$$i' = \text{real rate of interest.}$$

The relationship between these rates is

$$1 + i = (1 + r)(1 + i').$$

Solving for the real rate gives

$$1 + i' = \frac{1 + i}{1 + r}.$$

Thus,

$$i' = \frac{1 + i}{1 + r} - 1.$$

Equivalently,

$$i' = \frac{i - r}{1 + r}.$$

Real rate formula

$$\boxed{1+i=(1+r)(1+i')}$$

$$\boxed{i'=\frac{i-r}{1+r}}$$

Interpretation

The real rate is not exactly $i-r$.
The approximation

$$i'\approx i-r$$

can be useful when the rates are small, but the exact formula is

$$i'=\frac{i-r}{1+r}.$$

Nominal versus Real Interest

In this context, some people call i the **nominal rate of interest** because it is the rate measured in dollars before adjusting for inflation.

The true inflation-adjusted rate is i'.

This use of the word *nominal* is different from the earlier meaning of nominal rates such as

$$i^{(12)}=12\%$$

convertible monthly.

Two meanings of nominal

There are two different uses of the word *nominal*:

- A nominal rate convertible m-thly, such as $i^{(12)}$, describes a compounding frequency.
- A nominal rate before inflation describes a return measured in dollars rather than purchasing power.

Connection with Geometric Annuities

Geometric annuities can be interpreted as real-rate problems.

If payments grow at rate r, and the investment earns rate i, then the effective discounting rate after adjusting for the payment growth is

$$j=\frac{i-r}{1+r}.$$

Thus,

$$(G\ddot{a})_{\overline{n}|i,r}=\ddot{a}_{\overline{n}|j},$$

where

$$j=\frac{i-r}{1+r}.$$

This is the same structure as the real rate formula.

Example

Y

ou expect inflation to be 2% next year and to grade linearly to 0% by the fifth year. If your investment returns 8% per year, find the real rate of interest earned during the third year.

Solution.

The inflation rates by year are

$$2\%, \quad 1.5\%, \quad 1\%, \quad 0.5\%, \quad 0\%.$$

The third year is the period from time 2 to time 3, so the inflation rate during the third year is

$$r = 1\% = 0.01.$$

The nominal investment return is

$$i = 8\% = 0.08.$$

The real rate i' satisfies

$$1 + i = (1 + r)(1 + i').$$

Thus,

$$1.08 = (1.01)(1 + i').$$

So,

$$1 + i' = \frac{1.08}{1.01}.$$

Therefore,

$$i' = \frac{1.08}{1.01} - 1.$$

Equivalently,

$$i' = \frac{0.08 - 0.01}{1.01}.$$

Hence,

$$i' = 0.0693.$$

The real rate of interest earned during the third year is

$$\boxed{6.93\%}.$$

Exercise 5.2.14

The current real rate of interest is 4%. The expected annual inflation rate over the next several years is 5%.

Given the following expected nominal cash flows, calculate the net present value of the investment using the nominal rate of interest.

Year	0	1	2
Cash Flow	−300	160	160

Solution.
Let

$$i' = \text{real rate of interest,}$$

$$r = \text{inflation rate,}$$

and

$$i = \text{nominal rate of interest.}$$

We are given

$$i' = 0.04 \qquad \text{and} \qquad r = 0.05.$$

The relationship between the nominal rate, the inflation rate, and the real rate is

$$1 + i = (1 + r)(1 + i').$$

Substitute the given values:

$$1 + i = (1.05)(1.04).$$

Thus,

$$1 + i = 1.092.$$

Therefore,

$$i = 0.092.$$

So the nominal rate of interest is

$$9.2\%.$$

Now discount the nominal cash flows using the nominal rate $i = 0.092$.
The net present value is

$$NPV = -300 + \frac{160}{1.092} + \frac{160}{(1.092)^2}.$$

Therefore,

$$NPV = -19.30.$$

So the net present value of the investment is

$$\boxed{-19.30}.$$

5.3 Duration

5.3.1 Price as a Function of Yield

The price of a bond is the present value of all its future cash flows.

If the bond has cash flows CF_t at times t, and if the yield rate is i, its price is the present value of those cash flows (see formula box below).

Price as present value of cash flows

$$\boxed{P = \sum_t CF_t(1+i)^{-t}}$$

This formula shows an important relationship:

$$\boxed{\text{When the yield rate increases, the bond price decreases.}}$$

In other words, bond prices and yield rates move in opposite directions.

C

alculate the price of a 100 ten-year zero-coupon bond with an annual effective yield rate of 8%.
Solution.
A zero-coupon bond pays only one cash flow at maturity.
Here, the bond pays

$$100$$

at time 10.
Thus,

$$P_{0.08} = 100(1.08)^{-10}.$$

Therefore,

$$P_{0.08} = 46.32.$$

Now suppose the yield rate increases from 8% to 9%.
The new price is

$$P_{0.09} = 100(1.09)^{-10}.$$

Thus,

$$P_{0.09} = 42.24.$$

So the price decreases when the yield increases.

$$\boxed{P_{0.09} < P_{0.08}}$$

Price Sensitivity

Price sensitivity

Price sensitivity is the percentage change in the price of an asset caused by a change in the yield rate.

If the yield rate changes from i_0 to i_1, price sensitivity is measured by the relative change in price.

Price sensitivity

$$\boxed{\text{price sensitivity} = \frac{P_{i_1} - P_{i_0}}{P_{i_0}}}$$

C

alculate the price sensitivity of the 100 ten-year zero-coupon bond when the yield rate changes from 8% to 9%.
Solution.

From the previous example,

$$P_{0.08} = 46.32$$

and

$$P_{0.09} = 42.24.$$

Therefore,

$$\text{price sensitivity} = \frac{42.24 - 46.32}{46.32}.$$

Thus,

$$\text{price sensitivity} = -0.088.$$

So the price decreases by approximately

$$\boxed{8.8\%}.$$

Effect of Maturity

Cash flows further in the future are more sensitive to yield changes.

To see this, compare a ten-year zero-coupon bond with a twenty-year zero-coupon bond.

U

se the same change in yield rate, from 8% to 9%, to calculate the price sensitivity of a 100 twenty-year zero-coupon bond.
Solution.
At 8%, the price is

$$100(1.08)^{-20}.$$

At 9%, the price is

$$100(1.09)^{-20}.$$

Thus, the price sensitivity is

$$\frac{100(1.09)^{-20} - 100(1.08)^{-20}}{100(1.08)^{-20}}.$$

Therefore,

$$\text{price sensitivity} = -0.168.$$

So the twenty-year bond decreases by approximately

$$\boxed{16.8\%}.$$

This is larger in magnitude than the ten-year bond's price sensitivity of -8.8%.

Maturity and sensitivity

The longer the time until a cash flow is paid, the more sensitive its present value is to changes in the yield rate.
This is why long-term bonds are usually more sensitive to interest rate changes than short-term bonds.

Effect of the Current Yield Level

Price sensitivity also depends on the current level of the yield rate.

For the same change in yield, a bond is usually more sensitive when the starting yield rate is lower.

F

or the 100 ten-year zero-coupon bond, calculate the price sensitivity when the yield rate changes from 2% to 3%.
Solution.
At 2%, the price is

$$100(1.02)^{-10}.$$

At 3%, the price is

$$100(1.03)^{-10}.$$

Thus, the price sensitivity is

$$\frac{100(1.03)^{-10} - 100(1.02)^{-10}}{100(1.02)^{-10}}.$$

Therefore,

$$\text{price sensitivity} = -0.093.$$

So the price decreases by approximately

$$\boxed{9.3\%}.$$

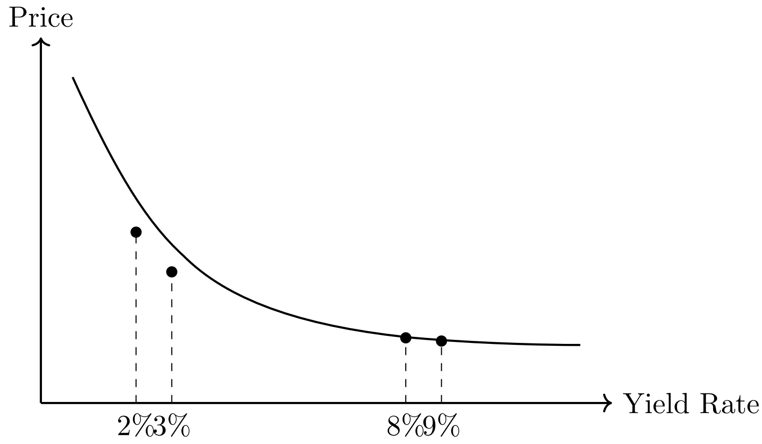
Recall that the price sensitivity from 8% to 9% was only −8.8%.
Thus, for this same one-percentage-point increase in yield, the bond is slightly more sensitive when the yield rate starts at 2% than when it starts at 8%.

Yield level and sensitivity

The lower the yield rate, the more sensitive future cash flows are to changes in the yield rate.

Shape of the Price-Yield Curve

The relationship between bond price and yield is decreasing and curved.



The curve is steeper at lower yield rates and flatter at higher yield rates.

This means that the same increase in yield can cause a larger percentage price change when yields are low.

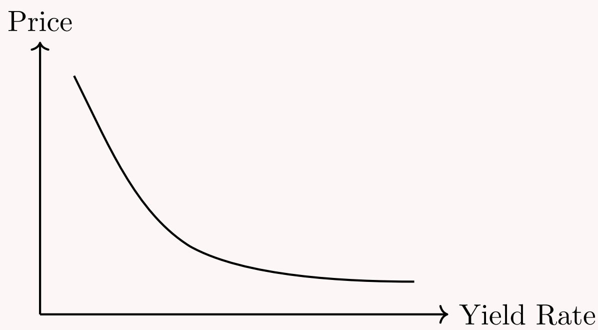
Exercise 5.3.1

True or false: when interest rates decrease, bond prices go down.
Solution.
False.
Bond prices and interest rates are inversely related. When interest rates decrease, bond prices increase.
The price of a bond is the present value of its future cash flows:

$$P = \sum_t CF_t(1+i)^{-t}.$$

If i decreases, the discount factor $(1+i)^{-t}$ increases. Therefore, the present value of the cash flows increases.
So a decrease in interest rates causes bond prices to go up.

$$\boxed{\text{False}}$$



Exercise 5.3.2

Which of the following bonds is the most price sensitive to changes in interest rates?

A. 5-year zero-coupon bond
B. 10-year zero-coupon bond
C. 20-year zero-coupon bond
D. They are equally sensitive

Solution.
Cash flows further into the future are more sensitive to changes in interest rates.
A zero-coupon bond has only one cash flow, paid at maturity. Therefore, the longer the maturity, the more sensitive the bond price is to changes in interest rates.
Among the choices, the longest maturity is the 20-year zero-coupon bond.
Thus, the most price sensitive bond is the

$$\boxed{\text{20-year zero-coupon bond}}.$$

The correct answer is

$$\boxed{\text{C}}.$$

Exercise 5.3.3

Let x be the price change for a 20-year zero-coupon bond when interest rates change from 5% to 4%.
Let y be the price change for the same bond when interest rates change from 15% to 14%.
True or false:

$$|x| > |y|.$$

Solution.
True.
The same bond is being considered in both cases. The yield decreases by one percentage point in both cases.
However, bond prices are more sensitive to interest rate changes when the current yield rate is lower.
A change from 5% to 4% occurs at a lower yield level than a change from 15% to 14%.
Therefore, the price change is larger in absolute value when the yield changes from 5% to 4%.
Thus,

$$|x| > |y|.$$

So the statement is

$$\boxed{\text{True}}.$$

Exercise 5.3.4

Infinite Life's only liability is due in 5 years. The liability is 5000.
Calculate the price sensitivity of the liability for an increase in yield from an annual effective rate of 6% to 7%.

A. -0.0481 B. -0.0459
C. 0.0459 D. 0.0470
E. 0.0481

Solution.

The liability is a single payment of 5000 due in 5 years.
At yield 6%, the price of the liability is

$$P_{0.06} = 5000v_{0.06}^5.$$

Thus,

$$P_{0.06} = 5000(1.06)^{-5} = 3736.29.$$

At yield 7%, the price of the liability is

$$P_{0.07} = 5000v_{0.07}^5.$$

Thus,

$$P_{0.07} = 5000(1.07)^{-5} = 3564.93.$$

Price sensitivity is the percentage change in price:

$$\text{Price Sensitivity} = \frac{P_{0.07} - P_{0.06}}{P_{0.06}}.$$

Therefore,

$$\text{Price Sensitivity} = \frac{3564.93 - 3736.29}{3736.29}.$$

So,

$$\text{Price Sensitivity} = -0.0459.$$

The correct answer is

$$\boxed{\text{B.} \ -0.0459}.$$

5.3.2 Macaulay Duration

We have seen that price sensitivity depends on when a cash flow occurs.

A cash flow paid far in the future is more sensitive to changes in the yield rate than a cash flow paid soon.

This leads to the following question:

> How can we measure the price sensitivity of a collection of cash flows occurring at different times?

This is the motivation for **Macaulay duration**.

Macaulay duration

The **Macaulay duration** of a set of cash flows is the weighted average time until the cash flows are received, where the weights are based on the present values of the cash flows.

For a single cash flow at time t, the Macaulay duration is simply

$$\text{MacD} = t.$$

For multiple cash flows, the duration becomes a present-value-weighted average.

Let

$$A_t = \text{present value of the cash flow occurring at time } t.$$

Then

Both common forms of Macaulay duration (using $\sum_t A_t or the price P) are summarized in the formula box below.$

Macaulay duration

$$\boxed{\text{MacD} = \frac{\sum_t tA_t}{\sum_t A_t} = \frac{\sum_t tA_t}{P}}$$

Interpretation

Macaulay duration is measured in years.
It represents the average timing of the cash flows, weighted by present value.
Cash flows with larger present values receive larger weights in the average.

Example: Two Cash Flows

A

n investment has payments of 100 at time 5 and time 10. The annual effective yield rate is 8%.
Find the Macaulay duration.
Solution.
The present value of the cash flow at time 5 is

$$A_5 = 100(1.08)^{-5} = 68.06.$$

The present value of the cash flow at time 10 is

$$A_{10} = 100(1.08)^{-10} = 46.32.$$

The time 5 cash flow receives more weight because its present value is larger.
Therefore,

$$\text{MacD} = 5\left(\frac{68.06}{68.06 + 46.32}\right) + 10\left(\frac{46.32}{68.06 + 46.32}\right).$$

Equivalently,

$$\text{MacD} = \frac{5(68.06) + 10(46.32)}{68.06 + 46.32}.$$

Thus,

$$\text{MacD} = 7.025.$$

So the duration is

$$\boxed{7.025 \text{ years}}.$$

This value is closer to 5 than to 10, because the earlier cash flow has a larger present value.

Effect of coupons and yield

Because Macaulay duration is a weighted average, all else equal:

- larger coupons reduce duration, because more value is paid earlier;
- higher yields reduce duration, because later cash flows are discounted more heavily.

Duration of a Level Annuity-Immediate

For a level annuity-immediate, the payments are equal and occur at regular intervals.

F

ind the duration of a ten-year annuity-immediate with annual payments of 5. The first payment occurs one year from now, and the annual effective interest rate is 3%.
Solution.
The cash flows are

$$5, 5, 5, \dots, 5$$

at times

$$1, 2, 3, \dots, 10.$$

The price of the annuity is

$$P = 5a_{\overline{10}|0.03}.$$

The numerator of Macaulay duration is

$$\sum_{t=1}^{10} tA_t = \sum_{t=1}^{10} t(5v^t) = 5(Ia)_{\overline{10}|0.03}.$$

Therefore,

$$\text{MacD} = \frac{5(Ia)_{\overline{10}|0.03}}{5a_{\overline{10}|0.03}}.$$

The factor 5 cancels:

$$\text{MacD} = \frac{(Ia)_{\overline{10}|0.03}}{a_{\overline{10}|0.03}}.$$

Thus,

$$\text{MacD} = 5.256.$$

So the duration is

$$\boxed{5.256 \text{ years}}.$$

Duration of a level annuity-immediate

For an n-year level annuity-immediate with m payments per year, the Macaulay duration is

$$\boxed{\mathrm{MacD} = \frac{(Ia)_{\overline{mn}|j}}{m a_{\overline{mn}|j}}}$$

where

$$j = \frac{i^{(m)}}{m}.$$

Duration of a Par Bond

A bond priced at par has coupon rate equal to yield rate, assuming the bond is redeemable at par.

C

alculate the duration of a 10-year bond with annual coupons of 5%, priced at par.
Solution.
Because the bond is priced at par, the coupon rate equals the yield rate:

$$i = 0.05.$$

Let P be the par value of the bond.
The annual coupon is

$$0.05P.$$

At maturity, the bond also pays the redemption value P.
The Macaulay duration is

$$\mathrm{MacD} = \frac{\sum_{k=1}^{10} k(0.05P)v^k + 10Pv^{10}}{P}.$$

Simplify:

$$\mathrm{MacD} = 0.05(Ia)_{\overline{10}|0.05} + 10v^{10}.$$

Using the annuity identity,

$$\mathrm{MacD} = \ddot{a}_{\overline{10}|0.05}.$$

Therefore,

$$\mathrm{MacD} = 8.107.$$

So the duration is

$$\boxed{8.107 \text{ years}}.$$

Shortcut for a par bond

The Macaulay duration of an n-year par bond with m-thly coupons is

$$\boxed{\text{MacD of par bond} = \ddot{a}^{(m)}_{\overline{n}|i}}$$

Equivalently,

$$\boxed{\text{MacD of par bond} = \frac{1}{m}\ddot{a}_{\overline{mn}|j}}$$

where

$$j = \frac{i^{(m)}}{m}.$$

Key idea

Macaulay duration turns a collection of cash flows into one average time.
This average time is useful because it helps summarize how sensitive the price is to changes in interest rates.

Exercise 5.3.5

A 20-year bond pays semiannual coupons of 7.4% and is priced at par.
Calculate the duration.
Solution.
Since the bond is priced at par, we can use the shortcut for the Macaulay duration of a par bond.
The bond has semiannual coupons, so

$$m = 2.$$

The term is 20 years, so the total number of coupon periods is

$$20(2) = 40.$$

The semiannual yield rate is

$$j = \frac{0.074}{2} = 0.037.$$

For an n-year par bond with m-thly coupons, the Macaulay duration is

$$\text{MacD} = \frac{1}{m}\ddot{a}_{\overline{mn}|j}.$$

Therefore,

$$\text{MacD} = \frac{1}{2}\ddot{a}_{\overline{40}|0.037}.$$

Thus,

$$\text{MacD} = 10.737.$$

So the duration is

$$\boxed{10.737 \text{ years}}.$$

Exercise 5.3.6

A three-year bond with 8% annual coupons has a redemption value of 105% of the par value and is priced to yield an annual effective rate of 6%.
Calculate the duration.
Solution.
The bond is not priced at par, so we cannot use the par bond duration shortcut.
Since duration is unaffected by scaling, let the par value be

$$F = 1.$$

The annual coupon is

$$0.08.$$

The redemption value is

$$1.05.$$

Therefore, the cash flows are

$$0.08 \quad \text{at time } 1,$$

$$0.08 \quad \text{at time } 2,$$

and

$$0.08 + 1.05 = 1.13 \quad \text{at time } 3.$$

The annual effective yield rate is

$$i = 0.06,$$

so

$$v = \frac{1}{1.06}.$$

The Macaulay duration is

$$\text{MacD} = \frac{\sum tA_t}{\sum A_t},$$

where A_t is the present value of the cash flow at time t.
Thus,

$$\text{MacD} = \frac{1(0.08)v + 2(0.08)v^2 + 3(1.13)v^3}{0.08v + 0.08v^2 + 1.13v^3}.$$

Substituting $v = v_{0.06}$, we get

$$\text{MacD} = \frac{1(0.08)v_{0.06} + 2(0.08)v_{0.06}^2 + 3(1.13)v_{0.06}^3}{0.08v_{0.06} + 0.08v_{0.06}^2 + 1.13v_{0.06}^3}.$$

Therefore,

$$\text{MacD} = 2.7972.$$

So the duration is

$$\boxed{2.7972 \text{ years}}.$$

Exercise 5.3.7

An annuity-immediate makes monthly payments of 120 for 15 years. The constant force of interest is 0.05.
Calculate the duration of the annuity.
Solution.

The annuity makes monthly payments for 15 years, so the total number of payments is

$$12(15) = 180.$$

Since the force of interest is constant with

$$\delta = 0.05,$$

the effective monthly interest rate is

$$j = e^{0.05/12} - 1.$$

The first payment occurs at time 1/12, the second at time 2/12, and so on.
For a level annuity-immediate with m payments per year, the Macaulay duration is

$$\text{MacD} = \frac{(Ia)_{\overline{mn}|j}}{m a_{\overline{mn}|j}}.$$

Here,

$$m = 12, \qquad n = 15, \qquad mn = 180.$$

Thus,

$$\text{MacD} = \frac{(Ia)_{\overline{180}|j}}{12 a_{\overline{180}|j}}.$$

Equivalently, using the payment amount 120,

$$\text{MacD} = \frac{10(Ia)_{\overline{180}|j}}{120 a_{\overline{180}|j}},$$

because

$$120\left(\frac{k}{12}\right) = 10k.$$

Substituting

$$j = e^{0.05/12} - 1,$$

we get

$$\text{MacD} = \frac{100278.91}{15164.21}.$$

Therefore,

$$\text{MacD} = 6.613.$$

So the duration of the annuity is

$$\boxed{6.613 \text{ years}}.$$

Exercise 5.3.8

You are given a 1000 par value 20-year bond with 4% annual coupons and a maturity value of 1000.
Calculate the duration of this bond at 5% interest.

A. 5.5 B. 8.9

C. 13.7 D. 20.0

E. 24.0

Solution.
The annual coupon payment is

$$Fr = 1000(0.04) = 40.$$

The maturity value is

$$C = 1000.$$

The annual effective yield rate is

$$i = 0.05,$$

so

$$v = \frac{1}{1.05}.$$

Since the coupon rate is 4% but the yield rate is 5%, the bond is not priced at par. Therefore, we cannot use the par bond shortcut.
We use the general Macaulay duration formula:

$$\text{MacD} = \frac{\sum tA_t}{P},$$

where A_t is the present value of the cash flow at time t, and P is the bond price.
The bond price is

$$P = 40a_{\overline{20}|0.05} + 1000v_{0.05}^{20}.$$

The numerator of the Macaulay duration is

$$40(Ia)_{\overline{20}|0.05} + 20(1000)v_{0.05}^{20}.$$

Therefore,

$$\text{MacD} = \frac{40(Ia)_{\overline{20}|0.05} + 20000v_{0.05}^{20}}{40a_{\overline{20}|0.05} + 1000v_{0.05}^{20}}.$$

Thus,

$$\text{MacD} = 13.68.$$

Rounded to one decimal place,

$$\text{MacD} = 13.7.$$

The correct answer is

$$\boxed{\text{C. } 13.7}.$$

Exercise 5.3.9

A perpetuity-due with level payments has a duration of 10 at an effective interest rate of i.
Calculate i.

A. 7% B. 8%

C. 9% D. 10%

E. 11%

Solution.
Assume the level payment is 1. Since this is a perpetuity-due, payments occur at times

$$0, 1, 2, 3, \ldots$$

The Macaulay duration is

$$\text{MacD} = \frac{\sum tA_t}{\sum A_t},$$

where A_t is the present value of the cash flow at time t.
The denominator is the present value of a perpetuity-due:

$$\sum A_t = \ddot{a}_{\overline{\infty}|} = 1 + \frac{1}{i}.$$

The numerator is the time-weighted present value of the cash flows:

$$\sum tA_t = 0(1) + 1v + 2v^2 + 3v^3 + \cdots .$$

This is the present value of an increasing perpetuity-immediate:

$$\sum tA_t = (Ia)_{\overline{\infty}|} = \frac{1}{i} + \frac{1}{i^2}.$$

Therefore,

$$\text{MacD} = \frac{(Ia)_{\overline{\infty}|}}{\ddot{a}_{\overline{\infty}|}} = \frac{\frac{1}{i} + \frac{1}{i^2}}{1 + \frac{1}{i}}.$$

Simplify:

$$\text{MacD} = \frac{\frac{1+i}{i^2}}{\frac{1+i}{i}} = \frac{1}{i}.$$

We are given that the duration is 10, so

$$10 = \frac{1}{i}.$$

Thus,

$$i = 0.10.$$

Therefore,

$$i = 10\%.$$

The correct answer is

$$\boxed{\text{D. } 10\%}.$$

Exercise 5.3.10

Kylie bought a 7-year, 5000 par value bond with an annual coupon rate of 7.6% paid semiannually. She bought the bond with no premium or discount.
Calculate the Macaulay duration of this bond with respect to the yield rate on the bond.

A. 5.16 B. 5.35

C. 5.56 D. 5.77

E. 5.99

Solution.
Since Kylie bought the bond with no premium or discount, the bond is priced at par. Therefore, we can use the shortcut for the Macaulay duration of a bond priced at par. The bond pays coupons semiannually, so

$$m = 2.$$

The bond has a term of 7 years, so the total number of coupon periods is

$$7(2) = 14.$$

The annual coupon rate is 7.6%, paid semiannually. Since the bond is priced at par, the yield rate per six-month period is

$$j = \frac{0.076}{2} = 0.038.$$

For a par bond with semiannual coupons, the Macaulay duration is

$$\text{MacD} = \frac{1}{2}\ddot{a}_{\overline{14}|0.038}.$$

Thus,

$$\text{MacD} = 5.555.$$

Rounded to two decimal places,

$$\text{MacD} = 5.56.$$

The correct answer is

$$\boxed{\text{C. } 5.56}.$$

Alternative calculation without the shortcut.
The semiannual coupon payment is

$$Fr = 5000\left(\frac{0.076}{2}\right) = 190.$$

The Macaulay duration is

$$\text{MacD} = \frac{\sum tA_t}{P}.$$

Because time is measured in years, the k-th coupon is paid at time $k/2$. Therefore,

$$\text{MacD} = \frac{\sum_{k=1}^{14}\left(\frac{k}{2}\right)190(1.038)^{-k} + \left(\frac{14}{2}\right)5000(1.038)^{-14}}{\sum_{k=1}^{14}190(1.038)^{-k} + 5000(1.038)^{-14}}.$$

Since the bond is priced at par, the denominator is 5000. Hence,

$$\text{MacD} = \frac{95(Ia)_{\overline{14}|0.038} + 35000(1.038)^{-14}}{5000}.$$

Thus,

$$\text{MacD} = \frac{7013.09 + 20763.72}{5000} = 5.555.$$

So again,

$$\boxed{\text{MacD} = 5.56}.$$

5.3.3 Modified Duration – Part 1

Macaulay duration helps us summarize the timing of a collection of cash flows.

However, Macaulay duration is not exactly the quantity we need to estimate price sensitivity. To estimate the percentage change in price caused by a change in yield, we use **modified duration**.

Recall that price sensitivity is defined as

$$\%\Delta P = \frac{P_{i_1} - P_{i_0}}{P_{i_0}}.$$

The price of a bond is a decreasing function of the yield rate. When the yield rate increases, the price decreases. When the yield rate decreases, the price increases.

The idea behind modified duration is to approximate the price-yield curve using a tangent line at the current yield rate.

Let

$$P_i$$

denote the price of the bond when the yield rate is i.

Suppose the current yield rate is i_0, and the yield changes to i_1. Then

$$\Delta i = i_1 - i_0.$$

Using a tangent line approximation,

$$P_{i_1} - P_{i_0} \approx \left.\frac{dP}{di}\right|_{i=i_0} (i_1 - i_0).$$

Therefore,

$$P_{i_1} - P_{i_0} \approx \left.\frac{dP}{di}\right|_{i=i_0} \Delta i.$$

Dividing both sides by P_{i_0}, we get

$$\frac{P_{i_1} - P_{i_0}}{P_{i_0}} \approx \frac{\left.\frac{dP}{di}\right|_{i=i_0}}{P_{i_0}} \Delta i.$$

Since bond prices and yield rates are inversely related, the derivative $\frac{dP}{di}$ is negative. Finance convention defines modified duration as a positive number:

$$\text{ModD} = -\frac{\left.\frac{dP}{di}\right|_{i=i_0}}{P_{i_0}}.$$

Modified duration

The **modified duration** of a bond is

$$\boxed{\text{ModD} = -\frac{\frac{dP}{di}}{P}}$$

where P is the current price of the bond and i is the yield rate.

Thus,

$$\%\Delta P \approx -\Delta i \cdot \text{ModD}.$$

Equivalently,

$$\Delta P \approx -\Delta i \cdot \text{ModD} \cdot P.$$

Modified duration approximation

$$\boxed{\%\Delta P \approx -\Delta i \cdot \text{ModD}}$$

$$\boxed{\Delta P \approx -\Delta i \cdot \text{ModD} \cdot P}$$

Sign interpretation

If $\Delta i > 0$, then yields increase. The formula gives

$$\%\Delta P < 0,$$

so the bond price decreases.
If $\Delta i < 0$, then yields decrease. The formula gives

$$\%\Delta P > 0,$$

so the bond price increases.

The Interest Rate Used

For the formulas above, the interest rate i can be expressed in different forms, such as an annual effective yield rate or a nominal annual yield rate.

The important point is consistency.

If the modified duration is calculated with respect to a nominal annual yield rate, then Δi must also be measured as a change in the nominal annual yield rate.

If the modified duration is calculated with respect to an effective rate per period, then Δi must also be measured as a change in the effective rate per period.

Consistency

The form of the yield rate used in modified duration must match the form of the yield change.

Do not mix a modified duration based on nominal annual yield with a yield change measured per coupon period.

Example

A

two-year 1000 par value bond with semiannual coupons of 8% is priced to yield 8% convertible semiannually.
The modified duration of the bond using the nominal annual yield is 1.815.
Estimate the price using modified duration if the yield changes to 7.8%.
Solution.
The current nominal annual yield rate is

$$i_0 = 0.08.$$

The new nominal annual yield rate is

$$i_1 = 0.078.$$

Thus,

$$\Delta i = i_1 - i_0 = 0.078 - 0.08 = -0.002.$$

Using modified duration,

$$\%\Delta P \approx -\Delta i \cdot \text{ModD}.$$

Substitute:

$$\%\Delta P \approx -(-0.002)(1.815).$$

Therefore,

$$\%\Delta P \approx 0.00363.$$

Since the bond is initially priced at par,

$$P_{i_0} = 1000.$$

So the estimated new price is

$$1000(1 + 0.00363) = 1003.63.$$

Therefore, the modified duration estimate is

$$\boxed{1003.63}.$$

The actual new price is found by valuing the bond at the new semiannual yield rate.
The new nominal annual yield is 7.8% convertible semiannually, so the semiannual yield rate is

$$\frac{0.078}{2} = 0.039.$$

The semiannual coupon payment is

$$1000\left(\frac{0.08}{2}\right) = 40.$$

There are 4 semiannual coupons, so the actual price is

$$P = 40a_{\overline{4}|0.039} + 1000v_{0.039}^{4}.$$

Thus,

$$P = 1003.64.$$

The modified duration estimate is very close:

$$1003.63 \approx 1003.64.$$

Tangent approximation

Modified duration uses a tangent line approximation to the price-yield curve.
Because the bond price-yield curve is curved, modified duration is only an approximation.
It is very useful for small yield changes, but it becomes less accurate for larger yield changes.

Exercise 5.3.11

A five-year bond with annual coupons of 100 and a redemption amount of 1500 is priced at 1608.24 to yield 5%.
Your coworker uses the bond's modified duration to estimate the change in price if the yield rises to 6%. The coworker's estimated change in price is 76.58.
Determine the estimated price of the bond at a 6% yield. Is this estimated price larger or smaller than the actual price at 6%?
Solution.
An increase in yield means a decrease in the bond price.
The estimated change in price is 76.58, so the estimated price is

$$1608.24 - 76.58 = 1531.66.$$

Thus, the estimated price is

$$\boxed{1531.66}.$$

Now calculate the actual price at a 6% yield.
The bond pays annual coupons of 100 for 5 years and a redemption amount of 1500 at maturity. Therefore,

$$P_{0.06} = 100a_{\overline{5}|0.06} + 1500v_{0.06}^{5}.$$

So,

$$P_{0.06} = 1542.12.$$

Therefore, the modified duration estimate is

$$1531.66,$$

while the actual price is

$$1542.12.$$

Hence, the estimated price is smaller than the actual price.

$$\boxed{\text{The estimated price is smaller than the actual price.}}$$

This estimate is not as close as in examples with smaller yield changes because Δi is larger here.

Exercise 5.3.12

A five-year bond with annual coupons of 100 and a redemption amount of 1500 is priced at 1608.24 to yield 5%.
Your coworker uses the bond's modified duration to estimate the change in price if the yield rises to 6%. The coworker's estimated change in price is 76.58.
Determine the modified duration of the bond using the effective annual yield.
Solution.
The modified duration approximation is

$$\%\Delta P \approx -\Delta i \cdot \text{ModD}.$$

The yield rises from 5% to 6%, so

$$\Delta i = 0.06 - 0.05 = 0.01.$$

Since the yield increases, the price decreases. Therefore, the estimated change in price is

$$\Delta P = -76.58.$$

The percentage change in price is

$$\%\Delta P = \frac{\Delta P}{P_{0.05}} = \frac{-76.58}{1608.24}.$$

Thus,

$$\frac{-76.58}{1608.24} \approx -0.01 \cdot \text{ModD}.$$

Solving for ModD, we get

$$\text{ModD} = \frac{76.58}{1608.24(0.01)}.$$

Therefore,

$$\text{ModD} = 4.762.$$

So the modified duration of the bond is

$$\boxed{4.762}.$$

Exercise 5.3.13

Which of the following are true?

1. Bond prices and yields are inversely related.
2. Cash flows further into the future are more sensitive to changes in interest rates.
3. Modified duration is a measure of price sensitivity.

A. 1 and 2 only
B. 1 and 3 only
C. 2 and 3 only
D. 1, 2 and 3
E. The correct answer is not given by A, B, C, or D

Solution.
Statement 1 is true.
Bond prices and yields are inversely related. The price of a bond is the present value of its future cash flows:

$$P = \sum_t CF_t(1+i)^{-t}.$$

As the yield rate i increases, the discount factors $(1+i)^{-t}$ decrease. Therefore, the bond price decreases.
Statement 2 is true.
A cash flow further in the future is more sensitive to changes in interest rates because it is discounted over a longer period:

$$PV = CF(1+i)^{-t}.$$

The larger t is, the more strongly a change in i affects the present value.
Statement 3 is true.
Modified duration is used to estimate the percentage change in price caused by a change in yield:

$$\%\Delta P \approx -\Delta i \cdot \text{ModD}.$$

Thus, modified duration is a measure of price sensitivity.
Therefore, statements 1, 2, and 3 are all true.
The correct answer is

$$\boxed{\text{D. } 1, 2, \text{ and } 3}.$$

Exercise 5.3.14

The modified duration of a bond is 6. Given a yield rate of 4%, the price of the bond is 105.
Estimate the price of the bond if the yield rate is 3% instead of 4%.

A. 98.70
B. 99.00
C. 109.50
D. 111.00
E. 111.30

Solution.
We use the modified duration approximation:

$$\%\Delta P \approx -\Delta y \cdot \text{ModD}.$$

We are given

$$\text{ModD} = 6.$$

The yield changes from 4% to 3%, so

$$\Delta y = 0.03 - 0.04 = -0.01.$$

Therefore,

$$\%\Delta P \approx -(-0.01)(6).$$

Thus,

$$\%\Delta P \approx 0.06.$$

So the bond price increases by approximately 6%.
The original price is

$$P_{0.04} = 105.$$

Therefore, the estimated new price is

$$P_{0.03} \approx 105(1 + 0.06).$$

So,

$$P_{0.03} \approx 105(1.06) = 111.30.$$

The correct answer is

$$\boxed{\text{E. } 111.30}.$$

5.3.4 Modified Duration – Part 2

In the previous subsection, we used modified duration to estimate price changes:

$$\%\Delta P \approx -\Delta i \cdot \text{ModD}.$$

Now we want to calculate modified duration directly.

Recall that modified duration is defined by

$$\text{ModD} = -\frac{\frac{dP}{di}}{P}.$$

The negative sign is included so that modified duration is usually positive, since bond prices and yields move in opposite directions.

Calculating Modified Duration

Consider a two-year 1000 par value bond with semiannual coupons of 8%, priced to yield 8% convertible semiannually.

The semiannual coupon is

$$1000\left(\frac{0.08}{2}\right) = 40.$$

There are 4 semiannual periods, so the cash flows are

$$40, \quad 40, \quad 40, \quad 1040.$$

Let j denote the nominal annual yield rate convertible semiannually. Then the semiannual yield rate is

$$\frac{j}{2}.$$

The price as a function of j is

$$P_j = 40\left(1+\frac{j}{2}\right)^{-1} + 40\left(1+\frac{j}{2}\right)^{-2} + 40\left(1+\frac{j}{2}\right)^{-3} + 1040\left(1+\frac{j}{2}\right)^{-4}.$$

Differentiate with respect to j:

$$\frac{dP_j}{dj} = -40\left(1+\frac{j}{2}\right)^{-2}\left(\frac{1}{2}\right) - 80\left(1+\frac{j}{2}\right)^{-3}\left(\frac{1}{2}\right)$$

$$-120\left(1+\frac{j}{2}\right)^{-4}\left(\frac{1}{2}\right) - 4160\left(1+\frac{j}{2}\right)^{-5}\left(\frac{1}{2}\right).$$

At $j = 0.08$, the semiannual rate is 0.04, and the bond is priced at par:

$$P_{0.08} = 1000.$$

Therefore,

$$-\left.\frac{dP_j}{dj}\right|_{j=0.08} = 20(1.04)^{-2} + 40(1.04)^{-3} + 60(1.04)^{-4} + 2080(1.04)^{-5}.$$

Thus,

$$-\left.\frac{dP_j}{dj}\right|_{j=0.08} = 1814.95.$$

Hence,

$$\text{ModD} = -\frac{\frac{dP_j}{dj}}{P_j} = \frac{1814.95}{1000} = 1.815.$$

So the modified duration is

$$\boxed{1.815}.$$

Relationship Between Modified Duration and Macaulay Duration

Let CF_t be the cash flow at time t, where t is measured in years.

Let i be a nominal annual yield rate convertible m-thly.

Then the price is

$$P_i = \sum_t CF_t\left(1+\frac{i}{m}\right)^{-mt}.$$

Let

$$A_t = CF_t\left(1+\frac{i}{m}\right)^{-mt}$$

be the present value of the cash flow at time t.

Differentiate the price with respect to i:

$$\frac{dP_i}{di} = \sum_t -mt\cdot CF_t\left(1+\frac{i}{m}\right)^{-mt-1}\cdot\frac{1}{m}.$$

Since the m's cancel,

$$\frac{dP_i}{di} = -\sum_t t\cdot CF_t\left(1+\frac{i}{m}\right)^{-mt-1}.$$

Factor out one extra discount factor:

$$-\frac{dP_i}{di} = \sum_t t\cdot CF_t\left(1+\frac{i}{m}\right)^{-mt}\left(1+\frac{i}{m}\right)^{-1}.$$

But

$$A_t = CF_t\left(1+\frac{i}{m}\right)^{-mt},$$

so

$$-\frac{dP_i}{di} = \left(1+\frac{i}{m}\right)^{-1}\sum_t tA_t.$$

Let

$$v = \left(1+\frac{i}{m}\right)^{-1}$$

be the discount factor for one interest conversion period.

Then

$$-\frac{dP_i}{di} = v\sum_t tA_t.$$

Therefore,

$$\text{ModD} = -\frac{\frac{dP_i}{di}}{P_i} = \frac{v\sum_t tA_t}{\sum_t A_t}.$$

But

$$\text{MacD} = \frac{\sum_t tA_t}{\sum_t A_t}.$$

Hence,

$$\boxed{\text{ModD} = v \cdot \text{MacD}}$$

where v is the discount factor for one interest conversion period.

Modified duration and Macaulay duration

If i is a nominal annual yield rate convertible m-thly, then

$$\boxed{\text{ModD} = v \cdot \text{MacD}}$$

where

$$\boxed{v = \left(1 + \frac{i}{m}\right)^{-1}}$$

and time t is measured in years.

Why is it called modified duration?

Modified duration is Macaulay duration multiplied by one discount factor.
That is why it is called *modified* duration:

$$\text{ModD} = v \cdot \text{MacD}.$$

Continuous Compounding

If the yield rate is compounded continuously, then the price is written as

$$P_\delta = \sum_t CF_t e^{-\delta t}.$$

Differentiating with respect to δ,

$$\frac{dP_\delta}{d\delta} = -\sum_t tCF_t e^{-\delta t}.$$

Thus,

$$-\frac{\frac{dP_\delta}{d\delta}}{P_\delta} = \frac{\sum_t tA_t}{\sum_t A_t}.$$

Therefore, when interest is compounded continuously,

$$\boxed{\text{ModD} = \text{MacD}}$$

Continuous compounding

Under continuous compounding, there is no extra per-period discount factor.
So modified duration and Macaulay duration are equal.

Example

A

three-year bond with 8% annual coupons has a redemption value of 105% of the par value and is priced to yield an annual effective rate of 6%.
Calculate the modified duration.
Solution.
The par value does not matter because it cancels out in the duration calculation.
Let the par value be

$$100.$$

Then the annual coupon is

$$8.$$

The redemption value is

$$105.$$

Therefore, the cash flows are

$$8 \quad \text{at time } 1,$$

$$8 \quad \text{at time } 2,$$

and

$$8 + 105 = 113 \quad \text{at time } 3.$$

The yield rate is annual effective:

$$i = 0.06.$$

Thus,

$$v = \frac{1}{1.06}.$$

The present value of the cash flows is

$$\sum A_t = 8a_{\overline{3}|0.06} + 105v_{0.06}^3.$$

So,

$$\sum A_t = 109.54.$$

Now compute the numerator for modified duration:

$$v\sum tA_t = 1(8)(1.06)^{-2} + 2(8)(1.06)^{-3} + 3(113)(1.06)^{-4}.$$

Thus,

$$v\sum tA_t = 289.07.$$

Therefore,

$$\text{ModD} = \frac{v\sum tA_t}{\sum A_t} = \frac{289.07}{109.54}.$$

Hence,

$$\text{ModD} = 2.639.$$

So the modified duration is

$$\boxed{2.639}.$$

Exercise 5.3.15

A thirty-year bond with 8% coupons payable quarterly is priced to yield $x\%$ convertible quarterly.
The modified duration is 8.392 and the Macaulay duration is 8.644.
Determine $x\%$.
Solution.
We use the relationship between modified duration and Macaulay duration:

$$\text{ModD} = v \cdot \text{MacD},$$

where v is the discount factor for one interest conversion period.
Since the yield is convertible quarterly, the period is one quarter. Therefore,

$$v = \left(1 + \frac{x\%}{4}\right)^{-1}.$$

We are given

$$\text{ModD} = 8.392$$

and

$$\text{MacD} = 8.644.$$

Thus,

$$8.392 = \left(1 + \frac{x\%}{4}\right)^{-1}(8.644).$$

Divide both sides by 8.644:

$$\left(1 + \frac{x\%}{4}\right)^{-1} = \frac{8.392}{8.644}.$$

Therefore,

$$1 + \frac{x\%}{4} = \frac{8.644}{8.392}.$$

So,

$$\frac{x\%}{4} = \frac{8.644}{8.392} - 1.$$

Hence,

$$x\% = 4\left(\frac{8.644}{8.392} - 1\right).$$

This gives

$$x\% \approx 0.1201.$$

Therefore,

$$x\% \approx 12\%.$$

So the nominal annual yield rate convertible quarterly is

$$\boxed{12\%}.$$

Exercise 5.3.16

A thirty-year bond with 8% coupons payable quarterly is priced to yield 12% convertible quarterly.
Calculate the modified duration.
Solution.
Let the par value be

$$1000.$$

The quarterly coupon is

$$1000\left(\frac{0.08}{4}\right) = 20.$$

The effective yield rate per quarter is

$$j = \frac{0.12}{4} = 0.03.$$

There are

$$30(4) = 120$$

quarterly coupon periods.
The bond price is

$$P = 20a_{\overline{120}|0.03} + 1000v_{0.03}^{120}.$$

Thus,

$$P = 676.27.$$

To calculate modified duration, use

$$\text{ModD} = \frac{v\sum tA_t}{\sum A_t},$$

where t is measured in years and v is the discount factor per quarter.
Here,

$$v = v_{0.03} = \frac{1}{1.03}.$$

The coupon at quarter k occurs at time $k/4$ years. Therefore,

$$t \cdot CF_t = \frac{k}{4}(20) = 5k.$$

The maturity payment occurs at time 30, so its time-weighted cash flow is

$$30(1000) = 30000.$$

Hence,

$$v\sum tA_t = v\left(5(Ia)_{\overline{120}|0.03} + 30000v_{0.03}^{120}\right).$$

Therefore,

$$v\sum tA_t = 5675.21.$$

So,

$$\text{ModD} = \frac{5675.21}{676.27}.$$

Thus,

$$\text{ModD} = 8.392.$$

The modified duration is

$$\boxed{8.392}.$$

Exercise 5.3.17

Regarding modified duration, which of the following are true?

1. It is the first derivative of price divided by price.
2. It is the percentage increase in price resulting from a 100 basis point increase in yield.
3. For bonds, it is not a function of par value.

A. 1 and 2 only

B. 1 and 3 only

C. 2 and 3 only

D. 1, 2 and 3

E. The correct answer is not given by A, B, C, or D

Solution.
Statement 1 is false.
Modified duration is not simply the first derivative of price divided by price. It is the negative of the first derivative of price divided by price:

$$\text{ModD} = -\frac{\frac{dP}{di}}{P}.$$

The negative sign is needed because bond prices and yields are inversely related.
Statement 2 is false.
Modified duration approximates the percentage change in price using

$$\%\Delta P \approx -\Delta i \cdot \text{ModD}.$$

A 100 basis point increase in yield means

$$\Delta i = 0.01.$$

Then

$$\%\Delta P \approx -0.01 \cdot \text{ModD}.$$

So the price decreases, not increases.
Modified duration can be interpreted as approximately the percentage increase in price resulting from a 100 basis point decrease in yield.
Statement 3 is true.
For bonds, modified duration is not a function of par value. If the par value is multiplied by a constant, then all cash flows and the bond price are multiplied by that same constant. Therefore, the factor cancels in the duration calculation.
Thus, only statement 3 is true.
Since this answer is not listed among A, B, C, or D, the correct answer is

$$\boxed{\text{E}}.$$

Exercise 5.3.18

A 10-year bond with quarterly coupons of 8% is priced at par.
Calculate the modified duration used to estimate changes in the yield rate convertible quarterly.

A. 6.459 B. 6.710
C. 6.839 D. 6.976
E. 7.105

Solution.
Since the bond is priced at par, the coupon rate equals the yield rate, assuming the redemption value equals the face value.
Therefore, the yield rate is

$$i^{(4)} = 8\%$$

convertible quarterly.
The effective quarterly yield rate is

$$j = \frac{0.08}{4} = 0.02.$$

The bond has quarterly coupons for 10 years, so the total number of coupon periods is

$$10(4) = 40.$$

For a par bond with quarterly coupons, the Macaulay duration is

$$\text{MacD} = \frac{1}{4}\ddot{a}_{\overline{40}|0.02}.$$

Thus,

$$\text{MacD} = 6.976.$$

Modified duration is related to Macaulay duration by

$$\text{ModD} = v \cdot \text{MacD},$$

where v is the discount factor for one interest conversion period.
Here,

$$v = \frac{1}{1.02}.$$

Therefore,

$$\text{ModD} = \frac{6.976}{1.02}.$$

So,

$$\text{ModD} = 6.839.$$

The correct answer is

$$\boxed{\text{C. } 6.839}.$$

Exercise 5.3.19

Krishna buys an n-year 1000 bond at par. The Macaulay duration is 7.959 years using an annual effective interest rate of 7.2%.

Calculate the estimated price of the bond, using the first-order modified approximation, if the interest rate rises to 8.0%.

A. 940.60 B. 942.88

C. 944.56 D. 947.03

E. 948.47

Solution.
First find the modified duration.
Since the yield rate is annual effective, the discount factor for one period is

$$v = \frac{1}{1+i}.$$

Thus,

$$\text{ModD} = v \cdot \text{MacD}.$$

Equivalently,

$$\text{ModD} = \frac{\text{MacD}}{1+i}.$$

Using $i = 0.072$, we get

$$\text{ModD} = \frac{7.959}{1.072}.$$

Therefore,

$$\text{ModD} = 7.424.$$

The interest rate rises from 7.2% to 8.0%, so

$$\Delta i = 0.080 - 0.072 = 0.008.$$

Using the first-order modified duration approximation,

$$\%\Delta P \approx -\Delta i \cdot \text{ModD}.$$

Thus,

$$\%\Delta P \approx -(0.008)(7.424).$$

So,

$$\%\Delta P \approx -0.0594.$$

Since the bond is bought at par,

$$P_0 = 1000.$$

Therefore, the estimated new price is

$$P_1 \approx 1000(1 - 0.0594).$$

Hence,

$$P_1 \approx 940.60.$$

The correct answer is

$$\boxed{\text{A. } 940.60}.$$

5.3.5 Portfolio Duration

For a portfolio of bonds or investments, duration can be calculated in two equivalent ways.

1. Determine the total cash flow of the entire portfolio at each time t, and then use the Macaulay duration or modified duration summation formula.
2. Compute the price-weighted average of the Macaulay durations or modified durations of the portfolio components.

If a portfolio contains assets with prices $P_1, P_2, \ldots, P_n$ and Macaulay durations $\mathrm{MacD}_1, \mathrm{MacD}_2, \ldots, \mathrm{MacD}_n$, then

$$\mathrm{MacD}_p = \frac{P_1\mathrm{MacD}_1 + P_2\mathrm{MacD}_2 + \cdots + P_n\mathrm{MacD}_n}{P_1 + P_2 + \cdots + P_n}.$$

Similarly, for modified duration,

$$\mathrm{ModD}_p = \frac{P_1\mathrm{ModD}_1 + P_2\mathrm{ModD}_2 + \cdots + P_n\mathrm{ModD}_n}{P_1 + P_2 + \cdots + P_n}.$$

Portfolio duration

$$\boxed{\mathrm{MacD}_p = \frac{\sum_k P_k\mathrm{MacD}_k}{\sum_k P_k}}$$

$$\boxed{\mathrm{ModD}_p = \frac{\sum_k P_k\mathrm{ModD}_k}{\sum_k P_k}}$$

Key idea

Portfolio duration is a weighted average of the durations of the assets in the portfolio. The weights are based on price, not face value.

Weighted Average Example

Y

ou are given the following information about a portfolio of three bonds:

Bond	Price	Duration
A	X	4 years
B	$2X$	6 years
C	$\frac{1}{2}X$	Y years

The duration of the entire portfolio is 8 years.
Find Y.
Solution.
Use the price-weighted average formula:

$$\mathrm{MacD}_p = \frac{P_A\mathrm{MacD}_A + P_B\mathrm{MacD}_B + P_C\mathrm{MacD}_C}{P_A + P_B + P_C}.$$

Substitute the given values:

$$8 = \frac{X(4) + 2X(6) + \frac{1}{2}X(Y)}{X + 2X + \frac{1}{2}X}.$$

Simplify the denominator:

$$X + 2X + \frac{1}{2}X = 3.5X.$$

Simplify the numerator:

$$X(4) + 2X(6) + \frac{1}{2}XY = 4X + 12X + \frac{1}{2}XY.$$

Thus,

$$8 = \frac{16X + \frac{1}{2}XY}{3.5X}.$$

Cancel X:

$$8 = \frac{16 + \frac{1}{2}Y}{3.5}.$$

Multiply both sides by 3.5:

$$28 = 16 + \frac{1}{2}Y.$$

Therefore,

$$12 = \frac{1}{2}Y.$$

So,

$$Y = 24.$$

Thus,

$$\boxed{Y = 24}.$$

Passage of Time

Duration changes as time passes.

As future cash flows move closer, the duration generally decreases. However, immediately after a cash flow is paid, duration can jump upward because the nearest cash flow has just disappeared from the remaining cash flows.

A

three-year bond with 8% annual coupons has a redemption value of 105% of the par value and is priced to yield an annual effective rate of 6%.
The duration at time 0 is 2.7972.
Calculate the duration at time 1, both just before and just after the coupon payment.
Solution.

Let the par value be F.
The annual coupon is

$$0.08F.$$

The redemption value is

$$1.05F.$$

Thus, the final cash flow at time 3 is

$$0.08F + 1.05F = 1.13F.$$

The annual effective yield rate is

$$i = 0.06,$$

so

$$v = v_{0.06}.$$

Duration just before the coupon payment at time 1.
Just before the coupon payment at time 1, the remaining cash flows are:

$$0.08F \quad \text{immediately,}$$

$$0.08F \quad \text{one year later,}$$

and

$$1.13F \quad \text{two years later.}$$

Therefore,

$$\text{MacD(before)} = \frac{0(0.08F) + 1(0.08F)v + 2(1.13F)v^2}{0.08F + 0.08Fv + 1.13Fv^2}.$$

Substituting $v = v_{0.06}$,

$$\text{MacD(before)} = 1.7972.$$

Notice that

$$\text{MacD(before)} = 2.7972 - 1.$$

Thus,

$$\boxed{\text{MacD(before)} = 1.7972}.$$

Duration just after the coupon payment at time 1.
Immediately after the coupon payment, the time 1 coupon has already been paid, so it is no longer part of the remaining cash flows.
The remaining cash flows are:

$$0.08F \quad \text{one year later,}$$

and

$$1.13F \quad \text{two years later.}$$

Therefore,

$$\text{MacD(after)} = \frac{1(0.08F)v + 2(1.13F)v^2}{0.08Fv + 1.13Fv^2}.$$

Substituting $v = v_{0.06}$,

$$\text{MacD(after)} = 1.9302.$$

Thus,

$$\boxed{\text{MacD(after)} = 1.9302}.$$

Immediately after a cash flow is paid, the duration goes up because the nearest cash flow has been removed from the remaining cash flows.

Passage of time

Between cash flow dates, duration decreases as time passes.
Immediately after a cash flow is paid, duration may jump upward because the earliest remaining cash flow has been removed.

Exercise 5.3.20

A portfolio contains three bonds, all of which have par value 1000.
The first bond has a two-year term and no coupons. The second bond is a three-year bond with no coupons. The third bond is a two-year 6% bond with semiannual coupons.
If each bond is purchased to yield 5% convertible semiannually, find the modified duration of the portfolio.
Solution.
The nominal annual yield rate is 5% convertible semiannually, so the effective yield rate per half-year is

$$j = \frac{0.05}{2} = 0.025.$$

Thus,

$$v = \frac{1}{1.025}.$$

Bond I.
The first bond is a two-year zero-coupon bond. Since coupons are semiannual, two years corresponds to 4 half-year periods.
Therefore,

$$P_I = 1000v^4.$$

So,

$$P_I = 1000(1.025)^{-4} = 905.95.$$

Since this is a zero-coupon bond with maturity at time 2, its Macaulay duration is

$$\text{MacD}_I = 2.$$

Bond II.

The second bond is a three-year zero-coupon bond. Three years corresponds to 6 half-year periods.
Therefore,

$$P_{II} = 1000v^6.$$

So,

$$P_{II} = 1000(1.025)^{-6} = 862.30.$$

Since this is a zero-coupon bond with maturity at time 3, its Macaulay duration is

$$\text{MacD}_{II} = 3.$$

Bond III.
The third bond is a two-year 6% bond with semiannual coupons.
The semiannual coupon is

$$1000\left(\frac{0.06}{2}\right) = 30.$$

The cash flows are

$$30, 30, 30, 1030$$

at times

$$0.5, 1, 1.5, 2.$$

The price is

$$P_{III} = 30v + 30v^2 + 30v^3 + 1030v^4.$$

So,

$$P_{III} = 1018.81.$$

The Macaulay duration of Bond III is

$$\text{MacD}_{III} = \frac{0.5(30)v + 1(30)v^2 + 1.5(30)v^3 + 2(1030)v^4}{P_{III}}.$$

Thus,

$$\text{MacD}_{III} = \frac{0.5(30)v + 1(30)v^2 + 1.5(30)v^3 + 2(1030)v^4}{1018.81}.$$

Therefore,

$$\text{MacD}_{III} = 1.9152.$$

Now compute the portfolio Macaulay duration using a price-weighted average:

$$\text{MacD}_P = \frac{P_I\text{MacD}_I + P_{II}\text{MacD}_{II} + P_{III}\text{MacD}_{III}}{P_I + P_{II} + P_{III}}.$$

Substitute:

$$\text{MacD}_P = \frac{2P_I + 3P_{II} + 1.9152P_{III}}{P_I + P_{II} + P_{III}}.$$

Thus,

$$\text{MacD}_P = 2.2784.$$

Finally, modified duration is

$$\text{ModD}_P = v \cdot \text{MacD}_P.$$

Since

$$v = \frac{1}{1.025},$$

we get

$$\text{ModD}_P = \frac{2.2784}{1.025}.$$

Therefore,

$$\text{ModD}_P = 2.2228.$$

So the modified duration of the portfolio is

$$\boxed{2.2228}.$$

Exercise 5.3.21

A 5000 loan is being repaid over a two-year period with level payments at the end of each quarter. The interest rate is 8% convertible quarterly.
Find the jump in the duration of the loan following the 5th payment.

Solution.

The nominal annual interest rate is 8% convertible quarterly, so the effective quarterly interest rate is

$$j = \frac{0.08}{4} = 0.02.$$

Thus,

$$v = \frac{1}{1.02}.$$

The loan is repaid over 2 years with quarterly payments, so there are

$$2(4) = 8$$

payments.
We want the jump in duration following the 5th payment. Just before the 5th payment, there are four remaining payments: one immediately, and then three more at times

$$0.25, \quad 0.50, \quad 0.75.$$

Let the level payment be R. The value of R is not needed because it cancels in the duration formula.

Duration just before the 5th payment.

Just before the 5th payment, the remaining cash flows occur at times

$$0, \quad 0.25, \quad 0.50, \quad 0.75.$$

Therefore,

$$\text{MacD(before)} = \frac{0 \cdot R + 0.25Rv + 0.50Rv^2 + 0.75Rv^3}{R(1 + v + v^2 + v^3)}.$$

Cancel R:

$$\text{MacD(before)} = \frac{0.25v + 0.50v^2 + 0.75v^3}{1 + v + v^2 + v^3}.$$

Substituting $v = 1/1.02$, we get

$$\text{MacD(before)} = 0.369.$$

Duration just after the 5th payment.

Immediately after the 5th payment, the payment at time 0 has been made. The remaining cash flows occur at times

$$0.25, \quad 0.50, \quad 0.75.$$

Therefore,

$$\text{MacD(after)} = \frac{0.25Rv + 0.50Rv^2 + 0.75Rv^3}{R(v + v^2 + v^3)}.$$

Cancel R:

$$\text{MacD(after)} = \frac{0.25v + 0.50v^2 + 0.75v^3}{v + v^2 + v^3}.$$

Substituting $v = 1/1.02$, we get

$$\text{MacD(after)} = 0.497.$$

Thus, the jump in duration is

$$0.497 - 0.369 = 0.128.$$

Therefore, the jump in duration is

$$\boxed{0.128}.$$

Exercise 5.3.22

Sam buys an eight-year, 5000 par bond with an annual coupon rate of 5%, paid annually. The bond sells for 5000.
Let d_1 be the Macaulay duration just before the first coupon is paid. Let d_2 be the Macaulay duration just after the first coupon is paid.
Calculate

$$\frac{d_1}{d_2}.$$

A. 0.91 B. 0.93

C. 0.95 D. 0.97

E. 1.00

Solution.
Since the bond is priced at par, the yield rate equals the coupon rate:

$$i = 0.05.$$

The annual coupon is

$$5000(0.05) = 250.$$

Let

$$v = \frac{1}{1.05}.$$

Just before the first coupon is paid, the coupon due immediately has time 0, and the remaining cash flows occur at times $1, 2, \ldots, 7$.
Thus,

$$d_1 = \frac{0(250) + 1(250v) + 2(250v^2) + \cdots + 7(5250v^7)}{250 + 250a_{\overline{7}|0.05} + 5000v^7}.$$

Just after the first coupon is paid, the immediate coupon is no longer part of the remaining cash flows. The remaining cash flows occur at times $1, 2, \ldots, 7$.
Thus,

$$d_2 = \frac{1(250v) + 2(250v^2) + \cdots + 7(5250v^7)}{250a_{\overline{7}|0.05} + 5000v^7}.$$

The numerators of d_1 and d_2 are the same, since the immediate coupon in d_1 is multiplied by 0.
Therefore,

$$\frac{d_1}{d_2} = \frac{250a_{\overline{7}|0.05} + 5000v^7}{250 + 250a_{\overline{7}|0.05} + 5000v^7}.$$

Since the bond is priced at par, immediately after the first coupon is paid, the remaining seven-year bond has value

$$250a_{\overline{7}|0.05} + 5000v^7 = 5000.$$

Therefore,

$$\frac{d_1}{d_2} = \frac{5000}{250 + 5000}.$$

So,

$$\frac{d_1}{d_2} = \frac{5000}{5250} = 0.9524.$$

Rounded to two decimal places,

$$\frac{d_1}{d_2} = 0.95.$$

The correct answer is

$$\boxed{\text{C. } 0.95}.$$

5.3.6 Approximation Using Duration

We have already used modified duration to estimate how the price of a bond changes when the yield rate changes.

The modified duration approximation is a linear approximation:

$$\%\Delta P \approx -\Delta i \cdot \text{ModD}.$$

In other words, modified duration was built to play the role of a constant of proportionality in a tangent-line approximation of the price-yield curve.

However, Macaulay duration can also be used to approximate the new price.

Approximation Using Macaulay Duration

Suppose the original annual effective yield rate is i_0, and the yield rate changes to i_1.

If the original price is P_{i_0}, then the Macaulay duration approximation gives

$$\boxed{P_{i_1} \approx P_{i_0} \left(\frac{1 + i_0}{1 + i_1} \right)^{\text{MacD}}}$$

where MacD is evaluated at the original yield rate i_0.

This also gives the percentage price change approximation:

$$\boxed{\%\Delta P \approx \left(\frac{1+i_0}{1+i_1}\right)^{\text{MacD}} - 1}$$

Important

This formula is written using annual effective yield rates.
If the yield rates are nominal rates convertible m-thly, convert them to annual accumulation factors before using the formula.

Example

A

two-year 1000 par value bond with semiannual coupons of 8% is priced to yield 8% convertible semiannually.
The duration of the bond is

$$\text{MacD} = 1.887545517.$$

Estimate the price using the Macaulay duration approximation if the yield changes to 7.8% convertible semiannually.
Solution.
Since the bond is priced at par,

$$P_{i_0} = 1000.$$

The original nominal annual yield rate is 8% convertible semiannually, so the semiannual yield rate is

$$0.04.$$

Thus the annual accumulation factor is

$$(1.04)^2.$$

The new nominal annual yield rate is 7.8% convertible semiannually, so the new semiannual yield rate is

$$0.039.$$

Thus the new annual accumulation factor is

$$(1.039)^2.$$

Using the Macaulay duration approximation,

$$P_{i_1} \approx P_{i_0} \left(\frac{1+i_0}{1+i_1}\right)^{\text{MacD}}.$$

In terms of the annual accumulation factors,

$$P_{i_1} \approx 1000\left(\frac{1.04^2}{1.039^2}\right)^{1.887545517}.$$

Therefore,

$$P_{i_1} \approx 1003.638244.$$

So the estimated price using Macaulay duration is

$$\boxed{1003.638244}.$$

Now calculate the actual new price.
The semiannual coupon payment is

$$1000\left(\frac{0.08}{2}\right) = 40.$$

There are four semiannual coupon payments, and the new semiannual yield rate is 0.039. Therefore,

$$P = 40a_{\overline{4}|0.039} + 1000v_{0.039}^4.$$

Thus,

$$P = 1003.638466.$$

The Macaulay duration approximation is extremely close to the actual price.

Comparison with Modified Duration

For the same example, the modified duration approximation gives

$$P \approx 1003.629895$$

when a less-rounded value of modified duration is used.

The actual price is

$$1003.638466.$$

Thus, both approximations are very accurate because the yield change is small.

However, in this example, the Macaulay duration approximation is slightly more accurate.

Comparison of methods

When all future cash flows are positive, the Macaulay duration approximation usually performs better than the modified duration approximation.
The Macaulay duration approximation often gives about one more trustworthy decimal place than the modified duration approximation.
If there is only one future cash flow, such as a zero-coupon bond, the Macaulay duration approximation gives the exact new price.

Summary

The modified duration approximation is

$$\boxed{\%\Delta P \approx -\Delta i \cdot \text{ModD}}$$

and the Macaulay duration approximation is

$$\boxed{P_{i_1} \approx P_{i_0} \left(\frac{1+i_0}{1+i_1}\right)^{\text{MacD}}}$$

Modified duration gives a simple linear approximation.

Macaulay duration gives a multiplicative approximation based on annual accumulation factors.

Exercise 5.3.23

A five-year bond with annual coupons of 100 and a redemption amount of 1500 is priced at 1608.24 to yield 5%.
The duration of this bond is 4.435.
Estimate the percent change in price that would occur if the yield rose to 5.5%.
Solution.
We are given:

$$P_{i_0} = 1608.24, \qquad i_0 = 0.05, \qquad i_1 = 0.055, \qquad \text{MacD} = 4.435.$$

Method 1: Direct percent change formula.
Using the Macaulay duration approximation for the percent change in price,

$$\%\Delta P \approx \left(\frac{1+i_0}{1+i_1}\right)^{\text{MacD}} - 1.$$

Substitute the given values:

$$\%\Delta P \approx \left(\frac{1.05}{1.055}\right)^{4.435} - 1.$$

Therefore,

$$\%\Delta P \approx -0.02085.$$

So the estimated percent change in price is

$$\boxed{-0.02085}.$$

Equivalently, the bond price is estimated to decrease by approximately

$$\boxed{2.085\%}.$$

Method 2: Estimate the new price first.
The more important formula to remember is the Macaulay duration approximation for the price:

$$P_{i_1} \approx P_{i_0} \left(\frac{1+i_0}{1+i_1}\right)^{\text{MacD}}.$$

Substitute the given values:

$$P_{i_1} \approx 1608.24 \left(\frac{1.05}{1.055}\right)^{4.435}.$$

Thus,

$$P_{i_1} \approx 1574.71.$$

Now compute the percent change manually:

$$\%\Delta P = \frac{P_{i_1} - P_{i_0}}{P_{i_0}}.$$

So,

$$\%\Delta P = \frac{1574.71 - 1608.24}{1608.24}.$$

Therefore,

$$\%\Delta P \approx -0.02085.$$

Both methods give the same estimated percent change:

$$\boxed{-0.02085}.$$

Exercise 5.3.24

A four-year bond with semiannual coupons is priced at 1179.25 to yield 5% convertible semiannually.
Suppose the bond's price rises to 1199.30, implying a new nominal yield of 4.5% convertible semiannually.
Estimate the Macaulay duration of the bond prior to the interest rate change.
Solution.
We use the Macaulay duration approximation:

$$P_{i_1} \approx P_{i_0} \left(\frac{1+i_0}{1+i_1}\right)^{\text{MacD}},$$

where i_0 and i_1 are annual effective yield rates.
The original yield is 5% convertible semiannually, so the semiannual rate is

$$\frac{0.05}{2} = 0.025.$$

Thus the original annual accumulation factor is

$$(1.025)^2.$$

The new yield is 4.5% convertible semiannually, so the semiannual rate is

$$\frac{0.045}{2} = 0.0225.$$

Thus the new annual accumulation factor is

$$(1.0225)^2.$$

Substitute into the approximation:

$$1199.30 \approx 1179.25 \left(\frac{1.025^2}{1.0225^2}\right)^{\text{MacD}}.$$

Divide both sides by 1179.25:

$$\frac{1199.30}{1179.25} \approx \left(\frac{1.025^2}{1.0225^2}\right)^{\text{MacD}}.$$

Take natural logarithms:

$$\ln\left(\frac{1199.30}{1179.25}\right) \approx \text{MacD} \cdot \ln\left(\frac{1.025^2}{1.0225^2}\right).$$

Therefore,

$$\text{MacD} \approx \frac{\ln\left(\frac{1199.30}{1179.25}\right)}{\ln\left(\frac{1.025^2}{1.0225^2}\right)}.$$

Thus,

$$\text{MacD} \approx 3.452.$$

The estimated Macaulay duration is

$$\boxed{3.452}.$$

Exercise 5.3.25

A 3-year bond with semiannual coupons has a duration of 2.7342 years. The current yield on the bond is 6% convertible semiannually.
Use a Macaulay duration approximation to estimate the new interest rate, convertible semiannually, that would give a price increase of 5%.

A. 4.17% B. 4.57%

C. 4.97% D. 5.37%

E. 5.77%

Solution.
We use the Macaulay duration approximation:

$$P_{i_1} \approx P_{i_0}\left(\frac{1+i_0}{1+i_1}\right)^{\text{MacD}}.$$

This gives the percent change approximation:

$$\%\Delta P \approx \left(\frac{1+i_0}{1+i_1}\right)^{\text{MacD}} - 1.$$

The current yield is 6% convertible semiannually, so the semiannual yield rate is

$$\frac{0.06}{2} = 0.03.$$

Therefore, the annual effective accumulation factor is

$$(1.03)^2 = 1.0609.$$

Thus,

$$1 + i_0 = 1.0609.$$

The price increase is 5%, so

$$\%\Delta P = 0.05.$$

Using MacD = 2.7342, we have

$$0.05 \approx \left(\frac{1.0609}{1+i_1}\right)^{2.7342} - 1.$$

Therefore,

$$1.05 \approx \left(\frac{1.0609}{1+i_1}\right)^{2.7342}.$$

Taking both sides to the power 1/2.7342,

$$1.05^{1/2.7342} \approx \frac{1.0609}{1+i_1}.$$

So,

$$1+i_1 \approx \frac{1.0609}{1.05^{1/2.7342}}.$$

Thus,

$$i_1 \approx 0.042137.$$

This is the new interest rate expressed as an annual effective rate.
We need the new nominal annual rate convertible semiannually. If this rate is $i^{(2)}$, then

$$1+i_1 = \left(1+\frac{i^{(2)}}{2}\right)^2.$$

Therefore,

$$i^{(2)} = 2\left((1+i_1)^{1/2} - 1\right).$$

Substitute $i_1 = 0.042137$:

$$i^{(2)} \approx 2\left((1.042137)^{1/2} - 1\right).$$

Hence,

$$i^{(2)} \approx 0.0417.$$

So the new yield rate convertible semiannually is

$$\boxed{4.17\%}.$$

The correct answer is

$$\boxed{\text{A. } 4.17\%}.$$

Exercise 5.3.26

Bob owns a 10-year bond with annual coupons, which has a present value of P_0. Suppose the yield rate changes, and Bob estimates the new price P_1 using:

- duration, giving an estimated new value of P_1^{MacD}, and
- modified duration, giving an estimated new value of P_1^{ModD}.

Which of the following relationships can possibly be true?

A. $P_1^{\text{MacD}} < P_1^{\text{ModD}} < P_1$

B. $P_1^{\text{ModD}} < P_1^{\text{MacD}} < P_1$

C. $P_1 < P_1^{\text{MacD}} < P_1^{\text{ModD}}$

D. $P_1^{\text{MacD}} < P_1 < P_1^{\text{ModD}}$

E. More than one of the above could be true

Solution.
We know two important facts.
First, the modified duration approximation always underestimates the new price:

$$P_1^{\text{ModD}} < P_1.$$

Second, when all future cash flows are positive, the Macaulay duration approximation is closer to the actual new price than the modified duration approximation.
Since Bob owns a bond with annual coupons, all future cash flows are positive.
Therefore, the Macaulay duration approximation must be closer to P_1 than the modified duration approximation.
Putting these two facts together, there are only two possible arrangements:

$$P_1^{\text{ModD}} < P_1^{\text{MacD}} < P_1,$$

or

$$P_1^{\text{ModD}} < P_1 < P_1^{\text{MacD}}.$$

The second arrangement is not among the answer choices.
The only listed relationship that can possibly be true is

$$P_1^{\text{ModD}} < P_1^{\text{MacD}} < P_1.$$

Therefore, the correct answer is

$$\boxed{\text{B}}.$$

5.4 Convexity

5.4.1 Convexity

We have already seen two ways to measure price sensitivity.

First, we can calculate the percentage change in price directly:

$$\%\Delta P = \frac{P_{i_1} - P_{i_0}}{P_{i_0}}.$$

Second, we can estimate it using modified duration:

$$\%\Delta P \approx -\Delta i \cdot \text{ModD}.$$

This modified duration approximation comes from using a linear approximation to the price-yield curve. In other words, we approximate the true curve by a tangent line.

However, the price-yield curve is not a straight line. It is curved. So we can improve the approximation by using a quadratic approximation instead of a linear approximation.

This is the idea behind convexity.

Linear and Quadratic Approximations

A linear approximation at $i = i_0$ uses the tangent line:

$$P_i - P_{i_0} \approx P'_{i_0}(i - i_0).$$

This approximation matches the original price curve at $i = i_0$ in two ways:

- it has the same value P_{i_0};
- it has the same first derivative at $i = i_0$.

A quadratic approximation adds a second-order term:

$$P_i - P_{i_0} \approx P'_{i_0}(i - i_0) + \frac{1}{2}P''_{i_0}(i - i_0)^2.$$

This approximation matches the original price curve at $i = i_0$ in three ways:

- it has the same value;
- it has the same first derivative;
- it has the same second derivative.

The second derivative lets the approximation curve like the true price-yield curve, which usually gives a better estimate.

Let

$$\Delta i = i_1 - i_0.$$

Then the quadratic approximation gives

$$P_{i_1} \approx P_{i_0} + P'_{i_0}\Delta i + \frac{1}{2}P''_{i_0}(\Delta i)^2.$$

Therefore,

$$\%\Delta P = \frac{P_{i_1} - P_{i_0}}{P_{i_0}} \approx \frac{P'_{i_0}\Delta i + \frac{1}{2}P''_{i_0}(\Delta i)^2}{P_{i_0}}.$$

So,

$$\%\Delta P \approx \Delta i\left(\frac{P'_{i_0}}{P_{i_0}}\right) + \frac{(\Delta i)^2}{2}\left(\frac{P''_{i_0}}{P_{i_0}}\right).$$

Since

$$\text{ModD} = -\frac{P'_{i_0}}{P_{i_0}},$$

we can rewrite the approximation as

$$\%\Delta P \approx -\Delta i \cdot \text{ModD} + \frac{(\Delta i)^2}{2}\left(\frac{P''_{i_0}}{P_{i_0}}\right).$$

This new term is called convexity.

Convexity

The convexity of a set of cash flows is defined by

$$\boxed{\text{Convexity} = \frac{P''_{i_0}}{P_{i_0}}}$$

where P''_{i_0} is the second derivative of price with respect to the yield rate, evaluated at the original yield rate i_0.

Using convexity, the second-order approximation becomes

$$\boxed{\%\Delta P \approx -\Delta i \cdot \text{ModD} + \frac{(\Delta i)^2}{2} \cdot \text{Convexity}}$$

Equivalently, the dollar change in price can be estimated by

$$\boxed{\Delta P \approx -P_{i_0}\Delta i \cdot \text{ModD} + P_{i_0}\frac{(\Delta i)^2}{2} \cdot \text{Convexity}}$$

Key idea

Modified duration gives the linear part of the approximation.
Convexity gives the quadratic correction.
Since $(\Delta i)^2$ is always nonnegative, the convexity adjustment is usually positive for ordinary bonds.

Calculating Convexity

Let CF_t be the cash flow at time t, where t is measured in years.

Let i be the nominal annual yield rate convertible m-thly. Then the price is

$$P_i = \sum_t CF_t \left(1 + \frac{i}{m}\right)^{-mt}.$$

Let

$$A_t = CF_t \left(1 + \frac{i}{m}\right)^{-mt}$$

be the present value of the cash flow at time t.

Then

$$P_i = \sum_t A_t.$$

The first derivative is

$$\frac{dP_i}{di} = \sum_t -t \cdot CF_t \left(1 + \frac{i}{m}\right)^{-mt-1}.$$

The second derivative is

$$\frac{d^2P_i}{di^2} = \sum_t t\left(t + \frac{1}{m}\right) CF_t \left(1 + \frac{i}{m}\right)^{-mt-2}.$$

Now let

$$v = \left(1 + \frac{i}{m}\right)^{-1}$$

be the discount factor for one interest conversion period.

Since

$$A_t = CF_t \left(1 + \frac{i}{m}\right)^{-mt},$$

we get

$$\frac{d^2 P_i}{di^2} = \sum_t t\left(t + \frac{1}{m}\right) A_t v^2.$$

Therefore, convexity can be calculated using the standard weighted-sum formula summarized below.

where

$$A_t = \text{present value of the cash flow at time } t,$$

$$t = \text{time in years},$$

and

$$v = \left(1 + \frac{i}{m}\right)^{-1}.$$

Convexity formula

If i is a nominal annual yield rate convertible m-thly, then

$$\boxed{\text{Convexity} = \frac{\sum_t t\left(t + \frac{1}{m}\right) A_t v^2}{\sum_t A_t}}$$

where t is measured in years and v is the per-period discount factor.

Special case

If the yield rate is annual effective, then $m = 1$. Therefore,

$$\text{Convexity} = \frac{\sum_t t(t+1) A_t v^2}{\sum_t A_t}.$$

Exercise 5.4.1

A three-year bond with 8% annual coupons has a redemption value of 105% of the par value and is priced to yield an annual effective rate of 6%.
Calculate the convexity.
Solution.
The par value does not matter because it cancels in the convexity calculation. Let the par value be

100.

Then the annual coupon is

$$100(0.08) = 8.$$

The redemption value is

$$100(1.05) = 105.$$

Thus, the cash flows are

$$8 \quad \text{at time } 1,$$

$$8 \quad \text{at time } 2,$$

and

$$8 + 105 = 113 \quad \text{at time } 3.$$

The yield rate is annual effective, so

$$i = 0.06.$$

Since the yield is annual effective, we use $m = 1$. Therefore, the convexity formula is

$$\text{Convexity} = \frac{\sum_t t(t+1)A_t v^2}{\sum_t A_t}.$$

Here,

$$v = \frac{1}{1.06}.$$

The denominator is the bond price:

$$\sum_t A_t = 8(1.06)^{-1} + 8(1.06)^{-2} + 113(1.06)^{-3}.$$

Thus,

$$\sum_t A_t = 109.54.$$

The numerator is

$$\sum_t t(t+1)A_t v^2.$$

So,

$$\sum_t t(t+1)A_t v^2 = 1(2)8(1.06)^{-3} + 2(3)8(1.06)^{-4} + 3(4)113(1.06)^{-5}.$$

Therefore,

$$\sum_t t(t+1)A_t v^2 = 1064.75.$$

Hence,

$$\text{Convexity} = \frac{1064.75}{109.54}.$$

So,

$$\text{Convexity} = 9.72.$$

The convexity is

$$\boxed{9.72}.$$

Exercise 5.4.2

Which of the following are true?

1. The duration of the portfolio is the weighted average of the duration of each asset in the portfolio, weighted by the price of each asset.
2. Using the modified duration to estimate the change in price for an asset always underestimates the price.
3. The convexity of a fixed cash flow asset is always negative.

A. 1 and 2 only   B. 1 and 3 only

C. 2 and 3 only   D. 1, 2 and 3

E. The correct answer is not given by A, B, C, or D

Solution.
Statement 1 is true.
The duration of a portfolio is the price-weighted average of the durations of the assets in the portfolio:

$$\text{Duration}_P = \frac{\sum_k P_k \text{Duration}_k}{\sum_k P_k}.$$

Therefore, the weights are based on the price of each asset.
Statement 2 is true.
The modified duration approximation uses a linear approximation to the price-yield curve:

$$\%\Delta P \approx -\Delta i \cdot \text{ModD}.$$

Because the price-yield curve is convex, this linear approximation underestimates the actual new price.
Statement 3 is false.
For a fixed cash flow asset with positive cash flows, convexity is positive. The convexity formula is

$$\text{Convexity} = \frac{\sum_t t\left(t + \frac{1}{m}\right) A_t v^2}{\sum_t A_t}.$$

If all cash flows are positive, then all present values A_t are positive. Therefore, the convexity is positive, not negative.
Thus, only statements 1 and 2 are true.
The correct answer is

$$\boxed{\text{A. 1 and 2 only}}.$$

Exercise 5.4.3

Calculate the convexity of a 4-year annuity-immediate with payments of 100, 100, 200, and 300. The annual effective yield rate is 6%.

A. 11.320   B. 12.320

C. 13.320   D. 14.320

E. 15.320

Solution.

The payments are made at the end of each year, so the cash flows are

$$100, \quad 100, \quad 200, \quad 300$$

at times

$$1, \quad 2, \quad 3, \quad 4.$$

The annual effective yield rate is

$$y = 0.06.$$

The price is the present value of the cash flows:

$$P = 100v + 100v^2 + 200v^3 + 300v^4,$$

where

$$v = \frac{1}{1.06}.$$

Equivalently,

$$P = 100(1+y)^{-1} + 100(1+y)^{-2} + 200(1+y)^{-3} + 300(1+y)^{-4}.$$

At $y = 0.06$,

$$P = 588.89.$$

Convexity is defined by

$$\text{Convexity} = \frac{P''}{P}.$$

Differentiate the price function:

$$P' = -100(1+y)^{-2} - 200(1+y)^{-3} - 600(1+y)^{-4} - 1200(1+y)^{-5}.$$

Differentiate again:

$$P'' = 200(1+y)^{-3} + 600(1+y)^{-4} + 2400(1+y)^{-5} + 6000(1+y)^{-6}.$$

At $y = 0.06$,

$$P'' = 6666.36.$$

Therefore,

$$\text{Convexity} = \frac{6666.36}{588.89}.$$

Thus,

$$\text{Convexity} \approx 11.320.$$

The correct answer is

$$\boxed{\text{A. } 11.320}.$$

5.4.2 Macaulay Convexity and Portfolios

We have seen that modified duration and convexity naturally occur together.

Macaulay duration also has a partner: Macaulay convexity.

Recall that when interest is compounded continuously, modified duration equals Macaulay duration. This gives the alternate expression

$$\mathrm{MacD} = -\frac{\frac{d}{d\delta}P_\delta}{P_\delta},$$

where δ is the force of interest.

By analogy, we define Macaulay convexity by

$$\boxed{\mathrm{MacC} = \frac{\frac{d^2}{d\delta^2}P_\delta}{P_\delta}}$$

This leads to the useful formula

$$\boxed{\mathrm{MacC} = \frac{\sum_t t^2 A_t}{\sum_t A_t}}$$

where t is measured in years and A_t is the present value of the cash flow at time t.

Macaulay convexity

Macaulay convexity is a present-value-weighted average of t^2.
Compare this with Macaulay duration:

$$\mathrm{MacD} = \frac{\sum_t tA_t}{\sum_t A_t}.$$

Why the formula works

Suppose the price is written using the force of interest δ:

$$P_\delta = \sum_t CF_t e^{-\delta t}.$$

Let

$$A_t = CF_t e^{-\delta t}.$$

Then

$$P_\delta = \sum_t A_t.$$

Differentiate once:

$$\frac{d}{d\delta}P_\delta = \sum_t -tCF_t e^{-\delta t}.$$

Differentiate again:

$$\frac{d^2}{d\delta^2}P_\delta = \sum_t t^2 CF_t e^{-\delta t}.$$

Since

$$A_t = CF_t e^{-\delta t},$$

we get

$$\frac{d^2}{d\delta^2}P_\delta = \sum_t t^2 A_t.$$

Therefore,

$$\text{MacC} = \frac{\frac{d^2}{d\delta^2} P_\delta}{P_\delta} = \frac{\sum_t t^2 A_t}{\sum_t A_t}.$$

Convexity versus Macaulay convexity

Convexity cannot be converted from Macaulay convexity as easily as modified duration can be converted from Macaulay duration.
So we usually refer to ordinary convexity simply as "convexity," rather than "modified convexity."

Macaulay Convexity Example

F

ind the Macaulay convexity of a 3-year increasing annuity with annual payments.
The first payment of 1000 is due in six months, and each subsequent payment increases by 1000.
The effective annual interest rate is 4%.

Solution.

The cash flows are

$$1000, \quad 2000, \quad 3000$$

at times

$$0.5, \quad 1.5, \quad 2.5.$$

The effective annual interest rate is

$$i = 0.04.$$

The present-value denominator is

$$\sum_t A_t = 1000(1.04)^{-0.5} + 2000(1.04)^{-1.5} + 3000(1.04)^{-2.5}.$$

Thus,

$$\sum_t A_t = 5586.12.$$

The numerator for Macaulay convexity is

$$\sum_t t^2 A_t.$$

Therefore,

$$\sum_t t^2 A_t = 0.5^2(1000)(1.04)^{-0.5} + 1.5^2(2000)(1.04)^{-1.5} + 2.5^2(3000)(1.04)^{-2.5}.$$

Thus,

$$\sum_t t^2 A_t = 21486.83.$$

Hence,

$$\text{MacC} = \frac{21486.83}{5586.12}.$$

So,

$$\text{MacC} = 3.846.$$

Therefore, the Macaulay convexity is

$$\boxed{3.846}.$$

Convexity of a Portfolio

We use the same methods for portfolio convexity as we use for portfolio duration.

There are two methods.

1. Determine the total cash flow of the entire portfolio at each time t, and then use the convexity or Macaulay convexity summation formula.
2. Compute the price-weighted average of the convexity or Macaulay convexity values of the portfolio components.

If a portfolio contains assets with prices $P_1, P_2, \ldots, P_n$ and convexities $\text{Conv}_1, \text{Conv}_2, \ldots, \text{Conv}_n$,

then

$$\boxed{\mathrm{Conv}_P = \frac{\sum_k P_k \mathrm{Conv}_k}{\sum_k P_k}}$$

Similarly,

$$\boxed{\mathrm{MacC}_P = \frac{\sum_k P_k \mathrm{MacC}_k}{\sum_k P_k}}$$

Portfolio convexity

Portfolio convexity is a price-weighted average of the convexities of the individual assets. The weights are based on price, not par value.

Portfolio Convexity Example

A

portfolio contains three bonds, all of which have par value 1000.
The first bond has a two-year term and no coupons. The second bond is a three-year bond with no coupons. The third bond is a two-year 6% bond with semiannual coupons.
If each bond is purchased to yield 5% convertible semiannually, find the convexity of the portfolio.
Solution.
The nominal annual yield rate is 5% convertible semiannually, so the effective yield rate per half-year is

$$j = \frac{0.05}{2} = 0.025.$$

Thus,

$$v = \frac{1}{1.025}.$$

The bond prices are

$$P_I = 1000v^4 = 905.95,$$

$$P_{II} = 1000v^6 = 862.30,$$

and

$$P_{III} = 30v + 30v^2 + 30v^3 + 1030v^4 = 1018.81.$$

For Bond I, the only cash flow occurs at time 2. Since the yield is convertible semiannually, $m = 2$. Thus,

$$\mathrm{Conv}(I) = 2\left(2 + \frac{1}{2}\right)v^2.$$

So,

$$\mathrm{Conv}(I) = 4.759.$$

For Bond II, the only cash flow occurs at time 3. Therefore,

$$\mathrm{Conv}(II) = 3\left(3 + \frac{1}{2}\right)v^2.$$

So,

$$\mathrm{Conv}(II) = 9.994.$$

For Bond III, the cash flows are

$$30, \quad 30, \quad 30, \quad 1030$$

at times

$$0.5, \quad 1, \quad 1.5, \quad 2.$$

Thus,

$$\text{Conv}(III) = \frac{0.5(1)30v^3 + 1(1.5)30v^4 + 1.5(2)30v^5 + 2(2.5)1030v^6}{1018.81}.$$

Therefore,

$$\text{Conv}(III) = 4.491.$$

Now use the price-weighted average:

$$\text{Conv}_P = \frac{4.759P_I + 9.994P_{II} + 4.491P_{III}}{P_I + P_{II} + P_{III}}.$$

Substitute the prices:

$$\text{Conv}_P = \frac{4.759(905.95) + 9.994(862.30) + 4.491(1018.81)}{905.95 + 862.30 + 1018.81}.$$

Thus,

$$\text{Conv}_P = 6.281.$$

Therefore, the convexity of the portfolio is

$$\boxed{6.281}.$$

Exercise 5.4.4

A three-year bond with 8% annual coupons has a redemption value of 105% of the par value and is priced to yield an annual effective rate of 6%.
Calculate the Macaulay convexity.
Solution.
The par value does not matter because it cancels in the Macaulay convexity calculation. Let the par value be

$$100.$$

The annual coupon is

$$100(0.08) = 8.$$

The redemption value is

$$100(1.05) = 105.$$

Thus, the cash flows are

$$8 \quad \text{at time } 1,$$

$$8 \quad \text{at time } 2,$$

and

$$8 + 105 = 113 \quad \text{at time } 3.$$

The annual effective yield rate is

$$i = 0.06.$$

Therefore,

$$v = \frac{1}{1.06}.$$

Macaulay convexity is

$$\text{MacC} = \frac{\sum_t t^2 A_t}{\sum_t A_t},$$

where A_t is the present value of the cash flow at time t.
The denominator is

$$\sum_t A_t = 8v + 8v^2 + 113v^3.$$

Thus,

$$\sum_t A_t = 109.54.$$

The numerator is

$$\sum_t t^2 A_t = 1^2(8v) + 2^2(8v^2) + 3^2(113v^3).$$

So,

$$\sum_t t^2 A_t = 889.92.$$

Therefore,

$$\text{MacC} = \frac{889.92}{109.54}.$$

Hence,

$$\text{MacC} = 8.124.$$

The Macaulay convexity is

$$\boxed{8.124}.$$

Exercise 5.4.5

We construct a portfolio of zero-coupon bonds:

Term (years)	1	2	3
Price of 100 par value zero-coupon bond	94.12	88.17	82.55

The portfolio has 1000 par value of 1-year bonds, 2500 par value of 2-year bonds, and 2000 par value of 3-year bonds.
Find the Macaulay convexity of the portfolio.
Solution.
We use the price-weighted average of the Macaulay convexities.
Since the quoted prices are for 100 par value zero-coupon bonds, the prices of the portfolio positions are:

$$P_I = 10(94.12) = 941.20,$$

$$P_{II} = 25(88.17) = 2204.25,$$

and

$$P_{III} = 20(82.55) = 1651.00.$$

For a zero-coupon bond, there is only one cash flow. Therefore, the Macaulay convexity is simply

$$\text{MacC} = t^2.$$

Thus,

$$\text{MacC}(I) = 1^2 = 1,$$

$$\text{MacC}(II) = 2^2 = 4,$$

and

$$\text{MacC}(III) = 3^2 = 9.$$

Now compute the portfolio Macaulay convexity using the price-weighted average:

$$\text{MacC}_P = \frac{P_I \text{MacC}(I) + P_{II} \text{MacC}(II) + P_{III} \text{MacC}(III)}{P_I + P_{II} + P_{III}}.$$

Substitute the values:

$$\text{MacC}_P = \frac{1P_I + 4P_{II} + 9P_{III}}{P_I + P_{II} + P_{III}}.$$

Therefore,

$$\text{MacC}_P = \frac{1(941.20) + 4(2204.25) + 9(1651.00)}{941.20 + 2204.25 + 1651.00}.$$

So,

$$\text{MacC}_P = \frac{24617.20}{4796.45}.$$

Hence,

$$\text{MacC}_P = 5.132.$$

The Macaulay convexity of the portfolio is

$$\boxed{5.132}.$$

Exercise 5.4.6

You are given the following information about a bond:

1. The term-to-maturity is two years.
2. The bond has a 9% annual coupon rate, paid semiannually.
3. The annual yield to maturity is 8% convertible semiannually.
4. The par value is $100.

Calculate the Macaulay convexity of the bond.

A	Less than 4
B	At least 4, but less than 4.5
C	At least 4.5, but less than 5
D	At least 5, but less than 5.5
E	At least 5.5

Solution.
The coupon rate is 9% annually, paid semiannually. Therefore, each semiannual coupon is

$$\frac{0.09}{2}(100) = 4.5.$$

The annual yield rate is 8% convertible semiannually, so the semiannual yield rate is

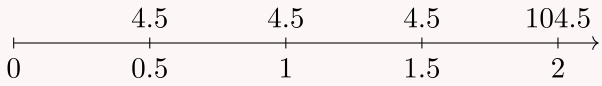
$$j = \frac{0.08}{2} = 0.04.$$

Since the bond matures in two years, there are four semiannual payment dates. The cash flows are:

$$4.5, \quad 4.5, \quad 4.5, \quad 104.5,$$

at times

$$0.5, \quad 1, \quad 1.5, \quad 2.$$



Let

$$v = \frac{1}{1.04}.$$

The price of the bond is

$$P = 4.5v + 4.5v^2 + 4.5v^3 + 104.5v^4.$$

Numerically,

$$P = 4.5(1.04)^{-1} + 4.5(1.04)^{-2} + 4.5(1.04)^{-3} + 104.5(1.04)^{-4}.$$

Thus,

$$P = 101.81495.$$

For Macaulay convexity, we use

$$\text{MacC} = \frac{\sum_t t^2 A_t}{\sum_t A_t},$$

where A_t is the present value of the cash flow paid at time t, and t is measured in years. Therefore,

$$\text{MacC} = \frac{0.5^2(4.5v) + 1^2(4.5v^2) + 1.5^2(4.5v^3) + 2^2(104.5v^4)}{4.5v + 4.5v^2 + 4.5v^3 + 104.5v^4}.$$

Substituting $v = (1.04)^{-1}$,

$$\text{MacC} = \frac{0.5^2(4.5)(1.04)^{-1} + 1^2(4.5)(1.04)^{-2} + 1.5^2(4.5)(1.04)^{-3} + 2^2(104.5)(1.04)^{-4}}{101.81495}.$$

The numerator is

$$371.55147.$$

Hence,

$$\text{MacC} = \frac{371.55147}{101.81495} = 3.64928.$$

Therefore, the Macaulay convexity is less than 4.

$$\boxed{\text{A}}$$

5.5 Immunization

5.5.1 Redington Immunization

Immunization is a strategy used to protect a portfolio of assets against changes in interest rates. The goal is to choose asset cash flows so that they are sufficient to meet a given liability, even if interest rates change slightly.

Let P_A denote the present value of the assets, and let P_L denote the present value of the liabilities. The surplus is defined by

$$S = P_A - P_L.$$

Redington Immunization (conditions)

A portfolio satisfies the Redington immunization conditions at a given interest rate if:

$$\boxed{P_A = P_L}, \qquad \boxed{\text{MacD}_A = \text{MacD}_L}, \qquad \boxed{\text{MacC}_A > \text{MacC}_L}.$$

The conditions require (i) equal present values, (ii) equal Macaulay durations, and (iii) asset cash flows that are more convex than the liabilities.

Interpretation

Redington immunization protects against small changes in interest rates. It does not guarantee protection against large changes in interest rates or against nonparallel shifts in the yield curve.

Example: Redington Immunization with Zero-Coupon Bonds

A company has a single liability of \$10,000 due in 5 years. The company can buy 3-year and 7-year zero-coupon bonds to cover this liability. Assume that the annual effective yield rate is 5%.
The present value of the liability is

$$P_L = 10{,}000v^5,$$

where

$$v = \frac{1}{1.05}.$$

Thus,

$$P_L = 10{,}000(1.05)^{-5} = 7{,}835.26.$$

Since the liability is a single payment due in 5 years, its Macaulay duration is

$$\text{MacD}_L = 5.$$

Let P_3 be the amount invested in the 3-year zero-coupon bond, and let P_7 be the amount invested in the 7-year zero-coupon bond.
The present value condition is

$$P_3 + P_7 = 7{,}835.26.$$

The duration condition is

$$\frac{3P_3 + 7P_7}{P_3 + P_7} = 5.$$

Solving the duration equation,

$$3P_3 + 7P_7 = 5P_3 + 5P_7,$$

so

$$2P_7 = 2P_3.$$

Therefore,

$$P_3 = P_7.$$

Using the present value condition,

$$P_3 = P_7 = \frac{7{,}835.26}{2} = 3{,}917.63.$$

Now check the convexity condition. The Macaulay convexity of the liability is

$$\text{MacC}_L = 5^2 = 25.$$

The Macaulay convexity of the assets is

$$\text{MacC}_A = \frac{3^2 P_3 + 7^2 P_7}{P_3 + P_7}.$$

Since $P_3 = P_7$,

$$\text{MacC}_A = \frac{9P_3 + 49P_3}{2P_3} = 29.$$

Thus,

$$\text{MacC}_A = 29 > 25 = \text{MacC}_L\,.$$

Therefore, all three Redington immunization conditions are satisfied.

Exercise 5.5.1

True or False: Redington immunization protects a surplus against any changes in the interest rate.

A	True
B	False

Solution.
The statement is false.
Redington immunization protects a surplus only against small changes in the interest rate.
It is based on local conditions around the current interest rate.
The three Redington immunization conditions are:

$$P_A = P_L,$$

$$\text{MacD}_A = \text{MacD}_L,$$

and

$$\text{MacC}_A > \text{MacC}_L\,.$$

These conditions imply that the surplus

$$S(i) = P_A(i) - P_L(i)$$

has a local minimum at the current interest rate. Therefore, for small changes in the interest rate, the surplus should not decrease.
However, Redington immunization does not guarantee protection against large changes in interest rates. If the interest rate change is large, the surplus may still decline.
Thus, the correct answer is:

$$\boxed{\text{B. False}}$$

Exercise 5.5.2

Infinite Life buys a 5-year zero-coupon bond that will mature for \$1,000 to immunize two liability cash flows of X and Y. The liability X occurs in 3 years, and the liability Y occurs in 10 years.
Find X and Y such that Conditions 1 and 2 of Redington immunization are satisfied if the effective annual rate of interest is 6%.

Solution.
Let

$$v = \frac{1}{1.06}.$$

The asset is a 5-year zero-coupon bond that matures for \$1,000. Therefore, its present value is

$$P_A = 1000v^5.$$

The two liability cash flows are X at time 3 and Y at time 10. Therefore, the present value of the liabilities is

$$P_L = Xv^3 + Yv^{10}.$$

Condition 1 of Redington immunization requires

$$P_A = P_L.$$

Thus,

$$1000v^5 = Xv^3 + Yv^{10}.$$

Condition 2 requires the Macaulay duration of the assets to equal the Macaulay duration of the liabilities:

$$\text{MacD}_A = \text{MacD}_L\,.$$

Since the asset is a 5-year zero-coupon bond,

$$\text{MacD}_A = 5.$$

The Macaulay duration of the liabilities is

$$\text{MacD}_L = \frac{3Xv^3 + 10Yv^{10}}{Xv^3 + Yv^{10}}.$$

Therefore,

$$5 = \frac{3Xv^3 + 10Yv^{10}}{Xv^3 + Yv^{10}}.$$

Multiplying both sides by $Xv^3 + Yv^{10}$,

$$5(Xv^3 + Yv^{10}) = 3Xv^3 + 10Yv^{10}.$$

Expanding and simplifying,

$$5Xv^3 + 5Yv^{10} = 3Xv^3 + 10Yv^{10},$$

so
$$2Xv^3 = 5Yv^{10}.$$
Hence,
$$Xv^3 = \frac{5}{2}Yv^{10}.$$
Now substitute this into Condition 1:
$$1000v^5 = Xv^3 + Yv^{10}.$$
Since $Xv^3 = \frac{5}{2}Yv^{10}$, we get
$$1000v^5 = \frac{5}{2}Yv^{10} + Yv^{10}.$$
Thus,
$$1000v^5 = \frac{7}{2}Yv^{10}.$$
Solving for Y,
$$Y = \frac{2}{7} \cdot 1000 \cdot v^{-5}.$$
Since $v^{-5} = (1.06)^5$,
$$Y = \frac{2000}{7}(1.06)^5.$$
Therefore,
$$Y = 382.35.$$
Now use
$$2Xv^3 = 5Yv^{10}.$$
Dividing by $2v^3$,
$$X = \frac{5}{2}Yv^7.$$
Substituting $Y = 382.35$,
$$X = \frac{5}{2}(382.35)(1.06)^{-7}.$$
Therefore,
$$X = 635.71.$$
Thus, the liabilities must be
$$\boxed{X = 635.71}$$
and
$$\boxed{Y = 382.35}.$$

Exercise 5.5.3

Infinite Life buys a 5-year zero-coupon bond that will mature for \$1,000 to immunize two liability cash flows of X and Y. The liability X occurs in 3 years and the liability Y occurs in 10 years.
Find X and Y such that Conditions 1 and 2 of Redington immunization are satisfied if the effective annual rate of interest is 6%.
If Infinite Life buys the bond and guarantees the payments of X and Y, has it achieved Redington immunization?
Solution.

Let
$$v = \frac{1}{1.06}.$$
The asset is a 5-year zero-coupon bond that matures for \$1,000. Therefore,
$$P_A = 1000v^5.$$
The liabilities are X due at time 3 and Y due at time 10. Therefore,
$$P_L = Xv^3 + Yv^{10}.$$
Condition 1 of Redington immunization requires
$$P_A = P_L.$$
Thus,
$$1000v^5 = Xv^3 + Yv^{10}.$$
Condition 2 requires
$$\text{MacD}_A = \text{MacD}_L\,.$$
Since the asset is a 5-year zero-coupon bond,
$$\text{MacD}_A = 5.$$
The Macaulay duration of the liabilities is
$$\text{MacD}_L = \frac{3Xv^3 + 10Yv^{10}}{Xv^3 + Yv^{10}}.$$
Therefore,
$$5 = \frac{3Xv^3 + 10Yv^{10}}{Xv^3 + Yv^{10}}.$$
Multiplying both sides by $Xv^3 + Yv^{10}$,
$$5(Xv^3 + Yv^{10}) = 3Xv^3 + 10Yv^{10}.$$
Simplifying,
$$2Xv^3 = 5Yv^{10}.$$
Hence,
$$Xv^3 = \frac{5}{2}Yv^{10}.$$
Substitute this into Condition 1:
$$1000v^5 = \frac{5}{2}Yv^{10} + Yv^{10}.$$
Thus,
$$1000v^5 = \frac{7}{2}Yv^{10}.$$
Solving for Y,
$$Y = \frac{2}{7}(1000)v^{-5}.$$
Since $v^{-5} = (1.06)^5$,
$$Y = \frac{2000}{7}(1.06)^5 = 382.35.$$

Now use

$$2Xv^3 = 5Yv^{10}.$$

Dividing by $2v^3$,

$$X = \frac{5}{2}Yv^7.$$

Therefore,

$$X = \frac{5}{2}(382.35)(1.06)^{-7} = 635.71.$$

So the liability cash flows are

$$\boxed{X = 635.71}$$

and

$$\boxed{Y = 382.35}.$$

Now check Condition 3 of Redington immunization.
Condition 3 requires

$$\text{MacC}_A > \text{MacC}_L\,.$$

Since the asset is a 5-year zero-coupon bond,

$$\text{MacC}_A = 5^2 = 25.$$

The Macaulay convexity of the liabilities is

$$\text{MacC}_L = \frac{3^2Xv^3 + 10^2Yv^{10}}{Xv^3 + Yv^{10}}.$$

Substituting $X = 635.71$, $Y = 382.35$, and $v = (1.06)^{-1}$,

$$\text{MacC}_L = \frac{9(635.71)v^3 + 100(382.35)v^{10}}{635.71v^3 + 382.35v^{10}}.$$

This gives

$$\text{MacC}_L = 35.$$

Therefore,

$$\text{MacC}_A = 25$$

and

$$\text{MacC}_L = 35.$$

Since

$$25 \not> 35,$$

Condition 3 is not satisfied.
Thus, Infinite Life has not achieved Redington immunization.

$$\boxed{\text{No, Redington immunization has not been achieved.}}$$

Exercise 5.5.4

Regarding Redington immunization, which of the following is not true?

A	Assumes a flat yield curve.
B	Assumes only parallel shifts in the yield curve.
C	Protects the surplus from small shifts in the yield curve.
D	The third condition is: $P''_A = P''_L$.
E	All of the above are true.

Solution.
Redington immunization is a local immunization strategy. It assumes a flat yield curve and protects the surplus against small parallel shifts in the yield curve.
Let
$$S(i) = P_A(i) - P_L(i),$$
where $P_A(i)$ is the present value of the assets and $P_L(i)$ is the present value of the liabilities. The Redington immunization conditions are:
$$P_A = P_L,$$
$$P'_A = P'_L,$$
and
$$P''_A > P''_L.$$
Equivalently, the surplus satisfies:
$$S(i) = 0,$$
$$S'(i) = 0,$$
and
$$S''(i) > 0.$$
The third condition is not
$$P''_A = P''_L.$$
Instead, it is
$$P''_A > P''_L.$$
Therefore, the statement that the third condition is $P''_A = P''_L$ is not true.

$$\boxed{\text{D}}$$

Exercise 5.5.5

A company has liabilities of \$573 due at the end of year 2 and \$701 due at the end of year 5.
A portfolio comprises two zero-coupon bonds, Bond A and Bond B.
Determine which portfolio produces a Redington immunization of the liabilities using an annual effective interest rate of 7.0%.

A	Bond A: 1-year, current price \$500; Bond B: 6-years, current price \$500
B	Bond A: 1-year, current price \$572; Bond B: 6-years, current price \$428
C	Bond A: 3-years, current price \$182; Bond B: 4-years, current price \$1092
D	Bond A: 3-years, current price \$637; Bond B: 4-years, current price \$637
E	Bond A: 3.5-years, current price \$1000; Bond B: not used

Solution.

Let

$$v = \frac{1}{1.07}.$$

The present value of the liabilities is

$$P_L = 573v^2 + 701v^5.$$

Numerically,

$$P_L = 573(1.07)^{-2} + 701(1.07)^{-5} \approx 1000.$$

Condition 1 of Redington immunization requires

$$P_A = P_L.$$

Since the current price of each bond is its present value, the total current price of the asset portfolio must be approximately \$1000. Choices C and D both have total current price

$$182 + 1092 = 1274$$

and

$$637 + 637 = 1274,$$

so they are eliminated.

Now compute the Macaulay duration of the liabilities:

$$\text{MacD}_L = \frac{2(573v^2) + 5(701v^5)}{573v^2 + 701v^5}.$$

Using $v = (1.07)^{-1}$,

$$\text{MacD}_L \approx 3.5.$$

For a zero-coupon bond, the Macaulay duration is simply the maturity of the bond. Also, the Macaulay duration of a portfolio is the price-weighted average of the Macaulay durations of its components.

For choice A,

$$\text{MacD}_A = \frac{500}{1000}(1) + \frac{500}{1000}(6) = 3.5.$$

So choice A satisfies Condition 2.

For choice B,

$$\text{MacD}_A = \frac{572}{1000}(1) + \frac{428}{1000}(6) = 3.14.$$

So choice B does not satisfy Condition 2.

Choice E has a single 3.5-year zero-coupon bond, so

$$\text{MacD}_A = 3.5.$$

Thus, after the first two conditions, choices A and E remain.

Now check Condition 3:

$$\text{MacC}_A > \text{MacC}_L\,.$$

The Macaulay convexity of the liabilities is

$$\text{MacC}_L = \frac{2^2(573v^2) + 5^2(701v^5)}{573v^2 + 701v^5}.$$

Numerically,

$$\text{MacC}_L \approx 14.5.$$

For a zero-coupon bond, the Macaulay convexity is the square of its maturity. For a portfolio, Macaulay convexity is the price-weighted average of the Macaulay convexities of its components.
For choice A,

$$\text{MacC}_A = \frac{500}{1000}(1^2) + \frac{500}{1000}(6^2).$$

Thus,

$$\text{MacC}_A = \frac{1}{2}(1) + \frac{1}{2}(36) = 18.5.$$

Since

$$18.5 > 14.5,$$

choice A satisfies Condition 3.
For choice E,

$$\text{MacC}_A = 3.5^2 = 12.25.$$

Since

$$12.25 < 14.5,$$

choice E does not satisfy Condition 3.
Therefore, the portfolio that produces a Redington immunization is

$$\boxed{\text{A}}$$

Exercise 5.5.6

Which of the following are ways to state the three conditions of Redington immunization?

1.
$$\begin{aligned} &\text{A. } PV(\text{Assets}) = PV(\text{Liabilities}) \\ &\text{B. } \text{Duration}(\text{Assets}) = \text{Duration}(\text{Liabilities}) \\ &\text{C. } \text{Convexity}(\text{Assets}) > \text{Convexity}(\text{Liabilities}) \end{aligned}$$

2.
$$\begin{aligned} &\text{A. } P_A(y) = P_L(y) \\ &\text{B. } P_A'(y) = P_L'(y) \\ &\text{C. } P_A''(y) > P_L''(y) \end{aligned}$$

3.
$$\begin{aligned} &\text{A. } S(y) = 0 \\ &\text{B. } S'(y) = 0 \\ &\text{C. } S''(y) > 0 \end{aligned}$$

A	1 and 2 only
B	1 and 3 only
C	2 and 3 only
D	1, 2 and 3
E	The correct answer is not given by A, B, C, or D

Solution.
The classic Redington immunization conditions are:

$$PV(\text{Assets}) = PV(\text{Liabilities}),$$

$$\text{Duration(Assets)} = \text{Duration(Liabilities)},$$

and

$$\text{Convexity(Assets)} > \text{Convexity(Liabilities)}.$$

Therefore, statement 1 is a valid way to state the three conditions.
Now let $P_A(y)$ be the price of the assets as a function of the yield rate y, and let $P_L(y)$ be the price of the liabilities. Redington immunization can also be stated using derivatives:

$$P_A(y) = P_L(y),$$

$$P'_A(y) = P'_L(y),$$

and

$$P''_A(y) > P''_L(y).$$

The first condition matches present values. The second condition matches the first-order sensitivity to changes in the yield rate, which corresponds to matching durations. The third condition requires the asset price function to have greater curvature than the liability price function.
Therefore, statement 2 is also valid.
Finally, define the surplus function by

$$S(y) = P_A(y) - P_L(y).$$

Redington immunization requires the surplus to have a local minimum at the current yield rate. This means:

$$S(y) = 0,$$

$$S'(y) = 0,$$

and

$$S''(y) > 0.$$

Thus, statement 3 is also valid.
Therefore, all three statements are valid ways to state the Redington immunization conditions.

$\boxed{\text{D}}$

5.5.2 Full Immunization and Dedication

Redington immunization protects the surplus against small parallel changes in the yield curve. However, it has some important limitations.

Weaknesses of Redington Immunization

Redington immunization relies on several simplifying assumptions:

1. The yield curve is assumed to be flat.
2. Interest rate changes are assumed to be parallel shifts of the yield curve.
3. Liability cash flows may not be completely known in advance.
4. Asset cash flows with the required maturities may not be available.

Therefore, Redington immunization is useful, but it does not always provide perfect protection in practice.

Dedication

Dedication is immunization by exact matching.
Each asset cash flow is chosen so that it exactly matches a liability cash flow at the same time.

If an asset cash flow exactly matches a liability cash flow at the same time, then any change in interest rates affects both cash flows in the same way. Therefore, their present values change by the same amount.

In this case, the liability is protected against any interest rate change, not only small parallel shifts.

Dedication Principle

If the asset cash flows exactly match the liability cash flows at every time, then

$$CF_{A,t} = CF_{L,t}$$

for each liability payment time t.
Therefore,

$$P_A(i) = P_L(i)$$

for any interest rate i.

Practical Limitations

Dedication is a strong form of immunization, but it can be difficult to implement in practice:

1. The timing or size of liability cash flows may be uncertain.
2. Assets that exactly match the liability cash flows may not be available.
3. The investor may have to accept lower returns to obtain exact matching.
4. Asset cash flows may not be perfectly predictable.

Dedication Example

A company has a liability stream over four years. Suppose the liability cash flows are

$$100, \quad 150, \quad 200, \quad 250$$

at times

$$1, \quad 2, \quad 3, \quad 4.$$

The company can buy annual coupon bonds. To dedicate the assets to the liabilities, the company starts from the last liability and works backward.
Suppose a 4-year bond has annual coupons and matures at time 4. Its cash flows can be used to help cover the liabilities at times $1, 2, 3$, and 4.
After choosing enough 4-year bonds to cover the liability at time 4, the remaining earlier liabilities are reduced by the coupons received from the 4-year bonds.

Then the company chooses 3-year bonds to cover the remaining liability at time 3, then 2-year bonds to cover the remaining liability at time 2, and finally 1-year bonds to cover the remaining liability at time 1.
This backward process constructs an asset portfolio whose cash flows exactly match the liability cash flows.
Therefore, the liabilities are fully immunized by dedication.

Exercise 5.5.7

A life insurance company has written a policy that pays \$100,000 one year after the death of the policyholder.
What makes this liability difficult to immunize?
Solution.
The liability is difficult to immunize because the timing of the liability is uncertain.
The insurance company knows the amount that will be paid:

$$100{,}000.$$

However, it does not know when the policyholder will die. Since the payment is made one year after death, the payment date is also uncertain.
This creates a problem for exact matching. In a dedication strategy, asset cash flows are chosen so that they exactly match liability cash flows at the same times. But here, the company does not know the exact time at which the liability cash flow will occur.
Therefore, the company does not know when the asset cash flow should arrive.
The uncertain timing also makes it difficult to satisfy the duration condition in Redington or full immunization. The duration of the liability depends on the timing of the payment, but that timing is unknown.
There is another practical issue. The liability could occur far in the future. If the policyholder lives for many decades, then the payment may not be due for a very long time. It may be difficult to find assets with sufficiently long maturities to match this liability.
Thus, this liability is difficult to immunize because:

1. the timing of the liability is uncertain;
2. exact matching is difficult because the payment date is unknown;
3. the duration of the liability is uncertain;
4. assets with very long maturities may not be available.

$$\boxed{\text{The timing of the liability is uncertain.}}$$

Exercise 5.5.8

A company must pay L one year from now and $2L$ two years from now.
The company buys two bonds to match the liability cash flows. The first bond is a \$1,000 one-year bond. The second bond is a \$2,500 two-year bond. Both bonds pay annual coupons at a rate r.
Calculate L.
Solution.
The company wants to exactly match the liability cash flows.

The liabilities are:

$$L \quad \text{at time } 1,$$

and

$$2L \quad \text{at time } 2.$$

Now consider the asset cash flows.
The one-year bond has face value \$1,000, so it pays principal plus coupon at time 1:

$$1000 + 1000r.$$

The two-year bond has face value \$2,500, so it pays a coupon at time 1:

$$2500r,$$

and principal plus coupon at time 2:

$$2500 + 2500r.$$

Therefore, at time 2, exact matching gives

$$2L = 2500 + 2500r.$$

At time 1, exact matching gives

$$L = (1000 + 1000r) + 2500r.$$

Thus,

$$L = 1000 + 3500r.$$

Now substitute this expression for L into the time 2 equation:

$$2(1000 + 3500r) = 2500 + 2500r.$$

Expanding,

$$2000 + 7000r = 2500 + 2500r.$$

Subtracting $2500r$ and 2000 from both sides,

$$4500r = 500.$$

Hence,

$$r = \frac{500}{4500} = \frac{1}{9}.$$

Now substitute $r = \frac{1}{9}$ into

$$L = 1000 + 3500r.$$

We get

$$L = 1000 + 3500\left(\frac{1}{9}\right).$$

Therefore,

$$L = 1388.89.$$

$$\boxed{L = 1388.89}$$

Dedication versus Redington Immunization

Dedication does not necessarily satisfy the strict third Redington condition

$$\mathrm{MacC}_A > \mathrm{MacC}_L\,.$$

Under exact cash flow matching, the asset and liability cash flows are identical at every time, so

$$P_A(i) = P_L(i)$$

for every interest rate i. Therefore,

$$\mathrm{MacD}_A = \mathrm{MacD}_L$$

and

$$\mathrm{MacC}_A = \mathrm{MacC}_L\,.$$

The strict convexity condition is not needed because the surplus is identically zero for all interest rates.

Exercise 5.5.9

Which of the following are true?

1. Full immunization protects the surplus for interest rate changes of any size.
2. Full immunization does not assume a flat yield curve like Redington immunization does.
3. If full immunization is achieved, then Redington immunization is achieved.

A	1 and 2 only
B	1 and 3 only
C	2 and 3 only
D	1, 2 and 3
E	The correct answer is not given by A, B, C, or D

Solution.
Statement 1 is true.
Full immunization is designed to protect the surplus against interest rate changes of any size, not only small changes. This is stronger than Redington immunization, which protects only locally against small changes in the interest rate.
However, this statement should be understood under the usual immunization framework, where interest rate changes are modeled as changes in a single yield rate or as parallel shifts.
Statement 2 is false.
Full immunization still relies on a simplified interest rate framework, such as a flat yield curve or parallel shifts in the yield curve. The method that avoids the need for a flat yield curve or parallel shifts is dedication, also called exact cash flow matching.
Under dedication, asset cash flows exactly match liability cash flows at each time. Therefore,

$$CF_{A,t} = CF_{L,t}$$

for every relevant time t, so the equality of present values holds under any set of discount factors.

Statement 3 is true.
Full immunization is stronger than Redington immunization. Redington immunization protects the surplus only for small changes in the interest rate, while full immunization protects the surplus for changes of any size under the standard interest rate model.
Therefore, if full immunization is achieved, then Redington immunization is also achieved.
Thus, statements 1 and 3 are true, while statement 2 is false.

$\boxed{\text{B}}$

Exercise 5.5.10

Which of the following is not a problem with dedication?

A	Requires rebalancing.
B	Asset cash flows are not always easy to predict.
C	Assets may not be available to exactly match all the liabilities.
D	Liability cash flows are often just estimates.
E	A higher yield might be available.

Solution.
Dedication is immunization by exact cash flow matching. The goal is to choose asset cash flows so that they exactly match the liability cash flows at the same times.
That is, dedication aims to satisfy

$$CF_{A,t} = CF_{L,t}$$

for each relevant time t.
If the asset and liability cash flows are known and exactly matched, then the strategy does not normally require continuous rebalancing. This is one of the advantages of dedication.
The other choices describe real practical problems with dedication.
Asset cash flows may not always be easy to predict. For example, some bonds may be callable, and loans may be prepaid.
Assets may also not be available with the exact maturities and cash flow amounts needed to match all liabilities.
Liability cash flows may also be uncertain or based on estimates, especially in insurance contexts.
Finally, dedication may force the investor to choose assets that match the liabilities exactly, even if another investment opportunity offers a higher yield.
Therefore, the statement that is not a problem with dedication is:

$\boxed{\text{A}}$

Exercise 5.5.11

A company's expected liabilities are as follows, assuming end-of-year payments:

Year	1	2	3
Payments	\$100	\$200	\$300

The company can buy the following bonds in any face amount:

Maturity	**Coupon Rate**
1 year	5%
2 years	6%
3 years	0%

Assume a flat yield curve with a yield-to-maturity of 6%. The company uses cash flow matching to determine how much of each bond to buy.
Calculate how much the company spends on 1-year bonds.

A	83.66
B	84.46
C	88.68
D	89.85
E	94.34

Solution.
Let F_1, F_2, and F_3 denote the face amounts of the 1-year, 2-year, and 3-year bonds, respectively.
Since the company is using cash flow matching, we work backward from the final liability. At time 3, the company must pay \$300. The 3-year bond is a zero-coupon bond, so it pays only its face amount at maturity. Therefore,

$$F_3 = 300.$$

At time 2, the company must pay \$200. The 2-year bond has a coupon rate of 6%, so it pays

$$1.06F_2$$

at time 2. Therefore,

$$1.06F_2 = 200.$$

Solving,

$$F_2 = \frac{200}{1.06} = 188.68.$$

At time 1, the company must pay \$100. The 1-year bond pays

$$1.05F_1$$

at time 1, and the 2-year bond pays a coupon of

$$0.06F_2$$

at time 1.
Thus,

$$1.05F_1 + 0.06F_2 = 100.$$

Substitute $F_2 = 188.68$:

$$1.05F_1 + 0.06(188.68) = 100.$$

Therefore,

$$1.05F_1 + 11.32 = 100.$$

So

$$1.05F_1 = 88.68,$$

and

$$F_1 = \frac{88.68}{1.05} = 84.46.$$

This is the face amount of the 1-year bond. However, the question asks how much the company spends on 1-year bonds.
The 1-year bond has yield-to-maturity 6%. Therefore, the price paid for the 1-year bond is the present value of its time 1 cash flow:

$$\text{Price of 1-year bond} = \frac{1.05F_1}{1.06}.$$

Substituting $F_1 = 84.46$,

$$\text{Price of 1-year bond} = \frac{1.05(84.46)}{1.06}.$$

Thus,

$$\text{Price of 1-year bond} = 83.66.$$

Therefore, the company spends

$$\boxed{83.66}$$

on 1-year bonds.